

**RANDOM DATA:
ANALYSIS AND
MEASUREMENT
PROCEDURES**

RANDOM DATA: ANALYSIS AND MEASUREMENT PROCEDURES

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PREFACE

This book is an extensive revision and replacement for the authors' early book, *Measurement and Analysis of Random Data*, 1966. Approximately 50 percent of the original material has been rewritten or deleted and replaced by new material. These changes reflect the technical advances that have taken place in the last five years as well as an increased awareness of pertinent matters gained through the further experience of the authors. Specifically, a broader discussion appears on statistical errors in random data analysis. An entirely new chapter has been introduced to integrate the general requirements for data acquisition, recording, preparation, qualification, and processing. The discussions of digital data analysis procedures have been greatly expanded to cover the more recent analysis techniques made feasible by the availability of fast Fourier transform algorithms. Discussions of transient and multidimensional random processes are now included. Finally, a number of illustrative examples involving actual physical data have been added to support theoretical developments. The illustrations are largely restricted to aerospace and automotive applications since these are the fields of most recent concern to the authors. The general techniques, however, are applicable to data common in many other fields including meteorology, oceanography, seismology, communications, nuclear processes, and biomedical research.

The emphasis in this new book is on the practical aspects of random data analysis and measurement procedures, with special attention to the interrelationships of the various technical disciplines involved. As before, the book is written with the primary intent of providing a convenient reference for practicing engineers and scientists. The secondary intent of providing a specialized textbook for students has been augmented by the addition of problem sets at the end of each chapter. The reader is assumed to have a basic knowledge of probability theory, statistics, and transform methods of applied mathematics.

Summaries of chapter contents appear at the beginning of each chapter. In brief, Chapters 1 through 4 present a review of basic theoretical background material needed for the developments in later chapters. Basic descriptive properties of random data are outlined in Chapter 1 while physical system response properties are reviewed in Chapter 2. Pertinent mathematical

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and statistical theory is summarized in Chapters 3 and 4. This review material is followed in Chapters 5 and 6 by extensive developments and formulations of input-output relationships and statistical errors in measured data. Chapter 7 outlines the overall procedures for random data acquisition and processing. Detailed procedures for analog and digital data analysis are presented in Chapters 8 and 9. The final Chapter 10 discusses some advanced ideas and procedures relevant to nonstationary, transient, and multidimensional data.

We wish to acknowledge the many contributions to this book by former associates in Measurement Analysis Corporation and Digitek Corporation. We also thank those government agencies, industrial companies, and individuals who supported our work. A special appreciation is given to Engineering Extension, University of California, Los Angeles, and to other organizations, who sponsored our presentation of short courses on this subject matter. Our final thanks extends to Teresa Piersol and Lucinda Bendat for their help in preparing the manuscript.

Los Angeles, California
July 1971

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GLOSSARY OF SYMBOLS

a, b	Sample Regression Coefficients, Arbitrary Constants
$b[\]$	Bias Error of []
B	Cyclical Frequency Bandwidth
c	Mechanical Damping Coefficient, Arbitrary Constant
C	Electrical Capacitance
C_{xy}	Covariance
$C_x(\tau)$	Autocovariance Function
$C_{xy}(\tau)$	Cross-Covariance Function
$C(t_1, t_2)$	Nonstationary Covariance Function
$C_{xy}(f)$	Coincident Spectral Density Function (One-Sided)
D	Time Displacement Range
$e(t)$	Potential Difference
$E[\]$	Expected Value of []
f	Cyclical Frequency
F	Cyclical Frequency Range, Statistical F Variable
$F(t)$	Mechanical Forcing Function
$G_x(f)$	Power Spectral Density Function Defined for Non-Negative Frequencies Only (One-Sided)
$G_{xy}(f)$	Cross-Spectral Density Function Defined for Non-Negative Frequencies Only (One-Sided)
h	Sampling Interval
$h(\tau)$	Weighting Function (Unit Impulse Response Function)
$H(f)$	Frequency Response Function
$ H(f) $	Gain Factor
$\text{Im}[\]$	Imaginary Part of []
j	$\sqrt{-1}$, Index
k	Mechanical Spring Constant, Index
K	RC Filter Time Constant, Number of Class Intervals
ℓ	Number of Frequency Components
L	Electrical Inductance, Length
m	Mechanical Mass, Maximum Number of Lag Values
m_f	Modulation Index
n	Degrees-of-Freedom
N	Sample Size
$p(x)$	Probability Density Function

$p(x, y)$	Joint Probability Density Function
$P(x)$	Probability Distribution Function
$P(x, y)$	Joint Probability Distribution Function
Prob[]	Probability that []
q	Number of Inputs, Number of Sample Records
$q(t)$	Electrical Charge
$Q_{xy}(f)$	Quadrature Spectral Density Function (One-Sided)
r	Number of Runs, Arbitrary Factor
r_{xy}	Sample Correlation Coefficient
R	Electrical Resistance
R_s	Analyzer Scan Rate
$R_x(\tau)$	Autocorrelation Function
$R_{xy}(\tau)$	Cross-Correlation Function
$R(t_1, t_2)$	Nonstationary Correlation Function
Re []	Real Part of []
s	Sample Standard Deviation
s^2	Sample Variance
s_{xy}^2	Sample Covariance
$S_x(f)$	Power Spectral Density Function Defined for Both Positive and Negative Frequencies (Two-Sided)
$S_{xy}(f)$	Cross Spectral Density Function Defined for Both Positive and Negative Frequencies (Two Sided)
$S(f_1, f_2)$	Generalized (Nonstationary) Spectral Density Function
S/N	Signal to Noise Ratio
t	Time Variable, Student t Variable
T	Observation Time, Averaging Time
T_s	Analysis Time
u_n	Raw Data Values
$u(x, t)$	Space and Time Dependent Variable
V	Voltage Range
Var []	Variance of []
W	Amplitude Window Width
$x(t), y(t)$	Time Dependent Variables
$\bar{x}$	Sample Mean Value of x
$\overline{ x }$	Mean Absolute Value (Average Rectified Value) of x
X	Amplitude of Sinusoidal $x(t)$
$X(f)$	Fourier Transform of $x(t)$
z	Standardized Normal Variable
$\{ \cdot \}$	Absolute Value of []
$\{ \cdot \}^{\wedge}$	Estimate of []
α	A Small Probability, Level of Significance
β	Probability of a Type II error

$\gamma^2(f)$	Coherence Function
$\delta(\)$	Delta Function
Δ	Small Increment
ϵ	Normalized Error
ζ	Mechanical Damping Ratio
θ	Phase Angle
$\theta_{xy}(f)$	Argument of $G_{xy}(f)$
μ	Mean Value
ρ	Correlation Coefficient
$\rho(\tau)$	Correlation Function Coefficient
σ	Standard Deviation
σ^2	Variance
τ	Time Displacement
$\phi(f)$	Phase Factor
Φ	Arbitrary Statistical Parameter
χ^2	Statistical Chi-Square Variable
Ψ	Root Mean Square Value
Ψ^2	Mean Square Value
λ	Wavelength

CHAPTER 1

BASIC DESCRIPTIONS OF PHYSICAL DATA

Any observed data representing a physical phenomenon can be broadly classified as being either deterministic or nondeterministic. Deterministic data are those that can be described by an explicit mathematical relationship. For example, consider a rigid body which is suspended from a fixed foundation by a linear spring, as shown in Figure 1.1. Let m be the mass of the body (assumed to be inelastic) and k be the spring constant of the spring (assumed to be massless). Suppose the body is displaced from its position of equilibrium by a distance X , and released at time $t = 0$. From either basic laws of mechanics or repeated observations, it can be established that the following relationship will apply.

$$x(t) = X \cos \sqrt{\frac{k}{m}} t \quad t \geq 0 \quad (1.1)$$

Equation (1.1) defines the exact location of the body at any instant of time in the future. Hence the physical data representing the motion of the mass are deterministic.

There are many physical phenomena in practice which produce data that can be represented with reasonable accuracy by explicit mathematical relationships. For example, the motion of a satellite in orbit about the earth, the potential across a condenser as it discharges through a resistor, the vibration response of an unbalanced rotating machine, or the temperature of water as heat is applied, are all basically deterministic. However, there are many other physical phenomena which produce data that are not deterministic. For example, the height of waves in a confused sea, the acoustic pressures generated by air rushing through a pipe, or the electrical

2 BASIC DESCRIPTIONS OF PHYSICAL DATA

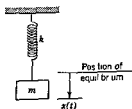


Figure 1.1 Simple spring mass system

output of a noise generator represent data which cannot be described by explicit mathematical relationships. There is no way to predict an exact value at a future instant of time. These data are random in character and must be described in terms of probability statements and statistical averages rather than by explicit equations.

The classification of various physical data as being either deterministic or random might be debated in many cases. For example, it might be argued that no physical data in practice can be truly deterministic since there is always a possibility that some unforeseen event in the future might influence the phenomenon producing the data in a manner that was not originally considered. On the other hand, it might be argued that no physical data are truly random since exact mathematical descriptions might be possible if a sufficient knowledge of the basic mechanisms of the phenomenon producing the data were known. In practical terms, the decision as to whether or not physical data are deterministic or random is usually based upon the ability to reproduce the data by controlled experiments. If an experiment producing specific data of interest can be repeated many times with identical results (within the limits of experimental error), then the data can generally be considered deterministic. If an experiment cannot be designed which will produce identical results when the experiment is repeated, then the data must usually be considered random in nature.

Various special classifications of deterministic and random data will now be discussed. Note that the classifications are selected from an analysis viewpoint and do not necessarily represent the most suitable classifications from other possible viewpoints. Further note that physical data are usually thought of as being functions of time and will be discussed in such terms for convenience. However, any other variable can replace time as required.

1.1 CLASSIFICATIONS OF DETERMINISTIC DATA

Data representing deterministic phenomena can be categorized as being either periodic or nonperiodic. Periodic data can be further categorized as

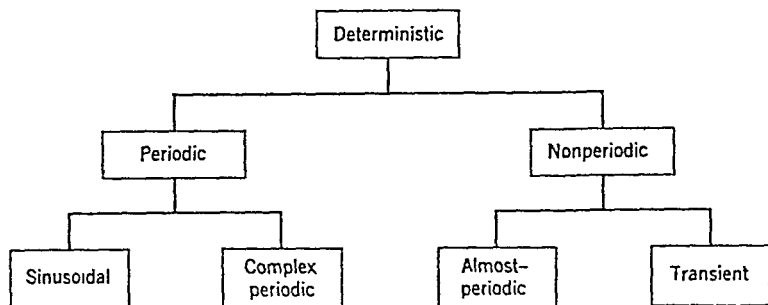


Figure 1.2 Classifications of deterministic data.

being either sinusoidal or complex periodic. Nonperiodic data can be further categorized as being either “almost-periodic” or transient. These various classifications of deterministic data are schematically illustrated in Figure 1.2. Of course, any combination of these forms may also occur. For purposes of review, each of these types of deterministic data along with physical examples will be briefly discussed.

1.1.1 Sinusoidal Periodic Data

Sinusoidal data are those types of periodic data which can be defined mathematically by a time-varying function of the form

$$x(t) = X \sin (2\pi f_0 t + \theta) \quad (1.2)$$

where X = amplitude

f_0 = cyclical frequency in cycles per unit time

θ = initial phase angle with respect to the time origin in radians

$x(t)$ = instantaneous value at time t

The sinusoidal time history described by Equation (1.2) is usually referred to as a sine wave. When analyzing sinusoidal data in practice, the phase angle θ is often ignored. For this case

$$x(t) = X \sin 2\pi f_0 t \quad (1.3)$$

Equation (1.3) can be pictured by a time history plot or by an amplitude-frequency plot (frequency spectrum), as illustrated in Figure 1.3.

The time interval required for one full fluctuation or cycle of sinusoidal data is called the period T_p . The number of cycles per unit time is called

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the frequency f_0 . The frequency and period are related by

$$T_p = \frac{1}{f_0} \quad (1.4)$$

Note that the frequency spectrum in Figure 1.3 is composed of an amplitude component at a specific frequency, as opposed to a continuous plot of amplitude versus frequency. Such spectra are called *discrete spectra* or *line spectra*.

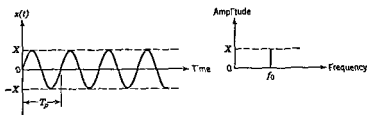


Figure 1.3 Time history and spectrum of sinusoidal data

There are many examples of physical phenomena which produce approximately sinusoidal data in practice. The voltage output of an electrical alternator is one example, the vibratory motion of an unbalanced rotating weight is another. Sinusoidal data represent one of the simplest forms of time varying data from the analysis viewpoint.

1.1.2 Complex Periodic Data

Complex periodic data are those types of periodic data which can be defined mathematically by a time varying function whose waveform exactly repeats itself at regular intervals such that

$$x(t) = x(t \pm nT_p) \quad n = 1, 2, 3, \quad (1.5)$$

As for sinusoidal data, the time interval required for one full fluctuation is called the *period* T_p . The number of cycles per unit time is called the *fundamental frequency* f_1 . A special case for complex periodic data is clearly sinusoidal data where $f_1 = f_0$.

With few exceptions in practice, complex periodic data may be expanded into a Fourier series according to the following formula

$$x(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos 2\pi n f_1 t + b_n \sin 2\pi n f_1 t) \quad (1.6)$$

where

$$f_1 = \frac{1}{T_p}$$

$$a_n = \frac{2}{T_p} \int_0^{T_p} x(t) \cos 2\pi n f_1 t \, dt \quad n = 0, 1, 2, \dots$$

$$b_n = \frac{2}{T_p} \int_0^{T_p} x(t) \sin 2\pi n f_1 t \, dt \quad n = 1, 2, 3, \dots$$

An alternate way to express the Fourier series for complex periodic data is as follows.

$$x(t) = X_0 + \sum_{n=1}^{\infty} X_n \cos(2\pi n f_1 t - \theta_n) \quad (1.7)$$

where $X_0 = a_0/2$

$$X_n = \sqrt{a_n^2 + b_n^2} \quad n = 1, 2, 3, \dots$$

$$\theta_n = \tan^{-1}(b_n/a_n) \quad n = 1, 2, 3, \dots$$

In words, Equation (1.7) says that complex periodic data consist of a static component, X_0 , and an infinite number of sinusoidal components called harmonics, which have amplitudes X_n and phases θ_n . The frequencies of the harmonic components are all integral multiples of f_1 .

When analyzing periodic data in practice, the phase angles θ_n are often ignored. For this case, Equation (1.7) can be characterized by a discrete spectrum as illustrated in Figure 1.4. Sometimes complex periodic data will include only a few components. In other cases, the fundamental component may be absent. For example, suppose a periodic time history is formed by mixing three sine waves which have frequencies of 60, 75, and 100 cps. The highest common divisor is 5 cps, so the period of the resulting periodic data

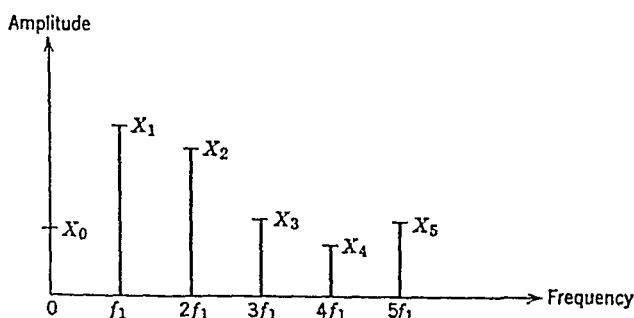


Figure 1.4 Spectrum of complex periodic data.

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is $T_0 = 0.2$ second. Hence when expanded into a Fourier series, all values of X_n are zero except for $n = 12$, $n = 15$, and $n = 20$.

Physical phenomena which produce complex periodic data are far more common than those which produce simple sinusoidal data. In fact, the classification of data as being sinusoidal is often only an approximation for data which are actually complex. For example, the voltage output from an electrical alternator may actually display, under careful inspection, some small contributions at higher harmonic frequencies. In other cases, intense harmonic components may be present in periodic physical data. For example, the vibration response of a multicylinder reciprocating engine will usually display considerable harmonic content.

1.1.3 Almost Periodic Data

In the previous section, it is noted that periodic data can generally be reduced to a series of sine waves with commensurately related frequencies. Conversely, the data formed by summing two or more commensurately related sine waves will be periodic. However, the data formed by summing two or more sine waves with arbitrary frequencies generally will not be periodic.

More specifically, the sum of two or more sine waves will be periodic only when the ratios of all possible pairs of frequencies form rational numbers. This indicates that a fundamental period exists which will satisfy the requirements of Equation (1.5). Hence

$$x(t) = X_1 \sin(2t + \theta_1) + X_2 \sin(3t + \theta_2) + X_3 \sin(7t + \theta_3)$$

is periodic since $\frac{2}{3}$, $\frac{2}{7}$, and $\frac{3}{7}$ are rational numbers (the fundamental period is $T_0 = 1$). On the other hand,

$$x(t) = X_1 \sin(2t + \theta_1) + X_2 \sin(3t + \theta_2) + X_3 \sin(\sqrt{50}t + \theta_3)$$

is not periodic since $2/\sqrt{50}$ and $3/\sqrt{50}$ are not rational numbers (the fundamental period is infinitely long). The resulting time history in this case will have an "almost periodic" character, but the requirements of Equation (1.5) will not be satisfied for any finite value of T_0 .

Based upon these discussions, almost periodic data are those types of nonperiodic data which can be defined mathematically by a time varying function of the form

$$x(t) = \sum_{n=1}^{\infty} X_n \sin(2\pi f_n t + \theta_n) \quad (1.8)$$

where $f_n/f_m \neq$ rational number in all cases. Physical phenomena producing almost periodic data frequently occur in practice when the effects of two or

more unrelated periodic phenomena are mixed. A good example is the vibration response in a multiple engine propeller airplane when the engines are out of synchronization.

An important property of almost-periodic data is as follows. If the phase angles θ_n are ignored, Equation (1.8) can be characterized by a discrete frequency spectrum similar to that for complex periodic data. The only difference is that the frequencies of the components are not related by rational numbers, as illustrated in Figure 1.5.

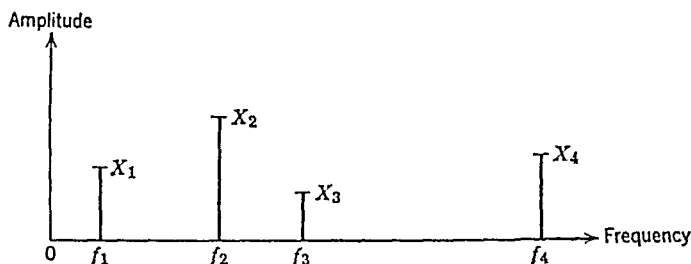


Figure 1.5 Spectrum of almost-periodic data.

1.1.4 Transient Nonperiodic Data

Transient data are defined as all nonperiodic data other than the almost-periodic data discussed in Section 1.1.3. In other words, transient data include all data not previously discussed which can be described by some suitable time-varying function. Three simple examples of transient data are given in Figure 1.6.

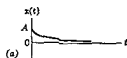
Physical phenomena which produce transient data are numerous and diverse. For example, the data in Figure 1.6(a) could represent the temperatures of water in a kettle (relative to room temperature) after the flame is turned off. The data in Figure 1.6(b) might represent the free vibration of a damped mechanical system after an excitation force is removed. The data in Figure 1.6(c) could represent the stress in an end-loaded cable which breaks at time c .

An important characteristic of transient data, as opposed to periodic and almost-periodic data, is that a discrete spectral representation is not possible. However, a continuous spectral representation for transient data can be obtained in most cases from a Fourier integral given by

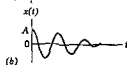
$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt \quad (1.9)$$

8 BASIC DESCRIPTIONS OF PHYSICAL DATA

$$x(t) = \begin{cases} A e^{-at} & t \geq 0 \\ 0 & t < 0 \end{cases}$$



$$x(t) = \begin{cases} A e^{-at} \cos bt & t \geq 0 \\ 0 & t < 0 \end{cases}$$



$$x(t) = \begin{cases} A & c \geq t \geq 0 \\ 0 & c < t < 0 \end{cases}$$

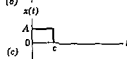


Figure 1 6 Illustrations of transient data.

The Fourier spectrum $X(f)$ is generally a complex number which can be expressed in complex polar notation as

$$X(f) = |X(f)| e^{-j\theta(f)}$$

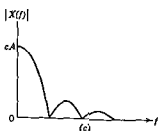
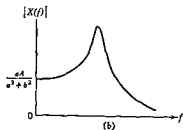
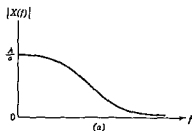


Figure 1 7 Spectra of transient data

Here, $|X(f)|$ is the magnitude of $X(f)$ and $\theta(f)$ is the argument. In terms of the magnitude $|X(f)|$, the Fourier spectra of the three transient time histories in Figure 1.6 are as presented in Figure 1.7.

1.2 CLASSIFICATIONS OF RANDOM DATA

As discussed earlier, data representing a random physical phenomenon cannot be described by an explicit mathematical relationship because each observation of the phenomenon will be unique. In other words, any given observation will represent only one of many possible results which might have occurred. For example, assume the output voltage from a thermal noise generator is recorded as a function of time. A specific voltage time history record will be obtained, as shown in Figure 1.8. However, if a second thermal noise generator of identical construction and assembly is operated simultaneously, a different voltage time history record would result. In fact, every thermal noise generator which might be constructed would produce a different voltage time history record, as illustrated in Figure 1.8. Hence the voltage time history for any one generator is merely one example of an infinitely large number of time histories which might have occurred.

A single time history representing a random phenomenon is called a *sample function* (or a *sample record* when observed over a finite time interval).

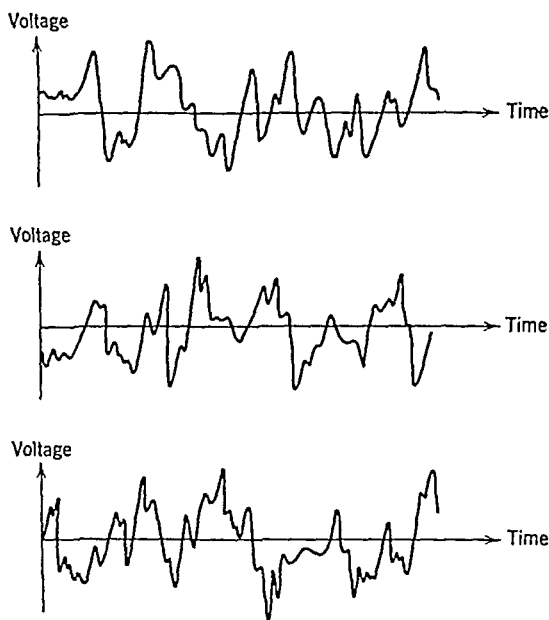


Figure 1.8 Sample records of thermal noise generator outputs.

10 BASIC DESCRIPTIONS OF PHYSICAL DATA

The collection of all possible sample functions which the random phenomenon might have produced is called a *random process* or a *stochastic process*. Hence a sample record of data for a random physical phenomenon may be thought of as one physical realization of a random process.

Random processes may be categorized as being either stationary or nonstationary. Stationary random processes may be further categorized as being either ergodic or nonergodic. Nonstationary random processes may be further categorized in terms of specific types of nonstationary properties. These various classifications of random processes are schematically illustrated in Figure 1.9. The meaning and physical significance of these various

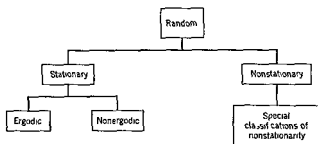


Figure 1.9 Classifications of random data

types of random processes will now be discussed in broad terms. More analytical definitions and developments are presented in Chapters 3 and 10.

1.2.1 Stationary Random Processes

When a physical phenomenon is considered in terms of a random process, the properties of the phenomenon can hypothetically be described at any instant of time by computing average values over the collection of sample functions which describe the random process. For example, consider the collection of sample functions (also called the *ensemble*) which forms the random process illustrated in Figure 1.10. The *mean value* (first moment) of the random process at some time t_1 can be computed by taking the instantaneous value of each sample function of the ensemble at time t_1 , summing the values, and dividing by the number of sample functions. In a similar manner, a correlation (joint moment) between the values of the random process at two different times (called the *autocorrelation function*) can be computed by taking the ensemble average of the product of instantaneous values at two

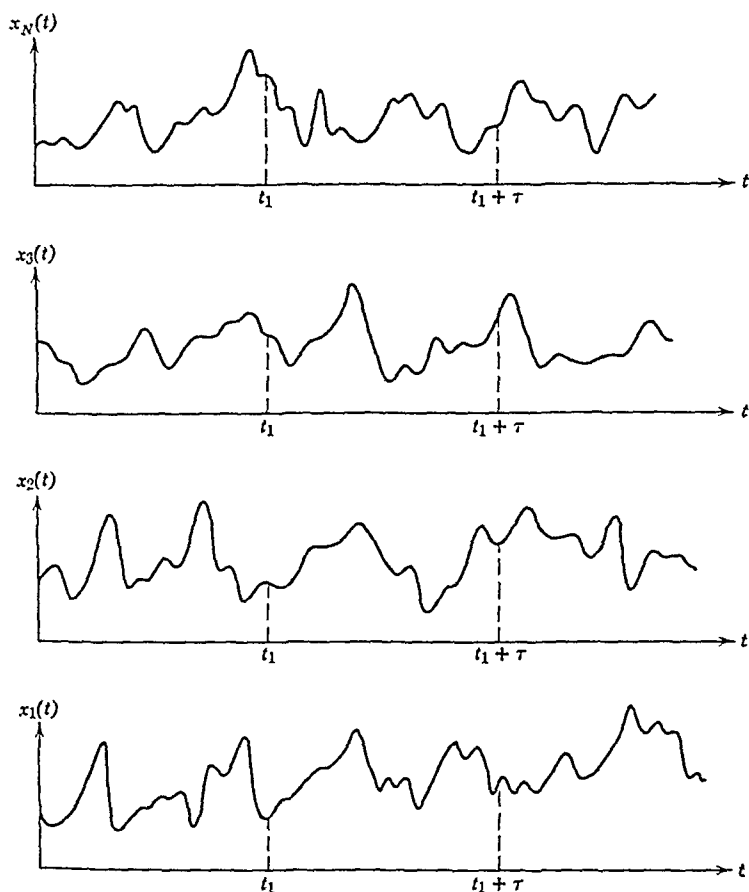


Figure 1.10 Ensemble of sample functions forming random process.

times, t_1 and $t_1 + \tau$. That is, for the random process $\{x(t)\}$, where the symbol $\{ \}$ is used to denote an ensemble of sample functions, the mean value $\mu_x(t_1)$ and the autocorrelation function $R_x(t_1, t_1 + \tau)$ are given by

$$\mu_x(t_1) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{k=1}^N x_k(t_1) \quad (1.10a)$$

$$R_x(t_1, t_1 + \tau) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{k=1}^N x_k(t_1) x_k(t_1 + \tau) \quad (1.10b)$$

where the final summation assumes each sample function is equally likely.

For the general case where $\mu_x(t_1)$ and $R_x(t_1, t_1 + \tau)$ defined in Equation (1 10) vary as time t_1 varies, the random process $\{x(t)\}$ is said to be *non-stationary*. For the special case where $\mu_x(t_1)$ and $R_x(t_1, t_1 + \tau)$ do not vary as time t_1 varies, the random process $\{x(t)\}$ is said to be *weakly stationary* or *stationary in the wide sense*. For weakly stationary random processes, the mean value is a constant and the autocorrelation function is dependent only upon the time displacement τ . That is, $\mu_x(t_1) = \mu_x$ and $R_x(t_1, t_1 + \tau) = R_x(\tau)$.

An infinite collection of higher-order moments and joint moments of the random process $\{x(t)\}$ could also be computed to establish a complete family of probability distribution functions describing the process. For the special case where all possible moments and joint moments are time invariant, the random process $\{x(t)\}$ is said to be *strongly stationary* or *stationary in the strict sense*. For many practical applications, verification of weak stationarity will justify an assumption of strong stationarity.

1 2 2 Ergodic Random Processes

The previous section discusses how the properties of a random process can be determined by computing ensemble averages at specific instants of time. In most cases, however, it is also possible to describe the properties of a stationary random process by computing time averages over specific sample functions in the ensemble. For example, consider the k th sample function of the random process illustrated in Figure 1 10. The mean value $\mu_x(k)$ and the autocorrelation function $R_x(\tau, k)$ of the k th sample function are given by

$$\mu_x(k) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T x_k(t) dt \quad (1 11a)$$

$$R_x(\tau, k) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T x_k(t) x_k(t + \tau) dt \quad (1 11b)$$

If the random process $\{x(t)\}$ is stationary, and $\mu_x(k)$ and $R_x(\tau, k)$ defined in Equation (1 11) do not differ when computed over different sample functions, the random process is said to be *ergodic*. For ergodic random processes, the time averaged mean value and autocorrelation function (as well as all other time averaged properties) are equal to the corresponding ensemble averaged values. That is, $\mu_x(k) = \mu_x$ and $R_x(\tau, k) = R_x(\tau)$. Note that only stationary random processes can be ergodic.

Ergodic random processes are clearly an important class of random processes since all properties of ergodic random processes can be determined by performing time averages over a single sample function. Fortunately, in practice, random data representing stationary physical phenomena are generally ergodic. It is for this reason that the properties of stationary random phenomena can be measured properly, in most cases, from a single observed time history record.

1.2.3 *Nonstationary Random Processes*

Nonstationary random processes include all random processes which do not meet the requirements for stationarity defined in Section 1.2.1. Unless further restrictions are imposed, the properties of a nonstationary random process are generally time-varying functions which can be determined only by performing instantaneous averages over the ensemble of sample functions forming the process. In practice, it is often not feasible to obtain a sufficient number of sample records to permit the accurate measurement of properties by ensemble averaging. This fact has tended to impede the development of practical techniques for measuring and analyzing nonstationary random data.

In many cases, the nonstationary random data produced by actual physical phenomena can be classified into special categories of nonstationarity which simplify the measurement and analysis problem. For example, some types of random data might be described by a nonstationary random process $\{y(t)\}$ where each sample function is given by $y(t) = A(t)x(t)$. Here, $x(t)$ is a sample function from a stationary random process $\{x(t)\}$ and $A(t)$ is a deterministic multiplication factor. In other words, the data might be represented by a nonstationary random process consisting of sample functions with a common deterministic time trend. If nonstationary random data fit a specific model of this type, ensemble averaging is not always needed to describe the data. The various desired properties can sometimes be estimated from a single sample record, as is true for ergodic stationary data.

1.2.4 *Stationary Sample Records*

The concept of stationarity, as defined and discussed in Section 1.2.1, relates to the ensemble averaged properties of a random process. In practice, however, data in the form of individual time history records of a random phenomenon are frequently referred to as being stationary or nonstationary. A slightly different interpretation of stationarity is involved here. When a single time history record is referred to as being stationary, it is generally meant that the properties computed over short time intervals do not vary "significantly" from one interval to the next. The word significantly is used here to mean that observed variations are greater than would be expected due to normal statistical sampling variations.

To help clarify this point, consider a single sample record $x_k(t)$ obtained from the k th sample function of a random process $\{x(t)\}$. Assume a mean value and autocorrelation function are obtained by time averaging over a short interval T with a starting time of t_1 as follows.

$$\mu_x(t_1, k) = \frac{1}{T} \int_{t_1}^{t_1+T} x_k(t) dt \quad (1.12a)$$

$$R_x(t_1, t_1 + \tau, k) = \frac{1}{T} \int_{t_1}^{t_1+T} x_k(t) x_k(t + \tau) dt \quad (1.12b)$$

For the general case where the sample properties defined in Equation (1.12) vary significantly as the starting time t_1 varies, the individual sample record is said to be nonstationary. For the special case where the sample properties defined in Equation (1.12) do not vary significantly as the starting time t_1 varies, the sample record is said to be stationary. Note that a sample record obtained from an ergodic random process will be stationary. Furthermore, sample records from most physically interesting nonstationary random processes will be nonstationary. Hence if an ergodic assumption is justified (as it is for most actual stationary physical phenomena), verification of stationarity for a single sample record will effectively justify an assumption of stationarity and ergodicity for the random process from which the sample record is obtained.

1.3 BASIC DESCRIPTIVE PROPERTIES OF RANDOM DATA

Four main types of statistical functions are used to describe the basic properties of random data: (a) mean square values, (b) probability density functions, (c) autocorrelation functions, and (d) power spectral density functions. The mean square value furnishes a rudimentary description of the intensity of the data. The probability density function furnishes information concerning the properties of the data in the amplitude domain. The autocorrelation function and the power spectral density function furnish similar information in the time domain and frequency domain, respectively. It will be shown in Chapter 3 that, for stationary data, the power spectral density function technically supplies no new information over the autocorrelation function since the two are Fourier transform pairs. The two functions present the information in different formats, however, where one format is often more desirable than the other for specific applications.

These various descriptive properties of stationary random data will now be defined in broad terms. The discussions assume that the data in question are ergodic, so that the properties of the data can be determined from time averages of individual sample records.

1.3.1 Mean Square Values (Mean Values and Variances)

The general intensity of any random data may be described in rudimentary terms by a *mean square value*, which is simply the average of the squared values of the time history. In equation form, the mean square value Ψ_x^2 of a sample time history record $x(t)$ is given by

$$\Psi_x^2 = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T x^2(\tau) d\tau \quad (1.13)$$

The positive square root of the mean square value is called the root mean square or *rms value*.

It is often desirable to think of physical data in terms of a combination of a *static* or time-invariant component and a *dynamic* or fluctuating component. The static component may be described by a *mean value* which is simply the average of all values. In equation form, the mean value μ_x is given by

$$\mu_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T x(t) dt \quad (1.14)$$

The dynamic component may be described by a *variance* which is simply the mean square value about the mean. In equation form, the variance σ_x^2 is given by

$$\sigma_x^2 = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T [x(t) - \mu_x]^2 dt \quad (1.15)$$

The positive square root of the variance is called the *standard deviation*. By expanding Equation (1.15), it is seen that the variance is equal to the mean square value minus the square of the mean value. That is,

$$\sigma_x^2 = \Psi_x^2 - \mu_x^2 \quad (1.16)$$

1.3.2 Probability Density Functions

The probability density function of random data describes the probability that the data will assume a value within some defined range at any instant of time. Consider the sample time history record $x(t)$ illustrated in Figure 1.11. The probability that $x(t)$ assumes a value within the range between x and $(x + \Delta x)$ may be obtained by taking the ratio of T_x/T , where T_x is the total

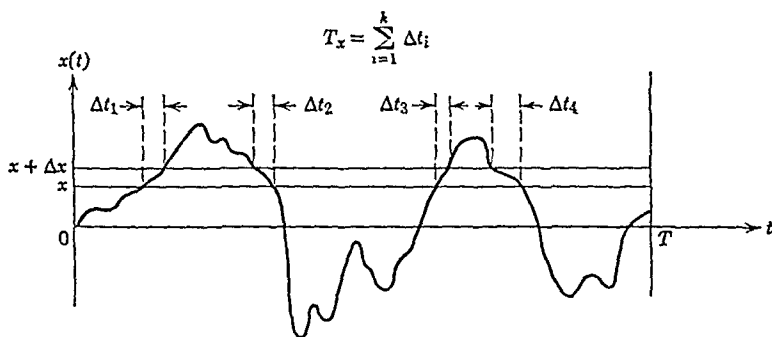


Figure 1.11 Probability measurement.

amount of time that $x(t)$ falls inside the range $(x, x + \Delta x)$ during an observation time T . This ratio will approach an exact probability description as T approaches infinity. In equation form,

$$\text{Prob} [x < x(t) \leq x + \Delta x] = \lim_{T \rightarrow \infty} \frac{T_x}{T} \quad (1.17)$$

For small Δx , a *probability density function* $p(x)$ can be defined as follows:

$$\text{Prob} [x < x(t) \leq x + \Delta x] \approx p(x) \Delta x \quad (1.18)$$

More precisely,

$$p(x) = \lim_{\Delta x \rightarrow 0} \frac{\text{Prob} [x < x(t) \leq x + \Delta x]}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{1}{\Delta x} \left[\lim_{T \rightarrow \infty} \frac{T_x}{T} \right] \quad (1.19)$$

The probability density function $p(x)$ is always a real valued, non negative function.

The probability that the instantaneous value $x(t)$ is less than or equal to some value x is defined by $P(x)$, which is equal to the integral of the probability density function from minus infinity to x . This function $P(x)$ is known as the *probability distribution function*, or *cumulative probability distribution function*, and should not be confused with the probability density function $p(x)$. Specifically

$$P(x) = \text{Prob} [x(t) \leq x] = \int_{-\infty}^x p(\xi) d\xi \quad (1.20)$$

The distribution function $P(x)$ is bounded by zero and one, since the probability of $x(t)$ being less than $-\infty$ is clearly zero while the probability of $x(t)$ being less than $+\infty$ is unity (a certainty). The probability that $x(t)$ falls inside any range (x_1, x_2) is given by

$$P(x_2) - P(x_1) = \text{Prob} [x_1 < x(t) \leq x_2] = \int_{x_1}^{x_2} p(x) dx \quad (1.21)$$

In terms of the probability density function $p(x)$, the mean value of $x(t)$ is given by

$$\mu_x = \int_{-\infty}^{\infty} x p(x) dx \quad (1.22)$$

In words, the mean value is a weighted linear sum of $x(t)$ over all values of x . Similarly, the mean square value is given by

$$\Psi_x^2 = \int_{-\infty}^{\infty} x^2 p(x) dx \quad (1.23)$$

which is a weighted linear sum of $x^2(t)$ over all values of x .

ILLUSTRATIONS To help clarify the practical significance of probability density functions, consider four examples of sample time history records

which might occur in practice: (a) sine wave, (b) sine wave plus random noise, (c) narrow-band random noise, and (d) wide-band random noise. Typical time history records for each of these examples are presented in Figure 1.12. In all cases, the mean value is assumed to equal zero ($\mu_x = 0$) for convenience.

Note that a sine wave is usually thought of as being deterministic since

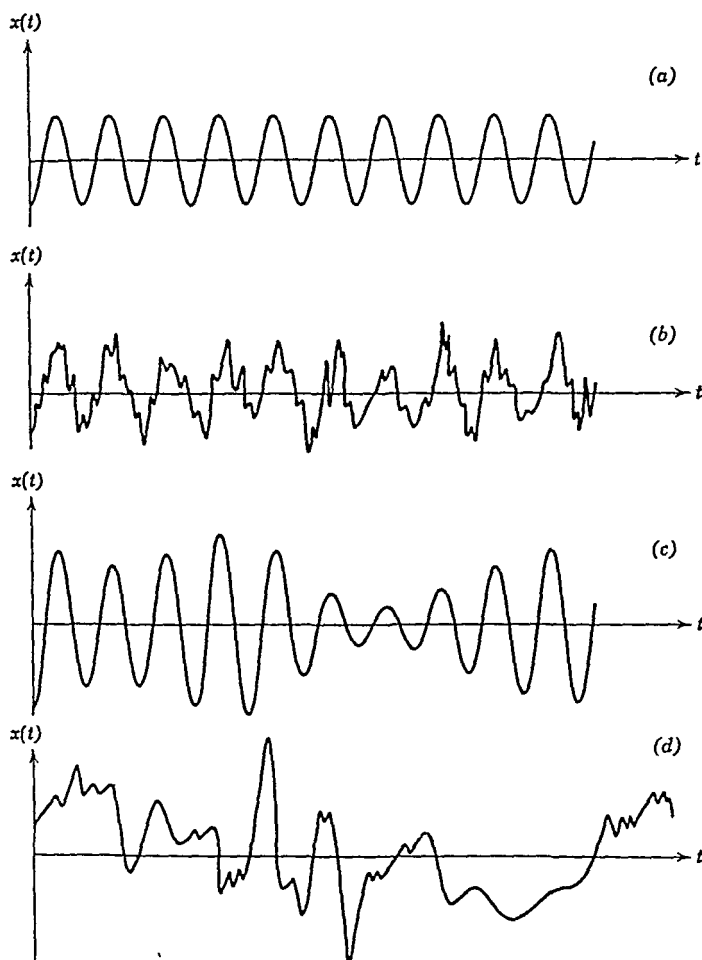


Figure 1.12 Four special time histories. (a) Sine wave. (b) Sine wave plus random noise. (c) Narrow-band random noise. (d) Wide-band random noise.

it can be described in detail by the equation $x(t) = X \sin(2\pi f_0 t + \theta)$. However, a sine wave may also be thought of as a sample function from a random process $\{x(t)\} = \{X \sin(2\pi f_0 t + \theta_k)\}$, where the initial phase angle θ_k for each sample function $x_k(t)$ is a random variable. Such an interpretation is made here to justify describing a sine wave in probabilistic terms.

A typical plot of probability density versus instantaneous value $\{p(x)$ versus $x\}$ for each of the four examples is presented in Figure 1-13. The dish-shaped probability density plot for a sine wave, as illustrated in Figure 1-13(a), is defined by the following equation:

$$p(x) = \begin{cases} (\pi\sqrt{X^2 - x^2})^{-1} & |x| < X \\ 0 & |x| \geq X \end{cases} \quad (1-24)$$

The bell-shaped probability density plots indicated in Figure 1-13(c) and (d) are typical of either narrow or wide band random data. These probability density plots would ideally be of the classical Gaussian form as follows:

$$p(x) = (\sigma_x \sqrt{2\pi})^{-1} e^{-x^2/2\sigma_x^2} \quad (1-25)$$

The probability density plot for a sine wave plus random noise takes on the predominant characteristics of both as illustrated in Figure 1-13(b). The four examples in Figure 1-13 illustrate a definite trend in the probability density plot going from the sine wave case to the wide band random noise case.

APPLICATIONS The principal application for a probability density function measurement of physical data is to establish a probabilistic description for the instantaneous values of the data. However, from Figure 1-13, it is seen that the probability density function could also be used to distinguish between sinusoidal and random data. Furthermore, various nonlinear physical effects will often exhibit their presence in the probability density function to one who has sufficient experience in such interpretations.

1.3.3 Autocorrelation Functions

The autocorrelation function of random data describes the general dependence of the values of the data at one time on the values at another time. Consider the sample time history record $x(t)$ illustrated in Figure 1-14. An estimate for the autocorrelation between the values of $x(t)$ at times t and $t + \tau$ may be obtained by taking the product of the two values and averaging over the observation time T . The resulting average product will approach an exact autocorrelation function as T approaches infinity. In equation form,

$$R_x(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T x(t) x(t + \tau) dt \quad (1-26)$$

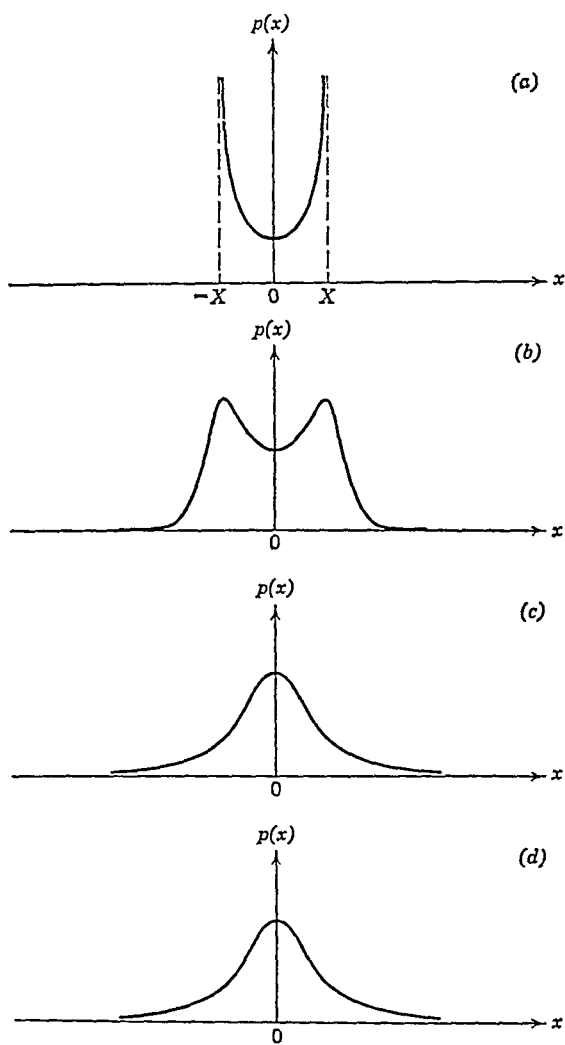


Figure 1.13 Probability density function plots. (a) Sine wave. (b) Sine wave plus random noise. (c) Narrow-band random noise. (d) Wide-band random noise.

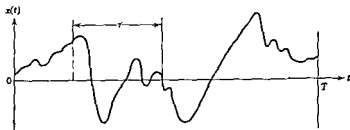


Figure 1.14 Autocorrelation measurement

The quantity $R_x(\tau)$ is always a real valued even function with a maximum at $\tau = 0$, and may be either positive or negative. In equation form,

$$R_x(-\tau) = R_x(\tau) \quad (1.27)$$

$$R_x(0) \geq |R_x(\tau)| \quad \text{for all } \tau \quad (1.28)$$

In terms of the autocorrelation function, the mean value of $x(t)$ is given (excluding such special cases as sine waves) by

$$\mu_x = \sqrt{R_x(\infty)} \quad (1.29)$$

In words, the mean value of $x(t)$ is equal to the positive square root of the autocorrelation as the time displacement becomes very long. Similarly, the mean square value of $x(t)$ is given by

$$\Psi_x^2 = R_x(0) \quad (1.30)$$

That is, the mean square value is equal to the autocorrelation at zero time displacement.

ILLUSTRATIONS A typical plot of autocorrelation versus time displacement $[R_x(\tau)$ versus $\tau]$ for each of the four time histories in Figure 1.12 is presented in Figure 1.15. These plots are called *autocorrelograms*. The cosine-type autocorrelogram for a sine wave, as illustrated in Figure 1.15(a), is defined by the equation

$$R_x(\tau) = \frac{X^2}{2} \cos 2\pi f_0 \tau \quad (1.31)$$

The important feature of this autocorrelogram is that it persists periodically over all time displacements with the same period as the underlying sine wave, but phase angle information is lost.

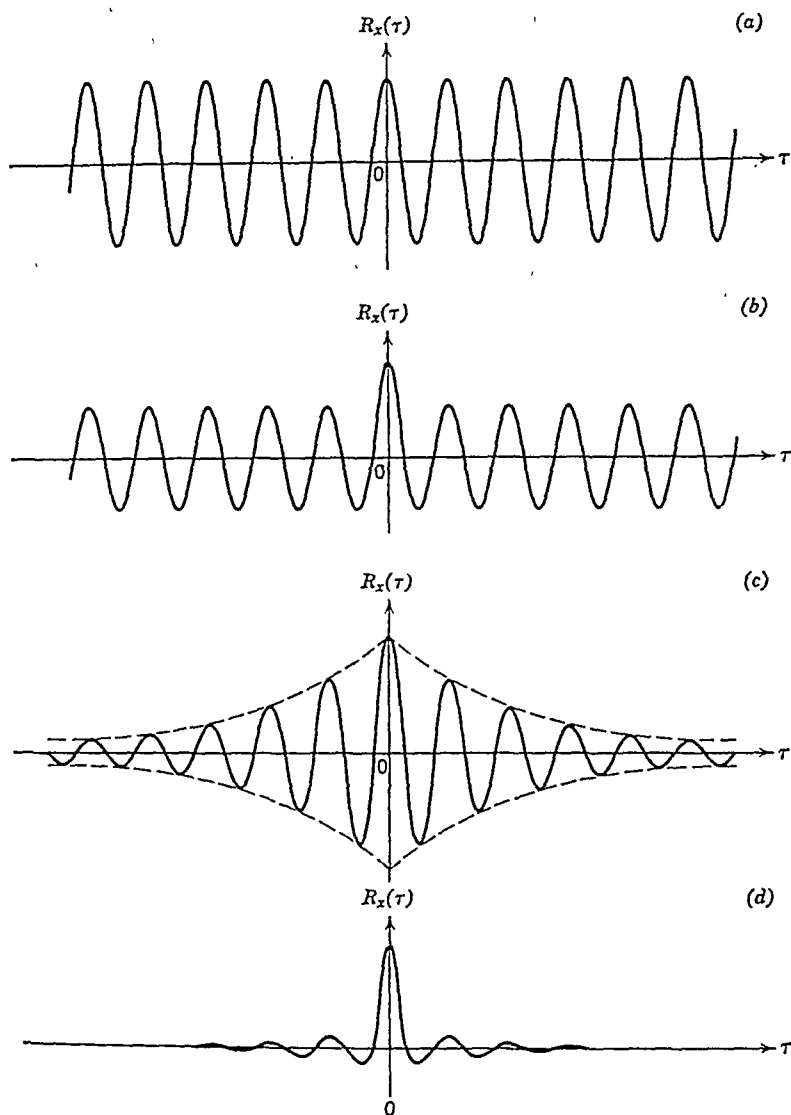


Figure 1.15 Autocorrelation function plots (autocorrelograms). (a) Sine wave. (b) Sine wave plus random noise. (c) Narrow-band random noise. (d) Wide-band random noise.

The sharply peaked autocorrelogram which diminishes rapidly to zero, as illustrated in Figure 1 15(d), is typical of wide band random data with a zero mean value (if the mean value were not zero the autocorrelogram would approach a value of μ_x^2). For the limiting case of hypothetical white noise (random data with energy distributed uniformly over all frequencies), the autocorrelogram is a Dirac delta function at zero time displacement ($\tau = 0$).

The autocorrelogram for the sine wave plus random noise is simply the sum of the autocorrelograms for the sine wave and random noise separately as illustrated in Figure 1 15(b). On the other hand, the autocorrelogram for the narrow band random noise in Figure 1 15(c) appears like a decaying version of a sine wave autocorrelogram. An important feature, however, is that this autocorrelogram will diminish to zero for large time displacements (assuming $\mu_x = 0$). The four examples in Figure 1 15 illustrate a definite trend in the autocorrelogram going from the sine wave case to the wide-band random noise case, just as was true for the probability density function.

APPLICATIONS The principal application for an autocorrelation function measurement of physical data is to establish the influence of values at any time over values at a future time. Because a sine wave, or any other deterministic data, will have an autocorrelation function which persists over all time displacements as opposed to random data which diminishes to zero for large time displacements (assuming $\mu_x = 0$), an autocorrelation measurement clearly provides a tool for detecting deterministic data which might be masked in a random background. There are other less obvious applications for autocorrelation functions, but these are generally better interpreted from its Fourier transform, the power spectral density function, which is discussed next.

1.3.4 Power Spectral Density Functions

The power spectral density function (also called the autospectral density function) of random data describes the general frequency composition of the data in terms of the spectral density of its mean square value. The mean square value of a sample time history record in a frequency range between f and $f + \Delta f$ may be obtained by filtering the sample record with a band pass filter having sharp cutoff characteristics, and computing the average of the squared output from the filter. This average squared value will approach an exact mean square value as the observation time T approaches infinity. In equation form,

$$\Psi_x^2(f, \Delta f) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T x^2(t, f, \Delta f) dt \quad (1.32)$$

where $x(t, f, \Delta f)$ is that portion of $x(t)$ in the frequency range from f to

$f + \Delta f$. For small Δf , a *power spectral density function* $G_x(f)$ can be defined such that

$$\Psi_x^2(f, \Delta f) \approx G_x(f) \Delta f \quad (1.33)$$

More precisely,

$$G_x(f) = \lim_{\Delta f \rightarrow 0} \frac{\Psi_x^2(f, \Delta f)}{\Delta f} = \lim_{\Delta f \rightarrow 0} \frac{1}{(\Delta f)} \left[\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T x^2(t, f, \Delta f) dt \right] \quad (1.34)$$

The quantity $G_x(f)$ is always a real-valued, non-negative function.

An important property of the power spectral density function lies in its relationship to the autocorrelation function. Specifically, for stationary data, the two functions are related by a Fourier transform as follows.

$$G_x(f) = 2 \int_{-\infty}^{\infty} R_x(\tau) e^{-j2\pi f\tau} d\tau = 4 \int_0^{\infty} R_x(\tau) \cos 2\pi f\tau d\tau \quad (1.35)$$

The second equality exists because $R_x(\tau)$ is an even function of τ .

In terms of the power spectral density function, the mean value of $x(t)$ is given by

$$\mu_x = \left[\int_{0^-}^{0^+} G_x(f) df \right]^{1/2} \quad (1.36)$$

where 0^- means the lower limit of integration, zero, is approached from below, and 0^+ means the upper limit of integration, also zero, is approached from above. In words, the mean value of $x(t)$ will appear in $G_x(f)$ as a Dirac delta function at zero frequency. The mean value is equal to the positive square root of the area under this delta function.

The mean square value of $x(t)$ is given by

$$\Psi_x^2 = \int_0^{\infty} G_x(f) df \quad (1.37)$$

where it is understood that the lower limit of integration is approached from below in order to include Eq. (1.36). Hence the mean square value is equal to the total area under a plot of the power spectral density function versus frequency.

ILLUSTRATIONS. A typical plot of power spectral density versus frequency [$G_x(f)$ versus f] for each of the four time histories in Figure 1.12 is presented in Figure 1.16. These plots are called *power spectra*. The discrete power spectrum for a sine wave, as illustrated in Figure 1.16(a), is defined by

$$G_x(f) = \frac{X^2}{2} \delta(f - f_0) \quad (1.38)$$

where $\delta(f - f_0)$ denotes a delta function at $f = f_0$. In words, the power spectral density function of a sine wave is infinitely large at the frequency of

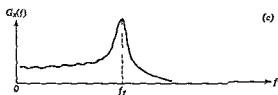
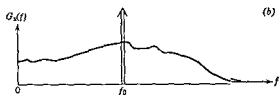
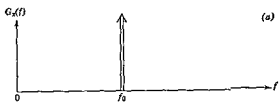


Figure 1.16 Power spectral density plots (power spectra) (a) Sine wave (b) Sine wave plus random noise (c) Narrow band random noise (d) Wide band random noise

the sine wave and zero at all other frequencies. However, the integral of the power spectrum over any frequency range that includes the sinusoidal frequency has a finite value equal to the mean square value ($X^2/2$) of the sine wave.

The relatively smooth and broad power spectrum illustrated in Figure 1.16(d) gives rise to the descriptive term "wide-band" for this type of random data. For the hypothetical case of white noise, this power spectrum is uniform over all frequencies by definition. The power spectrum of the sine wave plus random noise is simply the sum of the power spectra for the sine wave and random noise separately, as illustrated in Figure 1.16(b). On the other hand, the power spectrum of the narrow-band noise in Figure 1.16(c) is sharply peaked as for a sine wave (hence the term "narrow-band"), but still smoothly continuous as for random noise. Once again, the four examples in Figure 1.16 illustrate a definite trend in the power spectrum going from the sine wave case to the wide-band noise case.

APPLICATIONS. The principal application for a power spectral density function measurement of physical data is to establish the frequency composition of the data which, in turn, bears important relationships to the basic characteristics of the physical system involved. For example, consider an electrical circuit with a frequency response function of $H(f)$. Assume a stationary random signal with a power spectral density function of $G_x(f)$ is applied to the input of the electrical circuit. The output from the electrical circuit will be a stationary random signal with a power spectral density function given by

$$G_v(f) = |H(f)|^2 G_x(f) \quad (1.39)$$

It follows from Equation (1.39) that measurement or knowledge of any two quantities enables one to estimate the third quantity. Note that only the magnitude of the frequency response function is involved in the relationship. Determination of phase information requires a cross-spectra analysis, which is discussed later.

1.4 JOINT PROPERTIES OF RANDOM DATA

The statistical functions discussed in Section 1.3 are useful to describe the properties of data from individual random processes. It is often desirable to describe certain common or joint properties of different data from two or more random processes. For example, assume one is interested in the height of waves at various points on the ocean's surface. The average properties of the wave height at each point could be described using the statistical functions

discussed in Section 1.3. However, there may be additional important information in similar joint statistical functions which can be computed for the wave heights at two points on the ocean.

Three main types of statistical functions are used to describe the joint properties of sample records from two random processes: (a) joint probability density functions, (b) cross-correlation functions, and (c) cross spectral density functions. These three functions are effectively extensions of the basic formulations used to describe the properties of individual sample records. They supply information concerning joint properties in the amplitude domain, time domain, and frequency domain, respectively.

These various joint descriptive properties for two sets of stationary random data will now be defined in broad terms. Once again, the discussions will assume ergodicity so that the joint time averaged properties of single pairs of sample time history records can be considered.

1.4.1 Joint Probability Density Functions

The joint probability density function of two random sample records describes the probability that both sample records will simultaneously assume values within some defined pair of ranges at any instant of time. Consider the pair of time history records $x(t)$ and $y(t)$ illustrated in Figure 1.17. The probability that $x(t)$ assumes a value within the range between x

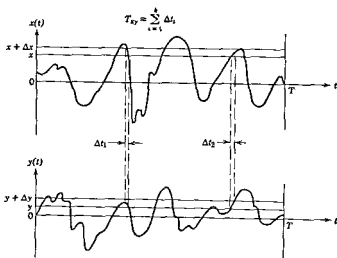


Figure 1.17 Joint probability measurement

and $(x + \Delta x)$ while $y(t)$ simultaneously assumes a value within the range between y and $(y + \Delta y)$ may be obtained by taking the ratio $T_{x,y}/T$, where $T_{x,y}$ is the total amount of time that $x(t)$ and $y(t)$ simultaneously fall inside the ranges $(x, x + \Delta x)$ and $(y, y + \Delta y)$, respectively, during an observation time T . This ratio will approach an exact probability description as T approaches infinity. The joint probability density function $p(x, y)$ can then be defined by

$$p(x, y) = \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{1}{(\Delta x)(\Delta y)} \left[\lim_{T \rightarrow \infty} \frac{T_{x,y}}{T} \right] \quad (1.40)$$

The joint probability density function $p(x, y)$ is always a real-valued, non-negative function.

The probability that the instantaneous values $x(t)$ and $y(t)$ are less than or equal to some values x and y is defined by the joint probability distribution function $P(x, y)$ as follows:

$$P(x, y) = \text{Prob} [x(t) \leq x, y(t) \leq y] = \int_{-\infty}^x \int_{-\infty}^y p(\xi, \eta) d\xi d\eta \quad (1.41)$$

When the two phenomena in question are statistically independent, the joint probability density function is given by

$$p(x, y) = p(x)p(y) \quad (1.42)$$

That is, the joint probability density function for such problems is given by the product of the individual probability density functions.

A typical plot of joint probability density versus instantaneous values $[p(x, y)$ versus x and $y]$ for a pair of random time history records is illustrated in Figure 1.18. Note that the plot has three dimensions. The volume under the joint probability density plot bounded by the values x_1, x_2, y_1, y_2 is equal to the probability that $x(t)$ and $y(t)$ will simultaneously have values within those ranges at any given time. Obviously, the total volume under the plot is equal to unity since the probability of the two sample records simultaneously having any value must be one.

APPLICATIONS. The principal application for a joint probability density function measurement is to establish a probabilistic description for an event which is associated with two sets of correlated random data. For example, the prediction of collisions between two adjacent elastic structures which are vibrating randomly, but with some correlation, is a direct application for a joint probability density function. In some applications, it is desirable to relate the probability of occurrence of two events to the probability of occurrence of one of these events. These questions of "conditional" probabilities require knowledge of the joint probability density function. Special applications occur in establishing the expected number of zero crossings or

$$\text{Prob}\{x_1 < x \leq x_2, y_1 < y \leq y_2\} = \int_{y_1}^{y_2} \int_{x_1}^{x_2} p(x, y) dx dy = P_{1,2}$$

$$\text{Prob}[-\infty < x < \infty, -\infty < y < \infty] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} p(x, y) dx dy = 1$$

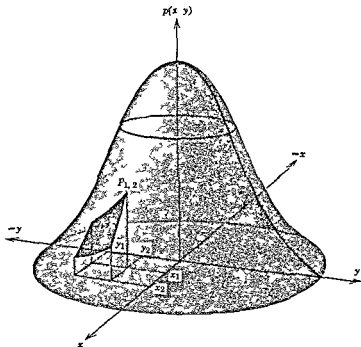


Figure 1.18 Typical joint probability density plot

threshold crossings of physical data and in predicting peak value distributions or extreme value properties

1.4.2 Cross Correlation Functions

The cross-correlation function of two sets of random data describes the general dependence of the values of one set of data on the other. Consider the pair of time history records $x(t)$ and $y(t)$ illustrated in Figure 1.19. An estimate for the cross correlation function of the values of $x(t)$ at time t and

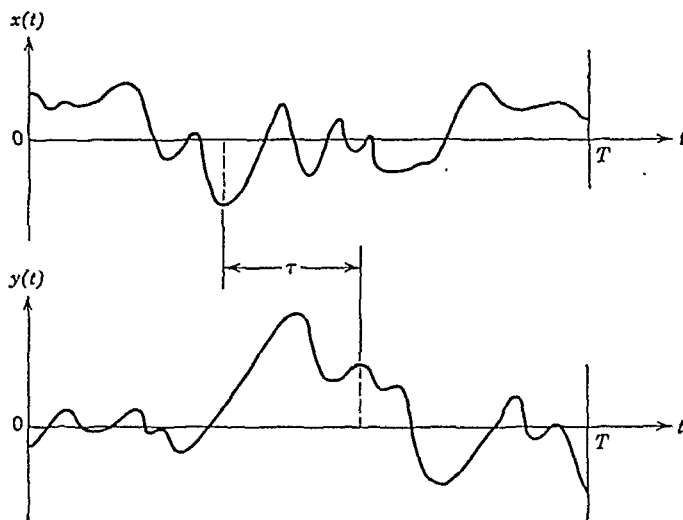


Figure 1.19 Cross-correlation measurement.

$y(t)$ at time $t + \tau$ may be obtained by taking the average product of the two values over the observation time T , exactly as is done for autocorrelation functions in Section 1.3.3. The resulting average product will approach an exact *cross-correlation function* as T approaches infinity. That is,

$$R_{xy}(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T x(t) y(t + \tau) dt \quad (1.43)$$

The function $R_{xy}(\tau)$ is always a real-valued function which may be either positive or negative. Furthermore, $R_{xy}(\tau)$ does not necessarily have a maximum at $\tau = 0$, nor is $R_{xy}(\tau)$ an even function as was true for autocorrelation functions. However, $R_{xy}(\tau)$ does display symmetry about the ordinate when x and y are interchanged. That is,

$$R_{xy}(-\tau) = R_{yx}(\tau) \quad (1.44)$$

Two useful relationships which bound the absolute value of the cross-correlation function are

$$|R_{xy}(\tau)|^2 \leq R_x(0)R_y(0) \quad (1.45)$$

$$|R_{xy}(\tau)| \leq \frac{1}{2}[R_x(0) + R_y(0)] \quad (1.46)$$

When $R_{xy}(\tau) = 0$, $x(t)$ and $y(t)$ are said to be uncorrelated. If $x(t)$ and $y(t)$

are statistically independent, then $R_{xy}(\tau) = 0$ for all time displacements, assuming either $x(t)$ or $y(t)$ have a zero mean value. If the mean values of both $x(t)$ and $y(t)$ are not zero, the cross-correlation function would have a value of $\mu_x\mu_y$ for all time displacements in this case.

A typical plot of the cross correlation versus time displacement [$R_{xy}(\tau)$ versus τ] for a pair of random time history records is illustrated in Figure 1.20. This plot is called a *cross correlogram*. Note that the plot will sometimes display sharp peaks which indicate the existence of correlation between $x(t)$ and $y(t)$ for specific time displacements.

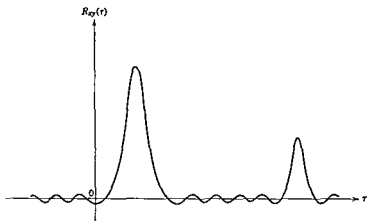


Figure 1.20 Typical cross correlation plot (cross-correlogram)

APPLICATIONS Cross-correlation function measurements have many important applications including the following:

Measurement of Time Delays Suppose one is interested in determining the time required for a signal to pass through a given system. Assuming the system is linear, a cross-correlation measurement between the input and output may yield such time delay information directly. As the output from the system is displaced in time relative to the input, the cross-correlation function will peak at that time displacement equal to the time required for the signal to pass through the system. This is true because the average product of two linearly related signals will always be a maximum when the time displacement between the signals is zero. Hence a system time delay can sometimes be established directly by noting the time displacement associated with an observed peak in the cross-correlogram between input and output.

It should be mentioned that this technique often fails in practice because the transmission velocity and/or path through the system may be frequency-sensitive. In such cases, the cross-correlogram may not display a distinct peak. This problem can be analyzed by using cross-spectra information, to be discussed later.

Determination of Transmission Paths. The use of cross-correlation function measurements to establish time delays leads directly to another important application, the determination of transmission paths. Consider the case of a linear system where an input may follow two or more different paths to produce an observed output. Assume the specific path followed through the system is of interest. For example, the operation of heavy factory machinery will often induce undesirable noise and vibration in neighboring office areas, where the energy might be transferred in several ways through the building structures or acoustically through the air. Before the noise and vibration can be effectively controlled, it is necessary to establish the exact transmission path. Such problems can often be solved by cross-correlating the input and output for the system in question. Since each transmission path through the system will generally be associated with a different delay time, a separate peak will occur in the cross-correlogram for each path which contributes significantly to the output. If the expected time delays associated with the various possible paths can be calculated, these expected delays can then be compared to the measured time displacements of peaks in the cross-correlogram to identify the paths contributing significantly to the output.

Detection and Recovery of Signals in Noise. A third application of cross-correlation function measurements is the detection and recovery of a signal buried in extraneous noise, where the signal is not necessarily of a periodic form. It has been noted in Section 1.3.3 that an autocorrelation function is useful for this application when the signal is periodic. However, an autocorrelation analysis cannot extract a random signal from extraneous noise. For this case, a cross-correlation function is needed. Specifically, if a noise-free replica of the signal (either random or periodic) which one wishes to detect is available, then a cross-correlation of the signal plus noise with a stored replica of the signal alone will extract the correlation function of the signal. Furthermore, for the case of periodic signals, the cross-correlation function will provide a greater output signal-to-noise ratio than will the autocorrelation function for any given input signal-to-noise ratio and sample record length.

1.4.3 Cross-Spectral Density Functions

The concept of a cross-spectral density function of two sets of random data evolves directly from the cross-correlation function. As the power spectral density function of a single time history record is the Fourier transform of the

autocorrelation function, so the cross spectral density function of a pair of time history records is the Fourier transform of the cross correlation function. Because a cross correlation function is not an even function the cross-spectral density function is generally a complex number such that

$$G_{xy}(f) = C_{xy}(f) - j Q_{xy}(f) \quad (1.47)$$

where the real part $C_{xy}(f)$, is called the *coincident spectral density function* and the imaginary part, $Q_{xy}(f)$, is called the *quadrature spectral density function*.

In direct frequency terms, the coincident spectral density function can be thought of as the average product of $x(t)$ and $y(t)$ within a narrow frequency interval between f and $f + \Delta f$, divided by the frequency interval. The quadrature spectral density is the same except that either $x(t)$ or $y(t)$, not both, is shifted in time sufficiently to produce a 90-degree phase shift at frequency f .

It is convenient to express the cross spectral density function in complex polar notation such that

$$G_{xy}(f) = |G_{xy}(f)| e^{-j\theta_{xy}(f)} \quad (1.48)$$

where the magnitude $|G_{xy}(f)|$ and the angle $\theta_{xy}(f)$ are related to $C_{xy}(f)$ and $Q_{xy}(f)$ by

$$|G_{xy}(f)| = \sqrt{C_{xy}^2(f) + Q_{xy}^2(f)} \quad (1.49)$$

$$\theta_{xy}(f) = \tan^{-1} \left[\frac{Q_{xy}(f)}{C_{xy}(f)} \right] \quad (1.50)$$

Another useful relationship is

$$|G_{xy}(f)|^2 \leq G_x(f) G_y(f) \quad (1.51)$$

When applying cross spectral density information to physical problems, it is often desirable to use a real-valued quantity given by

$$\gamma_{xy}^2(f) = \frac{|G_{xy}(f)|^2}{G_x(f) G_y(f)} \leq 1 \quad (1.52)$$

where $\gamma_{xy}^2(f)$ is called the *coherence function*. When $\gamma_{xy}^2(f) = 0$ at a particular frequency, $x(t)$ and $y(t)$ are said to be incoherent at that frequency, which is simply another word for uncorrelated. If $x(t)$ and $y(t)$ are statistically independent, then $\gamma_{xy}^2(f) = 0$ for all frequencies. When $\gamma_{xy}^2(f) = 1$ for all f , then $x(t)$ and $y(t)$ are said to be fully coherent.

A typical plot of the cross spectral density function versus frequency [$G_{xy}(f)$ versus f] for a pair of random time history records is illustrated in Figure 1.21. This plot is called a *cross-spectrum*. Note that the plot consists of two parts which give a magnitude and phase.

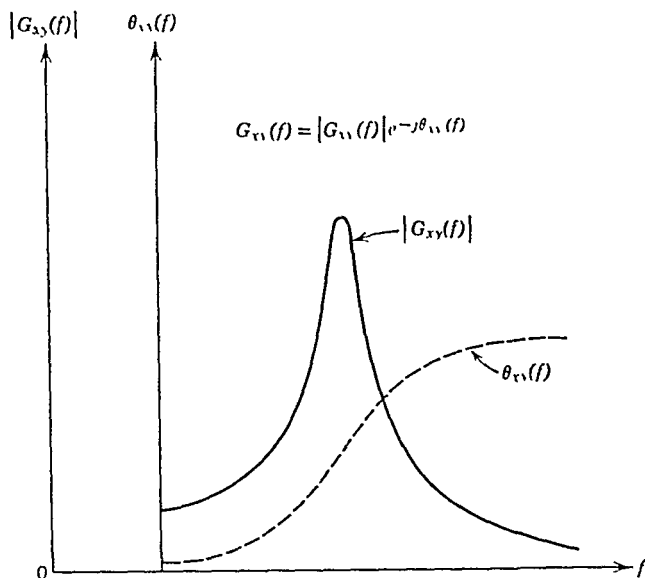


Figure 1.21 Typical cross-spectral density plot (cross-spectrum).

APPLICATIONS. Cross-spectral density function measurements have many applications similar to cross-correlation function measurements. A few of these applications follow.

Measurement of Frequency Response Functions. This fundamental application for a cross-spectral density function measurement evolves from an important relationship which it bears to the basic characteristics of the physical system involved. For example, consider an electrical circuit with a frequency response function of $H(f)$. Assume a stationary random signal with a power spectral density function of $G_x(f)$ is applied to the input of the electrical circuit. As noted in Section 1.3.4, the output from the electrical circuit will be a stationary random signal with a power spectrum given by Equation (1.39). However, another important relationship exists. The cross-spectrum between input and output signals will be

$$G_{xy}(f) = H(f) G_x(f) \quad (1.53)$$

The relationship in Equation (1.53) permits the complete frequency response function of linear systems to be evaluated from cross-spectra measurements by the relationship $H(f) = G_{xy}(f)/G_x(f)$. Confidence statements about the accuracy of measuring $H(f)$ may be made by calculating the associated coherence function. These matters are developed in detail in Chapters 5 and 6.

Measurement of Time Delays Another possible application for a cross-spectral density function measurement is the determination of time delays, considered in Section 1.4.2 as an application for cross correlation function measurements. In this case, the phase angle $\theta_{xy}(f)$ of the cross-spectrum between the input and output for a system constitutes the phase shift through the system at frequency f . Hence the time delay through the system at any frequency f will be given by $\tau = \theta_{xy}(f)/2\pi f$. Note that the cross spectral density function measurement permits the determination of time delays as a function of frequency, which is not available directly from a cross correlation function measurement.

Linear Prediction and Filter Theory Many applications for power spectral and cross spectral density information occur in optimum linear prediction and filtering. It is necessary in these problems to determine an optimum linear filter according to some established criterion which will transmit and predict desired signal information while rejecting undesired noise information. The various power spectra and cross spectra between input signals and noise, and output signals and noise, determine the optimum filter.

Consider the situation where one desires to obtain an estimate of the output $y(t)$ by a linear operation on the input $x(t)$. Suppose the measurement of $y(t)$ is distorted by extraneous noise $n(t)$ which is independent of $x(t)$. Then the optimum weighting function according to a mean square error criterion, if physical realizability is waived temporarily, has the form

$$H_{y|x}(f) = \frac{G_{xy}(f)}{G_x(f)} \quad (1.54)$$

where the bar notation in $y|x$ indicates that $y(t)$ is determined, given $x(t)$.

The minimum mean square error for the optimum system is

$$R_e(0) = \int_0^\infty G_y(f)[1 - \gamma_{xy}^2(f)] df \quad (1.55)$$

where $\gamma_{xy}^2(f)$ is the coherence function between $x(t)$ and $y(t)$, as defined by Equation (1.52). Observe that if the coherence function is unity for all frequencies, then $y(t)$ can be estimated by a linear operation on $x(t)$ with zero mean square error. The result in Equation (1.55) indicates in practice how closely an actual system approximates the optimum system.

PROBLEMS

In problems 1 through 4 state which properties are always true

1. A stationary random process must
 - (a) be discrete
 - (b) be continuous

- (c) be ergodic.
 - (d) have ensemble averaged properties which are independent of time.
 - (e) have time averaged properties which are equal to the ensemble averaged properties.
2. An ergodic random process must
- (a) be discrete.
 - (b) be continuous.
 - (c) be stationary.
 - (d) have ensemble averaged properties which are independent of time.
 - (e) have time averaged properties which are equal to the ensemble averaged properties.
3. A single sample function can be used to find all statistical properties of a random process if the process is
- (a) deterministic.
 - (b) ergodic.
 - (c) stationary.
 - (d) all of the above.
4. The autocorrelation function of a stationary random process
- (a) must decrease as $|\tau|$ increases.
 - (b) is a function of the time difference only.
 - (c) must approach a constant as $|\tau|$ increases.
 - (d) must always be non-negative.
5. How do the answers to question 4 change if the stationary random process is restricted not to include any periodic components?
6. Given a stationary random process with zero spectral density at $f = 0$, prove that the autocorrelation function of the process is such that $\int_{-\infty}^{\infty} R(\tau) d\tau = 0$.
7. Given a stationary random process with an autocorrelation function of $R(\tau) = Ae^{-a|\tau|}$ where $a > 0$, determine the
- (a) mean value μ .
 - (b) mean square value Ψ^2 .
 - (c) power spectral density function $G(f)$.
8. Assuming the random process in question 7 is Gaussian, write the equation for the probability density function of the process.
9. What is the general functional form for the autocorrelation function of a
- (a) sine wave?
 - (b) square wave?
 - (c) triangular wave?

- 10 If the joint probability density function of two random processes satisfies $p(x, y) = p(x)p(y)$, then which of the following properties are always true
- (a) the cross-correlation function between the two processes is zero for all delay values τ
 - (b) the cross spectral density function between the two processes is zero for all frequencies f
 - (c) the coherence function between the two processes is zero for all frequencies f

CHAPTER 2

REVIEW OF PHYSICAL SYSTEM RESPONSE PROPERTIES

Before the measurement and analysis of random physical data is discussed in greater detail, it is desirable to clarify some pertinent concepts and fundamental definitions related to the dynamic behavior of physical systems. This chapter reviews the theoretical formulas for describing the response characteristics of ideal systems, and illustrates the basic ideas for simple physical examples.

2.1 CONSTANT PARAMETER LINEAR SYSTEMS

An ideal system is one that has *constant parameters* and is *linear* between two clearly defined points of interest called the input or excitation point and the output or response point. A system has constant parameters if all fundamental properties of the system are invariant with respect to time. For example, a simple passive electrical circuit would be a constant parameter system if the values for the resistance, capacitance, and inductance of all elements did not change from one time to another. A system is linear if the response characteristics are additive and homogeneous. The term additive means that the output to a sum of inputs is equal to the sum of the outputs produced by each input individually. The term homogeneous means that the output produced by a constant times the input is equal to the constant times the output produced by the input alone. In equation form, if $f(x)$ represents the output to an input x , then the system is linear if for any two inputs

x_1 , x_2 , and constant c ,

$$f(x_1 + x_2) = f(x_1) + f(x_2) \quad \text{additive property} \quad (2\ 1a)$$

$$f(cx) = cf(x) \quad \text{homogeneous property} \quad (2\ 1b)$$

The constant parameter assumption is reasonably valid for many physical systems in practice. For example, the fundamental properties of an electrical circuit or a mechanical structure will usually not display significant changes over any time interval of practical interest. There are, of course, exceptions. The value of an electrical resistor may change owing to a high temperature exposure, or the stiffness of a structure may change due to fatigue damage caused by continual vibration. Furthermore, some physical systems are designed to have time-varying parameters which are fundamental to the desired purpose of the system. Electronic communication systems are an obvious example. However, such conditions are generally special cases which can be clearly identified in practice.

A linearity assumption for real systems is somewhat more critical. All physical systems will display nonlinear response characteristics under sufficiently extreme input conditions. For example, an electrical capacitor will ultimately arc as the applied voltage is increased and, hence, will no longer pass a current which is directly proportional to the applied voltage, or a metal cable will ultimately break as the applied load is increased and, hence, will no longer display a strain which is proportional to the applied load. To make the problem more difficult, common nonlinearities usually occur gradually rather than abruptly at one point. For example, the load-strain relationship for the metal cable would actually start deviating from a linear relationship long before the final abrupt break occurs. Nevertheless the response characteristics for many physical systems may be assumed to be linear, at least over some limited range of inputs, without involving unreasonable errors.

2.2 BASIC DYNAMIC CHARACTERISTICS

The dynamic characteristics of a constant parameter linear system can be described by a *weighting function* $h(\tau)$, which is defined as the output of the system at any time to a unit impulse input applied a time τ before. The usefulness of the weighting function as a description of the system is due to the following fact. For any arbitrary input $x(t)$, the system output $y(t)$ is given by the *convolution integral*

$$y(t) = \int_{-\infty}^{\infty} h(\tau) x(t - \tau) d\tau \quad (2\ 2)$$

That is, the value of the output $y(t)$ is given as a weighted linear (infinite) sum over the entire history of the input $x(t)$.

In order for a constant parameter linear system to be *physically realizable*, it is necessary that the system respond only to past inputs. This implies that

$$h(\tau) = 0 \quad \text{for } \tau < 0 \quad (2.3)$$

Hence, for physical systems, the effective lower limit of integration in Equation (2.2) is zero rather than minus infinity.

A constant parameter linear system is said to be *stable* if every possible bounded input function produces a bounded output function. From Equation (2.2),

$$|y(t)| = \left| \int_{-\infty}^{\infty} h(\tau) x(t - \tau) d\tau \right| \leq \int_{-\infty}^{\infty} |h(\tau)| |x(t - \tau)| d\tau \quad (2.4)$$

When the input $x(t)$ is bounded, there exists some finite constant A such that

$$|x(t)| \leq A \quad \text{for all } t \quad (2.5)$$

It follows from Equation (2.4) that

$$|y(t)| \leq A \int_{-\infty}^{\infty} |h(\tau)| d\tau \quad (2.6)$$

Hence if the constant parameter linear weighting function $h(\tau)$ is absolutely integrable, that is,

$$\int_{-\infty}^{\infty} |h(\tau)| d\tau < \infty \quad (2.7)$$

then the output will be bounded and the system is stable.

A constant parameter linear system can also be characterized by a *transfer function* $H(p)$, which is defined as the Laplace transform of $h(\tau)$. That is,

$$H(p) = \int_0^{\infty} h(\tau) e^{-p\tau} d\tau \quad p = a + jb \quad (2.8)$$

The criterion for stability of a constant parameter linear system (assumed to be physically realizable) takes on an interesting form when considered in terms of the transfer function $H(p)$. Specifically, if $H(p)$ has no poles in the right half of the complex p plane or on the imaginary axis (no poles where $a \geq 0$), then the system is stable. Conversely, if $H(p)$ has at least one pole in the right half of the complex p plane or on the imaginary axis, then the system is unstable.

An important property of constant parameter linear systems is frequency preservation. Specifically, consider a constant parameter linear system with a weighting function $h(\tau)$. From Equation (2.2), for any arbitrary input $x(t)$,

the n th derivative of the output $y(t)$ with respect to time is given by

$$\frac{d^n y(t)}{dt^n} = \int_{-\infty}^{\infty} h(\tau) \frac{d^n x(t - \tau)}{dt^n} d\tau \quad (2.9)$$

Now, assume the input $x(t)$ is sinusoidal, that is,

$$x(t) = X \sin(2\pi f t + \theta) \quad (2.10)$$

The second derivative of $x(t)$ is

$$\frac{d^2 x(t)}{dt^2} = -4\pi^2 f^2 x(t) \quad (2.11)$$

It follows from Equation (2.9) that the second derivative for the output $y(t)$ must be

$$\frac{d^2 y(t)}{dt^2} = -4\pi^2 f^2 y(t) \quad (2.12)$$

Thus $y(t)$ must also be sinusoidal with the same frequency as $x(t)$. This result shows that a constant parameter linear system cannot cause any frequency translation but can only modify the amplitude and phase of an applied input.

2.3 FREQUENCY RESPONSE FUNCTIONS

If a constant parameter linear system is physically realizable and stable, then the dynamic characteristics of the system can be described by a *frequency response function* $H(f)$, which is defined as the Fourier transform of $h(\tau)$. That is,

$$H(f) = \int_0^{\infty} h(\tau) e^{j2\pi f \tau} d\tau \quad (2.13)$$

Note that the lower limit of integration is zero rather than minus infinity since $h(\tau) = 0$ for $\tau < 0$. The frequency response function is simply a special case of the transfer function where, in the exponent $p = a + jb$, $a = 0$ and $b = 2\pi f$. For physically realizable and stable systems, the frequency response function may replace the transfer function with no loss of useful information.

An important relationship for the frequency response function of constant parameter linear systems is obtained by taking the Fourier transform of both sides of Equation (2.2). Letting $X(f)$ be the Fourier transform of an input $x(t)$ and $Y(f)$ be the Fourier transform of the resulting output $y(t)$, it follows from Equation (2.2) that

$$Y(f) = H(f) X(f) \quad (2.14)$$

Hence, in terms of the frequency response function of a system and the

Fourier transforms of the input and output, the convolution integral in Equation (2.2) reduces to the simple algebraic expression in Equation (2.14).

The frequency response function is generally a complex valued quantity which may be conveniently thought of in terms of a magnitude and an associated phase angle. This can be done by writing $H(f)$ in complex polar notation as follows,

$$H(f) = |H(f)| e^{-j\phi(f)} \quad (2.15)$$

The absolute value $|H(f)|$ is called the system *gain factor* and the associated phase angle $\phi(f)$ is called the system *phase factor*. In these terms, the frequency response function takes on a direct physical interpretation as follows. Assume a system is subjected to a sinusoidal input (hypothetically existing over all time) with a frequency f producing an output which, as illustrated in Section 2.2, will also be sinusoidal with the same frequency. The ratio of the output amplitude to the input amplitude is equal to the gain factor $|H(f)|$ of the system, and the phase shift between the output and input is equal to the phase factor $\phi(f)$ of the system.

From physical realizability requirements, the frequency response function, the gain factor, and the phase factor of a constant parameter linear system satisfy the following symmetry properties.

$$\begin{aligned} H(-f) &= H^*(f) \\ |H(-f)| &= |H(f)| \\ \phi(-f) &= -\phi(f) \end{aligned} \quad (2.16)$$

Furthermore, if one system described by $H_1(f)$ is followed by a second system described by $H_2(f)$, and there is no loading or feedback between the two systems, then the overall system may be described by $H(f)$ where

$$\begin{aligned} H(f) &= H_1(f) H_2(f) \\ |H(f)| &= |H_1(f)| |H_2(f)| \\ \phi(f) &= \phi_1(f) + \phi_2(f) \end{aligned} \quad (2.17)$$

Thus, on cascading two systems where there is no loading or feedback, the gain factors multiply and the phase factors add.

It is important to note that the frequency response function $H(f)$ of a constant parameter linear system is a function of only frequency, and is not a function of either time or the system excitation. If the system were nonlinear, $H(f)$ would also be a function of the applied input. If the parameters of the system were not constant, $H(f)$ would also be a function of time.

2.4 ILLUSTRATIONS OF FREQUENCY RESPONSE FUNCTIONS

A clearer understanding of the frequency response function of common physical systems will be afforded by considering some examples. The examples chosen involve simple mechanical and electrical systems because these particular physical systems are generally easier to visualize. The analogous characteristics relating mechanical and electrical systems to other physical systems are noted.

2.4.1 Mechanical Systems

Assume a simple mechanical structure can be represented by a lumped parameter system consisting of a mass, a spring, and a dashpot, where the motion of the mass is restricted to translation in only one direction as shown in Figure 2.1. In this figure k is a spring constant in pounds/inch, c is a

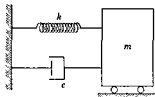


Figure 2.1 Simple mechanical system

viscous damping coefficient in pound seconds/inch and m is a mass in pound-seconds²/inch.

Before a frequency response function can be determined, it is necessary to define the input and output parameters of interest. There are a number of possibilities for the system in Figure 2.1, as will be illustrated now.

FORCE INPUT AND DISPLACEMENT OUTPUT Assume the input of interest is a force applied to the mass, and the output of interest is the resulting displacement of the mass as illustrated in Figure 2.2. Here, $F(t)$ is an applied force in pounds and $y(t)$ is the resulting output displacement of the mass in inches.

The first step toward establishing an appropriate frequency response function for this system is to determine the equation of motion. This is accomplished by using the relationship from basic mechanics that the sum of all forces acting on the mass must equal zero, as follows:

$$F(t) + F_k(t) + F_c(t) + F_m(t) = 0 \quad (2.18)$$

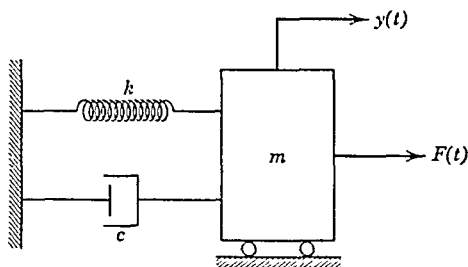


Figure 2.2 Mechanical system with force input.

where

$$F_k(t) = -k y(t) = \text{spring force} \quad (2.18a)$$

$$F_c(t) = -c \dot{y}(t) = \text{damping force} \quad (2.18b)$$

$$F_m(t) = -m \ddot{y}(t) = \text{inertial force} \quad (2.18c)$$

$$\dot{y}(t) = \frac{dy(t)}{dt} = \text{velocity}$$

$$\ddot{y}(t) = \frac{d^2y(t)}{dt^2} = \text{acceleration}$$

Hence the equation of motion for this system is

$$m \ddot{y}(t) + c \dot{y}(t) + k y(t) = F(t) \quad (2.19)$$

In Section 2.3, the frequency response function is defined as the Fourier transform of the output of the system to a unit impulse. For this case, the output of the system is the displacement $y(t)$ whose Fourier transform is given by

$$Y(f) = \int_0^{\infty} y(t) e^{-j2\pi f t} dt = H(f) \quad (2.20)$$

It follows that

$$\dot{Y}(f) = j2\pi f H(f) \quad (2.20a)$$

$$\ddot{Y}(f) = -(2\pi f)^2 H(f) \quad (2.20b)$$

Now, by taking the Fourier transform of both sides of Equation (2.19), and noting that the Fourier transform for a unit impulse force $F(t) = \delta(t)$ is unity, the following result is obtained.

$$[-(2\pi f)^2 m + j2\pi f c + k] H(f) = 1 \quad (2.21a)$$

Thus

$$H(f)_{f-a} = [k - (2\pi f)^2 m + j2\pi f c]^{-1} \quad (2.21b)$$

where the subscript $f-d$ is added to indicate that this particular $H(f)$ relates a force input to a displacement output

It is desirable to write Equation (2 21) in a different form by introducing two definitions

$$\zeta = \frac{c}{2\sqrt{km}} \quad (2\ 22a)$$

$$f_n = \frac{1}{2\pi} \sqrt{\frac{k}{m}} \quad (2\ 22b)$$

The term ζ in Equation (2 22a) is a dimensionless quantity called the *damping ratio*. The term f_n in Equation (2 22b) is called the *undamped natural frequency* and has the units of cycles per unit time. When these definitions are substituted into Equation (2 21), the following result is obtained

$$H(f)_{f-d} = \frac{1/k}{1 - (f/f_n)^2 + j2\zeta f/f_n} \quad (2\ 23)$$

Writing Equation (2 23) in complex polar notation gives the frequency response function in terms of a gain factor $|H(f)|$ and a phase factor $\phi(f)$ as follows

$$H(f) = |H(f)| e^{-j\phi(f)} \quad (2\ 24)$$

where

$$|H(f)|_{f-d} = \frac{1/k}{\sqrt{[1 - (f/f_n)^2]^2 + [2\zeta f/f_n]^2}} \quad (2\ 24a)$$

$$\phi(f)_{f-d} = \tan^{-1} \left[\frac{2\zeta f/f_n}{1 - (f/f_n)^2} \right] \quad (2\ 24b)$$

Note that $|H(f)|_{f-d}$ has units of $1/k$ or inches/pound. This particular function is sometimes called a *magnification function*.

Plots of $|H(f)|_{f-d}$ and $\phi(f)_{f-d}$ as defined in Equation (2 24) are presented in Figure 2 3. Two characteristics of these plots are of particular interest. First, the gain factor has a peak at some frequency less than f_n for all cases where $\zeta \leq 1/\sqrt{2}$. The frequency at which this peak gain factor occurs is called the *resonant frequency* of the system. Specifically, it can be shown by minimizing the denominator of $|H(f)|_{f-d}$ in Equation (2 24a) that the resonant frequency, denoted by f_r , is given by

$$f_r = f_n \sqrt{1 - 2\zeta^2} \quad \zeta^2 \leq 0.5 \quad (2\ 25)$$

and that the peak value of the gain factor which occurs at the resonant frequency is given by

$$|H(f_r)|_{f-d} = \frac{1/k}{2\zeta \sqrt{1 - \zeta^2}} \quad \zeta^2 < 0.5 \quad (2\ 26)$$

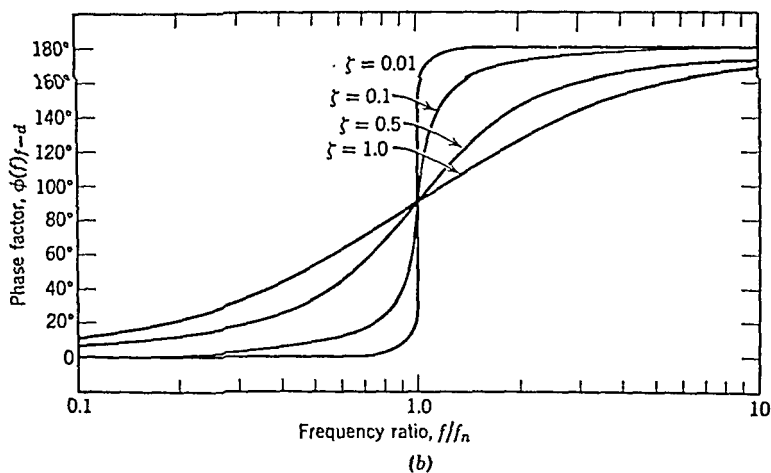
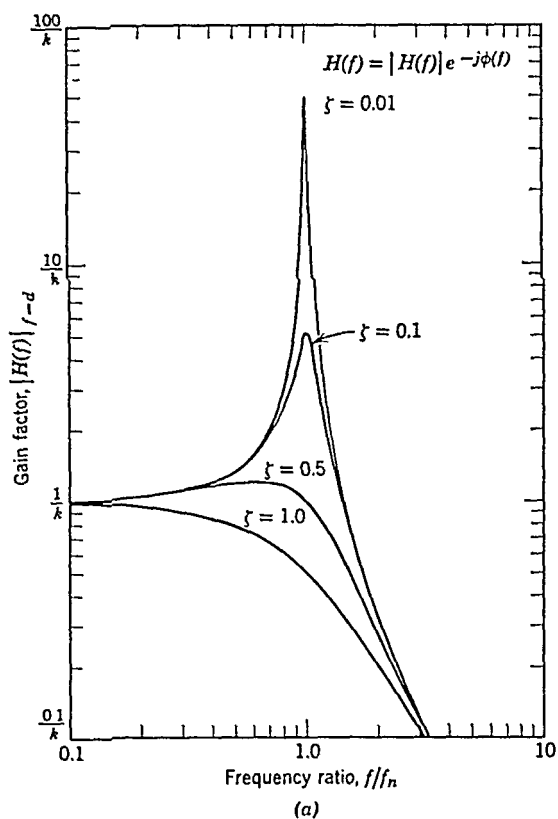


Figure 2.3 Frequency response function of mechanical system with force input.
 (a) Gain factor. (b) Phase factor.

Second, the phase factor varies from 0 degrees for frequencies much less than f_n to 180 degrees for frequencies much greater than f_n . The exact manner in which $\phi(f)$ varies between these phase angle limits depends upon the damping ratio ζ . However, for all values of ζ , the phase $\phi(f)_{f \rightarrow \infty} = 90$ degrees for $f = f_n$.

Referring to Section 2.3, the frequency response function $H(f)_{f \rightarrow \infty}$ may be interpreted as follows. Assume the applied force in Figure 2.2 is sinusoidal such that $F(t) = F_0 \sin 2\pi f t$. Then the output displacement would be given by

$$y(t) = F_0 |H(f)|_{f \rightarrow \infty} \sin [2\pi f t - \phi(f)_{f \rightarrow \infty}] \quad (2.27)$$

This particular interpretation actually provides another technique for determining the frequency response function. Specifically, one can solve for the output of a system to a sinusoidal input, and determine the frequency response function from the amplitude change and phase shift between the output and input. This will now be illustrated for the present problem.

The output of the system in Figure 2.2 to a sinusoidal input will be given by the particular solution of Equation (2.19) where $F(t)$ is sinusoidal, that is,

$$m \ddot{y}(t) + c \dot{y}(t) + k y(t) = F_0 \sin 2\pi f t = \text{Im} [F_0 e^{j2\pi f t}] \quad (2.28)$$

where $\text{Im} []$ means the imaginary part of $[]$. Now, assume a solution to Equation (2.28) in the general form of a sinusoidal output as follows

$$y(t) = Y \sin (2\pi f t - \phi) = \text{Im} [Y e^{j(2\pi f t - \phi)}] \quad (2.29)$$

When Equation (2.29) is substituted into Equation (2.28), the following relationship is obtained

$$\text{Im} [(- (2\pi f)^2 m + j2\pi f c + k) Y e^{j(2\pi f t - \phi)}] = \text{Im} [F_0 e^{j2\pi f t}] \quad (2.30)$$

The particular solution to Equation (2.28) is given from Equations (2.29) and (2.30) as follows

$$y(t) = \text{Im} \left[\frac{F_0 e^{j2\pi f t}}{k - (2\pi f)^2 m + j2\pi f c} \right] \quad (2.31)$$

Using the definitions from Equation (2.22) and converting to a trigonometric form, the output $y(t)$ becomes

$$y(t) = \frac{F_0 \sin [2\pi f t - \phi(f)]}{k \sqrt{[1 - (f/f_n)^2]^2 + [2\zeta f/f_n]^2}} \quad (2.32)$$

where

$$\phi(f) = \tan^{-1} \left[\frac{2\zeta f/f_n}{1 - (f/f_n)^2} \right]$$

Hence the output is changed in amplitude by a factor equal to the gain

factor defined in Equation (2.24a), and shifted in phase by a factor equal to the phase factor defined in Equation (2.24b).

FOUNDATION DISPLACEMENT INPUT AND DISPLACEMENT OUTPUT. Now consider a different case, where the input of interest is a motion of the foundation and the output of interest is the displacement of the mass, as illustrated in Figure 2.4. Here, $x(t)$ is an applied foundation displacement in inches

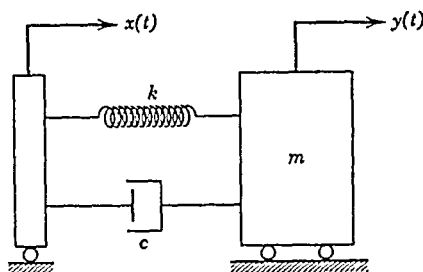


Figure 2.4 Mechanical system with foundation motion input.

measured from a mean foundation position, and $y(t)$ is the resulting output displacement of the mass in inches measured from the position of equilibrium.

As before, the equation of motion for the system can be determined from basic principles as follows.

$$F_k(t) + F_c(t) + F_m(t) = 0 \quad (2.33)$$

where $F_k(t) = -k[y(t) - x(t)] = \text{spring force} \quad (2.33a)$

$$F_c(t) = -c[\dot{y}(t) - \dot{x}(t)] = \text{damping force} \quad (2.33b)$$

$$F_m(t) = -m\ddot{y}(t) = \text{inertial force} \quad (2.33c)$$

Hence the equation of motion for this system is

$$m \ddot{y}(t) + c \dot{y}(t) + k y(t) = k x(t) + c \dot{x}(t) \quad (2.34)$$

Once again the frequency response function of the system will be given by the Fourier transform of the response displacement $y(t)$ for a unit impulse foundation displacement $x(t) = \delta(t)$. By taking the Fourier transform of both sides of Equation (2.34), and noting that $\dot{X}(f) = j2\pi f X(f)$, the following result is obtained.

$$[-(2\pi f)^2 m + j2\pi f c + k] Y(f) = [k + j2\pi f c] \quad (2.35)$$

Thus

$$Y(f) = H(f)_{d-a} = \frac{k + j2\pi f c}{k - (2\pi f)^2 m + j2\pi f c} \quad (2.36)$$

where the subscript $d-d$ means that this particular $H(f)$ relates a displacement input to a displacement output

Using the definitions from Equation (2.22), the result in Equation (2.36) may be written as

$$H(f)_{d-d} = \frac{1 + j2\zeta f/f_n}{1 - (f/f_n)^2 + j2\zeta f/f_n} \quad (2.37)$$

In complex polar notation, Equation (2.37) reduces to the following gain factor and phase factor

$$H(f) = |H(f)| e^{-j\phi(f)} \quad (2.38)$$

where

$$|H(f)|_{d-d} = \left(\frac{1 + [2\zeta f/f_n]^2}{[1 - (f/f_n)^2]^2 + [2\zeta f/f_n]^2} \right)^{1/2} \quad (2.38a)$$

$$\phi(f)_{d-d} = \tan^{-1} \left[\frac{2\zeta(f/f_n)^2}{1 - (f/f_n)^2 + 4\zeta^2(f/f_n)^2} \right] \quad (2.38b)$$

Note that $H(f)_{d-d}$ is dimensionless. This particular function is often called a *transmissibility function*. Plots of $|H(f)|_{d-d}$ and $\phi(f)_{d-d}$ are presented in Figure 2.5. Note that the gain factor displays a single peak similar to the example for a force input illustrated in Figure 2.3. However, the details of the gain factor as well as the phase factor in Figure 2.5 are quite different from the factors in Figure 2.3.

OTHER INPUT AND OUTPUT COMBINATIONS The previous two examples indicate how two different frequency response functions are applicable to the same simple mechanical system, depending upon the type of input to be considered. Actually, a different frequency response function is generally required for every different combination of input and output parameters which might be desired. For example, the relative displacement output $z(t) = y(t) - x(t)$ to a foundation displacement input $x(t)$ might be of interest for some applications while the absolute acceleration output $\ddot{y}(t)$ to a foundation velocity input $\dot{x}(t)$ would be appropriate for others. A slightly different frequency response function would be required for each case. To illustrate this point, the various possible gain factors of the simple mechanical system in Figure 2.1 for twenty one different combinations of input and output parameters are presented in Table 2.1.

2.4.2 Electrical Systems

Assume a simple electrical circuit can be represented by a lumped parameter system consisting of an inductor, a resistor, and a capacitor. Further assume

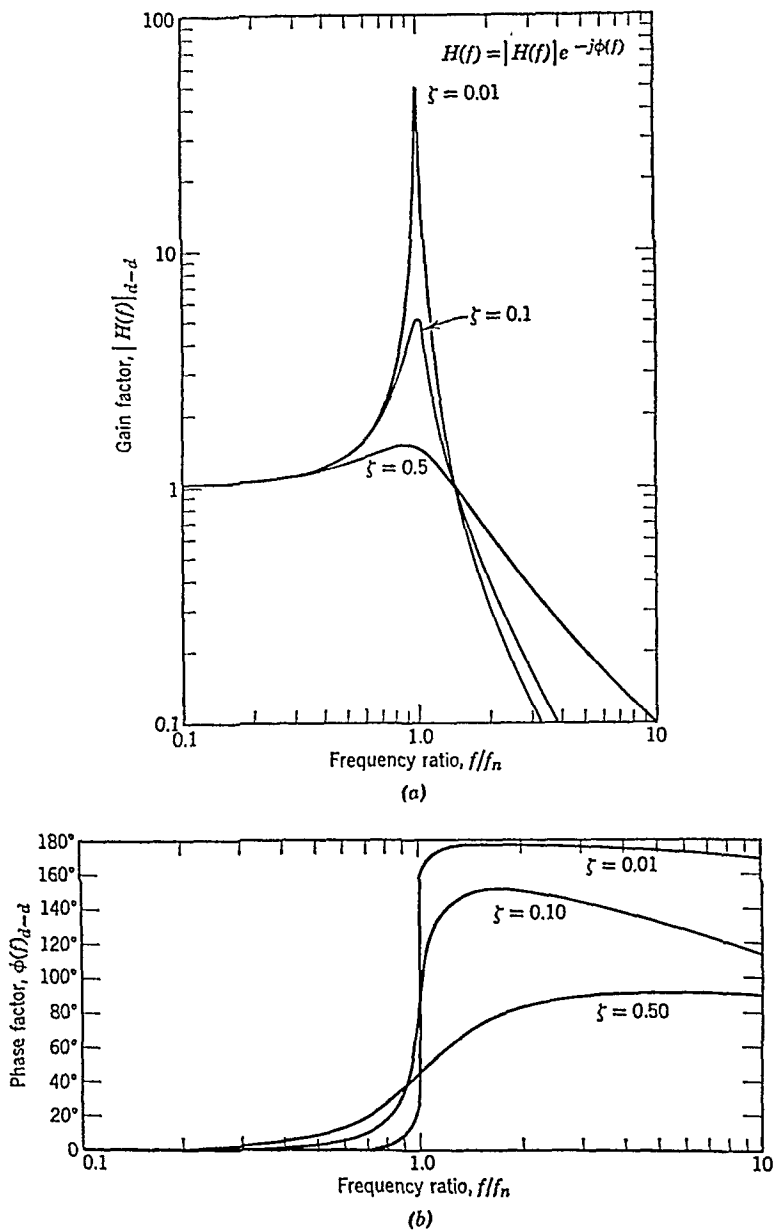


Figure 2.5 Frequency response function of mechanical system with foundation motion input. (a) Gain factor. (b) Phase factor.

Table 2.1

Summary of Gain Factors for Simple Mechanical System

Values for the Gain Factor $ H(f) $ of a Simple Mechanical System as a Function of the Input and Output Parameters		Foundation Motion Input (see Fig. 2.4 for model)			Force Input (see Fig. 2.2 for model)
		Displacement $x(t)$ in	Velocity $\dot{x}(t)$ in /sec	Acceleration $\ddot{x}(t)$ in /sec ²	Force (in displacement units) $x(t) = F(t)/k$ in
In terms of displacement output in	Absolute displacement $y(t)$	$\frac{D_1}{D_2}$	$\frac{D_1}{2\pi f D_2}$	$\frac{D_1}{4\pi^2 f^2 D_2}$	$\frac{1}{D_2}$
	Relative displacement $z(t) =$ $y(t) - x(t)$	$\frac{f^2}{f_n^2 D_2}$	$\frac{f}{2\pi f_n^2 D_2}$	$\frac{1}{4\pi^2 f_n^2 D_2}$	
In terms of velocity output in /sec	Absolute velocity $\dot{y}(t)$	$\frac{2\pi f D_1}{D^2}$	$\frac{D_1}{D_2}$	$\frac{D_1}{2\pi f D_2}$	$\frac{2\pi f}{D_2}$
	Relative velocity $\dot{z}(t) =$ $\dot{y}(t) - \dot{x}(t)$	$\frac{2\pi f^3}{f_n^2 D_2}$	$\frac{f^2}{f_n^2 D_2}$	$\frac{f}{2\pi f_n^2 D_2}$	
In terms of acceleration output in /sec ²	Absolute acceleration $\ddot{y}(t)$	$\frac{4\pi^2 f^2 D_1}{D^2}$	$\frac{2\pi f D_1}{D^2}$	$\frac{D_1}{D_2}$	$\frac{4\pi^2 f^2}{D_2}$
	Relative acceleration $\ddot{z}(t) =$ $\ddot{y}(t) - \ddot{x}(t)$	$\frac{4\pi^2 f^4}{f_n^2 D_2}$	$\frac{2\pi f^3}{f_n^2 D_2}$	$\frac{f^2}{f_n^2 D_2}$	
$D_1 = \sqrt{1 + [2\zeta(f/f_n)]^2}$		$D_2 = \sqrt{[1 - (f/f_n)^2]^2 + [2\zeta(f/f_n)]^2}$			
$f_n = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$		$\zeta = \frac{c}{2\sqrt{km}}$			

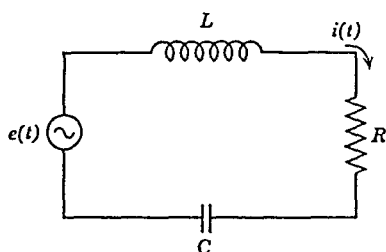


Figure 2.6 Electrical system with voltage input.

that the input to the system is a potential difference as shown in Figure 2.6. In this figure, C is a capacitance in farads, R is a resistance in ohms, L is an inductance in henries, $e(t)$ is an applied potential in volts, and $i(t)$ is the resulting current in amperes. Note that $i(t) = dq(t)/dt$ where $q(t)$ is charge in coulombs.

Assume the input of interest is an applied voltage and the output of interest is the resulting charge. As for the case of mechanical systems in Section 2.4.1, the first step toward establishing a proper frequency response function is to determine the differential equation describing the system. From basic circuit theory, the sum of all potential differences across the circuit elements must equal zero. That is,

$$e(t) + e_C(t) + e_R(t) + e_L(t) = 0 \quad (2.39)$$

where

$$e_C(t) = -\frac{1}{C} q(t) = \text{potential difference across capacitor} \quad (2.39a)$$

$$e_R(t) = -R \dot{q}(t) = \text{potential difference across resistor} \quad (2.39b)$$

$$e_L(t) = -L \ddot{q}(t) = \text{potential difference across inductor} \quad (2.39c)$$

Hence the differential equation for this system is

$$L \ddot{q}(t) + R \dot{q}(t) + \frac{1}{C} q(t) = e(t) \quad (2.40)$$

Note the similarity between Equation (2.40) and the equation of motion for a force excited mechanical system given by Equation (2.19). Using the same analysis procedures outlined in Section 2.4.1, it follows directly that the frequency response function of this simple electrical system is

$$H(f)_{e-q} = \left[\frac{1}{C} - (2\pi f)^2 L + j2\pi f R \right]^{-1} \quad (2.41)$$

where the subscript $e - q$ means that this particular $H(f)$ relates a voltage input to a charge output. Note that $H(f)_{e-q}$ has the units of coulombs/volt.

The plot for $H(f)_{e-e}$ would be identical to the plot for the mechanical frequency response function $H(f)_{f-e}$ presented in Figure 2.3, where the damping ratio ζ and the undamped natural frequency f_n of the electrical circuit are given as follows

$$\zeta = \frac{R}{2} \sqrt{\frac{C}{L}} \quad (2.42a)$$

$$f_n = \frac{1}{2\pi} \sqrt{\frac{1}{LC}} \quad (2.42b)$$

It should now be clear that a direct analogy may be made between mechanical and electrical systems as presented in Table 2.2

Table 2.2
Analogous Terms for Mechanical and Electrical Systems

Input	Electrical System with a Voltage Input	Mechanical System with a Force Input
	Voltage, $e(t)$	Force, $F(t)$
Output	Charge, $q(t)$ Current, $i(t) = dq/dt$	Displacement, $y(t)$ Velocity, $v(t) = dy/dt$
Constant parameters	Inductance, L Resistance, R Capacitance, C	Mass, m Damping, c Compliance, $1/k$

A more common frequency response function for electrical systems is one which relates a voltage input to a current output. This particular frequency response function is given by

$$H(f)_{e-i} = \left[R + j \left(2\pi f L - \frac{1}{2\pi f C} \right) \right]^{-1} \quad (2.43)$$

where $H(f)_{e-i}$ has the units of amperes/volt. The reciprocal of Equation (2.43), which may be denoted by $H(f)_{i-e}$, is called an *impedance function*

$$H(f)_{i-e} = R + j \left(2\pi f L - \frac{1}{2\pi f C} \right) \quad (2.44)$$

Note that the mechanical analogy to Equation (2.44) is given from Table 2.2 by $H(f)_{v-f}$ as follows.

$$H(f)_{v-f} = c + j\left(2\pi fm - \frac{k}{2\pi f}\right) \quad (2.45)$$

The function in Equation (2.45) is often called a mechanical impedance function because of its analogy to the common electrical impedance function.

2.4.3 Other Systems

By the same analytical procedures outlined in Section 2.4.1, an appropriate frequency response function can be developed, at least in theory, for any clearly defined stable physical system (that is, any system whose dynamic behavior can be described by an explicit equation). Moreover, the frequency response functions of different physical systems will often display analogous parameters, just as illustrated for mechanical and electrical systems in Section 2.4.2. A summary of analogous characteristics for several common physical systems is presented in Table 2.3.

Table 2.3
Analogous Characteristics for Several Physical Systems

System	Input	Output	Constant Parameters		
Electrical	Voltage	Current	Inductance	Resistance	Capacitance
Mechanical (translational)	Force	Velocity	Mass	Damping	Compliance
Mechanical (rotational)	Torque	Angular velocity	Moment of inertia	Angular damping	Angular compliance
Acoustical	Pressure	Particle velocity	Inertance (acoustical mass)	Acoustical damping	Acoustical capacitance
Thermo	Temperature	Heat flow	—	Thermo resistance	Thermo capacitance
Magnetic	Magneto- motive force	Flux	—	Reluctance	—

2.5 PRACTICAL CONSIDERATIONS

The analytical determination of frequency response functions of physical systems has been illustrated in Section 2.4. To facilitate the development and

clarification of basic ideas, examples were limited to simple mechanical and electrical systems. It should not be implied from these examples that the analytical determination of frequency response functions of physical systems is always so easy.

Consider, for example, a mechanical system in the form of a continuous elastic structure where the various parameters (mass, damping, and stiffness) are distributed rather than lumped as hypothetically assumed for the examples in Section 2.4.1. Such a mechanical system would have many different possible input and output points which might be of interest. Furthermore, the frequency response function of each input-output combination would generally display many peaks representing many resonant frequencies, as opposed to a single resonance as illustrated for the examples in Section 2.4.1. For relatively uncomplicated continuous structures such as beams, plates and shells, appropriate frequency response functions may still be established with reasonable accuracy by direct analytical procedures. For more complicated structures, however, considerable effort may be required to obtain even crude approximations for desired frequency response functions.

Because of these problems, frequency response functions of physical systems are often determined, or at least confirmed, by empirical procedures. The most straightforward empirical approach is to subject the system to a sinusoidal input, and measure the output amplitude and phase as the input frequency is varied. From Section 2.3, the ratio of the output to input amplitudes at any given frequency equals the gain factor, and the phase of the output relative to the input at any given frequency equals the phase factor.

Unfortunately, situations occur where it is not convenient to contrive a sinusoidal version of the input and/or apply it to the system of interest. For example, consider the problem of evaluating the dynamic response of a naval vessel to ocean wave loads, an aircraft to atmospheric gust loads, or a building structure to sonic boom pressure loads. The size of the systems in question and the complex distributed character of their appropriate input loads make the empirical determination of frequency response functions by application of controlled sinusoidal inputs difficult. Useful information might be obtained by applying a sinusoidal version of simulated loads to scale models of manageable size. On the other hand, it is sometimes more favorable to calculate the desired frequency response functions from input/output data measured during exposure of the physical system to its natural environment. The natural environments for the above noted examples are either random or transient, not sinusoidal. However, frequency response functions can be established by analysis of outputs to random and transient inputs, as will be developed in detail in Chapters 5 and 10.

PROBLEMS

- Write a single equation which defines the required conditions for linearity of physical systems.
- To define the dynamic properties of a system by a single-valued weighting function $h(\tau)$, which of the following requirements apply? The system must
 - have constant parameters.
 - be linear.
 - be physically realizable.
 - be stable.
- Prove that a physical system with a weighting function $h(\tau) = Ae^{a\tau}$ is unstable if $a > 0$.
- Determine the weighting function of the force excited mechanical system shown in Figure 2.2.
- Determine the frequency response function of a physical system with a weighting function $h(\tau) = Ae^{-a\tau}$ where $a > 0$.
- Assume the mechanical system shown in Figure 2.1 has a spring constant of $k = 10$ lb/inch, a viscous damping coefficient of $c = 0.2$ lb-sec/inch, and a mass of $m = 0.1$ lb-sec²/inch. Determine the
 - undamped natural frequency f_n .
 - damping ratio ζ .
 - force excited resonant frequency f_r .
 - peak value of the gain factor $|H(f)|$.
- Assume the mass of the mechanical system shown in Figure 2.1 is displaced from its position of equilibrium and then released. Prove that the time between crossings of the position of equilibrium in the resulting oscillation is $T = \frac{1}{2}[f_n\sqrt{1 - \zeta^2}]^{-1}$ for $\zeta^2 \leq 1.0$.
- Prove that the resonant frequency of the force excited mechanical system shown in Figure 2.2 is $f_r = f_n\sqrt{1 - 2\zeta^2}$; $\zeta^2 \leq 0.5$, as stated in Equation (2.25).
- The half power point bandwidth of a resonant physical system is defined as $B_r = f_2 - f_1$ where $|H(f_1)|^2 = |H(f_2)|^2 = \frac{1}{2}|H(f_r)|^2$. Given the force excited mechanical system shown in Figure 2.2, prove that $B_r \approx 2\zeta f_r$ for small ζ .
- Draw the electrical analog circuit for the mechanical system with foundation motion input shown in Figure 2.4. Determine the values of the analogous circuit parameters in terms of the mechanical parameters k , c , and m .

CHAPTER 3

REVIEW OF STATIONARY RANDOM PROCESSES THEORY

This chapter, an extension of Chapter 1, presents a review of basic mathematical methods for analyzing stationary random data. Elementary and advanced concepts from stationary random processes theory are discussed from first principles to form a proper foundation for the application of these ideas to analysis problems. The material includes probability fundamentals for random variables, stationary random processes, ergodic random processes, Gaussian random processes, linear transformations, and sampling theorems.

3.1 PROBABILITY FUNDAMENTALS FOR RANDOM VARIABLES

3.1.1 *One Random Variable*

The underlying concept in probability theory is that of a *set*, defined as a collection of objects (also called points or elements) about which it is possible to determine whether any particular object is a member of the set. In particular, the possible outcomes of an experiment (or a measurement) represent a set of *points* called the *sample space*. These points may be grouped together in various ways, called *events*, and under suitable conditions *probability functions* may be assigned to each. These probabilities always lie between zero and one, the probability of an impossible event being zero and the probability of the certain event being one. Sample spaces are either finite or infinite.

Consider a sample space of points representing the possible outcomes of a particular experiment (or measurement). A *random variable* $x(k)$ is a set function defined for points k from the sample space; that is, a random variable $x(k)$ is a real number between $-\infty$ and $+\infty$ which is associated to each sample point k that might occur. Stated another way, the random outcome of an experiment, indexed by k , can be represented by a real number $x(k)$, called the random variable. All possible experimental events which might occur constitute a completely additive class of sets, and a probability measure may be assigned to each event.

Let $x(k)$ denote a random variable of interest. Then for any fixed value of x , the random event $x(k) \leq x$ is defined as the set of possible outcomes k such that $x(k) \leq x$. In terms of the underlying probability measure in the sample space, one may define a *probability distribution function* $P(x)$ as the probability which is assigned to the set of points k satisfying the desired inequality $x(k) \leq x$. Observe that the set of points k satisfying $x(k) \leq x$ is a subset of the totality of all points k which satisfy $x(k) \leq \infty$. In notation form,

$$P(x) = \text{Prob } [x(k) \leq x] \quad (3.1)$$

Clearly,

$$P(a) \leq P(b) \quad \text{if } a \leq b \quad (3.2)$$

$$P(-\infty) = 0 \quad P(\infty) = 1 \quad (3.3)$$

If the random variable assumes a continuous range of values (which will be assumed hereafter), then a (first-order) *probability density function* $p(x)$ may be defined by the differential relation

$$p(x) = \lim_{\Delta x \rightarrow 0} \left[\frac{\text{Prob } [x < x(k) \leq x + \Delta x]}{\Delta x} \right] \quad (3.4)$$

It follows that

$$p(x) \geq 0 \quad (3.5)$$

$$\int_{-\infty}^{\infty} p(x) dx = 1 \quad (3.6)$$

$$P(x) = \int_{-\infty}^x p(\xi) d\xi \quad \frac{dP(x)}{dx} = p(x) \quad (3.7)$$

To handle discrete cases like Example 3.1, the probability density function $p(x)$ is permitted to include delta functions.

Example 3.1. Discrete Distribution. Suppose an experiment consists of tossing a single coin where the two possible outcomes, called heads and tails, are assumed to occur with equal probability ($\frac{1}{2}$). The random variable $x(k)$ for this example takes on only two discrete values, x (heads) and x (tails), to

which arbitrary real numbers may be assigned. Specifically, let x (heads) = a and x (tails) = b where a and b are real numbers with, say $b > a$. With these choices for $x(k)$, it follows that the probability distribution function is

$$P(x) = \begin{cases} 0 & x < a \\ \frac{1}{2} & a \leq x < b \\ 1 & x \geq b \end{cases}$$

and the probability density function is given by

$$p(x) = \frac{1}{2}\delta(x - a) + \frac{1}{2}\delta(x - b)$$

where $\delta(x - a)$ and $\delta(x - b)$ are delta functions

Example 3.2. Uniform (Rectangular) Distribution Suppose an experiment consists of choosing a point at random in the interval $[a, b]$, including the end points. A continuous random variable $x(k)$ for this example may be defined by the numerical value of the chosen point. The corresponding probability distribution function becomes

$$P(x) = \begin{cases} 0 & x < a \\ \frac{x - a}{b - a} & a \leq x \leq b \\ 1 & x > b \end{cases}$$

Hence the probability density function is given by

$$\begin{aligned} p(x) &= (b - a)^{-1} & a \leq x \leq b \\ &= 0 & \text{otherwise} \end{aligned}$$

This concludes Example 3.2

Assume the random variable $x(k)$ may take on values in the range $-\infty$ to $+\infty$. The *mean value* (also called *expected value* or *average value*) of $x(k)$ is obtained by an appropriate limiting operation when each value assumed by $x(k)$ is multiplied by its probability of occurrence. This gives

$$E[x(k)] = \int_{-\infty}^{\infty} x p(x) dx = \mu_x \quad (3.8)$$

where $E[\]$ represents the expected value over the index k of the term inside the brackets. Similarly, the expected value of any real single-valued continuous function $g(x)$ of the random variable $x(k)$ is given by

$$E[g(x(k))] = \int_{-\infty}^{\infty} g(x) p(x) dx \quad (3.9)$$

where $p(x)$ is the probability density function associated with $x(k)$. In particular, for $g(x) = x^2$, the *mean square value* of $x(k)$ is given by

$$E[x^2(k)] = \int_{-\infty}^{\infty} x^2 p(x) dx = \Psi_x^2 \quad (3.10)$$

The *variance* of $x(k)$ is defined by the mean square value of $x(k)$ about its mean value. Here, $g(x) = (x - \mu_x)^2$, and

$$E[(x(k) - \mu_x)^2] = \int_{-\infty}^{\infty} (x - \mu_x)^2 p(x) dx = \Psi_x^2 - \mu_x^2 = \sigma_x^2 \quad (3.11)$$

By definition, the *standard deviation* of $x(k)$, denoted by σ_x , is the positive square root of the variance. The standard deviation is measured in the same units as the mean value.

CHANGE OF VARIABLE. Suppose that $x(k)$ is a random variable with probability density function $p(x)$, and that $g(x)$ is any real single-valued continuous function of x . The probability density function $p(g)$ associated with the random variable $g(x(k)) = g(k)$ may be determined in terms of the probability density function $p(x)$ of $x(k)$ for cases where the derivative dg/dx exists and is not equal to zero. To allow for multivalued inverse functions, assume that the inverse function $x(g)$ is a real n -valued function of g , where n is an integer. For these cases,

$$\begin{aligned} \frac{\text{Prob } [g < g(x(k)) \leq g + \Delta g]}{\Delta g} &= \frac{n \text{ Prob } [x < x(k) \leq x + \Delta x]}{\Delta g} \\ &= \frac{n \text{ Prob } [x < x(k) \leq x + \Delta x]}{\Delta x} \cdot \frac{\Delta x}{\Delta g} \end{aligned} \quad (3.12)$$

Hence in the limit when $dg/dx \neq 0$,

$$p(g) = n p(x) \left| \frac{dx}{dg} \right| = \frac{n p(x)}{|dg/dx|} \quad (3.13)$$

When using this formula, it is necessary to replace the variable x on the right-hand side by its equivalent g .

Example 3.3. Sine Wave Distribution. A sine wave of fixed amplitude X and fixed frequency f_0 may be considered to be a random variable if its initial phase angle $\theta = \theta(k)$ is a random variable. In particular, consider t to be fixed at some value t_0 , and let the sine wave random variable be represented by

$$x(k) = x(\theta) = X \sin [2\pi f_0 t_0 + \theta(k)]$$

Suppose that $\theta(k)$ has a uniform probability density function $p(\theta)$ given by

$$p(\theta) = (2\pi)^{-1} \quad 0 \leq \theta \leq 2\pi$$

$$= 0 \quad \text{otherwise}$$

What is the sine wave probability density function $p(x)$ of $x(k)$?

For this example, the direct function $x(\theta)$ is single-valued, but the inverse function $\theta(x)$ is double-valued. From Equation (3.13), with θ replacing x , and x replacing g ,

$$p(x) = \frac{2p(\theta)}{dx/d\theta} \quad \text{for} \quad \frac{dx}{d\theta} \neq 0$$

where

$$\frac{dx}{d\theta} = X \cos(2\pi f_0 t_0 + \theta) = X \sqrt{1 - \sin^2(2\pi f_0 t_0 + \theta)} = \sqrt{X^2 - x^2}$$

Thus

$$p(x) = (\pi \sqrt{X^2 - x^2})^{-1} \quad |x| < X$$

$$= 0 \quad |x| \geq X \quad (3.14a)$$

which is the same as Equation (1.24). The corresponding sine wave probability distribution function is given by

$$P(x) = 0 \quad x < -X$$

$$= \int_{-X}^x p(\xi) d\xi = \frac{1}{\pi} \left(\frac{\pi}{2} + \sin^{-1} \frac{x}{X} \right) \quad -X \leq x \leq X \quad (3.14b)$$

$$= 1 \quad x > X$$

Example 3.4 Tchebycheff Inequality for Unknown Distribution Suppose that $x(k)$ is an arbitrary random variable with mean value μ_x , mean square value Ψ_x^2 and variance σ_x^2 . Suppose that its probability density function, which may be unknown, is $p(x)$. Then

$$\Psi_x^2 = \int_{-\infty}^{\infty} x^2 p(x) dx \geq \int_{|x| > \epsilon} x^2 p(x) dx \geq \epsilon^2 \int_{|x| > \epsilon} p(x) dx$$

since the integrand is non negative and since $x^2 \geq \epsilon^2$ at every point in the right hand region of integration. This proves

$$\text{Prob } [|x(k)| > \epsilon] = \int_{|x| > \epsilon} p(x) dx \leq \frac{\Psi_x^2}{\epsilon^2} \quad (3.15)$$

Replace now $x(k)$ by $x(k) - \mu_x$. Then Ψ_x^2 is replaced by σ_x^2 , and Eq. (3.15) becomes

$$\text{Prob } [|x(k) - \mu_x| \geq \epsilon] \leq \frac{\sigma_x^2}{\epsilon^2} \quad (3.15a)$$

In particular, if $\epsilon = c\sigma_x$, then

$$\text{Prob } [|x(k) - \mu_x| \geq c\sigma_x] \leq \frac{1}{c^2} \quad (3.15b)$$

which is equivalent to

$$\text{Prob } [|x(k) - \mu_x| \leq c\sigma_x] \geq 1 - \frac{1}{c^2} \quad (3.15c)$$

Any of the forms of Equation (3.15) are known as the *Tchebycheff inequality*.

3.1.2 Two Random Variables

Consider next two random variables $x(k)$ and $y(k)$ where k represents points in a suitable sample space. Let $P(x)$ and $P(y)$ be two distinct distribution functions associated with $x(k)$ and $y(k)$, respectively. The *joint probability distribution function* $P(x, y)$ is defined to be the probability which is associated with the subset of points k in the sample space satisfying simultaneously both of the inequalities $x(k) \leq x$ and $y(k) \leq y$. The totality of all points k satisfies the inequalities $x(k) \leq \infty$ and $y(k) \leq \infty$. In notation form,

$$P(x, y) = \text{Prob } [x(k) \leq x \text{ and } y(k) \leq y] \quad (3.16)$$

Clearly,

$$P(-\infty, y) = 0 = P(x, -\infty) \quad P(\infty, \infty) = 1 \quad (3.17)$$

As before, assuming the random variables to be continuous, the *joint probability density function* $p(x, y)$ is defined by

$$p(x, y) = \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \left[\frac{\text{Prob } [x < x(k) \leq x + \Delta x \text{ and } y < y(k) \leq y + \Delta y]}{\Delta x \Delta y} \right] \quad (3.18)$$

It follows that

$$p(x, y) \geq 0 \quad (3.19)$$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} p(x, y) dx dy = 1 \quad (3.20)$$

$$P(x, y) = \int_{-\infty}^y \int_{-\infty}^x p(\xi, \eta) d\xi d\eta \quad \frac{\partial}{\partial y} \left[\frac{\partial P(x, y)}{\partial x} \right] = p(x, y) \quad (3.21)$$

The probability density functions of $x(k)$ and $y(k)$ individually are obtained

from the joint probability density function by

$$\begin{aligned} p(x) &= \int_{-\infty}^{\infty} p(x, y) dy \\ p(y) &= \int_{-\infty}^{\infty} p(x, y) dx \end{aligned} \quad (3.22)$$

Now if

$$p(x, y) = p(x)p(y) \quad (3.23)$$

then the two random variables $x(k)$ and $y(k)$ are said to be *statistically independent*. It follows for statistically independent variables that

$$P(x, y) = P(x)P(y) \quad (3.24)$$

The *expected value* of any real single valued continuous function $g(x, y)$ of the two random variables $x(k)$ and $y(k)$ is given by

$$E[g(x, y)] = \iint_{-\infty}^{\infty} g(x, y) p(x, y) dx dy \quad (3.25)$$

For example, if $g(x, y) = (x(k) - \mu_x)(y(k) - \mu_y)$ where μ_x and μ_y are the mean values of $x(k)$ and $y(k)$, respectively, this defines the *covariance* C_{xy} between $x(k)$ and $y(k)$. That is

$$\begin{aligned} C_{xy} &= E[(x(k) - \mu_x)(y(k) - \mu_y)] = E[x(k)y(k)] - E[x(k)]E[y(k)] \\ &= \iint_{-\infty}^{\infty} (x - \mu_x)(y - \mu_y) p(x, y) dx dy \end{aligned} \quad (3.26)$$

Note that $C_{xx} = \sigma_x^2$, the variance of $x(k)$, as defined in Equation (3.11)

A simple relation exists between the covariance of $x(k)$ and $y(k)$ and the standard deviations of $x(k)$ and $y(k)$ as expressed by the inequality

$$|C_{xy}| \leq \sigma_x \sigma_y \quad (3.27)$$

Thus the magnitude of the covariance between $x(k)$ and $y(k)$ is less than or equal to the product of the standard deviation of $x(k)$ multiplied by the standard deviation of $y(k)$. This is proved later in Section 3.2.1

It follows from the above result that the normalized quantity

$$\rho_{xy} = \frac{C_{xy}}{\sigma_x \sigma_y} \quad (3.28)$$

known as the *correlation coefficient*, will lie between -1 and $+1$. Random variables $x(k)$ and $y(k)$ whose correlation coefficient is zero are said to be *uncorrelated*. This concept is quite distinct from the previous definition of

independent random variables. Note that if $x(k)$ and $y(k)$ are independent random variables, then, from Equation (3.23),

$$\begin{aligned} E[x(k) y(k)] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy p(x, y) dx dy \\ &= \int_{-\infty}^{\infty} x p(x) dx \int_{-\infty}^{\infty} y p(y) dy = E[x(k)] E[y(k)] \quad (3.29) \end{aligned}$$

Hence C_{xy} and, in turn, ρ_{xy} equal zero so that *independent random variables are also uncorrelated*. The converse statement is not true in general; that is, *uncorrelated random variables are not necessarily independent*. However, for physically important situations involving two or more normally (Gaussian) distributed random variables, being mutually uncorrelated does imply independence. This is proved later in Section 3.1.3.

Example 3.5. Distribution for Sum of Two Random Variables. Suppose $x(k)$ and $y(k)$ are two random variables with a joint probability density function $p(x, y)$. Determine the probability density function $p(z)$ of the sum random variable

$$z(k) = x(k) + y(k) \quad (3.30)$$

For each fixed value of x , the corresponding $y = z - x$. This gives

$$p(x, y) = p(x, z - x) \quad (3.31)$$

For each fixed value of z , the values of x may range from $-\infty$ to ∞ . Hence,

$$p(z) = \int_{-\infty}^{\infty} p(x, z - x) dx \quad (3.32)$$

which shows that the desired sum probability density function requires knowledge of the input joint probability density function. If $x(k)$ and $y(k)$ are independent random variables with probability density functions $p_1(x)$ and $p_2(y)$, respectively, then $p(x, y) = p_1(x)p_2(y) = p_1(x)p_2(z - x)$, and

$$p(z) = \int_{-\infty}^{\infty} p_1(x)p_2(z - x) dx \quad (3.33)$$

3.1.3 Gaussian (Normal) Distribution

A random variable $x(k)$ is said to follow a *Gaussian* (or *normal*) distribution if its probability density function is given by

$$p(x) = (b\sqrt{2\pi})^{-1} \exp \left[-\frac{(x - a)^2}{2b^2} \right] \quad (3.34)$$

where a is any real constant and b is any positive constant. It may be verified that a and b constitute the mean value and standard deviation of the random variable $x(k)$ since

$$E[x(k)] = \int_{-\infty}^{\infty} x p(x) dx = a = \mu_x$$

$$E[(x(k) - a)^2] = \int_{-\infty}^{\infty} (x - a)^2 p(x) dx = b^2 = \sigma_x^2$$

Thus the normal probability density function should be expressed by

$$p(x) = (\sigma_x \sqrt{2\pi})^{-1} \exp \left[-\frac{(x - \mu_x)^2}{2\sigma_x^2} \right] \quad (3.35a)$$

The normal probability distribution function is, by definition,

$$P(x) = (\sigma_x \sqrt{2\pi})^{-1} \int_{-\infty}^x \exp \left[-\frac{(\xi - \mu_x)^2}{2\sigma_x^2} \right] d\xi \quad (3.35b)$$

The importance of the normal distribution in physical problems may be attributed in part to the *Central Limit Theorem* [Refs. 30, 33], which asserts that this distribution will result quite generally from the sum of a large number of independent random variables acting together. To be a bit more specific, let $x_1(k)$, $x_2(k)$, ..., $x_N(k)$ be N mutually independent random variables whose individual distributions are not specified and may be different. Let μ_i and σ_i^2 be the mean value and variance of each random variable $x_i(k)$, $i = 1, 2, \dots, N$. Consider the sum random variable

$$x(k) = \sum_{i=1}^N a_i x_i(k)$$

where a_i are arbitrary fixed constants. Now, the mean value μ_x and variance σ_x^2 become, using the later result of Equation (3.137),

$$\mu_x = E[x(k)] = E \left[\sum_{i=1}^N a_i x_i(k) \right] = \sum_{i=1}^N a_i E[x_i(k)] = \sum_{i=1}^N a_i \mu_i$$

$$\sigma_x^2 = E[(x(k) - \mu_x)^2] = E \left[\sum_{i=1}^N a_i (x_i(k) - \mu_i) \right]^2 = \sum_{i=1}^N a_i^2 \sigma_i^2$$

The last expression is a result of the mutual independence of $x_i(k)$ with $x_j(k)$ for $i \neq j$. The Central Limit Theorem states that under fairly common conditions, the sum random variable $x(k)$ will be normally distributed as $N \rightarrow \infty$ with the above mean value μ_x and variance σ_x^2 .

Consider now N random variables $x_1(k), x_2(k), \dots, x_N(k)$ which may be correlated. Denote their respective mean values, variances, and covariances by

$$\begin{aligned}\mu_i &= E[x_i(k)] \\ \sigma_i^2 &= E[(x_i(k) - \mu_i)^2] \\ C_{ij} &= E[(x_i(k) - \mu_i)(x_j(k) - \mu_j)] \quad C_{ii} = \sigma_i^2\end{aligned}$$

Their joint distribution is said to follow an N -dimensional Gaussian (normal) distribution if the associated N -fold probability density function is given by

$$p(x_1, x_2, \dots, x_N) = \frac{\exp \left[\left(\frac{-1}{2|C|} \right) \sum_{i,j=1}^N |C_{ij}| (x_i - \mu_i)(x_j - \mu_j) \right]}{(2\pi)^{N/2} |C|^{1/2}} \quad (3.36)$$

where C is the covariance matrix of the C_{ij} defined below, $|C|$ is the determinant of C , and $|C_{ij}|$ is the cofactor of C_{ij} in determinant $|C|$. To be explicit,

$$C = \begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1N} \\ C_{21} & C_{22} & \cdots & C_{2N} \\ \vdots & \vdots & & \vdots \\ \vdots & \vdots & & \vdots \\ C_{N1} & C_{N2} & \cdots & C_{NN} \end{bmatrix}$$

and the cofactor $|C_{ij}|$ of any element C_{ij} is defined to be the determinant of order $N - 1$ formed by omitting the i th row and j th column of C , multiplied by $(-1)^{i+j}$.

The outstanding feature of the N -dimensional normal distribution is that all of its properties are determined solely from knowledge of the various mean values μ_i and covariances C_{ij} . For $N = 1$, this function reduces to

$$p(x_1) = (\sigma_1 \sqrt{2\pi})^{-1} \exp \left[-\frac{(x_1 - \mu_1)^2}{2\sigma_1^2} \right] \quad (3.36a)$$

which is the normal probability density function defined previously in Equation (3.35).

For $N = 2$, there results the joint normal probability density function

$$p(x_1, x_2) = \frac{\exp \left\{ \left[\frac{-1}{2(1 - \rho_{12}^2)} \right] \left[\left(\frac{x_1 - \mu_1}{\sigma_1} \right)^2 - 2\rho_{12} \left(\frac{x_1 - \mu_1}{\sigma_1} \right) \left(\frac{x_2 - \mu_2}{\sigma_2} \right) + \left(\frac{x_2 - \mu_2}{\sigma_2} \right)^2 \right] \right\}}{2\pi\sigma_1\sigma_2\sqrt{1 - \rho_{12}^2}} \quad (3.36b)$$

Table 3 1
Special Probability Density Functions

Type	Probability Density Function
Discrete	 $p(x) = A \delta(x - a) + B \delta(x - b) + \dots + N \delta(x - n)$ where $A + B + \dots + N = 1$
Uniform (rectangular)	 $p(x) = (b - a)^{-1}, \quad a \leq x \leq b, \quad \text{otherwise zero}$
Sine wave	 $p(x) = \left(\pi \sqrt{X^2 - x^2} \right)^{-1}, \quad x < X, \quad \text{otherwise zero}$
Gaussian (normal)	 $p(x) = \left(\sigma_x \sqrt{2\pi} \right)^{-1} e^{-(x - \mu_x)^2 / 2\sigma_x^2}$
Rayleigh	 $p(x) = \frac{x}{c^2} e^{-x^2/2c^2}, \quad x \geq 0, \quad \text{otherwise zero}$
Maxwell	 $p(x) = \frac{x^2}{c^3} \sqrt{\frac{2}{\pi}} e^{-x^2/2c^2}, \quad x \geq 0, \quad \text{otherwise zero}$
Truncated	 Assume original $p_1(x)$ defined over $(-\infty, \infty)$ Truncated $p(x) = C p_1(x) \quad a \leq x \leq b, \quad \text{otherwise zero}$ where $\int_{-\infty}^{\infty} p(x) dx = C \int_a^b p_1(x) dx = 1$
Clipped	 Assume original $p_1(x)$ defined over $(-\infty, \infty)$ Clipped $p(x) = p_1(x) \quad a < x < b$ $= A \delta(x - a) \quad x = a$ $= B \delta(x - b) \quad x = b$ $= 0 \quad x < a \text{ or } x > b$ where $\int_{-\infty}^{\infty} p(x) dx = \int_a^b p_1(x) dx + A + B = 1$

where $\rho_{12} = C_{12}/\sigma_1\sigma_2$ is the correlation coefficient between $x_1(k)$ and $x_2(k)$. Observe that when $x_1(k)$ and $x_2(k)$ are uncorrelated so that $\rho_{12} = 0$, one obtains

$$p(x_1, x_2) = p(x_1)p(x_2)$$

which shows that $x_1(k)$ and $x_2(k)$ are also independent. This result is not true for arbitrary distributions.

Similar formulas may be written down for higher-order cases where $N = 3, 4, 5, \dots$. For arbitrary N , it follows quite easily that if all different pairs of normally distributed random variables are mutually uncorrelated (that is, $\rho_{ij} = 0$ whenever $i \neq j$), then these random variables are mutually independent in the probability sense. That is,

$$p(x_1, x_2, \dots, x_N) = p(x_1)p(x_2) \cdots p(x_N)$$

The importance of the N -dimensional normal distribution in physical problems, analogous to the common one-dimensional normal distribution, is due in part to the *Multidimensional Central Limit Theorem*, [Refs. 30, 33]. This theorem yields the result that the vector sum of a large number of mutually independent N -dimensional random variables approaches an N -dimensional normal distribution under fairly general conditions.

A summary of equations for some special probability density functions which are used in theoretical studies is presented in Table 3.1.

3.2 STATIONARY RANDOM PROCESSES

A random process $\{x_k(t)\}$, $-\infty < t < \infty$ (also called a *time series* or *stochastic process*), denoted by the symbol $\{ \}$, is an ensemble of real-valued (or complex-valued) functions which can be characterized through its probability structure. For convenience, the variable t will be interpreted as time in the following discussion. Each particular function $x_k(t)$, where t is variable and k is fixed, is called a sample function. In practice, a sample function (or some time history record of finite length from a sample function) may be thought of as the observed result of a single experiment. The possible number of experiments represents a sample space of index k which may be countable or uncountable. For any number N and any fixed times $t_1, t_2, \dots, t_N$, the quantities $x_k(t_1), x_k(t_2), \dots, x_k(t_N)$, represent N random variables over the index k . It is required that there exist a well-defined N -dimensional probability distribution function for every N . An ensemble of sample functions forming a random process is illustrated in Figure 1.10.

A particular sample function $x_k(t)$, in general, would not be suitable for representing the entire random process $\{x_k(t)\}$ to which it belongs. Under certain conditions to be described later, however, it turns out that for the special class of ergodic random processes, it is possible to derive desired

statistical information about the entire random process from appropriate analysis of a single arbitrary sample function. For the situation of a pair of random processes $\{x_k(t)\}$ and $\{y_k(t)\}$, the corresponding problem is to estimate joint statistical properties of the two random processes from proper analysis of an arbitrary pair of sample functions $x_k(t)$ and $y_k(t)$.

Consider two arbitrary random processes $\{x_k(t)\}$ and $\{y_k(t)\}$. The first statistical quantities of interest are the ensemble *mean values* at arbitrary fixed values of t , where $x_k(t)$ and $y_k(t)$ are random variables over the index k . These are defined as in Equation (3.8) by

$$\begin{aligned}\mu_x(t) &= E[x_k(t)] \\ \mu_y(t) &= E[y_k(t)]\end{aligned}\quad (3.37)$$

In general, these mean values are different at different times, and must be calculated separately for every t of interest. That is,

$$\begin{aligned}\mu_x(t_1) &\neq \mu_x(t_2) & \text{if } t_1 \neq t_2 \\ \mu_y(t_1) &\neq \mu_y(t_2) & \text{if } t_1 \neq t_2\end{aligned}\quad (3.38)$$

The next statistical quantities of interest are the *covariance functions* at arbitrary fixed values of $t_1 = t$ and $t_2 = t + \tau$. These are defined by

$$C_x(t, t + \tau) = E[(x_k(t) - \mu_x(t))(x_k(t + \tau) - \mu_x(t + \tau))]\quad (3.39)$$

$$C_y(t, t + \tau) = E[(y_k(t) - \mu_y(t))(y_k(t + \tau) - \mu_y(t + \tau))]$$

$$C_{xy}(t, t + \tau) = E[(x_k(t) - \mu_x(t))(y_k(t + \tau) - \mu_y(t + \tau))]\quad (3.40)$$

In general, these quantities are different for different combinations of t_1 and t_2 . Observe that at $\tau = 0$ ($t_1 = t_2 = t$),

$$C_x(t, t) = E[(x_k(t) - \mu_x(t))^2] = \sigma_x^2(t)\quad (3.41)$$

$$C_y(t, t) = E[(y_k(t) - \mu_y(t))^2] = \sigma_y^2(t)$$

$$C_{xy}(t, t) = E[(x_k(t) - \mu_x(t))(y_k(t) - \mu_y(t))] = C_{xy}(t)\quad (3.42)$$

Thus the covariance functions $C_x(t, t)$ and $C_y(t, t)$ represent the ordinary variances of $\{x_k(t)\}$ and $\{y_k(t)\}$ at a fixed value of t , while $C_{xy}(t, t)$ represents the covariance between $\{x_k(t)\}$ and $\{y_k(t)\}$. As before, different results would generally be obtained for different values of t .

Other statistical quantities can be defined over the ensemble which involve fixing three or more times. The probability structure of the random processes is thus described in finer and finer detail. However, if $\{x_k(t)\}$ and $\{y_k(t)\}$ form a two-dimensional Gaussian distribution at a fixed value of t , then $\{x_k(t)\}$ and $\{y_k(t)\}$ are separately Gaussian. The mean values and covariance functions

listed above then provide a complete description of the underlying probability structure. For this reason, the main emphasis in this chapter is concerned only with these two statistical quantities and their relationships to spectral density functions.

If the mean values $\mu_x(t)$ and $\mu_y(t)$, together with the covariance functions $C_x(t, t + \tau)$, $C_y(t, t + \tau)$, and $C_{xy}(t, t + \tau)$, yield the same results for all fixed values of t (that is, are independent of time translations), then the random processes $\{x_k(t)\}$ and $\{y_k(t)\}$ are said to be *weakly stationary*. If all possible probability distributions involving $\{x_k(t)\}$ and $\{y_k(t)\}$ are independent of time translations, then the processes are said to be *strongly stationary*. Since the mean values and covariance functions are consequences only of the first- and second-order probability distributions, it follows that the class of strongly stationary random processes is a subclass of the class of weakly stationary random processes. For Gaussian random processes, however, weak stationarity implies strong stationarity since all possible probability distributions may be derived from the mean values and covariance functions. Thus, for Gaussian random processes, these two stationary concepts coincide.

3.2.1 Correlation (Covariance) Functions

For stationary random processes $\{x_k(t)\}$ and $\{y_k(t)\}$, which will be considered henceforth in this chapter, the mean values become constants independent of t . That is, for all t ,

$$\begin{aligned}\mu_x &= E[x_k(t)] = \int_{-\infty}^{\infty} x p(x) dx \\ \mu_y &= E[y_k(t)] = \int_{-\infty}^{\infty} y p(y) dy\end{aligned}\tag{3.43}$$

where $p(x)$ and $p(y)$ are the probability density functions associated with the random variables $x_k(t)$ and $y_k(t)$, respectively. The covariance functions of stationary random processes are also independent of t .

For arbitrary fixed t and τ , define

$$R_x(\tau) = E[x_k(t) x_k(t + \tau)]\tag{3.44}$$

$$R_y(\tau) = E[y_k(t) y_k(t + \tau)]$$

$$R_{xy}(\tau) = E[x_k(t) y_k(t + \tau)]\tag{3.45}$$

where R is introduced instead of C to distinguish between these special expressions and those presented in Equations (3.39) and (3.40). For nonzero mean values, R is different from C . The quantities $R_x(\tau)$ and $R_y(\tau)$ are called the *autocorrelation functions* of $\{x_k(t)\}$ and $\{y_k(t)\}$, respectively, whereas $R_{xy}(\tau)$ is called the *cross-correlation function* between $\{x_k(t)\}$ and $\{y_k(t)\}$.

A necessary and sufficient condition that $R_x(\tau)$ be the autocorrelation function of a weakly stationary random process $\{x_k(t)\}$ is that $R_x(\tau) = R_x(-\tau)$, and that $R_x(\tau)$ be a non-negative definite function, [Ref. 15]. One can prove also that $R_x(\tau)$ will be a continuous function of τ if it is continuous at the origin. Similarly the cross-correlation function $R_{xy}(\tau)$ will be continuous for all τ if $R_x(\tau)$ or $R_y(\tau)$ is continuous at the origin.

For a pair of stationary random processes $\{x_k(t)\}$ and $\{y_k(t)\}$, the joint probability density function $p(x_1, x_2)$ of the pair of random variables, $x_1 = x_k(t)$ and $x_2 = x_k(t + \tau)$, is independent of t , and the joint probability density function $p(y_1, y_2)$ associated with the pair of random variables $y_1 = y_k(t)$ and $y_2 = y_k(t + \tau)$ is independent of t . This is also true for the joint probability density function $p(x_1, y_2)$ associated with the pair of random variables $x_1 = x_k(t)$ and $y_2 = y_k(t + \tau)$. In terms of these probability density functions

$$\begin{aligned} R_x(\tau) &= \iint_{-\infty}^{\infty} x_1 x_2 p(x_1, x_2) dx_1 dx_2 \\ R_y(\tau) &= \iint_{-\infty}^{\infty} y_1 y_2 p(y_1, y_2) dy_1 dy_2 \\ R_{xy}(\tau) &= \iint_{-\infty}^{\infty} x_1 y_2 p(x_1, y_2) dx_1 dy_2 \end{aligned} \quad (3.46)$$

For arbitrary values of μ_x and μ_y , the covariance functions are related to the correlation functions by the equations

$$\begin{aligned} C_x(\tau) &= R_x(\tau) - \mu_x^2 \\ C_y(\tau) &= R_y(\tau) - \mu_y^2 \\ C_{xy}(\tau) &= R_{xy}(\tau) - \mu_x \mu_y \end{aligned} \quad (3.47)$$

Thus correlation functions are identical with covariance functions when the mean values are zero. Note that, by definition, two stationary random processes are uncorrelated if $C_{xy}(\tau) = 0$ for all τ . This occurs, from Equation (3.47), whenever $R_{xy}(\tau) = \mu_x \mu_y$ for all τ . Hence, the two processes will be uncorrelated when $R_{xy}(\tau) = 0$ for all τ only if also either μ_x or μ_y equals zero.

From the stationary hypothesis, it follows that the autocorrelation functions $R_x(\tau)$ and $R_y(\tau)$ are even functions of τ . That is,

$$\begin{aligned} R_x(-\tau) &= R_x(\tau) \\ R_y(-\tau) &= R_y(\tau) \end{aligned} \quad (3.48)$$

while the cross-correlation function is neither odd nor even, but satisfies the relation

$$R_{xy}(-\tau) = R_{yx}(\tau) \quad (3.49)$$

Examples of autocorrelation and cross-correlation functions are given in Figures 1.15 and 1.20, and in Table 3.2.

CROSS-CORRELATION INEQUALITY. An upper bound for the cross-correlation function is given by the inequality

$$|R_{xy}(\tau)|^2 \leq R_x(0) R_y(0) \quad (3.50)$$

which may be proved as follows. For any real constants a and b , the expected value

$$E[(a x(t) + b y(t + \tau))^2] \geq 0$$

since only non-negative quantities are being considered. This is equivalent to

$$a^2 R_x(0) + 2ab R_{xy}(\tau) + b^2 R_y(0) \geq 0$$

Hence, assuming $b \neq 0$,

$$\left(\frac{a}{b}\right)^2 R_x(0) + 2\left(\frac{a}{b}\right) R_{xy}(\tau) + R_y(0) \geq 0$$

This is a quadratic equation in a/b without real different roots since one side is non-negative. Therefore, the discriminant of this quadratic equation in a/b must be nonpositive. That is,

$$\text{Discriminant} = 4R_{xy}^2(\tau) - 4R_x(0) R_y(0) \leq 0$$

Thus

$$R_{xy}^2(\tau) = |R_{xy}(\tau)|^2 \leq R_x(0) R_y(0)$$

as stated in Equation (3.50). This completes the proof.

By considering $x(t) - \mu_x$ and $y(t + \tau) - \mu_y$ instead of $x(t)$ and $y(t + \tau)$, the same proof gives the cross-covariance inequality

$$|C_{xy}(\tau)|^2 \leq C_x(0) C_y(0) \quad (3.51)$$

When $x(t) = y(t)$, one obtains $|R_x(\tau)| \leq R_x(0)$ and $|C_x(\tau)| \leq C_x(0)$ for all τ . It follows from Equation (3.51) that a *correlation function coefficient (normalized cross-covariance function)* may be defined by

$$\rho_{xy}(\tau) = \frac{C_{xy}(\tau)}{\sqrt{C_x(0) C_y(0)}} \quad (3.52)$$

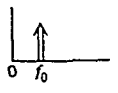
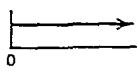
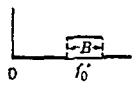
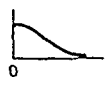
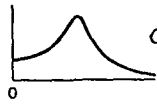
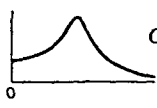
and satisfies, for all τ ,

$$-1 \leq \rho_{xy}(\tau) \leq 1 \quad (3.53)$$

Table 3 2
Special Autocorrelation Functions

Type	Autocorrelation Function	
Constant		$R_x(\tau) = c^2$
Sine wave		$R_x(\tau) = \frac{X^2}{2} \cos 2\pi f_0 \tau$
White noise		$R_x(\tau) = a \delta(\tau)$
Low pass white noise		$R_x(\tau) = a B \left(\frac{\sin 2\pi B \tau}{2\pi B \tau} \right)$
Band pass white noise		$R_x(\tau) = a B \left(\frac{\sin \pi B \tau}{\pi B \tau} \right) \cos 2\pi f_0 \tau$
Exponential		$R_x(\tau) = e^{-a \tau }$
Exponential cosine		$R_x(\tau) = e^{-a \tau } \cos 2\pi f_0 \tau$
Exponential cosine exponential sine		$R_x(\tau) = e^{-a \tau } (b \cos 2\pi f_0 \tau + c \sin 2\pi f_0 \tau)$

Table 3.3
Special Power Spectral Density Functions

Type	(One-Sided) Power Spectral Density Function
Constant	 $G_x(f) = c^2 \delta(f)$
Sine wave	 $G_x(f) = \frac{X^2}{2} \delta(f - f_0)$
White noise	 $G_x(f) = a, f \geq 0; \quad \text{otherwise zero}$
Low-pass white noise	 $G_x(f) = a, \quad 0 \leq f \leq B; \quad \text{otherwise zero}$
Band-pass white noise	 $G_x(f) = a, \quad 0 < f_0 - (B/2) \leq f \leq f_0 + (B/2) \quad \text{otherwise zero}$
Exponential	 $G_x(f) = \frac{4a}{a^2 + 4\pi^2 f^2}$
Exponential cosine	 $G_x(f) = 2a \left[\frac{1}{a^2 + 4\pi^2 (f + f_0)^2} + \frac{1}{a^2 + 4\pi^2 (f - f_0)^2} \right]$
Exponential cosine, exponential sine	 $G_x(f) = \frac{2ab + 4\pi c(f + f_0)}{a^2 + 4\pi^2 (f + f_0)^2} + \frac{2ab - 4\pi c(f - f_0)}{a^2 + 4\pi^2 (f - f_0)^2}$

If the mean values μ_x and μ_y are zero, then $\rho_{xy}(\tau)$ becomes

$$\rho_{xy}(\tau) = \frac{R_{xy}(\tau)}{\sqrt{R_x(0) R_y(0)}} \quad (3.54)$$

The function $\rho_{xy}(\tau)$ measures the degree of linear dependence between $\{x_k(t)\}$ and $\{y_k(t)\}$ for a displacement of τ in $\{y_k(t)\}$ relative to $\{x_k(t)\}$.

In summary, the correlation properties of stationary random processes $\{x_k(t)\}$, $\{y_k(t)\}$ may be described by the four functions $R_x(\tau)$, $R_y(\tau)$, $R_{xy}(\tau)$, and $R_{yx}(\tau)$. These need be calculated only for values of $\tau \geq 0$, since symmetry properties from Equations (3.48) and (3.49) yield results for $\tau < 0$.

Example 3.6. Autocorrelation Function of Sine Wave Process Suppose $\{x_k(t)\} = \{X \sin [2\pi f_0 t + \theta(k)]\}$ is a sine wave process in which X and f_0 are constants and $\theta(k)$ is a random variable with a uniform probability density function $p(\theta)$ over $(0, 2\pi)$. Determine the autocorrelation function $R_x(\tau)$.

Here, for any fixed value of t , the random variable

$$x_k(t) = X \sin [2\pi f_0 t + \theta(k)] = x_1(\theta)$$

$$x_k(t + \tau) = X \sin [2\pi f_0(t + \tau) + \theta(k)] = x_2(\theta)$$

From Equation (3.44),

$$R_x(\tau) = E[x_k(t) x_k(t + \tau)] = E[x_1(\theta) x_2(\theta)]$$

with

$$p(\theta) = (2\pi)^{-1} \quad 0 \leq \theta \leq 2\pi \quad \text{otherwise zero}$$

Hence

$$\begin{aligned} R_x(\tau) &= \frac{X^2}{2\pi} \int_0^{2\pi} \sin(2\pi f_0 t + \theta) \sin[2\pi f_0(t + \tau) + \theta] d\theta \\ &= \frac{X^2}{2} \cos 2\pi f_0 \tau \end{aligned} \quad (3.55)$$

giving the autocorrelation function of a sine wave stationary random process, as stated in Equation (1.31)

Example 3.7 Autocorrelation Function of Rectangular Wave Process. Consider a physical situation where a sample function $x_k(t)$ from a random rectangular wave process $\{x_k(t)\}$ is restricted so as to assume only values of c or $-c$, where the number of changes of sign in an interval $(t, t + \tau)$ occur at random and independent times with an average density of λ . Assume also that what happens inside an interval $(t, t + \tau)$ is independent of what happens outside the interval. Define

$$A_n = \text{Event [exactly } n \text{ changes of sign fall inside } (t, t + \tau)]$$

This type of physical situation follows a Poisson distribution [Ref. 1], where the probability of event A_n is

$$P(A_n) = \frac{(\lambda |\tau|)^n}{n!} e^{-\lambda |\tau|}$$

Determine the autocorrelation function of $\{x_k(t)\}$.

The autocorrelation function may be calculated as follows. An individual product term $x_k(t) x_k(t + \tau)$ equals c^2 if $x_k(t)$ and $x_k(t + \tau)$ are of the same sign, and it equals $-c^2$ if they are of opposite sign. The total probability for c^2 is given by the sum $P(A_0) + P(A_2) + P(A_4) + \dots$, and the total probability for $-c^2$ is given by the sum $P(A_1) + P(A_3) + P(A_5) + \dots$. Hence

$$\begin{aligned} R_x(\tau) &= E[x_k(t) x_k(t + \tau)] = c^2 \sum_{n=0}^{\infty} (-1)^n P(A_n) \\ &= c^2 e^{-\lambda |\tau|} \sum_{n=1}^{\infty} (-1)^n \frac{(\lambda |\tau|)^n}{n!} = c^2 e^{-2\lambda |\tau|} \end{aligned} \quad (3.56)$$

Example 3.8. Autocorrelation Function of Sum of Two Stationary Processes. Assume that a random process $\{y_k(t)\}$ is the sum of two stationary processes $\{x_{1,k}(t)\}$ and $\{x_{2,k}(t)\}$ such that each sample function

$$y_k(t) = a_1 x_{1,k}(t) + a_2 x_{2,k}(t)$$

where a_1 and a_2 are constants. Assume also that $\{x_{1,k}(t)\}$ and $\{x_{2,k}(t)\}$ may be correlated. Determine the autocorrelation function $R_y(\tau)$.

From Equation (3.44), one obtains

$$\begin{aligned} R_y(\tau) &= E[y_k(t) y_k(t + \tau)] \\ &= E[(a_1 x_{1,k}(t) + a_2 x_{2,k}(t))(a_1 x_{1,k}(t + \tau) + a_2 x_{2,k}(t + \tau))] \\ &= a_1^2 E[x_{1,k}(t) x_{1,k}(t + \tau)] + a_1 a_2 E[x_{1,k}(t) x_{2,k}(t + \tau)] \\ &\quad + a_1 a_2 E[x_{2,k}(t) x_{1,k}(t + \tau)] + a_2^2 E[x_{2,k}(t) x_{2,k}(t + \tau)] \\ &= a_1^2 R_{x_1}(\tau) + a_1 a_2 [R_{x_1 x_2}(\tau) + R_{x_2 x_1}(\tau)] + a_2^2 R_{x_2}(\tau) \end{aligned}$$

Thus the sum autocorrelation function requires knowledge of the input cross-correlation functions as well as their autocorrelation functions.

3.2.2 Spectral Density Functions

Spectral density functions are defined in the practical context of filtering, squaring, and averaging operations on sample records in Sections 1.3.4 and 1.4.3. The approach described in those sections leads to *one-sided* spectral density functions which are defined for frequencies over $(0, \infty)$ and denoted by $G(f)$. A more direct method of defining spectral density functions is in

terms of Fourier transforms of the correlation functions discussed in Section 3.2.1. This approach yields *two-sided* spectral density functions which are defined for frequencies $(-\infty, \infty)$ and denoted by $S(f)$. Specifically, let

$$S(f) = \int_{-\infty}^{\infty} R(\tau) e^{-j2\pi f\tau} d\tau \quad (3.57)$$

which will exist if $R(\tau)$ exists and if

$$\int_{-\infty}^{\infty} |R(\tau)| d\tau < \infty \quad (3.58)$$

The inverse Fourier transform of $S(f)$ gives

$$R(\tau) = \int_{-\infty}^{\infty} S(f) e^{j2\pi f\tau} df \quad (3.59)$$

It will now be assumed that the auto and cross correlation functions $R_x(\tau)$, $R_y(\tau)$, and $R_{xy}(\tau)$ exist, as defined in Equations (3.44) and (3.45). The *two-sided* power and cross spectral density functions are then given by

$$S_x(f) = \int_{-\infty}^{\infty} R_x(\tau) e^{-j2\pi f\tau} d\tau \quad (3.60)$$

$$S_y(f) = \int_{-\infty}^{\infty} R_y(\tau) e^{-j2\pi f\tau} d\tau$$

$$S_{xy}(f) = \int_{-\infty}^{\infty} R_{xy}(\tau) e^{-j2\pi f\tau} d\tau \quad (3.61)$$

To handle practical problems, both $R(\tau)$ and $S(f)$ are permitted to include delta functions. The results shown in Equations (3.60) and (3.61), and in equivalent expressions which follow, are known as the "Wiener-Khinchine" relations.

From the symmetry properties of stationary correlation functions, given in Equations (3.48) and (3.49), it follows that

$$S_x(-f) = S_x(f) \quad (3.62)$$

$$S_y(-f) = S_y(f)$$

$$S_{xy}(-f) = S_{xy}^*(f) = S_{yx}(f) \quad (3.63)$$

These equations state that two-sided power spectral density functions are real, non-negative, and even functions of f , whereas cross-spectral density

functions are complex-valued functions of f . The relations of Equation (3.60) may be simplified to

$$\begin{aligned} S_x(f) &= \int_{-\infty}^{\infty} R_x(\tau) \cos 2\pi f\tau \, d\tau = 2 \int_0^{\infty} R_x(\tau) \cos 2\pi f\tau \, d\tau \\ S_y(f) &= \int_{-\infty}^{\infty} R_y(\tau) \cos 2\pi f\tau \, d\tau = 2 \int_0^{\infty} R_y(\tau) \cos 2\pi f\tau \, d\tau \end{aligned} \quad (3.64)$$

Conversely,

$$\begin{aligned} R_x(\tau) &= 2 \int_0^{\infty} S_x(f) \cos 2\pi f\tau \, df \\ R_y(\tau) &= 2 \int_0^{\infty} S_y(f) \cos 2\pi f\tau \, df \end{aligned} \quad (3.65)$$

The *one-sided* power spectral density functions, $G_x(f)$ and $G_y(f)$, where f varies only over $(0, \infty)$, are defined by

$$\begin{aligned} G_x(f) &= 2S_x(f) \quad 0 \leq f < \infty \quad \text{otherwise zero} \\ G_y(f) &= 2S_y(f) \quad 0 \leq f < \infty \quad \text{otherwise zero} \end{aligned} \quad (3.66)$$

These are the quantities measured by direct filtering procedures in practice. For mathematical calculations, however, the use of $S_x(f)$ and $S_y(f)$ defined over $(-\infty, \infty)$ and exponentials with imaginary exponents often simplifies the analysis. It is important to be able to deal properly with both of these representations, and both will be used in this book. See Figure 3.1 for a graphical illustration of the relationship.

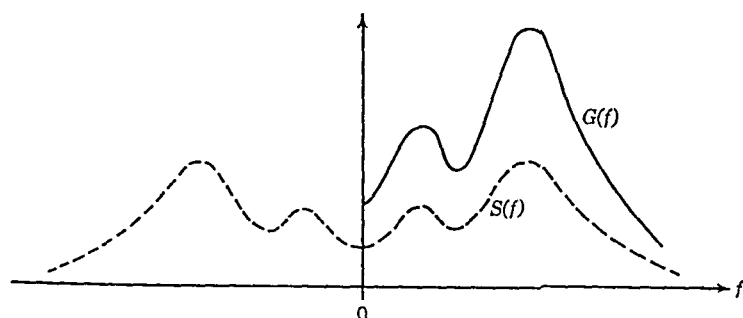


Figure 3.1 One-sided and two-sided spectral density functions.

In terms of the one sided power spectral density functions $G_x(f)$ and $G_y(f)$, the correspondence with the stationary correlation functions $R_x(\tau)$ and $R_y(\tau)$ becomes,

$$\begin{aligned} G_x(f) &= 4 \int_0^{\infty} R_x(\tau) \cos 2\pi f\tau \, d\tau \quad 0 \leq f < \infty \\ G_y(f) &= 4 \int_0^{\infty} R_y(\tau) \cos 2\pi f\tau \, d\tau \quad 0 \leq f < \infty \end{aligned} \quad (3.67)$$

Conversely,

$$\begin{aligned} R_x(\tau) &= \int_0^{\infty} G_x(f) \cos 2\pi f\tau \, df \\ R_y(\tau) &= \int_0^{\infty} G_y(f) \cos 2\pi f\tau \, df \end{aligned} \quad (3.68)$$

The one sided cross-spectral density function, $G_{xy}(f)$, where f varies only over $(0, \infty)$ is defined by

$$G_{xy}(f) = 2S_{xy}(f) \quad 0 \leq f < \infty \quad \text{otherwise zero} \quad (3.69)$$

From Equation (3.61),

$$G_{xy}(f) = 2 \int_{-\infty}^{\infty} R_{xy}(\tau) e^{-j2\pi f\tau} \, d\tau = C_{xy}(f) - jQ_{xy}(f) \quad (3.70)$$

where $C_{xy}(f)$ is called the coincident spectral density function (*co-spectrum*), and $Q_{xy}(f)$ is called the quadrature spectral density function (*quad spectrum*). In terms of $C_{xy}(f)$ and $Q_{xy}(f)$, the cross correlation function

$$R_{xy}(\tau) = \int_0^{\infty} [C_{xy}(f) \cos 2\pi f\tau + Q_{xy}(f) \sin 2\pi f\tau] \, df \quad (3.71)$$

From Equations (3.70) and (3.49), one obtains

$$\begin{aligned} C_{xy}(f) &= 2 \int_0^{\infty} [R_{xy}(\tau) + R_{yx}(\tau)] \cos 2\pi f\tau \, d\tau = C_{yx}(-f) \\ Q_{xy}(f) &= 2 \int_0^{\infty} [R_{xy}(\tau) - R_{yx}(\tau)] \sin 2\pi f\tau \, d\tau = -Q_{yx}(-f) \end{aligned} \quad (3.72)$$

Thus $C_{xy}(f)$ is a real valued even function of f , whereas $Q_{xy}(f)$ is a real-valued odd function of f . In terms of $G_{xy}(f)$ and $G_{yx}(f)$,

$$\begin{aligned} C_{xy}(f) &= \frac{1}{2} [G_{xy}(f) + G_{yx}(f)] \\ Q_{xy}(f) &= \frac{j}{2} [G_{xy}(f) - G_{yx}(f)] \end{aligned} \quad (3.73)$$

An alternative way to describe $S_{xy}(f)$ is by the complex polar form

$$S_{xy}(f) = |S_{xy}(f)| e^{-j\theta_{xy}(f)} \quad -\infty < f < \infty \quad (3.74)$$

containing an absolute magnitude and phase angle. Similarly,

$$G_{xy}(f) = |G_{xy}(f)| e^{-j\theta_{xy}(f)} \quad 0 \leq f < \infty \quad (3.75)$$

In terms of $C_{xy}(f)$ and $Q_{xy}(f)$,

$$\begin{aligned} |G_{xy}(f)| &= \sqrt{C_{xy}^2(f) + Q_{xy}^2(f)} \\ \theta_{xy}(f) &= \tan^{-1} \left[\frac{Q_{xy}(f)}{C_{xy}(f)} \right] \end{aligned} \quad (3.76)$$

A cross-spectrum inequality analogous to the cross-correlation inequality of Equation (3.50) is proved shortly. Examples of power spectral and cross-spectral density functions are displayed in Figures 1.16 and 1.21, and in Table 3.3.

When dealing with delta functions at the origin, it is convenient to let the lower limit of integration, zero, be approached from below. In particular, for $R(\tau) = c^2$, this allows the corresponding $G(f)$ to be represented by $G(f) = c^2 \delta(f)$. For this situation, $S(f)$ is also given by $S(f) = c^2 \delta(f)$ showing that the factor of 2 in Equations (3.66) and (3.69) should not be applied to delta functions at the origin. Equivalent relations for delta functions and constants are as follows.

$$\begin{aligned} R(\tau) &= a \delta(\tau) + b \\ S(f) &= a + b \delta(f) \\ G(f) &= 2a + b \delta(f) \end{aligned} \quad (3.77)$$

In summary, the spectral properties of stationary random processes $\{x_k(t)\}$ and $\{y_k(t)\}$ may be described by the three functions $S_x(f)$, $S_y(f)$, $S_{xy}(f)$, or by the four functions $S_x(f)$, $S_y(f)$, $C_{xy}(f)$, $Q_{xy}(f)$, which need be calculated only for $f \geq 0$, since the symmetry properties from Equations (3.62), (3.63), and (3.72) yield results for $f < 0$.

CROSS-SPECTRUM INEQUALITY. Consider any two filtered records $x(t)$ and $y(t)$ where each is passed through ideal narrow-band filters of arbitrary center frequency f_0 and narrow bandwidth Δf . It follows that the spectral density functions of the filtered records can be represented by $G_x(f)$ and $G_y(f)$ where

$$\begin{aligned} G_x(f) &\neq 0 & \text{only for} & & f_0 - (\Delta f/2) \leq f \leq f_0 + (\Delta f/2) \\ G_y(f) &\neq 0 & \text{only for} & & f_0 - (\Delta f/2) \leq f \leq f_0 + (\Delta f/2) \end{aligned} \quad (3.78)$$

Also the cross spectral density function between these filtered records can be represented by $G_{xy}(f)$ where

$$G_{xy}(f) \neq 0 \quad \text{only for} \quad f_0 - (\Delta f/2) \leq f \leq f_0 + (\Delta f/2) \quad (3.79)$$

Now the cross correlation function inequality of Equation (3.50) at the value $\tau = 0$ can be applied to these filtered records. At $\tau = 0$ this gives

$$|R_{xy}(0)|^2 \leq R_x(0) R_y(0) \quad (3.80)$$

From Equations (3.68) and (3.78)

$$R_x(0) = \int_0^\infty G_x(f) df = \int_{f_0 - (\Delta f/2)}^{f_0 + (\Delta f/2)} G_x(f) df \approx G_x(f_0) \Delta f \quad (3.81)$$

Similarly

$$R_y(0) \approx G_y(f_0) \Delta f \quad \text{and} \quad R_{xy}(0) \approx G_{xy}(f_0) \Delta f \quad (3.82)$$

The results of Equations (3.81) and (3.82) improve as Δf tends to zero. Substitution of Equations (3.81) and (3.82) into Equation (3.80) then yields

$$|G_{xy}(f_0) \Delta f|^2 \leq [G_x(f_0) \Delta f][G_y(f_0) \Delta f]$$

which simplifies to

$$|G_{xy}(f_0)|^2 \leq G_x(f_0) G_y(f_0)$$

Since the choice of f_0 is arbitrary it follows that for any f

$$|G_{xy}(f)|^2 \leq G_x(f) G_y(f) \quad (3.83a)$$

which may also be written as

$$|S_{xy}(f)|^2 \leq S_x(f) S_y(f) \quad (3.83b)$$

Observe that the cross spectrum inequality of Equation (3.83) is a more powerful result than the cross correlation inequality of Equation (3.50). This is because the magnitude of $|G_{xy}(f)|$ is bounded at any value of f in terms of the associated values of $G_x(f)$ and $G_y(f)$ at this same value of f . However the magnitude of $|R_{xy}(\tau)|$ is bounded at any value of τ only in terms of the associated values $R_x(0)$ and $R_y(0)$ at $\tau = 0$. From Equation (3.83) a normalized cross spectral density function known as the coherence function may be defined. This is discussed in Chapter 5.

Example 3.9 Fourier Transforms of Constants and Periodic Functions

The cases of a constant and a periodic function can be given special treatment by using delta functions. If $x_k(t) = 1$ a constant then the Fourier transform relation

$$1 = \int_{-\infty}^{\infty} X_k(f) e^{2\pi i f t} df \quad (3.84a)$$

implies that

$$X_k(f) = \delta(f) = \int_{-\infty}^{\infty} e^{-j2\pi ft} dt \quad (3.84b)$$

where $\delta(f)$ is the *delta function* which has the properties

$$\delta(f) = 0 \quad \text{for } f \neq 0 \quad \delta(0) = \infty \quad (3.85a)$$

$$\int_{-\epsilon}^{\epsilon} \delta(f) df = 1 \quad \text{for any } \epsilon > 0 \quad \delta(-f) = \delta(f) \quad (3.85b)$$

$$\int_{-\epsilon}^{\epsilon} F(f) \delta(f) df = F(0) \quad \text{for any function } F(f) \quad (3.85c)$$

If $x_k(t) = e^{j2\pi f_0 t}$, a complex-valued periodic function, then

$$e^{j2\pi f_0 t} = \int_{-\infty}^{\infty} X_k(f) e^{j2\pi ft} df$$

Hence

$$1 = \int_{-\infty}^{\infty} X_k(f) e^{j2\pi(f-f_0)t} df = \int_{-\infty}^{\infty} X_k(f+f_0) e^{j2\pi ft} df$$

which yields

$$X_k(f+f_0) = \delta(f)$$

equivalent to

$$X_k(f) = \delta(f-f_0)$$

Now, for a cosine function

$$x_c(t) = \cos 2\pi f_0 t = \frac{1}{2}(e^{j2\pi f_0 t} + e^{-j2\pi f_0 t}) \quad (3.86a)$$

It follows that its Fourier transform is

$$X_c(f) = \frac{1}{2}[\delta(f-f_0) + \delta(f+f_0)] \quad (3.86b)$$

Similarly, a sine function

$$x_s(t) = \sin 2\pi f_0 t = \frac{1}{2j}(e^{j2\pi f_0 t} - e^{-j2\pi f_0 t}) \quad (3.87a)$$

has a Fourier transform

$$X_s(f) = \frac{1}{2j}[\delta(f-f_0) - \delta(f+f_0)] \quad (3.87b)$$

Example 3.10. Bandwidth Limited White Noise. By definition, bandwidth limited white noise is a random process with a constant power spectral density function defined by

$$\begin{aligned} G_x(f) &= a & 0 \leq f_0 - (B/2) \leq f \leq f_0 + (B/2) \\ &= 0 & \text{otherwise} \end{aligned} \quad (3.88a)$$

where f_0 is the center frequency, B is the bandwidth, and a is a constant. From Equation (3.68), it follows that the associated autocorrelation function is

$$R_x(\tau) = \int_{f_0 - (B/2)}^{f_0 + (B/2)} a \cos 2\pi f \tau \, df = a B \left(\frac{\sin \pi B \tau}{\pi B \tau} \right) \cos 2\pi f_0 \tau \quad (3.88b)$$

For the special case where $f_0 = B/2$, these equations become

$$\begin{aligned} G_x(f) &\approx a & 0 \leq f \leq B \\ &= 0 & \text{otherwise} \end{aligned} \quad (3.89a)$$

and

$$R_x(\tau) = a B \left(\frac{\sin 2\pi B \tau}{2\pi B \tau} \right) \quad (3.89b)$$

In a theoretical limiting case when $G_x(f) = a$ for all frequencies, the process is called *white noise*. For this case, $S_x(f) = (a/2)$ and $R_x(\tau) = (a/2) \delta(\tau)$.

Example 3.11 Power Spectrum of Sum of Two Stationary Processes
Determine the one-sided power spectral density function of the sum of the two stationary random processes considered in Example 3.8. From Example 3.8, the sum autocorrelation function

$$R_y(\tau) = a_1^2 R_{x_1}(\tau) + a_1 a_2 [R_{x_1 x_2}(\tau) + R_{x_2 x_1}(\tau)] + a_2^2 R_{x_2}(\tau)$$

It follows from Equations (3.67) and (3.70) that

$$G_y(f) = a_1^2 G_{x_1}(f) + a_1 a_2 [G_{x_1 x_2}(f) + G_{x_2 x_1}(f)] + a_2^2 G_{x_2}(f)$$

Thus the sum power spectral density function requires knowledge of the input cross spectral density functions as well as their power spectral density functions.

3.2.3 Spectral Density via Finite Fourier Transforms

Alternate equivalent ways to define spectral density functions will now be developed. Consider a pair of associated sample records $x_k(t)$ and $y_k(t)$ from stationary random processes $\{x_k(t)\}$ and $\{y_k(t)\}$. For a finite time interval $0 \leq t \leq T$ define

$$S_{xy}(f, T, k) = \frac{1}{T} X_k^*(f, T) Y_k(f, T) \quad (3.90)$$

where

$$\begin{aligned} X_k(f, T) &= \int_0^T x_k(t) e^{-j2\pi f t} dt \\ Y_k(f, T) &= \int_0^T y_k(t) e^{-j2\pi f t} dt \end{aligned} \quad (3.91)$$

The quantities $X_k(f, T)$ and $Y_k(f, T)$ represent finite Fourier transforms of $x_k(t)$ and $y_k(t)$ respectively, and $X_k^*(f, T)$ is the complex conjugate of $X_k(f, T)$. These finite-range Fourier transforms will exist for general stationary records, whereas their infinite-range Fourier transforms would not exist since the stationary data theoretically persist forever.

A common mistake made by many people is now to use analogies from periodic data to define the cross-spectral density function by

$$S_{xy}(f, k) = \lim_{T \rightarrow \infty} S_{xy}(f, T, k) \quad (3.92)$$

This is an *unsatisfactory* definition for general stationary random data because the estimate of $S_{xy}(f, k)$ by $S_{xy}(f, T, k)$ does not improve in the statistical sense of consistency (defined in Section 4.1) as T tends to infinity. Also, observe that the left-hand side is still a function of the index k . The correct way to define $S_{xy}(f)$ is by the expression

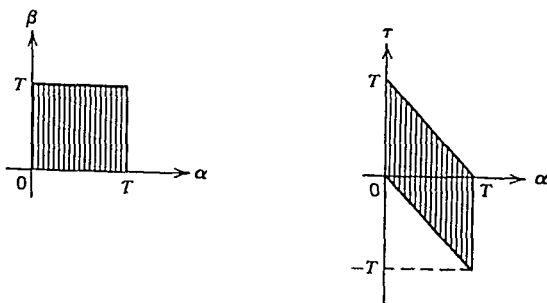
$$S_{xy}(f) = \lim_{T \rightarrow \infty} E[S_{xy}(f, T, k)] \quad (3.93)$$

where $E[S_{xy}(f, T, k)]$ is, of course, the expected value operation over the ensemble index k in question. This result of Equation (3.93) will now be proven.

Using different variables of integration to avoid confusion, Equation (3.90) becomes

$$\begin{aligned} S_{xy}(f, T, k) &= \frac{1}{T} \int_0^T x_k(\alpha) e^{j2\pi f \alpha} d\alpha \int_0^T y_k(\beta) e^{-j2\pi f \beta} d\beta \\ &= \frac{1}{T} \int_0^T \int_0^T x_k(\alpha) y_k(\beta) e^{-j2\pi f(\beta - \alpha)} d\alpha d\beta \end{aligned} \quad (3.94)$$

Now, change the region of integration from (α, β) to (α, τ) where $\tau = \beta - \alpha$, $d\tau = d\beta$. This changes the limits of integration as shown in the sketch below.



Proceed by integrating in the order (α, τ) instead of the order (α, β) . This leads to

$$\int_0^T \int_0^T dx d\beta = \int_{-T}^0 \int_{-\tau}^T dx d\tau + \int_0^T \int_0^{T-\tau} dx d\tau \quad (3.95)$$

As a check, it is readily verified that both sides of Equation (3.95) yield the value T^2 . Hence this change of the region of integration allows Equation (3.94) to take the form

$$S_{xy}(f, T, k) = \int_{-T}^0 \left[\frac{1}{T} \int_{-\tau}^T x_k(\alpha) y_k(\alpha + \tau) d\alpha \right] e^{-j2\pi f\tau} d\tau \\ + \int_0^T \left[\frac{1}{T} \int_0^{T-\tau} x_k(\alpha) y_k(\alpha + \tau) d\alpha \right] e^{-j2\pi f\tau} d\tau \quad (3.96)$$

By definition, the cross-correlation function $R_{xy}(\tau)$ is given by the expected value

$$R_{xy}(\tau) = E[x_k(\alpha) y_k(\alpha + \tau)] \quad (3.97)$$

The expected value of both sides of Equation (3.96) then gives

$$E[S_{xy}(f, T, k)] \\ = \int_{-T}^0 \left[\frac{1}{T} \int_{-\tau}^T R_{xy}(\tau) d\alpha \right] e^{-j2\pi f\tau} d\tau + \int_0^T \left[\frac{1}{T} \int_0^{T-\tau} R_{xy}(\tau) d\alpha \right] e^{-j2\pi f\tau} d\tau \\ = \int_{-T}^T \left(1 - \frac{|\tau|}{T} \right) R_{xy}(\tau) e^{-j2\pi f\tau} d\tau \quad (3.98)$$

In the limit as T tends to infinity, it follows that

$$\lim_{T \rightarrow \infty} E[S_{xy}(f, T, k)] = \int_{-\infty}^{\infty} R_{xy}(\tau) e^{-j2\pi f\tau} d\tau \quad (3.99)$$

This is the desired result of Equation (3.93) since the right-hand side of Equation (3.99) is $S_{xy}(f)$ as previously defined in Equation (3.61).

Observe that when $S(f)$ is replaced by its corresponding $G(f)$, the following formulas are obtained as special cases

$$G_{xy}(f) = 2 \lim_{T \rightarrow \infty} \frac{1}{T} E[X_k^*(f, T) Y_k(f, T)] \quad (3.100)$$

$$G_x(f) = 2 \lim_{T \rightarrow \infty} \frac{1}{T} E[|X_k(f, T)|^2] \quad (3.101)$$

$$G_y(f) = 2 \lim_{T \rightarrow \infty} \frac{1}{T} E[|Y_k(f, T)|^2]$$

where $X_k(f, T)$ and $Y_k(f, T)$ are defined in Equation (3.91). These formulas are estimated in fast Fourier transform digital computer procedures for the computation of spectral density functions, discussed later in Chapter 9.

3.2.4 Spectral Density via Filtering-Squaring-Averaging

The equivalence of the practical definition for spectral density functions in terms of filtering, squaring, and averaging operations, as given in Chapter 1, is also in order here. In particular, consider the power spectrum case where

$$G_x(f) = 2 \lim_{T \rightarrow \infty} \frac{1}{T} E[|X_k(f, T)|^2] \quad (3.102)$$

Although not obvious, the term

$$|X_k(f, T)|^2 = \left[\int_0^T x_k(t) \cos 2\pi f t \, dt \right]^2 + \left[\int_0^T x_k(t) \sin 2\pi f t \, dt \right]^2 \quad (3.103)$$

acts as a filter to help obtain the mean square value in any $x_k(t)$ associated with a narrow frequency band around the center frequency f . To derive this result, define for each $x_k(t)$ in $\{x_k(t)\}$

$$\begin{aligned} x_k(t, T) &= x_k(t) & 0 \leq t \leq T \\ &= 0 & \text{otherwise} \end{aligned} \quad (3.104)$$

Then the mean square value of a particular $x_k(t)$ may be determined by

$$\Psi_x^2(k) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T x_k^2(t) \, dt = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\infty}^{\infty} x_k^2(t, T) \, dt \quad (3.105)$$

By Parseval's Theorem, if $F(f)$ is the Fourier transform of $f(t)$, then

$$\int_{-\infty}^{\infty} f^2(t) \, dt = \int_{-\infty}^{\infty} |F(f)|^2 \, df \quad (3.106)$$

a relation which is an easy exercise to prove. Hence, since $X_k(f, T)$ is the Fourier transform for $x_k(t, T)$, namely,

$$X_k(f, T) = \int_0^T x_k(t) e^{-j2\pi f t} \, dt = \int_{-\infty}^{\infty} x_k(t, T) e^{-j2\pi f t} \, dt \quad (3.107)$$

it follows that

$$\Psi_x^2(k) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\infty}^{\infty} |X_k(f, T)|^2 \, df = 2 \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^{\infty} |X_k(f, T)|^2 \, df \quad (3.108)$$

Now, the expected value of $\Psi_x^2(k)$ over all possible records $x_k(t)$ in $\{x_k(t)\}$ yields the familiar formula

$$\Psi_x^2 = E[\Psi_x^2(k)] = \int_0^\infty G_x(f) df \quad (3.109)$$

where $G_x(f)$ is defined by Equation (3.102)

Next, suppose that $x_k(t)$ is passed through a narrow band filter of center frequency f_c and bandwidth Δf such that the frequency response function $H(f)$ of the filter is

$$\begin{aligned} H(f) &= 1 \quad \text{for } 0 \leq f_c - (\Delta f/2) \leq f \leq f_c + (\Delta f/2) \\ &= 0 \quad \text{otherwise} \end{aligned} \quad (3.110)$$

Then, the Fourier transform of the filter output is given by $H(f) X_k(f, T)$ instead of by $X_k(f, T)$. Hence in place of Equation (3.108), the mean square value of a particular filtered $x_k(t)$ becomes

$$\Psi_x^2(f_c, \Delta f, k) = 2 \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^\infty |H(f)|^2 |X_k(f, T)|^2 df \quad (3.111)$$

The expected value of both sides of Eq. (3.111) gives

$$\Psi_x^2(f_c, \Delta f) = \int_0^\infty |H(f)|^2 G_x(f) df = \int_{f_c - (\Delta f/2)}^{f_c + (\Delta f/2)} G_x(f) df \quad (3.112)$$

In words, Equation (3.112) states that $G_x(f)$ is the rate of change of mean square value with frequency. Furthermore, the term $|X_k(f, T)|^2$ must be acting as a filter on $x_k(t)$ which passes only a narrow band of frequency components in a certain range, and then squares these outputs prior to the final desired averaging operations. This is precisely the basis on which analog power spectral density analyzers are designed, as discussed in Chapter 8. A similar treatment can be carried out to prove the equivalence of digital and analog cross spectral density methods covered in this book.

3.3 ERGODIC RANDOM PROCESSES

Consider two weakly stationary random processes $\{x_k(t)\}$ and $\{y_k(t)\}$ with two arbitrary sample functions $x_k(t)$ and $y_k(t)$. These stationary random processes are said to be *weakly ergodic* if the mean values and covariance (correlation) functions, which are defined by certain *ensemble averages* in Section 3.2.1, may be calculated by performing corresponding *time averages* on the arbitrary pair of sample functions. In this way the underlying statistical structure of the weakly stationary random processes may be determined quite simply from an available sample pair without the need for collecting a considerable amount of data.

To be more specific, the mean values of the individual sample functions $x_k(t)$ and $y_k(t)$, when computed by a time average, may be represented by

$$\begin{aligned}\mu_x(k) &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T x_k(t) dt \\ \mu_y(k) &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T y_k(t) dt\end{aligned}\quad (3.113)$$

Observe that the answer is no longer a function of t since t has been averaged out. In general, however, the answer is a function of the particular sample function chosen, as denoted by the index k .

The cross-covariance function and cross-correlation function between $x_k(t)$ and $y_k(t + \tau)$, when computed by a time average, are defined by the expression

$$\begin{aligned}C_{xy}(\tau, k) &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T [x_k(t) - \mu_x(k)][y_k(t + \tau) - \mu_y(k)] dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T x_k(t) y_k(t + \tau) dt - \mu_x(k) \mu_y(k) \\ &= R_{xy}(\tau, k) - \mu_x(k) \mu_y(k)\end{aligned}\quad (3.114)$$

The autocovariance functions and autocorrelation functions are defined by

$$\begin{aligned}C_x(\tau, k) &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T [x_k(t) - \mu_x(k)][x_k(t + \tau) - \mu_x(k)] dt \\ &= R_x(\tau, k) - \mu_x^2(k) \\ C_y(\tau, k) &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T [y_k(t) - \mu_y(k)][y_k(t + \tau) - \mu_y(k)] dt \\ &= R_y(\tau, k) - \mu_y^2(k)\end{aligned}\quad (3.115)$$

These quantities should now be compared with the previously defined ensemble mean values μ_x , μ_y , and ensemble covariance functions $C_x(\tau)$, $C_y(\tau)$, $C_{xy}(\tau)$ for stationary random processes developed in Section 3.2.1. If it turns out that, independent of k ,

$$\begin{aligned}\mu_x(k) &= \mu_x \\ \mu_y(k) &= \mu_y\end{aligned}\quad (3.116)$$

$$\begin{aligned}C_x(\tau, k) &= C_x(\tau) \\ C_y(\tau, k) &= C_y(\tau) \\ C_{xy}(\tau, k) &= C_{xy}(\tau)\end{aligned}\quad (3.117)$$

then the random processes $\{x_k(t)\}$ and $\{y_k(t)\}$ are said to be *weakly ergodic*. If all ensemble averaged statistical properties of $\{x_k(t)\}$ and $\{y_k(t)\}$, not just the means and covariances, are deducible from corresponding time averages, then the random processes are said to be *strongly ergodic*. Thus strong ergodicity implies weak ergodicity but not conversely. No distinction between these concepts exists for Gaussian random processes.

For an arbitrary random process to be ergodic it must first be stationary. Each sample function must then be representative of all the others in the sense described above so that it does not matter which particular sample function is used in the time-averaging calculations.

Example 3.12 Nonergodic Stationary Random Process A simple example of a nonergodic stationary random process follows. Consider a hypothetical random process $\{x_k(t)\}$ composed of sinusoidal sample functions such that

$$\{x_k(t)\} = \{X(k) \sin [2\pi ft + \theta(k)]\}$$

Let the amplitude $X(k)$ and the phase angle $\theta(k)$ be random variables which take on a different set of values for each sample function. If $\theta(k)$ is uniformly distributed, the properties of the process computed over the ensemble at specific times will be independent of time, hence the process is stationary. However, the properties computed by time averaging over individual sample functions are not always the same. For example, the autocovariance (or autocorrelation) function for each sample function is given here by

$$C_x(\tau, k) = \frac{X^2(k)}{2} \sin 2\pi f\tau$$

Since $X(k)$ is a function of k , $C_x(\tau, k) \neq C_x(\tau)$. Hence the random process is nonergodic. This concludes Example 3.12.

SUFFICIENT CONDITIONS FOR ERGODICITY There are two important classes of random processes which one can state in advance will be ergodic. The first ergodic class is the class of stationary Gaussian random processes whose power spectral density functions are absolutely continuous [Ref. 15], that is, no delta functions appear in the power spectra corresponding to infinite mean square densities at discrete frequencies. The second ergodic class (a special case of the first class) is the class of stationary Gaussian Markov processes. A Markov process is one whose relationship to the past does not extend beyond the immediately preceding observation. The autocorrelation function of a stationary Gaussian Markov process may be shown to be of a simple exponential form, [Ref. 15].

Sufficient conditions for a random process to be ergodic are as follows.

I. A sufficient condition for an arbitrary random process to be weakly ergodic is that it be weakly stationary, and that the time averages $\mu_x(k)$ and $C_x(\tau, k)$ be the same for all sample functions k .

The proof of this result is as follows. By definition,

$$\mu_x(k) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T x_k(t) dt$$

By hypothesis, $\mu_x(k)$ is independent of k . Hence the expected value over k is the same as an individual estimate, namely,

$$E[\mu_x(k)] = \mu_x(k)$$

Also, as will be proved in Equation (3.137), expected values commute with linear operations. Hence

$$\begin{aligned} E[\mu_x(k)] &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T E[x_k(t)] dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \mu_x dt = \mu_x \end{aligned}$$

The assumption of weak stationarity is used in setting $E[x_k(t)] = \mu_x$. Thus

$$\mu_x(k) = \mu_x$$

Similarly,

$$C_x(\tau, k) = C_x(\tau)$$

since the hypothesis that $C_x(\tau, k)$ is independent of k yields

$$E[C_x(\tau, k)] = C_x(\tau, k)$$

while the stationary hypothesis yields

$$E[C_x(\tau, k)] = C_x(\tau)$$

This completes the proof.

II. A sufficient condition for a Gaussian random process to be ergodic is that it be weakly stationary, and the autocovariance function $C_x(\tau)$ has the following four integrable properties.

$$\begin{aligned} \int_{-\infty}^{\infty} |C_x(\tau)| d\tau < \infty & \quad \int_{-\infty}^{\infty} C_x^2(\tau) d\tau < \infty \\ \int_{-\infty}^{\infty} |\tau C_x(\tau)| d\tau < \infty & \quad \int_{-\infty}^{\infty} |\tau| C_x^2(\tau) d\tau < \infty \end{aligned} \tag{3.118a}$$

The four conditions of Equation (3 118a) can be replaced by the single requirement that

$$\frac{1}{T} \int_{-T}^T |C_x(\tau)| d\tau \rightarrow 0 \quad \text{as} \quad T \rightarrow \infty \quad (3 118b)$$

The proof of this result is contained in later Sections 6 2 1 and 6 2 2, where it is shown that mean value and autocorrelation function estimates produced by time averages are independent of the particular sample record when Equation (3 118) is satisfied. Result II then follows from the preceding Result I. In practice, these conditions are frequently satisfied, justifying the assumption of ergodicity.

3 4 GAUSSIAN RANDOM PROCESSES

The formal definition of a Gaussian random process is as follows. A random process $\{x_k(t)\}$ is said to be a Gaussian random process if, for every set of fixed times $\{t_n\}$, the random variables $x_k(t_n)$ follow a multidimensional normal distribution as defined by Equation (3 36). Gaussian random processes are quite prevalent in physical problems, and often may be mathematically predicted by the multidimensional Central Limit Theorem. Also, it can be shown that if a Gaussian process undergoes a linear transformation, then the output will still be a Gaussian process. This property is quite important in various theoretical and practical applications of random process theory.

Consider a time history $x(t)$ which is a sample function from an ergodic Gaussian random process with zero mean value. Note that the index k is no longer needed since the properties of any one sample function will be representative of all other sample functions. From the ergodic property, the timewise behavior of $x(t)$ over a long period of time will exhibit the same statistical characteristics as corresponding ensemble averages at various fixed times. As a consequence, it follows that the probability density function associated with the instantaneous values of $x(t)$ that will occur over a long time interval is given by the Gaussian probability density function with zero mean value, as follows

$$p(x) = (\sigma_x \sqrt{2\pi})^{-1} e^{-x^2/2\sigma_x^2} \quad (3 119)$$

The variance σ_x^2 when $x(t)$ has zero mean is determined by

$$\begin{aligned} \sigma_x^2 &= E[x^2(t)] = \int_{-\infty}^{\infty} x^2 p(x) dx \quad \text{independent of } t \\ &\approx \frac{1}{T} \int_0^T x^2(t) dt \quad \text{for large } T \\ &= \int_{-\infty}^{\infty} S_x(f) df = 2 \int_0^{\infty} S_x(f) df = \int_0^{\infty} G_x(f) df \end{aligned} \quad (3 120)$$

Thus the Gaussian probability density function $p(x)$ is completely characterized through knowledge of $S_x(f)$ or $G_x(f)$ since they alone determine σ_x . This important result places knowledge of $S_x(f)$ or $G_x(f)$ at the forefront of much work in analysis of random records. It should be noted that no restriction is placed on the shape of the power spectral density function or its associated autocorrelation function.

If the mean value of $x(t)$ is not zero, then the underlying probability density function is given by the general Gaussian formula

$$p(x) = (\sigma_x \sqrt{2\pi})^{-1} e^{-(x-\mu_x)^2/2\sigma_x^2} \quad (3.121)$$

where the mean value

$$\begin{aligned} \mu_x &= E[x(t)] = \int_{-\infty}^{\infty} x p(x) dx && \text{independent of } t \\ &\approx \frac{1}{T} \int_0^T x(t) dt && \text{for large } T \end{aligned} \quad (3.122)$$

and the variance

$$\sigma_x^2 = E[x(t) - \mu_x]^2 = E[x^2(t)] - \mu_x^2 \quad (3.123)$$

The *central moments* for any probability distribution are defined by

$$m_n' = E[(x - \mu_x)^n] = \int_{-\infty}^{\infty} (x - \mu_x)^n p(x) dx \quad (3.124)$$

where the prime is used to denote central moments as opposed to ordinary moments. For the Gaussian case, where $p(x)$ is given by Equation (3.121),

$$\begin{aligned} m_n' &= 0 && \text{if } n \text{ is an odd integer} \\ m_{2n}' &= 1 \cdot 3 \cdot 5 \cdots (2n-1) \sigma_x^{2n} && \text{for } n = 0, 1, 2, 3, \dots \end{aligned} \quad (3.125)$$

Thus

$$m_0' = 1 \quad m_2' = \sigma_x^2 \quad m_4' = 3\sigma_x^4 \quad \text{etc.}$$

Assume $\{x(t)\}$ is a stationary Gaussian random process where the index k is omitted for simplicity in notation. Consider the two random variables $x_1 = x(t)$ and $x_2 = x(t + \tau)$ at an arbitrary pair of fixed times t and $t + \tau$. Assume that x_1 and x_2 follow a two-dimensional (joint) Gaussian distribution with zero means and equal variances σ_x^2 . By definition, then

$$\sigma_x^2 = E[x^2(t)] = E[x^2(t + \tau)] = \int_{-\infty}^{\infty} x^2 p(x) dx \quad (3.126)$$

$$R_x(\tau) = E[x(t) x(t + \tau)] = \rho_x(\tau) \sigma_x^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 x_2 p(x_1, x_2) dx_1 dx_2 \quad (3.127)$$

where the quantity $\rho_x(\tau)$ is defined by Equation (3.127) as $R_x(\tau)/\sigma_x^2$. Here, letting $\rho = \rho_x(\tau)$ and $\mu = 0$, the joint Gaussian probability density function is given by a special case of Equation (3.36b) as

$$p(x_1, x_2) = \left(2\pi\sigma_x^2\sqrt{1-\rho^2}\right)^{-1} \exp\left[\frac{-1}{2\sigma_x^2(1-\rho^2)}(x_1^2 - 2\rho x_1x_2 + x_2^2)\right] \quad (3.128)$$

One can prove directly that

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} p(x_1, x_2) dx_1 dx_2 = 1 \quad (3.129)$$

Also, it follows that

$$p(x_1) = \int_{-\infty}^{\infty} p(x_1, x_2) dx_2 = (\sigma_x\sqrt{2\pi})^{-1} e^{-x_1^2/2\sigma_x^2} \quad (3.130)$$

with a similar result for $p(x_2)$. By definition, the two variables x_1 and x_2 are uncorrelated if $\rho = 0$. When $\rho = 0$, Equation (3.128) becomes

$$p(x_1, x_2) = p(x_1)p(x_2) \quad (3.131)$$

Consider four random variables x_1, x_2, x_3 , and x_4 which follow a four-dimensional Gaussian distribution with possibly different nonzero mean values. A useful relation which can be derived from Equation (3.36), [Ref 30], is

$$\begin{aligned} E[x_1x_2x_3x_4] = & E[x_1x_2]E[x_3x_4] + E[x_1x_3]E[x_2x_4] \\ & + E[x_1x_4]E[x_2x_3] - 2\mu_{x_1}\mu_{x_2}\mu_{x_3}\mu_{x_4} \end{aligned} \quad (3.132)$$

Note how the fourth order quantity in question is the product of its possible different second order quantities. In particular, if $x_1 = x(u)$, $x_2 = y(u + \tau)$, $x_3 = x(v)$, $x_4 = y(v + \tau)$, with the stationary cross-correlation function

$$R_{xy}(\tau) = E[x(t)y(t + \tau)] \quad (3.133)$$

and equal nonzero mean values μ_x , then

$$\begin{aligned} E[x(u)y(u + \tau)x(v)y(v + \tau)] \\ = & R_{xy}(\tau) + R_x(v - u)R_y(v - u) \\ & + R_{xy}(v - u + \tau)R_{xy}(v - u - \tau) - 2\mu_x^2 \end{aligned} \quad (3.134)$$

3.5 LINEAR TRANSFORMATIONS AND SAMPLING THEOREMS

3.5.1 Linear Transformations of Random Processes

The dynamic behavior of representative linear physical systems has been discussed in practical terms in Chapter 2. It will be helpful at this time to consider very briefly the mathematical properties of linear transformations of random processes. This background will be assumed in Chapter 5 to develop important input-output relationships for linear systems subjected to random inputs.

Consider an arbitrary random process $\{x_k(t)\}$. An operator A which transforms a sample function $x_k(t)$ into another function $y_k(v)$ may be written as

$$y_k(v) = A[x_k(t)] \quad (3.135)$$

where A denotes a functional operation on the term inside the brackets $[]$. The argument v may or may not be the same as t . For example, if the operation in question is differentiation, then $v = t$ and $y_k(t)$ will be a sample function from the derivative random process $\{\dot{x}_k(t)\}$, assuming of course that the derivative exists. A different example is when the operation in question is integration between definite limits. Here, $v \neq t$, and $y_k(v)$ will be a random variable over the index k , determined by $x_k(t)$ and the definite limits. The operator A can take many different forms. In the following, the sample space index k will be omitted for simplicity in notation.

The operator A is said to be *linear* and *time invariant* (constant parameter) if, for any set of admissible values $x_1, x_2, \dots, x_N$ and constants $a_1, a_2, \dots, a_N$, it follows that

$$A\left[\sum_{i=1}^N a_i x_i\right] = \sum_{i=1}^N a_i A[x_i] \quad (3.136)$$

In words, the operation is both additive and homogeneous. The admissible values here may be different sample functions at the same t , or they may be different values from the same sample function at different t .

For any linear operation where all quantities exist, the procedure of taking expected values of random variables is commutative with the linear operation. That is, for fixed t and v ,

$$E[y(v)] = E[A[x(t)]] = A[E[x(t)]] \quad (3.137)$$

This result is proved easily, as follows. Assume $x(t)$ takes on N discrete values $x_1, x_2, \dots, x_N$, and $y(v)$ takes on N corresponding discrete values $y_1, y_2, \dots$,

y_v , where $y_v = A[x_v]$. Then

$$E\{y(v)\} = \frac{1}{N} \sum_{i=1}^N y_i = \frac{1}{N} \sum_{i=1}^N A\{x_i\} \quad \text{and} \quad E\{x(t)\} = \frac{1}{N} \sum_{i=1}^N x_i$$

Now since A is a linear operator,

$$\frac{1}{N} \sum_{i=1}^N A\{x_i\} = A\left[\frac{1}{N} \sum_{i=1}^N x_i\right] = A\{E\{x(t)\}\}$$

Hence

$$E\{y(v)\} = A\{E\{x(t)\}\}$$

The continuous case follows by letting N approach infinity and using an appropriate convergence criterion such as Equation (3.140a). This completes the proof.

A basic result whose proof evolves directly from definitions is as follows: *If $x(t)$ is from a weakly (strongly) stationary random process and if the operator A is linear and time invariant then $y(v) = A\{x(t)\}$ will form a weakly (strongly) stationary random process.* Another result of special significance, proved in References 30, 41 is as follows: *If $x(t)$ follows a Gaussian distribution and the operator A is linear, then $y(v) = A\{x(t)\}$ will also follow a Gaussian distribution.*

An integral transformation of any particular sample function $x(t)$ from an arbitrary random process $\{x(t)\}$ is defined by

$$I = \int_a^b x(t) \phi(t) dt \quad (3.138)$$

where $\phi(t)$ is an arbitrary given function for which the integral exists. For any given $\phi(t)$ and limits (a, b) , the quantity I is a random variable which depends on the particular sample function $x(t)$. In order to investigate statistical properties of the random variable I , it is customary to break up the integration interval (a, b) into subintervals Δt and consider the approximation linear sum

$$I_N = \sum_{i=1}^N x(i \Delta t) \phi(i \Delta t) \Delta t \quad (3.139)$$

Convergence of I_N to I may now be defined by different criteria. The sequence $\{I_N\}$ is said to converge to I

1. In the mean square sense if

$$\lim_{N \rightarrow \infty} E\{|I_N - I|^2\} = 0 \quad (3.140a)$$

2. In probability if for every $\epsilon > 0$

$$\lim_{N \rightarrow \infty} \text{Prob}\{|I_N - I| \geq \epsilon\} = 0 \quad (3.140b)$$

From the Tchebycheff inequality of Equation (3.15), it follows directly that convergence in the mean square sense implies convergence in probability. In practice, most integral expressions involving random variables exist by assuming convergence in the mean square sense.

3.5.2 Sampling Theorems for Random Records

Data representing continuous random processes must often be processed and analyzed in terms of discrete sample values. Such problems will be considered in Chapters 7 and 9. Two pertinent sampling theorems are now presented as background for this later material.

Suppose a sample random time history record $x(t)$ from a random process $\{x_\lambda(t)\}$ exists only for the time interval from 0 to T seconds, and is zero at all other times. Its Fourier transform is

$$X(f) = \int_0^T x(t) e^{-j2\pi ft} dt \quad (3.141)$$

Assume that $x(t)$ is continually repeated to obtain a periodic time function with a period of T seconds. The fundamental frequency increment is $f = 1/T$. By a Fourier series expansion

$$x(t) = \sum_{-\infty}^{\infty} A_n e^{j2\pi nt/T} \quad (3.142)$$

where

$$A_n = \frac{1}{T} \int_0^T x(t) e^{-j2\pi nt/T} dt \quad (3.143)$$

From Equation (3.141),

$$X\left(\frac{n}{T}\right) = \int_0^T x(t) e^{-j2\pi nt/T} dt = T A_n \quad (3.144)$$

Thus $X(n/T)$ determines A_n and, therefore, $x(t)$ at all t . In turn, this determines $X(f)$ for all f . This result is the *sampling theorem* in the *frequency domain*. The fundamental frequency increment $1/T$ is called a *Nyquist co-interval*.

Suppose that a Fourier transform $X(f)$ of some sample random time history record $x(t)$ exists only over a frequency interval from $-B$ to B cps, and is zero at all other frequencies. The actual realizable frequency band ranges from 0 to B cps. The inverse Fourier transform yields

$$x(t) = \int_{-B}^B X(f) e^{j2\pi ft} df \quad (3.145)$$

Assume that $X(f)$ is continually repeated in frequency to obtain a periodic

frequency function with a period of $2B$ cps. The fundamental time increment is $t = 1/2B$. Now

$$X(f) = \sum_{-\infty}^{\infty} C_n e^{j\pi n f / B} \quad (3.146)$$

where

$$C_n = \frac{1}{2B} \int_{-B}^B X(f) e^{j\pi n f / B} df \quad (3.147)$$

From Equation (3.145),

$$x\left(\frac{n}{2B}\right) = \int_{-B}^B X(f) e^{j\pi n f / B} df = 2B C_n \quad (3.148)$$

Thus $x(n/2B)$ determines C_n and, hence, $X(f)$ at all f . In turn, this determines $x(t)$ for all t . This result is the *sampling theorem in the time domain*. The fundamental time increment $1/2B$ is called a *Nyquist interval*.

Assume that the sample record $x(t)$ exists only for the time interval from 0 to T seconds, and suppose also that its Fourier transform $X(f)$ exists only in a frequency interval from $-B$ to B cps. This dual assumption is not theoretically possible because of an *uncertainty principle* [Ref. 1]. In practice, however, it may be closely approximated with finite time intervals and band-pass filters. Assuming $x(t)$ and $X(f)$ are so restricted in their time and frequency properties, only a finite number of discrete samples of $x(t)$ or $X(f)$ are required to describe $x(t)$ completely for all t . From Equation (3.144), by sampling $X(f)$ at Nyquist co-interval points $1/T$ apart on the frequency scale from $-B$ to B , the number of discrete samples required to describe $x(t)$ is

$$N = \frac{2B}{1/T} = 2BT \quad (3.149)$$

From Equation (3.148), by sampling $x(t)$ at Nyquist interval points $1/2B$ apart on the time scale from 0 to T , it follows that

$$N = \frac{T}{1/2B} = 2BT \quad (3.150)$$

Thus the same number of discrete samples are required when sampling the Nyquist co-intervals on the frequency scale, or when sampling the Nyquist intervals on the time scale.

PROBLEMS

1. Consider the following statements as they may apply to the probability density function $p(x)$, the autocorrelation function $R_x(\tau)$, and the power spectral density function $G_x(f)$, of a stationary random process:
 - (a) must be an even function

- (b) must be non-negative
- (c) must be bounded by its value at zero
- (d) can determine the mean value of the process
- (e) can determine the variance of the process
- (f) can determine the probability density function
- (g) can determine the autocorrelation function
- (h) can determine the power spectral density function

As three separate sets of questions, which of the above statements are always true for every $p(x)$, every $R_x(\tau)$, or every $G_x(f)$.

2. Prove Equations (3.48) and (3.49).
3. Prove Equations (3.62) and (3.63).
4. Prove Parseval's Theorem, Equation (3.105), and explain the derivation of Equation (3.108).
5. By substituting Equations (3.146) and (3.148) into (3.145), show that $x(t)$ can be determined from the $x(n/2B)$ by the formula

$$x(t) = \sum_{-\infty}^{\infty} x\left(\frac{n}{2B}\right) \left(\frac{\sin 2\pi B \left(t - \frac{n}{2B}\right)}{2\pi B \left(t - \frac{n}{2B}\right)} \right)$$

6. If a random variable x has a probability density function

$$p(x) = \frac{1}{2} \delta(x+3) + \frac{1}{4} \delta(x-2) + \frac{1}{8\sqrt{2\pi}} e^{-x^2/2}$$

determine

- (a) the mean value and variance.
 - (b) the probability that $x > 1$.
7. Given $R_x(\tau) = 25e^{-4|\tau|} \cos 2\pi f\tau + 16$, determine
 - (a) the mean value and variance.
 - (b) the associated one-sided power spectral density function.
 8. Which of the following functions are valid expressions for a one-sided power spectral density function?

$$G_1(f) = \frac{f^2 + 9}{(f^2 + 4)(f + 1)^2}$$

$$G_2(f) = \frac{f^2 + 1}{f^4 + 5f^2 + 6}$$

$$G_3(f) = \frac{f^2 + 4}{f^4 - 4f^2 + 3}$$

$$G_4(f) = \frac{\exp[-jf^2]}{f^2 + 2}$$

- 9 Given a two-sided power spectral density function

$$S_x(f) = 8 \delta(f) + 20 \left(1 - \frac{|f|}{10} \right) \quad f \leq 10$$

$$= 0 \quad |f| > 10$$

determine

- (a) the mean value and variance
 - (b) the associated autocorrelation function
- 10 Which of the following properties are always true for two ergodic random processes?
- (a) $R_{xy}(\infty) = \mu_x \mu_y$
 - (b) $R_{xy}(0) = 0$ implies $\mu_x = 0$ or $\mu_y = 0$
 - (c) $R_{xy}(\tau) = 0$ when $R_x(\tau) = 0$ or $R_y(\tau) = 0$
 - (d) $|R_{xy}(\tau)|^2 \leq R_x(\tau) R_y(\tau)$
 - (e) $G_{xy}(0) = 0$ when $\mu_x = 0$ or $\mu_y = 0$
 - (f) $|G_{xy}(f)|^2 \leq G_x(0) G_y(0)$
 - (g) $G_{xy}(f) = 0$ when $G_x(f) = 0$ or $G_y(f) = 0$
 - (h) $G_x(\infty) = G_y(\infty) = G_{xy}(\infty) = 0$
 - (i) $p(x, y) = p(x) p(y)$ implies $R_{xy}(\tau) = 0$
 - (j) $p(x, y) = 0$ implies $G_{xy}(f) = 0$

CHAPTER 4

REVIEW OF STATISTICAL PRINCIPLES

Pertinent concepts of random process theory and its applications to the description and analysis of random data are discussed in Chapter 3. Beyond these basic ideas of random process theory, the evaluation of random data often involves certain numerical manipulations where statistical techniques are required. This chapter will review and illustrate various statistical ideas which have wide applications to commonly occurring data evaluation problems. The intent is to supply the reader with a minimum background in terminology and certain techniques of applied statistics which are relevant to later discussions and to general applications.

4.1 SAMPLE VALUES AND PARAMETER ESTIMATION

Consider a random variable x , as defined in Section 3.1.1, where the index k of the sample space is omitted for simplicity in notation. Further consider the two basic parameters of x which specify its central tendency and dispersion; namely, the mean value and variance, respectively. From Equations (3.8) and (3.11), the mean value and variance are given by

$$\mu_x = E[x] = \int_{-\infty}^{\infty} x p(x) dx \quad (4.1)$$

$$\sigma_x^2 = E[(x - \mu_x)^2] = \int_{-\infty}^{\infty} (x - \mu_x)^2 p(x) dx \quad (4.2)$$

where $p(x)$ is the probability density function of the variable x . These two parameters of x cannot, of course, be precisely determined in practice since

an exact knowledge of the probability density function will not generally be available. Hence one must be content with estimates of the mean value and variance based upon a finite number of observed values.

One possible method (there are others) for estimating the mean value and variance of x based upon N independent observations would be as follows:

$$\bar{x} = \hat{\mu}_x = \frac{1}{N} \sum_{i=1}^N x_i \quad (4.3)$$

$$s_b^2 = \hat{\sigma}_x^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2 \quad (4.4)$$

Here, $\bar{x}$ and s_b^2 are the *sample mean* and *sample variance*, respectively. The hats (^) over $\hat{\mu}_x$ and $\hat{\sigma}_x^2$ indicate that these sample values are being used as *estimators* for the mean value and variance of x . The subscript on s_b^2 means that this is a biased variance estimate (to be discussed later). The number of observations used to compute the estimates (sample values) is called the *sample size*.

The specific sample values in Equations (4.3) and (4.4) are not the only quantities which might be used to estimate the mean value and variance of a random variable x . For example, reasonable estimates of the mean value and variance would also be obtained by dividing the summations in Equations (4.3) and (4.4) by $N - 1$ instead of N . Estimators are never clearly right or wrong since they are defined somewhat arbitrarily. Nevertheless, certain estimators can be judged as being "good" estimators, or "better" estimators than others.

Three principal factors can be used to establish the quality or "goodness" of an estimator. First, it is desirable that the expected value of the estimator be equal to the parameter being established. That is,

$$E[\hat{\Phi}] = \Phi \quad (4.5)$$

where $\hat{\Phi}$ is an estimator for the parameter Φ . If this is true, the estimator is said to be *unbiased*. Second, it is desirable that the mean square error of the estimator be smaller than for other possible estimators. That is,

$$E[(\hat{\Phi}_1 - \Phi)^2] \leq E[(\hat{\Phi}_2 - \Phi)^2] \quad (4.6)$$

where Φ_1 is the estimator of interest and $\hat{\Phi}_2$ is any other possible estimator. If this is true, the estimator is said to be more *efficient* than other possible estimators. Third, it is desirable that the estimator approach the parameter being estimated with a probability approaching unity as the sample size becomes large. That is, for any $\epsilon > 0$,

$$\lim_{N \rightarrow \infty} \text{Prob} \{ |\hat{\Phi} - \Phi| \geq \epsilon \} = 0 \quad (4.7a)$$

If this is true, the estimator is said to be *consistent*. It follows from the Tchebycheff inequality of Equation (3.15) that a sufficient (but not necessary) condition to meet the requirements of Equation (4.7a) is given by

$$\lim_{N \rightarrow \infty} E[(\hat{\Phi} - \Phi)^2] = 0 \quad (4.7b)$$

Note that the requirements stated in Equation (4.7) are simply convergence requirements in (a) probability and (b) the mean square sense, as previously stated in Equation (3.140).

Consider the example of the mean value estimator given by Equation (4.3). The expected value of the sample mean $\bar{x}$ is

$$E[\bar{x}] = E\left[\frac{1}{N} \sum_{i=1}^N x_i\right] = \frac{1}{N} E\left[\sum_{i=1}^N x_i\right] = \frac{1}{N} (N\mu_x) = \mu_x \quad (4.8)$$

Hence from Equation (4.5), the estimator $\hat{\mu}_x = \bar{x}$ is unbiased. The mean square error of the sample mean $\bar{x}$ is given by

$$E[(\bar{x} - \mu_x)^2] = E\left[\left(\frac{1}{N} \sum_{i=1}^N x_i - \mu_x\right)^2\right] = \frac{1}{N^2} E\left[\left(\sum_{i=1}^N (x_i - \mu_x)\right)^2\right]$$

From Section 3.1.2, since the observations x_i are independent, the cross product terms in the last expression will have an expected value of zero. It then follows that

$$E[(\bar{x} - \mu_x)^2] = \frac{1}{N^2} E\left[\sum_{i=1}^N (x_i - \mu_x)^2\right] = \frac{1}{N^2} (N\sigma_x^2) = \frac{\sigma_x^2}{N} \quad (4.9)$$

Hence from Equation (4.7b), the estimator $\hat{\mu}_x = \bar{x}$ is consistent. It can be shown that the estimator is also efficient.

Now consider the example of the variance estimator given by Equation (4.4). The expected value of the sample variance s_b^2 is

$$E[s_b^2] = E\left[\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2\right] = \frac{1}{N} E\left[\sum_{i=1}^N (x_i - \bar{x})^2\right]$$

However,

$$\begin{aligned} \sum_{i=1}^N (x_i - \bar{x})^2 &= \sum_{i=1}^N (x_i - \mu_x + \mu_x - \bar{x})^2 \\ &= \sum_{i=1}^N (x_i - \mu_x)^2 - 2(\bar{x} - \mu_x) \sum_{i=1}^N (x_i - \mu_x) + \sum_{i=1}^N (\bar{x} - \mu_x)^2 \\ &= \sum_{i=1}^N (x_i - \mu_x)^2 - 2(\bar{x} - \mu_x)N(\bar{x} - \mu_x) + N(\bar{x} - \mu_x)^2 \\ &= \sum_{i=1}^N (x_i - \mu_x)^2 - N(\bar{x} - \mu_x)^2 \end{aligned} \quad (4.10)$$

Since $E[(x_i - \mu_x)^2] = \sigma_x^2$ and $E[(x - \mu_x)^2] = \sigma_x^2/N$, it follows that

$$E[s_0^2] = \frac{1}{N} (N\sigma_x^2 - \sigma_x^2) = \frac{(N-1)}{N} \sigma_x^2 \quad (4.11)$$

Hence the estimator $\hat{\sigma}_x^2 = s_0^2$ is *biased*. Although the sample variance s_0^2 is a biased estimator for σ_x^2 , it is a consistent and efficient estimator.

From the results in Equation (4.11), it is clear that an unbiased estimator for σ_x^2 may be obtained by computing a slightly different sample variance as follows:

$$s^2 = \hat{\sigma}_x^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2 \quad (4.12)$$

The quantity defined in Equation (4.12) is an *unbiased* estimator for σ_x^2 , as well as an efficient and consistent estimator. For this reason, the sample variance defined in Equation (4.12) is often considered a "better" estimator than the sample variance given by Equation (4.4). The sample variance defined in Equation (4.12) will be used henceforth as an estimator for the variance of a random variable.

4.2 IMPORTANT PROBABILITY DISTRIBUTION FUNCTIONS

Examples of several theoretical probability distribution functions are given in Section 3.1. The most important of these distribution functions from the viewpoint of applied statistics is the Gaussian (*normal*) distribution. There are three other distribution functions associated with normally distributed random variables that have wide applications as statistical tools. These are the χ^2 distribution, the t distribution, and the F distribution. Each of these three, along with the normal distribution, will now be defined and discussed. Applications for each as an analysis tool will be covered in later sections.

4.2.1 Normal Distribution

The probability density and distribution functions of a normally distributed random variable x are defined by Equation (3.35) in Section 3.1.3. A more convenient form of the normal distribution, however, is obtained by using the standardized variable z given by

$$z = \frac{x - \mu_x}{\sigma_x} \quad (4.13)$$

When Equation (4.13) is substituted into Equation (3.35), standardized normal density and distribution functions with zero mean and unit variance

($\mu_z = 0$; $\sigma_z^2 = 1$) are obtained as follows

$$p(z) = (\sqrt{2\pi})^{-1} e^{-z^2/2} \quad (4.14a)$$

$$P(z) = (\sqrt{2\pi})^{-1} \int_{-\infty}^z e^{-\xi^2/2} d\xi \quad (4.14b)$$

It is desirable for later applications to denote the value of z which corresponds with a specific probability $P(z) = 1 - \alpha$, by z_α . That is,

$$P(z_\alpha) = \int_{-\infty}^{z_\alpha} p(z) dz = \text{Prob} [z \leq z_\alpha] = 1 - \alpha \quad (4.15a)$$

or

$$1 - P(z_\alpha) = \int_{z_\alpha}^{\infty} p(z) dz = \text{Prob} [z > z_\alpha] = \alpha \quad (4.15b)$$

The value of z_α which satisfies Equation (4.15) is called the 100α percentage point.

The density function $p(z)$ is unimodal, monotonic about the mode, and symmetric with inflection points at ± 1 , as illustrated in Figure 4.1(a). The corresponding distribution function $P(z)$ is illustrated in Figure 4.1(b). A limited tabulation is presented for the ordinates of the standardized normal density function in Table A.1, and for the areas under the standardized normal density function in Table A.2 in the Appendix.

4.2.2 Chi-Square Distribution

Let $z_1, z_2, z_3, \dots, z_n$ be n independent random variables, each of which has a normal distribution with zero mean and unit variance. Let a new random variable be defined as

$$\chi_n^2 = z_1^2 + z_2^2 + z_3^2 + \dots + z_n^2 \quad (4.16)$$

The random variable χ_n^2 is the chi-square variable with n degrees of freedom. The number of *degrees of freedom* n represents the number of independent or "free" squares entering into the expression. From Reference 21, the probability density function of χ_n^2 is given by

$$p(\chi^2) = [2^{n/2} \Gamma(n/2)]^{-1} (\chi^2)^{(n/2)-1} e^{-\chi^2/2} \quad \chi^2 \geq 0 \quad (4.17)$$

where $\Gamma(n/2)$ is the gamma function. The corresponding distribution function of χ_n^2 , given by the integral of Equation (4.17) from minus infinity to a specific value of χ_n^2 , is called the *chi-square distribution with n degrees of freedom*. The 100α percentage point of the χ^2 distribution will be denoted by $\chi_{n;\alpha}^2$. That is,

$$\int_{\chi_{n;\alpha}^2}^{\infty} p(\chi^2) d\chi^2 = \text{Prob} [\chi_n^2 > \chi_{n;\alpha}^2] = \alpha \quad (4.18)$$

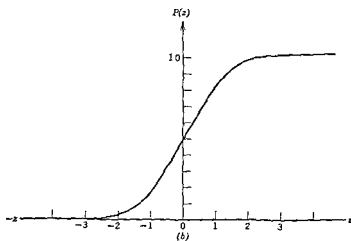
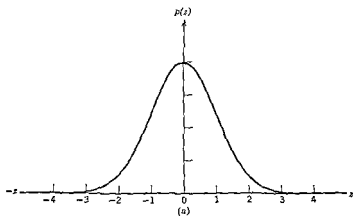


Figure 4.1 Standardized normal probability density and distribution functions
 (a) Probability density function (b) Probability distribution function

The mean value and variance of the variable χ_n^2 are

$$E[\chi_n^2] = \mu_{\chi^2} = n \quad (4.19)$$

$$E[(\chi_n^2 - \mu_{\chi^2})^2] = \sigma_{\chi^2}^2 = 2n \quad (4.20)$$

The density function $p(\chi^2)$ is monotonic for $n = 1$ and 2 , and unimodal, monotonic about the mode, and nonsymmetric for $n > 2$, as illustrated in Figure 4.2(a). The corresponding distribution function $P(\chi^2)$ is shown in Figure 4.2(b). A limited tabulation of percentage points for the χ^2 distribution function is presented in Table A.3 in the Appendix.

Several features of the chi-square distribution should be noted. First, the chi-square distribution is actually a special case of the more general gamma distribution function. Second, the square root of chi-square with two degrees of freedom (χ^2) constitutes an important special case called the *Rayleigh distribution function*. The Rayleigh distribution has wide applications to two-dimensional target problems and is also the limiting distribution function of *peak values* for a narrow-band Gaussian random signal as the bandwidth approaches zero. Third, the square root of chi-square with three degrees of freedom (χ_3) constitutes another important special case which is called the *Maxwell distribution function*. The Maxwell distribution has applications to three-dimensional target problems. Fourth, a chi-square distribution approaches a normal distribution as the number of degrees of freedom becomes large. Specifically, for $n > 30$, the quantity $\sqrt{2\chi_n^2}$ is distributed approximately as a normal variable with a mean of $\mu = \sqrt{2n-1}$ and a variance of $\sigma^2 = \text{unity}$.

4.2.3 Student t Distribution

Let y and z be independent random variables such that y has a χ_n^2 distribution function and z has a normal distribution function with zero mean and unit variance. Let a new random variable be defined as

$$t_n = \frac{z}{\sqrt{y/n}} \quad (4.21)$$

The random variable t_n is the Student t variable with n degrees of freedom. From Reference 21, the probability density function of t_n is given by

$$p(t) = \frac{\Gamma((n+1)/2)}{\sqrt{\pi n} \Gamma(n/2)} \left[1 + \frac{t^2}{n} \right]^{-(n+1)/2} \quad (4.22)$$

The corresponding distribution function of t_n , given by the integral of Equation (4.22) from minus infinity to a specific value of t_n , is called the *Student t distribution with n degrees of freedom*. The 100α percentage point of

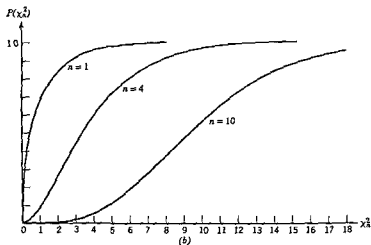
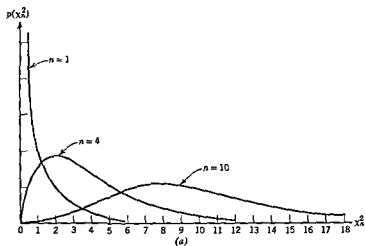


Figure 4.2 Chi square probability density and distribution functions (a) Probability density function (b) Probability distribution function

the t distribution will be denoted by $t_{n;\alpha}$. That is,

$$\int_{t_{n;\alpha}}^{\infty} p(t) dt = \text{Prob} [t_n > t_{n;\alpha}] = \alpha \quad (4.23)$$

The mean value and variance of the variable t_n are

$$E[t_n] = \mu_t = 0 \quad \text{for} \quad n > 1 \quad (2.24)$$

$$E[(t_n - \mu_t)^2] = \sigma_t^2 = \frac{n}{n-2} \quad \text{for} \quad n > 2 \quad (4.25)$$

The density function $p(t)$ is unimodal, monotonic about the mode, and symmetric, as illustrated in Figure 4.3(a). The corresponding distribution function $P(t)$ is illustrated in Figure 4.3(b). A limited tabulation of percentage points for the t distribution function is presented in Table A.4 in the Appendix. It should be noted that the t distribution approaches a standardized normal distribution as the number of degrees of freedom n becomes large.

4.2.4 The F Distribution

Let y_1 and y_2 be independent random variables such that y_1 has a χ^2 distribution function with n_1 degrees of freedom, and y_2 has a χ^2 distribution function with n_2 degrees of freedom. Let a new random variable be defined as

$$F_{n_1, n_2} = \frac{y_1/n_1}{y_2/n_2} = \frac{y_1 n_2}{y_2 n_1} \quad (4.26)$$

The random variable F_{n_1, n_2} is the F variable with n_1 and n_2 degrees of freedom. From Reference 21, the probability density function of F_{n_1, n_2} is given by

$$p(F) = \frac{\Gamma[(n_1 + n_2)/2] (n_1/n_2)^{n_1/2} F^{(n_1/2)-1}}{\Gamma(n_1/2) \Gamma(n_2/2) [1 + (n_1 F/n_2)]^{(n_1+n_2)/2}} \quad F \geq 0 \quad (4.27)$$

The corresponding distribution function of F_{n_1, n_2} , given by the integral of Equation (4.27) from minus infinity to a specific value of F_{n_1, n_2} , is called the F distribution with n_1 and n_2 degrees of freedom. The 100α percentage point of the F distribution will be denoted by $F_{n_1, n_2; \alpha}$. That is,

$$\int_{F_{n_1, n_2; \alpha}}^{\infty} p(F) dF = \text{Prob} [F_{n_1, n_2} > F_{n_1, n_2; \alpha}] = \alpha \quad (4.28)$$

The mean value and variance of F_{n_1, n_2} are

$$E[F_{n_1, n_2}] = \mu_F = \frac{n_2}{n_2 - 2} \quad \text{for} \quad n_2 > 2 \quad (4.29)$$

$$E[(F_{n_1, n_2} - \mu_F)^2] = \sigma_F^2 = \frac{2n_2^2(n_1 + n_2 - 2)}{n_1(n_2 - 2)^2(n_2 - 4)} \quad \text{for} \quad n_2 > 4 \quad (4.30)$$

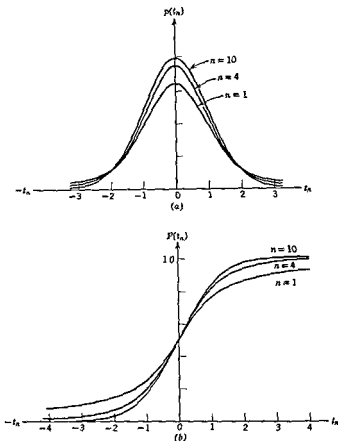


Figure 4.3 Student t probability density and distribution functions (a) Probability density function (b) Probability distribution function

The density function $p(F)$ is unimodal, monotonic about the mode, and nonsymmetric for $n_2 > 2$, as illustrated in Figure 4.4(a). The corresponding distribution function $P(F)$ is illustrated in Figure 4.4(b). A limited tabulation of percentage points for the F distribution function is presented in Tables A.5(a), (b), and (c) in the Appendix. It should be noted that the statistic t_n^2 , the square of the variable defined in Equation (4.21), has an F distribution with $n_1 = 1$ and $n_2 = n$ degrees of freedom.

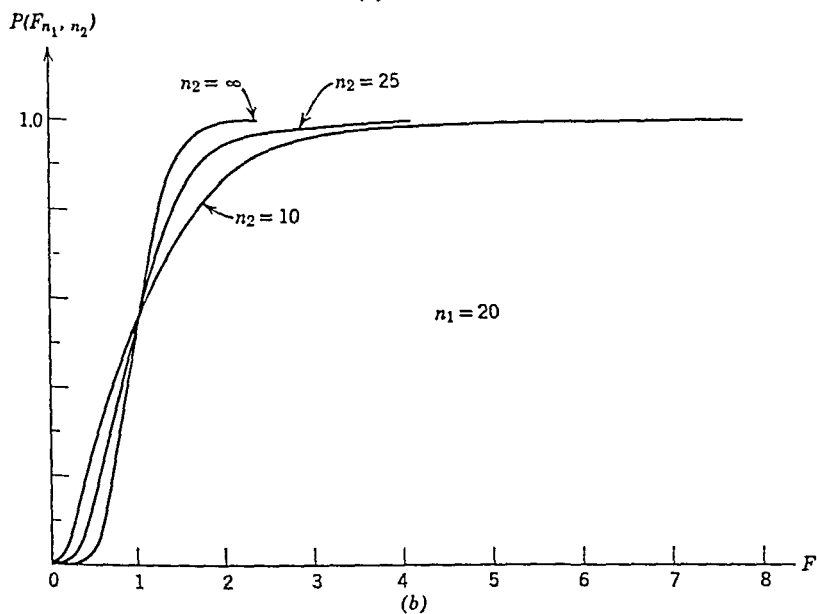
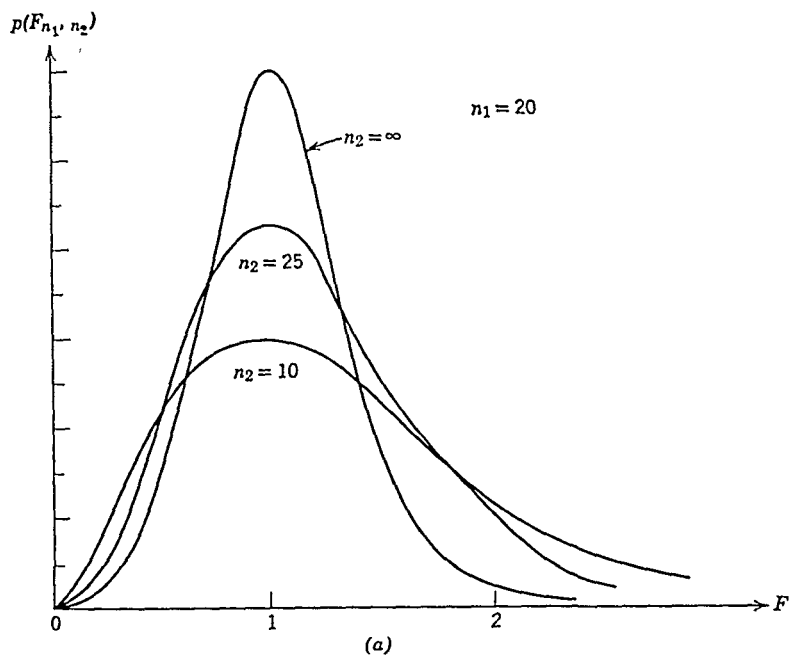


Figure 4.4 F probability density and distribution functions for special case where $n_1 = 20$. (a) Probability density function. (b) Probability distribution function.

4.3 SAMPLING DISTRIBUTIONS AND ILLUSTRATIONS

Consider a random variable x with a probability distribution function $P(x)$. Let $x_1, x_2, \dots, x_N$ be a sample of N observed values of x . Any quantity computed from these sample values will also be a random variable. For example, consider the mean value $\bar{x}$ of the sample. If a series of different samples of size N were selected from the same random variable x , the value of $\bar{x}$ computed from each sample would generally be different. Hence, $\bar{x}$ is also a random variable with a probability distribution function $P(\bar{x})$. This probability distribution function is called the *sampling distribution* of $\bar{x}$.

Some of the more common sampling distributions which often arise in practice will now be considered. These involve the probability distribution functions defined and discussed in Section 4.2. The use of these sampling distributions to establish confidence intervals and perform hypothesis tests is illustrated in Sections 4.4 through 4.8.

4.3.1 Distribution of Sample Mean with Known Variance

Consider the mean value of a sample of N independent observations from a random variable x as follows

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i \quad (4.31)$$

First consider the case where the random variable x is normally distributed with a mean value of μ_x and a known variance of σ_x^2 . From Section 3.5.1, the sampling distribution of the sample mean $\bar{x}$ will also be normally distributed. From Equation (4.8), the mean value of the sampling distribution of $\bar{x}$ is

$$\mu_{\bar{x}} = \mu_x \quad (4.32)$$

and from Equation (4.9) the variance of the sampling distribution of $\bar{x}$ is

$$\sigma_{\bar{x}}^2 = \frac{\sigma_x^2}{N} \quad (4.33)$$

Hence, from Equation (4.13) the following sampling distribution applies for the sample mean $\bar{x}$

$$\frac{(\bar{x} - \mu_x)\sqrt{N}}{\sigma_x} = z \quad (4.34)$$

where z has a standardized normal distribution, as defined in Section 4.2.1

It follows that a probability statement concerning future values of the sample mean may be made as follows.

$$\text{Prob} \left[\bar{x} > \left(\frac{\sigma_x^2}{\sqrt{N}} + \mu_x \right) \right] = \alpha \quad (4.35)$$

Now, consider the case where the random variable x is not normally distributed. From the practical implications of the Central Limit Theorem, the following result occurs. As the sample size N becomes large, the *sampling distribution of the sample mean $\bar{x}$ approaches a normal distribution regardless of the distribution of the original variable x* . In practical terms, a normality assumption for the sampling distribution of $\bar{x}$ becomes reasonable in many cases for $N > 4$ and quite accurate in most cases for $N > 10$. Hence for reasonably large sample sizes, Equation (4.34) applies to the sampling distribution of $\bar{x}$ computed for any random variable x , regardless of its probability distribution function.

4.3.2 Distribution of Sample Variance

Consider the variance of a sample of N independent observations from a random variable x as follows.

$$s^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2 \quad (4.36)$$

If the variable x is normally distributed with a mean of μ_x and a variance of σ_x^2 , it is readily shown using Equations (4.10), (4.16), and (4.34) that

$$\begin{aligned} \sum_{i=1}^N (x_i - \bar{x})^2 &= \sum_{i=1}^N (x_i - \mu_x)^2 - N(\bar{x} - \mu_x)^2 \\ &= \sigma_x^2 \sum_{i=1}^N z_i^2 - \frac{N\sigma_x^2}{N} z^2 = \sigma_x^2 \sum_{i=1}^{N-1} z_i^2 \\ &= \sigma_x^2 \chi_n^2 \quad n = N - 1 \end{aligned}$$

where χ_n^2 has a chi-square distribution with $n = N - 1$ degrees of freedom, as defined in Section 4.2.2. Hence the sampling distribution of the sample variance s^2 is given by

$$\frac{ns^2}{\sigma_x^2} = \chi_n^2 \quad n = N - 1 \quad (4.37)$$

It follows that a probability statement concerning future values of the sample variance s^2 may be made as follows.

$$\text{Prob} \left[s^2 > \frac{\sigma_x^2 \chi_{n;\alpha}^2}{n} \right] = \alpha \quad (4.38)$$

4.3.3 *Distribution of Sample Mean with Unknown Variance*

Consider the mean value of a sample of N independent observations from a random variable x , as given by Equation (4.31). If the variable x is normally distributed with a mean value of μ_x and an unknown variance, it is seen from Equations (4.21) and (4.37) that

$$\frac{(\bar{x} - \mu_x)}{s/\sqrt{N}} = \frac{\sigma_x z/\sqrt{N}}{\sqrt{\sigma_x^2 \chi_n^2/n}/\sqrt{N}} = \frac{z}{\sqrt{\chi_n^2/n}} = t_n$$

where t_n has a Student t distribution with $n = N - 1$ degrees of freedom, as defined in Section 4.2.3. Hence the sampling distribution of the sample mean $\bar{x}$ when σ_x^2 is unknown is given by

$$\frac{(x - \mu_x)\sqrt{N}}{s} = t_n \quad n = N - 1 \quad (4.39)$$

It follows that a probability statement concerning future values of the sample mean $\bar{x}$ may be made as follows

$$\text{Prob} \left[x > \left(\frac{st_{n,\alpha}}{\sqrt{N}} + \mu_x \right) \right] = \alpha \quad (4.40)$$

4.3.4 *Distribution of Ratio of Two Sample Variances*

Consider the variances of two samples, one consisting of N_x independent observations of a random variable x , and the other consisting of N_y observations of a random variable y , as given by Equation (4.36). If the variable x is normally distributed with a mean value of μ_x and a variance of σ_x^2 , and the variable y is normally distributed with a mean value of μ_y and a variance σ_y^2 , it is seen from Equations (4.26) and (4.37) that

$$\frac{s_x^2/\sigma_x^2}{s_y^2/\sigma_y^2} = \frac{\sigma_x^2 \chi_{n_x}^2/n_x \sigma_x^2}{\sigma_y^2 \chi_{n_y}^2/n_y \sigma_y^2} = F_{n_x, n_y}$$

where F_{n_x, n_y} has an F distribution with $n_x = N_x - 1$ and $n_y = N_y - 1$ degrees of freedom, as defined in Section 4.2.4. Hence the sampling distribution of the ratio of the sample variances s_x^2 and s_y^2 is given by

$$\frac{s_x^2/\sigma_x^2}{s_y^2/\sigma_y^2} = F_{n_x, n_y} \quad \begin{matrix} n_x = N_x - 1 \\ n_y = N_y - 1 \end{matrix} \quad (4.41)$$

It follows that a probability statement concerning future values of the ratio of the sample variances s_x^2 and s_y^2 may be made as follows

$$\text{Prob} \left[\frac{s_x^2}{s_y^2} > \frac{\sigma_x^2}{\sigma_y^2} F_{n_x, n_y, \alpha} \right] = \alpha \quad (4.42)$$

Note that if the two samples are obtained from the same random variable, $x = y$, then Equation (4.41) reduces to

$$\frac{s_1^2}{s_2^2} = F_{n_1, n_2} \quad \begin{matrix} n_1 = N_1 - 1 \\ n_2 = N_2 - 1 \end{matrix} \quad (4.43)$$

4.4 CONFIDENCE INTERVALS

The use of sample values as estimators for parameters of random variables is discussed in Section 4.1. However, those procedures result only in point estimates for a parameter of interest; no indication is provided as to how closely a sample value estimates the parameter. A more meaningful procedure for estimating parameters of random variables involves the estimation of an interval, as opposed to a single point value, which will include the parameter being estimated with a known degree of uncertainty. For example, consider the case where the sample mean $\bar{x}$ computed from N independent observations of a random variable x is being used as an estimator for the mean value μ_x . It is usually more desirable to estimate μ_x in terms of some interval, such as $\bar{x} \pm d$, where there is a specified uncertainty that μ_x falls within that interval. Such intervals can be established if the sampling distribution of the estimator in question is known.

Continuing with the example of a mean value estimate, it is shown in Section 4.3 that probability statements can be made concerning the value of a sample mean $\bar{x}$ as follows.

$$\text{Prob} \left[z_{1-\alpha/2} < \frac{(\bar{x} - \mu_x)\sqrt{N}}{\sigma_x} \leq z_{\alpha/2} \right] = 1 - \alpha \quad (4.44)$$

The above probability statement is technically correct *before* the sample has been collected and $\bar{x}$ has been computed. After the sample has been collected, however, the value of $\bar{x}$ is a fixed number rather than a random variable. Hence it can be argued that the probability statement in Equation (4.44) no longer applies since the quantity $(\bar{x} - \mu_x)\sqrt{N}/\sigma_x$ either *does* or *does not* fall within the indicated limits. In other words, after a sample has been collected, a technically correct probability statement would be as follows.

$$\text{Prob} \left[z_{1-\alpha/2} < \frac{(\bar{x} - \mu_x)\sqrt{N}}{\sigma_x} \leq z_{\alpha/2} \right] = \begin{cases} 0 \\ 1 \end{cases} \quad (4.45)$$

Whether the correct probability is zero or unity is usually not known. However, as the value of α becomes small (as the interval between $z_{1-\alpha/2}$ and $z_{\alpha/2}$ becomes wide), one would tend to guess that the probability is more likely to be unity than zero. In slightly different terms, if many different

samples were repeatedly collected and a value of $\bar{x}$ were computed for each sample, one would tend to expect the quantity in Equation (4.45) to fall within the noted interval for about $1 - \alpha$ of the samples. In this context, a statement can be made about an interval within which one would expect to find the quantity $(x - \mu_x)\sqrt{N}/\sigma_x$ with a small degree of uncertainty. Such statements are called *confidence statements*. The interval associated with a confidence statement is called a *confidence interval*. The degree of trust associated with the confidence statement is called the *confidence coefficient*.

For the case of the mean value estimate, a confidence interval can be established for the mean value μ_x based upon the sample value $\bar{x}$ by rearranging terms in Equation (4.45) as follows:

$$\left[\bar{x} - \frac{\sigma_x z_{\alpha/2}}{\sqrt{N}} \leq \mu_x < \bar{x} + \frac{\sigma_x z_{\alpha/2}}{\sqrt{N}} \right] \quad (4.46a)$$

Furthermore, if σ_x is unknown, a confidence interval can still be established for the mean value μ_x based upon the sample values $\bar{x}$ and s by rearranging terms in Equation (4.39) as follows:

$$\left[\bar{x} - \frac{st_{n, \alpha/2}}{\sqrt{N}} \leq \mu_x < \bar{x} + \frac{st_{n, \alpha/2}}{\sqrt{N}} \right] \quad n = N - 1 \quad (4.46b)$$

Equation (4.46) uses the fact that $z_{1-\alpha/2} = -z_{\alpha/2}$ and $t_{n, 1-\alpha/2} = -t_{n, \alpha/2}$. The confidence coefficient associated with the intervals is $1 - \alpha$. Hence the confidence statement would be as follows: "The true mean value μ_x falls within the noted interval with a confidence coefficient of $1 - \alpha$," or, in more common terminology, "with a confidence of $100(1 - \alpha)$ percent." Similar confidence statements can be established for any parameter estimates where proper sampling distributions are known. For example, from Equation (4.37), a $1 - \alpha$ confidence interval for the variance σ_x^2 based upon a sample variance s^2 from a sample of size N is:

$$\left[\frac{n s^2}{\chi_{n, \alpha/2}^2} \leq \sigma_x^2 < \frac{n s^2}{\chi_{n, 1-\alpha/2}^2} \right] \quad n = N - 1 \quad (4.47)$$

Example 4.1 Illustration of Confidence Intervals Assume a sample of $N = 31$ independent observations are collected from a normally distributed random variable x with the following results:

60	61	47	56	61	63
65	69	54	59	43	61
55	61	56	48	67	65
60	58	57	62	57	58
53	59	58	61	67	62
54					

Determine a 90 percent confidence interval for the mean value and variance of the random variable x .

From Equation (4.46), a $1 - \alpha$ confidence interval for the mean value μ_x based on the sample mean $\bar{x}$ and the sample variance s^2 for a sample size of $N = 31$ is given by

$$\left[\left(\bar{x} - \frac{st_{30;\alpha/2}}{\sqrt{31}} \right) \leq \mu_x < \left(\bar{x} + \frac{st_{30;\alpha/2}}{\sqrt{31}} \right) \right]$$

From Table A.4, for $\alpha = 0.10$, $t_{30;\alpha/2} = t_{30;0.05} = 1.697$, so the interval reduces to

$$[\bar{x} - 0.3048s] \leq \mu_x < (\bar{x} + 0.3048s)$$

From Equation (4.47), a $1 - \alpha$ confidence interval for the variance σ_x^2 based on the sample variance s^2 for a sample size of $N = 31$ is given by

$$\left[\frac{30s^2}{\chi_{30;\alpha/2}^2} \leq \sigma_x^2 < \frac{30s^2}{\chi_{30;1-\alpha/2}^2} \right]$$

From Table A.3, for $\alpha = 0.10$, $\chi_{30;\alpha/2}^2 = \chi_{30;0.05}^2 = 43.77$ and $\chi_{30;1-\alpha/2}^2 = \chi_{30;0.95}^2 = 18.49$, so the interval reduces to

$$[0.6854s^2 \leq \sigma_x^2 < 1.622s^2]$$

It now remains to calculate the sample mean and variance, and substitute these values into the interval statements. From Equation (4.3), the sample mean is

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i = 58.61$$

From Equation (4.12), the sample variance is

$$s^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2 = \frac{1}{N-1} \left\{ \sum_{i=1}^N x_i^2 - N(\bar{x})^2 \right\} = 33.43$$

Hence the 90 percent confidence intervals for the mean value and variance of the random variable x are as follows.

$$\begin{aligned} [56.85 \leq \mu_x < 60.37] \\ [22.91 \leq \sigma_x^2 < 54.22] \end{aligned}$$

4.5 HYPOTHESIS TESTS

Consider the case where a given estimator $\hat{\Phi}$ is computed from a sample of N independent observations of a random variable x . Assume that there is reason to believe that the true parameter Φ being estimated has a specific value Φ_0 . Now, even if $\Phi = \Phi_0$, the sample value $\hat{\Phi}$ will probably not come

out exactly equal to Φ_0 because of the sampling variability associated with $\hat{\Phi}$. Hence the following question arises: If it is hypothesized that $\Phi = \Phi_0$, how much difference between $\hat{\Phi}$ and Φ_0 must occur before the hypothesis should be rejected as being invalid? This question can be answered in statistical terms by considering the probability of any noted difference between $\hat{\Phi}$ and Φ_0 based upon the sampling distribution of $\hat{\Phi}$. If the probability of a given difference is small, the difference would be considered significant and the hypothesis that $\Phi = \Phi_0$ would be rejected. If the probability of a given difference is not small, the difference would be accepted as normal statistical variability and the hypothesis that $\Phi = \Phi_0$ would be accepted.

The preceding discussion outlines the simplest form of a statistical procedure called hypothesis testing. To clarify the general technique, assume that a sample value $\hat{\Phi}$, which is an estimate of a parameter Φ , has a probability density function of $p(\hat{\Phi})$. Now, if a hypothesis that $\Phi = \Phi_0$ is true, then

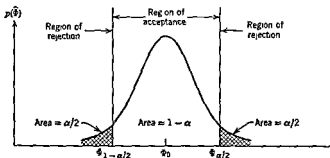


Figure 4.5 Acceptance and rejection regions for hypothesis tests

$p(\hat{\Phi})$ would have a mean value of Φ_0 as illustrated in Figure 4.5. The probability that $\hat{\Phi}$ would fall below the lower level $\Phi_{1-\alpha/2}$ is

$$\text{Prob} [\hat{\Phi} \leq \Phi_{1-\alpha/2}] = \int_{-\infty}^{\Phi_{1-\alpha/2}} p(\hat{\Phi}) d\hat{\Phi} = \frac{\alpha}{2} \quad (4.48a)$$

The probability that $\hat{\Phi}$ would fall above the upper value $\Phi_{\alpha/2}$ is

$$\text{Prob} [\hat{\Phi} > \Phi_{\alpha/2}] = \int_{\Phi_{\alpha/2}}^{\infty} p(\hat{\Phi}) d\hat{\Phi} = \frac{\alpha}{2} \quad (4.48b)$$

Hence the probability that $\hat{\Phi}$ would be outside the range between $\Phi_{1-\alpha/2}$

and $\Phi_{\alpha/2}$ is α . Now let α be small so that it is very unlikely that $\hat{\Phi}$ would fall outside the range between $\Phi_{1-\alpha/2}$ and $\Phi_{\alpha/2}$. If a sample were collected and a value of $\hat{\Phi}$ were computed which in fact fell outside the range between $\Phi_{1-\alpha/2}$ and $\Phi_{\alpha/2}$, there would be strong reason to question the original hypothesis that $\Phi = \Phi_0$ since such a value for $\hat{\Phi}$ would be very unlikely if the hypothesis were true. Hence the hypothesis that $\Phi = \Phi_0$ would be rejected. On the other hand, if the value for $\hat{\Phi}$ fell within the range between $\Phi_{1-\alpha/2}$ and $\Phi_{\alpha/2}$, there would be no strong reason to question the original hypothesis. Hence the hypothesis that $\Phi = \Phi_0$ would be accepted.

The small probability α used for the hypothesis test is called the *level of significance* of the test. The range of values of $\hat{\Phi}$ for which the hypothesis will be rejected is called the *region of rejection* or *critical region*. The range of values of $\hat{\Phi}$ for which the hypothesis will be accepted is called the *region of acceptance*. The simple hypothesis test outlined above is called a *two-sided test* because, if the hypothesis is not true, the value of Φ could be either greater or less than Φ_0 . Hence it is necessary to test for significant differences between Φ and Φ_0 in both directions. In other cases a *one-sided test* might be sufficient. For example, let it be hypothesized that $\Phi \geq \Phi_0$. For this case, the hypothesis would be false only if Φ were less than Φ_0 . Thus the test would be performed using the lower side of the probability density function $p(\hat{\Phi})$.

Two possible errors can occur when a hypothesis test is performed. First, the hypothesis might be rejected when in fact it is true. This possible error is called a *Type I Error*. Second, the hypothesis might be accepted when in fact it is false. This possible error is called a *Type II Error*. From Figure 4.5, a Type I error would occur if the hypothesis were true and $\hat{\Phi}$ fell in the region of rejection. It follows that the probability of a Type I Error is equal to α , the level of significance of the test.

In order to establish the probability of a Type II Error, it is necessary to specify some deviation of the true parameter Φ from the hypothesized parameter Φ_0 which one desires to detect. For example, assume that the true parameter actually has a value of either $\Phi = \Phi_0 + d$ or $\Phi = \Phi_0 - d$, as illustrated in Figure 4.6. If it is hypothesized that $\Phi = \Phi_0$ when in fact $\Phi = \Phi_0 \pm d$, the probability that $\hat{\Phi}$ would fall inside the acceptance region between $\Phi_{1-\alpha/2}$ and $\Phi_{\alpha/2}$ is β . Hence the probability of a Type II Error is β for detecting a difference of $\pm d$ from the hypothesized value Φ_0 .

The probability $1 - \beta$ is called the *power of the test*. Clearly, for any given sample size N , the probability of a Type I Error can be reduced by reducing the level of significance α . However, this will increase the probability β of a Type II Error (reduce the power of the test). The only way to reduce both α and β is to increase the sample size N for the estimate $\hat{\Phi}$. These ideas form the basis for selecting the necessary sample sizes for statistical experiments.

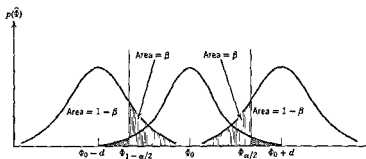


Figure 4.6 Type II error regions for hypothesis tests

Example 4.2 Illustration of Hypothesis Test Design Assume there is reason to believe that the mean value of a random variable x is $\mu_x = 10$. Further assume that the variance of x is known to be $\sigma_x^2 = 4$. Determine the proper sample size to test the hypothesis that $\mu_x = 10$ at the 5 percent level of significance where the probability of a Type II Error is to be 5 percent for detecting a difference of 10 percent from the hypothesized value. Determine the region of acceptance to be used for the test.

An unbiased estimate for μ_x is given by the sample mean value $\bar{x}$ as defined in Equation (4.3). The appropriate sampling distribution of $\bar{x}$ is given by Equation (4.34) as

$$\bar{x} = \frac{\sigma_x}{\sqrt{N}} z + \mu_x$$

where z is normally distributed with zero mean and unit variance. Note that this sampling distribution of $\bar{x}$ is precise if x is normally distributed and is still a good approximation if x is not normally distributed.

The upper and lower limits of the acceptance region for the hypothesis test are as follows

$$\text{upper limit} = \frac{\sigma_x}{\sqrt{N}} z_{\alpha/2} + \mu_x$$

$$\text{lower limit} = \frac{\sigma_x}{\sqrt{N}} z_{1-\alpha/2} + \mu_x$$

Now if the true mean value were in fact $\mu_x = \mu_x \pm d$, a Type II Error would occur with probability β if the sample value $\bar{x}$ fell below the upper limit or above the lower limit. In terms of the sampling distributions of

$\bar{x}$ given mean values $\mu_x' = \mu_x + d$ and $\mu_x' = \mu_x - d$,

$$\text{upper limit} = \frac{\sigma_x}{\sqrt{N}} z_{1-\beta} + \mu_x + d$$

$$\text{lower limit} = \frac{\sigma_x}{\sqrt{N}} z_{\beta} + \mu_x - d$$

Hence the following equalities apply.

$$\frac{\sigma_x}{\sqrt{N}} z_{\alpha/2} + \mu_x = \frac{\sigma_x}{\sqrt{N}} z_{1-\beta} + \mu_x + d$$

$$\frac{\sigma_x}{\sqrt{N}} z_{1-\alpha/2} + \mu_x = \frac{\sigma_x}{\sqrt{N}} z_{\beta} + \mu_x - d$$

These relationships both reduce to

$$z_{\alpha/2} = z_{1-\beta} + \frac{\sqrt{N}}{\sigma_x} d = -z_{\beta} + \frac{\sqrt{N}}{\sigma_x} d$$

It follows that the required sample size is given by

$$N = \left[\frac{\sigma_x(z_{\alpha/2} + z_{\beta})}{d} \right]^2$$

For the specific values in this example ($\sigma_x = 2$, $z_{\alpha/2} = 1.96$, $z_{\beta} = 1.645$, $d = 0.1(10) = 1$), the required sample size is

$$N = 52$$

The region of acceptance for the hypothesis test will be

$$\text{upper limit} = \frac{\sigma_x}{\sqrt{N}} z_{\alpha/2} + \mu_x = 10.54$$

$$\text{lower limit} = \frac{\sigma_x}{\sqrt{N}} z_{1-\alpha/2} + \mu_x = 9.46$$

4.6 CHI-SQUARE GOODNESS-OF-FIT TEST

A special type of hypothesis test which is often used to test the equivalence of a probability density function of sampled data to some theoretical density function is called the chi-square goodness-of-fit test. The general procedure involves the use of a statistic with an approximate chi-square distribution as a measure of the discrepancy between an observed probability density function

and the theoretical density function. A hypothesis of equivalence is then tested by studying the sampling distribution of this statistic.

To be more specific, consider a sample of N independent observations from a random variable x with a probability density function of $p(x)$. Let the N observations be grouped into K intervals, called *class intervals*, which together form a *frequency histogram*. The number of observations falling within the i th class interval is called the *observed frequency* in the i th class, and will be denoted by f_i . The number of observations which would be expected to fall within the i th class interval if the true probability density function of x were $p_0(x)$ is called the *expected frequency* in the i th class interval, and will be denoted by F_i . Now, the discrepancy between the observed frequency and the expected frequency within each class interval is given by $f_i - F_i$. To measure the total discrepancy for all class intervals, the squares of the discrepancies in each interval are summed to obtain the sample statistic

$$X^2 = \sum_{i=1}^K \frac{(f_i - F_i)^2}{F_i} \quad (4.49)$$

The distribution for X^2 is approximately the same as for χ_n^2 discussed in Section 4.2.2. The number of degrees of freedom n in this case is equal to K minus the number of different independent linear restrictions imposed on the observations. There is one such restriction due to the fact that the frequency in the last class interval is determined once the frequencies in the first $K - 1$ class intervals are known. If the comparison is made by fitting the expected theoretical density function to the frequency histogram for the observed data, then one additional constraint results from each independent parameter of the theoretical density function which must be computed to make the fit. For example, if the expected theoretical density function is a normal density function with unknown mean and variance, then two additional constraints are involved, since two parameters (a mean and variance) must be computed to fit a normal density function. Hence for the common case where the chi square goodness-of-fit test is used as a test for normality, the number of degrees of freedom for X^2 in Equation (4.49) is $n = K - 3$.

Having established the proper degrees of freedom for X^2 , a hypothesis test may be performed as follows. Let it be hypothesized that the variable x has a probability density function of $p(x) = p_0(x)$. After grouping the sampled observations into K class intervals and computing the expected frequency for each interval assuming $p(x) = p_0(x)$, compute X^2 as indicated in Equation (4.49). Since any deviation of $p(x)$ from $p_0(x)$ will cause X^2 to increase, a one-sided (upper tail) test is used. The region of acceptance is

$$X^2 \leq \chi_{n, \alpha}^2 \quad (4.50)$$

where the value of $\chi^2_{n;\alpha}$ is available from Table A.3. If the sample value X^2 is greater than $\chi^2_{n;\alpha}$, the hypothesis that $p(x) = p_0(x)$ is rejected at the α level of significance. If X^2 is less than or equal to $\chi^2_{n;\alpha}$, the hypothesis is accepted at the α level of significance.

There are two basic ways to apply the chi-square goodness-of-fit test. The first way is to select class intervals in a manner which will provide equal expected frequencies within each interval. Excluding a uniform distribution hypothesis, this procedure will result in different interval widths from one class interval to another. The second way is to select class intervals of equal width. Again except for the uniform distribution hypothesis, this procedure will result in different expected frequencies from one class interval to another. If the equal frequency procedure is used, Reference 51 suggests that the minimum number of class intervals for a test at the $\alpha = 0.05$ level of significance should be as given in Table 4.1. If the equal interval width procedure is used, about 2.5 class intervals per standard deviation should suffice in most cases. However, it is desirable to have an expected frequency of at least five in each class interval [References 7 and 21]. Satisfaction of this goal will often require the pooling of class intervals in regions of low probability density.

Table 4.1
Minimum Number of Class Intervals for $\alpha = 0.05$

N	200	400	600	800	1000	1500	2000
K	16	20	24	27	30	35	39

Example 4.3. Illustration of Test for Normality. A sample of $N = 200$ independent observations of the digitized output of a thermal noise generator are presented in Table 4.2. The sample values are presented in increasing order for convenience. Test the noise generator output for normality by performing a chi-square goodness-of-fit test at the $\alpha = 0.05$ level of significance. Use an equal frequency procedure with $K = 16$ class intervals.

The calculations required to perform the test are summarized in Table 4.3. For an equal frequency test with $K = 16$ class intervals, the probability associated with each interval must be $P = 1/K = 0.0625$. The appropriate interval limits for the normal distribution hypothesis are established by finding the required value of z_α for each interval limit from Table A.2. These standardized interval limits are then converted to interval limits in terms of the noise generator output x by using the sample mean and variance of the data, as computed by Equations (4.3) and (4.12). The observed frequencies are now

Table 4.2

Sample Observations Arranged in Increasing Order

-7.6	-3.8	-2.5	-1.6	-0.7	0.2	1.1	2.0	3.4	4.6
-6.9	-3.8	-2.5	-1.6	-0.7	0.2	1.1	2.1	3.5	4.8
-6.6	-3.7	-2.4	-1.6	-0.6	0.2	1.2	2.3	3.5	4.8
-6.4	-3.6	-2.3	-1.5	-0.6	0.3	1.2	2.3	3.6	4.9
-6.4	-3.5	-2.3	-1.5	-0.5	0.3	1.3	2.3	3.6	5.0
-6.1	-3.4	-2.3	-1.4	-0.5	0.3	1.3	2.4	3.6	5.2
-6.0	-3.4	-2.2	-1.4	-0.4	0.4	1.3	2.4	3.7	5.3
-5.7	-3.4	-2.2	-1.2	-0.4	0.4	1.4	2.5	3.7	5.4
-5.6	-3.3	-2.1	-1.2	-0.4	0.5	1.5	2.5	3.7	5.6
-5.5	-3.2	-2.1	-1.2	-0.3	0.5	1.5	2.6	3.7	5.9
-5.1	-3.2	-2.0	-1.1	-0.3	0.6	1.6	2.6	3.8	6.1
-4.8	-3.1	-2.0	-1.1	-0.2	0.6	1.6	2.6	3.8	6.3
-4.8	-3.0	-1.0	-1.0	-0.2	0.7	1.6	2.7	3.9	6.3
-4.6	-3.0	-1.9	-1.0	-0.2	0.8	1.7	2.8	4.0	6.5
-4.4	-2.9	-1.8	-1.0	-0.1	0.9	1.8	2.8	4.2	6.9
-4.4	-2.9	-1.8	-0.9	-0.0	0.9	1.8	2.9	4.2	7.1
-4.3	-2.9	-1.8	-0.9	0.0	1.0	1.8	3.1	4.3	7.2
-4.1	-2.7	-1.7	-0.8	0.1	1.0	1.9	3.2	4.3	7.4
-4.0	-2.6	-1.7	-0.8	0.1	1.1	1.9	3.2	4.4	7.9
-3.8	-2.6	-1.6	-0.7	0.2	1.1	2.0	3.3	4.4	9.0

determined by superimposing the interval limits on the observed data, as indicated in Figure 4.2. The final step is to compute the normalized squared deviation of the observed frequency from the expected frequency for each class interval and sum to obtain the result $X^2 = 3.36$. Noting that the appropriate degrees of freedom is $n = K - 3 = 13$, the acceptance region for the test is found in Table A.3 to be $X^2 \leq \chi_{13, 0.05}^2 = 22.36$. Hence, the hypothesis of normality is accepted at the $\alpha = 0.05$ level of significance.

4.7 RUN TEST

The various procedures for establishing confidence intervals and performing hypothesis tests using sample data, as discussed in Sections 4.4 and 4.5, assume that the random variables being sampled are normally distributed. In practice, the procedures are probably usable with reasonable accuracy for many random variables which are not normally distributed. However, there is no clear measure as to how much a random variable may deviate from normality before the sampling distributions discussed in Section 4.4 are no longer valid. This problem can often be avoided by using statistical procedures which do not assume a specific distribution function for the original

Table 4.3
Calculations for Goodness-of-Fit Test

Interval Number	Upper Limit of Interval			f	F	$ F - f $	$\frac{(F - f)^2}{F}$
	α	z_α	$x = sz + \bar{x}$				
1	0.9375	-1.53	-4.79	13	12.5	0.5	0.02
2	0.8750	-1.15	-3.54	11	12.5	1.5	0.18
3	0.8125	-0.89	-2.68	14	12.5	1.5	0.18
4	0.7500	-0.67	-1.95	14	12.5	1.5	0.18
5	0.6875	-0.49	-1.35	15	12.5	2.5	0.50
6	0.6250	-0.32	-0.79	12	12.5	0.5	0.02
7	0.5625	-0.16	-0.26	12	12.5	0.5	0.02
8	0.5000	0	0.27	12	12.5	0.5	0.02
9	0.4375	0.16	0.79	10	12.5	2.5	0.50
10	0.3750	0.32	1.33	14	12.5	1.5	0.18
11	0.3125	0.49	1.89	10	12.5	2.5	0.50
12	0.2500	0.67	2.49	10	12.5	2.5	0.50
13	0.1875	0.89	3.22	12	12.5	0.5	0.02
14	0.1250	1.15	4.08	15	12.5	2.5	0.50
15	0.0625	1.53	5.33	13	12.5	0.5	0.02
16	0			13	12.5	0.5	0.02
				200	200		3.36
$N = 200 \quad \bar{x} = 0.27 \quad s = 3.31$				$n = K - 3 = 13$		$X^2 = 3.36$	

random variable of interest. Such procedures are called *distribution-free* or *nonparametric* procedures. One of the best-known distribution-free procedures used for data evaluation is the chi-square goodness-of-fit test, which is discussed in Section 4.6. Another distribution free test which is valuable for data evaluation techniques, discussed in Chapter 7, is the *run test*.

Consider a sequence of N observed values of a random variable x where each observation is classified into one of two mutually exclusive categories, which may be identified simply by plus (+) or minus (-). The simplest example would be a sequence of flips of a coin where each observation is either a head (+) or a tail (-). A second example might be a sequence of measured values x_i , $i = 1, 2, 3, \dots, N$, with a mean value $\bar{x}$, where each observation is $x_i \geq \bar{x}$ (+) or $x_i < \bar{x}$ (-). A third example might be a simultaneous sequence of two sets of measured values x_i and y_i , $i = 1, 2, 3, \dots, N$, where each observation is $x_i \geq y_i$ (+) or $x_i < y_i$ (-). In any case,

the sequence of plus and minus observations might be as follows

$\underbrace{++}_1$ $-$ $\underbrace{++}_3$ $-$ $\underbrace{+++}_5$ $-$ $+$ $\underbrace{--}_8$ $+$ $\underbrace{--}_{10}$ $+$ $\underbrace{---}_{12}$

A run is defined as a sequence of identical observations that is followed and preceded by a different observation or no observation at all. In this example there are $r = 12$ runs in the sequence of $N = 20$ observations.

The number of runs which occur in a sequence of observations gives an indication as to whether or not results are independent random observations of the same random variable. Specifically, if a sequence of N observations are independent observations of the same random variable, that is, the probability of a (+) or (-) result does not change from one observation to the next, then the sampling distribution of the number of runs in the sequence is a random variable r with a mean value and variance as follows:

$$\mu_r = \frac{2N_1N_2}{N} + 1 \quad (4.51)$$

$$\sigma_r^2 = \frac{2N_1N_2(2N_1N_2 - N)}{N^2(N - 1)} \quad (4.52)$$

Here, N_1 is the number of (+) observations and N_2 is the number of (-) observations. For the special case where $N_1 = N_2 = N/2$, Equations (4.51) and (4.52) reduce to

$$\mu_r = \frac{N}{2} + 1 \quad (4.53)$$

$$\sigma_r^2 = \frac{N(N - 2)}{4(N - 1)} \quad (4.54)$$

A limited tabulation of 100% percentage points for the distribution function of runs is presented in Table A.6 in the Appendix.

Perhaps the most direct application of runs to data evaluation problems involves the testing of a single sequence of observations for independence. Assume there is reason to suspect an underlying trend in a sequence of observations, that is, there is reason to believe that the probability of a (+) or (-) is changing from one observation to the next. The existence of a trend can be tested for as follows. Let it be hypothesized that there is no trend by assuming that the sequence of N observations are independent observations of the same random variable. Then, assuming the number of (+) observations equals the number of (-) observations, the number of runs in the sequence will have a sampling distribution as given in Table A.6.

The hypothesis can be tested at any desired level of significance α by comparing the observed runs to the interval between $r_{n;1-\alpha/2}$ and $r_{n;\alpha/2}$ where $n = N/2$. If the observed runs fall outside the interval, the hypothesis would be rejected at the α level of significance. Otherwise, the hypothesis would be accepted.

Example 4.4. Illustration of Run Test. Assume a sequence of $N = 20$ observations of a random variable produces results as noted below.

(1) 5.5	(6) 5.7	(11) 6.8	(16) 5.4
(2) 5.1	(7) 5.0	(12) 6.6	(17) 6.8
(3) 5.7	(8) 6.5	(13) 4.9	(18) 5.8
(4) 5.2	(9) 5.4	(14) 5.4	(19) 6.9
(5) 4.8	(10) 5.8	(15) 5.9	(20) 5.5

Determine if the observations are independent by testing the runs which occur in the variation of the observations about their median value. Perform the test at the $\alpha = 0.05$ level of significance.

By visual inspection of the data, it is seen that $x = 5.6$ is the median value of the 20 observations. Let all observations with a value greater than 5.6 be identified by (+) and all with a value less than 5.6 be identified by (-). The result is

<u>--</u>	+	<u>--</u>	+	-	+	-	<u>+++</u>	<u>--</u>	+	-	<u>+++</u>	-
1	2	3	4	5	6	7	8	9	10	11	12	13

Hence there are 13 runs represented by the sequence of 20 observations. Let it be hypothesized that the observations are independent. The acceptance region for this hypothesis is

$$r_{10,1-\alpha/2} < r \leq r_{10,\alpha/2}$$

From Table A.6, for $\alpha = 0.05$, $r_{10,1-\alpha/2} = r_{10,0.975} = 6$ and $r_{10,\alpha/2} = r_{10,0.025} = 15$. The hypothesis is accepted since $r = 13$ falls within the range between 6 and 15. That is, there is no reason to question that the observations are independent, which means there is no evidence of an underlying trend.

4.8 CORRELATION AND REGRESSION PROCEDURES

Techniques of correlation and regression analysis are fundamental to much of the material developed in this book. The concept of correlation between two random variables has already been introduced in Chapter 3 and will be expanded upon in Chapters 5 and 6. The concept of linear regression is basic to the techniques of frequency response function estimation from input-output

data, as formulated in Chapters 5 and 6. The material in these Chapters, however, is developed in a frequency domain context which may obscure associations with more familiar classical presentations. Hence a brief review of correlation and regression concepts from the viewpoint of elementary statistics may be helpful as an introduction to this later material.

4.8.1 Linear Correlation

For a wide class of problems, a matter of primary interest is whether or not two or more random variables are interrelated. For example, is there a relationship between cigarette smoking and life expectancy, or between measured aptitude and academic success. In an engineering context, such problems often reduce to detecting a relationship between some assumed excitation and an observed response of a physical system of interest. The existence of such interrelationships and their relative strength can be measured in terms of a correlation coefficient ρ as defined in Section 3.1.2. For the simple case of two random variables x and y , the correlation coefficient is given by Equation (3.28) as

$$\rho_{xy} = \frac{C_{xy}}{\sigma_x \sigma_y} \quad (4.55)$$

where C_{xy} is the covariance of x and y as defined in Equation (3.26).

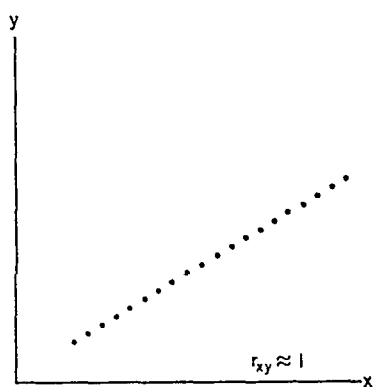
Now assume the random variables x and y are sampled to obtain N pairs of observed values. The correlation coefficient may be estimated from the sample data by

$$\begin{aligned} r_{xy} = \hat{\rho}_{xy} &= \frac{s_{xy}}{s_x s_y} = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\left[\sum_{i=1}^N (x_i - \bar{x})^2 \sum_{i=1}^N (y_i - \bar{y})^2 \right]^{1/2}} \\ &= \frac{\sum_{i=1}^N x_i y_i - N \bar{x} \bar{y}}{\left[\left(\sum_{i=1}^N x_i^2 - N \bar{x}^2 \right) \left(\sum_{i=1}^N y_i^2 - N \bar{y}^2 \right) \right]^{1/2}} \end{aligned} \quad (4.56)$$

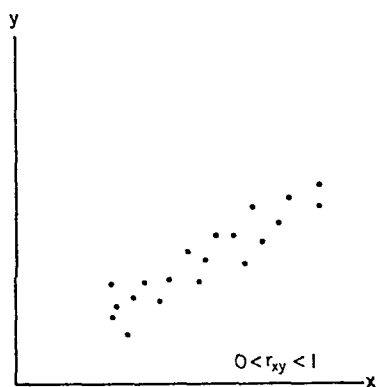
Like ρ_{xy} , the sample correlation coefficient r_{xy} will lie between -1 and $+1$, and will have a bounding value only when the observations display a perfect linear relationship. A nonlinear relationship and/or data scatter, whether it be due to measurement errors or imperfect correlation of the variables, will force the value of r_{xy} towards zero, as illustrated in Figure 4.7.

To evaluate the accuracy of the estimate r_{xy} , it is convenient to work with a particular function of r_{xy} given by

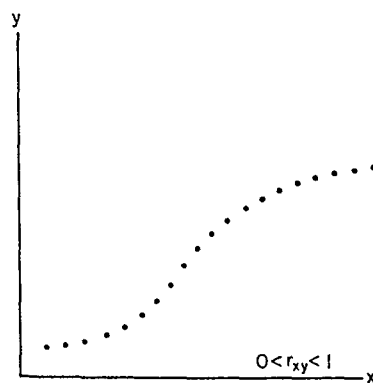
$$w = \frac{1}{2} \ln \left[\frac{1 + r_{xy}}{1 - r_{xy}} \right] \quad (4.57)$$



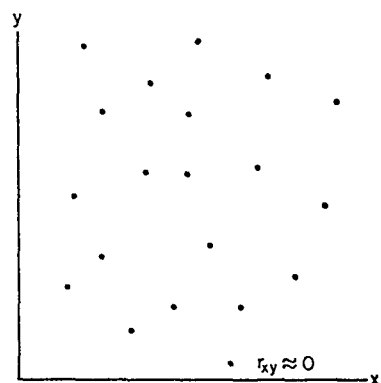
(a)



(b)



(c)



(d)

Figure 4.7 Illustration of varying degrees of correlation. (a) Perfect linear correlation. (b) Moderate linear correlation. (c) Nonlinear correlation. (d) No correlation.

From Reference 7, the random variable w has an approximately normal distribution with a mean and variance of

$$\mu_w = \frac{1}{2} \ln \left[\frac{1 + \rho_{xy}}{1 - \rho_{xy}} \right] \quad (4.58)$$

$$\sigma_w^2 = \frac{1}{N - 3} \quad (4.59)$$

Using the above relationships, confidence intervals for ρ_{xy} based upon a sample estimate r_{xy} may be readily establishing as outlined in Section 4.4

Because of the variability of correlation estimates, it is usually desirable to verify that a nonzero value of the sample correlation coefficient indeed reflects the existence of a statistically significant correlation between the variables of interest. This may be accomplished by testing the hypothesis that $\rho_{xy} = 0$, where a significant correlation is indicated if the hypothesis is rejected. From Equations (4.58) and (4.59), the sampling distribution of w given $\rho_{xy} = 0$ is normal with a mean of $\mu_w = 0$ and a variance of $\sigma_w^2 = 1/(N - 3)$. Hence, the acceptance region for the hypothesis of zero correlation is given by

$$-z_{\alpha/2} \leq \frac{\sqrt{N - 3}}{2} \ln \left[\frac{1 + r_{xy}}{1 - r_{xy}} \right] < z_{\alpha/2} \quad (4.60)$$

where z is the standardized normal variable. Values outside the above interval would constitute evidence of statistical correlation at the α level of significance.

Example 4.5 Illustration of Linear Correlation Analysis. The heights and weights of $N = 25$ male university students selected at random are presented in Table 4.4. Is there reason to believe that the height and weight of male students are correlated at the $\alpha = 0.05$ level of significance?

Table 4.4

Height and Weight Data for Male Students

$x = \text{height in inches}$												$y = \text{weight in pounds}$	
x	70	74	70	65	69	73	72	69	72	76	74	72	
y	140	210	148	145	182	165	155	170	174	155	185	185	
x	68	70	71	68	73	65	73	74	64	72	72	67	73
y	165	220	185	180	170	135	175	180	150	170	165	145	170

Let x be height and y be weight. From the data in Table 4.4, the following values needed in Equation (4.56) are calculated.

$$\begin{aligned}\sum_{i=1}^N x_i y_i &= 299,056 & \sum_{i=1}^N x_i^2 &= 124,986 & \sum_{i=1}^N y_i^2 &= 723,604 \\ \bar{x} &= \frac{1}{N} \sum_{i=1}^N x_i = \frac{1766}{25} = 70.64 & \bar{y} &= \frac{1}{N} \sum_{i=1}^N y_i = \frac{4224}{25} = 168.96\end{aligned}$$

Substituting the above values into Equation (4.56) yields the estimated correlation coefficient as follows.

$$\begin{aligned}r_{xy} &= \frac{299,056 - (25)(70.64)(168.96)}{[(124,986 - 25(70.64)^2)(723,604 - 25(168.96)^2)]^{1/2}} \\ &= 0.44\end{aligned}$$

From Equation (4.57), the quantity $w = 0.472$; thus $\sqrt{N-3} w = 2.21$. Now using Equation (4.60), the hypothesis that $\rho_{xy} = 0$ is rejected at the 5% level of significance since $\sqrt{N-3} w = 2.21$ falls outside the acceptance region bounded by $\pm z_{\alpha/2} = \pm 1.96$. Hence, there is reason to believe that significant correlation exists between the height and weight of male students.

4.8.2 Linear Regression

Correlation analysis can establish the degree to which two or more random variables are interrelated. Beyond this, however, a model for the relationship may be desired so that predictions can be made for one variable based upon specific values of other variables. For instance, a significant linear relationship between the height and weight of male university students is indicated by the correlation analysis of data presented in Example 4.5. A logical second step would be to further evaluate the relationship so that the weight of students can be predicted for any given height. Procedures for dealing with problems of this type come under the heading of regression analysis.

Consider the simple case of two correlated random variables x and y . Referring again to Example 4.5, x might be student height and y student weight. A linear relationship between the two variables would suggest that for a given value of x , a value of y would be predicted by

$$\tilde{y} = A + Bx \quad (4.61)$$

where A and B are the intercept and slope, respectively, of a straight line. For the case of data which display perfect linear correlation ($r_{xy} = 1$), the predicted value $\tilde{y}_i$ would always equal the observed value y_i for any given x_i . In practice, however, data usually do not display a perfect linear relationship. There generally is some scatter due to extraneous random effects, and

perhaps distortion due to nonlinearities, as illustrated in Figure 4.7. Nevertheless, if a linear relationship is assumed and unlimited data are available, appropriate values of A and B can be determined which will predict the expected value of y_i for any given x_i . That is, $\hat{y}_i$ will not necessarily equal the observed value y_i associated with the corresponding x_i , but it will be an average for all such observed values.

The accepted procedure for determining the coefficients in Equation (4.61) is to select those values of A and B which minimize the sum of the squared deviations of the observed values from the predicted values of y . This procedure is called a *least squares fit*. Specifically, noting that the deviation of the observed values from the predicted values is

$$y_i - \hat{y}_i = y_i - (A + Bx_i) \quad (4.62)$$

it follows that the sum of the squared deviations is given by

$$Q = \sum_{i=1}^N (y_i - A - Bx_i)^2 \quad (4.63)$$

Hence, a least squares fit is provided by those values of A and B which make

$$\frac{\partial Q}{\partial A} = \frac{\partial Q}{\partial B} = 0 \quad (4.64)$$

In practice, the available data will be limited to a sample of N pairs of observed values for x and y . This means that Equation (4.64) will yield only estimates of A and B , to be denoted by a and b , respectively. Substituting Equation (4.63) into Equation (4.64) and solving for the estimates of A and B yields

$$a = \bar{y} - b\bar{x} \quad (4.65a)$$

$$b = \frac{\sum_{i=1}^N (x_i - \bar{x})y_i}{\sum_{i=1}^N (x_i - \bar{x})^2} = \frac{\sum_{i=1}^N x_i y_i - N\bar{x}\bar{y}}{\sum_{i=1}^N x_i^2 - N\bar{x}^2} \quad (4.65b)$$

These estimates can now be used to write a prediction model for y given x as follows

$$\hat{y} = a + bx = (\bar{y} - b\bar{x}) + bx = \bar{y} + b(x - \bar{x}) \quad (4.66)$$

The straight line defined by Equation (4.66) is called the *linear regression line* for y on x . By switching the dependent and independent variables in

Equation (4.65), a regression line for x on y could also be calculated. Specifically,

$$\hat{x} = \bar{x} + b'(y - \bar{y}) \quad (4.67a)$$

where

$$b' = \frac{\sum_{i=1}^N x_i y_i - N \bar{x} \bar{y}}{\sum_{i=1}^N y_i^2 - N \bar{y}^2} \quad (4.67b)$$

Comparing the product of Equations (4.65b) and (4.67b) to Equation (4.56), it is seen that the slopes of the regression lines for y on x and x on y are related to the sample correlation coefficient of x and y by

$$r_{xy} = [bb']^{1/2} \quad (4.68)$$

Now consider the accuracy of the estimates a and b given by Equation (4.65). Assuming a normal distribution of y given x , it is shown in Reference 7 that a and b are unbiased estimates of A and B , respectively, with sampling distributions related to the *Student t* distribution as follows.

$$\frac{a - A}{\left(\frac{1}{N} + \frac{\bar{x}^2}{\sum_{i=1}^N (x_i - \bar{x})^2} \right)^{1/2}} = s_{y|x} t_{N-2} \quad (4.69)$$

$$\frac{b - B}{\left(\frac{1}{\sum_{i=1}^N (x_i - \bar{x})^2} \right)^{1/2}} = s_{y|x} t_{N-2} \quad (4.70)$$

Of particular interest is the sampling distribution of $\hat{y}$ associated with a specific value of $x = x_0$. This is given by

$$\frac{\hat{y} - \tilde{y}}{\left(\frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^N (x_i - \bar{x})^2} \right)^{1/2}} = s_{y|x} t_{N-2} \quad (4.71)$$

In Equations (4.69) through (4.71), the term $s_{y|x}$ is the sample standard deviation of the observed values of y_i about the prediction $\hat{y}_i = a + bx_i$, and is given by

$$s_{y|x} = \left[\frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{N - 2} \right]^{1/2} \quad (4.72)$$

The above relationships provide a basis for establishing confidence intervals for A , B , and $\hat{y}$ based upon the estimates a , b , and $\hat{y}$

Example 4 6 Illustration of Linear Regression Analysis Using the data presented in Table 4 4 for Example 4 5 determine a regression line which will provide a linear prediction for the average weight of male university students as a function of their height Determine a 95% confidence interval for the average weight of male students who are 70 inches tall

As in Example 4 5, let x be height and y be weight The values needed to determine the slope and intercept of the regression line for y on x have already been calculated in Example 4 5 Substituting these values into Equation (4 65) yields the estimated slope and intercept as follows

$$b = \frac{299.056 - (25)(70.64)(168.96)}{168.96 - (25)(70.64)^2} = 2.85$$

$$a = 168.96 - (2.85)(70.64) = -32.3$$

Hence the regression line estimating the average weight of male university students given height is

$$\hat{y} = -32.3 + 2.85x$$

which yields an estimated weight of $\hat{y} = 167.2$ lbs for a height of $x = 70$ inches

To establish a confidence interval for the average weight $\hat{y}$ based upon the estimate $\hat{y} = 167.2$ lbs, it is necessary to calculate $s_{y|x}$ given by Equation (4 72) A more convenient equation for $s_{y|x}$ from the computational viewpoint is

$$s_{y|x} = \left[\frac{1}{N-2} \left(\sum_{i=1}^N (y_i - \bar{y})^2 - \frac{\left[\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y}) \right]^2}{\sum_{i=1}^N (x_i - \bar{x})^2} \right) \right]^{1/2}$$

where the terms in the above expression are further simplified for computational purposes by noting that

$$\sum_{i=1}^N (y_i - \bar{y})^2 = \sum_{i=1}^N y_i^2 - N\bar{y}^2 \quad \sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^N x_i y_i - N\bar{x}\bar{y}$$

Substitution of the appropriate values into these expressions yields

$$s_{y|x} = \left[\frac{1}{23} \left(9917 - \frac{(673)^2}{236} \right) \right]^{1/2} = 18.65$$

Then, from Equation (4 71), a 95% confidence interval for the average weight

of male university students with a height of 70 inches is

$$\begin{aligned}\hat{y} \pm s_{y|x} t_{N-2; \alpha/2} & \left[\frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^N (x_i - \bar{x})^2} \right]^{1/2} \\ & = 167.2 \pm (18.65) t_{23; 0.025} \left[\frac{1}{25} + \frac{(70 - 70.64)^2}{236} \right]^{1/2} \\ & = 167.2 \pm 7.9 = 159.3 \text{ to } 175.1 \text{ lbs.}\end{aligned}$$

This concludes Example 4.6.

The techniques of correlation and regression analysis are readily extended for applications involving more than two random variables. As noted earlier, such extensions are fundamental to the analysis of multiple input-output problems developed in Chapters 5 and 6. Hence, further discussions of this subject are deferred to those chapters.

PROBLEMS

1. Given a random variable x with a probability density function of

$$p(x) = \frac{1}{2\sqrt{2\pi}} e^{-(x-1)^2/8}$$

What are the mean value and variance of x ?

2. Given two independent random variables, x and y , with mean values of μ_x and μ_y , and variances of σ_x^2 and σ_y^2 , determine the
 - (a) mean value of the product xy
 - (b) variance of the difference $x - y$.
3. Given the random variable $y = cx$ where c is a constant and x is a random variable with a mean value and variance of μ_x and σ_x^2 , respectively, prove the following relationships are true.
 - (a) $\mu_y = c\mu_x$
 - (b) $\sigma_y^2 = c^2\sigma_x^2$
4. Given four independent standardized normally distributed random variables, z_1 , z_2 , z_3 , and z_4 , define the distribution functions of the following combinations of these variables. For each case, specify the associated degrees of freedom or mean value and variance, as appropriate.
 - (a) $z_1^2 + z_2^2 + z_3^2 + z_4^2$
 - (b) $z_1 + z_2 - z_3 - z_4$

$$(c) \frac{z_1}{\{(z_1^2 + z_2^2 + z_3^2)/3\}^{1/2}}$$

$$(d) \frac{[z_1^2 + z_2^2 + z_3^2]/3}{z_4^2}$$

- 5 What distribution function would be used to establish confidence intervals for the following parameters of two independent normally distributed random variables x and y ?
- interval for μ_x based upon a sample mean $\bar{x}$ and known variance σ_x^2
 - interval for σ_x^2/σ_y^2 based upon a ratio of sample variances s_x^2/s_y^2
 - interval for σ_x^2 based upon a sample variance s_x^2
 - interval for μ_x based upon a sample mean $\bar{x}$ and sample variance s_x^2
- 6 Assume a chi square goodness of fit test is performed using $N = 200$ sample values divided into $K = 16$ class intervals. How many degrees of freedom are associated with the resulting test statistic $X^2 \approx \chi_n^2$ for the following cases?
- testing the hypothesis that the sampled random variable has a normal distribution with a known mean value and variance, μ and σ^2
 - testing the hypothesis that the sampled random variable has an F distribution with unknown degrees of freedom
- 7 Given a sample of N independent observations of a random variable x with a known mean value of zero, an *efficient* estimator for the variance of x is

$$s^2 = \frac{1}{N} \sum_{i=1}^N x_i^2$$

- prove the above estimator is unbiased
 - write an expression relating the above estimator to a chi square variable with the appropriate degrees of freedom specified
 - what is the variance of the above estimator?
(hint the variance of χ_n^2 is $2n$)
- 8 The normalized standard error (coefficient of variation) ϵ_r of a parameter estimate $\hat{\Phi}$ is defined as the ratio of the standard deviation of the estimate to the expected value of the estimate that is $\epsilon_r = \sigma_{\hat{\Phi}}/\mu_{\hat{\Phi}}$. Determine the normalized standard error of a variance estimate s^2 computed from $N = 200$ sample observations using Equation (4.12)
- 9 A correlation study is performed using a sample of $N = 7$ pairs of observations $(x_1, y_1, x_2, y_2, \dots, x_7, y_7)$. A sample correlation coefficient of $r_{xy} = 0.77$ is calculated. Test the hypothesis that ρ_{xy} is greater than zero at the $\alpha = 0.01$ level of significance

10. Assume the sample mean values of two correlated random variables are $\bar{x} = 1$ and $\bar{y} = 2$. Further assume that the sample correlation coefficient is $r_{xy} = 0.5$. If the regression line for y on x is given by $\hat{y} = 1 + x$,
- (a) what is the slope b' of the regression line for x on y ?
 - (b) what is the equation for the regression line for x on y ; ($\hat{x} = a' + b'y$)?

CHAPTER 5

INPUT-OUTPUT RELATIONSHIPS FOR PHYSICAL SYSTEMS

This chapter is concerned with the theory and application of input output relationships for both single input and multiple input problems. Unless stated otherwise, it is assumed in all the analytical developments that inputs are records from stationary random processes and that systems are constant parameter linear systems.

5.1 SINGLE-INPUT LINEAR SYSTEMS

Consider a constant parameter linear system with a weighting function $h(\tau)$ and frequency response function $H(f)$, as defined and discussed in Chapter 2. Assume the system is subjected to a well defined single input $x(t)$, which is from a stationary random process $\{x(t)\}$, as illustrated in Figure 5.1. The output $y(t)$ will then belong to a stationary random process $\{y(t)\}$. Various important relations among the input, the system, and the output will now be developed. Note that the index k of the sample space is omitted for simplicity in notation.

As stated earlier by Equations (2.2) and (2.3), the output $y(t)$ is given by

$$y(t) = \int_0^{\infty} h(\tau) x(t - \tau) d\tau \quad (5.1)$$

If desired, the lower limit may be set equal to $-\infty$ instead of 0 to facilitate changes of variables that will be made later. However, it is understood that $h(\tau) = 0$ for $\tau < 0$.

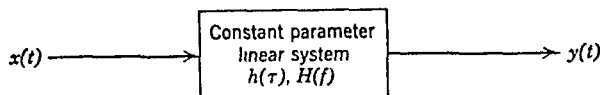


Figure 5.1 Single-input linear system.

For a pair of times t and $t + \tau$, the product $y(t) y(t + \tau)$ is given by

$$y(t) y(t + \tau) = \iint_0^{\infty} h(\xi) h(\eta) x(t - \xi) x(t + \tau - \eta) d\xi d\eta \quad (5.2)$$

The expected value of Equation (5.2) yields

$$R_y(\tau) = \iint_0^{\infty} h(\xi) h(\eta) R_x(\tau + \xi - \eta) d\xi d\eta \quad (5.3)$$

Note that this result is independent of t . This general result shows how to obtain the output stationary autocorrelation function from an input stationary autocorrelation function and the system weighting function entirely from operations in the time domain.

The stationary cross-correlation function $R_{xy}(\tau)$ between the input $x(t)$ and the output $y(t + \tau)$ is derived from the relation

$$x(t) y(t + \tau) = \int_0^{\infty} h(\xi) x(t) x(t + \tau - \xi) d\xi \quad (5.4)$$

Here, the expected value of Equation (5.4) gives

$$R_{xy}(\tau) = \int_0^{\infty} h(\xi) R_x(\tau - \xi) d\xi \quad (5.5)$$

The transformation of Equations (5.3) and (5.5) to a complex-valued frequency domain by taking Fourier transforms yields the important power spectra and cross-spectra relations

$$S_y(f) = |H(f)|^2 S_x(f) \quad (5.6)$$

and

$$S_{xy}(f) = H(f) S_x(f) \quad (5.7)$$

These formulas follow directly from definitions, together with suitable substitutions and algebraic manipulations. Note that Equation (5.6) contains only the gain factor $|H(f)|$, while Equation (5.7) is actually a pair of equations containing both the gain factor and the phase factor. In terms of one-sided

spectral density functions $G_x(f)$, $G_y(f)$, and $G_{xy}(f)$, which exist only for $f \geq 0$, Equations (5.6) and (5.7) become

$$G_y(f) = |H(f)|^2 G_x(f) \quad (5.8)$$

$$G_{xy}(f) = H(f) G_x(f) \quad (5.9)$$

From Equations (2.15) and (3.75), Equation (5.9) is equivalent to

$$|G_{xy}(f)| e^{-j\theta_{xy}(f)} = |H(f)| e^{-j\phi(f)} G_x(f) \quad (5.10)$$

Thus

$$|G_{xy}(f)| = |H(f)| G_x(f) \quad (5.11)$$

$$\theta_{xy}(f) = \phi(f) \quad (5.12)$$

Equation (5.8) shows how the output spectral density function $G_y(f)$ can be determined from the input spectral density function $G_x(f)$ and the system gain factor $|H(f)|$. It follows that the output mean square value is given by

$$\Psi_y^2 = \int_0^\infty G_y(f) df = \int_0^\infty |H(f)|^2 G_x(f) df \quad (5.13)$$

Equation (5.8) also permits the input spectral density function to be determined from the output spectral density function and system gain factor, or the system gain factor to be determined from the input and output spectral density functions. However, Equation (5.8) does not permit the determination of the complete frequency response function $H(f)$ for the system since this relationship does not include the system phase factor $\phi(f)$.

In order to determine the complete frequency response function of the system, one must acquire cross-spectral density function information. The relation of Equation (5.9) provides a convenient interpretation of the cross-spectral density function in terms of $H(f)$ and $G_x(f)$. A special relationship involving cross spectral densities which would determine the phase factor $\phi(f)$ of the system alone is

$$\frac{G_{xy}(f)}{G_{yx}(f)} = \frac{H(f)}{H^*(f)} = e^{-j2\phi(f)} \quad (5.14)$$

where $H^*(f)$ is the complex conjugate of $H(f)$. Note that Equation (5.14) actually determines $2\phi(f)$ instead of $\phi(f)$.

The results of Equations (5.8) and (5.9) provide the basis for many physical applications of random process theory. Figure 5.2 illustrates how an input power spectrum, described by $G_x(f)$, is modified in passing through a linear system, described by $H(f)$.

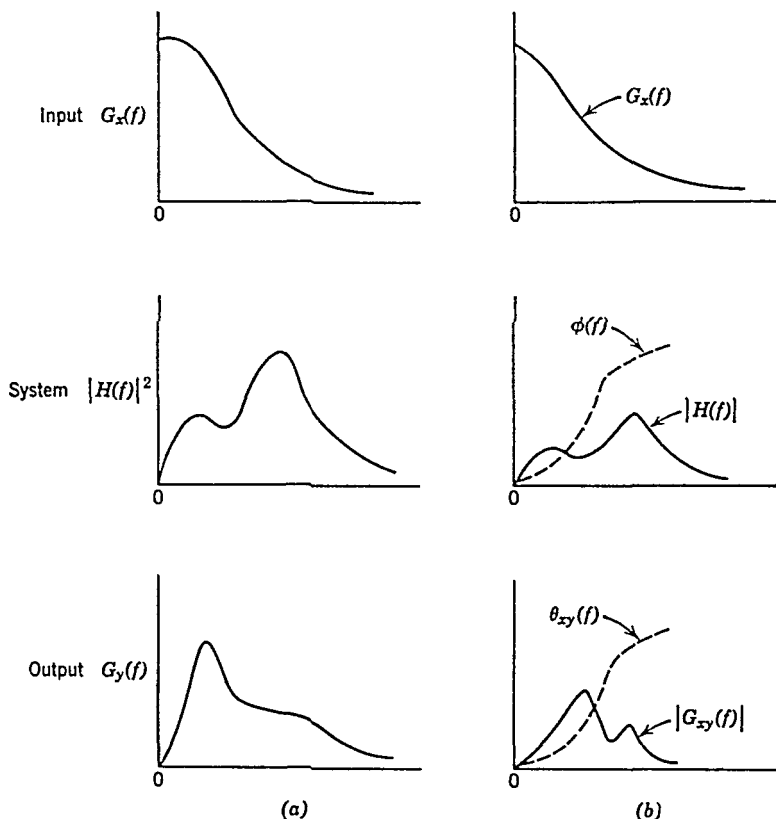


Figure 5.2 Illustration of input-output spectral relationships for linear systems. (a) Power spectrum. (b) Cross-spectrum.

Example 5.1. Response Properties of Low-Pass Filter to White Noise. Assume white noise is applied to the input of a low-pass RC filter with a time constant $K = RC$. Determine the output power spectral density function, the output mean square value, and the output autocorrelation function.

The frequency response function of the low-pass RC filter is

$$H(f) = (1 + j2\pi Kf)^{-1} = |H(f)| e^{-j\phi(f)}$$

corresponding to a weighting function of

$$h(\tau) = \begin{cases} \frac{1}{K} e^{-\tau/K} & \tau \geq 0 \\ 0 & \tau < 0 \end{cases}$$

Here

$$|H(f)| = \{1 + (2\pi Kf)^2\}^{-1/2}$$

$$\phi(f) = \tan^{-1}(2\pi Kf)$$

From Equations (5.11) and (5.13), if $G_x(f)$ is white noise where $G_x(f) = a$, a constant for all $f \geq 0$,

$$G_y(f) = |H(f)|^2 G_x(f) = \frac{a}{1 + (2\pi Kf)^2} \quad 0 \leq f < \infty$$

$$\Psi_y^2 = \int_0^\infty G_y(f) df = \int_0^\infty \frac{a}{1 + (2\pi Kf)^2} df = \frac{a}{4K} \quad (5.15)$$

Equation (3.68) yields

$$R_y(\tau) = \int_0^\infty G_y(f) \cos 2\pi f\tau df = \frac{a}{4K} e^{-|\tau|/K} \quad (5.16)$$

Example 5.2. Response Properties of Low-Pass Filter to Sine Wave Assume a sine wave with power spectral density function $G_x(f) = (X^2/2) \delta(f - f_0)$ where $f_0 > 0$ is applied to the low-pass RC filter of Example 5.1. Determine the output power spectral density function, the output mean square value, and the output autocorrelation function.

For this problem,

$$G_y(f) = |H(f)|^2 G_x(f) = \frac{(X^2/2) \delta(f - f_0)}{1 + (2\pi Kf)^2}$$

$$\Psi_y^2 = \int_0^\infty G_y(f) df = \frac{X^2/2}{1 + (2\pi Kf_0)^2} \quad (5.17)$$

$$R_y(\tau) = \int_0^\infty G_y(f) \cos 2\pi f\tau df = \frac{X^2/2}{1 + (2\pi Kf_0)^2} \cos 2\pi f_0\tau \quad (5.18)$$

Example 5.3 Force Input-Displacement Output System Determine the output power spectral density function, the output autocorrelation function, and the output mean square value when the input is white noise, for the force input and displacement output system of Figure 2.2. These results apply also to other analogous systems, as discussed in Chapter 2.

Assume $G_x(f) = a$. Then, from Equation (2.24a) or Table 2.1, when the force is expressed in displacement units, that is, $x(t) = F(t)/k$, the output power spectral density function becomes

$$G_y(f) = |H(f)|^2 G_x(f) = \frac{a}{[1 - (f/f_n)^2]^2 + (2\zeta f/f_n)^2} \quad 0 \leq f < \infty$$

The corresponding output autocorrelation function is given by

$$R_y(\tau) = \frac{a\pi f_n e^{-2\pi f_n \zeta |\tau|}}{4\zeta} \left[\cos(2\pi f_n \sqrt{1 - \zeta^2} \tau) + \frac{\zeta}{\sqrt{1 - \zeta^2}} \sin(2\pi f_n \sqrt{1 - \zeta^2} |\tau|) \right] \quad (5.19)$$

The output mean square value is

$$\Psi_y^2 = \int_0^\infty G_y(f) df = R_y(0) = \frac{a\pi f_n}{4\zeta} \quad (5.20)$$

Example 5.4. Displacement Input-Displacement Output System. Determine the output power spectral density function, the output autocorrelation function, and the output mean square value when the input is white noise, for the displacement input and displacement output system of Figure 2.4. These results apply also to other analogous systems, as discussed in Chapter 2.

Assume $G_x(f) = a$. Then, from Equation (2.38a), or Table 2.1, the output power spectral density function becomes

$$G_y(f) = |H(f)|_{a=a}^2 = \frac{a[1 + (2\zeta f/f_n)^2]}{[1 - (f/f_n)^2]^2 + (2\zeta f/f_n)^2}$$

The corresponding output autocorrelation function is given by

$$R_y(\tau) = \frac{a\pi f_n(1 + 4\zeta^2)}{4\zeta} e^{-2\pi f_n \zeta |\tau|} \left[\cos(2\pi f_n \sqrt{1 - \zeta^2} \tau) + \frac{\zeta(1 - 4\zeta^2)}{\sqrt{1 - \zeta^2}(1 + 4\zeta^2)} \sin(2\pi f_n \sqrt{1 - \zeta^2} |\tau|) \right] \quad (5.21)$$

The output mean square value is

$$\Psi_y^2 = \int_0^\infty G_y(f) df = R_y(0) = \frac{a\pi f_n(1 + 4\zeta^2)}{4\zeta} \quad (5.22)$$

The importance of exponential-cosine and exponential-sine autocorrelation functions for many physical problems is apparent from the last two examples.

5.2 ORDINARY COHERENCE FUNCTIONS

Assuming $G_x(f)$ and $G_y(f)$ are both different from zero and do not contain delta functions, the coherence function between the input $x(t)$ and the output $y(t)$ is a real-valued quantity defined by

$$\gamma_{xy}^2(f) = \frac{|G_{xy}(f)|^2}{G_x(f) G_y(f)} = \frac{|S_{xy}(f)|^2}{S_x(f) S_y(f)} \quad (5.23)$$

where the G 's are the one sided spectra and the S s are the two-sided theoretical spectra defined previously. From Equation (3.83), it follows that the coherence function satisfies for all f

$$0 \leq \gamma_{xy}^2(f) \leq 1 \quad (5.24)$$

To eliminate delta functions at the origin, mean values different from zero should be removed from the data before applying these last two results. Note that the coherence function is analogous to the square of the correlation function coefficient $\rho_{xy}^2(\tau)$ defined by Equations (3.52) or (3.54).

For a constant parameter linear system, Equations (5.8) and (5.11) apply and may be substituted into Equation (5.23) to obtain

$$\gamma_{xy}^2(f) = \frac{|H(f)|^2 G_x^2(f)}{G_x(f) |H(f)|^2 G_x(f)} = 1 \quad (5.25)$$

Hence for the ideal case of a constant parameter linear system with a single clearly defined input and output, the coherence function will be unity. If $x(t)$ and $y(t)$ are completely unrelated, the coherence function will be zero. If the coherence function is greater than zero but less than unity, one or more of three possible situations exist:

- Extraneous noise is present in the measurements
- The system relating $x(t)$ and $y(t)$ is not linear
- $y(t)$ is an output due to an input $x(t)$ as well as to other inputs

For linear systems, the coherence function $\gamma_{xy}^2(f)$ can be interpreted as the fractional portion of the mean square value at the output $y(t)$ which is contributed by the input $x(t)$ at frequency f . Conversely, the quantity $[1 - \gamma_{xy}^2(f)]$ is a measure of the mean square value of $y(t)$ not accounted for by $x(t)$ at frequency f .

Example 5.5 Effect of Extraneous Noise. Assume a measured input $x(t)$ and a measured output $y(t)$ are composed of true signals $u(t)$ and $v(t)$ and uncorrelated noise components $n(t)$ and $m(t)$, respectively, as shown in

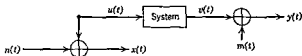


Figure 5.3 Single input system with measurement noise

Figure 5.3. The measured input and output are given by

$$x(t) = u(t) + n(t)$$

$$y(t) = v(t) + m(t)$$

The measured spectral density relations are, under ideal conditions,

$$G_x(f) = G_u(f) + G_n(f)$$

$$G_y(f) = G_v(f) + G_m(f)$$

$$G_{xy}(f) = G_{uv}(f)$$

For this problem, the desired coherence function is

$$\gamma_{uv}^2(f) = \frac{|G_{uv}(f)|^2}{G_u(f) G_v(f)} \quad (5.26)$$

The measured coherence function will be

$$\begin{aligned} \gamma_{xy}^2(f) &= \frac{|G_{xy}(f)|^2}{G_x(f) G_y(f)} = \frac{|G_{uv}(f)|^2}{[G_u(f) + G_n(f)][G_v(f) + G_m(f)]} \\ &= \frac{\gamma_{uv}^2(f)}{1 + (N_1/G_1) + (N_2/G_2) + (N_1/G_1)(N_2/G_2)} < \gamma_{uv}^2(f) \end{aligned} \quad (5.27)$$

where

$$N_1 = G_n(f) \quad G_1 = G_u(f)$$

$$N_2 = G_m(f) \quad G_2 = G_v(f)$$

Equation (5.27) illustrates the behavior that would be expected when uncorrelated extraneous noise is present in both the input and output measurements. Observe that the measured coherence function will theoretically always be less than the desired coherence function whenever the noise terms N_1 and/or N_2 are greater than zero. Further note that the contributions of uncorrelated inputs other than the measured input $x(t)$ will appear as uncorrelated extraneous noise at the output.

Example 5.6. Physical Illustration of Coherence Measurement. Consider an airplane flying through a patch of atmospheric turbulence, as illustrated in Figure 5.4. Let the input $x(t)$ be vertical gust velocity in ft/sec as measured with a probe extending forward of the airplane, and the output $y(t)$ be vertical acceleration in g's measured with an accelerometer at the center of gravity of the airplane. The resulting power spectra and coherence function for actual data of this type are presented in Figure 5.5. In this problem, the spectral data were computed over a frequency range from 0.1 to 4.0 cps with a resolution bandwidth of 0.05 cps and a record length of about 10 minutes.

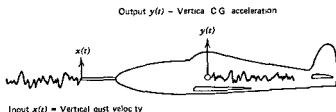


Figure 5.4 Airplane flying through atmospheric turbulence

From Figure 5.5(a), it is seen that the input gust velocity and output airplane acceleration display a relatively strong coherence of 0.8 to 0.9 over the frequency range from about 0.3 to about 2.0 cps. Below and above this range, however, the coherence function diminishes. At the lower frequencies, the vertical acceleration of the airplane is increasingly due to maneuver loads induced through the control system by the pilot, rather than due to atmospheric turbulence loads. Hence, the loss of coherence at the lower frequencies probably reflects contributions to the output $y(t)$ from inputs other than the measured input $x(t)$. At the higher frequencies, the low pass filtering characteristics of the airplane response plus the decaying nature of the input power spectrum cause the output power spectrum to fall off sharply, as indicated in Figure 5.5(b). On the other hand, the noise floor for the data acquisition and recording equipment generally does not fall off with increasing frequency. Hence, the diminishing coherence at the higher frequencies probably results from the contributions of extraneous measurement noise. This concludes Example 5.6.

For applications of coherence functions to problems of estimating linear frequency response functions, the coherence function may be considered to be the ratio of two different measures of the square of the system gain factor. From Equation (5.8), one measure is given by

$$|H(f)|_1^2 = \frac{G_y(f)}{G_x(f)} \quad (5.28)$$

From Equation (5.11), the second measure is given by

$$|H(f)|_2^2 = \frac{|G_{xy}(f)|^2}{G_x^2(f)} \quad (5.29)$$

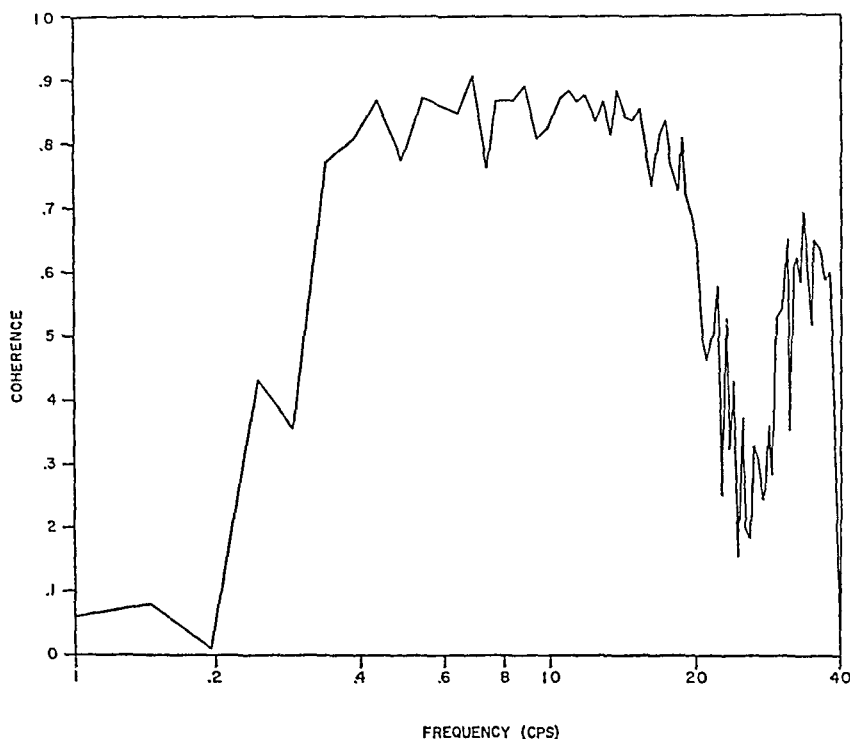


Figure 5.5(a) Coherence function between gust velocity and response acceleration. These data resulted from studies funded by the NASA Langley Research Center, Hampton, Virginia, under Contract NAS 1-8538.

Now, their ratio gives the coherence function as follows.

$$\frac{|H(f)|_2^2}{|H(f)|_1^2} = \frac{|G_{xy}(f)|^2}{G_x(f) G_y(f)} = \gamma_{xy}^2(f) \quad (5.30)$$

In practice, measured values of Equation (5.30) will be between zero and unity. The gain factor estimate of Equation (5.28), based upon power spectra calculations of input and output, will be a biased estimate for all cases except when the coherence function equals unity. The gain factor estimate of Equation (5.29), however, based upon the input power spectrum and the cross-spectrum between input and output, will be a biased estimate if extraneous noise is present at the input, but will be an unbiased estimate (assuming very long record lengths) if extraneous noise is present at the output only. In particular, Equation (5.29) provides an unbiased estimate of the

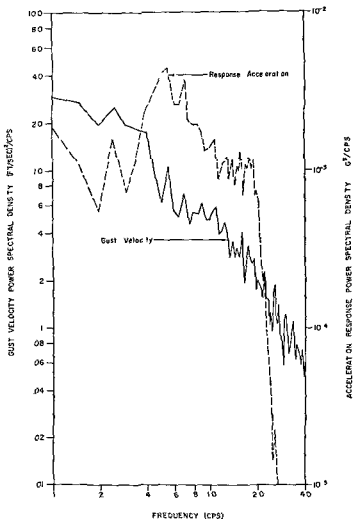


Figure 5.5(b) Power spectra of gust velocity and response acceleration. The data resulted from studies funded by the NASA Langley Research Center, Hampton, Virginia, under Contract NAS 1-8538.

frequency response function gain factors in multiple input problems when the inputs are uncorrelated. These matters are discussed further in Chapter 6, where it is shown how the accuracy of frequency response function estimates increases as the coherence function approaches unity.

5.3 MULTIPLE-INPUT LINEAR SYSTEMS

Constant parameter linear systems responding to multiple inputs from stationary random processes will now be considered. It will be assumed that q inputs are occurring with a single output being measured. For convenience, two-sided spectral quantities $S(f)$ will be used in the developments instead of one-sided spectral quantities $G(f)$. All the resulting formulas will be valid when the S 's are replaced by their corresponding G 's.

5.3.1 Autocorrelation and Power Spectra Relations

Consider q constant parameter linear systems with q clearly defined inputs $x_i(t)$, $i = 1, 2, 3, \dots, q$, and one measured output $y(t)$, as illustrated in Figure 5.6. The output may be considered as the sum of the q outputs $y_i(t)$, $i = 1, 2, 3, \dots, q$. That is,

$$y(t) = \sum_{i=1}^q y_i(t) \quad (5.31)$$

where $y_i(t)$ is defined as that part of the output which is produced by the i th input when all the other inputs are zero.

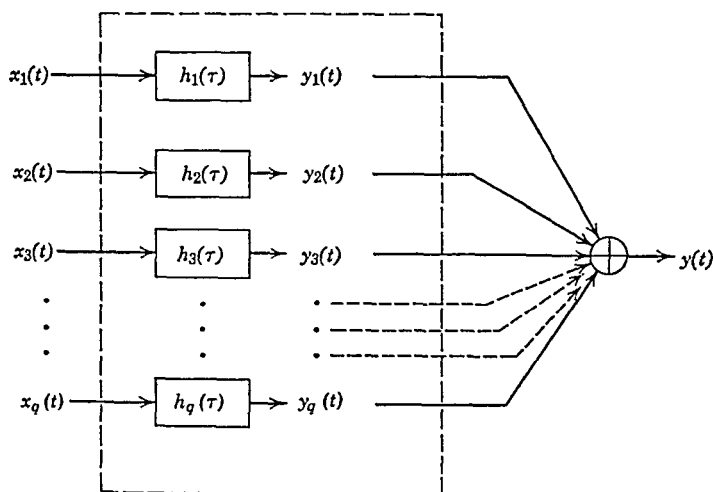


Figure 5.6 Multiple-input linear system.

Let the function $h_i(\tau)$ be defined as the weighting function which is associated with the input $x_i(t)$. Hence, from Equation (5.1), $y_i(t)$ is given by

$$y_i(t) = \int_0^\infty h_i(\tau) x_i(t - \tau) d\tau \quad (5.32)$$

and the total output $y(t)$ is given by

$$y(t) = \sum_{i=1}^q \int_0^\infty h_i(\tau) x_i(t - \tau) d\tau \quad (5.33)$$

Assume next that the inputs $x_i(t)$ are sample functions from different stationary random processes with mean values of μ_i . Here, independent of t ,

$$\mu_i = E[x_i(t)] \quad (5.34)$$

where each $x_i(t)$ is defined over an appropriate sample space. Recalling that the expected value operator is linear, the expected value of $y(t)$ then is obtained as

$$\begin{aligned} \mu_y = E[y(t)] &= E\left[\sum_{i=1}^q \int_0^\infty h_i(\tau) x_i(t - \tau) d\tau\right] \\ &= \sum_{i=1}^q \int_0^\infty h_i(\tau) E[x_i(t - \tau)] d\tau \\ &= \sum_{i=1}^q \mu_i \int_0^\infty h_i(\tau) d\tau = \sum_{i=1}^q \mu_i H_i(0) \end{aligned} \quad (5.35)$$

Equation (5.35) is a general result for arbitrary μ_i . If each $\mu_i = 0$, then $\mu_y = 0$. Nonzero mean values should be removed in practice before carrying out the following analysis. This simplifies greatly the resulting formulas and their interpretation.

The autocorrelation function $R_y(\tau)$ will now be computed. Assuming that the process is stationary, one finds

$$\begin{aligned} R_y(\tau) &= E[y(t) y(t + \tau)] = E\left[\sum_i y_i(t) \sum_j y_j(t + \tau)\right] \\ &= E\left[\sum_i \sum_j \int_0^\infty h_i(\xi) x_i(t - \xi) d\xi \int_0^\infty h_j(\eta) x_j(t + \tau - \eta) d\eta\right] \\ &= \sum_i \sum_j \int_0^\infty \int_0^\infty h_i(\xi) h_j(\eta) E[x_i(t - \xi) x_j(t + \tau - \eta)] d\xi d\eta \end{aligned}$$

$$= \sum_{i=1}^q \sum_{j=1}^q \int_0^{\infty} \int_0^{\infty} h_i(\xi) h_j(\eta) R_{ij}(\xi - \eta + \tau) d\xi d\eta \quad (5.36)$$

In Equation (5.36), $R_{ij}(\tau)$ is defined as

$$R_{ij}(\tau) = R_{x_i x_j}(\tau) = E[x_i(t) x_j(t + \tau)] \quad (5.37)$$

Equation (5.36) is a general result for correlated inputs. If it is assumed that all the various inputs are *mutually uncorrelated* and have *zero means*, then

$$R_{ij}(\tau) = \begin{cases} R_i(\tau) & \text{when } i = j \\ 0 & \text{when } i \neq j \end{cases} \quad (5.38)$$

In this case, Equation (5.36) simplifies to

$$R_v(\tau) = \sum_{i=1}^q \int_0^{\infty} \int_0^{\infty} h_i(\xi) h_i(\eta) R_i(\xi - \eta + \tau) d\xi d\eta \quad (5.39)$$

The power spectral density relations will now be computed. For stationary random processes, the two-sided power spectral density function $S_v(f)$ is given as the Fourier transform of the autocorrelation function $R_v(\tau)$. Thus the Fourier transform of Equation (5.36) yields the output power spectral density function as follows.

$$\begin{aligned} S_v(f) &= \int_{-\infty}^{\infty} e^{-j2\pi f\tau} R_v(\tau) d\tau \\ &= \int_{-\infty}^{\infty} e^{-j2\pi f\tau} \left(\sum_{i=1}^q \sum_{j=1}^q \int_0^{\infty} \int_0^{\infty} h_i(\xi) h_j(\eta) R_{ij}(\xi - \eta + \tau) d\xi d\eta \right) d\tau \quad (5.40) \end{aligned}$$

To simplify Equation (5.40), insert the factor $e^{-j2\pi f(\xi-\eta)} e^{j2\pi f(\xi-\eta)} = 1$. The equation then becomes

$$\begin{aligned} S_v(f) &= \sum_{i=1}^q \sum_{j=1}^q \int_{-\infty}^{\infty} \int_0^{\infty} \int_0^{\infty} h_i(\xi) e^{j2\pi f\xi} h_j(\eta) e^{-j2\pi f\eta} \\ &\quad \times R_{ij}(\xi - \eta + \tau) e^{-j2\pi f(\xi-\eta+\tau)} d\xi d\eta d\tau \quad (5.41) \end{aligned}$$

Now make the change of the variable $t = \xi - \eta + \tau$ and $dt = d\tau$. Equation (5.41) may be factored to give the following general result for correlated inputs.

$$\begin{aligned} S_v(f) &= \sum_{i=1}^q \sum_{j=1}^q \int_0^{\infty} h_i(\xi) e^{j2\pi f\xi} d\xi \int_0^{\infty} h_j(\eta) e^{-j2\pi f\eta} d\eta \int_{-\infty}^{\infty} R_{ij}(t) e^{-j2\pi ft} dt \\ &= \sum_{i=1}^q \sum_{j=1}^q H_i^*(f) H_j(f) S_{ij}(f) \quad (5.42) \end{aligned}$$

In Equation (5.42), $S_{iy}(f)$ represents the cross-spectral density function between the inputs $x_i(t)$ and $x_j(t)$, that is, $S_{iy}(f) = S_{x_i x_j}(f)$. One-sided spectra $G_{iy}(f)$ and $G_{ij}(f)$ can replace $S_{iy}(f)$ and $S_{ij}(f)$ since the common factor of two cancels out. For the uncorrelated case where the autocorrelation function is given by Equation (5.38), Equation (5.42) becomes

$$S_y(f) = \sum_{i=1}^q |H_i(f)|^2 S_{ii}(f) \quad (5.43)$$

5.3.2 Cross-Correlation and Cross-Spectra Relations

The cross spectral relations for the system in Figure 5.6 are obtained by calculating the cross spectral density function $S_{xy}(f) = S_{iy}(f)$ of the output $y(t)$ with one of the inputs, say $x_i(t)$. The general result for correlated inputs is

$$S_{iy}(f) = \sum_{j=1}^q H_j(f) S_{ij}(f) \quad (5.44)$$

This equation is derived in a manner similar to the previous derivation of $S_y(f)$, as follows. For each $i = 1, 2, \dots, q$,

$$\begin{aligned} S_{iy}(f) &= \int_{-\infty}^{\infty} e^{-j2\pi f\tau} R_{iy}(\tau) d\tau \\ &= \sum_{j=1}^q \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} e^{-j2\pi f\xi} h_j(\xi) d\xi \right) e^{-j2\pi f(\tau-\xi)} R_{ij}(\tau-\xi) d\xi d\tau \\ &= \sum_{j=1}^q H_j(f) S_{ij}(f) \end{aligned} \quad (5.45)$$

The above result is arrived at utilizing the fact that the cross-correlation function $R_{iy}(\tau)$ is given by

$$\begin{aligned} R_{iy}(\tau) &= E[x_i(t) y(t+\tau)] = E\left[x_i(t) \sum_{j=1}^q \int_0^{\infty} x_j(t+\tau-\xi) h_j(\xi) d\xi\right] \\ &= \sum_{j=1}^q \int_0^{\infty} h_j(\xi) E[x_i(t) x_j(t+\tau-\xi)] d\xi \\ &= \sum_{j=1}^q \int_0^{\infty} h_j(\xi) R_{ij}(\tau-\xi) d\xi \end{aligned} \quad (5.46)$$

If it is assumed that the inputs are *mutually uncorrelated* and have zero means as per Equation (5.38) then Equation (5.46) reduces to

$$R_{iy}(\tau) = \int_0^{\infty} h_i(\xi) R_{ii}(\tau-\xi) d\xi \quad (5.47)$$

Hence the cross-spectral input/output formula becomes

$$S_{iv}(f) = H_i(f) S_i(f) \quad (5.48)$$

Equation (5.48) is a useful and interesting result since it implies that the frequency response characteristics of the structural path associated with the input $x_i(t)$ can be measured by means of the cross-spectrum whether or not the other inputs are active, as long as the inputs are mutually uncorrelated. This result solves the problem of how to measure the individual frequency response functions for uncorrelated inputs. The statistical accuracy of the measurements can be determined from a knowledge of various coherence functions, as discussed in Chapter 6.

5.3.3 Special Case of Two Inputs

Consider a two-input linear system as pictured in Figure 5.7 where $x_1(t)$ and $x_2(t)$ may be correlated. Assume one desires to determine the frequency

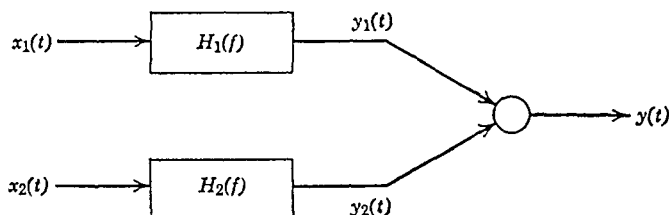


Figure 5.7 Two-input linear system.

response functions of the systems from spectral density information. For this case, Equation (5.44) states

$$\begin{aligned} S_{1v}(f) &= H_1(f) S_{11}(f) + H_2(f) S_{12}(f) \\ S_{2v}(f) &= H_1(f) S_{21}(f) + H_2(f) S_{22}(f) \end{aligned} \quad (5.49)$$

The solutions for $H_1(f)$ and $H_2(f)$, assuming $\gamma_{12}^2(f) \neq 1$, are

$$H_1(f) = \frac{S_{1v}(f) \left[1 - \frac{S_{12}(f) S_{2v}(f)}{S_{22}(f) S_{1v}(f)} \right]}{S_{11}(f) [1 - \gamma_{12}^2(f)]} \quad (5.50)$$

$$H_2(f) = \frac{S_{2v}(f) \left[1 - \frac{S_{21}(f) S_{1v}(f)}{S_{11}(f) S_{2v}(f)} \right]}{S_{22}(f) [1 - \gamma_{12}^2(f)]} \quad (5.51)$$

where

$$\gamma_{12}^2(f) = \frac{\{S_{12}(f)\}^2}{S_{11}(f) S_{22}(f)}$$

For the special case when $\gamma_{12}^2(f) = 0$, the terms $S_{12}(f)$ and $S_{21}(f)$ are zero also, and Equations (5.50) and (5.51) reduce to the usual relations

$$H_1(f) = \frac{S_{12}(f)}{S_{11}(f)} \quad (5.52)$$

$$H_2(f) = \frac{S_{21}(f)}{S_{22}(f)} \quad (5.53)$$

The case when $\gamma_{12}^2(f) = 1$ must be handled separately. A coherence function of unity between $x_1(t)$ and $x_2(t)$ implies complete linear dependence. Hence one would consider a linear system existing between them as illustrated in Figure 5.8. The implication is that the first input $x_1(t)$ is actually

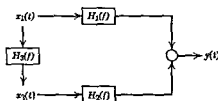


Figure 5.8 Example of fully coherent inputs

taking two different paths to arrive at the output $y(t)$. For this situation, a single frequency response function $H(f)$ will relate $y(t)$ to $x_1(t)$ namely,

$$H(f) = H_1(f) + H_2(f) H_3(f)$$

In a general case where $\gamma_{12}^2(f) \neq 0$ or 1, the denominator in Equation (5.50) is the spectrum of the residual process $\{\Delta x_1(t)\}$ resulting from $\{x_1(t)\}$ after a linear least squares prediction of $\{x_1(t)\}$ from $\{x_2(t)\}$ has been subtracted from $\{x_1(t)\}$. Similarly, the numerator in Equation (5.50) is the cross spectrum of the residual processes $\{\Delta y(t)\}$ and $\{\Delta x_1(t)\}$ where $\{\Delta y(t)\}$ is the residual process resulting from $\{y(t)\}$ after a linear least squares prediction of $\{y(t)\}$ from $\{x_2(t)\}$ has been subtracted from $\{y(t)\}$. These statements follow by analogy from basic regression analysis theory as discussed previously in Chapter 4 and in the next section.

For an arbitrary two-input linear system where the inputs may be correlated, $S_{1v}(f)$ and $S_{2v}(f)$ are given by Equation (5.49), whereas $S_{vv}(f)$ is given by Equation (5.42). Specifically,

$$S_{vv}(f) = |H_1(f)|^2 S_{11}(f) + H_1^*(f) H_2(f) S_{12}(f) \\ + H_2^*(f) H_1(f) S_{21}(f) + |H_2(f)|^2 S_{22}(f) \quad (5.54)$$

The ordinary coherence functions between each input and the output are

$$\gamma_{1v}^2(f) = \frac{|H_1(f) S_{11}(f) + H_2(f) S_{12}(f)|^2}{S_{11}(f) S_{vv}(f)} \\ \gamma_{2v}^2(f) = \frac{|H_1(f) S_{21}(f) + H_2(f) S_{22}(f)|^2}{S_{22}(f) S_{vv}(f)} \quad (5.55)$$

5.4 PARTIAL AND MULTIPLE COHERENCE FUNCTIONS

Definitions of residual random variables and partial coherence functions from mean square error criteria are developed in Sections 5.4.1 and 5.4.2 for the special case of two-input linear systems. Extensions of these ideas are based on similar mean square error criteria. General definitions of multiple coherence functions are developed in Sections 5.4.3 and 5.4.4.

5.4.1 Residual Random Variables

Consider any two real-valued stationary random processes $\{x(t)\}$ and $\{y(t)\}$. For fixed t , the quantities $x(t)$ and $y(t)$ are random variables over the sample space in question. To remove the need to subtract off mean values, assume that the mean values are zero. A linear prediction $\hat{y}(t)$ of $y(t)$ from $x(t)$ can be described by the following equation, where the lower limit may be set equal to $-\infty$ instead of to 0 as a matter of convenience when desired.

$$\hat{y}(t) = \int_0^\infty h_0(\tau) x(t - \tau) d\tau \quad (5.56)$$

Here, $h_0(\tau)$ is a weighting function to be determined. Following the basic ideas of least squares prediction outlined in Section 4.8.2, $h_0(\tau)$ is chosen so that the mean square error ϵ^2 given by

$$\epsilon^2 = E[\{y(t) - \hat{y}(t)\}^2] = E\left[\left\{y(t) - \int_0^\infty h_0(\tau) x(t - \tau) d\tau\right\}^2\right] \\ = R_{yy}(0) - 2 \int_0^\infty h_0(\tau) R_{xy}(\tau) d\tau + \iint_0^\infty h_0(\tau) h_0(\nu) R_{xx}(\tau - \nu) d\nu d\tau \quad (5.57)$$

is a minimum over all possible choices of $h_0(\tau)$. This condition is satisfied by that $h_0(\tau)$ which makes

$$\frac{\partial \epsilon^2}{\partial h_0(\tau)} = 0 \quad (5.58)$$

This leads to the convolution integral result,

$$R_{xy}(\tau) = \int_0^\infty h_0(v) R_{xx}(\tau - v) dv \quad (5.59)$$

which, upon taking Fourier transforms of both sides, is equivalent to

$$S_{xy}(f) = H_0(f) S_{xx}(f) \quad (5.60)$$

The similarity of Equation (5.60) to Equation (5.48) should be noted.

Now, the *residual (conditioned) random variable* $\Delta y(t)$ resulting from $y(t)$, after a linear least squares prediction of $y(t)$ from $x(t)$ has been subtracted from $y(t)$, is denoted by

$$\Delta y(t) = y(t) - \hat{y}(t) = y(t) - \int_0^\infty h_0(\tau) x(t - \tau) d\tau \quad (5.61)$$

Then the autocorrelation function $R_{\Delta y \Delta y}(\tau)$ is determined by

$$R_{\Delta y \Delta y}(\tau) = E[\Delta y(t) \Delta y(t + \tau)] \quad (5.62)$$

which after substitution and simplification yields

$$R_{\Delta y \Delta y}(\tau) = R_{yy}(\tau) - \int_0^\infty h_0(v) R_{yx}(\tau - v) dv \quad (5.63)$$

The Fourier transform of Equation (5.63) leads to

$$S_{\Delta y \Delta y}(f) = S_{yy}(f) - H_0(f) S_{yx}(f) \quad (5.64)$$

Use of Equation (5.60) then gives

$$\begin{aligned} S_{\Delta y \Delta y}(f) &= S_{yy}(f) - \frac{S_{xy}(f) S_{yx}(f)}{S_{xx}(f)} \\ &= S_{yy}(f) [1 - \gamma_{xy}^2(f)] \end{aligned} \quad (5.65)$$

where

$$\gamma_{xy}^2(f) = \frac{|S_{xy}(f)|^2}{S_{xx}(f) S_{yy}(f)} \quad (5.66)$$

is the ordinary coherence function between x and y . The quantity $S_{\Delta y \Delta y}(f)$ in Equation (5.65) is called the *residual (unconditioned) spectral density function*

of $\Delta y(t)$. An accepted notation for $S_{\Delta y \Delta y}(f)$ which points out the least-squares prediction of $y(t)$ from $x(t)$ is

$$S_{\Delta y \Delta y}(f) = S_{y y \cdot x}(f) \quad (5.67)$$

Consider again the two-input linear system of Figure 5.7, where the inputs $x_1(t)$ and $x_2(t)$ may be correlated. Let $\hat{y}(t)$ be a linear least-squares prediction of $y(t)$ from $x_2(t)$ as given by

$$\hat{y}(t) = \int_0^\infty c(\tau) x_2(t - \tau) d\tau \quad (5.68)$$

Similarly, let $\hat{x}_1(t)$ be a linear least-squares prediction of $x_1(t)$ from $x_2(t)$ as given by

$$\hat{x}_1(t) = \int_0^\infty b(\tau) x_2(t - \tau) d\tau \quad (5.69)$$

The weighting functions $c(\tau)$ and $b(\tau)$ must be suitably determined as developed in Equations (5.56) through (5.60). Thus they must satisfy the relations

$$S_{2y}(f) = C(f) S_{22}(f) \quad (5.70)$$

$$S_{21}(f) = B(f) S_{22}(f) \quad (5.71)$$

Now, define the residual random variables

$$\Delta x_1(t) = x_1(t) - \hat{x}_1(t) \quad (5.72)$$

$$\Delta y(t) = y(t) - \hat{y}(t) \quad (5.73)$$

which result by subtracting from $x_1(t)$ and $y(t)$ their respective linear least-squares predictions from $x_2(t)$. To keep this particular operation in mind, the following notation will be used for the associated residual spectral density functions.

$$\begin{aligned} S_{\Delta x_1 \Delta x_1}(f) &= S_{11 \cdot 2}(f) \\ S_{\Delta y \Delta y}(f) &= S_{y y \cdot 2}(f) \end{aligned} \quad (5.74)$$

$$S_{\Delta x_1 \Delta y}(f) = S_{1y \cdot 2}(f)$$

From Equation (5.65), it follows that

$$S_{11 \cdot 2}(f) = S_{11}(f)[1 - \gamma_{12}^2(f)] \quad (5.75)$$

$$S_{y y \cdot 2}(f) = S_{yy}(f)[1 - \gamma_{y2}^2(f)] \quad (5.76)$$

where

$$\gamma_{12}^2(f) = \frac{|S_{12}(f)|^2}{S_{11}(f) S_{22}(f)} \quad (5.77)$$

$$\gamma_{y2}^2(f) = \frac{|S_{y2}(f)|^2}{S_{yy}(f) S_{22}(f)} \quad (5.78)$$

The derivation of $S_{1y \cdot 2}(f)$ requires a little more effort.

Starting with Equations (5 72) and (5 73) the *residual (conditional) cross-correlation function* between $\Delta x_1(t)$ and $\Delta y(t)$ is given by

$$R_{1y\ 2}(\tau) = E[\Delta x_1(t) \Delta y(t + \tau)] \quad (5 79)$$

Substitutions and simplifications now lead to

$$R_{1y\ 2}(\tau) = R_{1y}(\tau) - \int_0^\infty c(v) R_{12}(\tau - v) dv \quad (5 80)$$

Then, the Fourier transform of both sides gives

$$S_{1y\ 2}(f) = S_{1y}(f) - C(f) S_{12}(f) \quad (5 81)$$

which yields, with the aid of Equation (5 70), the *residual (conditioned) cross spectral density function* between $\Delta x_1(t)$ and $\Delta y(t)$ as follows

$$S_{1y\ 2}(f) = S_{1y}(f) \left[1 - \frac{S_{12}(f) S_{2y}(f)}{S_{22}(f) S_{1y}(f)} \right] \quad (5 82)$$

Examinations of Equations (5 75) and (5 82) in relation to the frequency response function $H_1(f)$ of Equation (5 50) reveals the interesting result that

$$H_1(f) = \frac{S_{1y\ 2}(f)}{S_{11\ 2}(f)} \quad (5 83)$$

5 4 2 Partial Coherence Functions

The *partial coherence function* between $x_1(t)$ and $y(t)$, when $x_2(t)$ is removed from both $x_1(t)$ and $y(t)$ by linear least squares prediction is now defined as the ordinary coherence function between $\Delta x_1(t)$ and $\Delta y(t)$. It is determined by the formula

$$\gamma_{1y\ 2}^2(f) = \frac{|S_{1y\ 2}(f)|^2}{S_{11\ 2}(f) S_{yy\ 2}(f)} \quad (5 84)$$

This partial coherence function satisfies the usual inequality

$$0 \leq \gamma_{1y\ 2}^2(f) \leq 1 \quad (5 85)$$

Equations (5 84) and (5 85) are direct extensions of Equations (5 23) and (5 24). Under ideal conditions, assuming $H_1(f) \neq 0$ and $\gamma_{12}^2(f) \neq 1$, it can be shown from previous equations that

$$S_{1y\ 2}(f) = H_1(f) S_{11}(f) [1 - \gamma_{12}^2(f)] \quad (5 86)$$

$$S_{11\ 2}(f) = S_{11}(f) [1 - \gamma_{12}^2(f)] \quad (5 87)$$

$$S_{yy\ 2}(f) = |H_1(f)|^2 S_{11}(f) [1 - \gamma_{12}^2(f)] \quad (5 88)$$

Hence the partial coherence function of Equation (5 84) becomes

$$\gamma_{1y\ 2}^2(f) = 1 \quad \text{for all } f \quad (5 89)$$

It should be clear from these developments that the partial coherence function will reveal the existence of a linear relationship between $\Delta x_1(t)$ and $\Delta y(t)$ even when such a relationship is not apparent from the ordinary coherence function between $x_1(t)$ and $y(t)$. Furthermore, this result occurs regardless of whether or not the two inputs are correlated.

The ordinary coherence functions between $x_1(t)$ and $y(t)$, and between $x_2(t)$ and $y(t)$, are defined by

$$\gamma_{1y}^2(f) = \frac{|S_{1y}(f)|^2}{S_{11}(f) S_{yy}(f)} \quad (5.90)$$

$$\gamma_{2y}^2(f) = \frac{|S_{2y}(f)|^2}{S_{22}(f) S_{yy}(f)} \quad (5.91)$$

These two ordinary coherence functions may or may not be unity, depending upon the particular situation. Both erroneous high ordinary coherence and erroneous low ordinary coherence can occur in practice. Consider the following situations.

For the special case when the two inputs are uncorrelated, $S_{12}(f)$ and $\gamma_{12}(f) = 0$. Here, the ordinary coherence functions become

$$\gamma_{1y}^2(f) = \frac{|H_1(f)|^2 S_{11}(f)}{|H_1(f)|^2 S_{11}(f) + |H_2(f)|^2 S_{22}(f)} \quad (5.92)$$

$$\gamma_{2y}^2(f) = \frac{|H_2(f)|^2 S_{22}(f)}{|H_1(f)|^2 S_{11}(f) + |H_2(f)|^2 S_{22}(f)} \quad (5.93)$$

Thus, for the uncorrelated two-input case, the sum

$$\gamma_{1y}^2(f) + \gamma_{2y}^2(f) = 1 \quad (5.94)$$

proving that each of these ordinary coherence functions will be strictly less than unity if the other is nonzero. Note that the effect of $x_2(t)$ in computing the ordinary coherence function $\gamma_{1y}^2(f)$ between $x_1(t)$ and $y(t)$ is to appear as a noise component at the output $y(t)$. This brings about an erroneous low ordinary coherence reading of less than unity which hides the true linear relationship between $x_1(t)$ and $y(t)$. However, as Equation (5.89) shows, when the partial coherence function between $x_1(t)$ and $y(t)$ is computed, the effects of $x_2(t)$ are removed from both $x_1(t)$ and $y(t)$, and the true coherence of unity is computed.

An example of erroneous high ordinary coherence is shown in Figure 5.9. Assume that a coherence function value near unity is computed between the two variables $x_1(t)$ and $y(t)$. One would be inclined to believe that there is a physical linear system relating these two variables as input and output. But suppose there is a third variable $x_2(t)$ which is highly coherent with $x_1(t)$ and also passes through a linear system to make up $y(t)$. In this type of situation,

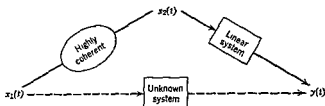
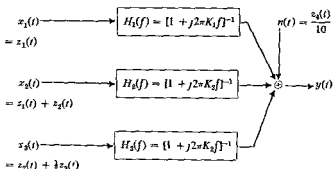


Figure 5.9 Illustration of erroneous high coherence

the high coherence computed between $x_1(t)$ and $y(t)$ might only be a reflection of the fact that $x_2(t)$ is highly coherent with $x_1(t)$, and $x_2(t)$ is related via a linear system to $y(t)$. In reality there may be no direct physical system between $x_1(t)$ and $y(t)$ at all. If the partial coherence function is computed between $x_1(t)$ and $y(t)$ in this situation, it might turn out to be a very small number near zero. Both the ordinary and the partial coherence between $x_2(t)$ and $y(t)$, however, would prove to be near unity.

Example 5.7. Illustration of Erroneous High Coherence. Consider a three input single output system with output measurement noise and low pass RC filter transmission paths, as detailed in Figure 5.10. Note that input $x_2(t)$ is correlated with both inputs $x_1(t)$ and $x_3(t)$. How coherent is the input $x_2(t)$ with the output $y(t)$ through the indicated transmission path?



$z_1(t)$, $z_2(t)$, $z_3(t)$, and $z_4(t)$ are four independent normally distributed random time histories with means of zero and variances of unity.

Figure 5.10 Three-input linear system with output measurement noise

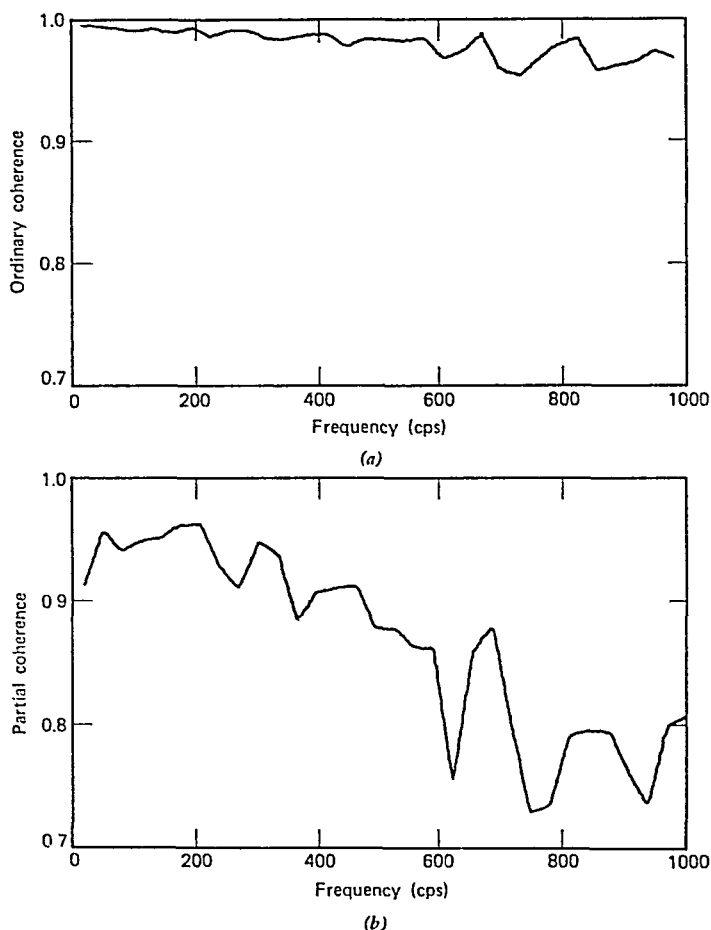


Figure 5.11 Coherence functions between input $x_2(t)$ and output for system in Figure 5.10. (a) Ordinary coherence function. (b) Partial coherence function.

To answer this question, the ordinary and partial coherence functions between input $x_2(t)$ and the output $y(t)$ were computed over a frequency range from near zero to about 1000 cps with a resolution bandwidth of 32 cps and a record length of 1 second. The results are presented in Figure 5.11. From Figure 5.11 (a), it is seen that the ordinary coherence function between $x_2(t)$ and $y(t)$ is quite strong over the entire frequency range, varying from near unity at low frequencies to about 0.97 at the higher frequencies.

However, this is due in part to the fact that both $x_1(t)$ and $x_2(t)$ are coherent with $x_3(t)$ and also contribute directly to the output $y(t)$ through separate paths. When the partial coherence function between $x_2(t)$ and $y(t)$ is computed, a somewhat reduced indication of coherence (as low as 0.75) results, as shown in Figure 5.11(b).

5.4.3 Multiple Coherence Functions

For the multiple input problem illustrated in Figure 5.6, let the output $y(t)$ include additive noise $z(t)$ defined by

$$z(t) = y(t) - \sum_{i=1}^q \int_0^{\infty} h_i(\tau) x_i(t - \tau) d\tau \quad (5.95)$$

where $z(t)$ is a residual noise term to account for deviations in the linear nature of the model as well as unknown extraneous components, if any, which may be present. The quantity $z(t)$ is similar to the residual random variable $\Delta y(t)$ in Equation (5.61) insofar as it indicates how closely an actual output $y(t)$ can be predicted from linear operations using only the inputs $x_i(t)$, $i = 1, 2, \dots, q$, when noise is present. Observe that $y(t)$ in Equation (5.95) is *not* the same as the $y(t)$ in Equation (5.33).

The autocorrelation function of the residual stationary random process $\{z(t)\}$ is found by calculating

$$\begin{aligned} R_{zz}(\tau) &= E[z(t) z(t + \tau)] \\ &= R_{yy}(\tau) - \sum_{i=1}^q \int_0^{\infty} h_i(\beta) R_{yz}(\tau - \beta) d\beta - \sum_{i=1}^q \int_0^{\infty} h_i(\alpha) R_{yz}(\tau + \alpha) d\alpha \\ &\quad + \sum_{i=1}^q \sum_{j=1}^q \iint_0^{\infty} h_i(\alpha) h_j(\beta) R_{ij}(\tau + \alpha - \beta) d\beta d\alpha \end{aligned} \quad (5.96)$$

But, as shown in Equation (5.46), a desired condition for the h_i terms is

$$R_{iz}(\tau) = \sum_{j=1}^q \int_0^{\infty} h_j(\beta) R_{ij}(\tau - \beta) d\beta \quad (5.97)$$

This condition results as in the derivation of Equation (5.59), when the weighting functions $h_i(\beta)$ are chosen to minimize the mean square error between the actual output $y(t)$ and its linear approximation using only the $x_i(t)$. This condition also follows directly from Equation (5.95) when the residual noise term $z(t)$ is assumed to be uncorrelated with the $x_i(t)$, without any consideration of mean square errors. Conversely, Equation (5.97) implies that $z(t)$ and the $x_i(t)$ are uncorrelated. Hence by specifying Equation (5.97), it follows that Equation (5.96) becomes

$$R_{zz}(\tau) = R_{yy}(\tau) - \sum_{i=1}^q \int_0^{\infty} h_i(\beta) R_{ii}(\tau - \beta) d\beta \quad (5.98)$$

The Fourier transform of both sides of Equation (5.98) now gives for the spectral density function of the residual variables,

$$\begin{aligned} S_{zz}(f) &= S_{yy}(f) - \sum_{i=1}^q H_i(f) S_{vi}(f) \\ &= S_{yy}(f)[1 - \gamma_{y,x}^2(f)] \end{aligned} \quad (5.99)$$

where $\gamma_{y,x}^2(f)$ is defined by

$$\gamma_{y,x}^2(f) = \frac{\sum_{i=1}^q H_i(f) S_{vi}(f)}{S_{yy}(f)} = 1 - \frac{S_{zz}(f)}{S_{yy}(f)} \quad (5.100)$$

The quantity $\gamma_{y,x}^2(f)$ is called the *multiple coherence function* between the output process $y(t)$ and all the input processes $x_i(t)$; $i = 1, 2, \dots, q$. It satisfies the inequality

$$0 \leq \gamma_{y,x}^2(f) \leq 1 \quad \text{for all } f \quad (5.101)$$

Equation (5.101) can be derived directly from Equation (5.100) by setting $S_{zz}(f) \geq 0$ and by setting $S_{zz}(f) \leq S_{yy}(f)$, two properties which are valid for this model. To prove that $S_{zz}(f) \leq S_{yy}(f)$, it is sufficient to show that $z(t)$ and $x_i(t)$ are uncorrelated for each $i = 1, 2, \dots, q$. For then it will follow from Equation (5.95) that $S_{yy}(f)$ is the sum of $S_{zz}(f)$ plus a power spectral density function term (which is non-negative), and that all cross-spectrum terms will be zero. By definition, the cross-correlation function between $z(t)$ and $x_i(t)$ is

$$R_{zi}(\tau) = E[z(t)x_i(t + \tau)]$$

where $z(t)$ can be expressed from Equation (5.95) as

$$z(t) = y(t) - \sum_{j=1}^q \int_0^\infty h_j(\beta) x_j(t - \beta) d\beta$$

Then, since expected values commute with linear operations,

$$\begin{aligned} R_{zi}(\tau) &= E[y(t)x_i(t + \tau)] - \sum_{j=1}^q \int_0^\infty h_j(\beta) E[x_i(t + \tau)x_j(t - \beta)] d\beta \\ &= R_{yi}(\tau) - \sum_{j=1}^q \int_0^\infty h_j(\beta) R_{ij}(-\tau - \beta) d\beta \end{aligned}$$

But from Equation (5.97),

$$R_{yi}(\tau) = R_{iy}(-\tau) = \sum_{j=1}^q \int_0^\infty h_j(\beta) R_{ij}(-\tau - \beta) d\beta$$

Hence

$$R_{zi}(\tau) = 0 \quad (5.102)$$

proving the desired result.

The multiple coherence function defined in Equation (5 100) includes both ordinary and partial coherence functions defined previously. In particular, a partial coherence function may be considered as the multiple coherence function applied to residual (conditioned) random variables. Consider applying Equation (5 100) to a simple input output problem where $q = 1$. For this case from Equation (5 100)

$$\gamma_{yx}^2(f) = \frac{H_1(f) S_{y1}(f)}{S_{yy}(f)} \quad (5 103)$$

Also for this case the Fourier transform of Equation (5 97) yields

$$S_{1y}(f) = H_1(f) S_{11}(f) \quad (5 104)$$

Then eliminating $H_1(f)$ gives

$$\gamma_{yx}^2(f) = \frac{|S_{1y}(f)|^2}{S_{11}(f) S_{yy}(f)} \quad (5 105)$$

which is simply the usual ordinary coherence function. Thus the multiple coherence function contains the ordinary coherence function as a special case.

In general by taking Fourier transforms the frequency response functions $H_i(f)$ in Equation (5 100) can be determined from the relationship in Equation (5 97) by solving the system of equations

$$S_{iy}(f) = \sum_{j=1}^q H_j(f) S_{ij}(f) \quad i = 1, 2, \dots, q \quad (5 106)$$

Now consider the ideal case where the residual noise term $z(t)$ is zero. From Equation (5 42) the quantity $S_{yy}(f)$ in Equation (5 100) is given by

$$S_{yy}(f) = \sum_{i=1}^q \sum_{j=1}^q H_i^*(f) H_j(f) S_{ij}(f) \quad (5 107)$$

Substitution of the complex conjugate of Equation (5 106) into Equation (5 100) then proves

$$\gamma_{yx}^2(f) = \frac{\sum_{i=1}^q \sum_{j=1}^q H_i(f) H_j^*(f) S_{ij}(f)}{S_{yy}(f)} = 1 \quad (5 108)$$

since $S_{yy}(f) = S_{yy}^*(f)$ and the numerator is $S_{yy}^*(f)$. Thus the multiple coherence function will be unity under noise free ideal conditions when there is a true linear relationship occurring in the multiple input single output problem. This result of course is shown in a more direct way by merely setting $S_{zz}(f)$ in Equation (5 100) equal to zero.

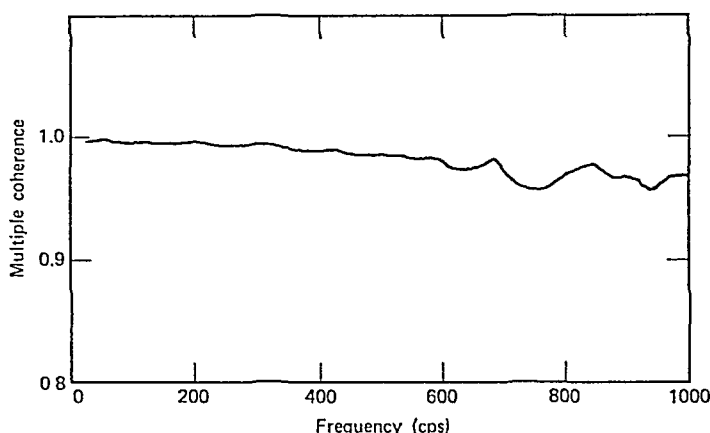


Figure 5.12 Multiple coherence function between the three inputs and the output for the system in Figure 5.10.

Further detailed theoretical developments for multiple coherence functions, as well as partial and other types of coherence functions, are presented in References 20 and 26. Practical interpretations of multiple coherence functions are discussed further in Chapter 6.

Example 5.8. Illustration of Multiple Coherence Measurement. Consider the three input-one output system previously discussed in Example 5.7 and detailed in Figure 5.10. The multiple coherence function computed between the output $y(t)$ and the three inputs $x_i(t)$; $i = 1, 2, 3$, is presented in Figure 5.12. Note that the multiple coherence function is quite high (greater than 0.95) over the entire frequency range, but does drop off slightly with increasing frequency. The only reason the multiple coherence is not unity over the entire frequency range is because of the measurement noise at the output. Since the measurement noise has a uniform power spectrum while the inputs are low pass filtered, the relative contribution of measurement noise in the total output signal increases with frequency. This fact is clearly indicated by the decrease in the multiple coherence function with frequency.

5.4.4 Matrix Formulation of Results

The preceding formulas can be expressed more concisely by the use of matrix notation and, in addition, some further results become more readily apparent. First, define a q -dimensional input vector

$$\mathbf{x}(t) = [x_1(t), x_2(t), \dots, x_q(t)] \quad (5.109)$$

Also define a q dimensional frequency response function vector

$$\mathbf{H}(f) = [H_1(f), H_2(f), \dots, H_q(f)] \quad (5.110)$$

Next define a q dimensional cross-spectrum vector of the output $y(t)$ with the inputs $x_i(t)$,

$$S_{xy}(f) = [S_{1y}(f) \ S_{2y}(f), \dots, S_{qy}(f)] \quad (5.111)$$

where

$$S_{iy}(f) = S_{xy}(f) \quad i = 1, 2, \dots, q \quad (5.112)$$

Finally, define the $q \times q$ spectral matrix of all the inputs $x_i(t)$ by

$$S_{xx}(f) = \begin{bmatrix} S_{11}(f) & S_{12}(f) & S_{1q}(f) \\ S_{21}(f) & S_{22}(f) & S_{2q}(f) \\ \vdots & \vdots & \vdots \\ S_{q1}(f) & S_{q2}(f) & S_{qq}(f) \end{bmatrix} \quad (5.113)$$

where

$$S_{ij}(f) = S_{x_i x_j}(f) \quad i, j = 1, 2, \dots, q \quad (5.114)$$

The result of Equation (5.42) may be rewritten in matrix form as

$$S_{yy}(f) = \mathbf{H}(f) S_{xx}(f) \mathbf{H}^*(f) \quad (5.115)$$

where $\mathbf{H}^*(f)$ denotes the complex conjugate transpose vector. When fully written out, Equation (5.115) becomes

$$S_{yy}(f) = [H_1(f), H_2(f), \dots, H_q(f)] \times \begin{bmatrix} S_{11}(f) & S_{12}(f) & S_{1q}(f) \\ S_{21}(f) & S_{22}(f) & S_{2q}(f) \\ \vdots & \vdots & \vdots \\ S_{q1}(f) & S_{q2}(f) & S_{qq}(f) \end{bmatrix} \begin{bmatrix} H_1^*(f) \\ H_2^*(f) \\ \vdots \\ H_q^*(f) \end{bmatrix} \quad (5.116)$$

The column vector on the right is the transposed complex conjugate of $\mathbf{H}(f)$, that is,

$$\mathbf{H}^*(f) = \begin{bmatrix} H_1^*(f) \\ H_2^*(f) \\ \vdots \\ H_q^*(f) \end{bmatrix} \quad (5.117)$$

Note that $S_{yy}(f)$ in Equation (5.115) is still a scalar quantity but the other quantities are not. Equation (5.115) is also a proper representation of Equation (5.43) where the off-diagonal elements of the matrix $S_{xx}(f)$ become zero for uncorrelated inputs. It should be emphasized, however, that this result applies only to an ideal noise-free situation.

In place of Equation (5.44), the system of equations for $i = 1, 2, \dots, q$ may be written as the matrix equation

$$S_{xy}'(f) = S_{xx}(f) H'(f) \quad (5.118)$$

This is equivalent to

$$\begin{bmatrix} S_{1y}(f) \\ S_{2y}(f) \\ \vdots \\ S_{qy}(f) \end{bmatrix} = \begin{bmatrix} S_{11}(f) & S_{12}(f) & \cdots & S_{1q}(f) \\ S_{21}(f) & S_{22}(f) & \cdots & S_{2q}(f) \\ \vdots & \vdots & \cdots & \vdots \\ S_{q1}(f) & S_{q2}(f) & \cdots & S_{qq}(f) \end{bmatrix} \begin{bmatrix} H_1(f) \\ H_2(f) \\ \vdots \\ H_q(f) \end{bmatrix} \quad (5.119)$$

where the column vectors are the transposed row vectors of $S_{xy}(f)$ and $H(f)$. The matrix equation (5.118) may be solved for the transposed row vector $H'(f)$ if $S_{xy}(f)$ and $S_{xx}(f)$ have been measured or are known. This is, of course, a system of q simultaneous linear equations whose solution is

$$H'(f) = S_{xx}^{-1}(f) S_{xy}'(f) \quad (5.120)$$

where $S_{xx}^{-1}(f)$ represents the inverse matrix to $S_{xx}(f)$. Equation (5.120) gives each $H_i(f)$ as a function of the input/output cross-spectra $S_{iy}(f)$ and the input/input cross-spectra $S_{ii}(f)$, and holds *whether or not the various inputs are correlated*.

To derive further relationships regarding multiple coherence functions, define the $(q+1) \times (q+1)$ augmented spectral matrix of the output $y(t)$ with the inputs $x_i(t)$, namely,

$$S_{yxx}(f) = \begin{bmatrix} S_{yy}(f) & S_{y1}(f) & S_{y2}(f) & \cdots & S_{yq}(f) \\ S_{1y}(f) & S_{11}(f) & S_{12}(f) & \cdots & S_{1q}(f) \\ S_{2y}(f) & S_{21}(f) & S_{22}(f) & \cdots & S_{2q}(f) \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ S_{qy}(f) & S_{q1}(f) & S_{q2}(f) & \cdots & S_{qq}(f) \end{bmatrix} \quad (5.121)$$

The determinant $|S_{vxx}(f)|$ of this matrix will now be shown to be zero for all f in an ideal noise-free situation

From Equation (5.44), the $S_{iv}(f)$ terms in the first column of S_{vxx} , below the first row, can be expressed by

$$S_{iv}(f) = \sum_{j=1}^q H_j(f) S_{ij}(f) \quad i = 1, 2, \dots, q \quad (5.122)$$

showing that each $S_{iv}(f)$ term is a linear combination of the $S_{ij}(f)$ terms in the columns of its row. From Equation (5.42), which applies only to the ideal noise-free case, the remaining $S_{vv}(f)$ term in the first column, first row, can be expressed by

$$\begin{aligned} S_{vv}(f) &= \sum_{i=1}^q \sum_{j=1}^q H_i^*(f) H_j(f) S_{ij}(f) \\ &= \sum_{i=1}^q H_i^*(f) S_{iv}(f) = \sum_{i=1}^q H_i(f) S_{vi}(f) \end{aligned} \quad (5.123)$$

The latter relationships follow from substitution of Equation (5.44) into Equation (5.42), and using the fact that $S_{vv}(f) = S_{vv}^*(f)$. Thus $S_{vv}(f)$ is a linear combination of the $S_{vi}(f)$ terms in the columns of the first row. It follows that for all f in the noise-free case, the matrix S_{vxx} is such that the first column of terms is the result of linear combinations of terms from corresponding terms in the other columns. By the well-known theorem, proved in Reference 32, the determinant of this matrix must then be zero.

Now, consider the more general residual noise model discussed in Section 5.4.3. From Equation (5.99), the $S_{vv}(f)$ term becomes

$$S_{vv}(f) = \sum_{i=1}^q H_i(f) S_{vi}(f) + S_{xx}(f) \quad (5.124)$$

which is different from the expression given by Equation (5.123). However, as noted in Equation (5.106), there is no change in the $S_{vi}(f)$ terms since this is a specified condition which is imposed whether or not noise is present. Hence in this general situation, by using the result derived above for the noise-free case, the determinant $|S_{vxx}|$ of the matrix S_{vxx} can be computed by

$$|S_{vxx}(f)| = S_{xx}(f) |S_{xx}(f)| \quad (5.125)$$

where $|S_{xx}|$ is the determinant of the matrix S_{xx} . Clearly, the determinant $|S_{vxx}|$ must be zero when $S_{xx}(f) = 0$, as required for the noise-free case.

Equation (5.125) leads to a simple determinant formula for the multiple

coherence function defined previously in Equation (5.100), namely,

$$\gamma_{y,x}^2(f) = 1 - \frac{|S_{yx}(f)|}{S_{yy}(f) |S_{xx}(f)|} \quad (5.126)$$

As a check on this formula, note that when $q = 1$,

$$S_{yx} = \begin{bmatrix} S_{yy}(f) & S_{yx}(f) \\ S_{xy}(f) & S_{xx}(f) \end{bmatrix} \quad (5.127)$$

Here

$$|S_{yx}| = S_{xx}(f) S_{yy}(f) - |S_{xy}(f)|^2 \quad \text{and} \quad |S_{xx}(f)| = S_{xx}(f)$$

Then, substituting into Equation (5.126) and simplifying, it follows that

$$\gamma_{y,x}^2(f) = \frac{|S_{xy}(f)|^2}{S_{xx}(f) S_{yy}(f)} \quad (5.128)$$

which is in agreement with the ordinary coherence function definition.

An alternate equivalent way to define the multiple coherence function between $y(t)$ and all the $x_i(t)$ is by the formula

$$\gamma_{y,x}^2(f) = 1 - [S_{yy}(f) S^{\nu\nu}(f)]^{-1} \quad (5.129)$$

where $S^{\nu\nu}(f)$ denotes the first diagonal element of the inverse matrix $S_{yxx}^{-1}(f)$ associated with $S_{yxx}(f)$. This result agrees with Equation (5.126) because the inverse matrix $S_{yxx}^{-1}(f)$ is obtained by dividing the transposed adjoint matrix of $S_{yxx}(f)$ by its determinant $|S_{yxx}(f)|$. The adjoint matrix of $S_{yxx}(f)$ can be computed by substituting the cofactor of the element $S_{ii}(f)$, denoted by $\text{Cof } S_{ii}(f)$, for the element $S_{ii}(f)$. In matrix form

$$S_{yxx}^{-1}(f) = \frac{1}{|S_{yxx}(f)|} \begin{bmatrix} \text{Cof } S_{yy}(f) & \text{Cof } S_{y1}(f) & \text{Cof } S_{y2}(f) & \cdots & \text{Cof } S_{yq}(f) \\ \text{Cof } S_{y1}(f) & \text{Cof } S_{11}(f) & \text{Cof } S_{21}(f) & \cdots & \text{Cof } S_{q1}(f) \\ \text{Cof } S_{y2}(f) & \text{Cof } S_{12}(f) & \text{Cof } S_{22}(f) & \cdots & \text{Cof } S_{q2}(f) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \text{Cof } S_{yq}(f) & \text{Cof } S_{1q}(f) & \text{Cof } S_{2q}(f) & \cdots & \text{Cof } S_{qq}(f) \end{bmatrix}$$

Hence

$$S^{\nu\nu}(f) = \frac{\text{Cof } S_{yy}(f)}{|S_{yxx}(f)|} = \frac{|S_{xx}(f)|}{|S_{yxx}(f)|} \quad (5.130)$$

where Equation (5.121) shows that $\text{Cof } S_{yy}(f) = |S_{xx}(f)|$. Thus Equations (5.126) and (5.129) represent two different equivalent ways to compute the general multiple coherence function of Equation (5.100).

PROBLEMS

- 1 For the model of Figure 5.1, state how to compute $G_s(f)$ and $H(f)$ when $G_v(f)$ and $G_{zw}(f)$ are given, and the limitations in these formulas
- 2 Verify the results in Examples 5.1 and 5.2
- 3 Verify the results in Examples 5.3 and 5.4
- 4 Consider the model of Figure 5.3 where

$$\begin{aligned} G_s(f) &= \frac{1}{1 + 4f^2} & H(f) &= \frac{4}{1 - 9f^2 + j2f} \\ G_n(f) &= 5 & G_m(f) &= 0 \end{aligned}$$

Compute the following quantities

- (a) $G_v(f)$ and $G_{wv}(f)$
 - (b) $\gamma_{wv}^2(f)$ and $\gamma_{zw}^2(f)$
- 5 In a multiple input linear system problem where the various inputs have non zero means and are mutually uncorrelated, Equation (5.38) is replaced by

$$R_{ij}(\tau) = \begin{cases} R_i(\tau) & \text{when } i = j \\ \mu_i \mu_j & \text{when } i \neq j \end{cases}$$

For this case, show that Equations (5.43) and (5.48) become

$$\begin{aligned} S_y(f) &= \sum_{i=1}^q |H_i(f)|^2 S_i(f) + \sum_{i=1}^q \sum_{\substack{j=1 \\ (i \neq j)}}^q H_i(0) H_j(0) \mu_i \mu_j \delta(f) \\ S_{iy}(f) &= H_i(f) S_i(f) + \sum_{\substack{j=1 \\ (j \neq i)}}^q H_j(0) \mu_i \mu_j \delta(f) \end{aligned}$$

- 6 Consider the two input model of Figure 5.7 where

$$H_1(f) = \frac{2}{5 + jf} \quad H_2(f) = \frac{3}{5 + jf}$$

Assume the inputs satisfy

$$R_1(\tau) = 3\delta(\tau) \quad G_2(f) = 12 \quad G_{12}(f) = 8$$

Compute the following quantities

- (a) $\gamma_{12}^2(f)$
- (b) $R_y(\tau)$ and $G_y(f)$
- (c) $R_{1y}(\tau)$ and $G_{1y}(f)$
- (d) $\gamma_{1y}^2(f)$

7. Compute the partial coherence function using Equation (5.84) for the quantities in Problem 6.
8. Derive Equations (5.86) and (5.88).
9. Compute the multiple coherence function using Equation (5.100) for the quantities in Problem 6, assuming that an extraneous noise component $z(t)$ occurs where $S_z(f) = 19/(25 + f^2)$.
10. Verify Equation (5.125) for Problem 9.

CHAPTER 6

STATISTICAL ERRORS IN RANDOM DATA ANALYSIS

As noted in Chapter 4, the descriptive properties of a random variable cannot be precisely determined from sample data. Only estimates of the parameters of interest can be obtained from a finite sample of observations. The accuracy of certain basic parameter estimates is discussed in Chapter 4 for the case of data in the form of discrete independent observations of sample size N . In this chapter, the accuracy of parameter estimates is developed for data in the form of continuous time history records of record length T . It is assumed throughout that the data are sample records from continuous stationary (ergodic) random processes. Basic parameter estimates are discussed, as well as frequency response and coherence functions computed from spectral density estimates in single and multiple input problems. Attention in this chapter is restricted to those errors which are due solely to statistical considerations. Other errors associated with data acquisition and processing are covered in Chapter 7.

6.1 DEFINITION OF ERRORS

Referring to Section 4.1, the accuracy of parameter estimates based upon sample values can be described by a mean square error defined as

$$\text{mean square error} = E[(\hat{\Phi} - \Phi)^2] \quad (6.1)$$

where $\hat{\Phi}$ is an estimator for Φ . Expanding Equation (6.1) yields

$$\begin{aligned} E[(\hat{\Phi} - \Phi)^2] &= E[(\hat{\Phi} - E[\hat{\Phi}] + E[\hat{\Phi}] - \Phi)^2] \\ &= E[(\hat{\Phi} - E[\hat{\Phi}])^2] + 2E[(\hat{\Phi} - E[\hat{\Phi}])(E[\hat{\Phi}] - \Phi)] \\ &\quad + E[(E[\hat{\Phi}] - \Phi)^2] \end{aligned}$$

Note that the middle term in the above expression has a factor equal to zero, namely,

$$E[\hat{\Phi} - E[\hat{\Phi}]] = E[\hat{\Phi}] - E[\hat{\Phi}] = 0$$

Hence the mean square error reduces to

$$\text{mean square error} = E[(\hat{\Phi} - E[\hat{\Phi}])^2] + E[(E[\hat{\Phi}] - \Phi)^2] \quad (6.2)$$

In words, the mean square error is the sum of two parts. The first part is a variance term which describes the random portion of the error,

$$\text{Var} [\hat{\Phi}] = E[(\hat{\Phi} - E[\hat{\Phi}])^2] = E[\hat{\Phi}^2] - E^2[\hat{\Phi}] \quad (6.3)$$

and the second part is the square of a bias term which describes the systematic portion of the error,

$$b^2[\hat{\Phi}] = E[b^2[\hat{\Phi}]] = E[(E[\hat{\Phi}] - \Phi)^2] \quad (6.4)$$

Thus the mean square error is the sum of the variance of the estimate plus the square of the bias of the estimate, that is

$$E[(\hat{\Phi} - \Phi)^2] = \text{Var} [\hat{\Phi}] + b^2[\hat{\Phi}] \quad (6.5)$$

It is generally more convenient to describe the error of an estimate in terms which have the same engineering units as the parameter being estimated. This can be achieved by taking the positive square roots of the error terms in Equations (6.3) through (6.5). The square root of Equation (6.3) yields the standard deviation for the estimate, called the *standard error* (or *random error*) as follows

$$\text{standard error} = \sigma[\hat{\Phi}] = \sqrt{E[\hat{\Phi}^2] - E^2[\hat{\Phi}]} \quad (6.6)$$

The square root of Equation (6.4) gives the *bias error* directly as

$$\text{bias error} = b[\hat{\Phi}] = E[\hat{\Phi}] - \Phi \quad (6.7)$$

The square root of the sum of the squared errors, as given by Equation (6.5), defines the *root mean square* (rms) error as

$$\text{rms error} = \sqrt{E[(\hat{\Phi} - \Phi)^2]} = \sqrt{\sigma^2[\hat{\Phi}] + b^2[\hat{\Phi}]} \quad (6.8)$$

As a further convenience, it is often desirable to define the error of an estimate in terms of a fractional portion of the quantity being estimated. This is done by dividing the error by the quantity being estimated to obtain a

normalized error For $\Phi \neq 0$, the normalized standard, bias, and rms errors are given by

$$\text{normalized standard error} = \epsilon_s = \frac{\sigma[\hat{\Phi}]}{\Phi} = \frac{\sqrt{E[\hat{\Phi}^2] - E^2[\hat{\Phi}]}}{\Phi} \quad (6.9a)$$

$$\text{normalized bias error} = \epsilon_b = \frac{b[\hat{\Phi}]}{\Phi} = \frac{E[\hat{\Phi}]}{\Phi} - 1 \quad (6.9b)$$

$$\text{normalized rms error} = \epsilon = \frac{\sqrt{\sigma^2[\hat{\Phi}] + b^2[\hat{\Phi}]}}{\Phi} = \frac{\sqrt{E[(\hat{\Phi} - \Phi)^2]}}{\Phi} \quad (6.9c)$$

Note that the normalized standard error ϵ_s is often called the *coefficient of variation*.

In the following sections of this chapter, the errors associated with parameter estimates are derived in terms of the quantities defined by Equation (6.9). Given the sampling distributions of the estimators, these quantities can be extended to confidence interval statements as described in Section 4.4. With the exception of spectral density and frequency response function estimates, a normality assumption for the sampling distribution of the parameter estimates is usually acceptable when the normalized rms error is small. Spectral density and frequency response function estimates require special consideration because the normalized rms error for these estimates is often relatively large in practice. Along with the derivation of errors, the consistency of the various parameter estimates, as defined by Equation (4.7), is investigated.

6.2 MEAN AND MEAN SQUARE VALUE ESTIMATES

6.2.1 Mean Values

Suppose that a single sample time history record $x(t)$ from a stationary (ergodic) random process $\{x(t)\}$ exists over a finite time T . The mean value of $\{x(t)\}$ can be estimated by

$$\hat{\mu}_x = \frac{1}{T} \int_0^T x(t) dt \quad (6.10)$$

The true mean value is

$$\mu_x = E[x(t)] \quad (6.11)$$

and is independent of t since $\{x(t)\}$ is stationary. The expected value of the estimate $\hat{\mu}_x$ is

$$E[\hat{\mu}_x] = E\left[\frac{1}{T} \int_0^T x(t) dt\right] = \frac{1}{T} \int_0^T E[x(t)] dt = \frac{1}{T} \int_0^T \mu_x dt = \mu_x \quad (6.12)$$

since expected values commute with linear operations, as given by Equation (3.137). Hence $\hat{\mu}_x$ is an *unbiased estimate* of μ_x , independent of T . Since $\hat{\mu}_x$

is unbiased, the mean square error of the estimate $\hat{\mu}_x$ is equal to the variance,

$$\text{Var} [\hat{\mu}_x] = E[(\hat{\mu}_x - \mu_x)^2] = E[\hat{\mu}_x^2] - \mu_x^2 \quad (6.13)$$

where from Equation (6.10),

$$E[\hat{\mu}_x^2] = \frac{1}{T^2} \int_0^T \int_0^T E[x(\xi) x(\eta)] d\eta d\xi \quad (6.14)$$

Now the autocorrelation function $R_x(\tau)$ of a stationary random process $\{x(t)\}$ is defined by Equation (3.44) as

$$R_x(\tau) = E[x(t) x(t + \tau)] \quad (6.15)$$

From the stationary hypothesis, $R_x(\tau)$ is independent of t , and an even function of τ with a maximum at $\tau = 0$. It will be assumed that $R_x(\tau)$ is continuous and finite for all values of τ , and that all periodic components in $R_x(\tau)$ have been removed at the onset. The autocovariance function $C_x(\tau)$ is defined by Equation (3.47) as

$$C_x(\tau) = R_x(\tau) - \mu_x^2 \quad (6.16)$$

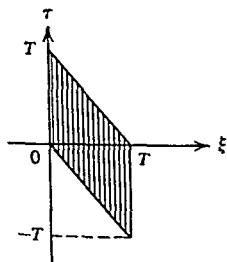
It turns out that whenever $\mu_x \neq 0$, it is more convenient to work with $C_x(\tau)$ than with $R_x(\tau)$. It will be assumed that $C_x(\tau)$ satisfies the integrable properties of Equation (3.118) so as to make $\{x(t)\}$ ergodic.

In terms of the autocovariance function $C_x(\tau)$, the variance (mean square error) from Equations (6.13) and (6.14) becomes

$$\begin{aligned} \text{Var} [\hat{\mu}_x] &= \frac{1}{T^2} \int_0^T \int_0^T C_x(\eta - \xi) d\eta d\xi = \frac{1}{T^2} \int_0^T \int_{-\xi}^{T-\xi} C_x(\tau) d\tau d\xi \\ &= \frac{1}{T} \int_{-T}^T \left(1 - \frac{|\tau|}{T}\right) C_x(\tau) d\tau \end{aligned} \quad (6.17)$$

The last expression occurs by reversing the orders of integration between τ and ξ and carrying out the ξ integration. This changes the limits of integration for τ and ξ as shown in the sketch below so that

$$\begin{aligned} \int_0^T \int_{-\xi}^{T-\xi} C_x(\tau) d\tau d\xi &= \int_{-T}^0 \int_{-\tau}^T C_x(\tau) d\xi d\tau + \int_0^T \int_0^{T-\tau} C_x(\tau) d\xi d\tau \\ &= \int_{-T}^0 (T + \tau) C_x(\tau) d\tau + \int_0^T (T - \tau) C_x(\tau) d\tau \\ &= \int_{-T}^T (T - |\tau|) C_x(\tau) d\tau \end{aligned}$$



Now on letting T tend to infinity Equation (6 17) becomes

$$\lim_{T \rightarrow \infty} T \text{Var} [\mu_x] = \int_{-\infty}^{\infty} C_x(\tau) d\tau < \infty \quad (6 18)$$

where Equation (3 118) is used which provides that $C_x(\tau)$ and $\tau C_x(\tau)$ are absolutely integrable over $(-\infty, \infty)$ to justify passage to the limit inside the integral sign. In particular Equation (6 18) shows that for large T where $|\tau| \ll T$ the variance is given by

$$\text{Var} [\mu_x] \approx \frac{1}{T} \int_{-\infty}^{\infty} C_x(\tau) d\tau \quad (6 19)$$

Hence when the integral is finite valued $\text{Var} [\mu_x]$ approaches zero as T approaches infinity proving that μ_x is a *consistent* estimate of μ_x .

Example 6 1 Variance of Mean Value Estimates of Bandwidth Limited White Noise Consider the important special case where $\{x(t)\}$ is bandwidth limited white noise with a mean value $\mu_x \neq 0$. Assume the power spectral density function is

$$\begin{aligned} G_x(f) &= \frac{1}{B} + \mu_x^2 \delta(f) & 0 \leq f \leq B \\ &= 0 & f > B \end{aligned} \quad (6 20)$$

where B is the bandwidth. The associated autocovariance function $C_x(\tau)$ is given by

$$\begin{aligned} C_x(\tau) &= \int_0^{\infty} G_x(f) \cos 2\pi f\tau df - \mu_x^2 \\ &= \int_0^B \frac{1}{B} \cos 2\pi f\tau df = \frac{\sin 2\pi B\tau}{2\pi B\tau} \end{aligned} \quad (6 21)$$

Note that $C_x(0) = 1$ and that $C_x(\tau) = 0$ for $\tau = 1/2B$. Thus points $1/2B$ apart are uncorrelated. (They will be statistically independent if $\{x(t)\}$ is Gaussian.) For this case Equation (6 19) yields the result

$$\text{Var} [\mu_x] \approx \frac{1}{T} \int_{-\infty}^{\infty} \left(\frac{\sin 2\pi B\tau}{2\pi B\tau} \right) d\tau = \frac{1}{2BT} \quad (6 22)$$

This is the variance of mean value estimates of bandwidth limited white noise with bandwidth B and sampling time T . Equation (6 22) requires that T be sufficiently large so that Equation (6 19) can replace Equation (6 17). Satisfactory conditions in practice are $T \geq 10 |\tau|$ and $BT \geq 5$.

For $C_x(0) \neq 1$, Equation (6.22) would be

$$\text{Var} [\hat{\mu}_x] \approx \frac{C_x(0)}{2BT} = \frac{\sigma_x^2}{2BT}$$

Hence, when $\mu_x \neq 0$, the normalized mean square error is given by

$$\epsilon^2 = \frac{\text{Var} [\hat{\mu}_x]}{\mu_x^2} \approx \frac{1}{2BT} \left(\frac{\sigma_x}{\mu_x} \right)^2 \quad (6.23)$$

The square root of Equation (6.23) gives the *normalized rms error* ϵ , which includes only a random error term since the bias error is zero.

6.2.2 Mean Square Values

As in Section 6.2.1, let $x(t)$ be a single sample time history record from a stationary (ergodic) random process $\{x(t)\}$. The mean square value of $\{x(t)\}$ can be estimated by time averaging over a finite time interval T as follows.

$$\hat{\Psi}_x^2 = \frac{1}{T} \int_0^T x^2(t) dt \quad (6.24)$$

The true mean square value is

$$\Psi_x^2 = E[x^2(t)] \quad (6.25)$$

and is independent of t since $\{x(t)\}$ is stationary. The expected value of the estimate $\hat{\Psi}_x^2$ is

$$E[\hat{\Psi}_x^2] = \frac{1}{T} \int_0^T E[x^2(t)] dt = \frac{1}{T} \int_0^T \Psi_x^2 dt = \Psi_x^2 \quad (6.26)$$

Hence $\hat{\Psi}_x^2$ is an *unbiased* estimate of Ψ_x^2 , independent of T .

The mean square error here is given by the variance

$$\begin{aligned} \text{Var} [\hat{\Psi}_x^2] &= E[(\hat{\Psi}_x^2 - \Psi_x^2)^2] = E[\hat{\Psi}_x^4] - \Psi_x^4 \\ &= \frac{1}{T^2} \int_0^T \int_0^T (E[x^2(\xi) x^2(\eta)] - \Psi_x^4) d\eta d\xi \end{aligned} \quad (6.27)$$

Assume now that $\{x(t)\}$ is a Gaussian random process with mean value $\mu_x \neq 0$. Then the expected value in Equation (6.27) takes the special form, derived from Equation (3.134), as follows.

$$E[x^2(\xi) x^2(\eta)] = 2(R_x^2(\eta - \xi) - \mu_x^4) + \Psi_x^4 \quad (6.28)$$

From the basic relationship in Equation (6.16),

$$R_x^2(\eta - \xi) - \mu_x^4 = C_x^2(\eta - \xi) + 2\mu_x^2 C_x(\eta - \xi) \quad (6.29)$$

Hence

$$\begin{aligned}\text{Var} [\hat{\Psi}_x^2] &= \frac{2}{T^2} \int_0^T \int_0^T (R_x^2(\eta - \xi) - \mu_x^4) d\eta d\xi \\ &= \frac{2}{T} \int_{-T}^T \left(1 - \frac{|\tau|}{T}\right) (R_x^2(\tau) - \mu_x^4) d\tau \\ &= \frac{2}{T} \int_{-T}^T \left(1 - \frac{|\tau|}{T}\right) (C_x^2(\tau) + 2\mu_x^2 C_x(\tau)) d\tau\end{aligned}\quad (6.30)$$

For large T , where $|\tau| \ll T$, the variance becomes

$$\text{Var} [\hat{\Psi}_x^2] \approx \frac{2}{T} \int_{-\infty}^{\infty} (C_x^2(\tau) + 2\mu_x^2 C_x(\tau)) d\tau \quad (6.31)$$

Thus $\hat{\Psi}_x^2$ is a *consistent* estimate of Ψ_x^2 since $\text{Var} [\hat{\Psi}_x^2]$ will approach zero as T approaches infinity assuming $C_x^2(\tau)$ and $C_x(\tau)$ are absolutely integrable over $(-\infty, \infty)$, as stated in Equation (3.118).

Example 6.2. Variance of Mean Square Value Estimates of Bandwidth Limited Gaussian White Noise Consider the important special case of bandwidth limited Gaussian white noise where

$$C_x(\tau) = C_x(0) \left(\frac{\sin 2\pi B\tau}{2\pi B\tau} \right) \quad (6.32)$$

For this case, Equation (6.31) shows that

$$\text{Var} [\hat{\Psi}_x^2] \approx \frac{C_x^2(0)}{BT} + \frac{2}{BT} \mu_x^2 C_x(0) \quad (6.33)$$

This is the variance of mean square value estimates of bandwidth limited Gaussian white noise with bandwidth B and sampling time T . Equation (6.33) requires that T be sufficiently large so that Equation (6.31) can replace Equation (6.30). Satisfactory conditions in practice are $T \geq 10|\tau|$ and $BT \geq 5$.

For $\mu_x = 0$, Equation (6.33) becomes

$$\text{Var} [\hat{\Psi}_x^2] \approx \frac{R_x^2(0)}{BT} = \frac{\Psi_x^4}{BT} \quad (6.34)$$

Hence when $\mu_x = 0$, the normalized mean square error is given by

$$\epsilon^2 = \frac{\text{Var} [\hat{\Psi}_x^2]}{\Psi_x^4} \approx \frac{1}{BT} \quad (6.35)$$

The square root of Equation (6.35) gives the normalized rms error ϵ . Note that this result also represents the normalized rms error of variance estimates when $\mu_x = 0$. Further note that, as for mean value estimates, ϵ includes only a random error term since the bias error is zero.

6.3 PROBABILITY DENSITY ESTIMATES

Consider a probability density measurement of a single sample time history record $x(t)$ from a stationary (ergodic) random process $\{x(t)\}$. The probability that $x(t)$ assumes particular amplitude values between $x - (W/2)$ and $x + (W/2)$ during a time interval T may be estimated by

$$\begin{aligned}\hat{P}[x, W] &= \widehat{\text{Prob}} \left[\left(x - \frac{W}{2} \right) \leq x(t) \leq \left(x + \frac{W}{2} \right) \right] \\ &= \frac{1}{T} \sum_i \Delta t_i = \frac{T_x}{T}\end{aligned}\quad (6.36)$$

where Δt_i is the time spent by $x(t)$ in this amplitude range during the i th entry into the range, and $T_x = \sum \Delta t_i$. The ratio T_x/T is the total fractional portion of the time spent by $x(t)$ in the range $[x - (W/2), x + (W/2)]$. It should be noted that T_x will usually be a function of the amplitude x . This estimated probability $\hat{P}[x, W]$ will approach the true probability $P[x, W]$ as T approaches infinity. Moreover this estimated probability is an unbiased estimate of the true probability. Hence

$$P[x, W] = E[\hat{P}[x, W]] = \lim_{T \rightarrow \infty} \hat{P}[x, W] = \lim_{T \rightarrow \infty} \frac{T_x}{T} \quad (6.37)$$

The probability density function $p(x)$ is defined by

$$p(x) = \lim_{W \rightarrow 0} \frac{P[x, W]}{W} = \lim_{\substack{T \rightarrow \infty \\ W \rightarrow 0}} \frac{\hat{P}[x, W]}{W} = \lim_{\substack{T \rightarrow \infty \\ W \rightarrow 0}} \hat{p}(x) \quad (6.38)$$

where

$$\hat{p}(x) = \frac{\hat{P}[x, W]}{W} = \frac{T_x}{TW} \quad (6.39)$$

is a sample estimate of $p(x)$. In terms of the probability density function $p(x)$, the probability of the time history $x(t)$ falling between any two values x_1 and x_2 is given by

$$\text{Prob} [x_1 \leq x \leq x_2] = \int_{x_1}^{x_2} p(x) dx \quad (6.40)$$

In particular

$$P[x, W] = \text{Prob} \left[x - \frac{W}{2} \leq x(t) \leq x + \frac{W}{2} \right] = \int_{x-(W/2)}^{x+(W/2)} p(\xi) d\xi \quad (6.41)$$

Then, from Equation (6.39),

$$E[\hat{p}(x)] = \frac{E[\hat{P}(x, W)]}{W} = \frac{P(x, W)}{W} = \frac{1}{W} \int_{x-(W/2)}^{x+(W/2)} p(\xi) d\xi \quad (6.42)$$

Thus for most $p(x)$,

$$E[\hat{p}(x)] \neq p(x) \quad (6.43)$$

proving that $\hat{p}(x)$ is generally a biased estimate of $p(x)$

The mean square error of the estimate $\hat{p}(x)$ is calculated from Equation (6.6) by

$$E[(\hat{p}(x) - p(x))^2] = \text{Var}[\hat{p}(x)] + b^2[\hat{p}(x)] \quad (6.44)$$

where $\text{Var}[\hat{p}(x)]$ is the variance of the estimate as defined by

$$\text{Var}[\hat{p}(x)] = E[(\hat{p}(x) - E[\hat{p}(x)])^2] \quad (6.45)$$

and $b[\hat{p}(x)]$ is the bias of the estimate as defined by

$$b[\hat{p}(x)] = E[\hat{p}(x) - p(x)] \quad (6.46)$$

6.3.1 Variance of the Estimate

To evaluate the variance of an estimate $\hat{p}(x)$, it is necessary to know the statistical properties of the time intervals Δt , which comprise T_x . Unfortunately, such time statistics for a random process are very difficult to obtain. However, the general form of an appropriate variance expression for $p(x)$ can be established by the following heuristic argument.

From Equation (6.39), the variance of $\hat{p}(x)$ is given by

$$\text{Var}[\hat{p}(x)] = \frac{1}{W^2} \text{Var}[\hat{P}(x, W)] \quad (6.47)$$

where $\hat{P}(x, W)$ is the estimate of a proportion $P(x, W)$. The variance for a proportion estimate based upon N independent sample values is given by, [Ref. 13],

$$\text{Var}[\hat{P}(x, W)] = \frac{P(x, W)(1 - P(x, W))}{N} \quad (6.48)$$

Substituting Equation (6.48) into Equation (6.47), and assuming $P(x, W) \approx Wp(x) \ll 1$, the variance of probability density estimates may be approximated by

$$\text{Var}[\hat{p}(x)] \approx \frac{p(x)}{NW} \quad (6.49)$$

where N is still to be determined. Now from the time domain sampling theorem derived in Section 3.5.2, a sample record $x(t)$ of bandwidth B and

length T can be completely reproduced with $N = 2BT$ discrete values. Of course, the N discrete values will not necessarily be statistically independent. Nevertheless, for any given stationary (ergodic) random process, each sample record will represent $n = N/c^2$ independent sample values (degrees of freedom), where c is a constant. Hence from Equation (6.49),

$$\text{Var} [\hat{p}(x)] \approx \frac{c^2 p(x)}{2BTW} \quad (6.50)$$

The constant c is dependent upon the autocorrelation function of the data and the sampling rate. For continuous bandwidth limited white noise, experimental studies indicate $c \approx 0.3$. If the bandwidth limited white noise is digitized to $N = 2BT$ discrete values, the experimental studies indicate $c = 1.0$ as would be expected from the results of Equation (6.21).

6.3.2 Bias of the Estimate

An expression will now be derived for the bias term of Equation (6.46). In terms of the true probability density function, Equation (6.42) shows that

$$E[\hat{p}(x)] = \frac{1}{W} \int_{x-(W/2)}^{x+(W/2)} p(\xi) d\xi \quad (6.51)$$

By expanding $p(\xi)$ in a Taylor series about the point $\xi = x$, and retaining only the first three terms,

$$p(\xi) \approx p(x) + (\xi - x) p'(x) + \frac{(\xi - x)^2}{2} p''(x) \quad (6.52)$$

From the two relations

$$\int_{x-(W/2)}^{x+(W/2)} (\xi - x) d\xi = 0 \quad (6.53a)$$

and

$$\int_{x-(W/2)}^{x+(W/2)} \frac{(\xi - x)^2}{2} d\xi = \frac{W^3}{24} \quad (6.53b)$$

It follows that

$$E[\hat{p}(x)] \approx p(x) + \frac{W^2}{24} p''(x) \quad (6.54)$$

Thus a first order approximation for the bias term is given by

$$b[\hat{p}(x)] \approx \frac{W^2}{24} p''(x) \quad (6.55)$$

where $p''(x)$ is the second derivative of $p(x)$ with respect to x .

6.3.3 Normalized rms Error

The total mean square error of the probability density estimate $\hat{p}(x)$ is given by Equation (6.44) as the sum of the variance defined in Equation (6.50) and the square of the bias defined in Equation (6.55). That is,

$$E[(\hat{p}(x) - p(x))^2] = \frac{c^2 p(x)}{2BTW} + \left[\frac{W^2 p''(x)}{24} \right]^2 \quad (6.56)$$

Hence, the normalized mean square error is approximated by

$$\epsilon^2 \approx \frac{c^2}{2BTW p(x)} + \frac{W^4 [p''(x)]^2}{576 [p(x)]} \quad (6.57)$$

The square root of Equation (6.57) gives the normalized rms error ϵ .

It is clear from Equation (6.57) that there are conflicting requirements on the window width W in probability density measurements. On the one hand, a large value of W is desirable to reduce the random error. On the other hand, a small value of W is needed to suppress the bias error. However, the total error will approach zero as $T \rightarrow \infty$ if W is restricted so that $W \rightarrow 0$ and $WT \rightarrow \infty$. In practice, values of $W \leq 0.2\sigma_x$ will usually limit the normalized bias error to less than one percent. This is true because of the $p''(x)$ term in the bias portion of the error given by Equation (6.57). Probability density functions of common (approximately Gaussian) random data do not display abrupt or sharp peaks which are indicative of a large second derivative.

6.3.4 Joint Probability Density Estimates

Joint probability density estimates for a pair of sample time history records, $x(t)$ and $y(t)$, from two stationary (ergodic) random processes $\{x(t)\}$ and $\{y(t)\}$, may be defined as follows. Analogous to Equation (6.36), let

$$\hat{P}[x, W_x, y, W_y] = \frac{T_{xy}}{T} \quad (6.58)$$

estimate the joint probability that $x(t)$ is inside the amplitude interval W_x centered at x while simultaneously $y(t)$ is inside the amplitude interval W_y centered at y . This is measured by the ratio T_{xy}/T where T_{xy} represents the amount of time that these two events coincide in time T . Clearly, T_{xy} will usually be a function of both x and y . This estimated joint probability will approach the true probability $P[x, W_x, y, W_y]$ as T approaches infinity, namely,

$$P[x, W_x, y, W_y] = \lim_{T \rightarrow \infty} \hat{P}[x, W_x, y, W_y] = \lim_{T \rightarrow \infty} \frac{T_{xy}}{T} \quad (6.59)$$

The joint probability density function $p(x, y)$ is now defined by

$$p(x, y) = \lim_{\substack{W_x \rightarrow 0 \\ W_y \rightarrow 0}} \frac{P[x, W_x; y, W_y]}{W_x W_y} = \lim_{\substack{T \rightarrow \infty \\ W_x \rightarrow 0 \\ W_y \rightarrow 0}} \frac{\hat{P}[x, W_x; y, W_y]}{W_x W_y} = \lim_{\substack{T \rightarrow \infty \\ W_x \rightarrow 0 \\ W_y \rightarrow 0}} \hat{p}(x, y) \quad (6.60)$$

where

$$\hat{p}(x, y) = \frac{\hat{P}[x, W_x; y, W_y]}{W_x W_y} = \frac{T_{x,y}}{T W_x W_y} \quad (6.61)$$

Assume that W_x and W_y are sufficiently small so that the bias errors are negligible. Then the mean square error associated with the estimate $\hat{p}(x, y)$ will be given by the variance of the estimate. As for first-order probability density estimates, this quantity is difficult to determine precisely by theoretical arguments alone. However, by using the same heuristic arguments which produced Equation (6.49), a general form for the variance can be approximated. Specifically, for the special case where $x(t)$ and $y(t)$ are both bandwidth limited white noise with identical bandwidths B ,

$$\text{Var} [\hat{p}(x, y)] \approx \frac{c^2 p(x, y)}{B T W_x W_y} \quad (6.62)$$

where c is an unknown constant.

6.4 CORRELATION FUNCTION ESTIMATES

Consider now two sample time history records $x(t)$ and $y(t)$, from two stationary (ergodic) random processes, $\{x(t)\}$ and $\{y(t)\}$. The next statistical quantities of interest are the stationary autocorrelation functions, $R_x(\tau)$ and $R_y(\tau)$, and the cross-correlation function $R_{xy}(\tau)$. To simplify the following derivation, the mean values μ_x and μ_y will be assumed to be zero. For continuous data, $x(t)$ and $y(t)$, which exist only over a time interval T , the sample cross-correlation estimate $\hat{R}_{xy}(\tau)$ can be defined by

$$\begin{aligned} \hat{R}_{xy}(\tau) &= \frac{1}{T - \tau} \int_0^{T-\tau} x(t) y(t + \tau) dt \quad 0 \leq \tau < T \\ &= \frac{1}{T - |\tau|} \int_{|\tau|}^T x(t) y(t + \tau) dt \quad -T < \tau \leq 0 \end{aligned} \quad (6.63)$$

To avoid use of absolute value signs, τ will be considered positive henceforth since a similar proof applies for negative τ . The sample autocorrelation

function estimates $\hat{R}_x(\tau)$ and $\hat{R}_y(\tau)$ are merely special cases when the two records coincide. That is,

$$\begin{aligned}\hat{R}_x(\tau) &= \frac{1}{T-\tau} \int_0^{T-\tau} x(t) x(t+\tau) dt & 0 \leq \tau < T \\ \hat{R}_y(\tau) &= \frac{1}{T-\tau} \int_0^{T-\tau} y(t) y(t+\tau) dt & 0 \leq \tau < T\end{aligned}\quad (6.64)$$

Thus by analyzing the cross-correlation function estimate, one derives results which are also applicable to the autocorrelation function estimates.

If the data exist for time $T + \tau$ instead of only for time T , then an alternative definition for $\hat{R}_{xy}(\tau)$ is

$$\hat{R}_{xy}(\tau) = \frac{1}{T} \int_0^T x(t) y(t+\tau) dt \quad 0 \leq \tau < T \quad (6.65)$$

This formula has a fixed integration time T instead of a variable integration time as in Equation (6.63), and is the way the correlation functions have been defined previously. Note that for either Equation (6.63) or Equation (6.65), mean square estimates of $x(t)$ or $y(t)$ are merely special cases when $\tau = 0$. For simplicity in notation, Equation (6.65) will be used in the following development instead of Equation (6.63). The same final results are obtained for both definitions, assuming the data exists for time $T + \tau$.

The expected value of the estimate $\hat{R}_{xy}(\tau)$ is given by

$$\begin{aligned}E[\hat{R}_{xy}(\tau)] &= \frac{1}{T} \int_0^T E[x(t) y(t+\tau)] dt \\ &= \frac{1}{T} \int_0^T R_{xy}(\tau) dt = R_{xy}(\tau)\end{aligned}\quad (6.66)$$

Hence $\hat{R}_{xy}(\tau)$ is an *unbiased* estimate of $R_{xy}(\tau)$, independent of T . The mean square error is given by the variance

$$\begin{aligned}\text{Var}[\hat{R}_{xy}(\tau)] &= E[(\hat{R}_{xy}(\tau) - R_{xy}(\tau))^2] = E[\hat{R}_{xy}^2(\tau)] - R_{xy}^2(\tau) \\ &= \frac{1}{T^2} \int_0^T \int_0^T (E[x(u) y(u+\tau) x(v) y(v+\tau)] - R_{xy}^2(\tau)) dv du\end{aligned}\quad (6.67)$$

At this point, in order to simplify the later mathematical analysis and to agree with many physical cases of interest, it will be assumed that the random processes $\{x(t)\}$ and $\{y(t)\}$ are jointly *Gaussian* for any set of fixed times. This restriction may be removed by substituting certain integrability conditions on the non-Gaussian parts of the random processes without altering

in any essential way the results to be derived. When $\{x(t)\}$ and $\{y(t)\}$ are jointly Gaussian, it follows that $\{x(t)\}$ and $\{y(t)\}$ are separately Gaussian.

For Gaussian stationary random processes with zero mean values, the fourth-order statistical expression is obtained from Equation (3.134) as follows.

$$\begin{aligned} E[x(u) y(u + \tau) x(v) y(v + \tau)] \\ = R_{xy}^2(\tau) + R_x(v - u) R_y(v - u) + R_{xy}(v - u + \tau) R_{yx}(v - u - \tau) \end{aligned} \quad (6.68)$$

Hence the variance expression may be written as

$$\begin{aligned} \text{Var} [\hat{R}_{xy}(\tau)] &= \frac{1}{T^2} \int_0^T \int_0^T (R_x(v - u) R_y(v - u) \\ &\quad + R_{xy}(v - u + \tau) R_{yx}(v - u - \tau)) dv du \\ &= \frac{1}{T} \int_{-T}^T \left(1 - \frac{|\xi|}{T}\right) (R_x(\xi) R_y(\xi) \\ &\quad + R_{xy}(\xi + \tau) R_{yx}(\xi - \tau)) d\xi \end{aligned} \quad (6.69)$$

The second expression occurs from the first by letting $\xi = v - u$, $d\xi = dv$, and then reversing the order of integration between ξ and u . Now,

$$\lim_{T \rightarrow \infty} T \text{Var} [\hat{R}_{xy}(\tau)] = \int_{-\infty}^{\infty} (R_x(\xi) R_y(\xi) + R_{xy}(\xi + \tau) R_{yx}(\xi - \tau)) d\xi < \infty \quad (6.70)$$

assuming $R_x(\xi) R_y(\xi)$ and $R_{xy}(\xi) R_{yx}(\xi)$ are absolutely integrable over $(-\infty, \infty)$. This proves that $\hat{R}_{xy}(\tau)$ is a *consistent* estimate of $R_{xy}(\tau)$, and for large T has a variance given by

$$\text{Var} [\hat{R}_{xy}(\tau)] \approx \frac{1}{T} \int_{-\infty}^{\infty} (R_x(\xi) R_y(\xi) + R_{xy}(\xi + \tau) R_{yx}(\xi - \tau)) d\xi \quad (6.71)$$

Several special cases of Equation (6.71) are worthy of note. For autocorrelation estimates, Equation (6.71) becomes

$$\text{Var} [\hat{R}_x(\tau)] \approx \frac{1}{T} \int_{-\infty}^{\infty} (R_x^2(\xi) + R_x(\xi + \tau) R_x(\xi - \tau)) d\xi \quad (6.72)$$

At the zero displacement point $\tau = 0$,

$$\text{Var} [\hat{R}_x(0)] \approx \frac{2}{T} \int_{-\infty}^{\infty} R_x^2(\xi) d\xi \quad (6.73)$$

The assumption that $R_x(\tau)$ approaches zero for large τ shows that

$$R_x^2(\xi) \gg R_x(\xi + \tau) R_x(\xi - \tau) \quad \text{for } \tau \gg 0 \quad (6.74)$$

Hence for large τ

$$\text{Var} [\hat{R}_x(\tau)] \approx \frac{1}{T} \int_{-\infty}^{\infty} R_x^2(\xi) d\xi \quad (6.75)$$

which is one half the value of Equation (6.73)

Example 6.3 Variance of Correlation Estimates of Bandwidth Limited White Noise For bandwidth limited white noise with a mean value $\mu_x = 0$ a bandwidth B and covering a time interval T the variance for all τ is conservatively given by

$$\text{Var} [\hat{R}_x(\tau)] \approx \frac{1}{2BT} (R_x^2(0) + R_x^2(\tau)) \quad (6.76)$$

This reduces to Equation (6.34) at the point $\tau = 0$. Similarly when both $x(t)$ and $y(t)$ are samples of length T from bandwidth limited white noise with mean values $\mu_x = \mu_y = 0$ and identical bandwidths B it follows that

$$\text{Var} [\hat{R}_{xy}(\tau)] \approx \frac{1}{2BT} (R_x(0) R_y(0) + R_{xy}^2(\tau)) \quad (6.77)$$

Equation (6.77) requires that T be sufficiently large such that Equation (6.71) can replace Equation (6.69). Satisfactory conditions in practice are $T \geq 10 |\tau|$ and $BT \geq 5$.

For $\mu_x = \mu_y = 0$ and $R_{xy} \neq 0$ the normalized mean square error is given by

$$\epsilon^2 = \frac{\text{Var} [\hat{R}_{xy}(\tau)]}{R_{xy}^2(\tau)} \approx \frac{1}{2BT} \left(1 + \frac{R_x(0) R_y(0)}{R_{xy}^2(\tau)} \right) \quad (6.78)$$

The square root of Equation (6.78) gives the normalized rms error ϵ which includes only a random error term since the bias error is zero if the record length is longer than $T + \tau$.

6.5 SPECTRAL DENSITY FUNCTION ESTIMATES

A schematic diagram of a general filter device for estimating the power spectral density function associated with a sample time history record $x(t)$ is displayed in Figure 6.1. The input sample record $x(t)$ is assumed to be averaged over a time interval T and to be drawn from a stationary (ergodic) random process with zero mean value. The tunable narrow band resolution filter is assumed to have a finite nonzero constant bandwidth B_c centered at ω .

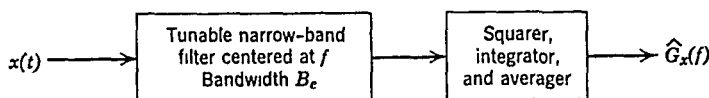


Figure 6.1 Constant bandwidth filter device for measuring power spectrum.

frequency f which may be varied over the frequency range of interest. This resolution bandwidth B_e should not be confused with the full bandwidth occupied by the record $x(t)$. It turns out that in order to obtain a consistent estimate of $G_x(f)$, one must introduce a filtering procedure which averages over a band of frequencies. The final estimate $\hat{G}_x(f)$ describes the time average of $x^2(t)$ in terms of its frequency components lying inside the frequency band $f - (B_e/2)$ to $f + (B_e/2)$, divided by the bandwidth B_e . Note that because a nonzero mean value corresponds to a discrete spectral component at zero frequency, the zero mean value assumption in the following developments is critical only when B_e includes $f = 0$. For all other cases where B_e does not include $f = 0$, the results to follow apply to data with arbitrary nonzero mean values.

The mean square value of $x(t)$ within the bandwidth B_e centered at f is estimated by

$$\hat{\Psi}_x^2(f, B_e) = \frac{1}{T} \int_0^T x^2(t, f, B_e) dt \quad (6.79)$$

where $x(t, f, B_e)$ represents the narrow-band filter output and T is the averaging time. It is shown in Section 6.2.2 that this estimated value will be an unbiased and consistent estimate of the true mean square value as T approaches infinity. Hence

$$\Psi_x^2(f, B_e) = E[\hat{\Psi}_x^2(f, B_e)] = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T x^2(t, f, B_e) dt \quad (6.80)$$

where $\Psi_x^2(f, B_e)$ is the mean square value of $x(t)$ associated with the filter bandwidth B_e centered at f .

The power spectral density function is defined by

$$G_x(f) = \lim_{B_e \rightarrow 0} \frac{\Psi_x^2(f, B_e)}{B_e} = \lim_{\substack{T \rightarrow \infty \\ B_e \rightarrow 0}} \frac{1}{B_e T} \int_0^T x^2(t, f, B_e) dt = \lim_{\substack{T \rightarrow \infty \\ B_e \rightarrow 0}} \hat{G}_x(f) \quad (6.81)$$

where

$$\hat{G}_x(f) = \frac{1}{B_e T} \int_0^T x^2(t, f, B_e) dt = \frac{\hat{\Psi}_x^2(f, B_e)}{B_e} \quad (6.82)$$

is the sample estimate of $G_x(f)$ determined by the procedure of Figure 6.1.

In terms of the true power spectral density function $G_x(f)$, the mean square value of $x(t)$ between any two frequency limits f_1 and f_2 is given by

$$\Psi_x^2(f_1, f_2) = \int_{f_1}^{f_2} G_x(f) df \quad (6.83)$$

In particular

$$\Psi_x^2(f, B_e) = \int_{f-(B_e/2)}^{f+(B_e/2)} G_x(\xi) d\xi \quad (6.84)$$

Equations (6.82) and (6.84) show that

$$E[\hat{G}_x(f)] = \frac{\Psi_x^2(f, B_e)}{B_e} = \frac{1}{B_e} \int_{f-(B_e/2)}^{f+(B_e/2)} G_x(\xi) d\xi \quad (6.85)$$

Thus for most $G_x(f)$

$$E[\hat{G}_x(f)] \neq G_x(f) \quad (6.86)$$

so that $\hat{G}_x(f)$ is generally a biased estimate of $G_x(f)$

The mean square error of the estimate $\hat{G}_x(f)$ is calculated from Equation (6.5) by

$$E[(\hat{G}_x(f) - G_x(f))^2] = \text{Var}[\hat{G}_x(f)] + b^2[\hat{G}_x(f)] \quad (6.87)$$

where $\text{Var}[\hat{G}_x(f)]$ is the variance of the estimate as defined by

$$\text{Var}[\hat{G}_x(f)] = E[(\hat{G}_x(f) - E[\hat{G}_x(f)])^2] \quad (6.88)$$

and $b[\hat{G}_x(f)]$ is the bias of the estimate as defined by

$$b[\hat{G}_x(f)] = E[\hat{G}_x(f) - G_x(f)] \quad (6.89)$$

6.5.1 Variance of the Estimate

The most direct way to arrive at a variance expression for power spectral density estimates is to apply the results derived in Section 6.2.2, as follows. From Equation (6.82), the estimate

$$B_e G_x(f) = \hat{\Psi}_x^2(f, B_e) \quad (6.90)$$

is an unbiased estimate of the mean square value of $x(t)$ within the bandwidth B_e centered at f . The true value is given by $\Psi_x^2(f, B_e) = B_e G_x(f)$, if $G_x(f)$ were constant over the bandwidth B_e . This will be approximately the case if B_e is sufficiently small. The result of Equation (6.34) applies to these estimates with $R_x(0) = B_e G_x(f)$. Hence

$$\text{Var}[B_e \hat{G}_x(f)] \approx \frac{B_e^2 G_x^2(f)}{B_e T} \quad (6.91)$$

But, since B_e is a constant,

$$\text{Var}[B_e \hat{G}_x(f)] = B_e^2 \text{Var}[\hat{G}_x(f)] \quad (6.92)$$

This gives for the variance of the estimate

$$\text{Var} [\hat{G}_x(f)] \approx \frac{G_x^2(f)}{B_e T} \quad (6.93)$$

Further discussion of the variance of power spectral density estimates is presented in Section 6.5.5.

6.5.2 Bias of the Estimate

An expression for the bias term of Equation (6.89) may be derived by the procedure used to derive the bias term for probability density estimates in Section 6.3.2. Specifically, by expanding $G_x(\xi)$ in a Taylor series about the point $\xi = f$, and retaining only the first three terms, it follows from Equation (6.85) that

$$E[\hat{G}_x(f)] \approx G_x(f) + \frac{B_e^2}{24} G_x''(f) \quad (6.94)$$

Thus the bias term is approximated by

$$b[\hat{G}_x(f)] \approx \frac{B_e^2}{24} G_x''(f) \quad (6.95)$$

where $G_x''(f)$ is the second derivative of $G_x(f)$ with respect to f , and is related to $R_x(\tau)$ by the expression

$$G_x''(f) = -8\pi^2 \int_{-\infty}^{\infty} \tau^2 R_x(\tau) e^{-j2\pi f\tau} d\tau \quad (6.96)$$

It should be emphasized that Equation (6.95) is only a first order approximation of the bias error which is applicable for cases where $B_e^2 G_x''(f) < G_x(f)$. Since power spectra in practice often display sharp peaks reflecting large second derivatives, Equation (6.95) may provide an inadequate measure of the bias error in some cases. Generally speaking, Equation (6.95) will exaggerate the degree of bias in estimates where $B_e^2 G_x''(f)$ is large. Bias errors are discussed further in Sections 6.5.5 and 6.9.

6.5.3 Normalized rms Error

The total mean square error of the power spectral density estimate $\hat{G}_x(f)$ is given by Equation (6.87) as the sum of the variance defined in Equation (6.93) and the square of the bias defined in Equation (6.95). That is,

$$E[(\hat{G}_x(f) - G_x(f))^2] \approx \frac{G_x^2(f)}{B_e T} + \left(\frac{B_e^2 G_x''(f)}{24} \right)^2 \quad (6.97)$$

Hence, the normalized mean square error is approximated by

$$\epsilon^2 = \frac{E[(\hat{G}_x(f) - G_x(f))^2]}{G_x^2(f)} \approx \frac{1}{B_e T} + \frac{B_e^4}{576} \left(\frac{G_x''(f)}{G_x(f)} \right)^2 \quad (6.98)$$

The square root of Equation (6.98) gives the normalized rms error ϵ .

Two important features of the error expression for power spectral density estimates should be noted. First, there are conflicting requirements on the resolution bandwidth B_e , namely a small value of B_e is needed to suppress the bias portion of the error while a large value of B_e is desired to reduce the random portion of the error. This is similar to the situation discussed in Section 6.3.3 for the window width W in probability density measurements. The problem here, however, is more critical since power spectra in practice often display sharp peaks (large second derivatives) which aggravate the bias error problem. Second, the random portion of the error includes only the resolution bandwidth B_e , and not the total data bandwidth B . Hence the random portion of the error is a function primarily of analysis parameters rather than unknown data parameters. This greatly enhances the practical value of Equation (6.98) in experimental design and data analysis. The above issues are pursued further in Section 6.9.

6.5.4 Cross-Spectral Density Estimates

The cross spectral density function of two stationary (ergodic) random processes $\{x(t)\}$ and $\{y(t)\}$ is usually estimated in terms of the co spectrum $C_{xy}(f)$ and the quad spectrum $Q_{xy}(f)$ where

$$G_{xy}(f) = C_{xy}(f) - j Q_{xy}(f) \quad (6.99)$$

Through an analysis similar to the derivation of Equation (6.93) it can be shown that the variance terms for estimates $\hat{C}_{xy}(f)$ and $\hat{Q}_{xy}(f)$ are bounded by

$$\begin{aligned} \text{Var} [\hat{C}_{xy}(f)] &\leq \frac{G_x(f) G_y(f)}{B_e T} \\ \text{Var} [\hat{Q}_{xy}(f)] &\leq \frac{G_x(f) G_y(f)}{B_e T} \end{aligned} \quad (6.100)$$

Furthermore the bias terms are approximated by

$$\begin{aligned} b[\hat{C}_{xy}(f)] &\approx \frac{B_e^2}{24} C_{xy}''(f) \\ b[\hat{Q}_{xy}(f)] &\approx \frac{B_e^2}{24} Q_{xy}''(f) \end{aligned} \quad (6.101)$$

The exact value of the variance terms in Equation (6.100) is dependent upon the coherence function $\gamma_{xy}^2(f)$. Since cross-spectra are usually measured only as an intermediate step in the estimation of other parameters (coherence functions or frequency response functions), further discussion of the estimation error problem is incorporated in the development of statistical errors for such other parameter estimates.

6.5.5 Estimates from Finite Fourier Transforms

In Section 6.5.1, an expression is derived for the variance of power spectral density estimates obtained by filtering, squaring, and averaging operations, as would be performed by analog instruments. One might also arrive at a variance expression for spectral density estimates obtained by direct Fourier transform operations, as would be accomplished on a digital computer using the fast Fourier transform techniques discussed in Chapter 9. A brief summary of this second approach may add insight into the spectral density estimation error problem.

Consider the power spectral density function of a stationary (ergodic) Gaussian random process $\{x(t)\}$, as defined in Equation (3.102). Specifically, given a sample record $x(t)$ of unlimited length T , the power spectrum is

$$G_x(f) = 2 \lim_{T \rightarrow \infty} \frac{1}{T} E[|X(f, T)|^2] \quad (6.102)$$

where $X(f, T)$ is the finite Fourier transform of $x(t)$, that is,

$$X(f, T) = \int_0^T x(t) e^{-j2\pi ft} dt \quad (6.103)$$

Now an estimate of $G_x(f)$ can be obtained by simply omitting the limiting and expectation operations in Equation (6.102). This will yield the estimate

$$\hat{G}_x(f) = \frac{2}{T} |X(f, T)|^2 \quad (6.104)$$

with the narrowest possible resolution $B_c = (1/T)$.

To determine the variance of this estimate, observe that the finite Fourier transform $X(f, T)$ is defined by a series of components at frequencies $f = k/T$; $k = 1, 2, 3$, etcetera. Further observe that $X(f, T)$ is a complex number where the real and imaginary parts, $X_R(f, T)$ and $X_I(f, T)$, can be shown to be uncorrelated random variables with zero means and equal variances. Since a Fourier transformation is a linear operation, $X_R(f, T)$ and $X_I(f, T)$ will be Gaussian random variables if $x(t)$ is Gaussian. It follows that the quantity

$$|X(f, T)|^2 = X_R^2(f, T) + X_I^2(f, T) \quad (6.105)$$

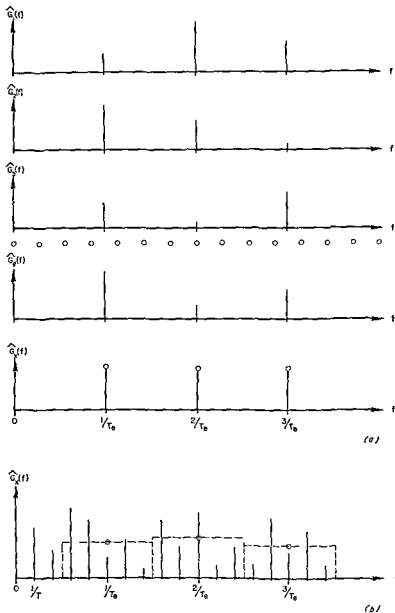


Figure 6.2 Smoothing procedures for spectral density estimates (a) Ensemble smoothing procedure (b) Frequency smoothing procedure

is the sum of the squares of two independent Gaussian variables. Hence from the definition of Equation (4.16) as applied in Equation (4.37), each frequency component of the estimate $\hat{G}(f, T)$ will have a sampling distribution given by

$$\frac{\hat{G}(f, T)}{G(f)} = \frac{\chi_2^2}{2} \quad (6.106)$$

where χ_2^2 is the chi-square variable with $n = 2$ degrees of freedom.

Note that the result in Equation (6.106) is independent of the record length T , that is, increasing the record length does not alter the distribution function defining the random error of the estimate. It only increases the number of spectral components in the estimate. If the record length is interpreted as a measure of the sample size for the estimate, this implies that Equation (6.104) produces an inconsistent estimate of power spectral density functions, as previously stated in Section 3.2.3. Furthermore, the random error of the estimate is substantial. Referring to Equations (4.19) and (4.20), the mean and variance of the chi-square variable are n and $2n$, respectively. Then the normalized standard error, which defines the random portion of the estimation error, is

$$\epsilon_r = \frac{\sigma[\hat{G}_x(f)]}{G_x(f)} = \frac{\sqrt{2n}}{n} = \sqrt{\frac{2}{n}} \quad (6.107)$$

For the case at hand, $n = 2$ so $\epsilon_r = 1$, which means that the standard deviation of the estimate is as great as the quantity being estimated. This would be an unacceptable random error for most applications.

In practice, the random error of an estimate produced by Equation (6.104) is reduced by smoothing the estimate further in one of two ways. The first way is to *smooth over an ensemble* of estimates. This can be done by computing individual estimates from q independent sample records, $x_i(t)$; $i = 1, 2, 3, \dots, q$, and then averaging the q estimates at each frequency of a spectral component as illustrated in Figure 6.2(a). The second way is to *smooth over frequency*. This can be done by averaging together the results for ℓ contiguous spectral components in the estimate from a single sample record as illustrated in Figure 6.2(b). In either case, the smoothing operation approximates the expectation operation in Equation (6.102) and, hence, makes the estimates consistent.

To further illustrate this point, let the resolution bandwidth for the spectral analysis be defined as $B_e = 1/T_e$. Then for the ensemble smoothing case, the total required record length is $T = qT_e$ which gives $q = T/T_e$. For frequency smoothing, since the spectral components are spaced at intervals of $\Delta f = 1/T$, it follows that $\ell = B_e/\Delta f = T/T_e$ as for ensemble smoothing. Hence for

$q = \ell$, the two procedures are equivalent. Furthermore, noting that $\chi_a^2 + \chi_b^2 = \chi_{a+b}^2$ and assuming that $G_x(f) = \text{constant}$ within the bandwidth B_x , the resulting estimates will be chi square variables with $n = 2\ell$ degrees of freedom. Since $\ell = B_x/\Delta f$ and $\Delta f = 1/T$, the following important relationship evolves

$$n = 2B_x T \quad (6.108)$$

Substituting Equation (6.108) into Equation (6.107) yields

$$\epsilon_r = \frac{\sigma[\hat{G}_x(f)]}{G_x(f)} = \frac{1}{\sqrt{B_x T}} \quad (6.109)$$

which agrees with the result derived in Section 6.5.1

Several important implications of the foregoing discussions should be re-emphasized. First, if a power spectrum is estimated by direct Fourier transform operations in accordance with Equation (6.104), a smoothing operation is required to obtain a consistent estimate. Second, if a smoothing operation is not performed, the normalized standard error of the estimate will be $\epsilon_r = 1$. Third, the sampling distribution of a smoothed estimate is approximately chi square with $n = 2B_x T$ degrees of freedom. Hence a $(1 - \alpha)$ confidence interval for a power spectral density function $G(f)$ based upon an estimate $\hat{G}(f)$ measured with a resolution bandwidth B_x and a record length T is given by

$$\left[\frac{n\hat{G}(f)}{\chi_{n-1-\alpha/2}^2} \leq G(f) < \frac{n\hat{G}(f)}{\chi_{n-1-\alpha/2}^2} \right] \quad n = 2B_x T \quad (6.110)$$

G(f)

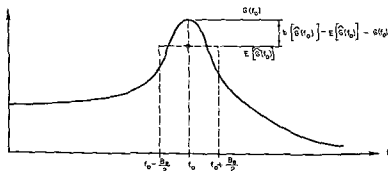


Figure 6.3 Illustration of bias error introduced by frequency smoothing of spectral density estimates

Now consider the bias error in power spectral density estimates obtained by smoothing the results of Equation (6.104). A bias error of the form derived in Section 6.5.2 is best illustrated by assuming the smoothing is accomplished over frequency. For this case, an estimate is obtained at f_0 by averaging the values of all spectral components in the frequency range $f_0 - B_e/2 \leq f \leq f_0 + B_e/2$, as illustrated in Figure 6.3. This gives rise to a bias error since $E[\hat{G}(f_0)] \neq G(f_0)$ except for the special case where $G''(f_0) = 0$. It is clear that the bias error becomes more severe as $G''(f_0)$ increases for a given B_e , or as B_e increases for a given $G''(f_0) \neq 0$.

6.6 COHERENCE FUNCTION ESTIMATES

Consider two sample time history records $x(t)$ and $y(t)$, from two stationary (ergodic) random processes, $\{x(t)\}$ and $\{y(t)\}$. Assume the sample records exist over a common time interval T and the power and cross-spectral density functions of the records are estimated with $n = 2B_e T$ degrees of freedom, as discussed in Section 6.5.5. From Equation (5.23) the coherence function $\gamma_{xy}^2(f)$ for the two random processes is estimated by

$$\hat{\gamma}_{xy}^2(f) = \frac{|\hat{G}_{xy}(f)|^2}{\hat{G}_x(f) \hat{G}_y(f)} \quad (6.111)$$

The problem now is to define the statistical accuracy of the estimate $\hat{\gamma}_{xy}^2(f)$.

Referring to Section 4.8.1, it is noted that the accuracy of a correlation coefficient estimate $r_{xy} = \hat{\rho}_{xy}$ can be conveniently evaluated in terms of a transformation variable w defined in Equation (4.57). The similarity of the estimate $\hat{\gamma}_{xy}(f)$, as given by the square root of Equation (6.111), to the correlation coefficient estimate $\hat{\rho}_{xy}$, as defined in Equation (4.56), is clear. Hence, it is not surprising that a similar transformation would be convenient for evaluating the accuracy of $\hat{\gamma}_{xy}(f)$. From Reference 16, empirical studies show that estimates of coherence functions in the range $0.35 \leq \gamma_{xy}^2(f) \leq 0.95$ based upon spectral density estimates with $n \geq 40$ degrees of freedom can be evaluated in terms of the transformation

$$w(f) = \frac{1}{2} \ln \left[\frac{1 + \hat{\gamma}_{xy}(f)}{1 - \hat{\gamma}_{xy}(f)} \right] = \tanh^{-1} \hat{\gamma}_{xy}(f) \quad (6.112)$$

where $w(f)$ has an approximately normal distribution with a mean and variance of

$$\mu_w(f) = (n - 2)^{-1} + \tanh^{-1} \gamma_{xy}(f) \quad (6.113)$$

$$\sigma_w^2 = (n - 2)^{-1} \quad n = 2B_e T \quad (6.114)$$

From the preceding equations, the $(1 - \alpha)$ confidence interval for a coherence function $\gamma_{xy}^2(f)$ based upon an estimate $\hat{\gamma}_{xy}^2(f)$ can be established

as follows. First, since $u(f)$ is normally distributed, Equation (4.15) applies to yield

$$\text{Prob} \left[z_{\alpha/2} < \frac{u(f) - \mu_u(f)}{\sigma_u} \leq z_{\alpha/2} \right] = 1 - \alpha \quad (6.115)$$

where z_α is the 100 α percentage point of the standardized normal distribution. In Equation (6.115), $\mu_u(f)$ can now be replaced by Equation (6.113). After rearranging terms, the resulting $(1 - \alpha)$ confidence interval is found to be

$$\begin{aligned} [\tanh \{u(f) - (n - 2)^{-1} - \sigma_u z_{\alpha/2}\} < \gamma_{xy}(f) \\ \leq \tanh \{u(f) - (n - 2)^{-1} + \sigma_u z_{\alpha/2}\}] \end{aligned} \quad (6.116)$$

The above result gives the $(1 - \alpha)$ confidence interval for $\gamma_{xy}(f)$ as a function of n , $\hat{\gamma}_{xy}^2(f)$, and α . The confidence limits for $\gamma_{xy}^2(f)$ are the square of the corresponding limits for $\gamma_{xy}(f)$.

Example 6.4 Confidence Interval for Coherence Estimate. Suppose a coherence function is measured from two sample records of length $T = 10$ seconds using a resolution bandwidth of $B_r = 5$ cps so that the resulting coherence estimate $\hat{\gamma}_{xy}^2(f)$ has $n = 100$ degrees of freedom. What are the 99 percent confidence limits for $\gamma_{xy}^2(f)$ if the measured values of $\hat{\gamma}_{xy}^2(f)$ are 0.40, 0.50, 0.60, 0.70, 0.80, or 0.90?

From Table A.2, for $\alpha/2 = 0.005$, $z_{\alpha/2} = 2.58$, from Equation (6.114), $\sigma_u = 0.10$. Hence, Equation (6.116) becomes

$$[\tanh \{u(f) - 0.268\} \leq \gamma_{xy}(f) \leq \tanh \{u(f) + 0.248\}]$$

with $u(f)$ determined by Equation (6.112). Table 6.1 shows the 99 percent confidence limits for $\gamma_{xy}^2(f)$ as a function of $\hat{\gamma}_{xy}^2(f)$ for this example. Similar results may be obtained for other values of n and α . This table concludes Example 6.4.

Table 6.1
99 Percent Confidence Limits for $\gamma^2(f)$ when $n = 100$

$\hat{\gamma}^2(f)$	0.4	0.5	0.6	0.7	0.8	0.9
Lower limit	0.19	0.29	0.41	0.54	0.68	0.83
Upper limit	0.58	0.66	0.74	0.81	0.88	0.94

Referring back to Equation (6.113), it should be noted that there is an inherent bias error in coherence function estimates which is dependent upon the degrees of freedom n . This bias error occurs in addition to the errors

introduced by bias in the spectral density estimates used to calculate $\hat{\gamma}_{xy}^2(f)$. Furthermore, it becomes quite severe as n becomes small. In particular, for the limiting case where $n = 2$, the estimate $\hat{\gamma}_{xy}(f) = 1$, independent of the actual coherence function value, as will be demonstrated shortly. The failure to note this important fact is the source of a common error in coherence function analysis. Specifically, coherence function estimates are often determined from spectral density estimates computed by direct Fourier transform procedures, as outlined in Section 6.5.5 and detailed in Chapter 9. The first estimate of spectral density provided by the direct Fourier transform approach is an $n = 2$ degree of freedom estimate, as defined by Equation (6.104). It is noted in Section 6.5.5 that this estimate must be smoothed, either by ensemble or frequency averaging, to suppress the random error to acceptable levels. However, when spectral density estimates are obtained to calculate coherence functions, one might be inclined to postpone the required smoothing operations on the spectral densities in favor of a single smoothing of the resulting coherence function estimate. Unfortunately, such an approach will produce a coherence function estimate of unity, even for the case of totally incoherent data.

The above fact is easily demonstrated as follows. From Equations (3.100) and (3.101), unsmoothed estimates for power and cross-spectra are given by

$$\hat{G}_{xy}(f) = \frac{2}{T} X^*(f, T) Y(f, T)$$

$$\hat{G}_x(f) = \frac{2}{T} |X(f, T)|^2 = \frac{2}{T} X^*(f, T) X(f, T)$$

$$\hat{G}_y(f) = \frac{2}{T} |Y(f, T)|^2 = \frac{2}{T} Y^*(f, T) Y(f, T)$$

where $X(f, T)$ and $Y(f, T)$ are finite Fourier transforms of $x(t)$ and $y(t)$ over the record length T . Omitting the functional notation, the coherence function estimate is then given by Equation (6.111) as

$$\hat{\gamma}_{xy}^2 = \frac{\hat{G}_{xy}^* \hat{G}_{xy}}{\hat{G}_x \hat{G}_y} = \frac{\left[\frac{2}{T} X^* Y \right]^* \left[\frac{2}{T} X^* Y \right]}{\left[\frac{2}{T} X^* X \right] \left[\frac{2}{T} Y^* Y \right]} = \frac{X Y^* X^* Y}{X^* X Y^* Y} = 1$$

where no restrictions have been placed on the actual coherence function $\gamma_{xy}^2(f)$.

This result can be explained heuristically by considering an analogy to classical correlation estimates, as discussed in Section 4.8.1. The sample coherence function, like the sample correlation coefficient, is effectively a

measure of how well a straight line can be fit to available data points. A two degree of freedom coherence estimate is analogous to a correlation coefficient estimate based upon two pairs of data values, x_1, y_1 and x_2, y_2 , defining two points in the xy plane. Independent of the actual relationship between x and y , it is clear that a straight line can always be fit without error to any two given data points. This suggests why perfect coherence must always be indicated for two degree of freedom estimates.

As a final point, it should be mentioned that the accuracy of estimates of the multiple coherence function, as defined in Section 5.4, can be described by relationships similar to those presented for the ordinary coherence function in Equations (6.112) through (6.114). To be more specific, consider a multiple input linear system with q inputs, where the multiple coherence function between the inputs $x_i(t)$, $i = 1, 2, 3, \dots, q$, and the input $y(t)$ is estimated from sample values of the terms in Equation (5.126) or (5.129). It is shown in Reference 16 that the transformation

$$w(f) = \frac{1}{2} \ln \left[\frac{1 + \hat{\gamma}_{y,x}(f)}{1 - \hat{\gamma}_{y,x}(f)} \right] = \tanh^{-1} \hat{\gamma}_{y,x}(f) \quad (6.117)$$

has an approximately normal distribution with a mean and variance of

$$\mu_w(f) = \frac{q}{(n - 2q)} + \tanh^{-1} \gamma_{y,x}(f) \quad (6.118)$$

$$\sigma_w^2 = \frac{q}{(n - 2q)} \quad n = 2B_s T \quad (6.119)$$

In the above expressions, $\hat{\gamma}_{y,x}(f)$ is the positive square root of the multiple coherence function estimate.

6.7 FREQUENCY RESPONSE FUNCTION ESTIMATES—SINGLE INPUT CASE

Consider a single input single output constant parameter linear system with a frequency response function of $H(f)$, as illustrated in Figure 6.4. Let the input $x(t)$ be a sample time history record from a stationary (ergodic) random process $\{x(t)\}$. It follows that the output $y(t)$ will be a sample record from a stationary (ergodic) random process $\{y(t)\}$. Assume the sample records exist over a common time interval T and the power and cross spectral density functions of the records are estimated with $n = 2B_s T$ degrees of freedom, as

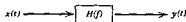


Figure 6.4 Single input linear system

discussed in Section 6.5.5. From Equation (5.9), the frequency response function of the linear system is estimated by

$$\hat{H}(f) = \frac{\hat{G}_{xy}(f)}{\hat{G}_x(f)} = |\hat{H}(f)| e^{-j\hat{\phi}(f)} \quad (6.120)$$

Note that Equation (6.120) includes both the gain factor estimate $|\hat{H}(f)|$ and the phase factor estimate $\hat{\phi}(f)$, as per Equations (5.11) and (5.12). The statistical accuracy of the estimate $\hat{H}(f)$ will now be investigated.

6.7.1 Bias Errors

The estimation of a frequency response function using Equation (6.120) will generally involve bias errors from a number of sources as follows:

1. bias inherent in the estimation procedure.
2. nonlinear and/or time varying system parameters.
3. bias in the power and cross-spectral density estimates used for the calculation.
4. measurement noise at the input (uncorrelated noise at the output does not cause bias).
5. contributions to the measured output from other inputs correlated with the measured input.

The first noted source of bias evolves from the fact that, in general,

$$E[\hat{H}(f)] = E\left[\frac{\hat{G}_{xy}(f)}{\hat{G}_x(f)}\right] \neq \frac{E[\hat{G}_{xy}(f)]}{E[\hat{G}_x(f)]} \quad (6.121)$$

Hence $E[\hat{H}(f)] \neq H(f)$. A detailed evaluation of this inherent bias in the estimation procedure is not pursued here since the resulting error is usually negligible compared to other bias and random errors which occur in practice. To be more specific, it can be shown that $E[\hat{H}(f)] \rightarrow H(f)$ as either $n = 2B_eT \rightarrow \infty$ or $\gamma_{xy}^2(f) \rightarrow 1$, assuming no other bias errors are present. As will be seen later, relatively large values of n and/or $\gamma_{xy}^2(f)$ are needed to suppress the random error of the estimate $\hat{H}(f)$ to an acceptable level. The bias error is generally much less than the random error for any given combination of n and $\gamma_{xy}^2(f)$.

The second indicated source of bias results from violations of the basic assumption that the system of interest is a constant parameter linear system. Even when the constant parameter assumption is valid, the linearity assumption will often be violated if the operating range of interest is sufficiently wide. It should be noted, however, that the application of Equation (6.120) to nonlinear systems will produce the best linear approximation (in the least squares sense) for the frequency response function under the specified input

conditions. This fact constitutes an important advantage to estimating frequency response functions from actual data rather than from laboratory or simulated data which do not reproduce the actual input conditions.

The next indicated source of bias is that introduced by bias errors in the spectral density estimates. As noted in Section 6.5.2, the bias in spectral density measurements can be quite significant at frequencies where spectral peaks occur. These errors are suppressed by obtaining properly resolved estimates of power and cross spectra, that is, making B_x sufficiently narrow to accurately define peaks in the spectra. This problem is discussed further in Section 6.9.

The fourth source of bias is measurement noise at the input, meaning extraneous noise in the measured $x(t)$ which does not actually pass through the system. To illustrate this error, let $x(t) = u(t) + n(t)$ where $u(t)$ is the input signal which actually passes through the system, and $n(t)$ is extraneous measurement noise. If all other bias errors are ignored, it follows from Equation (6.120) that

$$\frac{E\{\hat{H}(f)\}}{H(f)} = \left[\frac{G_{uv}(f)}{G_u(f) + G_n(f)} \right] \left[\frac{G_u(f)}{G_{uv}(f)} \right] = \left[1 + \frac{G_n(f)}{G_u(f)} \right]^{-1} \quad (6.122)$$

For example, if the spectral density of the noise in the input measurement were about 10 percent of the spectral density of the signal, the resulting frequency response function estimate would be biased downward by about 10 percent.

The final source of bias to be considered is that due to the contributions of other inputs which are correlated with the measured input. This error is readily illustrated for the simple case of one additional correlated input by the developments in Section 5.3.3. Specifically, let $x(t)$ be the measured input, $v(t)$ be a second unmeasured input, and $y(t)$ be the output. From Equation (5.50), the correct frequency response function is given by

$$H(f) = \frac{G_{xy}(f) \left[1 - \frac{G_{xv}(f) G_{vy}(f)}{G_v(f) G_{xx}(f)} \right]}{G_x(f) [1 - \gamma_{xv}^2(f)]} \quad (6.123)$$

Hence, if all other bias errors are ignored, it follows from Equations (6.120) and (6.123) that

$$\frac{E\{\hat{H}(f)\}}{H(f)} = \frac{[1 - \gamma_{xv}^2(f)]}{\left[1 - \frac{G_{xv}(f) G_{vy}(f)}{G_v(f) G_{xx}(f)} \right]} \quad (6.124)$$

Note that if the measured and unmeasured inputs $x(t)$ and $v(t)$ are uncorrelated, then $\gamma_{xv}^2(f) = G_{xv}(f) = 0$ and the right side of Equation (6.124)

reduces to unity. This demonstrates the fact that unmeasured inputs which are not correlated with the measured input cause no bias error since they appear at the output as uncorrelated extraneous noise. They do, however, contribute to the random error in frequency response function estimates, as is discussed in the next section.

6.7.2 Random Errors

To evaluate the random error associated with frequency response function estimates, consider the single input-single output system previously described where $x(t)$ is the input time history and $y(t)$ is the output time history. Assume the conditions necessary to make bias errors small, as outlined in Section 6.7.1. Let the residual extraneous noise at the output due to measurement noise and/or the contributions of other uncorrelated inputs be denoted by $z(t)$. Then, for an arbitrary pair of sample records, the quantity $z(t)$ has the same meaning previously used in Equation (5.95), that is

$$z(t) = y(t) - \int_0^{\infty} h(\tau) x(t - \tau) d\tau \quad (6.125)$$

Define finite Fourier transforms of $x(t)$, $y(t)$, and $z(t)$, as in Equation (3.91), by expressions of the form

$$X(f, T) = \int_0^T x(t) e^{-j2\pi ft} dt \quad (6.126)$$

Then

$$Z(f, T) \approx Y(f, T) - H(f) X(f, T) \quad (6.127)$$

Now, Equation (6.127) can be written in the form

$$Z(f, T) \approx \hat{Z}(f, T) + X(f, T)[\hat{H}(f, T) - H(f)] \quad (6.128)$$

where

$$\hat{Z}(f, T) = Y(f, T) - \hat{H}(f, T) X(f, T) \quad (6.129)$$

is the best estimate of the noise term at frequency f in the least squares sense when $\hat{H}(f, T)$ is computed from Equation (6.120). From Equations (6.128) and (6.129), it follows that

$$|Z(f, T)|^2 \approx |\hat{Z}(f, T)|^2 + |X(f, T)|^2 |\hat{H}(f, T) - H(f)|^2 \quad (6.130)$$

since the cross-product term vanishes by using the relationship

$$|X(f, T)|^2 \hat{H}(f, T) = X^*(f, T) Y(f, T) \quad (6.131)$$

Equation (6.131) is based upon Equation (6.120) together with Equations

(3 100) and (3 101), where spectral estimates are computed from

$$\begin{aligned}\hat{G}_{xy}(f, T) &= \frac{2}{T} X^*(f, T) Y(f, T) \\ \hat{G}_x(f, T) &= \frac{2}{T} |X(f, T)|^2\end{aligned}\quad (6 132)$$

Dividing Equation (6 130) by $T/2$ gives the approximate result

$$\hat{G}_x(f, T) \approx \hat{G}_x(f, T) + \hat{G}_x(f, T) |\hat{H}(f, T) - H(f)|^2 \quad (6 133)$$

The estimate $\hat{G}_x(f, T)$ as developed so far has only two degrees of freedom associated with it. This is due to the bandwidth $B_s = 1/T$ associated with the estimates given by Equation (6 132). To increase the number of degrees of freedom, it is necessary to smooth the estimate $\hat{G}_x(f, T)$ by averaging over either an ensemble of independent estimates or a larger bandwidth B_s , as discussed in Section 6 5 5. When this is done

$$\hat{G}_x(f) \approx \hat{G}_x(f) + \hat{G}_x(f) |\hat{H}(f) - H(f)|^2 \quad (6 134)$$

where the dependence on B_s and T is not shown to simplify the notation. Assume that this smoothing procedure is applied to the various spectral estimates so as to produce n degrees of freedom in the estimate $\hat{G}_x(f)$, where $n = 2B_sT$. Then

$$\frac{n\hat{G}_x(f)}{\hat{G}_x(f)} = \chi_n^2 \quad (6 135)$$

will be distributed as χ_n^2 with n degrees of freedom, as defined in Section 4 3 2. The two terms on the right hand side of Equation (6 134) are statistically independent. Hence, their number of degrees of freedom will be additive, that is

$$\frac{\chi_n^2}{n} = \frac{\chi_{n_1}^2}{n} + \frac{\chi_{n-n_1}^2}{n} \quad (6 136)$$

where n_1 is to be determined. Now the first term on the right hand side of Equation (6 134), namely $\hat{G}_x(f)$, must have the degrees of freedom associated with $\hat{G}_x(f)$ reduced only by the number of additional constraints imposed on $\hat{G}_x(f)$. Referring to Equations (6 127) and (6 129), it is seen that $\hat{Z}(f, T)$ is defined in terms of the estimate $\hat{H}(f, T)$ rather than the actual value $H(f)$. This constitutes two independent constraints on $\hat{Z}(f, T)$, one each for the real and imaginary parts used to estimate $\hat{H}(f, T)$. Hence, $\hat{G}_x(f)$ will have two less degrees of freedom than $\hat{G}_x(f)$, that is

$$\frac{n\hat{G}_x(f)}{\hat{G}_x(f)} = \chi_{n-2}^2 \quad (6 137)$$

is distributed as chi-square with $n - 2$ degrees of freedom. Since the degrees of freedom on the right-hand side of Equation (6.134) must sum to n , it follows that the second term on the right-hand side must have only 2 degrees of freedom, that is,

$$\frac{n\hat{G}_x(f) |\hat{H}(f) - H(f)|^2}{G_z(f)} = \chi_2^2 \quad (6.138)$$

is distributed as chi-square with 2 degrees of freedom.

The term $\hat{G}_z(f)$ is the smoothed estimate of $\hat{G}_z(f, T)$. From Equations (6.129) and (6.131)

$$\begin{aligned} |\hat{Z}(f, T)|^2 &= |Y(f, T) - \hat{H}(f) X(f, T)|^2 \\ &= |Y(f, T)|^2 - \frac{|X^*(f, T) Y(f, T)|^2}{|X(f, T)|^2} \end{aligned} \quad (6.139)$$

Then, using Equation (6.132), it follows that

$$\begin{aligned} \hat{G}_z(f, T) &= \frac{2}{T} |\hat{Z}(f, T)|^2 \\ &= \hat{G}_y(f, T) - \frac{|\hat{G}_{xy}(f, T)|^2}{\hat{G}_x(f, T)} \end{aligned} \quad (6.140)$$

and by smoothing,

$$\hat{G}_z(f) = \hat{G}_y(f) [1 - \hat{\gamma}_{xy}^2(f)] \quad (6.141)$$

where

$$\hat{\gamma}_{xy}^2(f) = \frac{|\hat{G}_{xy}(f)|^2}{\hat{G}_x(f) \hat{G}_y(f)} \quad (6.142)$$

is an estimate of the ordinary coherence function $\gamma_{xy}^2(f)$. The result of Equation (6.141) when substituted into Equation (6.137) yields

$$\frac{n\hat{G}_y(f) [1 - \hat{\gamma}_{xy}^2(f)]}{G_z(f)} = \chi_{n-2}^2 \quad (6.143)$$

As discussed in Section 4.2.4, the F distribution with n_1 and n_2 degrees of freedom is defined by

$$F_{n_1, n_2} = \frac{y_1/n_1}{y_2/n_2} \quad (6.144)$$

where $y_1 = \chi^2$ with n_1 degrees of freedom, and $y_2 = \chi^2$ with n_2 degrees of freedom. Hence, using Equations (6.138) and (6.143), it follows that

$$F_{2, n-2} = \frac{(n-2) \hat{G}_x(f) |\hat{H}(f) - H(f)|^2}{2\hat{G}_y(f) [1 - \hat{\gamma}_{xy}^2(f)]} \quad (6.145)$$

Equation (6 145) requires only further physical geometric interpretation to provide confidence intervals for $H(f)$ based upon sample estimates $\hat{H}(f)$. Observe that all terms on the right hand side of Equation (6 145) are measurable quantities. The occurrence of the F distribution is not surprising in view of the fact that $\hat{H}(f)$ in Equation (6 120) is the ratio of two spectral quantities, each of which is governed by a chi-square distribution.

From Equation (6 145), it follows that if $H(f)$ is unknown, then a confidence interval for $H(f)$ with confidence coefficient $1 - \alpha$ can be determined by a quantity $\hat{r}(f)$ such that

$$|\hat{H}(f) - H(f)|^2 \leq r^2(f)$$

$$r^2(f) = \frac{2}{(n-2)} F_{2, n-2, \alpha} [1 - \hat{\gamma}_{xy}^2(f)] \frac{\hat{G}_x(f)}{\hat{G}_y(f)} \quad (6 146)$$

where

$n = 2B_f T$ = number of degrees of freedom of each spectral estimate

$F_{2, n-2, \alpha}$ = 100 α percentage point of an F distribution with $n_1 = 2$ and $n_2 = n - 2$ degrees of freedom

$\hat{G}_x(f)$ = power spectrum estimate of the input $x(t)$

$\hat{G}_y(f)$ = power spectrum estimate of the output $y(t)$

$\hat{\gamma}_{xy}^2(f)$ = sample estimate of the ordinary coherence function between the input $x(t)$ and the output $y(t)$

Geometrically, Equation (6 146) describes a circle of radius $r(f)$ centered at $\hat{H}(f)$, as illustrated later in Figure 6 6. In terms of the gain factor estimate $|\hat{H}(f)|$ and the phase factor estimate $\hat{\phi}(f)$, the approximate $1 - \alpha$ confidence intervals for the actual gain factor $|H(f)|$ and phase factor $\phi(f)$ are now given at each frequency f by

$$\left[|\hat{H}(f)| - \hat{r}(f) \leq |H(f)| \leq |\hat{H}(f)| + \hat{r}(f) \right]$$

$$\left[\hat{\phi}(f) - \Delta\hat{\phi}(f) \leq \phi(f) \leq \hat{\phi}(f) + \Delta\hat{\phi}(f) \right] \quad (6 147)$$

where $\hat{r}(f)$ is the positive square root of $r^2(f)$ defined in Equation (6 146), and

$$\Delta\hat{\phi}(f) = \sin^{-1} \left[\frac{\hat{r}(f)}{|\hat{H}(f)|} \right] \quad (6 148)$$

From Equation (6 146), it is seen that the random error in frequency response function estimates, as measured by $\hat{r}(f)$, is heavily dependent upon

the degrees of freedom n and the sample coherence function $\hat{\gamma}_{xy}^2(f)$. Specifically, $\hat{r}(f) \rightarrow 0$ as either $n \rightarrow \infty$ or $\hat{\gamma}_{xy}^2(f) \rightarrow 1$. Caution must be exercised, however, when these results are interpreted for certain limiting cases. For example, if $\hat{\gamma}_{xy}^2(f) \approx \gamma_{xy}^2(f) = 0$, it follows that no physical system exists between the measurement points and no frequency response function can be defined. On the other hand, if $\gamma_{xy}^2(f)$ has a nonzero value, no matter how small, then a frequency response function can be estimated with any desired degree of accuracy given sufficient data to make n appropriately large. At the other extreme, if $\hat{\gamma}_{xy}^2(f) \approx \gamma_{xy}^2(f) = 1$, then Equation (6.146) suggests the random error will be zero independent of n . This is technically correct even for the limiting case of $n = 2$ since perfect coherence means $y(t)$ is simply a deterministic function of $x(t)$. Nevertheless, the estimation of frequency response functions with two degree of freedom spectral measurements should never be attempted in practice. If the assumption of perfect coherence were incorrect, the resulting frequency response function estimates would include severe errors. Furthermore, the lack of perfect coherence would not be indicated by the sample coherence function. As noted in Section 6.6, the sample coherence function computed from two degree of freedom spectral estimates is always unity, independent of the actual coherence function value.

Example 6.4. Confidence Interval for Gain Factor Estimate (Single Input).

From Equation (6.120), assume a gain factor of 1.25 is estimated at a given frequency f_0 between an input $x(t)$ and an output $y(t)$. Suppose that the respective power spectral density functions are measured as $\hat{G}_x(f_0) = 0.13$ and $\hat{G}_y(f_0) = 0.28$, with $\hat{\gamma}_{xy}^2 = 0.70$. Assume the number of degrees of freedom $n = 100$. Determine a 99 percent confidence interval for the true gain factor at this frequency, assuming the bias error is negligible.

For this example, $n_1 = 2$ and $n_2 = 98$ degrees of freedom. From Table A.5(c), for a $1 - \alpha = 0.99$ confidence interval,

$$F_{2,98,0.01} = 4.82$$

It follows from Equation (6.146) that

$$\hat{r}_i^2(f_0) = \left(\frac{2}{98}\right)(4.82)(0.3)\left(\frac{0.28}{0.13}\right) = 0.063$$

and

$$\hat{r}_i(f_0) = 0.25$$

Thus the 99 percent confidence interval for the gain factor is

$$1.00 \leq |H(f_0)| \leq 1.50$$

6.8 FREQUENCY RESPONSE FUNCTION ESTIMATES— MULTIPLE INPUT CASE

Now consider a multiple input single output constant parameter linear system with q clearly defined inputs $x_i(t)$ $i = 1, 2, \dots, q$ contributing to the output $y(t)$ through paths with frequency response functions of $H_i(f)$, $i = 1, 2, \dots, q$ as illustrated in Figure 6.5. Let $\hat{G}_{x_i}(f) = \hat{G}_{ix}(f)$ be an estimate of the power spectrum of the input $x_i(t)$, and $\hat{G}_{ii}(f)$ be an estimate of the

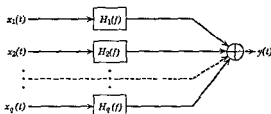


Figure 6.5 Multiple input linear system

cross spectrum between the inputs $x_i(t)$ and $x_j(t)$. Further let $\hat{G}_{iy}(f)$ be an estimate of the cross spectrum between the input $x_i(t)$ and the output $y(t)$. Then from Equation (5.120), the frequency response function between each input point and the output point can be estimated from the matrix relation

$$\begin{bmatrix} \hat{H}_1(f) \\ \hat{H}_2(f) \\ \vdots \\ \hat{H}_q(f) \end{bmatrix} = \begin{bmatrix} \hat{G}_{11}(f) & \hat{G}_{12}(f) & \dots & \hat{G}_{1q}(f) \\ \hat{G}_{21}(f) & \hat{G}_{22}(f) & \dots & \hat{G}_{2q}(f) \\ \vdots & \vdots & \ddots & \vdots \\ \hat{G}_{q1}(f) & \hat{G}_{q2}(f) & \dots & \hat{G}_{qq}(f) \end{bmatrix}^{-1} \begin{bmatrix} \hat{G}_{1y}(f) \\ \hat{G}_{2y}(f) \\ \vdots \\ \hat{G}_{qy}(f) \end{bmatrix} \quad (6.149)$$

For the special case where the inputs are incoherent, the square matrix reduces to a diagonal matrix, whose inverse is such that Equation (6.149) becomes a set of independent equations, as given by Equation (6.120). Hence Equation (6.149) represents the general form of the relationship needed to compute frequency response functions from measured input output data.

The frequency response function estimates provided by Equation (6.149) will generally involve bias errors of the same type and from the same sources

as the bias errors discussed for the single input problem in Section 6.7.1. Furthermore, the estimates will involve a random error similar in form to the random error developed in Section 6.7.2. For the multiple input case, the random error expression, from Reference 20, can be evaluated by

$$|\hat{H}_i(f) - H_i(f)|^2 \leq \hat{r}_i^2(f) \quad (6.150)$$

where

$$\hat{r}_i^2(f) = \frac{2q}{n - 2q} (F_{n_1, n_2; \alpha}) \frac{[1 - \hat{\gamma}_{v \cdot x}^2(f)] \hat{G}_v(f)}{[1 - \hat{\gamma}_{i \cdot x}^2(f)] \hat{G}_i(f)} \quad (6.151)$$

q = number of inputs (excluding output)

$n = 2B_e T$ = number of degrees of freedom of each spectral estimate

$F_{n_1, n_2, \alpha}$ = 100 α percentage point of an F distribution with $n_1 = 2q$ and $n_2 = n - 2q$ degrees of freedom

$\hat{G}_i(f)$ = power spectrum estimate of the input $x_i(t)$

$\hat{G}_v(f)$ = power spectrum estimate of the output $y(t)$

$\hat{\gamma}_{v \cdot x}^2(f)$ = sample estimate of the multiple coherence function between the output $y(t)$ and all the measured inputs

$\hat{\gamma}_{i \cdot x}^2(f)$ = sample estimate of the multiple coherence function between the input $x_i(t)$ and the other measured inputs excluding $x_i(t)$

Hence, the $(1 - \alpha)$ confidence intervals for the gain factor $|H_i(f)|$ and the phase factor $\phi_i(f)$ are given for each frequency f and each path $i = 1, 2, 3, \dots, q$, by

$$\left[|\hat{H}_i(f)| - \hat{r}_i(f) \leq |H_i(f)| \leq |\hat{H}_i(f)| + \hat{r}_i(f) \right] \quad (6.152)$$

$$\left[\hat{\phi}_i(f) - \Delta \hat{\phi}_i(f) \leq \phi_i(f) \leq \hat{\phi}_i(f) + \Delta \hat{\phi}_i(f) \right]$$

where the phase error $\Delta \hat{\phi}_i(f)$ is calculated from

$$\Delta \hat{\phi}_i(f) = \sin^{-1} \left[\frac{\hat{r}_i(f)}{|\hat{H}_i(f)|} \right] \quad (6.153)$$

Observe that Equation (6.151) includes Equation (6.146) as a special case by setting $q = 1$ and $\hat{\gamma}_{i \cdot x}^2(f) = 0$.

A polar diagram for the confidence region represented by Equation (6.152) is shown in Figure 6.6 at frequency f_0 . Different regions apply to each estimated frequency response function $\hat{H}_i(f_0)$ for which different $\hat{r}_i(f_0)$ and $\Delta \hat{\phi}_i(f_0)$ are calculated.

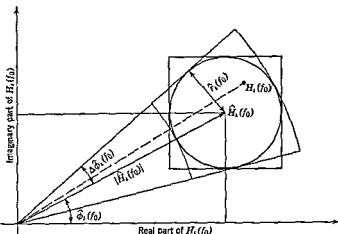


Figure 6.6 Confidence interval for frequency response function estimate

From Equations (6.151) through (6.153), the accuracy in estimating frequency response functions for multiple input linear system problems improves when

- 1 The number of degrees of freedom n increases for a given value of q
- 2 The multiple coherence estimate $\hat{\gamma}_{y,x}^2(f)$ approaches unity
- 3 The multiple coherence estimate $\hat{\gamma}_{i,x}^2(f)$ approaches zero
- 4 The input power spectral density estimate $G_i(f)$ increases
- 5 The output power spectral density estimate $\hat{G}_y(f)$ decreases

These five points concur with intuitive expectations. As n increases, the accuracy of all sample values in Equation (6.151) increases. Hence, one would anticipate a more accurate estimate of $H_i(f)$. A large value of multiple coherence between the output and all measured inputs suggests that the multiple input model is well defined with little extraneous noise. The interpretation here is the same as for a large value of ordinary coherence in the single input problem discussed in Section 6.7.2. A small value of multiple coherence between the input of interest and all other inputs suggests low redundancy in the data for this path and, hence, greater statistical accuracy in the resulting estimate. A high ratio of spectral density for the input of interest relative to the output indicates that this input is a more significant contributor to the output than other inputs. Thus, the signal to noise ratio

for that path would be superior, producing greater accuracy in the resulting estimate.

Example 6.5. Confidence Intervals for Gain and Phase Factor Estimates (Multiple Inputs). Consider an eight input-single output system. From Equation (6.149), assume a phase factor of 1.94 radians is estimated at a given frequency f_0 between an input $x_i(t)$ and the output $y(t)$, and the corresponding gain factor estimate is 3.20. Suppose the number of degrees of freedom $n = 100$ and the sample estimates $\hat{G}_i(f_0) = 0.11$, $\hat{G}_y(f_0) = 0.82$, $\hat{\gamma}_{y,x}^2(f_0) = 0.90$, and $\hat{\gamma}_{i,x}^2(f_0) = 0.85$. Determine a 95 percent confidence interval for the gain and phase factor at this frequency, assuming the bias error is negligible.

For this example, $q = 8$, $n_1 = 16$, $n_2 = 84$, and $\alpha = 0.05$. From Table A.5(a),

$$F_{16,84,0;05} = 1.77$$

It follows from Equation (6.151) that

$$\hat{r}_i^2(f_0) = \left(\frac{16}{84}\right)(1.77)\left(\frac{0.10}{0.15}\right)\left(\frac{0.82}{0.11}\right) = 1.67$$

and

$$\hat{r}_i(f_0) = 1.29$$

Then, from Equation (6.153),

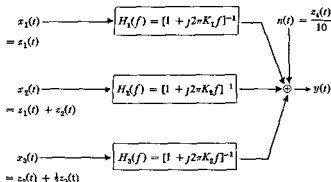
$$\Delta \hat{\phi}_i(f) = \sin^{-1} \left(\frac{1.29}{3.20} \right) = 0.42 \text{ radian}$$

Thus 95 percent confidence intervals for the gain and phase factors are given by Equation (6.152) as

$$\begin{bmatrix} 1.91 \leq |H(f_0)| \leq 4.49 \\ 1.52 \leq \phi(f_0) \leq 2.36 \end{bmatrix}$$

where the phase is measured in radians. This example illustrates how significant the errors in frequency response function estimates can be when there is relatively large coherence between the input of concern and all other inputs.

Example 6.6. Illustration of Frequency Response Function Estimate. Consider the three input-single output system previously discussed in Example 5.7 and repeated here as Figure 6.7. Note that input $x_2(t)$ is correlated with the other two inputs and that measurement noise $n(t)$ is present in the output $y(t)$. The three paths for the system are all ideal RC low pass filters.



$z_1(t)$, $z_2(t)$, $z_3(t)$, and $z_4(t)$ are four independent normally distributed random time histories with means of zero and variances of unity

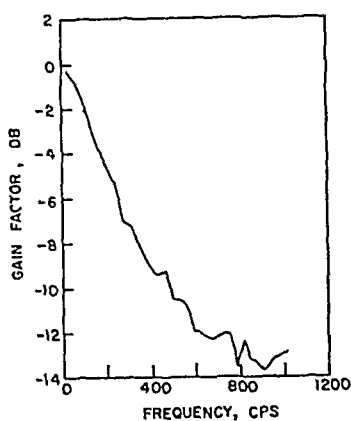
Figure 6.7 Three input linear system with output measurement noise

The frequency response function of the path between $x_2(t)$ and $y(t)$ was calculated using Equation (6.149) to obtain the results shown in Figure 6.8(a). The number of degrees of freedom for the estimate is $n = 64$ as detailed in Example 5.7, and the multiple coherence function varies from near unity to about 0.95 with increasing frequency as discussed in Example 5.8. The random error in the gain and phase factor estimates is clearly revealed by the random fluctuations in the plots.

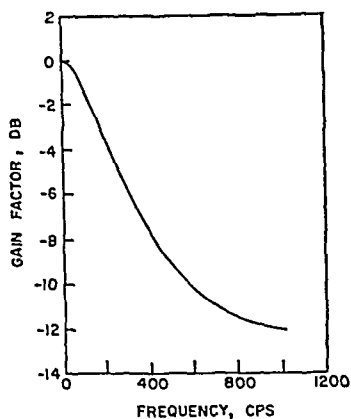
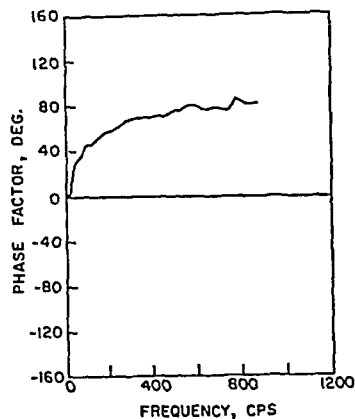
Now, Equation (6.151) suggests that the random error would be zero if the multiple coherence function between the output and all inputs were unity. This can be achieved in the system described in Figure 6.7 by setting the output measurement noise $n(t) = 0$. This was done and the frequency response function between $x_2(t)$ and $y(t)$ was recomputed with the results shown in Figure 6.8(b). Note that the plots display no random fluctuations indicative of random estimation error. Furthermore, the results concur with the theoretical frequency response function of the actual RC filter used in the system between $x_2(t)$ and $y(t)$.

6.9 RECORD LENGTH REQUIREMENTS

The error expressions derived in Sections 6.2 through 6.8 provide a basis for assessing the statistical accuracy of various parameter estimates after an experiment has been completed. It would be desirable if these error expressions could also be used to predict the accuracy of parameter estimates



(a)



(b)

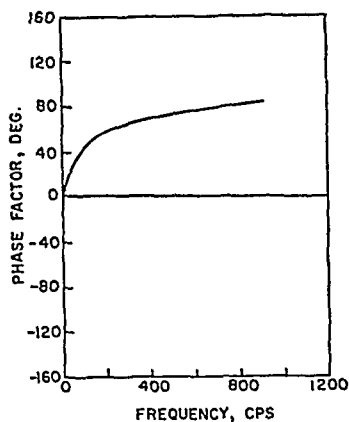


Figure 6.8 Gain and phase factor estimates between input $x_0(t)$ and output for system in Figure 6.7. (a) Gain and phase estimates with output measurement noise. (b) Gain and phase estimates with no output measurement noise.

for future experimental data—in particular, to establish the record length required to obtain a predetermined degree of accuracy. The error formulas do relate the record length T to an error for each parameter estimate. These relationships, however, generally include other factors which are unknown prior to data collection. Hence, it is not feasible to use them directly as a basis for record length selections. Certain assumptions based upon a prior

knowledge of the data properties and/or engineering judgement will also be required

For example, the variance of a mean square value estimate for Gaussian data is related to the record length T in terms of the covariance function and mean value of the data, as detailed in Equation (6.31). If one were prepared to assume that all data of interest were bandwidth limited Gaussian white noise with a bandwidth of B and a mean value of zero, then the record length required for a specified normalized rms error ϵ would be given by Equation (6.35) as $T = (B\epsilon^2)^{-1}$. Unfortunately, such an assumption is rarely justified in practice even as a first order approximation.

There is one parameter estimate where reasonable assumptions often can be made which provide a usable relationship between record length and error, namely, a power spectral density estimate. From Equation (6.98), it is seen that the random portion of the normalized rms error of power spectra estimates is a function only of the record length T and the resolution bandwidth B_r (assuming the spectral density is relatively smooth within B_r and the data is Gaussian). This implies that the record length needed to provide a specified normalized random error ϵ_r in power spectral density estimates can be predicted by $T = (B_r\epsilon_r^2)^{-1}$ where B_r is a known parameter of the analysis procedure rather than an unknown parameter of the data. The remaining problem then is to select an appropriate value for B_r .

Again referring to Equation (6.98), the resolution bandwidth B_r is a primary factor in the bias portion of the spectral density estimation error. Also important in the determination of bias, however, is $G_x''(f)$ which defines the "peakedness" of the power spectrum within the bandwidth B_r . Hence a limiting value for $G_x''(f)$ must be anticipated to relate the bias error ϵ_b to the bandwidth B_r . This is usually accomplished by an evaluation of the physics of the problem in question. For example, consider the special case of data which represent the response of a resonant physical system where the behavior of each resonance can be approximated by a simple second order system as described by Equation (2.19). If the spectrum of the system excitation is assumed to be approximately uniform over the bandwidth of any given resonance, then the response power spectrum near a resonance can be anticipated to have the form $G_x(f) = c |H(f)|^2$ where c is a constant and $H(f)$ is given in mechanical units by Equation (2.24). A bias error can now be calculated for spectral measurements at resonance frequencies in terms of a ratio of the resolution bandwidth B_r to the half power point bandwidth B_p of $H(f)$. One evaluation of this type presented in Reference 18 leads to the results in Figure 6.9. Note here that the bias error in spectral density estimates always is in the direction of reduced dynamic range, that is, densities of spectral peaks are underestimated and densities of spectral valleys are overestimated.

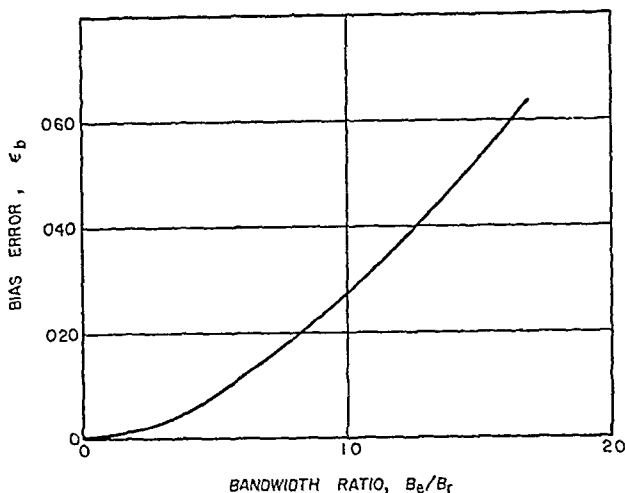


Figure 6.9 Bias error for second order system response.

Example 6.7 Record Length Selection for Power Spectrum Estimate. Suppose the power spectrum of the random vibration response of a structure is to be estimated with a maximum random error of $\epsilon_r = 0.1$ and a maximum bias error of $\epsilon_b = 0.1$. Assume that past experience and/or theoretical studies indicate the structure will have no resonant frequency below $f_r = 20$ cps and the damping ratio for resonances will be about $\zeta = 0.05$. How long a sample record of data should be collected?

From Equation (2.24a), the half power point bandwidth of a system resonance may be approximated by $B_r \approx 2\zeta f_r = 0.1f_r$. For the worst case where $f_r = 20$ cps, $B_r = 2$ cps. Then from Figure 6.9, a resolution bandwidth of $B_e \approx B_r/2 = 1$ cps will be required to provide a bias error of $\epsilon_b \leq 0.1$. Hence if the random error is also to be limited to $\epsilon_r \leq 0.1$, the required record length is $T = (B_e \epsilon_r^2)^{-1} = [(1.0)(0.01)]^{-1} = 100$ seconds. This concludes Example 6.7.

The foregoing discussions of record length selections for power spectral densities function estimates are particularly important for two reasons. First, the power spectrum is the single most important parameter of random data for many engineering applications. Second, power spectra estimates are generally the most demanding of the various parameter estimates considered in this chapter from the viewpoint of required record length for a given error. This latter fact is easily verified by comparing the error expression for power spectra estimates, as given by Equation (6.98), to the error expressions for other parameter estimates. It is seen that the denominator of the error terms

for other parameters estimates generally involve factors which are larger than the value B_e required for well resolved power spectra estimates. Even for frequency response function estimates, if the assumed linear model is relatively efficient (if the multiple coherence function between the inputs and the output is relatively high), the frequency response function estimate will often have less random error than the spectral density estimates used for its calculation. Hence, the record length requirements for suitably accurate power spectra estimates will usually be adequate for other parameter estimates.

PROBLEMS

1. Assume $x(t)$ is a record from a bandwidth limited Gaussian white noise process where $\mu_x = 0.10$, $\sigma_x = 0.20$, $C_x(0) = 1$, $B = 200$ cps and $T = 25$ seconds. Compute the normalized rms error for
 - (a) mean value estimates
 - (b) mean square value estimates
2. Consider the problem of estimating mean square values of general zero mean stationary random processes where the covariance function $C_x(\tau)$ is known. State how to determine the bandwidth B of an equivalent bandwidth limited Gaussian white noise process which gives the same normalized mean square error as would be obtained using $C_x(\tau)$.
3. Assume $x(t)$ is a record from a Gaussian ergodic process where $\mu_x = 0$, $\sigma_x = 1.2$, $B = 1000$ cps, $T = 10$ seconds.
 - (a) Compute the approximate normalized bias error of an estimate $\hat{\rho}(x)$ at $x = 0.5$ if the window width W satisfies $W = 0.2 \sigma_x$.
 - (b) Compute a conservative value for the normalized random error of the estimate $\hat{\rho}(x)$ at $x = 0.5$.
 - (c) What is the total normalized rms error here?
4. Consider the cross-correlation of two bandwidth limited white noise records $x(t)$ and $y(t)$ of bandwidth B and length T where

$$x(t) = s(t) + n_1(t) \quad \text{and} \quad y(t) = s(t) + n_2(t)$$

Assume that $s(t)$, $n_1(t)$, and $n_2(t)$ are mutually uncorrelated with zero mean values and mean square values $\Psi_s^2 = S$, $\Psi_{n_1}^2 = N_1$, and $\Psi_{n_2}^2 = N_2$.

- (a) Compute the normalized rms error of an estimate $\hat{R}_{xy}(\tau)$ at $\tau = 0$.
 - (b) Determine the result in (a) if $N_1 \gg S$ and $N_2 \gg S$.
 - (c) Determine the result in (a) if $N_1 = 0$ and $N_2 \gg S$.
 - (d) Discuss the physical significance of (b) versus (c).
5. Indicate how to choose B_e as a function of T so that the normalized mean square error of power spectral density estimates, as given by Equation

(6.98), will approach zero as T approaches infinity, independent of the nature of $G_x(f)$.

6. The quantity $\lambda(f) = |G_x(f)/G_x''(f)|^{1/2}$ is known as the "spectral bandwidth" of the random process at frequency f . Suppose $\lambda_{\max} =$ maximum value of $\lambda(f)$. Determine the requirements on B_e and T as a function of $\lambda_{\max}$ if the normalized bias error and the normalized random error of power spectral density estimates are each to be limited to at most 5 percent.
7. Consider a power spectral density function estimate $\hat{G}(f)$ computed from finite Fourier transform operations and the ensemble smoothing procedure illustrated in Figure 6.2(a). Suppose a record $x(t)$ of $T = 60$ seconds is divided into 12 independent contiguous parts. Compute the 95 percent confidence interval for the true value $G(f)$ when $\hat{G}(f) = 0.30$ units²/cps. What is the normalized standard error of this estimate?
8. Determine the overall bias error as per Equations (6.122) and (6.123) in a frequency response function estimate for a single input case when the measured input consists of an input signal $u(t)$ which passes through the system, corrupted by extraneous measurement noise $n(t)$ which does not pass through the system; and the measured output $y(t)$ includes a contribution from an unmeasured second input $v(t)$ which is correlated with $u(t)$ but not with $n(t)$.
9. Consider a single input-single output system where the measured coherence function $\hat{\gamma}_{xy}^2(f_0) = 0.75$ at frequency f_0 between an input $x(t)$ and an output $y(t)$. Suppose that the measured spectra are $\hat{G}_x(f_0) = 2.1$ and $\hat{G}_y(f_0) = 0.80$ for a certain number of degrees of freedom n . What should be the minimum value of n in order to have 95 percent confidence that the true gain factor will be within ± 10 percent of the measured gain factor, assuming the bias error is negligible.
10. Consider a two input-single output system where the multiple coherence function estimate $\hat{\gamma}_{y,x}^2(f_0) = 0.70$ at frequency f_0 between the output $y(t)$ and the two measured inputs $x_1(t)$ and $x_2(t)$. Suppose that $\hat{\gamma}_{1,2}^2(f_0) = 0.20$ between the two inputs, and that $\hat{G}_1(f_0) = 20$, $\hat{G}_2(f_0) = 30$, $\hat{G}_y(f_0) = 40$, $\hat{G}_{12}(f_0) = 15$, and $\hat{G}_{1y}(f_0) = \hat{G}_{2y}(f_0) = 16 - 12j$, where all spectral estimates are obtained using $T = 13$ seconds and $B_e = 8$ cps.
 - (a) Compute sample estimates of the gain and phase factor at f_0 between $x_2(t)$ and $y(t)$.
 - (b) Compute the 95 percent confidence interval for the gain and phase factor associated with the above estimates, assuming the bias error is negligible.

CHAPTER 7

GENERAL CONSIDERATIONS IN DATA ACQUISITION AND PROCESSING

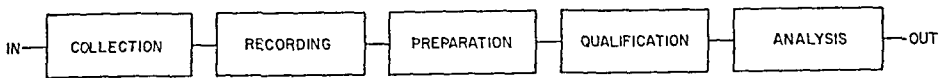
Appropriate techniques for the acquisition and processing of random data are heavily dependent upon the physical phenomenon represented by the data and the desired engineering goals of the processing. In broad terms, however, the required operations may be divided into five primary categories as follows:

- a Data collection
- b Data recording (including transmission)
- c Data preparation
- d Data qualification
- e Data analysis

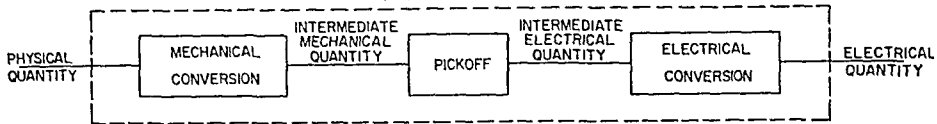
Each of these categories involves a number of sequential steps as schematically illustrated in Figure 7.1. The purpose of this chapter is to summarize basic considerations associated with each of these key steps. The emphasis throughout is on potential sources of error over and above the statistical errors inherent in sampling considerations, as developed in Chapter 6.

7.1 DATA COLLECTION

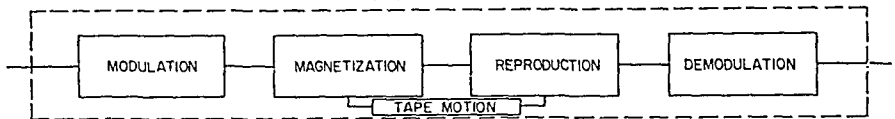
The primary element in data collection is the instrumentation transducer. In general terms, a transducer is any device which translates power from one form to another. In an engineering context, this usually means the translation



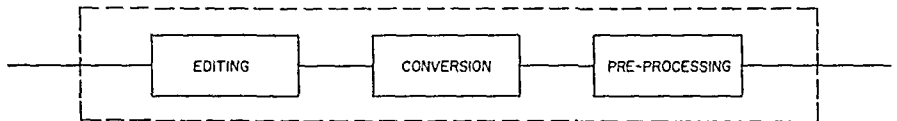
(a) COLLECTION



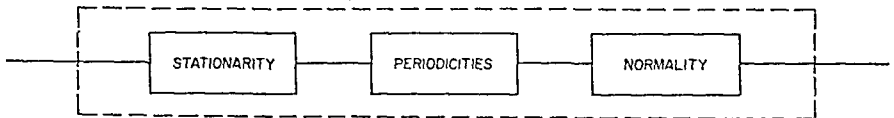
(b) RECORDING



(c) PREPARATION



(d) QUALIFICATION



(e) ANALYSIS

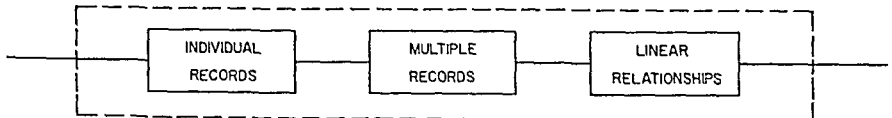


Figure 7.1 Key steps in data acquisition and processing.

of some measure of a physical phenomenon of interest into an analog signal with a calibrated relationship between the input and output quantities. Referring to Figure 7 1(a), this translation may involve up to three basic operations, (a) a mechanical conversion of the physical quantity of interest into an intermediate mechanical quantity, (b) a pickoff step which converts the intermediate mechanical quantity into an intermediate electrical quantity, and (c) an electrical conversion into a final electrical quantity, usually voltage. Some transducers may combine any two or all three of the operations, depending upon the physical quantity being measured and the specific nature of the transducer.

For example, a thermocouple, which is widely used as a temperature transducer, converts a difference in temperature directly to a difference in voltage without intermediate steps. On the other hand, a resistance thermometer, another commonly used temperature transducer, first converts temperature to a change in resistance, and then converts the resistance change to an electrical voltage change. For this case, there is a two-step operation which bypasses the conversion to an intermediate mechanical quantity. Seismic type motion transducers, which are often used to measure displacements, velocities, and accelerations, involve all three steps indicated in Figure 7 1(a). The input motion is converted to a relative displacement, which in turn is converted through a pick-off device to an intermediate electrical quantity such as change in resistance. Finally, the change in resistance is converted to an electrical voltage proportional to the input.

Ideally, the above operations would be accomplished without distortion or modification of the time history of the physical quantity being measured. In other words, if the input time history is $x(t)$ and the output time history is $y(t)$, the perfect transducer would provide an analog output, $y(t) = c x(t)$, where c is a simple calibration constant. Unfortunately, this ideal situation is difficult to achieve in practice. Gain and phase modifications as well as distortion producing nonlinearities are often inherent in the transducer operations. This fact makes the transducer a potential source of error in any data acquisition and processing program.

Example 7.1. Open Loop Seismic Accelerometer. Consider an idealized open loop seismic accelerometer, as schematically illustrated in Figure 7 2(a). The mechanical conversion in this transducer is achieved through a spring-dashpot supported seismic mass, where an input acceleration at the foundation is converted to a relative displacement of the mass. The pick off device is a potentiometer. The final electrical conversion is achieved by measuring the voltage across the potentiometer. Using the procedures outlined in Section 2 4 1, it is readily shown that the frequency response function

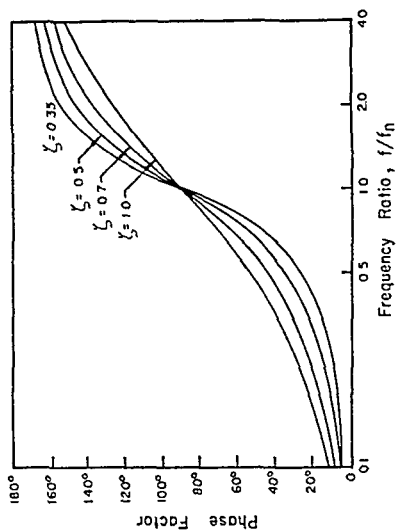
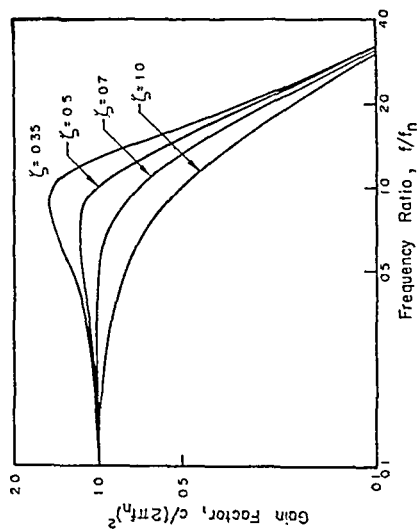
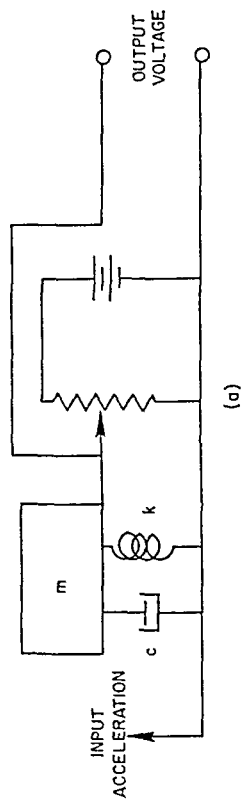


Figure 7.2 Schematic diagram and frequency response characteristics of open loop seismic accelerometer. (a) Schematic diagram of idealized accelerometer. (b) Accelerometer gain factor. (c) Accelerometer phase factor.

of this system between the input acceleration and the output voltage is given by

$$H(f) = \frac{c}{4\pi^2 f_n^2 [1 - (f/f_n)^2 + j2\zeta f/f_n]} \quad (7.1)$$

where c is a calibration constant, and ζ and f_n are the damping ratio and undamped natural frequency, respectively, as defined in Equation (2.22). The gain and phase factors of this system are plotted in Figures 7.2(b) and (c). Note that these gain and phase factors are identical in form to those for a simple mechanical system with force excitation, as plotted in Figure 2.3.

Consider first the gain factor given in Figure 7.2(b). It is seen that the accelerometer provides a relatively uniform gain for frequencies well below the undamped natural frequency of the transducer. For frequencies approaching the undamped natural frequency, the gain factor tends to peak and/or diminish depending upon the damping ratio. The widest frequency range with uniform gain is provided by a damping ratio of approximately $\zeta = 0.7$. The characteristics illustrated in Figure 7.2(b) limit the upper frequency range of seismic accelerometers to something less than the undamped natural frequency, depending upon the amount of error (deviation from a constant gain factor) which can be accepted, as summarized in Figure 7.3. The errors indicated for various conditions in Figure 7.3 might be either corrected by calibration or simply tolerated. Either way, it is essential that they be acknowledged to avoid unrecognized and perhaps serious magnitude errors in the final processed data.

Now consider the phase factor given in Figure 7.2(c). The accelerometer introduces a phase shift in the data which varies from zero degrees at zero frequency to 180 degrees at very high frequencies. The exact details of the phase shift are a function of the damping ratio. The important point here is that this phase shift corresponds to a time delay at any given frequency and, hence, can lead to phase distortion in the resulting data. Such distortion may cause errors in the analysis of single channel data of the complex periodic and transient form as well as multiple channel data of any type where joint properties are of interest. The single channel errors can be minimized by selecting a transducer with a phase factor $\phi(f)$ such that $\phi(f)/f = \text{constant}$. For the seismic accelerometer being considered, this will be provided by a damping ratio of either $\zeta \approx 0.7$ or $\zeta \approx 0$. The cross channel errors can be controlled by using transducers with matched phase factors for the acquisition of multiple channel data. This concludes Example 7.1.

In summary, considerable care is warranted in the selection and use of data acquisition transducers. When such transducers are procured from commercial sources, supporting literature is usually provided which specifies

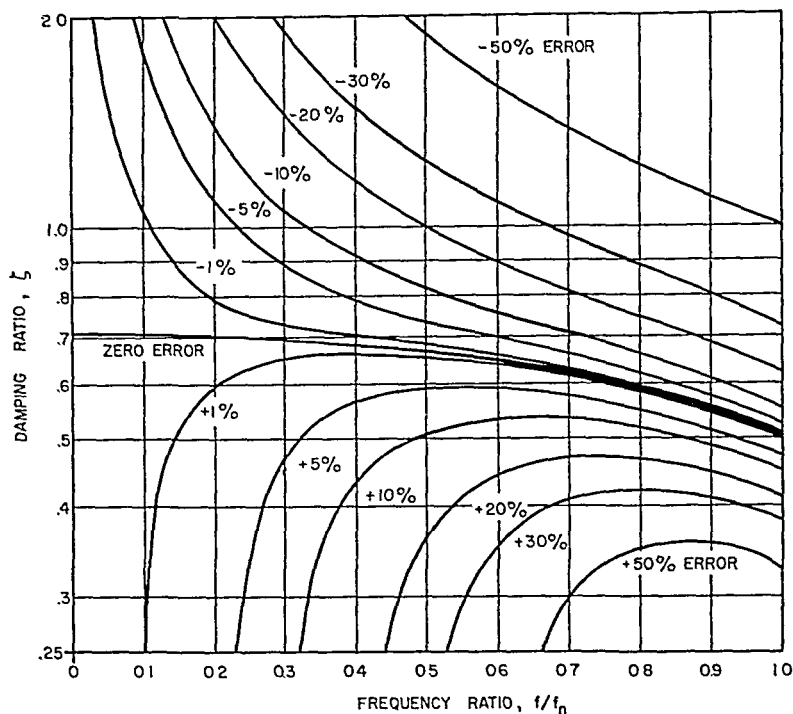


Figure 7.3 Frequency response gain errors for open loop seismic accelerometers.

the limitations on their use. These specifications are generally accurate, but it must be remembered that commercial manufacturers are not inclined to be pessimistic in the claims they make for their products. It is unwise to use a commercially procured transducer under conditions which exceed the limits of the manufacturers specifications unless their applicability to such use has been substantiated by appropriate studies. See References 14 and 35 for more detailed discussions of data collection and transducer systems.

7.2 DATA RECORDING

For some applications, it is possible to perform all desired data processing directly on the transducer signals in real time. For most applications, however, this is not practical and some form of storage (and perhaps remote transmission) of the transducer signals will be required. The most desirable and

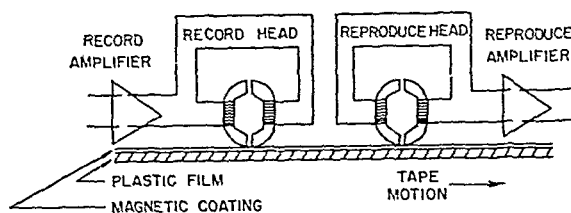
convenient type of data storage system is the magnetic tape recorder. Although other types of recorders could be used, the magnetic tape recorder has the advantage of being able to store large quantities of data and to reproduce them in electrical form. The most desirable way to transmit data signals is through electrical lines. There are obvious situations where this is not feasible, for example, retrieving data from a spacecraft in earth orbit. For such cases, radio transmission (telemetry) of the transducer signals is usually required.

Both magnetic tape recording and radio telemetry are major subjects within themselves and generally beyond the scope of this book. Basic material on these subjects is available from References 12, 22, 34, and 36. There are two phases of the subjects, however, which should be discussed since they directly relate to fundamental problems in data processing. As noted in Figure 7-1(b), these two phases are the magnetization reproduction operation in magnetic tape recording and the modulation-demodulation operation in both magnetic tape recording and telemetry.

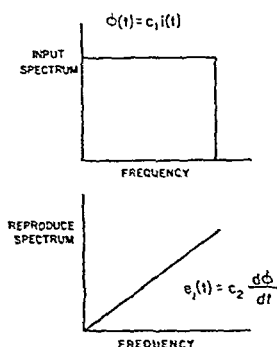
7.2.1 *Magnetization Reproduce Procedures*

The configuration for recording and reproducing information on magnetic tape for most commercially available recorders is as indicated in Figure 7-4(a). A moving plastic film with a magnetic coating slides over a record head. When an electrical current is passed through the record head, it generates a magnetic flux which magnetizes the particles on the tape. As the tape moves by the reproduce head, a signal is generated proportional to the rate of change of flux.

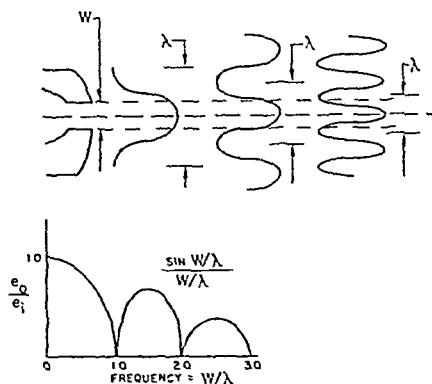
This type of record-reproduce system poses a number of troublesome problems, including the basic nonlinearity of the magnetization process, tape dropouts, and other side effects introduced by design compromises in the magnetic tape. From the viewpoint of basic recorder capabilities, however, the two most pertinent problems are as follows. First, the reproduce head responds to rate of change of flux on the magnetic tape and, hence, extracts a differentiated version of the input signal rather than a faithful reproduction of it. Second, the reproduce head averages the change of flux across the head gap. The first noted problem tends to limit the lower frequency response capabilities of the recorder as illustrated in Figure 7-4(b), while the second problem tends to limit the high frequency capabilities of the recorder as illustrated in Figure 7-4(c). Note that the high frequency response of the recorder can be improved by reducing the gap width, but this also reduces the sensitivity of the reproduce head. The gap depth may be reduced to bring the sensitivity back up, but this reduces head life. The sensitivity may also be increased by adding windings to the reproduce head, but this also attenuates the high frequency response.



(a)

 $\phi(t) = \text{MAGNETIC FLUX TIME HISTORY}$
 $e(t) = \text{VOLTAGE TIME HISTORY}$
 $i(t) = \text{CURRENT TIME HISTORY}$
 $c_1, c_2 = \text{CONSTANTS}$
 $W = \text{GAP WIDTH}$
 $\lambda = \text{WAVE LENGTH}$
 $e_o = \text{ACTUAL OUTPUT VOLTAGE}$
 $e_i = \text{IDEAL OUTPUT VOLTAGE}$


(b)



(c)

Figure 7.4 Basic principles and properties of magnetic tape recording. (a) Operating principles. (b) Recording frequency response. (c) Reproduce gap attenuation.

Commercial recorders counteract the above two problems by using equalizer networks to produce a resultant output with a uniform frequency response function over the widest possible frequency range. In quantitative terms, commercially available machines with a tape speed of 120 inches per second will provide a frequency response within ± 3 dB over a frequency range from four hundred to two million cps. The resulting signal to noise ratio, however, is only about 28 dB. Such record-reproduce capabilities are unsatisfactory for most instrumentation purposes for two reasons; first, the recorder is not capable of DC response and, second, the recorder has an inadequate signal to noise ratio. On the other hand, the recorder does provide

a generous frequency bandwidth. The problem then is to exchange bandwidth for the desirable features of DC response and improved signal to noise ratio. This is achieved by the modulation-demodulation procedure.

7.2.2 Modulation-Demodulation Procedures

The four most common modulation-demodulation procedures used for scientific data transmission and/or recording are frequency modulation, pulse amplitude modulation, pulse duration modulation, and pulse code modulation.

Frequency modulation (FM) is the most widely used modulation procedure for analog telemetry and tape recording. The basic approach in FM is to make the frequency of a carrier signal analogous to the amplitude of the input data signal, as indicated in Figure 7.5(a). With zero input, the modulated signal is simply the carrier frequency itself. With a full scale DC (static) input, the modulated signal is a pure tone at the upper deviation frequency. With a sinusoidal input, the modulated signal theoretically looks like a sine wave which varies in frequency. For the case of tape recording, the modulated signal actually looks more like a square wave with varying width because FM signals are usually recorded at saturation. An important parameter in FM is the ratio of the deviation of the carrier frequency (Δf) to the modulating data frequency (f_d). This ratio is called the *modulation index* (m_f), and is closely related to both the bandwidth capabilities and the output signal to noise ratio provided by FM transmission and recording. These relationships are summarized in Table 7.1, along with the basic advantages and disadvantages of FM.

Referring to Table 7.1, it is seen that two sets of bandwidth and signal to noise ratio formulas are presented for FM, one for $m_f > 5$ and the other for $m_f < 0.6$. These represent bounding values of the modulation index where theoretical results for bandwidth and signal to noise relationships can be obtained, as developed in Reference 22. In practice, the modulation index is often between these bounding values, for example, $m_f = 2.16$ is common for wideband FM tape recording. It is interesting to note that the signal to noise ratio provided by FM with $m_f = 2.16$ is generally about 6 dB greater than indicated by either of the bounding values in Table 7.1. This added improvement is probably due to the hard clipping of FM signals caused by saturation recording.

The other three modulation procedures involve sampling of the data signal wave form at discrete time intervals. In the case of pulse amplitude modulation (PAM), the sampling generates a train of pulses with amplitudes proportional to the signal level, as illustrated in Figure 7.5(b). Pulse duration modulation (PDM) consists of making the width of equal amplitude pulses analogous to the amplitude of the data signal. In simplified terms, this is

achieved by adding a triangular pulse wave form to the data signal and then clipping the composite signal, as indicated in Figure 7.5(c). The widths of the resulting pulses are proportional to the amplitude of the data signal at that time. In pulse code modulation (PCM), the amplitude of the signal wave form is sampled and converted to a series of pulses which are coded, usually into binary digits, as illustrated in Figure 7.5(d). The resulting bandwidth and signal to noise ratio properties associated with each of these pulse modulation procedures, as well as their relative advantages and disadvantages, are summarized in Table 7.1.

In practice, it often occurs that the number of transducer signals exceeds the number of transmission and/or recorder channels available. Hence a number of transducer signals must sometimes share a single transmission and/or recording channel. This is accomplished by some form of signal commutation or multiplexing. For those types of modulation which involve sampling, multiplexing is straightforward since coded data from several signals can easily be mixed onto one recorder channel. For frequency modulation, multiplexing must be accomplished by having the different data signals share the available frequency bandwidth of the telemetry or tape recorder channel. It is indicated in Table 7.1 that PAM and PCM are the most desirable forms of modulation from the viewpoint of multiplexing convenience. In all cases, however, the number of multiplexed channels influences the transmission bandwidth requirements.

7.2.3 *Time Base Errors in Tape Recording*

Beyond the limitations inherent in the magnetization-reproduce and modulation-demodulation operations, there are a number of miscellaneous problems in magnetic tape recording which must be considered. Among the more important of these problems are the errors associated with variations in the speed at which the tape passes over the record and/or reproduce heads. These are called time base errors, and may be either static or dynamic in character. Furthermore, they may affect either single-channel or multi-channel analyses.

The most commonly discussed single-channel dynamic time base error is flutter, which may be defined as a variation in tape velocity from the nominal. Flutter is particularly important in the case of frequency modulation where the time base or frequency is analogous to amplitude. The integral of flutter is the dynamic time base error itself. The magnitude of flutter for high quality commercial tape recorders is on the order of 0.25 percent. In general, the spectrum of the flutter is relatively flat out to some cutoff frequency, usually about 1000 cps for the higher tape speeds.

The most disturbing inter-channel dynamic time base error is dynamic skew, which may be defined as a variation in the angle at which the tape

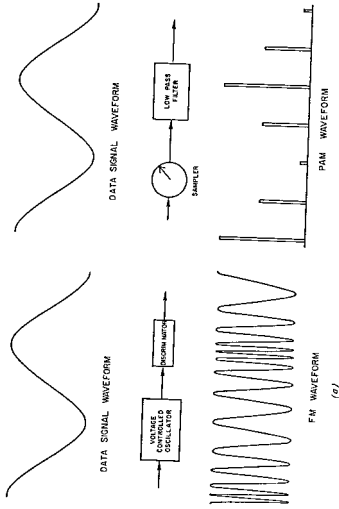
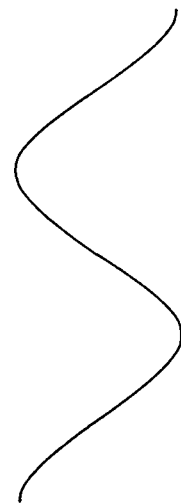
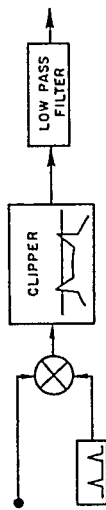


Figure 7.5 Basic principles of data modulation procedures (a) Frequency modulation (FM) (b) Pulse amplitude modulation (PAM)

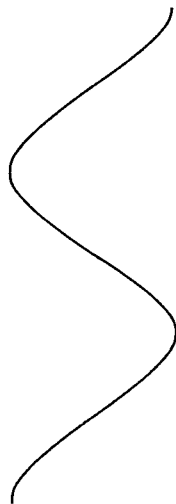


DATA SIGNAL WAVEFORM

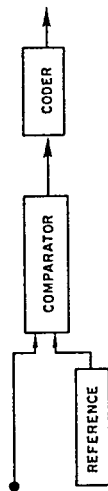


PDM WAVEFORM

(c)



DATA SIGNAL WAVEFORM



PCM WAVEFORM

(d)

Figure 7.5 (c) Pulse duration modulation (PDM). (d) Pulse code modulation (PCM).

Table 7.1

Basic Characteristics of Modulation Procedures

	Frequency Modulation	Pulse Amplitude Modulation	Pulse Duration Modulation	Pulse Code Modulation
Bandwidth Requirements	for $m_f > 5$ $B_r \approx 2m_f f_d$ for $m_f < 0.6$ $B_r \approx 2f_d$	$B_r = 2cf_d$	$B_r = 1/t_r$	$B_r = ncf_d$
Signal to Noise Relationships	for $m_f > 5$ $S_d/N_d = 3m_f^2 S/N$ for $m_f < 0.6$ $S_d/N_d = 3m_f^2 S/N$	$S_d/N_d = 2S/N_i$	$\frac{S_d}{N_d} = \left(\frac{2t_0^2 B_r^2}{f_d} \right) \frac{S}{N_i}$	$S_d/N_d = 2^n - 1$
Advantages	DC response good S/N improvement	DC response multiplexing convenience	DC response moderate S/N improvement	DC response very good S/N improvement multiplexing convenience
Disadvantages	tape speed sensitivity reduced data bandwidth	poor S/N improvement	tape speed sensitivity reduced data bandwidth	reduced data bandwidth

 m_f = modulation index c = number of multiplexed channels f_d = maximum frequency of data signal in cps n = number of pulses per code group in PCM B_r = required recorder (transmission) bandwidth in cps S/N = direct record signal to noise ratio t_r = rise time for PDM pulse in sec S_d/N_d = modulated signal to noise ratio t_0 = maximum time duration for PDM pulse in sec

passes over the head. This phenomenon clearly will affect inter channel analyses such as cross correlation and cross spectra. Dynamic skew problems can be minimized by recording data intended for joint analysis on adjacent record heads when feasible. For the case of data which will be analyzed digitally, further improvement can be obtained by mixing a clock pulse with the data signal on each channel, and then using this pulse to trigger the analog to digital sampling of the signal on each track.

In addition to fixed deviations in the tape speed, single channel static time base errors can be introduced by the tape itself. Temperature, humidity, and tension can change the dimensions of the tape and therefore the static time base. In general, with a high quality commercial tape recorder using 1 or 1.5 mil tape, the static time base error will be on the order of 0.1 to 0.25 percent.

Inter-channel static time base errors are generally due to manufacturing tolerances rather than tape speed variations. These tolerances include the angle at which the tape passes the heads (static skew) and the position of the heads (tolerances from stack to stack as well as from head to head on a given stack, and the azimuth alignment). An important point here is that the record and reproduce channels on most commercial tape recorders are each divided between two head stacks, that is, the odd numbered record tracks and the even numbered record tracks are on different heads. Therefore, closer tolerances from channel to channel can be maintained between tracks on the same head than between tracks on different heads. Inter-channel static time base errors are important when the characteristics of two or more channels are to be analyzed. The magnitude of the inter-channel time base errors depends on the particular equipment, but is generally on the order of 0.001 inch. The error in terms of time is therefore a function of the tape speed.

In conclusion, the selection and use of tape recording equipment for data recording warrants the same care suggested for data acquisition transducers in the final remarks of Section 7.1. See References 12 and 34 for more detailed discussions of magnetic tape recording principles and equipment.

7.3 DATA PREPARATION

The next key phase in data acquisition and processing is the preparation of raw data for detailed analysis. The raw data are usually supplied from a tape recorder in the form of voltage time histories (or as direct analog signals from the transducers, if the data are being analyzed on-line). A number of operations are needed at this point to make the voltage time histories suitable for detailed analysis. Referring back to Figure 7.1, the first of these operations is generally classified as data editing.

Data editing refers to those pre-analysis procedures which are designed to detect and eliminate spurious and/or degraded data signals which might have resulted from acquisition and recording problems such as excessive noise, signal dropout, loss of signal due to transducer malfunctions, etcetera. Editing can often be accomplished through visual inspection of the data time history signals by a talented analyst. In more elaborate data acquisition and processing systems, a specific instrument for quick-look evaluation might be employed. Real-time spectrum analyzers, to be discussed in Chapter 8, are popular for this application. It should be noted that the data editing step is more critical for the case of digital processing than for analog processing. This is true because once the data have been converted to a digital format, it is often difficult to detect even the most obvious errors in the original signal.

For the case of analog data processing, data preparation beyond editing usually includes only conversion into engineering units (calibration), and

perhaps the formation of tape loops to permit the data to be recirculated for continuous display to an analog analyzer. For the case of digital data analysis, a number of additional steps are required in the data preparation phase. These digital data preparation procedures may be classified as digitizing procedures and preprocessing procedures.

7.3.1 Digitization

Digitization consists of two separate and distinct operations, (a) sampling and (b) quantization. Sampling is the process of defining the instantaneous points at which the data are to be observed, while quantization is the conversion of data values at the sampling points into numerical form.

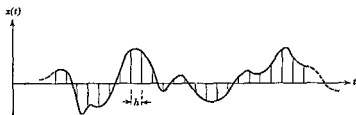


Figure 7.6 Sampling of a continuous record

SAMPLING Sampling for digital data analysis is usually performed at equally spaced intervals, as illustrated in Figure 7.6. The problem then is to determine an appropriate sampling interval h . On the one hand, sampling at points which are too close together will yield correlated and highly redundant data, and thus unnecessarily increase the labor and cost of calculations. On the other hand, sampling at points which are too far apart will lead to confusion between the low and high frequency components in the original data. This latter problem is called *aliasing*. It constitutes a potential source of error which does not arise in direct analog data processing, but is inherent in all digital processing which is preceded by an analog to digital conversion. Those who have viewed a motion picture of the classic western vintage have undoubtedly observed the apparent reversal in the direction of rotation of the stage coach wheels as the coach slows down or speeds up. That observation is a simple illustration of an aliasing error caused by the analog to digital conversion operation performed by a motion picture camera.

To be more explicit on this matter of aliasing, consider a continuous record which is sampled such that the time interval between sample values is

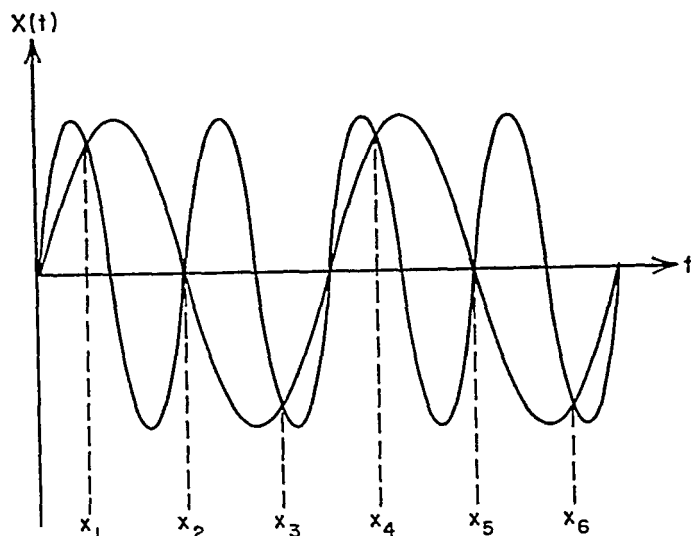


Figure 7.7 Illustration of aliasing problem.

h seconds, as shown in Figure 7.6. The sampling rate is then $1/h$ samples per second. However, at least two samples per cycle are required to define a frequency component in the original data. Hence, the highest frequency which can be defined by sampling at a rate of $1/h$ samples per second is $1/2h$ cps. Frequencies in the original data above $1/2h$ cps will be folded back into the frequency range from 0 to $1/2h$ cps, and be confused with data in this lower range, as illustrated in Figure 7.7. The cutoff frequency

$$f_c = \frac{1}{2h} \quad (7.2)$$

is called the *Nyquist frequency* or *folding frequency*. Actually, the folding of data frequencies into the frequency range from 0 to f_c occurs in an accordion-pleated fashion, as indicated in Figure 7.8. Specifically, for any frequency f in the range $0 \leq f \leq f_c$, the higher frequencies which are aliased with f are defined by

$$(2f_c \pm f), (4f_c \pm f), \dots, (2nf_c \pm f), \dots \quad (7.3)$$

To prove this fact, observe that for $t = 1/2f_c$,

$$\cos 2\pi f t = \cos 2\pi(2nf_c \pm f) \frac{1}{2f_c} = \cos \frac{\pi f}{f_c} \quad (7.4)$$

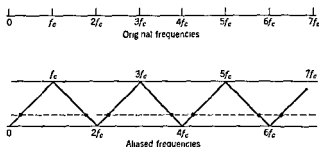


Figure 7.8 Folding about the Nyquist cutoff frequency f_c

Thus all data at frequencies $2nf_c \pm f$ have the same cosine function as data at frequency f when sampled at points $1/2f_c$ apart. For example, if $f_c = 100$ cps, then data at 30 cps would be aliased with data at frequencies 170 cps, 230 cps, 370 cps, 430 cps, and so forth. Similarly, the power at these higher confounding frequencies is aliased with the power in the lower frequencies. This occurs because for $t = 1/2f_c$, the power quantities $\sin^2(2\pi ft)$ and $\cos^2(2\pi ft)$ do not distinguish between a frequency f and frequencies $2nf_c \pm f$. Hence when the cutoff frequency f_c is as shown in Figure 7.9, a true power spectral density function as pictured in Figure 7.9(a) would be folded into the aliased power spectral density function as illustrated in Figure 7.9(b).

Two practical methods exist for handling this aliasing problem. The first method is to choose h sufficiently small so that it is physically unreasonable for data to exist above the associated cutoff frequency f_c . For example, assume that information below 1000 cps is desired from some given data. This means that $h \leq 0.50$ msec would technically be sufficient. However, further assume

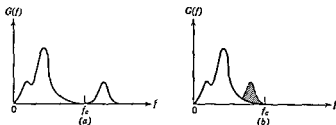


Figure 7.9 Aliased power spectrum due to folding (a) True spectrum (b) Aliased spectrum

there is reason to believe that the data may contain contributions up to perhaps 2000 cps. Then the Nyquist frequency should be established at $f_c \geq 2000$ cps which requires that $h \leq 0.25$ msec in order to avoid aliasing. In general, it is a good rule to select f_c to be one-and-a-half or two times greater than the maximum anticipated frequency. The second method is to filter the data prior to sampling so that information above a maximum frequency of interest is no longer contained in the filtered data. Then choosing f_c equal to the maximum frequency of interest will give accurate results for frequencies below f_c . The second method is preferred over the first to save on computing time and costs.

QUANTIZATION. Now consider the quantization operation. Since the magnitude of each data sample must be expressed by some fixed number of digits, only a fixed set of levels are available for approximating the infinite number of levels in the continuous data. No matter how fine the scale, a choice between two consecutive values will be required as illustrated in Figure 7.10. If the quantization is done properly, the true level will be approximated

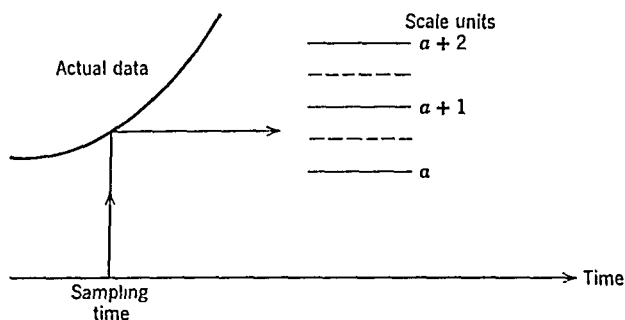


Figure 7.10 Illustration of quantization error.

by the quantizing level closest to it. The accuracy of the approximating process is a function of the number of available levels. Since most present day digitizers produce binary outputs compatible with computers, the number of levels may be described by the number of binary digits (bits) produced. Typical analog to digital conversion systems produce 6 to 16 bits, which corresponds to a range of 64 to 65,536 levels. For ideal conversion, the quantizing error will have a uniform probability distribution with a standard deviation of approximately $0.29 \Delta x$, where Δx is the quantizing increment.

This fact is easily shown as follows

Let $p(x)$ be the quantization error probability density function defined by

$$p(x) = 1 \quad -0.50 \leq x \leq 0.50 \\ = 0 \quad \text{otherwise}$$

The mean value of the error is clearly zero since $p(x)$ is symmetric about $x = 0$, and the variance of the error is

$$\sigma_x^2 = \int_{-\infty}^{\infty} (x - \mu_x)^2 p(x) dx = \int_{-0.5}^{0.5} x^2 dx = \frac{1}{12}$$

The standard deviation is

$$\sigma_x = \sqrt{\frac{1}{12}} \approx 0.29 \text{ scale unit} \quad (7.5)$$

This is the rms value of the quantization error, which may be considered as an rms noise on desired signals. For example, suppose the full range of a signal is quantized at 256 scale units. The peak signal to rms noise ratio in this case would be $(256 \Delta x)/(0.29 \Delta x) \approx 1000$, or about 60 dB.

In practice, the quantizing error is usually unimportant relative to other sources of error in the data acquisition and processing procedures. However, care must be exercised to assure that the range of the continuous data is set to occupy as much of the available quantizing range as possible. Otherwise, the resolution will be poor and the quantizing error could become significant.

PRACTICAL CONSIDERATIONS Commercial analog-to-digital converters (ADC) are usually either binary or BCD (binary coded decimal) systems. A binary system maps a continuous voltage onto a set of 2 numbers, while a BCD system maps the voltage onto a set of 10 numbers. Binary systems are simpler to build but usually require machine language programming to process data. Conversely, BCD systems, while more complicated to build and relatively inefficient, have the advantage that data may be read directly by a computer program written in a common language such as FORTRAN. Besides the sampling and quantization errors already discussed, other ADC errors of importance include

- 1 Aperture error—arising from the fact that the data sample is taken over a finite period of time rather than instantaneously
- 2 Jitter—arising from the fact that the time interval between samples can vary slightly in some random manner
- 3 Nonlinearities—arising from many sources such as misalignment of parts, bit dropouts, quantization spacing zero discontinuity, etc.

Typical digitization rates in available ADC equipment can vary from 100,000 sps to 250,000 sps. The limitations imposed by digital tape units and desired bit resolution often reduce the digitization rates in practice to something less

than 50,000 sps. These lower rates must be divided by the number of multiplexed channels being simultaneously digitized to obtain the individual data channel rate. Thus, for 10 multiplexed channels of data, the individual rates are typically less than 5000 sps. Of course, these are real time rates. The sampling rate relative to the data time base can be increased or decreased as desired by slowing down or speeding up the rate at which the data are input to the ADC equipment.

One final point concerning ADC equipment should be mentioned. When more than one channel of data are being simultaneously digitized, the time interval between the sampling of each channel introduces an inter-channel time base error in the digital data which can be significant at the higher frequencies. This problem may be avoided by the use of sample-hold circuits in the ADC equipment. These circuits simultaneously sample all input data channels and then hold the sampled signals until the ADC multiplexing switch passes the various channels.

7.3.2 *Preprocessing*

After data have been digitized, certain operations must be performed on the data prior to any actual analysis. The first such operation is reformatting. This step is theoretically trivial, but often takes considerable effort in practice. Present day computer operating systems are set up to accept standard formats; i.e., the types of formats generated by the operating systems themselves. Formats generated by analog to digital conversion systems are often non-standard and, thus, require machine language programs.

The next operation to be performed is that of conversion of the digitized data to physical units. Most digitizing procedures produce information in units that are related linearly to the true physical units. A common technique here is to digitize a reference or calibration signal at the same time the data are digitized. It is then possible to determine the relationship between digitized units and physical units directly.

The final step is to pre-qualify the data. This involves the detection and removal of isolated outliers, level shifts, trends, and other types of errors which may have occurred. Detection procedures for outliers usually involve a first difference check. The computation of running sample means will usually detect level shifts. Trend removal is discussed later in Section 9.1.3.

7.4 DATA QUALIFICATION

The correct procedures for analyzing random data, as well as interpreting the analyzed results, are strongly influenced by certain basic characteristics which may or may not be exhibited by the data. The three most important of these

basic characteristics are the stationarity of the data, the presence of periodicities in the data, and the normality of the data. Stationarity is of concern because the analysis procedures required for nonstationary data are generally more complicated than those which are appropriate for stationary data. Periodicities in the data should at least be identified to avoid erroneous interpretations of later results. The validity of an assumption that the data (excluding periodicities) have a Gaussian probability density function should be investigated since the normality assumption is vital to many analytical applications for random data. Referring back to Figure 7.1, qualification of sampled data in terms of these basic characteristics is indicated as a separate operation to be performed prior to detailed data analysis. In practice, however, it is often accomplished as an integral part of the data analysis phase. Practical considerations and procedures for such qualification will now be discussed.

7.4.1 *Test for Stationarity*

Perhaps the simplest way to evaluate the stationarity of sampled random data is to consider the physics of the phenomenon producing the data. If the basic physical factors which generate the phenomenon are time invariant, then stationarity of the resulting data generally can be accepted without further study. For example, consider the random data representing pressure fluctuations in the turbulent boundary layer generated by the flight of a high speed aircraft. If the aircraft is flying at constant altitude and airspeed with a fixed configuration, it would be reasonable to assume that the resulting pressure data are stationary. On the other hand, if the aircraft is rapidly changing altitude, airspeed, and/or configuration, then nonstationarities in the resulting pressure data would be anticipated.

In practice, data are often collected under circumstances which do not permit an assumption of stationarity based upon simple physical considerations. In such cases, the stationarity of the data must be evaluated by studies of available sample time history records. This evaluation might range from a visual inspection of the time histories by a talented analyst to detailed statistical tests of appropriate data parameters. In any case, there are certain important assumptions which must be made if the stationarity of data is to be ascertained from individual sample records. First, it must be assumed that any given sample record will properly reflect the nonstationary character of the random process in question. This is a reasonable assumption for those nonstationary random processes which involve deterministic trends, as discussed in Chapter 10. Second, it must be assumed that any given sample record is very long compared to the lowest frequency component in the data, excluding a nonstationary mean. In other words, the sample record must be long enough to permit nonstationary trends to be differentiated from the random fluctuations of the time history.

Beyond these basic assumptions, it is convenient (but not necessary) to further assume that any nonstationarity of interest will be revealed by time trends in the mean square value of the data. Of course, one can readily contrive a nonstationary random process with a stationary mean square value; for example, a process where each sample function is a constant amplitude sine wave with continuously increasing frequency and random initial phase. Nevertheless, such cases are unusual in practice because it is highly unlikely for nonstationary data to have a time varying autocorrelation function at any time displacement τ without the value at $\tau = 0$ varying. Since $R(0) = \Psi^2$, the mean square value will usually reveal a time varying autocorrelation. A similar argument applies to higher order properties.

With these assumptions in mind, the stationarity of random data can be tested by investigating a single record $x(t)$ as follows.

1. Divide the sample record into N equal time intervals where the data in each interval may be considered independent.
2. Compute a mean square value (or mean value and variance separately) for each interval and align these sample values in time sequence, as follows.

$$\overline{x_1^2}, \overline{x_2^2}, \overline{x_3^2}, \dots, \overline{x_N^2}$$

3. Test the sequence of mean square values for the presence of underlying trends or variations other than those due to expected sampling variations.

The final test of the sample values for nonstationary trends may be accomplished in many ways. If the sampling distribution of the sample values is known, various statistical tests discussed in Chapter 4 could be applied. However, as noted in Section 6.2.2, the sampling distribution of mean square values requires a detailed knowledge of the frequency composition of the data. Such knowledge is generally not available at the time one wishes to establish whether or not data are stationary. Hence a nonparametric approach which does not require a knowledge of the sampling distributions of data parameters is more desirable. One such nonparametric test which is applicable to this problem is outlined in Section 4.7. This is the run test which may be applied directly as follows.

Let it be hypothesized that the sequence of sample mean square values $(\overline{x_1^2}, \overline{x_2^2}, \overline{x_3^2}, \dots, \overline{x_N^2})$ are each independent sample values of a stationary random variable with a mean value of Ψ_x^2 . If this hypothesis is true, the variations in the sequence of sample values will be random and display no trends. Hence the number of runs in the sequence relative to, say, the median value, will be as expected for a sequence of independent random observations of the random variable, as presented in Table A.6. If the number of runs is significantly different from the expected number given in Table A.6, the hypothesis of stationarity would be rejected. Otherwise, the hypothesis would

be accepted. Note that the above testing procedure does not require a knowledge of either the frequency bandwidth of the data or the averaging time used to compute sample mean square values. Nor is it limited to a sequence of mean square values. It will work equally well on mean values, rms values, standard deviations, mean absolute values, or any other parameter estimate. Furthermore, it is not necessary for the data under investigation to be free of periodicities. Valid conclusions are obtained, even when periodicities are present, as long as the fundamental period is short compared to the averaging time used to compute sample values.

Example 7.2. Test for Stationarity. To illustrate application of the run test as a test for stationarity, consider the results summarized in Figure 7.11. The original data for this example consisted of a 480 second long sample record of gust velocity measured with a probe extending forward of a test aircraft flying through a patch of clear air turbulence. To evaluate the stationarity of the turbulence data, the sample record was divided into 20 segments of equal length (24 seconds each), and a sample standard deviation was computed for each segment. Note that the median standard deviation is about 1.12 ft/sec. Now let it be hypothesized that the data are stationary. From Table A.6, this hypothesis would be accepted at the $\alpha = 0.05$ level of significance if the number of runs observed in the sequence of standard deviations

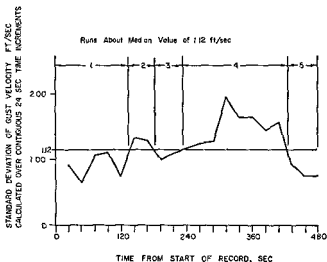


Figure 7 11 Test for stationarity of gust velocity data

relative to the median were at least 6 but not more than 15. From Figure 7.11, it is seen that only 5 runs occur in the sequence. Hence, the hypothesis of stationarity is rejected at the 5 percent level of significance, meaning the data should be considered nonstationary. This concludes Example 7.2.

In conclusion, an assumption of stationarity can often be supported (or rejected) by a simple nonparametric test of sample mean square values (or related sample parameters) computed from the available data. However, if one is not prepared to accept time invariance of mean square values as sufficient evidence of time invariance of autocorrelation functions, then tests can still be performed by further segmenting of the data in the frequency domain. Specifically, the data can be segmented into several contiguous frequency ranges by bandpass filtering, and the sample mean square values in each frequency interval can then be individually tested for time invariance. Since spectral density functions and correlation functions are Fourier transform pairs, time invariance of one directly implies time invariance of the other.

7.4.2 *Test for Periodicities*

Periodic and/or almost periodic components in otherwise random data will theoretically appear as delta functions in the power spectrum of the data. In practice, they will appear as sharp peaks in the power spectrum which might be confused with narrow-band random contributions. Hence it is desirable to identify the presence of periodic components so they will not be misinterpreted as narrow band random components with finite power spectral densities. If periodic components in the data are intense, their presence is usually obvious. However, less intense periodic components in random data may not be so obvious. The most effective procedures for detecting periodic components are those associated with the various analysis procedures which would be employed for random data analysis anyway. Hence in practical terms, a test for periodicities usually evolves from analysis procedures which would be performed assuming the data are random. Specifically, the presence of periodic components in otherwise random data may often be detected by visual inspection of a power spectral density function, an amplitude probability density function, and/or an autocorrelation function measured from stationary data, as discussed in Chapter 1.

Example 7.3. Spectrum of Sine Wave in Noise. To illustrate how a power spectrum can reveal the presence of a periodic component in otherwise random data, refer to Figure 7.12. In this example, the output of a thermal noise generator is mixed with a sinusoidal signal. The sinusoidal signal has an rms amplitude equal to one-twentieth that of the random signal. Figure 7.12(a), which was made using a relatively wide filter bandwidth, gives little or no

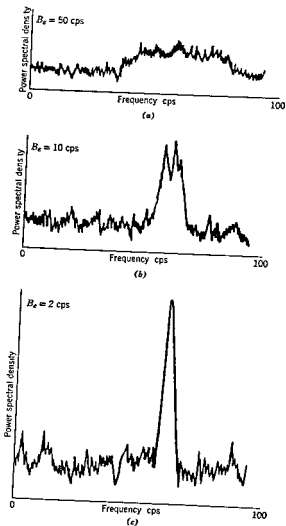


Figure 7.12 Measured power spectra of a sine wave in random noise

indication of the presence of the sinusoid. Figure 7.12(b), which was made using one-fifth of the previous filter bandwidth, indicates a possible sinusoid quite clearly. Figure 7.12(c), which was made using a filter bandwidth reduced by another factor of 5, gives a strong indication. This concludes Example 7.3.

Figure 7.12 illustrates how a highly resolved spectrum will reveal periodic components as sharp peaks, even when the periodicities are of relatively small intensity. A sharp peak in the power spectrum of sample data, however, may also represent narrow-band random data. These two cases can sometimes be distinguished from one another by repeating the power spectral density measurement with a narrower resolution filter bandwidth. If the measured spectral peak represents a sine wave, the indicated bandwidth of the peak will always be equal to the bandwidth of the analyzer filter, no matter how narrow the filter. Furthermore, the indicated spectral density will always increase in direct proportion to the reduction in filter bandwidth. This method of detection will clearly not work unless the resolution bandwidth of the analyzer filter is smaller than the bandwidth of possible narrow-band random data. It is also necessary that the available sample record be sufficiently long to make $B_e T \gg 1$. Otherwise, uncertainties in the spectral density measurements of the random portions of the data will produce spurious peaks which may obscure the peak caused by the sine wave.

The presence of sinusoidal components in otherwise random data may also be revealed by an amplitude probability density function of the data. The probability density plots for sinusoidal and random data are markedly different, as illustrated in Figure 1.13. A random signal will usually have a probability density function which at least resembles the familiar bell-shaped Gaussian characteristic, whereas a sine wave has a dish-shaped probability density function. The problem with this procedure is that the sinusoidal component must be fairly intense relative to the random portion of the data before its presence is obvious from the probability density plot. Generally speaking, the presence of the sine wave is difficult to identify if the variance of the sine wave is less than the variance of the random portion of the data. Furthermore, the presence of more than one sine wave can produce probability density results which are difficult to distinguish from random data. Of course, both these problems can be minimized by filtering the data to isolate single suspicious peaks for probability density analysis.

A final method for detecting sinusoidal components in otherwise random data is presented by an autocorrelogram. For random data without periodicities, the autocorrelation function will always approach a value equal to the square of the mean value as the time displacement becomes large. On the other hand, the autocorrelation function of a sine wave or collection of sine waves will continue to oscillate no matter how large the time displacement becomes. This fact is illustrated in Figure 1.15.

7.4.3 *Test for Normality*

Perhaps the most obvious way to test samples of stationary random data for normality is to measure the probability density function of the data and compare it to the theoretical normal distribution. If the sample record is sufficiently long to permit a measurement with small error compared to the deviations from normality, the lack of normality will be obvious. If the sampling distribution of the probability density estimate is known, various statistical tests for normality can be performed even when the random error is large. However, as for stationarity testing discussed in Section 7.4.1, a knowledge of the sampling distribution of probability density measurements requires frequency information for the data which may be difficult to obtain in practical cases. Hence a nonparametric test is desirable.

One of the most convenient nonparametric tests for normality is the chi-square goodness of fit test outlined in Section 4.6. This test must be applied to discrete sample values representing the random data of interest. Where the data are being analyzed by digital techniques, no problem exists since the data are already in the form of discrete sample values. For analog data analysis, however, the data must be converted into a digital form to apply the test. The details of applying the chi square goodness-of-fit test with a numerical illustration are presented in Example 4.3.

7.5 DATA ANALYSIS

The procedures for analyzing the properties of random data may be divided logically into two categories, the procedure for analyzing individual sample records, and the procedure for analyzing a collection of sample records given the properties of the individual records. Applicable data analysis procedures for these two categories are now outlined. This is followed by a summary of the relative merits of analog and digital data analysis techniques.

7.5.1 *Procedure for Analyzing Individual Records*

An overall procedure for analyzing the pertinent statistical properties of individual sample time history records is presented in Figure 7.13. Note that many of the suggested steps in the procedure might be omitted for some applications while additional steps would be required for other applications. Further note that the data qualification steps discussed in Section 7.4 are incorporated into the procedure to help clarify how these two parts of the overall data processing problem interact. Each block in Figure 7.13 will now be discussed.

MEAN AND MEAN SQUARE VALUE ANALYSIS The first step indicated by Block A is a mean and mean square value (or variance) measurement. This

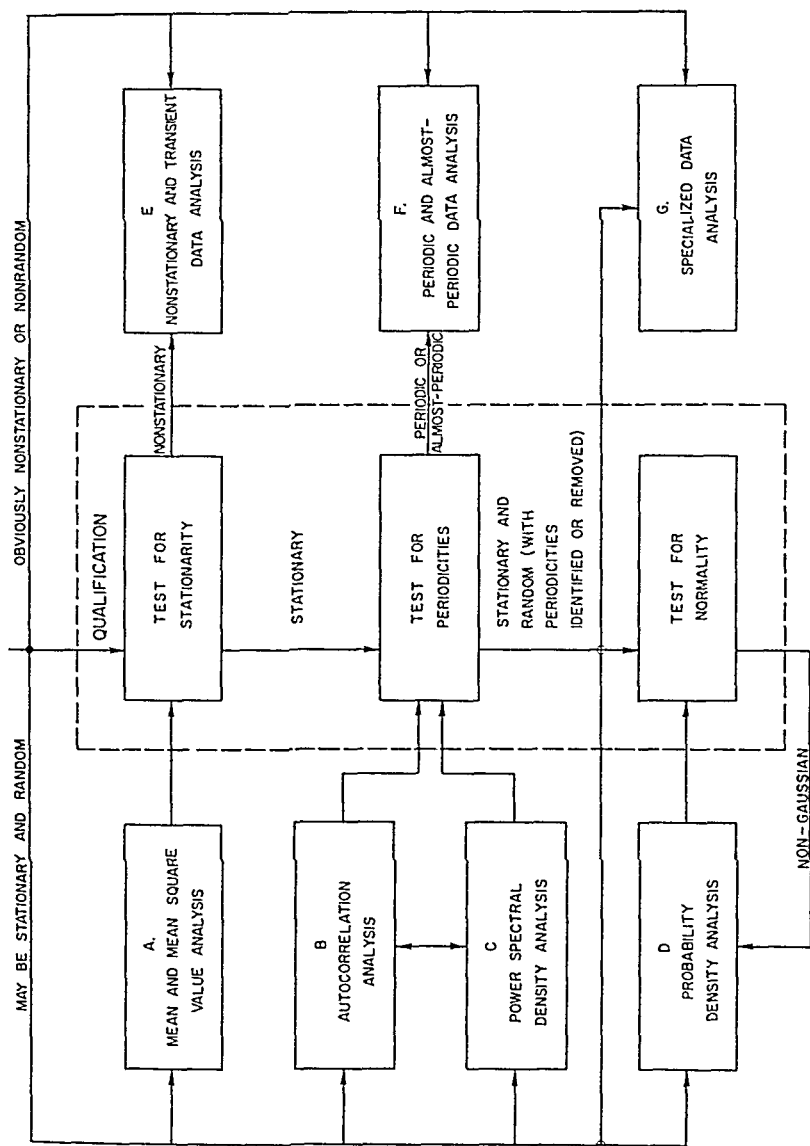


Figure 7.13 General procedure for analyzing individual sample records.

step is almost universally performed for one or more of three sound reasons. First, since the mean and mean square values are the basic measures of central tendency and dispersion, their calculation is generally required for even the most rudimentary applications. Second, the calculation of short time averaged mean and mean square value estimates provides a basis for evaluating the stationarity of the data, as indicated in Figure 7.13 and illustrated in Section 7.4.1. Third, mean and mean square value estimates can be extracted from other descriptive properties (probability density plots, correlograms, and/or power spectra) which might be measured later. The comparison of directly measured mean and mean square values estimates to the corresponding estimates extracted from other analyses provides an excellent method for checking the data analysis equipment or computer programs for correct operation.

Analog procedures for measuring mean and mean square values are presented in Section 8.1. Digital computer techniques are discussed in Section 9.1. The statistical accuracy of mean and mean square value estimates is developed in Section 6.2.

AUTOCORRELATION ANALYSIS The next suggested analysis is autocorrelation, as indicated by Block B. The autocorrelation function of stationary data is the inverse Fourier transform of the power spectral density function. Thus the determination of the autocorrelation function will technically not yield any new information over the power spectrum. There might be applications, however, where an autocorrelogram would present the desired information in a more convenient format. Autocorrelation functions can be useful as a tool for detecting periodicities in otherwise random data, as indicated in Figure 7.13 and discussed in Section 7.4.2. Furthermore, autocorrelation functions might be computed as an intermediate step in the calculation of power spectra estimates. This is an outdated but still sometimes employed technique for estimating power spectra, as discussed in Section 9.6.1.

Analog procedures for measuring autocorrelation functions are detailed in Section 8.3, and digital techniques are discussed in Section 9.5. The statistical accuracy of autocorrelation measurements is discussed in Section 6.4.

POWER SPECTRAL DENSITY ANALYSIS Perhaps the most important single descriptive characteristic of stationary random data is the power spectral density function, which defines the frequency composition of the data. For constant parameter linear physical systems, the output power spectrum is equal to the input power spectrum multiplied by the square of the gain factor of the system. Thus power spectra measurements can yield information concerning the dynamic characteristics of the system. The total area under a power spectrum is equal to the mean square value Ψ^2 . To be more general,

the mean square value of the data in any frequency range of concern is determined by the area under the power spectrum bounded by the limits of that frequency range. Obviously, the measurement of power spectra data, as indicated by Block C, will be valuable for many analysis objectives. Secondary applications include its use for the detection of periodicities, as illustrated in Section 7.4.2, and as an intermediate step in the calculation of autocorrelation functions.

The physical significance of the power spectral density function for input-output random data is shown clearly in Chapter 5. Analog procedures for measuring power spectra are detailed in Section 8.4, and digital techniques are discussed in Section 9.6. The statistical accuracy of power spectral density measurements is discussed in Section 6.5.

PROBABILITY DENSITY ANALYSIS. The last fundamental analysis included in the procedure is probability density analysis, as indicated by Block D. Probability density analysis is often omitted from a data analysis procedure because of the tendency to assume that all random phenomena are normally distributed. In some cases, however, random data may deviate substantially from the Gaussian form. If such deviations are detected by a test for normality, then the probability density function of the data must be measured to establish the actual probabilistic characteristics of the data. Furthermore, a probability density function estimate is sometimes used as a basis for a normality test, as indicated in Figure 7.13 and discussed in Section 7.4.3.

Analog procedures for measuring probability density functions are detailed in Section 8.2, and digital techniques are discussed in Section 9.4. The statistical accuracy of probability density measurements is discussed in Section 6.3.

NONSTATIONARY AND TRANSIENT DATA ANALYSIS. All of the analysis techniques discussed thus far apply only to sample records of stationary data. If the data are determined to be nonstationary during the qualification phase of the processing, then special analysis techniques will be required as indicated by Block E. The analysis of nonstationary and transient data is discussed in Chapter 10. Note that certain classes of nonstationary data can sometimes be analyzed using the same equipment or computer programs employed for stationary data analysis. However, the results of such analyses must be interpreted with caution as illustrated in Sections 10.1 and 10.2.

PERIODIC AND ALMOST-PERIODIC DATA ANALYSIS. If sinusoids due to periodic or almost-periodic contributions are detected in the data during the qualification phase, then special attention is warranted. Specifically, one of two approaches should be followed. First, the sinusoidal components might be isolated from the random portion of the data by filtering operations and

analyzed separately, as indicated by Block F. Second, the sinusoidal components might be analyzed along with the random portion of the data, and then simply accounted for in the results. For example, if a power spectrum is computed for data which include sinusoidal components, a delta function symbol might be superimposed on each spectral peak at the frequency of an identified sinusoid, and labeled with the mean square value of the sinusoid. The mean square value can be estimated from the spectral plot by multiplying the maximum indicated spectral density of the peak by the resolution band width used for the analysis. If this is not done, the physical significance of such spectral peaks might be misinterpreted, as discussed in Section 7.4.2.

SPECIALIZED DATA ANALYSIS Various other analyses of individual time history records are often required, depending upon the specific goals of the data processing. For example, studies of fatigue damage in mechanical systems usually involve the calculation of peak probability density functions of strain data, as discussed in Reference 10. An investigation of zero crossings or arbitrary level crossings might be warranted for certain communication noise problems, as discussed in Reference 1. Spectral descriptions other than power spectral density functions are sometimes desired, for example, the spectra for acoustic noise levels are commonly presented in terms of rms values in 1/1 or 1/3 octave frequency bands. Such specialized analyses, as indicated by Block G, must be established in the context of the specific engineering problem of concern.

7.5.2 *Procedure for Analyzing a Collection of Records*

The preceding section presented methods for analyzing each individual sample record from an experiment. A procedure for analyzing further pertinent statistical properties of a collection of sample records is presented in Figure 7.14. As for the analysis of individual sample records outlined in Figure 7.13, many of the suggested steps in Figure 7.14 might be omitted for some applications while additional steps would be required for others. Furthermore, the suggested steps assume the individual records are stationary. Each block in Figure 7.14 will now be discussed.

ANALYSIS OF INDIVIDUAL RECORDS This first step in the procedure is to analyze the pertinent statistical properties of the individual sample records, as outlined in Figure 7.13. Hence the applicable portions of Figure 7.13 constitute Block A in Figure 7.14.

TEST FOR CORRELATION The next step indicated by Block B is to determine whether or not the individual sample records are correlated. In many cases, this decision involves little more than a cursory evaluation of pertinent physical considerations. For example, if the collection of sample records

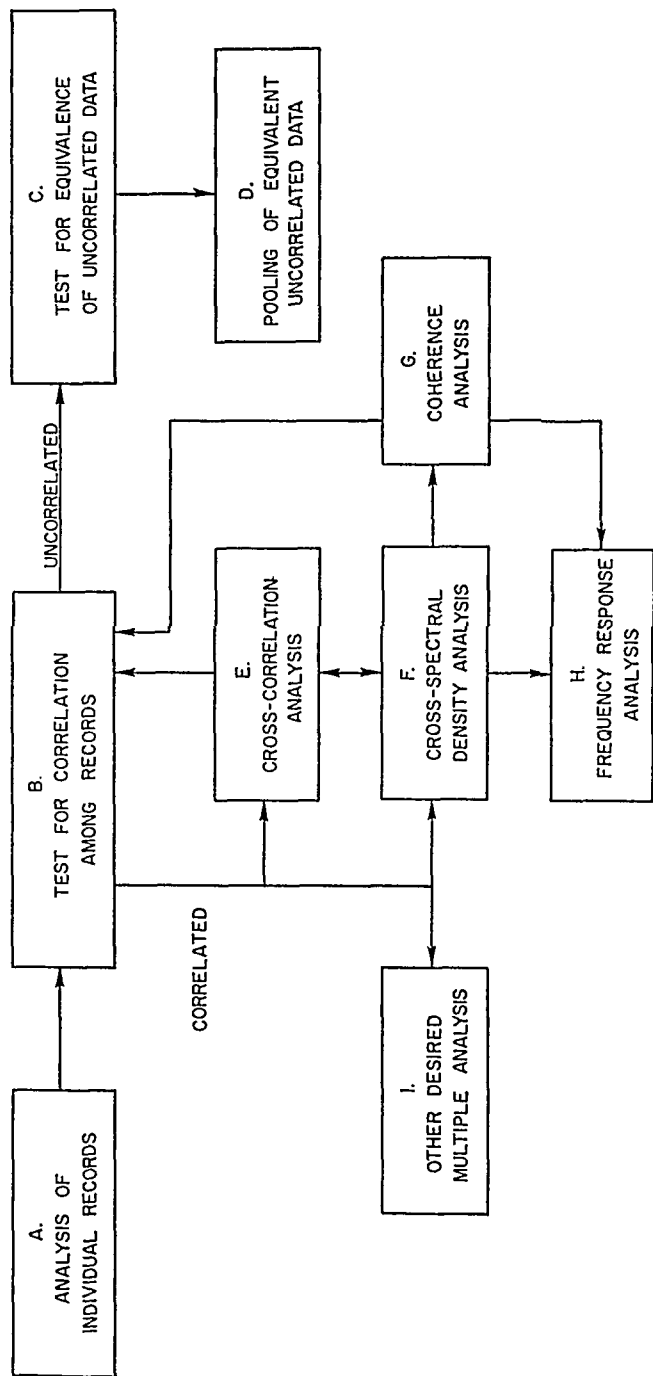


Figure 7.14 General procedure for analyzing a collection of sample records.

represents measurements of a physical phenomenon over widely separated time intervals then usually the individual records can be accepted as uncorrelated without further study. On the other hand, if the collection represents simultaneous measurements of the excitation and response of a physical system, then correlation would be anticipated. For those cases where a lack of correlation is not obvious from basic considerations a test for correlation among the sample records should be performed using cross correlation functions or coherence functions as indicated in Figure 7.14.

TEST FOR EQUIVALENCE OF UNCORRELATED DATA If sample records are found to be uncorrelated in Block B then these records should be tested for equivalent statistical properties as indicated by Block C. This is an important but often overlooked step in the analysis of random data. Far too often the analyzed results for a large number of sample records are presented as individual plots when in fact the results differ only by amounts which fall within the acceptable limits of random error. The formal presentation of such redundant data is usually of no value, and can actually be detrimental in several ways. First, large quantities of analyzed data will sometimes tend to overwhelm the user and unnecessarily complicate the interpretation of the results. Second, the unsophisticated user might interpret the statistical scatter in individual results as physically meaningful differences. Third, more accurate results could be presented for the equivalent data if they were pooled prior to plotting, as will be discussed next. Note that for most applications, an equivalence of power spectra is a sufficient criterion for equivalence of sampled data. A procedure for testing the equivalence of power spectra is presented in Section 7.5.3.

POOLING OF EQUIVALENT UNCORRELATED DATA The analyzed results for individual sample records which are found to represent equivalent data should be pooled together as indicated by Block D. This is done by computing appropriately weighted averages of the results for the individual records. For example, assume two power spectral density function estimates were computed from two uncorrelated sample records which now are found to represent equivalent data. If $\hat{G}_1(f)$ and $\hat{G}_2(f)$ were the original power spectra estimates with n_1 and n_2 degrees of freedom, respectively, a new pooled estimate for the power spectrum is given by

$$\hat{G}_s(f) = \frac{n_1 \hat{G}_1(f) + n_2 \hat{G}_2(f)}{n_1 + n_2} \quad \begin{matrix} n_1 = 2B_{e1}T_1 \\ n_2 = 2B_{e2}T_2 \end{matrix} \quad (7.6)$$

where $\hat{G}_s(f)$ has $n_s = n_1 + n_2$ degrees of freedom. Equation (7.6) may be

generalized for q estimates from uncorrelated but equivalent samples as follows

$$\hat{G}_p(f) = \frac{\sum_{i=1}^q n_i \hat{G}_i(f)}{\sum_{i=1}^q n_i} \quad n_i = 2B_{ei}T_i \quad (7.7)$$

The sampling distribution of the pooled estimate is then

$$\frac{\hat{G}_p(f)}{G(f)} = \frac{\chi_n^2}{n} \quad n = \sum_{i=1}^q n_i \quad (7.8)$$

where χ_n^2 has a chi-square distribution with n degrees of freedom as defined in Section 4.2.2.

It is clear from Equation (7.8) that the pooling operation produces a power spectrum estimate with a reduced random error. However, it also should be noted that the pooling operation generally will not suppress the systematic error (bias) in the power spectra estimates, as defined and discussed in Section 6.5. This fact often leads data analysts to reprocess sample records with equivalent statistical properties in a manner designed to reduce bias errors. For the case of power spectra estimates, the reprocessing might consist of a recomputation of power spectral density estimates from the original sample records using a greatly reduced resolution bandwidth to suppress the bias errors at the expense of increased random errors. The random errors in the individual estimates are then suppressed by the pooling operation.

Another approach which is sometimes employed with analog data analysis equipment is to splice together the original sample records in a nose to tail fashion to obtain one long sample record for reprocessing. This procedure can produce acceptable results, but it must be remembered that certain inherent limitations imposed by the length of the original records still apply. Specifically, if q sample records each of length T are spliced together to form a record of length qT , the lowest frequency which can be defined in the data is still $f = 1/T$, and not $f = 1/(qT)$.

CROSS-CORRELATION ANALYSIS. As for the case of autocorrelation and power spectral density functions, the cross-correlation and cross-spectral density functions are Fourier transform pairs. Hence the measurement of a cross-correlogram will technically not yield any new information over the cross-spectrum. However, it may present desired information in a more convenient format. An example is the measurement of a simple time delay between two measurement points. Therefore, cross-correlation analysis is included in the procedure as a separate step indicated by Block E. Note that a

cross-correlation estimate can be used as a test for correlation between two individual records, and as an intermediate step in the calculation of a cross-spectral density estimate

Analog procedures for measuring cross correlation functions are outlined in Section 8 5 2, and digital techniques are discussed in Section 9 7 2 The statistical accuracy of cross correlation measurements is discussed in Section 6 4

CROSS SPECTRAL DENSITY ANALYSIS The most important joint measurement for a collection of correlated sample records is the cross-spectral density analysis indicated by Block F Cross spectral density functions provide information concerning the linear relationships which might exist among the collection of sample records When interpreted into a physical context such information often leads directly to problem solutions as developed in Chapter 5

Analog procedures for measuring cross spectral density functions are covered in Section 8 5 3, and digital techniques in Section 9 7 3 The statistical accuracy of cross spectral density measurements is discussed in Section 6 5 4

COHERENCE FUNCTION ANALYSIS Block G indicates the calculation of coherence functions based upon power and cross-spectral density estimates Coherence functions of various types (ordinary, multiple, and partial) are valuable in several ways First, they can be used to test for correlation among the collection of sample records Second, they constitute a vital parameter in assessing the accuracy of frequency response function estimates Third, they can provide a direct solution for certain types of problems

The computation of coherence functions and their applications are detailed in Chapter 5 The statistical accuracy of coherence function estimates is discussed in Section 6 6 Their digital computation is outlined in Section 9 8

FREQUENCY RESPONSE FUNCTION ANALYSIS The ultimate goal in the analysis of a collection of sample records is often to establish linear relationships among the data represented by the various records The existence of such linear relationships can be detected from cross-correlation, cross spectral density, or coherence function estimates However, a meaningful description of the linear relationships is best provided by computing the frequency response functions of the relationships, as indicated by Block H

The computation of frequency response functions and their applications are developed in Chapter 5 The statistical accuracy of frequency response estimates is discussed in Sections 6 7 and 6 8 Digital computational procedures are outlined in Section 9 8

OTHER DESIRED MULTIPLE ANALYSIS Block I indicates other joint analyses of a collection of sample records needed to satisfy special data

processing goals. Included might be joint probability density and distribution functions as discussed in Section 3.1.

7.5.3 Test for Equivalence of Power Spectra

The previous section discussed the need to establish the possible equivalence of power spectra estimates computed from two or more statistically independent sample records. A useful technique for this application will now be described.

An estimate $\hat{G}(f)$ of a power spectral density function $G(f)$ will have a sampling distribution which is approximately normal if the number of degrees of freedom n is large, say $n \geq 30$. It is shown in Section 6.5 that the mean value and variance of the estimate are given by

$$E[\hat{G}(f)] \approx G(f) \quad (7.9)$$

$$\text{Var} [\hat{G}(f)] \approx \frac{2}{n} G^2(f) \quad (7.10)$$

Hence a $1 - \alpha$ confidence interval for $G(f)$ based on a measurement $\hat{G}(f)$ may be approximated by

$$[\hat{G}(f)(1 - z_{\alpha/2}\sqrt{2/n}) \leq G(f) \leq \hat{G}(f)(1 + z_{\alpha/2}\sqrt{2/n})] \quad (7.11)$$

where $z_{\alpha/2}$ is the $100\alpha/2$ percentage point of the standardized normal distribution. To arrive at Equation (7.11), it is assumed that $z_{\alpha/2}\sqrt{2/n} \ll 1$, so that

$$(1 \pm z_{\alpha/2}\sqrt{2/n})^{-1} \approx 1 \mp z_{\alpha/2}\sqrt{2/n} \quad (7.12)$$

A logarithmic transformation of the estimate $\hat{G}(f)$ to $\log \hat{G}(f)$ has the effect of producing a distribution which is closer to normal than the original distribution. The sample mean value and variance of $\log \hat{G}(f)$ become

$$E[\log \hat{G}(f)] \approx \log G(f) \quad (7.13)$$

$$\text{Var} [\log \hat{G}(f)] \approx 2/n \quad (7.14)$$

Thus the variance here is independent of frequency. Now, a $1 - \alpha$ confidence interval for $\log G(f)$ may be approximated by

$$[\log \hat{G}(f) - z_{\alpha/2}\sqrt{2/n} \leq \log G(f) \leq \log \hat{G}(f) + z_{\alpha/2}\sqrt{2/n}] \quad (7.15)$$

This result can be derived directly from Equation (7.11) to provide a heuristic explanation for Equations (7.13) and (7.14). This derivation uses the assumption that $z_{\alpha/2}\sqrt{2/n} \ll 1$, so that

$$\log (1 \pm z_{\alpha/2}\sqrt{2/n}) \approx \pm z_{\alpha/2}\sqrt{2/n} \quad (7.16)$$

Consider now two different power spectral density function estimates $\hat{G}_1(f)$ and $\hat{G}_2(f)$ obtained under different conditions for example from two different sample records or from two different parts of the same sample record. The problem is to decide whether or not these two power spectra are statistically equivalent over some frequency interval (f_a - f_b) of bandwidth $B = f_b - f_a$.

Assume each of the two power spectral density function estimates is based upon a resolution bandwidth B_r where N_f number of bandwidths are needed to cover the frequency range of interest. That is,

$$N_f = \frac{B}{B_r} \quad (7.17)$$

Further assume the degrees of freedom for each estimate are n_1 and n_2 , respectively, meaning that the averaging time (record length) for each estimate may be different even though the resolution bandwidth is the same. From Equations (7.13) and (7.14), the sampling distributions of the logarithm of the estimates in the i th bandwidth are approximated by

$$\begin{aligned} \log \hat{G}_1(f_i) &= y \left[\log G_1(f_i), \frac{2}{n_1} \right] \\ \log \hat{G}_2(f_i) &= y \left[\log G_2(f_i), \frac{2}{n_2} \right] \end{aligned} \quad (7.18)$$

where $y[\mu, \sigma^2]$ is a normally distributed random variable with a mean of μ and a variance of σ^2 . Now, if the two sample records in question have the same power spectral density function $G(f) = G_1(f) = G_2(f)$, it follows from Equations (7.18) that

$$\log \frac{\hat{G}_1(f_i)}{\hat{G}_2(f_i)} = y \left[0, \frac{2}{n_1} + \frac{2}{n_2} \right] \quad (7.19)$$

Hence the statistic

$$X^2 = \left[\frac{2}{n_1} + \frac{2}{n_2} \right]^{-1} \sum_{i=1}^{N_f} \left[\log \frac{\hat{G}_1(f_i)}{\hat{G}_2(f_i)} \right]^2 \quad (7.20)$$

has a chi-square distribution with N_f degrees of freedom. That is,

$$X^2 = \chi_n^2 \quad n = N_f \quad (7.21)$$

The result in Equation (7.20) provides a basis to test the hypothesis that $G_1(f) = G_2(f)$. The region of acceptance for the hypothesis test is

$$X^2 \leq \chi_{\alpha}^2 \quad n = N_f \quad (7.22)$$

where α is the level of significance for the test.

7.5.4 Analog Versus Digital Data Analysis

The design of a data acquisition and processing system must ultimately involve a decision as to whether the final data analysis is performed with analog, hybrid, or digital equipment. There are a number of options available, as outlined in Figure 7.15. Analog equipment includes all devices which perform desired calculations by direct operations on electrical signals representing continuous time history records. General purpose analog computers are

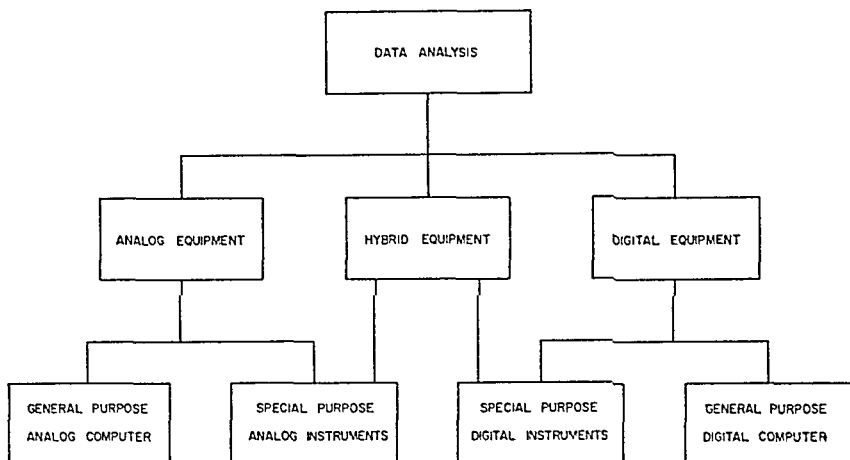


Figure 7.15 Equipment options for data analysis.

those analog devices which can be programmed to solve differential equations using networks involving operational amplifiers. Special purpose analog instruments are those analog devices which are designed to perform only limited types of operations—for example, a voltmeter or a crystal filter wave analyzer. Digital equipment includes all devices which perform desired calculations by numerical operations on discrete sample values representing either continuous or discrete time history records. As for analog equipment, digital equipment may be either general or special purpose. Hybrid equipment includes those devices which employ both analog and digital operations in an effort to combine the more desirable features of both.

General purpose analog computers have never been widely used for the type of data analysis discussed in this chapter because they are not well suited for such applications. Special purpose analog instruments are generally more efficient for random data analysis and, hence, have tended to prevail on the analog side. Indeed, prior to 1950, the vast majority of random data analysis

was performed using special purpose analog instruments. In the 1950 s, large scale general purpose digital computers with appropriate applications software became available for random data analysis in a practical way. This led to considerable discussion in the period between 1950 and 1965 on the relative merits of analog versus digital data analysis, meaning special purpose analog instruments as opposed to general purpose digital computers with appropriate software.

Many of the arguments during this period centered around academic issues such as analysis accuracy and speed. Yes, digital computers are more accurate than analog instruments, but the 2 or 3 percent machine error of an analog instrument is often acceptable when compared to the other errors in random data analysis evolving from statistical sampling considerations and pre-analysis data acquisition and processing operations. Yes, digital computers are generally faster for most desired operations, but speed can be achieved in analog processing by parallel analysis equipment. Other arguments were economic in nature. Yes, the capital investment and overhead is substantially less for analog analysis equipment, but the higher cost of a digital computer facility can be spread over many unrelated users (inventory control, payroll, scientific computing, etc.). In the end, the final decision between analog and digital analysis was usually governed by purely practical considerations.

Perhaps the most common of these practical considerations was simply equipment availability. Users in organizations which maintained a large scale digital computer facility, already justified for other applications, were inclined to make use of it for random data analysis as well. Conversely, users in organizations which did not maintain a computer facility were more inclined to procure and use specialized analog equipment. A second important practical consideration was user control and turn-around time. Because of their specialized nature, analog instruments were generally under the direct control of the user and, hence, instantly available as required. On the other hand, most organizations have a centralized computer facility under a separate jurisdiction from the users and, hence, available on a priority basis. The resulting turn-around time was often annoying to data analysts, particularly when their activities were of a research nature involving considerable trial and error type analysis.

Starting in 1965, many of the above considerations were sharply altered by a single technical development, namely, the introduction of algorithms for the fast computation of Fourier transformations on a digital computer. These algorithms, usually referred to as fast Fourier transform (FFT) procedures, have had a profound impact on random data analysis procedures and equipment because of their applicability to the calculation of spectral

density functions. By previous methods, the computation of Fourier transforms was a relatively time consuming operation on digital computers. As a result, power and cross-spectra were usually calculated by first computing an auto or cross-correlation function, and then computing Fourier transforms of truncated versions of the correlation functions. This provided estimates of spectral density functions, as defined by Equations (3.67) and (3.70), without the need for computing Fourier transforms of all sample values of the time history. Even then, the spectral density function calculations were the single most time consuming operation in random data analysis. With the introduction of FFT procedures, the computer time required for Fourier transform calculations was so sharply reduced that power and cross-spectra could be efficiently estimated from direct Fourier transforms of time history records, using the definitions of Equations (3.100) and (3.101). The net result was on the order of a 10 to 1 reduction in the overall computer time required for spectral analysis.

The reduced computer time for power and cross-spectra calculations provided by FFT procedures immediately increased the cost effectiveness of large scale digital computers for random data analysis. Even more important, however, is the fact that the FFT algorithms permit the calculation of spectral density estimates with greatly simplified digital circuitry. This factor along with the increased computation speed made the development and use of special purpose digital equipment economically feasible for random data analysis applications. Such special purpose digital instruments have since become commercially available. In most cases, they will compute not only spectral density functions, but also correlation and probability density functions as well as Fourier integrals of transients. Some even permit the programming of limited additional calculations as desired. They often incorporate their own analog to digital conversion system to accomplish the required digitization operations outlined in Section 7.3.1. Hence they can be used in the same way as analog instruments, and with the same advantages while still providing the features of speed and accuracy inherent in digital computers.

In parallel with the above noted events has been the development and use of hybrid data analysis instruments of various types. In many cases, these hybrid instruments have been home made and designed for limited applications. Some hybrid instruments for random data analysis, however, have been commercially marketed. Perhaps the most successful instruments of this type are spectrum analyzers which employ analog filters and detection networks, but use digital techniques to compress the time base of the data and, hence, the required analysis time. In other cases, the detection networks are digital as well.

Hybrid equipment of the type mentioned above are discussed along with fully analog data analysis equipment and procedures in Chapter 8. Digital computer techniques including FFT procedures are presented in Chapter 9. The material in Chapter 9 is appropriate for random data analysis by either general purpose or special purpose digital computers.

PROBLEMS

- 1 Suppose acceleration data are measured using a seismic accelerometer as illustrated in Figure 7.2 where $\zeta = 0.7$ and $f_n = 100$ cps. Assume the power spectral density function of the data is calculated over the frequency range from zero to 100 cps. Determine the magnitude of the bias error in the resulting power spectrum at 100 cps which can be attributed to the nonuniform gain factor of the accelerometer.
- 2 Assume the autocorrelation function of the acceleration data in Problem 1 is also calculated. Describe the nature of the error, if any, in the resulting autocorrelogram which can be attributed to the nonuniform phase factor of the accelerometer.
- 3 Suppose 10 random data signals are to be recorded over a frequency range from zero to 1000 cps using pulse code modulation (PCM) with 12 pulses per code group. If the 10 signals are to be multiplexed onto a single recorder channel,
 - (a) how much recorder bandwidth is required?
 - (b) what is the resulting signal to noise ratio of the recorded data?
- 4 If random data signals must be recorded on a magnetic tape machine which is known to exhibit excessive flutter, which type(s) of modulation would be preferable?
- 5 Assume power spectral density functions of sample time history records are to be calculated by digital procedures. If it is known that the data exist only in the frequency range below 500 cps,
 - (a) what is the minimum sampling rate which can be used in digitizing the original time history?
 - (b) is this minimum sampling rate also required for the calculation of autocorrelation functions?
 - (c) is this minimum sampling rate also required for the calculation of probability density functions?
- 6 In an analog to digital conversion operation on time history data, how many quantization levels are required if the resulting digital data are to have a potential dynamic range (peak signal to rms noise ratio) of 100:1 (40 dB)?

7. Suppose an otherwise random data signal includes a sine wave with an rms value of 1 volt. Assume the power spectral density function of the signal is measured using a resolution filter bandwidth of $B_c = 10$ cps. If the power spectral density of the random portion of the signal in the vicinity of the sine wave frequency is $G = 0.1$ volts²/cps, what will be the indicated power spectral density (on the average) at the frequency of the sine wave?
8. For the combined random-sinusoidal signal in Problem 7, what resolution bandwidth would be required to produce an indicated spectral density at the frequency of the sine wave which would be 10 times greater (on the average) than the spectral density of the random portion of the data.
9. Referring to the general data analysis procedure for single records presented in Figure 7.13, explain why a test for stationarity is suggested prior to a test for periodicities and/or normality.
10. Referring to the general data analysis procedure for multiple records presented in Figure 7.14, explain why a coherence function estimate should always accompany a frequency response function calculation.

CHAPTER 8

ANALOG DATA ANALYSIS PROCEDURES

Data acquisition and processing generally culminates with a data analysis effort as outlined in Section 7.5. The analysis might be accomplished with either analog or digital equipment, or a combination of both. This chapter describes the basic equipment and procedures employed for analog data processing. Included in the discussions are recommendations for the selection of analysis parameters. Note that these recommendations constitute only rules of thumb based upon the practical experience of the authors and others. Each such recommendation is accompanied by appropriate discussions and qualitative justifications. Nevertheless, the specific applications for the analyzed data or the specific characteristics of the actual instruments used for the analysis may often warrant deviations from the suggested procedures. Further note that all discussions assume the data to be analyzed are continuous voltage-time history signals representing sample records from stationary (ergodic) random processes. Procedures for nonstationary data analysis are considered in Chapter 10.

8.1 MEAN AND MEAN SQUARE VALUES

The most fundamental analog instrument used in random data analysis is the *voltmeter*, a term applied to a class of devices which detect the mean and/or mean square value of voltage signals. Voltmeters are widely used as independent instruments for the measurement of central tendency and dispersion of data signals. More important, they are incorporated in most other analog data analysis instruments to monitor input signal levels and detect output

parameter analogs. Hence consideration of their operating principles and accuracy is warranted as background for the discussions of more elaborate data analysis instruments.

8.1.1 Basic Instrument Requirements

MEAN VALUES. Given a sample voltage time history record $x(t)$ from a stationary random signal, the mean value μ_x of the signal may be estimated from Equation (6.10) as follows.

$$\hat{\mu}_x = \frac{1}{T} \int_0^T x(t) dt \quad (8.1)$$

In words, the mean value is estimated by simply averaging the instantaneous value of the signal over the sampling time T .

This operation is accomplished by a direct current voltmeter, which will be called a DC voltmeter for convenience. In general, a DC voltmeter consists of a simple averaging circuit followed by a readout device. The required average may be accomplished in one of two ways. The first approach is to feed the signal into a true integrating circuit (operational amplifier with feedback condenser) which computes, when divided by the sampling time, a one-number average value of the signal after a specific time interval T . This procedure, which will be called true averaging, is a direct functional representation of the mathematical averaging operation defined in Equation (8.1). The second approach is to feed the signal into a low-pass filter (usually of the RC type) which continuously smooths the signal fluctuations to produce, after three or four RC time constants have elapsed, a continuous estimate of the average value of the signal. This procedure, which will be called RC averaging, is a practical implementation of the mathematical averaging operation defined in Equation (8.1).

MEAN SQUARE VALUES. The mean square value Ψ_x^2 of the signal may be estimated from Equation (6.24) as follows.

$$\hat{\Psi}_x^2 = \frac{1}{T} \int_0^T x^2(t) dt \quad (8.2)$$

In words, the mean square value is estimated by the following operations: (1) squaring of the instantaneous value of the signal; (2) averaging the squared instantaneous value over the sampling time. An estimate of the root mean square (rms) value may be obtained by taking the square root of the measured mean square value.

The preceding operations are accomplished by a true mean square (or root mean square) voltmeter, which will be called a TMS voltmeter for convenience. One type of TMS voltmeter in current use instantaneously squares the voltage of the applied signal using a detector with a square law

characteristic. The squared voltage signal is time-averaged using either true averaging or RC averaging. When an rms value is desired, the required square root may be accomplished by a proper readout scale calibration. Other types of TMS voltmeters use detectors which compute rms values directly.

8.1.2 Practical Voltmeter Considerations

The mean value of a voltage signal is a measure of the static portion of the signal whereas the variance is a measure of the dynamic portion of the signal. The mean square value is a measure of both static and dynamic portions, as given by the previously derived relationship

$$\Psi_x^2 = \sigma_x^2 + \mu_x^2 \quad (8.3)$$

In many cases, it is necessary to measure the mean square value of a signal by measuring the static (mean value) and dynamic (variance) portions separately. This is true because voltmeters with desirable frequency response characteristics for high frequency measurements often do not possess DC response capabilities. The measurements produced by such instruments are really variance or standard deviation measurements, although they are usually referred to as mean square or rms measurements.

It should be mentioned that many commercial voltmeters have scales which are calibrated to read the rms voltage of a *sinusoidal* signal, but do not actually measure the rms voltage of the applied signal. These instruments usually consist of a simple rectifier followed by an equivalent low-pass smoothing filter such that a quantity proportional to either the mean absolute value (average rectified value) or peak value of the applied signal is measured. Note that the mean absolute value $\overline{|x|}$ is defined as follows

$$\overline{|x|} = \frac{1}{T} \int_0^T |x(t)| dt \quad (8.4)$$

For either a peak or mean absolute value measurement, a readout in terms of the rms value of a sine wave may be obtained by a proper readout scale calibration. This is easy to accomplish since the peak, mean absolute, and rms values of a sine wave have fixed relationships to one another. Assuming the mean value μ_x is zero, these relationships are as follows:

Instantaneous value	$x(t) = X \sin 2\pi ft$	
Peak value	$X = X$	
Mean absolute value	$\overline{ x } = \frac{2}{\pi} X$	(8.5)
Root mean square value	$\Psi_x = \frac{1}{\sqrt{2}} X$	

Commercial instruments often take advantage of these relationships because peak or mean absolute voltage detection circuits are much cheaper to build than true rms voltage detection circuits. Such instruments are usually called AC voltmeters to indicate that they are usable for only AC (sinusoidal) signals. These AC voltmeters will produce inaccurate readings if they are used to measure complex periodic signals or random signals.

8.1.3 Measurement Accuracy

Any voltage measurement will involve basic errors due to the instrument design capabilities and calibration techniques. Such instrument errors are a function of the initial quality of the instrument and its maintenance. However, the measurement of random signal data involves an additional statistical error as discussed in Section 6.2. This statistical error will generally include a bias term defining the systematic portion of the error and a variance term defining the random portion of the error. For both mean and mean square value estimates measured using Equations (8.1) and (8.2), it is shown in Section 6.2 that these measurements will be unbiased (the bias term equals zero). The variance term in both cases is a function of the spectral characteristics of the signal and the sampling time.

For the special case of bandwidth limited Gaussian white noise with a bandwidth of B cps, the statistical error of a mean value measurement is given in terms of the normalized rms error ϵ by Equation (6.23) as

$$\epsilon = \frac{\text{s.d. } [\hat{\mu}_x]}{\mu_x} \approx \frac{1}{\sqrt{2BT}} \left(\frac{\sigma_x}{\mu_x} \right) \quad (8.6)$$

For mean square value measurements where $\mu_x = 0$, the normalized rms error is given by Equation (6.35) as

$$\epsilon = \frac{\text{s.d. } [\hat{\Psi}_x^2]}{\Psi_x^2} \approx \frac{1}{\sqrt{BT}} \quad (8.7)$$

If a TMS voltmeter reads out in root mean square values instead of mean square values, it can be shown that the appropriate expression for the normalized rms error, assuming $\mu_x = 0$, is

$$\epsilon = \frac{\text{s.d. } [\hat{\Psi}_x]}{\Psi_x} \approx \frac{1}{2\sqrt{BT}} \quad (8.8)$$

which is one-half the error for mean square value measurements. In these error formulas, T is the equivalent true averaging time for the measurement.

8.1.4 Averaging Time

The averaging time T in Equations (8.6)–(8.8) is an idealized quantity. That is, T is the time for a true average as indicated by the integration in Equations (8.1)–(8.2). In actual practice, this averaging procedure is often accomplished by using a low pass RC smoothing filter, as mentioned in Section 8.1.1. Furthermore, the available data are usually limited to some sample record of finite length. These factors must be considered in determining an appropriate value for T in Equations (8.6)–(8.8).

First, consider a voltmeter which computes an average by performing an integration over a specific time interval T_a (true averaging). The integration time T_a is equivalent to the idealized averaging time T in Equations (8.6)–(8.8). However, the value of T can never be longer than the length of the sample time history record obtained for analysis. If the available sample record length is T_r seconds long, the statistical error of the resulting measurement can never be less than the value given by Equations (8.6)–(8.8) where $T = T_r$. In other words, nothing is to be gained by averaging over a given amount of information more than once. Hence, for the case of true averaging, the time T in Equations (8.6)–(8.8) should actually be

$$\begin{aligned} T &= T_a & \text{for } T_a \leq T_r \\ T &= T_r & \text{for } T_r < T_a \end{aligned} \quad (8.9)$$

where T_a is the true averaging (integration) time and T_r is the sample record length. Clearly, from Equation (8.9), the most efficient averaging procedure which will minimize the statistical error is to use a true averaging time which is equal to the length of the available sample record.

Now consider a voltmeter which computes an average by smoothing with a low pass RC filter having a time constant K (RC averaging). An RC averaging filter produces a continuous measurement which at any instant of time represents a time weighted average of all inputs that have gone before. For purposes of illustration, assume a mean square value is measured using RC averaging. Typical time histories of RC averaged mean square value measurements are illustrated for periodic and random signals in Figure 8.1.

As seen in Figure 8.1, RC averaging for a steady state periodic signal will produce an accurate measurement after a time interval equal to about four or five time constants has elapsed, assuming the time constant is longer than the period of the data. For stationary random signals, the measurement will continue to fluctuate after the original rise. These fluctuations are indicative of the statistical error of the measurement at any instant of time. The question is as follows: For any given averaging time constant and record length, what is the appropriate value of T in Equations (8.6)–(8.8) to define the error of the continuous RC averaged measurement at any instant of time? This question will now be answered for the case of mean square value measurements.

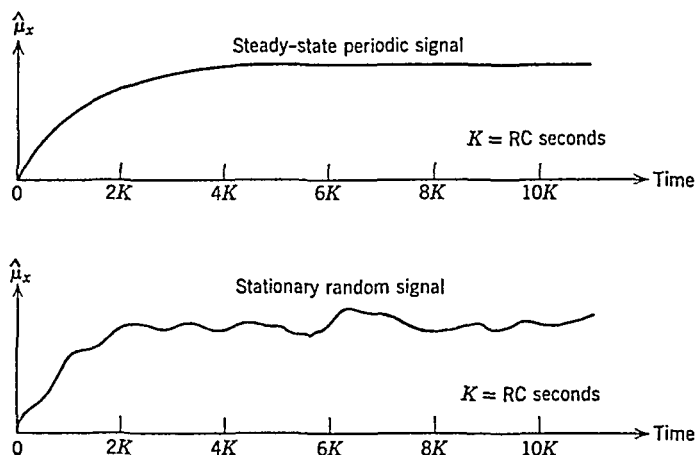


Figure 8.1 Time histories of RC averaged mean square value measurements.

First, consider the case where a mean square value of a random signal $y(t)$ is measured by integration over a time interval T as follows.

$$x(T) = \frac{1}{T} \int_0^T y^2(t) dt \quad (8.10)$$

The mean and variance of this mean square value measurement are given by

$$\begin{aligned} \mu_x &= E[x(T)] = R_y(0) \\ \sigma_x^2 &= E[x^2(T)] - \mu_x^2 = \frac{4}{T^2} \int_0^T (T - \tau) R_y^2(\tau) d\tau \end{aligned} \quad (8.11)$$

Now assume the autocorrelation function $R_y(\tau)$ is given by

$$R_y(\tau) = R_y(0) e^{-2B|\tau|} = \mu_x e^{-2B|\tau|} \quad (8.12)$$

Here, B is actually equal to an equivalent statistical bandwidth B_s to be defined later, but the subscript s will be dropped for clarity. The normalized mean square error of a measurement $x(T)$ is then

$$\begin{aligned} \epsilon_x^2 &= \frac{\sigma_x^2}{\mu_x^2} = \frac{4}{T^2} \int_0^T (T - \tau) e^{-4B|\tau|} d\tau \\ &= \frac{1}{BT} [1 - e^{-4BT}] + \frac{1}{4B^2T^2} [(4BT + 1) e^{-4BT} - 1] \end{aligned} \quad (8.13)$$

For the special case of interest where the BT product is relatively large compared to unity, Equation (8 13) reduces to

$$\epsilon^2 \approx \frac{1}{BT} \quad \text{for } BT \gg 1 \quad (8 14)$$

Now consider the case where a mean square value of a random signal $y(t)$ is measured by smoothing the instantaneous squared values with a low pass RC filter. Let the RC time constant for the smoothing filter be K and the available sample record length be T_r . The mean square value at time T_r is

$$z(K, T_r) = \int_0^{T_r} y^2(t) h(T_r - t) dt \quad (8 15)$$

where, for a low pass RC filter,

$$h(T_r - t) = \frac{1}{K} e^{-(T_r - t)/K} \quad \text{for } (T_r - t) \geq 0$$

The mean and variance of the mean square value $z(K, T_r)$ are given by

$$\mu_z = E[z(K, T_r)] = R_y(0)[1 - e^{-T_r/K}] \quad (8 16a)$$

$$\sigma_z^2 = E[z^2(K, T_r)] - \mu_z^2 = \frac{2}{K} \int_0^{T_r} R_y^2(\tau) [e^{-\tau/K} - e^{-2T_r/K} e^{\tau/K}] d\tau \quad (8 16b)$$

Now, assume as before that the autocorrelation function $R_y(\tau)$ is as given by Equation (8 12). The variance of the mean square value becomes

$$\begin{aligned} \sigma_z^2 &= \frac{2R_y^2(0)}{K} \int_0^{T_r} e^{-4B\tau} [e^{-\tau/K} - e^{-2T_r/K} e^{\tau/K}] d\tau \\ &= \frac{2R_y^2(0)}{K} \left[\frac{1 - e^{-4BT_r} e^{-T_r/K}}{4B + (1/K)} + \frac{e^{-4BT_r} e^{-T_r/K} - e^{2T_r/K}}{4B - (1/K)} \right] \end{aligned} \quad (8 17)$$

For the special case where the BK product is relatively large compared to unity, Equation (8 17) reduces to

$$\sigma_z^2 \approx \frac{R_y^2(0)}{2BK} [1 - e^{-2T_r/K}] \quad \text{for } 4BK \gg 1$$

Hence the normalized variance of a measurement of $z(K, T_r)$ is

$$\epsilon_z^2 = \frac{\sigma_z^2}{\mu_z^2} = \frac{1}{2BK} \left(\frac{1 - e^{-2T_r/K}}{[1 - e^{-T_r/K}]^2} \right) \quad \text{for } 4BK \gg 1 \quad (8 18)$$

Equation (8.18) defines the normalized variance of a continuous RC averaged mean square measurement for any given RC averaging time constant and record length. By comparing Equations (8.18) and (8.14), an equivalent true averaging time for an RC averaged measurement is obtained as follows

$$T = \frac{2K[1 - e^{-T_r/K}]^2}{[1 - e^{-2T_r/K}]} \quad (8.19)$$

It can be shown that the above development, when applied to mean value measurements rather than mean square value measurements, will produce the same result presented in Equation (8.19).

From Equation (8.19), it is seen that the value of T approaches $2K$ as T_r becomes very long compared to K . Furthermore, by expanding the exponentials, it can be shown that the value of T approaches T_r as K becomes very long compared to T_r . That is

$$\begin{aligned} T &\approx 2K && \text{for } K \ll T_r \\ T &\approx T_r && \text{for } T_r \ll K \end{aligned} \quad (8.20)$$

Note that if a mean or mean square value is measured using RC averaging, the resulting measurement generally will be biased as indicated by Equation (8.16a), which assumes the measurement is computed by viewing the data over one sample record length. This bias error will be substantial for the case where $T_r \leq K$. However, the error can be avoided by forming the sample record into a loop and recirculating the record to obtain a continuous presentation of the data signal. This will allow time for the RC averaging filter to respond fully. After four or five time constants have elapsed, the bias error in the measurement will be negligible. Note that the normalized error expression in Equation (8.18) still applies to the unbiased estimate obtained by recirculation. This is true because the recirculation effectively changes only a scale factor which does not change the normalized variance of an estimate.

From Equations (8.7) and (8.19), the minimum attainable normalized rms error of an RC averaged mean square value measurement is $\epsilon = 1/\sqrt{BT_r}$, which is exactly the same as for a true averaged measurement. Of course, this minimum error can be achieved by RC averaging only in the limit as $K \rightarrow \infty$. However, the error approaches the minimum value rapidly as K becomes larger than $T_r/2$. For example, if $K = T_r$, the normalized rms error would be $\epsilon = 1.04/\sqrt{BT_r}$, or only 4 percent greater than the minimum value. Hence very little is to be gained by using an RC averaging time constant which is longer than one sample record length. Note that all of these conclusions apply to mean value measurements as well.

8.2 PROBABILITY DENSITY FUNCTIONS

8.2.1 Basic Instrument Requirements

Given a sample voltage time history record $x(t)$ from a stationary random signal, the probability density function $p(x)$ of the signal may be estimated from Equation (6.39) as follows

$$\hat{p}(x) = \frac{T_x}{TW} \quad (8.21)$$

Here, T_x is the time spent by the signal $x(t)$ within a narrow voltage interval having a width of W volts and a center voltage of x volts. In words, the probability density function is estimated by the following operations

- 1 Amplitude filtering of the signal by a narrow amplitude window having a window width of W volts
- 2 Measurement of the total time spent by the signal within the window
- 3 Averaging of the time spent within the window over the sampling time
- 4 Division of the average portion of the time spent within the window by the window width W

As the center voltage of the window is moved, a plot of the probability density function versus voltage level is obtained.

The preceding operations are accomplished by an analog amplitude probability density analyzer, which will be called an APD analyzer for convenience. In general, an APD analyzer measures the time spent by a signal within some narrow voltage interval by use of an amplitude window (a pair of Schmidt triggers or similar circuit) which actuates an AND gate on the output of a clock pulse generator. Each time the input signal level from the sample record enters the window, the AND gate opens and passes clock pulses. When the signal level leaves the window, the AND gate closes. This train of pulses from the AND gate is then averaged over the total sampling time to obtain the average portion of time spent by the signal within the window. The required division by the window width W may be obtained by a proper scale calibration. As for voltmeters, the required time averaging may be accomplished by true averaging, or by RC averaging, as discussed in Section 8.1.1. The scan required to cover the voltage range of interest is usually accomplished by shifting the signal in voltage past a fixed amplitude window through mixing the signal and a DC bias voltage. A functional block diagram of an APD analyzer is shown in Figure 8.2. It should be mentioned that an APD analyzer might be constructed with a collection of contiguous amplitude windows which together cover the voltage range of interest. For this case, no scan would be needed to measure

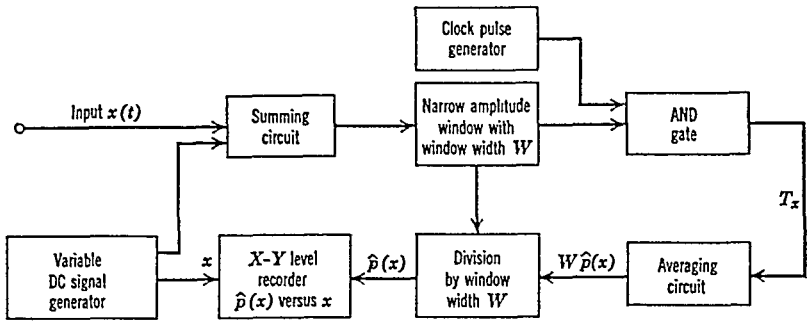


Figure 8.2 Functional block diagram of probability density analyzer.

a probability density plot. However, the single-window APD analyzer is far more common in practice.

8.2.2 Amplitude Resolution and Averaging Time

The statistical error in probability density estimates involves both a bias term and a variance term as given by Equation (6.57). In terms of controllable analysis parameters, the bias error is a function of the amplitude window width W ($\epsilon_b \rightarrow 0$ as $W \rightarrow 0$) while the random error is a function of the product WT ($\epsilon_r \rightarrow 0$ as $WT \rightarrow \infty$). As noted in Section 6.3.3, the bias error will usually be less than $\epsilon_b = 0.01$ for $W \leq 0.2\sigma_x$ assuming the probability density function is approximately Gaussian. Most commercial APD analyzers have a fixed window width of $W = 0.1\sigma_x$. Hence the bias error in analog probability density analysis is usually negligible compared to other sources of error.

Given a fixed amplitude window width W , the random error in probability density analysis is then controlled by the averaging time T . Commercial APD analyzers may perform the required time average by an integration over a specified time interval T_a (true averaging) or by continuous smoothing with a low-pass RC filter (RC averaging) as previously discussed in Section 8.1.4. In either case, a wide range of averaging time options are usually provided in the selectable controls of the analyzer. If true averaging is employed, the integration time should be set approximately equal to the sample record length. If RC averaging is employed, it is recommended that the time constant for the RC averaging filter be set equal to the sample record length. That is,

$$\begin{aligned} T_a &= T_r & \text{for true averaging} \\ K &= T_r & \text{for RC averaging} \end{aligned} \quad (8.22)$$

where T_a is the true averaging (integration) time in seconds, T_r is the sample

record length in seconds, and K is the RC averaging time constant in seconds

The suggested averaging times in Equation (8.22) assume the available sample record is of limited length T_r . There are sometimes situations where the available sample record is effectively unlimited. For example, the sample record of interest might be the output of a noise generator which can be permitted to operate as long as desired. For this case, the selection of an averaging time is governed by random error considerations detailed in Section 6.3.1

8.2.3 Scan Rate and Analysis Time

For a multiple window APD analyzer, the probability density at all signal levels of interest would be concurrently measured. This means that the analysis time would effectively be equal to the averaging time T . For a single window APD analyzer, however, the probability density at all signal levels of interest must be measured by scanning through the desired range of values with the single amplitude window while the data are continually displayed for analysis by recirculation of the sample record. The required scan may be accomplished either by discrete steps of the window or by a continuous sweep of the window. A stepped scan is often used with true averaging, whereas a continuous scan is normally used with RC averaging. In either case, if the scan rate is too fast, unnecessary errors will be introduced into the analysis.

For the case of true averaging, the scan rate should be sufficiently slow to permit all information at each signal level to be reflected in each average value computation. That is, the scan rate should be less than one window width per averaging time. Hence the basic scan rate limitation for true averaged probability density measurements is

$$R_s \leq \frac{W}{T_a} \quad (8.23a)$$

where T_a is the true averaging (integration) time in seconds, W is the amplitude window width in volts, and R_s is the scan rate in volts per second.

For the case of RC averaging, the scan rate must be sufficiently slow to allow the RC averaging filter to respond to abrupt changes in the probability density function, as illustrated in Figure 8.1. Otherwise, the RC averaging filter will tend to smooth or fair through abrupt probability density changes producing a bias error. For the worst case of a step change in the probability density function, this bias error will be less than 2 percent of reading after four RC time constants have elapsed. It follows that each incremental amplitude level must be viewed through the window for at least four RC time constants. If this worst-case error of less than 2 percent is to be

achieved. Hence a recommended conservative scan rate limitation for RC averaged probability density measurements is

$$R_s \leq \frac{W}{4K} \quad (8.23b)$$

where K is the RC averaging time constant in seconds and W is the amplitude window width in volts. Of course, a more rapid scan rate can be used if one is prepared to accept a greater bias error, or if variations in the probability density function are less abrupt. For example, experimental studies on a digital computer indicate that the maximum bias error induced by scanning a Gaussian probability density function from minus three standard deviations to plus three standard deviations is about 3 percent of reading for a scan rate of $R_s = W/2K$. The maximum error rises to about 10 percent of reading for a scan rate of $R_s = W/K$.

Assuming it is desired to minimize the statistical error, the averaging time relationships presented in Equation (8.22) can be substituted into Equation (8.23) to obtain the maximum scan rate as a function of the sample record length with the following results. For the case of true averaging, $R_s \leq W/T_r$ since $T_a = T_r$ for minimum statistical error. For the case of RC averaging, $R_s \leq W/4T_r$ since $K = T_r$ is recommended.

Now consider the total required analysis time. If the total voltage range for an APD analysis is V volts, then it follows from Equation (8.23) that the minimum analysis time would be

$$T_s \geq \begin{cases} \frac{T_a V}{W} & \text{for true averaging} \\ \frac{4KV}{W} & \text{for RC averaging} \end{cases} \quad (8.24a)$$

$$(8.24b)$$

Example 8.1. Probability Density Analysis Time. Assume the probability density function of a random signal with a mean value of zero and an rms value of 1 volt is to be estimated from a sample record of length $T_r = 10$ seconds over a range of values from -3 volts to $+3$ volts ($V = 6$ volts) using an APD analyzer with a window width of $W = 0.1$ volt. If true averaging is used, $T_a = 10$ seconds and the maximum scan rate is $R_s = 0.01$ volt per second. Hence the minimum analysis time would be $T_s = 600$ seconds, or 10 minutes. If RC averaging is used, $K = 10$ seconds and the maximum recommended scan rate is $R_s = 0.0025$ volt per second. Hence the minimum analysis time would be $T_s = 2400$ seconds or 40 minutes.

8.3 AUTOCORRELATION FUNCTIONS

8.3.1 Basic Instrument Requirements

Given a sample voltage time history record $x(t)$ from a stationary random signal, the autocorrelation function $R_x(\tau)$ of the signal may be estimated from Equation (6.65) as follows

$$\hat{R}_x(\tau) = \frac{1}{T} \int_0^T x(t) x(t + \tau) dt \quad (8.25)$$

In words, the autocorrelation function is estimated by the following operations

- 1 Delaying the signal by a time displacement equal to τ seconds, called the lag time
- 2 Multiplying the signal value at any instant by the value that had occurred τ seconds before
- 3 Averaging the instantaneous product value over the sampling time

As the lag time is moved, a plot of the autocorrelation function versus lag time (autocorrelogram) is obtained

The preceding operations are accomplished by an analog autocorrelation function analyzer, which will be called a correlator for convenience. Some correlators generate the required time displacement τ by use of a magnetic record reproduce system with a variable lag time between record and playback. In other cases, delay lines are used to generate the lag time. The input and output of the lag time generator are then multiplied and time averaged. The required time average may be accomplished either by true averaging or by RC averaging, as discussed in Section 8.1.1. The lag time is variable over a range from zero to the longest sampling times that are anticipated. Since the autocorrelation function is an even function, it is not necessary to make measurements with negative lag times. A functional block diagram of a correlator is shown in Figure 8.3. As for the case of APD analyzers discussed in Section 8.2.1, a correlator might be constructed with a collection of equally spaced lag time generators which together cover the time displacement range of interest. For this case, no lag time scan would be needed to measure the autocorrelogram. Note that the estimate $\hat{R}_x(\tau)$ actually obtained in Figure 8.3 is the average of $x(t - \tau)x(t)$ which equals the average of $x(t)x(t + \tau)$ for stationary signals.

8.3.2 Lag Time Resolution and Averaging Time

As seen from Equation (8.25), the autocorrelation function $R_x(\tau)$ must be estimated at various different lag times τ to obtain a correlogram. This may

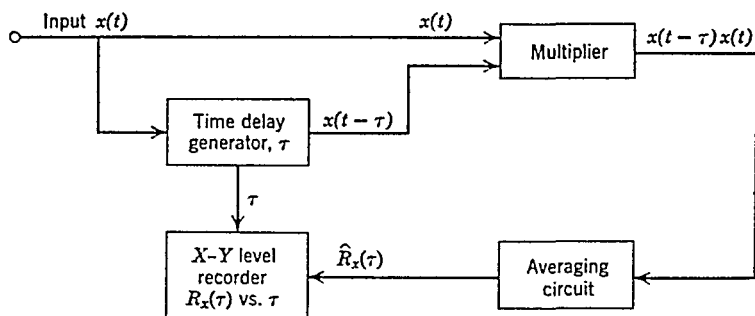


Figure 8.3 Functional block diagram of autocorrelation analyzer.

be accomplished in one of two ways. The first way is to measure a correlation value with the lag time fixed at a specific time displacement of interest. The lag time would then be changed to a new time displacement and the correlation value would again be measured. This stepped scan procedure is usually employed by correlators which incorporate an integration type averaging circuit so that the resulting correlogram will appear as a series of discrete correlation values measured at specific lag times. The second way is to measure correlation values while the lag time is continuously varied through the time displacement range of interest. This continuous scan procedure is normally employed by correlators which incorporate an RC averaging circuit, so that a continuous correlogram is obtained.

For the first method involving a stepped scan, the interval between the lag times determines how well the correlation function is resolved. As the interval is made smaller, the resolution of the measured correlogram will be improved. However, the number of correlation measurements required to cover a given time displacement range will be increased. Hence the selection of a lag time interval will always involve a compromise between measurement resolution and analysis time. The maximum permissible interval which can be employed for any given analysis is, of course, a function of the exact correlogram for the specific random signal being analyzed. For general applications, however, it can be said that at least two correlation values per cycle are needed to define data at any given frequency. This follows directly from the fact that correlation functions and power spectral density functions are Fourier transform pairs. Thus a minimum resolution criterion for autocorrelation measurements is given by $h \leq 1/(2f_m)$ where h is the interval between lag times in seconds and f_m is the maximum frequency of significant data in cps.

For the second method involving a continuous scan lag time, a slightly different interpretation of measurement resolution is required. This procedure

produces a continuous correlogram (assuming RC averaging is used) where correlation values appear to be defined for every lag time over the time displacement range of the analysis. However, the value at any given lag time is actually a time-weighted average of the correlation characteristics at all lag times that had been previously scanned. These effects tend to smooth out rapid changes in the correlogram and thus introduce a bias error which reduces the resolution of the measurement. As the lag time scan rate is made slower, the resolution of the measured correlogram will be improved. However, as for the case of measurements at discrete lag times, the analysis time will be increased producing the same requirement for a compromise between measurement resolution and analysis time. The maximum permissible scan rate which can be employed while maintaining acceptable resolution is discussed in Section 8.3.3.

The general discussions and relationships for averaging time presented for probability density measurements in Section 8.2.2 also apply to autocorrelation measurements. Specifically, the most practical approach to autocorrelation analysis is to select an averaging time which is as long as the available sample record so that minimum statistical errors are obtained for all correlation values. Hence the integration time for true averaging or the time constant for RC averaging should be as given in Equation (8.22). If the available sample record length is unlimited, the selection of an averaging time should be governed by random error considerations detailed in Section 6.4.

8.3.3 Scan Rate and Analysis Time

For a multiple lag time correlator, the autocorrelation functions at all time displacements of interest would be concurrently measured. This means that the analysis time would effectively be equal to the averaging time T . However, for a single lag time correlator, the autocorrelation function at various time displacements of interest must be measured by scanning with a single lag time generator while the sample record is continuously recirculated. If this scan is too fast, unnecessary errors will be introduced into the analysis.

For a stepped scan where true averaging is used, if all the information at each time delay is to be reflected in the measurement, the scan rate must clearly be limited to one lag time per averaging time. Hence the basic scan rate limitation for a stepped scan with true averaging is

$$R_s \leq \frac{h}{T_a} \quad (8.26a)$$

where T_a is the true averaging (integration) time in seconds, h is the interval between lag times in seconds, and R_s is the scan rate in seconds per second.

For a continuous scan where RC averaging is used, if abrupt changes in

the correlation function are to be accurately detailed for an equivalent resolution interval of h , it is suggested that the scan rate be limited to one resolution interval per four time constant for the reasons given in Section 8.2.3. Hence the recommended scan rate limitation for a continuous scan with RC averaging is

$$R_s \leq \frac{h}{4K} \quad (8.26b)$$

where K is the RC averaging time in seconds and h is the desired resolution interval in seconds.

Assuming it is desired to minimize the statistical error, the averaging time relationships presented in Equation (8.22) can be substituted into Equation (8.26) to obtain the maximum scan rate as a function of the sample record length with the following results. For the case of true averaging, $R_s \leq h/T_r$, since $T_a = T_r$ for minimum error. For the case of RC averaging, $R_s \leq h/4T_r$, since $K = T_r$ is recommended.

Now consider the total required analysis time. If the total lag time range for a correlation analysis is D seconds, then it follows from Equation (8.26) that the minimum analysis time would be

$$T_s \geq \begin{cases} T_a D/h & \text{for true averaging} \\ 4KD/h & \text{for RC averaging} \end{cases} \quad (8.27a)$$

$$(8.27b)$$

Example 8.2. Correlation Analysis Time. Assume the autocorrelation function of a random signal is to be estimated from a sample record of length $T_r = 10$ seconds over a time displacement range of 0 to 0.1 second ($D = 0.1$ second), where time displacements down to $h = 0.0025$ second are to be resolved. If true averaging is used, $T_a = 10$ seconds and the maximum scan rate is $R_s = 0.00025$ second/second. Hence the minimum analysis time would be $T_s = 400$ seconds or about 7 minutes. If RC averaging is used, $K = 10$ seconds and the recommended maximum scan rate is $R_s = 0.000625$ second/second. Hence the minimum analysis time would be $T_s = 1600$ seconds or about 27 minutes.

8.4 POWER SPECTRAL DENSITY FUNCTIONS

8.4.1 Basic Instrument Requirements

Given a sample voltage time history record $x(t)$ from a stationary random signal, the power spectral density function $G_x(f)$ of the signal may be estimated from Equation (6.82) as follows.

$$\hat{G}_x(f) = \frac{1}{B_e T} \int_0^T x^2(t, f, B_e) dt \quad (8.28)$$

Here, $x(t, f, B_e)$ is that portion of $x(t)$ passed by a narrow band pass filter with a bandwidth of B_e cps and a center frequency of f cps. In words, the power spectral density function is estimated by the following operations

- 1 Frequency filtering of the signal by a narrow band pass filter having a bandwidth of B_e cps
- 2 Squaring of the instantaneous value of the filtered signal
- 3 Averaging the squared instantaneous value over the sampling time
- 4 Division of the mean square output by the bandwidth B_e

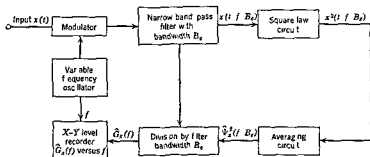


Figure 8.4 Functional block diagram of power spectral density analyzer

As the center frequency of the narrow band pass filter is moved, a plot of the power density function versus frequency (power spectrum) is obtained.

The preceding operations are accomplished by an analog power spectral density analyzer, which will be called a PSD analyzer for convenience. Most PSD analyzers perform the filtering operation by heterodyning the input data signal past a highly selective narrow band pass filter with a fixed center frequency. The mean square value computation is effectively equivalent to a TMS voltmeter measurement as discussed in Section 8.1.1. The required division by the bandwidth B_e may be obtained by a proper scale calibration. A functional block diagram of a PSD analyzer is shown in Figure 8.4. As for the case of APD analyzers and correlators discussed in Sections 8.2.1 and 8.3.1, a PSD analyzer can be constructed with a collection of contiguous narrow band pass filters which together cover the frequency range of interest. For this case, no frequency scan would be needed to measure a power spectrum. Such multiple filter PSD analyzers are widely used in practice.

It should be mentioned that a PSD analyzer is nothing more than a common harmonic wave analyzer with a TMS voltmeter circuit. The conventional harmonic wave analyzer usually employs a simple AC voltmeter circuit which computes a mean absolute value of the filtered signal. If one is prepared to make definite assumptions concerning the probability density function of the signal to be analyzed, the use of a conventional wave analyzer as a PSD analyzer is possible. To be more specific, assume the random signal to be analyzed has a Gaussian probability density function. Further assume a power spectrum is to be measured using an instrument which computes a mean absolute value as given by Equation (8.4). For a Gaussian density function with zero mean, the mean absolute value is

$$\overline{|x|} = \frac{2\sqrt{2}}{\pi} \hat{\Psi}_x \quad (8.29)$$

It follows that the power spectral density function may be estimated from Equation (8.28) as follows.

$$\hat{G}_x(f) = \frac{\pi^2}{8B_e} (\overline{|x(f, B_e)|})^2 \quad (8.30)$$

For a constant bandwidth analysis, the coefficient $\pi^2/(8B_e)$ is simply a calibration constant. The required square of the mean absolute value $\overline{|x|}$ may be obtained by an appropriate readout scale calibration. In fact, if the analyzer output is passed through a logarithmic converter, the squaring may be accomplished by a proper adjustment of the readout sensitivity.

8.4.2 Frequency Resolution and Averaging Time

The statistical error in power spectral density estimates involves both a bias term and a variance term as given by Equation (6.98). In terms of controllable analysis parameters, the bias error is a function of the resolution bandwidth B_e ($\epsilon_b \rightarrow 0$ as $B_e \rightarrow 0$) while the random error is a function of the product $B_e T$ ($\epsilon_r \rightarrow 0$ as $B_e T \rightarrow \infty$). An appropriate value for B_e should be established first. This selection is critical since the bias errors in spectral density estimates can be quite severe, as discussed in Section 6.9. Most commercial PSD analyzers offer a wide range of resolution bandwidth selections. Further flexibility in the selection of an appropriate bandwidth can be achieved by expanding or compressing the time base of the input data signal, to be discussed later.

Having fixed the resolution bandwidth B_e , the random error in the power spectral density analysis is then controlled by the averaging time T . As for the previously discussed data analysis instruments, PSD analyzers may perform the required time average by an integration over a specific time interval T_a (true averaging) or by continuous smoothing with a low-pass RC filter (RC

averaging) Many commercial PSD analyzers provide both types of averaging circuits with a wide range of selectable averaging time options. In either case, if the available sample record is of limited length T_r , the averaging time should be selected in accordance with the requirements of Equation (8 22). If the available sample record length is unlimited, the selection of an averaging time should be governed by random error considerations detailed in Sections 6 5 1 and 6 9.

8 4 3 Scan Rate and Analysis Time

The scan rate considerations for a power spectral density analysis are exactly the same as discussed for a probability density analysis in Section 8 2 3 except the frequency bandwidth B_e replaces the amplitude window width W . Hence the basic scan rate limitation for power spectral density measurements is

$$R_s \leq \begin{cases} B_e/T_a & \text{for true averaging} \end{cases} \quad (8 31a)$$

$$R_s \leq \begin{cases} B_e/4K & \text{for RC averaging} \end{cases} \quad (8 31b)$$

where T_a is the true averaging (integration) time in seconds, K is the RC averaging time constant in seconds, B_e is the analyzer filter bandwidth in cps, and R_s is the scan rate in cps per second.

Note that there is an additional scan rate restriction imposed by the required rise time of the narrow band pass analyzer filter. The rise time for an ideal band pass filter of bandwidth B_e is about $T_r = 1/B_e$ seconds. This means that the scan rate must be restricted to less than $(B_e/T_r) = B_e^2$ cps/second if the analyzer filter is to respond fully to abrupt changes in the power spectrum being analyzed. Comparing this relationship to the scan rate restriction given by Equation (8 31), it is seen that Equation (8 31) is always more restrictive if the $B_e T$ product is greater than unity. Since the $B_e T$ product in practice must be greater than unity if reasonable accuracy is to be achieved in the measurements, the restrictions given by Equation (8 31) will apply and the B_e^2 scan rate restriction can be ignored.

Assuming it is desired to minimize the statistical error, the averaging time relationships presented in Equation (8 22) can be substituted into Equation (8 31) to obtain the maximum scan rate as a function of the sample record length with the following results. For the case of true averaging, $R_s \leq B_e/T_r$ since $T_a = T_r$ for minimum error. For the case of RC averaging, $R_s \leq B_e/4T_r$ since $K = T_r$ to be within 4 percent of minimum statistical error.

Now consider the total required analysis time. If the total frequency range for a PSD analysis is F cps, then it follows from Equation (8 31) that the minimum analysis time would be as follows

$$T_s \geq \begin{cases} T_a F/B_e & \text{for true averaging} \end{cases} \quad (8 32a)$$

$$T_s \geq \begin{cases} 4KF/B_e & \text{for RC averaging} \end{cases} \quad (8 32b)$$

The results in Equation (8.32) assume that the data are analyzed with a constant resolution bandwidth B_e and a real time record length T_r . Neither situation is necessary or even desirable in practice. Quite to the contrary, changing the resolution bandwidth during the frequency scan and/or compressing the time base of the data can greatly reduce the required analysis time with no loss in statistical accuracy, as demonstrated in the next two sections.

8.4.4 Variable Resolution Bandwidth

The recommended averaging times in Equation (8.22) will produce measurements with the same statistical error at all frequencies, as long as the PSD analyzer filter bandwidth is not changed. There are situations in practice, however, where it is desirable to change the filter bandwidth during an analysis. As discussed in Section 6.9, the frequency response functions of physical systems often display sharp peaks with frequency bandwidths that are proportional to the center frequency of the peak. Because the response power spectrum for a physical system is proportional to the square of the frequency response function magnitude (gain factor), these constant percentage bandwidth peaks tend to appear in random signals representing the response of such systems. It follows that the bandwidth B_e for a power spectral density analysis can often be increased in direct proportion to the frequency of the analysis while still maintaining proper measurement resolution. The result will be increased measurement accuracy at higher frequencies. Another approach is to maintain a constant measurement accuracy by reducing the averaging time as the analysis frequency increases. Referring to Equation (8.32), it is seen that either approach will significantly reduce the total required analysis time.

Example 8.3. Power Spectrum Analysis Time. Assume the power spectrum of a random signal is to be estimated from a sample record length $T_r = 5$ seconds over a frequency range from 10 to 1000 cps ($F = 990$ cps) using a PSD analyzer with a filter bandwidth of $B_e = 5$ cps. If true averaging is used, $T_a = 5$ seconds and the maximum scan rate is $R_s = 1.0$ cps per second. Hence the minimum analysis time would be $T_s = 990$ seconds or 16.5 minutes. If RC averaging is used, $K = 5$ seconds and the maximum recommended scan rate is $R_s = 0.25$ cps per second. This gives a minimum analysis time of $T_s = 3960$ seconds or 66 minutes.

Consider the case where the analyzer filter bandwidth is changed during the analysis in a manner which maintains proper resolution. Let the bandwidth be changed to $B_e = 10$ cps when the center frequency reaches 100 cps, to $B_e = 20$ cps when the center frequency reaches 200 cps, and to $B_e = 50$ cps when the center frequency reaches 500 cps. The frequency range of the spectrum analysis has now been broken up into four subranges, each

scanned with a different bandwidth. Assuming the averaging time is not changed, and that true averaging is used, the maximum scan rate for each subrange is as follows

$$\begin{array}{lll} B_1 = 5 \text{ cps} & F_1 = 90 \text{ cps} & R_{s1} = 1.0 \text{ cps/sec} \\ B_2 = 10 \text{ cps} & F_2 = 100 \text{ cps} & R_{s2} = 2.0 \text{ cps/sec} \\ B_3 = 20 \text{ cps} & F_3 = 300 \text{ cps} & R_{s3} = 4.0 \text{ cps/sec} \\ B_4 = 50 \text{ cps} & F_4 = 500 \text{ cps} & R_{s4} = 10.0 \text{ cps/sec} \end{array}$$

The total analysis time would be

$$T_s = \frac{F_1}{R_{s1}} + \frac{F_2}{R_{s2}} + \frac{F_3}{R_{s3}} + \frac{F_4}{R_{s4}} = 90 + 50 + 75 + 50 = 265 \text{ seconds}$$

The total analysis time is now 265 seconds or less than 5 minutes, compared to 16.5 minutes for the analysis with a constant bandwidth of $B_s = 5$ cps.

Finally, consider the case where the analysis averaging time is changed when the bandwidth is changed. Assume that a normalized standard error of $\epsilon = 1/\sqrt{B_s T_a} = 0.20$ is acceptable for the entire analysis. Then, the maximum scan rate for each subrange is as follows

$$\begin{array}{lll} B_1 = 5 \text{ cps} & T_{a1} = 5 \text{ sec} & R_{s1} = 1.0 \text{ cps/sec} \\ B_2 = 10 \text{ cps} & T_{a2} = 2.5 \text{ sec} & R_{s2} = 4.0 \text{ cps/sec} \\ B_3 = 20 \text{ cps} & T_{a3} = 1.25 \text{ sec} & R_{s3} = 16.0 \text{ cps/sec} \\ B_4 = 50 \text{ cps} & T_{a4} = 0.5 \text{ sec} & R_{s4} = 100.0 \text{ cps/sec} \end{array}$$

The total analysis time would now be $T_s \approx 139$ seconds or only about 2.3 minutes. These examples clearly illustrate how proper analyzer filter bandwidth and averaging time selections can greatly reduce the time required to perform a power spectral density analysis.

8.4.5 Time Base Compression

In many cases, the upper frequency capability of an analog PSD analyzer may be substantially higher than the highest frequency of interest in the data to be analyzed. If this is true, the frequency range of the data can be expanded for analysis by speeding up the recirculation of the sample data record to compress the time base of the data. This will permit wider bandwidths and shorter averaging times to be used for the analysis. The net result is that, for any given resolution and statistical accuracy, the total analysis time will be decreased by the same factor that the data recirculation rate is increased.

For example, assume the recirculation speed is increased by a factor of r . All spectral peaks in the data will then be increased in bandwidth by a factor of r while the effective record length is reduced by the same factor. This means that the analyzer filter bandwidth required for a given bias error will be increased by a factor of r while the averaging time required for a given random error will be decreased by a factor of r ; that is,

$$B_e' = rB_e \quad T_a' = T_a/r \quad (8.33)$$

It follows from Equation (8.31) that the new scan rate limit becomes

$$R_s' = \frac{B_e'}{T_a'} = \frac{rB_e}{T_a/r} = r^2 R_s \quad (8.34)$$

showing the scan rate for equivalent statistical errors increases by the square of the time base compression factor. Of course, the frequency range F that must be scanned is also increased by a factor of r to $F' = rF$. From Equation (8.32), the final result is a total analysis time of

$$T_s' = T_a' F' / B_e' = (T_a/r)(rF)/(rB_e) = T_s/r \quad (8.35)$$

Hence, compressing the data time base by a factor of r also compresses the required analysis time for equivalent statistical errors by a factor of r .

Time base compression is clearly a valuable practical tool for analog power spectral density analysis and should be used whenever possible. It is also the basis for many so called "real time" spectrum analyzers. These instruments are usually hybrid devices which convert the input analog data record to a digital time series. The digital sequence is compressed in time by a factor of up to 10^6 and then reconverted to a continuous analog signal for analysis by a high speed scan over the expanded frequency range.

Time base compression or expansion can also be employed as a technique for varying the effective resolution bandwidth of an analysis. For example, assume it has been determined that a resolution bandwidth of $B_e = 1$ cps is needed to analyze a given sample record of data with an acceptable bias error. Further assume that the narrowest filter incorporated in the available spectrum analyzer is $B_e = 5$ cps. The desired resolution can be achieved by simply analyzing the data at $r = 5$ real time speed. From Equation (8.33), the required bandwidth of the resolution filter for this case would be $B_e' = 5B_e = 5$ cps as desired.

8.4.6 Bandpass Filter Characteristics

From Equation (8.28), it is seen that a knowledge of the PSD analyzer filter bandwidth is required to obtain accurate power spectral density measurements. Furthermore, from Equation (6.98), the value of the filter bandwidth influences the normalized standard error of the measurement. The bandwidth

B_p in these equations is an idealized quantity, that is, B_p is the bandwidth of a narrow band pass filter which is assumed to have a perfect rectangular band pass characteristic. The narrow band pass filters in PSD analyzers do not have such perfect characteristics. Hence, given an actual PSD analyzer filter, the following question arises: What is the proper value of the bandwidth in Equations (8.28) and (6.98)?

The frequency bandwidth of a narrow band-pass filter may be described in many different ways. It is appropriate here to define and clarify three different descriptions of bandwidth which are of interest to the power spectral density measurement problem. These three bandwidths are the *half-power point bandwidth* B_p , the *noise bandwidth* B_n , and the *equivalent statistical bandwidth* B_s . Given any linear filter with a frequency response function of $H(f)$, these three frequency bandwidths terms are defined mathematically as follows:

$$B_p = f_2 - f_1 \quad \text{where } |H(f_1)|^2 = |H(f_2)|^2 = \frac{1}{2} |H_m|^2 \quad (8.36a)$$

$$B_n = \frac{\int_0^\infty |H(f)|^2 df}{|H_m|^2} \quad (8.36b)$$

$$B_s = \frac{\left[\int_0^\infty |H(f)|^2 df \right]^2}{\int_0^\infty |H(f)|^4 df} \quad (8.36c)$$

$|H(f)|$ is the magnitude of the frequency response function (gain factor) at any frequency f and $|H_m|$ is the maximum value of the gain factor.

The half-power point bandwidth B_p , defined in Equation (8.36a), is the frequency interval between the upper and lower frequencies where the filter attenuates an applied signal by 3 dB below maximum transmissibility. This description of bandwidth is convenient because it is easy to measure, but is otherwise of little physical significance for random signal applications.

The noise bandwidth B_n , defined in Equation (8.36b), is the bandwidth of a hypothetical rectangular filter which would pass a signal with the same mean square value as the actual filter when the input is white noise. In other words, B_n is that measure of bandwidth which will satisfy the relationship $\Psi_x^2 = B G_x$ for filters which do not have ideal rectangular band pass characteristics. Hence B_n is clearly a convenient measure of bandwidth to use for normalizing narrow-band mean square value measurements as required for analog power spectral density measurements in accordance with Equation (8.28).

The equivalent statistical bandwidth B_s , defined in Equation (8.36c), is the bandwidth of a hypothetical rectangular filter which would pass a signal with the same mean square value statistical error as the actual filter when

the input is white noise. In other words, B_s is the description of filter bandwidth which is technically appropriate for the random error portion of the statistical error given by Equation (6.98).

For the case of an ideal filter with a rectangular bandpass characteristic, the three descriptions of bandwidth defined in Equation (8.36) are equal. For any other filter characteristic, however, the bandwidth terms will generally be different. Hence the bandpass filters in analog spectrum analyzers should be calibrated in terms of both the noise and statistical bandwidths as given by Equations (8.36b) and (8.36c).

Example 8.4. Calibration of Single-Tuned Filter. Consider the case of a narrow band-pass filter consisting of a single-tuned resonant circuit where the bandwidth is small compared to the filter center frequency. The gain factor of a single-tuned filter is given by

$$|H(f)| = \left[\frac{a^2}{a^2 + (f - f_r)^2} + \frac{a^2}{a^2 + (f + f_r)^2} \right]^{1/2} \quad a \ll f_r \quad (8.37)$$

Note that the maximum value for $|H(f)|$ occurs when $f \approx f_r$ and is approximately equal to unity, as shown below.

$$|H_m| \approx |H(f_r)| = \left[1 + \frac{a^2}{a^2 + 4f_r^2} \right]^{1/2} \approx 1 \quad \text{if } a \ll f_r \quad (8.38)$$

A typical plot of the gain factor versus frequency for a single-tuned band-pass filter is shown in Figure 8.5. This plot is for the special case where the half-power point bandwidth is one-tenth of the center frequency, that is, $B_r = 2a = 0.1f_r$.

From Equation (8.36a), the half-power point bandwidth of this filter is clearly

$$B_r = (f_r + a) - (f_r - a) = 2a \quad (8.39a)$$

From Equation (8.36b), the noise bandwidth is

$$B_n = \int_0^\infty \left[\frac{a^2}{a^2 + (f - f_r)^2} + \frac{a^2}{a^2 + (f + f_r)^2} \right] df = \pi a \quad (8.39b)$$

Finally, from Equation (8.36c), the equivalent statistical bandwidth is

$$B_s = \frac{\left\{ \int_0^\infty \left[\frac{a^2}{a^2 + (f - f_r)^2} + \frac{a^2}{a^2 + (f + f_r)^2} \right] df \right\}^2}{\int_0^\infty \left[\frac{a^2}{a^2 + (f - f_r)^2} + \frac{a^2}{a^2 + (f + f_r)^2} \right]^2 df} = \frac{(\pi a)^2}{\pi a/2} = 2\pi a \quad (8.39c)$$

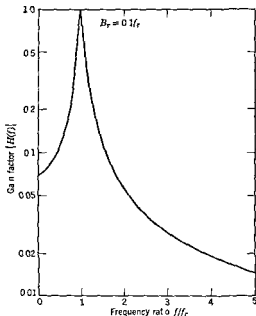


Figure 8.5 Gain factor of single-tuned band pass filter

Hence compared to the half power point bandwidth the noise bandwidth and equivalent statistical bandwidth of a single tuned filter are

$$B_n = \frac{\pi}{2} B_r \quad (8.40a)$$

$$B_s = \pi B_r \quad (8.40b)$$

The meaning of the preceding results is as follows. Assume a power spectral density function is to be measured using a PSD analyzer with a single tuned band pass filter having a half power point bandwidth of B_r cps. Then the proper estimate of power spectral density from Equation (8.28) will be obtained by using B_n for B_s . This gives

$$\hat{G}_x(f) = \frac{2}{\pi B_r T} \int_0^T x^2(t, f; B_r) dt \quad (8.41)$$

On the other hand, the precise expression for the normalized standard error of the measurement will be obtained by using B_s for B_e . This gives

$$\epsilon = \frac{1}{\sqrt{\pi B_s T}} \quad (8.42)$$

8.5 ANALYSIS OF JOINT DATA PROPERTIES

8.5.1 Joint Probability Density Functions

Given two sample voltage time history records $x(t)$ and $y(t)$ from two stationary random signals, the joint probability density function $p(x, y)$ of the two signals may be estimated from Equation (6.61) as follows.

$$\hat{p}(x, y) = \frac{T_{x,y}}{TW_x W_y} \quad (8.43)$$

$T_{x,y}$ is the time spent by the signals $x(t)$ and $y(t)$ simultaneously within the narrow voltage intervals having widths of W_x and W_y volts and center voltages of x and y volts, respectively. In words, the joint amplitude probability density function is estimated by the following operations.

1. Individual amplitude filtering of the two signals $x(t)$ and $y(t)$ by two narrow amplitude windows having widths of W_x and W_y volts.
2. Measurement of the total joint time spent by the two signals while they are simultaneously within the windows.
3. Averaging of the joint time spent within the windows over the sampling time.
4. Division of the average portion of time spent within the windows by the product of the window widths $W_x W_y$.

As the center voltage of each window is moved, a three-dimensional plot of the joint probability density function versus voltage levels x and y is obtained.

The preceding operations are accomplished by an analog joint probability density analyzer, which will be called a JPD analyzer for convenience. A JPD analyzer effectively consists of two simple APD analyzers as illustrated in Figure 8.2, except the amplitude windows in the two individual analyzers are followed by a single AND gate on the output of a clock pulse generator. When the input signal level from one sample record falls within the first amplitude window while the input signal level from a second sample record simultaneously falls within the second amplitude window, the AND gate opens and passes clock pulses indicating the joint time spent simultaneously within the windows. The train of pulses from the AND gate is then averaged

by either true averaging or RC averaging, as for the case of the simple APD analyzer

The amplitude resolution and averaging time considerations for joint probability density analysis are the same as discussed for probability density estimates in Section 8.2.2. Furthermore, the general limitations imposed upon the scan rate are the same as outlined in Section 8.2.3. For this case, however, there are two signals whose range of values must be scanned. In effect this means that the x window must be scanned through the entire range of interest for $x(t)$ while the y window is fixed at each of the numerous positions required to cover the entire range of interest for $y(t)$. If the total range of values of interest for $x(t)$ and $y(t)$ is V_x and V_y volts, respectively, the minimum analysis time is

$$T_a \geq \begin{cases} \frac{T_a V_x V_y}{W_x W_y} & \text{for true averaging} \\ \frac{4KV_x V_y}{W_x W_y} & \text{for RC averaging} \end{cases} \quad (8.44a) \quad (8.44b)$$

where T_a is the integration time in seconds, K is the RC averaging time constant in seconds, and W_x and W_y are the amplitude window widths in volts

8.5.2 Cross-Correlation Functions

Given two sample voltage time history records $x(t)$ and $y(t)$ from two stationary random signals, the cross-correlation function $R_{xy}(\tau)$ of the two signals may be estimated from Equation (6.65) as follows

$$\hat{R}_{xy}(\tau) = \frac{1}{T} \int_0^T x(t) y(t + \tau) dt \quad (8.45)$$

In words, the cross correlation function is estimated by the following operations

1. Delaying the signal $x(t)$ relative to the signal $y(t)$ by a time displacement equal to τ seconds, called the lag time
2. Multiplying the value of $y(t)$ at any instant by the value of $x(t)$ that had occurred τ seconds before
3. Averaging the instantaneous product value over the sampling time

As the lag time is moved, a plot of the cross-correlation function versus lag time (cross correlogram) is obtained

The preceding operations are accomplished by an analog cross-correlation function analyzer, which is exactly the same instrument as the correlator illustrated in Figure 8.3 except the direct input to the multiplier and the input

to the lag time generator are independent. Lag time resolution, averaging time, scan rate, and analysis time considerations for cross-correlation analysis are basically the same as discussed for autocorrelation analysis in Section 8.3.

8.5.3 Cross-Spectral Density Functions

Given two sample voltage time history records $x(t)$ and $y(t)$ from two stationary random signals, the cross-spectral density function $G_{xy}(f)$ of the two signals may be estimated as follows.

$$\hat{G}_{xy}(f) = \hat{C}_{xy}(f) - j \hat{Q}_{xy}(f) \quad (8.46)$$

$$\hat{C}_{xy}(f) = \frac{1}{B_e T} \int_0^T x(t, f, B_e) y(t, f, B_e) dt \quad (8.46a)$$

$$\hat{Q}_{xy}(f) = \frac{1}{B_e T} \int_0^T x(t, f, B_e) y^\circ(t, f, B_e) dt \quad (8.46b)$$

Here, $x(t, f, B_e)$ and $y(t, f, B_e)$ are those portions of $x(t)$ and $y(t)$, respectively, passed by narrow band-pass filters with bandwidths of B_e cps and center frequencies of f cps. The symbol $y^\circ(t, f, B_e)$ denotes a 90-degree phase shift from $y(t, f, B_e)$. In words, the cross-spectral density function is estimated by the following operations.

1. Individual frequency filtering of the two signals $x(t)$ and $y(t)$ by narrow band-pass filters having identical bandwidths of B_e cps and the same center frequency.
2. Multiplying the instantaneous values of the two filtered signals with no phase shift.
3. Multiplying the instantaneous values of the two filtered signals with one signal shifted 90 degrees out of phase compared to the other.
4. Averaging each of the two instantaneous product values over the sampling time.
5. Division of each of the two mean products by the bandwidth B_e .

As the center frequency of the two band-pass filters is moved, a plot of the real and imaginary parts of the cross-spectral density function versus frequency (cross-spectrum) is obtained.

The preceding operations are accomplished by an analog cross-spectral density analyzer. In general terms, a cross-spectral density analyzer consists of two PSD analyzers as illustrated in Figure 8.4, except that multipliers replace the squaring circuits and a 90-degree phase shift circuit is added. The narrow band-pass filters in the two component analyzers must be well matched to prevent unwanted phase shifts. Frequency resolution, averaging

time, scan rate, and analysis time considerations are exactly the same as discussed for power spectral density analysis in Section 8.4

8.5.4 Frequency Response and Coherence Functions

It is shown in Chapter 6 that the estimation of frequency response functions and coherence functions involves the calculation of ratios of power and cross-spectral density measurements. Furthermore, an evaluation of the statistical accuracy of the estimates requires the solution of multiple term error formulas. It generally is not convenient to perform arithmetic operations of this type with analog instruments. Hence, frequency response and coherence function measurements are usually performed by digital data analysis procedures as discussed in the next chapter.

PROBLEMS

1. Given a sinusoidal signal $x(t) = X \sin 2\pi ft$, prove that the mean absolute value of the signal is given by $\overline{|x|} = (2/\pi)X$, as stated in Equation (8.5).
2. Prove Equation (8.8) is valid for small errors ($\epsilon \ll 1$).
3. Assume the mean square value of a zero mean bandwidth limited white noise signal with a bandwidth of $B = 100$ cps is to be measured with a normalized rms error of $\epsilon = 0.10$.
 - (a) Determine the required averaging time if a true averaging circuit (integrator) is employed.
 - (b) Assuming the resulting measurement is $\hat{\Psi}^2 = 1$ volt², determine a 95 percent confidence interval for $\hat{\Psi}^2$.
4. Using Equation (8.19), prove that the effective averaging time for an RC averaged mean square value measurement is $T = 2K$ for the case where the record length is very much longer than the RC averaging time constant ($K \ll T_r$), as stated in Equation (8.20).
5. If an autocorrelation function is measured from a recirculated sample record of length T_r using Equation (8.25), a bias error will occur for nonzero values of τ . Explain the source of this error.
6. Assume a power spectrum is measured using a PSD analyzer with a mean absolute value detection circuit. If the analyzer is calibrated for Gaussian data as specified in Equation (8.30), determine the error in the calibration for the case where the data actually have a uniform probability density function over the range $\pm a$, that is,

$$p(x) = \begin{cases} \frac{1}{2a} & -a \leq x \leq a \\ 0 & a < x < -a \end{cases}$$

7. Assume a power spectral density analysis is to be performed using a constant resolution bandwidth of $B_c = 5$ cps over a frequency range from 10 to 1000 cps with a normalized standard error of $\epsilon_r = 0.10$. If the available PSD analyzer is a single filter (scan) type instrument with an upper frequency limit of 20,000 cps and a true averaging circuit (integrator), what is the minimum total time required for the analysis?
8. Draw a functional block diagram of a joint probability density analyzer.
9. Draw a functional block diagram of a cross-spectral density analyzer.
10. It is shown in Section 8.4.5 that time base compression can be used to reduce the required time for a power spectral density analysis. Will time base compression also reduce the required time for probability density and correlation analysis? Explain your answer.

CHAPTER 9

DIGITAL DATA ANALYSIS PROCEDURES

Distinct from the analog data analysis procedures in Chapter 8, this chapter is devoted to digital techniques for analyzing random data. Most formulas assume the data to be processed are discrete time series values representing sample records from stationary (ergodic) random processes. The discussions include methods for trend removal, digital filtering, computation of Fourier series, fast Fourier transforms, probability functions, correlation functions, spectral density functions, frequency response and coherence functions for both single input and multiple input linear systems. Important related technical considerations are discussed in Chapter 7.

9.1 PRE-PROCESSING OPERATIONS

A number of basic operations will now be described which might be carried out in digital form for single time history records. This listing is not exhaustive and should not be automatically applied to all records. Parts of this procedure which may not be needed for certain applications should be omitted, and other parts should be extended as required for special applications. This listing does describe, however, the main operations which should be available for general digital computer processing of random data.

9.1.1 *Sampling Considerations*

DIGITIZATION OF DATA Let a single continuous random record $u(t)$ be sampled at points $\Delta t = h$ apart. These points determine a Nyquist folding

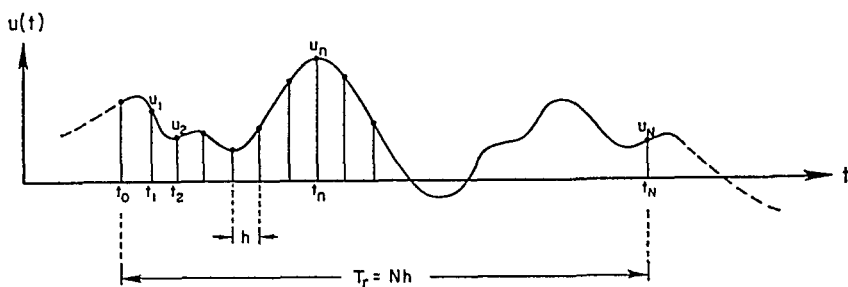


Figure 9.1 Sampling of a continuous record.

frequency f_c given by

$$f_c = \frac{1}{2\Delta t} = \frac{1}{2h} \quad (9.1)$$

as discussed in Section 7.3.1. Let

$$\{u_n\} \quad n = 1, 2, \dots, N \quad (9.2)$$

be the data values found at the N points

$$t_n = t_0 + nh \quad n = 1, 2, \dots, N \quad (9.3)$$

where the initial point t_0 is arbitrary and does not enter into the later formulas unless the sequence is assumed to start at $n = 0$ and end at $n = N - 1$.

SELECTION OF SAMPLE SIZE. The sample size N should be selected, if possible, on the basis of desired accuracies for various estimates. Rules for this selection are discussed in later sections of this chapter. In equation form, the original data values are

$$u_n = u(t_0 + nh) \quad n = 1, 2, \dots, N \quad (9.4)$$

Figure 9.1 illustrates this sampling procedure. Note that the record length T_r must satisfy $T_r = Nh$.

9.1.2 Arithmetic Quantities

CALCULATION OF THE MEAN VALUE. The sample mean value is given by

$$\bar{u} = \frac{1}{N} \sum_{n=1}^N u_n \quad (9.5)$$

where N is the number of data samples and u_n are the data values. The quantity $\bar{u}$ calculated here is an unbiased estimate of the mean value μ .

TRANSFORMATION OF DATA TO ZERO MEAN VALUE In order that subsequent formulas and calculations may be simplified, it is desirable at this time to transform the data to have a zero mean value. Define a new time history record $x(t) = u(t) - u$. Then, $x(t)$ has data values $\{x_n\}$ given by

$$x_n = x(t_0 + nh) = u_n - u \quad n = 1, 2, \dots, N \quad (9.6)$$

Note that $x = 0$. The reason for representing the original data values by $\{u_n\}$ instead of $\{x_n\}$ is to have the $\{x_n\}$ notation indicate a zero sample mean value. Subsequent formulas will now be stated in terms of the transformed data values $\{x_n\}$.

CALCULATION OF THE STANDARD DEVIATION The sample standard deviation is given by

$$s = \left[\sum_{n=1}^N \frac{(x_n)^2}{N-1} \right]^{1/2} \quad (9.7)$$

where N is the number of data samples and x_n are transformed data values with $x = 0$. The quantities s and s^2 calculated here are unbiased estimates of the standard deviation and variance, σ_x and σ_x^2 , respectively, as demonstrated in Section 4.1.

STANDARDIZATION TO UNIT STANDARD DEVIATION A further transformation on the data may be convenient at this time if the computer calculations are to be performed with fixed point, as opposed to floating point, arithmetic. Multiplying the transformed values x_n by $1/s$ yields

$$z_n = \frac{x_n}{s} \quad n = 1, 2, \dots, N \quad (9.8)$$

This results in final data with a sample mean of zero and a sample standard deviation of unity.

9.1.3 Trend Removal

A special correction may be needed to remove a trend in the data, defined as any frequency component whose period is longer than the record length. In particular, this type of component cannot be removed by highpass digital filtering as discussed in the next section. Hence some special trend removal technique must be applied. Least squares procedures can be employed for the removal of linear as well as higher order polynomial trends. An alternative method for removing linear trends only, which is not as accurate as least squares procedures, is the average slope method. Both procedures will now be described.

AVERAGE SLOPE METHOD. Let the original random record $u(t)$ be represented by

$$u(t) = \bar{u} + \bar{\alpha}_u \left(t - \frac{T_r}{2} \right) + x(t) \quad 0 \leq t \leq T_r \quad (9.9)$$

where $\bar{u}$ denotes the sample mean value of $u(t)$ over $(0, T_r)$, the parameter $\bar{\alpha}_u$ denotes the average slope of $u(t)$ with respect to t , and $x(t)$ is the corrected sample record with zero mean value and zero average slope. The previous analysis assumed $\bar{\alpha}_u = 0$. By integrating $u(t)$ over 0 to $T_r/3$, and subtracting from the integral over $2T_r/3$ to T_r , one can solve for $\bar{\alpha}_u$ as follows.

$$\bar{\alpha}_u = \frac{1}{(T_r/3)(2T_r/3)} \left[\int_{2T_r/3}^{T_r} u(t) dt - \int_0^{T_r/3} u(t) dt \right] \quad (9.10)$$

In terms of the data samples $\{u_n\}$, $n = 1, 2, \dots, N$, where $T_r = Nh$,

$$\bar{\alpha}_u = \frac{1}{hv(N-v)} \left[\sum_{n=N-v}^N u_n - \sum_{n=1}^v u_n \right] \quad (9.11)$$

where v is the largest integer less than or equal to $[N/3]$.

LEAST SQUARES METHOD. Consider now the application of least squares techniques as outlined previously in Section 4.8.2. As before, let $\{u_n\}$, $n = 1, 2, \dots, N$, be the data values sampled at an interval of h . Suppose it is desired to fit these data with a polynomial of degree K defined by

$$\hat{u}_n = \sum_{k=0}^K b_k (nh)^k \quad n = 1, 2, \dots, N \quad (9.12)$$

In a least squares fit, the set of coefficients $\{b_k\}$ are chosen so as to minimize the expression

$$Q(b) = \sum_{n=1}^N (u_n - \hat{u}_n)^2 = \sum_{n=1}^N \left[u_n - \sum_{k=0}^K b_k (nh)^k \right]^2 \quad (9.13)$$

which is always non-negative for any choice of $b = (b_0, b_1, \dots, b_K)$. The desired coefficients can be obtained by taking partial derivatives of Equation (9.13) with respect to b_ℓ and setting them equal to zero,

$$\frac{\partial Q}{\partial b_\ell} = \sum_{n=1}^N 2 \left[u_n - \sum_{k=0}^K b_k (nh)^k \right] [-(nh)^\ell] \quad (9.14)$$

This yields $K + 1$ equations of the form

$$\sum_{k=0}^K b_k \sum_{n=1}^N (nh)^{k+\ell} = \sum_{n=1}^N u_n (nh)^\ell \quad \ell = 0, 1, 2, \dots, K \quad (9.15)$$

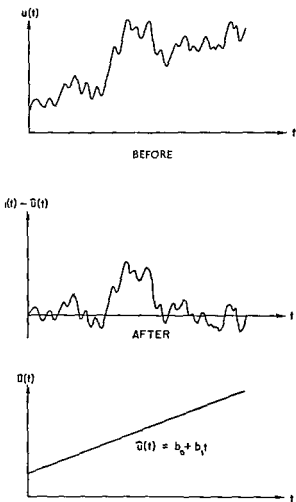


Figure 9.2 Illustration of linear trend removal

which now can be solved for the desired $\{b_k\}$. For example, when $K = 0$, Equation (9.15) becomes

$$b_0 \sum_{n=1}^N (nh)^0 = \sum_{n=1}^N u_n (nh)^0 \quad (9.16)$$

giving the result

$$b_0 = \frac{1}{N} \sum_{n=1}^N u_n \quad (9.17)$$

Note that b_0 is an estimate of the mean value μ .

For $K = 1$, Equation (9.15) becomes

$$b_0 \sum_{n=1}^N (nh)^\ell + b_1 \sum_{n=1}^N (nh)^{1+\ell} = \sum_{n=1}^N u_n (nh)^\ell \quad \ell = 0, 1 \quad (9.18)$$

giving the results

$$b_0 = \frac{2(2N+1) \sum_{n=1}^N u_n - 6 \sum_{n=1}^N nu_n}{N(N-1)} \quad (9.19)$$

$$b_1 = \frac{12 \sum_{n=1}^N nu_n - 6(N+1) \sum_{n=1}^N u_n}{hN(N-1)(N+1)} \quad (9.20)$$

providing a preferred least squares method to remove a linear trend in data over the method of Equation (9.11). Second and third order polynomials are also practical by this method. A case of linear trend removal is illustrated in Figure 9.2.

Trend removal is an important intermediate step in the digital processing of random data and should be given its due consideration. If trends are not eliminated in data, large distortions can occur in the later processing of correlation and spectral quantities. In particular, trends in data can completely nullify the estimation of low frequency spectral content. In some problems, knowledge of these trends may be a desired end result in itself. Caution is advised here, however, in that trend removal should be performed only if trends are physically expected or clearly apparent in the data.

9.2 DIGITAL FILTERING METHODS

Filtering of data may be desired for such purposes as data smoothing, separation of frequency components, and evaluation of properties in various frequency intervals. Figure 9.3 shows the nature of highpass and lowpass filtering for the case of a sine wave contaminated by high frequency noise. The highpass filter will pass the high frequency noise, while the lowpass filter will yield a smooth version of the underlying sine wave.

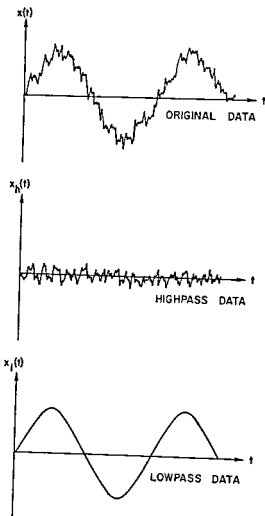


Figure 9 3 Illustration of highpass and lowpass filtering

The general relationship between the input $x(t)$ and the output $y(t)$ of a linear filter is given by the convolution integral of Equation (2.2), namely,

$$y(t) = \int_{-\infty}^{\infty} h(\tau) x(t - \tau) d\tau \quad (9.21)$$

where $h(\tau)$ is the weighting function of the filter. The frequency response function of the filter, $H(f)$, is the Fourier transform of $h(\tau)$ defined by

$$H(f) = \int_{-\infty}^{\infty} h(\tau) e^{-j2\pi f\tau} d\tau \quad (9.22)$$

In designing a digital filter, unlike an analog filter, it is not necessary for the filter to be physically realizable. That is, it is not required that $h(\tau)$ be zero for $\tau < 0$, since the data can be stored in the computer and then run backwards to filter the data in reverse order.

Examples of ideal gain factors $|H(f)|$ for the special cases of lowpass, highpass and bandpass filters are shown in Figure 9.4. The shape of digital filters, to be discussed shortly, which can approximate these ideal filters are shown in Figure 9.5. Programs can be written to obtain these digital filters where it is sufficient to specify simple parameters such as the frequency cutoff f_0 and desired rolloff characteristics.

9.2.1 Nonrecursive Digital Filters

The finite sum equivalent to Equation (9.21) for $t = k \Delta t$, $k = 1, 2, \dots$, M , can be expressed by a symmetric filter having the form

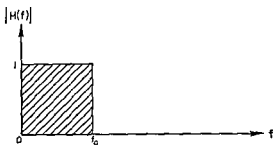
$$y_n = \sum_{k=1}^M h_k (x_{n+k} + x_{n-k}) \quad n = 1, 2, \dots, N \quad (9.23)$$

where $h_k = h_{-k}$. Note that Equation (9.23) involves future values of the input. For convenience, the sampling interval Δt is usually included in the filter weights. With a symmetric filter, the finite sum equivalent to Equation (9.22) gives a filter with zero phase characteristics, namely,

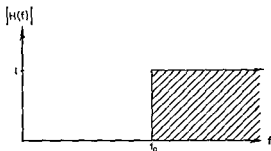
$$H(f) = 2 \sum_{k=1}^M h_k \cos(2\pi f k \Delta t) \quad (9.24)$$

Nonzero phase characteristics of physical interest can be obtained with non-symmetric filters. Note that Equations (9.23) and (9.24) have a total of M coefficients h_k , known as filter weights. The k th weight is given by the inverse Fourier transform of Equation (9.24) as follows.

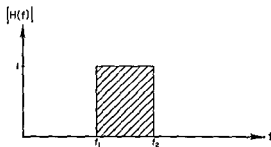
$$h_k = \int_{-\infty}^{\infty} H(f) \cos(2\pi f k \Delta t) df \quad (9.25)$$



(a)

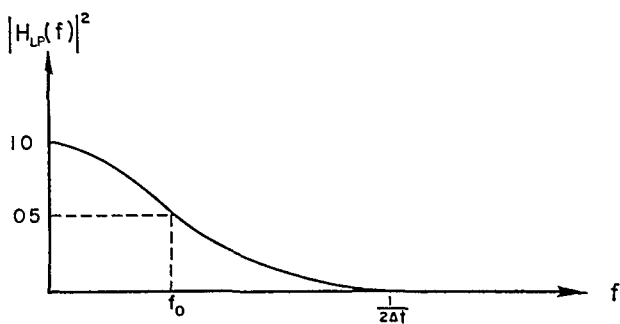


(b)

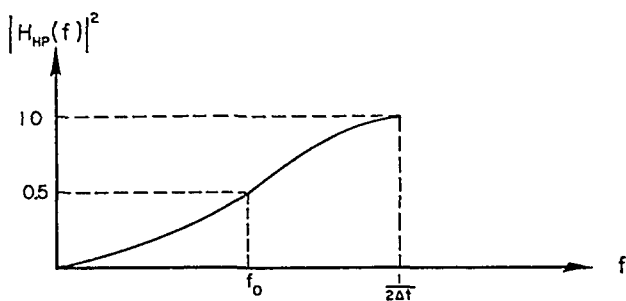


(c)

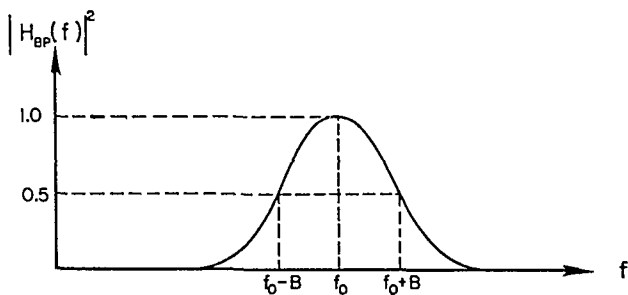
Figure 9.4 Characteristics of ideal filters (a) Lowpass (b) Highpass (c) Bandpass



(a)



(b)



(c)

Figure 9.5 Characteristics of digital filters. (a) Lowpass. (b) Highpass. (c) Bandpass.

Filters of this type whether symmetric or not, are called *nonrecursive digital filters* because their output is the result of a finite sum of input terms only

Example 9.1 Nonrecursive Lowpass Digital Filter As an example of this procedure, consider the filter weights for a symmetric nonrecursive lowpass filter where the ideal frequency response function $H(f)$ is defined by

$$\begin{aligned} H(f) &= 1 & -f_0 \leq f \leq f_0 \\ &= 0 & \text{otherwise} \end{aligned} \quad (9.26)$$

From Equation (9.25), the filter weights $\{h_k\}$ are

$$h_k = \int_{-f_0}^{f_0} \cos(2\pi f k \Delta t) df = \frac{\sin(2\pi f_0 k \Delta t)}{\pi k \Delta t} \quad (9.27)$$

Thus the filter weights are proportional to $(1/k)$ so that large values of k are required before these weights become small. In practice, this type of nonrecursive filter usually requires so many weights (100 or more) that it is not considered to be a very efficient method of filtering. Also, large truncation errors will occur between the desired $H(f)$ and that found by the h_k weights because of the abrupt transition in $H(f)$ from zero to one causing a "Gibbs" phenomena overshoot in the vicinity of the cutoff frequency. This concludes Example 9.1

Other types of nonrecursive filters have been developed [see Ref. 38] which can reduce effects of truncation errors by making the transition in $H(f)$ from one to zero less abrupt than given by Equation (9.26). These nonrecursive filters still usually require many weights to apply, whereas with recursive filters (discussed in the next section) useful results can be obtained with a very few weights.

9.2.2 Recursive Digital Filters

Recursive digital filters are filters where the output results not only from a finite sum of input terms but also by using previous outputs as inputs. Stated in engineering terms, this procedure is called *feedback*. A simple standard type of recursive filter is given by

$$y_n = cx_n + \sum_{k=1}^M h_k y_{n-k} \quad (9.28)$$

which uses M previous outputs and only one input. More general recursive filters involve M outputs and larger numbers of inputs. Equation (9.28) is illustrated in Figure 9.6. The triangles represent multiplication by the values

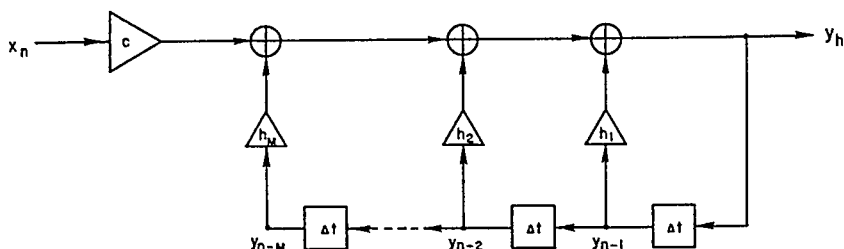


Figure 9.6 Schematic diagram for simple recursive filter.

shown within the triangles, the rectangles represent a delay of Δt from one point to another, and the circles with plus signs $\oplus$ represent summing operations.

The Fourier transform of Equation (9.28) yields the result

$$Y(f) = cX(f) + Y(f) \sum_{k=1}^M h_k e^{-j2\pi f k \Delta t} \quad (9.29)$$

where the summation involves a polynomial in powers of the exponential $\exp[-j2\pi f \Delta t]$. Replacing this exponential by the letter z leads to a procedure for analyzing digital filters in terms of what is called z -transform theory. From Equation (9.29), the frequency response function for the entire system is given by

$$H(f) = \frac{Y(f)}{X(f)} = \frac{c}{1 - \sum_{k=1}^M h_k e^{-j2\pi f k \Delta t}} \quad (9.30)$$

Studies of the properties of $H(f)$ are thus reduced to determining the location and nature of the poles in the denominator of this last result.

Example 9.2. Recursive Lowpass Digital Filter. As an example of this procedure, consider the recursive filter defined by

$$y_n = (1 - a)x_n + ay_{n-1} \quad (9.31)$$

where $a = \exp(-\Delta t/RC)$. This can behave like a lowpass RC filter in the continuous case. To verify this property, note that from Equation (9.30),

$$H(f) = \frac{1 - a}{1 - ae^{-j2\pi f \Delta t}} \quad (9.32)$$

The square of the filter gain factor is here given by

$$|H(f)|^2 = \frac{(1-a)^2}{(1+a^2) - 2a \cos 2\pi f \Delta t} \quad (9.33)$$

Observe that if $RC \gg \Delta t$, then $a = \exp(-\Delta t/RC) \approx 1 - (\Delta t/RC)$ and $(1-a) \approx \Delta t/RC$. Also if $2\pi f \Delta t \ll 1$, then $e^{-j2\pi f \Delta t}$ may be approximated by $1 - j2\pi f \Delta t$. For this situation

$$H(f) \approx \frac{1}{1 + j2\pi f RC}$$

and

$$|H(f)|^2 \approx \frac{1}{1 + (2\pi f RC)^2}$$

which are the usual lowpass RC filter results. This concludes Example 9.2

Recursive digital filters which give good approximations to Butterworth filters have been synthesized with the aid of Equation (9.30) by finding a set of numerical weights $\{h_k\}$ and a coefficient c such that the resulting $|H(f)|^2$ has the form

$$|H(f)|^2 = \frac{1}{1 + \left(\frac{\sin \pi f \Delta t}{\sin \pi f_0 \Delta t} \right)^{2M}} \quad 0 \leq f \leq \frac{1}{2\Delta t} \quad (9.34)$$

Note that $|H(f)|^2$ is equal to unity for f equal to zero, while at $f = f_0$, the quantity $|H(f)|^2$ is equal to one half. At $f = (1/2\Delta t)$, the Nyquist folding frequency, the quantity $|H(f)|^2$ approaches zero for large M . Thus, over the domain $[0, (1/2\Delta t)]$, the principal domain in the digital case, the filter described by Equation (9.34) acts like a lowpass Butterworth filter of the form

$$|H(f)|^2 = \frac{1}{1 + \left(\frac{f}{f_0} \right)^K} \quad (9.35)$$

where f_0 is the half power point and K determines the slope. Fuller mathematical details for the synthesis of recursive digital filters appear in References 17, 25, 39.

An important application of lowpass filters is *decimation of data* in order to decrease the amount of data for later analysis. By definition, an r th order decimation of sampled data consists of keeping only every r th point. Suppose, as usual, that a record $x(t)$ is sampled at points Δt apart and an r th order decimation performed. Then the new sampling rate is actually $\Delta t = r \Delta t$, and the new Nyquist folding frequency becomes $(1/2r \Delta t)$. Hence all the

information above $(1/2r \Delta t)$ will be aliased (folded) back into the interval $[0, (1/2r \Delta t)]$. To avoid aliasing, the original data should be filtered to remove information in the frequency range above $(1/2r \Delta t)$ by using a suitable analog lowpass filter.

9.3 FOURIER SERIES AND FAST FOURIER TRANSFORMS

Fourier series and Fourier transforms of data differ in their theoretical properties but not, for most practical purposes, in their digital computational details. This is because only a finite range Fourier series or transform can actually be computed with digitized data, and this finite range can always be considered as the period of an associated Fourier series. From the discussion in Section 3.2.3, one of the main reasons for the importance of fast Fourier transforms is that they can be used to provide estimates of desired spectral density and correlation functions. Before explaining the basis for methods to compute fast Fourier transforms, it is instructive to review a standard Fourier series procedure.

9.3.1 Standard Fourier Series Procedure

If a stationary sample record $x(t)$ is periodic with a period of T_p and a fundamental frequency $f_1 = 1/T_p$, then $x(t)$ can be represented by the Fourier series

$$x(t) = \frac{a_0}{2} + \sum_{q=1}^{\infty} (a_q \cos 2\pi q f_1 t + b_q \sin 2\pi q f_1 t) \quad (9.36)$$

where

$$\begin{aligned} a_q &= \frac{2}{T} \int_0^T x(t) \cos 2\pi q f_1 t \, dt & q = 0, 1, 2, \dots \\ b_q &= \frac{2}{T} \int_0^T x(t) \sin 2\pi q f_1 t \, dt & q = 1, 2, 3, \dots \end{aligned} \quad (9.37)$$

Assume a sample record $x(t)$ is of finite length $T_r = T_p$, the fundamental period of the data. Further assume that the record is sampled at an even number of N equally spaced points a distance h apart, where h has been selected to produce a sufficiently high cutoff frequency $f_c = 1/2h$. Consider the initial point of the record to be zero and denote the transformed data values, as before, by

$$x_n = x(nh) \quad n = 1, 2, \dots, N \quad (9.38)$$

Proceed now to calculate the finite version of a Fourier series which will pass through these N data values. For any point t in the interval $(0, T_p)$, the result is

$$x(t) = A_0 + \sum_{q=1}^{N/2} A_q \cos \left(\frac{2\pi q t}{T_p} \right) + \sum_{q=1}^{(N/2)-1} B_q \sin \left(\frac{2\pi q t}{T_p} \right) \quad (9.39)$$

At the particular points $t = nh$ $n = 1, 2, \dots, N$, where $T_p = Nh$,

$$x_n = x(nh) = A_0 + \sum_{q=1}^{N/2} A_q \cos \left(\frac{2\pi qn}{N} \right) + \sum_{q=1}^{(N/2)-1} B_q \sin \left(\frac{2\pi qn}{N} \right) \quad (9.40)$$

The coefficients A_q and B_q are given by

$$\begin{aligned} A_0 &= \frac{1}{N} \sum_{n=1}^N x_n = \bar{x} = 0 \\ A_q &= \frac{2}{N} \sum_{n=1}^N x_n \cos \frac{2\pi qn}{N} \quad q = 1, 2, \dots, \frac{N}{2} - 1 \\ A_{N/2} &= \frac{1}{N} \sum_{n=1}^N x_n \cos q\pi \\ B_q &= \frac{1}{N} \sum_{n=1}^N x_n \sin \frac{2\pi qn}{N} \quad q = 1, 2, \dots, \frac{N}{2} - 1 \end{aligned} \quad (9.41)$$

A digital computer program to obtain A_q and B_q involves the following steps

- 1 Evaluate $\theta = \frac{2\pi qn}{N}$ for fixed q and n
- 2 Compute $\cos \theta$, $\sin \theta$
- 3 Compute $x_n \cos \theta$, $x_n \sin \theta$
- 4 Accumulate both sums for $n = 1, 2, \dots, N$
- 5 Increment q and repeat operations

This procedure requires a total of approximately N^2 real multiply add operations

For large N , these standard digital computation methods for determining the coefficients A_q and B_q can be time and cost consuming since both time and cost are a function of N^2 . To greatly reduce these standard computational times, alternative methods have been proposed and developed, known as *Fast Fourier Transform* (FFT) procedures. These methods will now be discussed in some detail because of their importance in digital processing of random data.

9.3.2 Fast Fourier Transforms

An infinite range Fourier transform of a real valued or a complex valued record $x(t)$ is defined by the complex valued quantity

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-i2\pi ft} dt \quad (9.42)$$

Theoretically, as noted previously, this transform $X(f)$ will not exist for an $x(t)$ which is a representative member of a stationary random process when the infinite limits are used. However, by restricting the limits to a finite time interval of $x(t)$, say in the range $(0, T)$, then the finite-range Fourier transform will exist as defined by

$$X(f, T) = \int_0^T x(t) e^{-j2\pi ft} dt \quad (9.43)$$

Assume now that this $x(t)$ is sampled at N equally spaced points a distance h apart, where h has been selected to produce a sufficiently high cutoff frequency. Since $t_0 = 0$, the points $t_n = nh$ (it is convenient here to start with $n = 0$). Hence in place of Equation (9.38), let

$$x_n = x(nh) \quad n = 0, 1, 2, \dots, N-1 \quad (9.44)$$

Then, for arbitrary f , the discrete version of Equation (9.43) is

$$X(f, T) = h \sum_{n=0}^{N-1} x_n \exp [-j2\pi fnh] \quad (9.45)$$

The usual selection of discrete frequency values for the computation of $X(f, T)$ is

$$f_k = kf_0 = \frac{k}{T} = \frac{k}{Nh} \quad k = 0, 1, 2, \dots, N-1 \quad (9.46)$$

At these frequencies, the transformed values give the Fourier components defined by

$$X_k = \frac{X(f_k, T)}{h} = \sum_{n=0}^{N-1} x_n \exp \left[-j \frac{2\pi kn}{N} \right] \quad k = 0, 1, 2, \dots, N-1 \quad (9.47)$$

where h has been included with $X(f_k, T)$ to have a scale factor of unity before the summation. Note that results are unique only out to $k = N/2$ since the Nyquist cutoff frequency occurs at this point. Fast Fourier transform (FFT) methods are designed to compute these quantities, X_k , and can also be used to compute the coefficients A_q and B_q in Equation (9.41).

To simplify the notation further, let

$$W(u) = \exp \left[-j \frac{2\pi u}{N} \right] \quad (9.48)$$

Observe that $W(N) = 1$ and for all u and v ,

$$W(u+v) = W(u) W(v) \quad (9.49)$$

Also, let

$$X(k) = X_k \quad \text{and} \quad x(n) = x_n \quad (9.50)$$

Then Equation (9 47) becomes

$$X(k) = \sum_{n=0}^{N-1} x(n) W(kn) \quad k = 0, 1, 2, \dots, N-1 \quad (9 51)$$

Equations (9 47) and (9 51) should be studied so as to be easily recognized as the Fourier transform of $x(n)$ when $x(n)$ is expressed by a series of N terms. Such equations require a total of approximately N^2 complex multiply-add operations (where 1 complex multiply-add = 4 real multiply-adds) to compute all of the $X(k)$ terms involved.

BASIS FOR FFT PROCEDURES The fast Fourier transform procedures are now based upon decomposing N into its composite (nonunity) factors, and carrying out Fourier transforms over the smaller number of terms in each of the composite factors. In particular, if N is the product of p factors such that

$$N = \prod_{i=1}^p r_i = r_1 r_2 \dots r_p \quad (9 52)$$

where the r 's are all integers greater than unity, then as will be proved below, the $X(k)$ in Equation (9 51) can be found by computing in an iterative fashion the sum of p terms,

(N/r_1) Fourier transforms requiring $4r_1^2$ real operations each,

(N/r_2) Fourier transforms requiring $4r_2^2$ real operations each,

(9 53)

(N/r_p) Fourier transforms requiring $4r_p^2$ real operations each.

Hence the total number of real operations becomes

$$4(Nr_1 + Nr_2 + Nr_3 + \dots + Nr_p) = 4N \sum_{i=1}^p r_i \quad (9 54)$$

The resulting speed ratio of these FFT procedures to the standard method is then

$$\text{Speed Ratio} = \frac{N^2}{4N \sum_{i=1}^p r_i} = \frac{N}{4 \sum_{i=1}^p r_i} \quad (9 55)$$

Example 9 3 Speed Ratio for Powers of Two. If $N = 2^p$, then $\sum_{i=1}^p r_i = 2p = 2 \log_2 N$. In this case, the speed ratio by Equation (9 55) appears to be

$$\text{Speed Ratio} = \frac{N^2}{8Np} = \frac{N}{8p}$$

However, a doubling of the speed can be achieved in practice by noting that the values for $W(kn)$, when N is a power of 2, all turn out to be $+1$ or -1 , so as to yield a higher speed ratio of the order of

$$\text{Speed Ratio} = \frac{N}{4p} \quad (9.56)$$

This result is deemed to be a conservative estimate since a further speed improvement of at most two can be obtained by dividing a single record into two parts and computing as indicated later in Section 9.3.4. To be specific, if $N = 2^{13} = 8192$, Equation (9.56) gives a speed ratio of $(8192/52) \approx 158$.

DERIVATION OF GENERAL PROCEDURE. To derive the result stated in Equation (9.53), express the indices k and n in Equation (9.51) in the following way.

$$\begin{aligned} k &= \sum_{v=0}^{p-1} k_v \prod_{i=0}^v r_i, & \text{where } k_v &= 0, 1, 2, \dots, r_{v+1} - 1 \\ & & r_0 &= 1 \\ n &= \sum_{v=0}^{p-1} n_v \prod_{i=0}^v r_{m+1-i}, & \text{where } n_v &= 0, 1, 2, \dots, r_{p-v} - 1 \\ & & r_{p+1} &= 1 \end{aligned} \quad (9.57)$$

Note the indices k and n are replaced by new indices k_v and n_v as defined above. Equation (9.57) has the following interpretation

$$\begin{aligned} k &= k_0 + k_1 r_1 + k_2 r_1 r_2 + \dots + k_{p-1} (r_1 r_2 \dots r_{p-1}) \\ n &= n_0 + n_1 r_p + n_2 r_p r_{p-1} + \dots + n_{p-1} (r_p r_{p-1} \dots r_2) \end{aligned} \quad (9.57a)$$

where

$$\begin{aligned} k_0 &= 0, 1, 2, \dots, r_1 - 1 & n_0 &= 0, 1, 2, \dots, r_p - 1 \\ k_1 &= 0, 1, 2, \dots, r_2 - 1 & n_1 &= 0, 1, 2, \dots, r_{p-1} - 1 \\ \vdots & & \vdots & \\ \vdots & & \vdots & \\ \vdots & & \vdots & \\ k_{p-1} &= 0, 1, 2, \dots, r_p - 1 & n_{p-1} &= 0, 1, 2, \dots, r_1 - 1 \end{aligned} \quad (9.57b)$$

By fixing values of k_v and n_v in turn, it is a straight-forward exercise to verify that k and n will each vary from 0 to $N - 1$, where N is the product of all the r 's as per Equation (9.52). Equation (9.51) can now be rewritten as

$$\begin{aligned} X(k) &= X(k_0, k_1, \dots, k_{p-1}) \\ &= \sum_{n_0=0}^{r_p-1} \sum_{n_1=0}^{r_{p-1}-1} \dots \sum_{n_{p-2}=0}^{r_2-1} \sum_{n_{p-1}=0}^{r_1-1} x(n_0, n_1, \dots, n_{p-2}, n_{p-1}) W(kn) \end{aligned} \quad (9.58)$$

where

$$W(kn) = W(k[n_0 + n_1 r_p + \dots + n_{p-v}(r_p r_{p-1} \dots r_{v+1}) + \dots + n_{p-1}(r_p r_{p-1} \dots r_2)]) \quad (9.59)$$

with k given by Equation (9.57)

An alternative way to write k is as follows

$$k = (k_0 + k_1 r_1 + \dots + k_{v-1} r_1 r_2 \dots r_{v-1}) + (r_1 r_2 \dots r_v)(k_v + k_{v+1} r_{v+1} + \dots + k_{p-1} r_{v+1} r_{v+2} \dots r_{p-1}) \quad (9.60)$$

Hence, a representative term in Equation (9.59) is such that

$$\begin{aligned} & k n_{p-v}(r_p r_{p-1} \dots r_{v+1}) \\ &= (k_0 + k_1 r_1 + \dots + k_{v-1} r_1 r_2 \dots r_{v-1}) n_{p-v}(r_p r_{p-1} \dots r_{v+1}) \\ &+ N n_{p-v}(k_v + k_{v+1} r_{v+1} + \dots + k_{p-1} r_{v+1} r_{v+2} \dots r_{p-1}) \end{aligned} \quad (9.61)$$

Then, since W for any integer power of N is equal to unity, it follows for $v = 1, 2, \dots, p$, that

$$\begin{aligned} & W(k n_{p-v} r_p r_{p-1} \dots r_{v+1}) \\ &= W[(k_0 + k_1 r_1 + \dots + k_{v-1} r_1 r_2 \dots r_{v-1}) n_{p-v} r_p r_{p-1} \dots r_{v+1}] \quad (9.62) \end{aligned}$$

Equation (9.62) will now be interpreted. For $v = 1$, observe that

$$\begin{aligned} W(k n_{p-1} r_p r_{p-1} \dots r_2) &= W(k_0 n_{p-1} r_p r_{p-1} \dots r_2) \\ &= W\left(\frac{k_0 n_{p-1} N}{r_1}\right) = \exp\left[-j \frac{2\pi k_0 n_{p-1}}{r_1}\right] \quad (9.62a) \end{aligned}$$

This is the exponential required in a Fourier transform of $x(n_{p-1})$ as expressed by a series of r_1 terms, instead of N terms as shown in Equations (9.47) and (9.51). Note also that the indices k_0 and n_{p-1} each vary over the values $0, 1, 2, \dots, r_1 - 1$, thus requiring a total of r_1^2 multiply-add operations to compute each of the associated $X(k_0)$ which might be involved here. For $v = 2, 3, \dots, p$, Equation (9.62) with the aid of Equation (9.49) becomes the product of two factors,

$$\begin{aligned} & W[(k_0 + k_1 r_1 + \dots + k_{v-2} r_1 r_2 \dots r_{v-2}) n_{p-v} r_p r_{p-1} \dots r_{v+1}] \\ & \text{multiplied by } W(k_{v-1} r_1 r_2 \dots r_{v-1} n_{p-v} r_p r_{p-1} \dots r_{v+1}) \quad (9.62b) \end{aligned}$$

where only the second factor contains k_{v-1} . This second factor is the quantity

$$W\left(\frac{k_{v-1} n_{p-v} N}{r_v}\right) = \exp\left[-j \frac{2\pi k_{v-1} n_{p-v}}{r_v}\right] \quad (9.63)$$

which is the exponent required in a Fourier transform of $x(n_{p-v})$ as expressed by a series of r_v terms. Furthermore, the indices k_{v-1} and n_{p-v} each vary over the values $0, 1, 2, \dots, r_v - 1$. Hence, a total of r_v^2 multiply-add operations are needed to compute each of the associated $X(k_{v-1})$ which might be involved here.

THE FFT ALGORITHM. From the development in Equations (9.60) through (9.63), Equation (9.59) becomes

$$W(kn) = \prod_{v=1}^p T(k_0, k_1, \dots, k_{v-2}) W\left(\frac{k_{v-1} n_{p-v} N}{r_v}\right) \quad (9.64)$$

where

$$\begin{aligned} T(k_0, k_1, \dots, k_{v-2}) &= 1 \quad \text{for } v = 1 \\ &= W[(k_0 + k_1 r_1 + \dots + k_{v-2} r_1 r_2 \dots r_{v-2})(n_{p-v} r_p r_{p-1} \dots r_{v+1})] \\ &\quad \text{for } v = 2, 3, \dots, p \end{aligned} \quad (9.65)$$

Quantities such as the T terms in Equation (9.65) are called "*twiddle factors*" in Reference 19. The result of Equation (9.64) should now be substituted into Equation (9.58). Regrouping of terms then yields

$$\begin{aligned} X(k_0, k_1, \dots, k_{p-1}) &= \sum_{n_0=0}^{r_p-1} T(k_0, k_1, \dots, k_{p-2}) W\left(\frac{k_{p-1} n_0 N}{r_p}\right) \\ &\quad \text{times } \sum_{n_1=0}^{r_{p-1}-1} T(k_0, k_1, \dots, k_{p-3}) W\left(\frac{k_{p-2} n_1 N}{r_{p-1}}\right) \\ &\quad \text{times } \dots \dots \dots \quad (9.66) \\ &\quad \text{times } \sum_{n_{p-2}=0}^{r_2-1} T(k_0) W\left(\frac{k_1 n_{p-2} N}{r_2}\right) \\ &\quad \text{times } \sum_{n_{p-1}=0}^{r_1-1} x(n_0, n_1, \dots, n_{p-2}, n_{p-1}) W\left(\frac{k_0 n_{p-1} N}{r_1}\right) \end{aligned}$$

Thus, the desired Fourier transform can be computed in p successive iterative steps as shown by Equation (9.66). The principal remaining question of concern here is the understanding of these steps.

Consider the last innermost sum of terms in Equation (9.66). Let

$$A_1(k_0, n_0, n_1, \dots, n_{p-2}) = \sum_{n_{p-1}=0}^{r_1-1} x(n_0, n_1, \dots, n_{p-2}, n_{p-1}) W\left(\frac{k_0 n_{p-1} N}{r_1}\right) \quad (9.67)$$

Then, holding $n_0, n_1, \dots, n_{p-2}$ fixed for each of their possible values,

Equation (9 67) gives a total of (N/r_1) Fourier transforms of $x(n_{p-1})$ requiring r_1^2 operations each. At the next innermost sum of terms in Equation (9 66), let

$$A_2(k_0, k_1, n_0, n_1, \dots, n_{p-3}) \\ = \sum_{n_{p-2}=0}^{r_2-1} A_1(k_0, n_0, n_1, \dots, n_{p-2}) T(k_0) W\left(\frac{k_1 n_{p-2} N}{r_2}\right) \quad (9 68)$$

Here, holding $k_0, n_0, n_1, \dots, n_{p-3}$ fixed for each of their possible values, Equation (9 68) gives a total of (N/r_2) Fourier transforms of $x(n_{p-2})$ requiring r_2^2 operations each. Continuing in this way, at the v th step, where $v = 2, 3, \dots, p-1$, let

$$A_v(k_0, k_1, \dots, k_{v-1}, n_0, n_1, \dots, n_{p-v-1}) \\ = \sum_{n_{p-v}=0}^{r_v-1} A_{v-1}(k_0, k_1, \dots, k_{v-2}, n_0, n_1, \dots, n_{p-v}) T(k_0, k_1, \dots, k_{v-2}) \\ \text{times } W\left(\frac{k_{v-1} n_{p-v} N}{r_v}\right) \quad (9 69)$$

Here, holding $k_0, k_1, \dots, k_{v-2}, n_0, n_1, \dots, n_{p-v-1}$ fixed for each of their possible values gives a total of (N/r_v) Fourier transforms of $x(n_{p-v})$ requiring r_v^2 operations each. At the last step, Equation (9 66) yields

$$X(k_0, k_1, \dots, k_{p-1}) \\ = A_p(k_0, k_1, \dots, k_{p-1}) \\ = \sum_{n_0=0}^{r_p-1} A_{p-1}(k_0, k_1, \dots, k_{p-2}, n_0) T(k_0, k_1, \dots, k_{p-2}) W\left(\frac{k_{p-1} n_0 N}{r_p}\right) \quad (9 70)$$

Holding $k_0, k_1, \dots, k_{p-2}$ fixed for each of their possible values, this gives a total of (N/r_p) Fourier transforms of $x(n_0)$ requiring r_p^2 operations each. The sequence of steps in Equations (9 67) to (9 70) proves the result stated in Equation (9 53), making allowance for complex to real operations.

The formula derived in Equation (9 69) is a general *Fast Fourier Transform Algorithm*, and is the basis for many Fourier transform computational procedures discussed in the literature [Refs 3, 4, 8, 17, 19].

9.3.3 Cooley-Tukey Procedure

The Cooley-Tukey procedure described in Reference 8 is a special case of the general algorithm of Equation (9 69) that is appropriate for binary digital computers. In particular, it applies to those situations where the number of data samples N is a power of 2, namely,

$$N = 2^p \quad (9 71)$$

If necessary, zeros are added to the data sequence to satisfy this requirement. Here, the iterative procedures of Equation (9.66) become the sum of p terms, where every term involves $(N/2)$ Fourier transforms requiring 4 operations each. This gives a total of $2Np$ complex multiply-add operations.

It is desirable to write down the special equations which apply to this case because of their widespread importance. This will be done by substituting into previous results. Equation (9.57a) becomes

$$\begin{aligned} k &= k_0 + 2k_1 + 2^2k_2 + \cdots + 2^{p-1}k_{p-1} \\ n &= n_0 + 2n_1 + 2^2n_2 + \cdots + 2^{p-1}n_{p-1} \end{aligned} \quad (9.72)$$

where each k and n take on the values 0 and 1 only. Equation (9.63) becomes for all $v = 1, 2, \dots, p$,

$$W\left(\frac{k_{v-1} n_{p-v} N}{r_v}\right) = \exp(-j\pi k_{v-1} n_{p-v}) \quad (9.73)$$

which takes on the values 1 and -1 only. Equation (9.65) becomes

$$\begin{aligned} T(k_0, k_1, \dots, k_{v-2}) &= 1 \quad \text{for } v = 1 \\ &= W[(k_0 + 2k_1 + \cdots + 2^{v-2}k_{v-2})2^{p-v}n_{p-v}] \quad \text{for } v = 2, 3, \dots, p \end{aligned} \quad (9.74)$$

The Fourier transform iteration of Equation (9.66) can be expressed in this special case by the formula

$$\begin{aligned} X(k_0, k_1, \dots, k_{p-1}) \\ = \sum_{n_0=0}^1 \sum_{n_1=0}^1 \cdots \sum_{n_{p-2}=0}^1 \sum_{n_{p-1}=0}^1 x(n_0, n_1, \dots, n_{p-2}, n_{p-1}) W(kn) \end{aligned} \quad (9.75)$$

where

$$W(kn) = \prod_{v=1}^p T(k_0, k_1, \dots, k_{v-2}) \exp(-j\pi k_{v-1} n_{p-v}) \quad (9.76)$$

The first step in this iteration is now from Equation (9.67)

$$A_1(k_0, n_0, n_1, \dots, n_{p-2}) = \sum_{n_{p-1}=0}^1 x(n_0, n_1, \dots, n_{p-2}, n_{p-1}) \exp(-j\pi k_0 n_{p-1}) \quad (9.77)$$

The v th step, for $v = 2, 3, \dots, p-1$, becomes from Equation (9.69)

$$\begin{aligned} A_v(k_0, k_1, \dots, k_{v-1}, n_0, n_1, \dots, n_{p-v-1}) \\ = \sum_{n_{p-v}=0}^1 A_{v-1}(k_0, k_1, \dots, k_{v-2}, n_0, n_1, \dots, n_{p-v}) \\ \text{times } T(k_0, k_1, \dots, k_{v-2}) \exp(-j\pi k_{v-1} n_{p-v}) \end{aligned} \quad (9.78)$$

This result is called the *Cooley-Tukey* fast Fourier transform algorithm. The last step in the iteration, Equation (9.70), is

$$\begin{aligned} X(k_0, k_1, \dots, k_{p-1}) \\ = A_p(k_0, k_1, \dots, k_{p-1}) \\ = \sum_{n_0=0}^1 A_{p-1}(k_0, k_1, \dots, k_{p-2}, n_0) T(k_0, k_1, \dots, k_{p-1}) \exp(-j\pi k_{p-1} n_0) \end{aligned} \quad (9.79)$$

and completes the procedure for this special case. A fuller discussion of these matters appears in References 3, 4, 8, 17, 19.

9.3.4 Further Related Formulas

Fourier transforms of two real-valued records may be computed simultaneously by inserting one record $x(n)$ as the real part and one record $y(n)$ as the imaginary part of a complex record $z(n)$. In equation form, let

$$z(n) = x(n) + jy(n) \quad n = 0, 1, \dots, N-1 \quad (9.80)$$

The Fourier transform of $z(n)$ by Equation (9.47) is

$$Z(k) = \sum_{n=0}^{N-1} [x(n) + jy(n)] \exp \left[-j \frac{2\pi kn}{N} \right] \quad k = 0, 1, 2, \dots, N-1 \quad (9.81)$$

This can be computed by the fast Fourier transform procedures described. It is usually assumed in Equations (9.80) and (9.81) that N data points in $x(n)$ and $y(n)$ are transformed into N frequency points which are spaced $1/T$ apart. For these situations, the Nyquist cut off frequency occurs when $k = N/2$ so that for N even, unique results occur only for $k = 0, 1, 2, \dots, (N/2) - 1$. To obtain $X(k)$ and $Y(k)$, observe that

$$\exp \left[j \frac{2\pi n(N-k)}{N} \right] = \exp \left[-j \frac{2\pi nk}{N} \right] \quad (9.82)$$

since $\exp[j2\pi n] = 1$ for any n . Hence, if $Z^*(k)$ is the complex conjugate of $Z(k)$, then

$$Z^*(N-k) = \sum_{n=0}^{N-1} [x(n) - jy(n)] \exp \left[-j \frac{2\pi nk}{N} \right] \quad (9.83)$$

From Equations (9.81) and (9.83), it follows that

$$Z(k) + Z^*(N-k) = 2 \sum_{n=0}^{N-1} x(n) \exp \left[-j \frac{2\pi nk}{N} \right] = 2X(k)$$

$$Z(k) - Z^*(N-k) = 2j \sum_{n=0}^{N-1} y(n) \exp \left[-j \frac{2\pi nk}{N} \right] = 2jY(k)$$

Thus, the two real-valued records $x(n)$ and $y(n)$ have Fourier transforms $X(k)$ and $Y(k)$ given by

$$\begin{aligned} X(k) &= \frac{Z(k) + Z^*(N - k)}{2} \\ Y(k) &= \frac{Z(k) - Z^*(N - k)}{2j} \end{aligned} \quad k = 0, 1, \dots, N - 1 \quad (9.84)$$

As a last topic here, consider the computation of inverse Fourier transforms. The inverse Fourier transform of the $X(f)$ in Equation (9.42) is

$$x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi ft} df \quad (9.85)$$

This leads to the discrete inverse Fourier transform formula

$$x_n = \frac{1}{N} \sum_{k=0}^{N-1} X_k \exp \left[j \frac{2\pi kn}{N} \right] \quad n = 0, 1, 2, \dots, N - 1 \quad (9.86)$$

using the Fourier components X_k of Equation (9.47) that are computed by the FFT procedures. The constant $(1/N)$ in Equation (9.86) is a scale factor only and is not otherwise important. This inverse Fourier transform can be computed by the same FFT procedures previously described by merely interchanging k and n , and replacing x_n by X_k and $W(kn)$ by $W(-kn)$.

9.4 PROBABILITY DENSITY FUNCTIONS

Consider N data values $\{x_n\}$, $n = 1, 2, \dots, N$, from a transformed record $x(t)$ which is stationary with zero mean $\bar{x}$. It follows from Equation (6.39) that the probability density function of $x(t)$ can be estimated by

$$\hat{p}(x) = \frac{N_x}{NW} \quad (9.87)$$

where W is a narrow interval centered at x and N_x is the number of data values which fall within the range $x \pm W/2$. Hence an estimate $\hat{p}(x)$ is obtained digitally by dividing the full range of x into an appropriate number of equal width class intervals, tabulating the number of data values in each class interval, and dividing by the product of the class interval width W and the sample size N . Note that the estimate $\hat{p}(x)$ is not unique since it clearly is dependent upon the number of class intervals and their width selected for the analysis.

A formal statement of this procedure will now be given. Let K denote the number of class intervals selected to cover the entire range of the data values

from a to b . Then the width of each interval is given by

$$W = \frac{b - a}{K} \quad (9.88)$$

and the end point of the i th interval is defined by

$$d_i = a + iW \quad i = 0, 1, 2, \dots, K \quad (9.89)$$

Note that $d_0 = a$ and $d_K = b$. Now define a sequence of $K + 2$ numbers $\{N_i\}$, $i = 0, 1, 2, \dots, K + 1$, by the conditions

$$N_0 = [\text{number of } x \text{ such that } x \leq d_0]$$

$$N_i = [\text{number of } x \text{ such that } d_0 < x \leq d_i]$$

$$N_i = [\text{number of } x \text{ such that } d_{i-1} < x \leq d_i] \quad (9.90)$$

$$N_K = [\text{number of } x \text{ such that } d_{K-1} < x \leq d_K]$$

$$N_{K+1} = [\text{number of } x \text{ such that } x > d_K]$$

This procedure will sort out the N data values of x so that the number sequence $\{N_i\}$ satisfies

$$N = \sum_{i=0}^{K+1} N_i \quad (9.91)$$

One method of doing this sorting on a digital computer is to examine each value x_n , $n = 1, 2, \dots, N$, in turn as follows

- 1 If $x_n \leq a$, add the integer one to N_0 .
- 2 If $a < x_n \leq b$, compute $I = (x_n - a)/W$. Then, select i as the largest integer less than or equal to I , and add the integer one to N_i .
- 3 If $x_n > b$, add the integer one to N_{K+1} .

Four output forms for the sequence $\{N_i\}$ can be used. The first output is the histogram, which is simply the sequence $\{N_i\}$ without changes. The second output is the sample percentage of data in each class interval defined for $i = 0, 1, 2, \dots, K + 1$, by

$$\hat{P}_i = \text{Prob}[d_{i-1} < x \leq d_i] = \frac{N_i}{N} \quad (9.92)$$

The third output is the sequence of sample probability density estimates

$\{\hat{p}_i\}$ defined at the midpoints of the K class intervals in $[a, b]$ by

$$\hat{p}_i = \frac{\hat{P}_i}{W} = \left(\frac{N_i}{N}\right) \left(\frac{K}{b-a}\right) \quad i = 1, 2, \dots, K \quad (9.93)$$

The fourth output is the sequence of sample probability distribution estimates $\{\hat{P}(i)\}$ defined at the class interval end points where $i = 0, 1, 2, \dots, K+1$, by

$$\hat{P}(i) = \text{Prob} [-\infty < x \leq d_i] = \sum_{j=0}^i \hat{P}_j = W \sum_{j=0}^i \hat{p}_j \quad (9.94)$$

9.5 AUTOCORRELATION FUNCTIONS

Two methods will be discussed for the digital computation of autocorrelation function estimates. The first method is the standard approach of estimating the correlation function by direct computation of average products among the sample data values. The second method is the roundabout approach of first computing a power spectrum by direct Fourier transform procedures, and then computing the inverse Fourier transform of the power spectrum.

9.5.1 Autocorrelation Estimates via Direct Computations

For N data values $\{x_n\}$, $n = 1, 2, \dots, N$, from a transformed record $x(t)$ which is stationary with $\bar{x} = 0$, the estimated autocorrelation function at the displacement rh is defined by the formula

$$\hat{R}_r = \hat{R}_x(rh) = \frac{1}{N-r} \sum_{n=1}^{N-r} x_n x_{n+r} \quad r = 0, 1, 2, \dots, m \quad (9.95)$$

where r is the lag number, m is the maximum lag number, and $\hat{R}_r$ is the estimate of the true value R_r at lag r , corresponding to the displacement rh . Note that the maximum lag number m is related to the maximum time displacement of the estimate by

$$\tau_{\max} = \tau_m = mh \quad (9.96)$$

Further note that if a power spectral density estimate is computed by taking the Fourier transform of an autocorrelation estimate (as detailed in Section 9.6.1 to follow), then the resolution bandwidth B_e for the resulting spectral density estimate is related to the maximum lag number m by

$$B_e = \frac{1}{\tau_m} = \frac{1}{mh} \quad (9.97)$$

The sample autocorrelation function may also be defined at lag r by

$$\hat{R}_r = \hat{R}_x(rh) = \frac{1}{N} \sum_{n=1}^{N-r} x_n x_{n+r} \quad r = 0, 1, 2, \dots, m \quad (9.98)$$

where the division by N remains constant instead of changing to $N - r$ as in Equation (9.95). Use of this equation gives a biased estimate of the autocorrelation function. However, for N large and m small with respect to N , the values obtained by using Equation (9.98) differ very little from those obtained by use of Equation (9.95).

The autocorrelation function may take on negative as well as positive values. A normalized value for the autocorrelation function is obtained by dividing $\hat{R}_r$ by $\hat{R}_0$ where

$$\hat{R}_0 = \hat{R}_x(0) = \frac{1}{N} \sum_{n=1}^N (x_n)^2 = \bar{x}^2 \quad (9.99)$$

Note that the quantity $\hat{R}_0$ is a sample estimate of the true mean square value in the data. The quantity $\hat{R}_0$ is related to the sample variance s^2 by the relation

$$\hat{R}_0 = \left(\frac{N-1}{N} \right) s^2 \quad (9.100)$$

Thus for large N , there is negligible difference between $\hat{R}_0$ and s^2 . When $\hat{R}_r$ is normalized, one obtains the quantity $\hat{R}_r/\hat{R}_0$ which theoretically will be between plus and minus one, that is,

$$-1 \leq \frac{\hat{R}_r}{\hat{R}_0} \leq 1 \quad (9.101)$$

9.5.2 Autocorrelation Estimates via FFT Computations

A roundabout method for computing autocorrelation function estimates is to first compute by fast Fourier transform procedures the power spectral density functions and then to compute the inverse Fourier transforms of these results. The double use of fast Fourier transform (FFT) procedures to compute both the spectral density functions and the correlation functions can make this total operation more efficient than direct procedures, depending upon the maximum lag values desired. To be more specific, the straightforward implementation of Equation (9.95) requires approximately Nm real multiply-add operations for m lags of an autocorrelation function. If $N = 2^p$, the FFT roundabout method requires approximately $8Np$ real multiply-add operations. Hence the speed ratio is estimated by

$$\text{Speed Ratio} = \frac{Nm}{8Np} = \frac{m}{8p} \quad (9.102)$$

For example, if $N = 2^{13} = 8192$, and $m = 0.1N = 819$, then $8p = 104$, thus indicating that the FFT approach here will be approximately 8 times as fast.

Computation of spectra by FFT procedures, from which correlation functions can follow by computing the inverse FFT of the spectra, are discussed in Section 9.6.2. However, some care must be exercised in using this roundabout approach. It turns out that the usual correlation function is not obtained, but rather a "circular" correlation function defined by the relationship

$$R_x^c(rh) = \frac{(N-r)}{N} [R_x(rh) + R_x([N-1-r]h)] \quad (9.103)$$

where $R_x(rh)$ is defined by Equation (9.95) and

$$R_x([N-1-r]h) = R_x(rh) \quad r = 0, 1, 2, \dots, m \quad (9.104)$$

The two parts of Equation (9.103) are illustrated in Figure 9.7. In practice, for correlation functions which damp out rapidly, this effect is not of any concern for lags m less than, say, 20 percent of N . In any case, the problem can be avoided by adding zeros to the original data. The effect on the correlation function of adding zeros to the data is to spread apart the two portions of the circular correlation function. In particular, if N zeros are added to the original N data values, then the two portions will separate completely as shown in Figure 9.8. See Reference 47 for details.

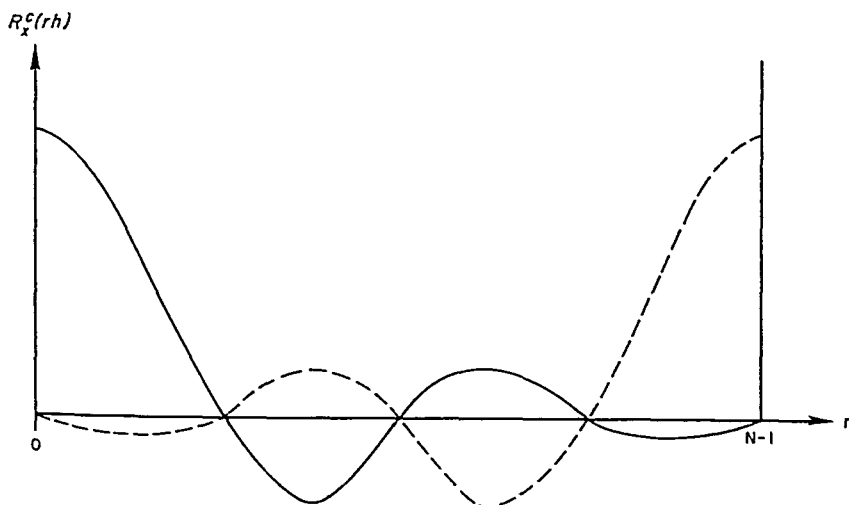


Figure 9.7 Illustration of circular correlation function.

In summary, the following steps are recommended to compute the autocorrelation function via FFT procedures. The available sample size for the record $x(t)$ is assumed to be $N = 2^p$.

- 1 Augment the N data values of x_n , where $n = 0, 1, \dots, N-1$, with N zeros to obtain a new x_n sequence of $2N$ terms
- 2 Compute the $2N$ -point FFT giving X_k for $k = 0, 1, \dots, 2N-1$, using Equation (9.75)
- 3 Compute the raw spectral estimates $\bar{G}_k$ for $k = 0, 1, \dots, 2N-1$, using Equation (9.138)
- 4 Compute the inverse FFT of $\bar{G}_k$ and then multiply by the scale factor $N/(N-r)$ to obtain $\hat{R}_r$ for $r = 0, 1, \dots, 2N-1$. This scale factor is required because of Equation (9.103)
- 5 Discard the last half of $\hat{R}_r$ to obtain results for $r = 0, 1, \dots, N-1$

9.6 POWER SPECTRAL DENSITY FUNCTIONS

For stationary random data, the two most often applied digital methods for computing power spectral density functions are (1) the "standard" or "Blackman-Tukey" method based on computing the power spectral density function via its Fourier transform relation to the autocorrelation function, and (2) the "direct Fourier transform" or "Cooley-Tukey" method based on computing the power spectral density function via a finite-range fast Fourier transform (FFT) of the original data. Although the direct Fourier transform procedure is the more desirable from the viewpoint of computational efficiency, the older standard method is still sometimes used. Hence it is included here for completeness.

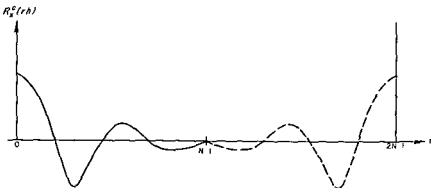


Figure 9.8 Effect on circular correlation function when N zeros are added

9.6.1 Power Spectra Estimates via Correlation Estimates

For sampled data from a transformed record $x(t)$ which is stationary with $\bar{x} = 0$, a raw estimate $\tilde{G}_x(f)$ of a true power spectral density function $G_x(f)$ is defined for an arbitrary f in the range $0 \leq f \leq f_c$ by

$$\tilde{G}_x(f) = 2h \left[\hat{R}_0 + 2 \sum_{r=1}^{m-1} \hat{R}_r \cos \left(\frac{\pi r f}{f_c} \right) + \hat{R}_m \cos \left(\frac{\pi m f}{f_c} \right) \right] \quad (9.105)$$

where h is the time interval between samples, $\hat{R}_r$ is the estimate of the auto-correlation function at lag r , m is the maximum lag number, $f_c = 1/2h$ is the cutoff frequency, and $\tilde{G}_x(f)$ is the raw estimate of true value $G_x(f)$ at frequency f . The function $\tilde{G}_x(f)$ defined by Equation (9.105) is often termed the *periodogram*. Note that $\tilde{G}_x(f)$ is obtained after first computing the auto-correlation estimates $\hat{R}_r$.

Equation (9.105) is a discrete approximation to the theoretical relation of Equation (3.67), namely,

$$G_x(f) = 4 \int_0^{\infty} R_x(\tau) \cos 2\pi f \tau \, d\tau = 2 \int_{-\infty}^{\infty} R_x(\tau) \cos 2\pi f \tau \, d\tau \quad (9.106)$$

where the maximum value for τ is given by $\tau_m = mh$. Thus

$$\begin{aligned} \tilde{G}_x(f) &= 2 \int_{-\tau_m}^{\tau_m} R_x(\tau) \cos 2\pi f \tau \, d\tau \\ &= 2 \int_{-\infty}^{\infty} u_{\tau_m}(\tau) R_x(\tau) \cos 2\pi f \tau \, d\tau \end{aligned} \quad (9.107)$$

where $u_{\tau_m}(\tau)$ is a simple *boxcar function* defined by

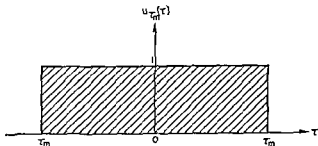
$$\begin{aligned} u_{\tau_m}(\tau) &= 0 & \tau < -\tau_m \\ &= 1 & -\tau_m \leq \tau \leq \tau_m \\ &= 0 & \tau > \tau_m \end{aligned} \quad (9.108)$$

See Figure 9.9 for an illustration. The Fourier transform of $u_{\tau_m}(\tau)$ gives

$$\begin{aligned} U_{\tau_m}(f) &= \int_{-\infty}^{\infty} u_{\tau_m}(\tau) e^{-j2\pi f \tau} \, d\tau \\ &= \int_{-\tau_m}^{\tau_m} e^{-j2\pi f \tau} \, d\tau = 2\tau_m \left(\frac{\sin 2\pi f \tau_m}{2\pi f \tau_m} \right) \end{aligned} \quad (9.109)$$

called a *window function*. Note that this $U_{\tau_m}(f)$ is an even function of f whose shape for positive f only is shown in Figure 9.10.

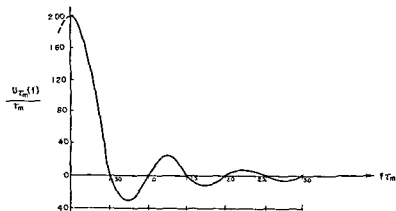
The raw estimate $\tilde{G}_x(f)$ is the convolution of the true $G_x(f)$ with the

Figure 9.9 Illustration of boxcar function $u_{\tau_m}(\tau)$

window function $U_{\tau_m}(f)$. The net effect of this convolution when computing an estimate $\hat{G}_x(f_0)$ at frequency f_0 is to move the main peak of the window function to f_0 , then to multiply the true $G_x(f)$ by the translated window function and finally to integrate over all frequencies. In equation form

$$\hat{G}_x(f) = \int_0^{\infty} G_x(\alpha) U_{\tau_m}(f - \alpha) d\alpha \quad (9.110)$$

Hence the boxcar weighting causes *leakage* by spreading the main lobe of the true power spectral density function and by adding an infinite number of smaller side lobes because of the truncation at τ_m . Half of these side lobes are negative. This introduces the possibility of computing an erroneous negative

Figure 9.10 Illustration of window function $U_{\tau_m}(f)$

power result, particularly since the height of the first two negative lobes is about one-fifth that of the main lobe. To suppress this leakage problem, it is necessary to modify the boxcar type of weighting in the time domain (or its equivalent Fourier transform operations in the frequency domain) so as to broaden the main lobe and decrease the negative effects of the side lobes. This can be accomplished by smoothing the raw estimates by techniques to be described shortly. Before considering these matters, however, a few more properties of the raw estimate should be noted.

Given the raw estimate $\tilde{G}_x(f)$, the total mean square value of the record in the frequency range $0 \leq f \leq f_c$ is defined by

$$\int_0^{f_c} \tilde{G}_x(f) df = \hat{R}_0 = \hat{R}_x(0) \quad (9.111)$$

The power spectral density function estimate $\tilde{G}_x(f)$ should not be confused with an alternative two-sided power spectral density function estimate $\tilde{S}_x(f)$ which is defined for negative as well as positive f by

$$\tilde{S}_x(f) = \tilde{S}_x(-f) = \frac{\tilde{G}_x(f)}{2}$$

Here

$$\int_{-f_c}^{f_c} \tilde{S}_x(f) df = \hat{R}_x(0)$$

It is recommended that the values of the function $\tilde{G}_x(f)$ be calculated only at the $m + 1$ special discrete frequencies where

$$f = \frac{kf_c}{m} \quad k = 0, 1, 2, \dots, m \quad (9.112)$$

This will provide $m/2$ independent spectral estimates since spectral estimates at points less than $2f_c/m$ apart will be correlated. At these discrete frequency points,

$$\tilde{G}_k = \tilde{G}_x\left(\frac{kf_c}{m}\right) = 2h \left[\hat{R}_0 + 2 \sum_{r=1}^{m-1} \hat{R}_r \cos\left(\frac{\pi rk}{m}\right) + (-1)^k \hat{R}_m \right] \quad (9.113)$$

The index k is called the *harmonic number*, and $\tilde{G}_k$ is the raw estimate of the power spectral density function at harmonic k , corresponding to the frequency $f = kf_c/m$. A convenient check formula which requires all of the $m + 1$ estimates of $\tilde{G}_k$ is

$$\hat{R}_x(0) = \frac{1}{2hm} \left[\frac{1}{2} \tilde{G}_0 + \sum_{k=1}^{m-1} \tilde{G}_k + \frac{1}{2} \tilde{G}_m \right] \quad (9.114)$$

LEAKAGE REDUCTION A final smooth estimate of the power spectral density may now be found by further frequency smoothing with a procedure called "Hanning" [Ref. 5]. Let $\hat{G}_k$ denote a *smooth* estimate at harmonic k , where the $\wedge$ replaces the $\sim$. Then at the $m+1$ frequencies $f = kf_c/m$, $k = 0, 1, 2, \dots, m$, one obtains

$$\begin{aligned}\hat{G}_0 &= 0.5\tilde{G}_0 + 0.5\tilde{G}_1 \\ \hat{G}_k &= 0.25\tilde{G}_{k-1} + 0.5\tilde{G}_k + 0.25\tilde{G}_{k+1} \quad k = 1, 2, \dots, m-1 \\ \hat{G}_m &= 0.5\tilde{G}_{m-1} + 0.5\tilde{G}_m\end{aligned} \quad (9.115)$$

An alternative equivalent way to obtain Equation (9.115) is in terms of the Hanning lag weighting function D_r defined by

$$\begin{aligned}D_r &= D(rh) = \frac{1}{2} \left(1 + \cos \frac{\pi r}{m} \right) \quad r = 0, 1, 2, \dots, m \\ &= 0 \quad r > m\end{aligned} \quad (9.116)$$

Note that $D_0 = 1$ and $D_m = 0$. Equation (9.116) may now be combined with Equation (9.113) to yield a condensed formula for the smooth estimates at harmonics $k = 0, 1, 2, \dots, m$. Specifically,

$$\hat{G}_k = \hat{G}_x \left(\frac{kf_c}{m} \right) = 2h \left[\hat{R}_0 + 2 \sum_{r=1}^{m-1} D_r \hat{R}_r \cos \left(\frac{\pi r k}{m} \right) \right] \quad (9.117)$$

where $\hat{R}_r$ is obtained from Equation (9.95) or Equation (9.98). To verify Equation (9.117), observe that for $k = 0$,

$$\begin{aligned}\hat{G}_0 &= 2h \left[\hat{R}_0 + 2 \sum_{r=1}^{m-1} D_r \hat{R}_r \right] \\ &= 2h \left[\hat{R}_0 + \sum_{r=1}^{m-1} \hat{R}_r \left(1 + \cos \frac{\pi r}{m} \right) \right] \\ &= 0.5\tilde{G}_0 + 0.5\tilde{G}_1\end{aligned}$$

Also, for $k = m$,

$$\begin{aligned}\hat{G}_m &= 2h \left[\hat{R}_0 + \sum_{r=1}^{m-1} \hat{R}_r \left(1 + \cos \frac{\pi r}{m} \right) \cos \pi r \right] \\ &= 2h \left[\hat{R}_0 + \sum_{r=1}^{m-1} \hat{R}_r \left[\cos \frac{\pi r(m-1)}{m} + \cos \pi r \right] \right] \\ &= 0.5\tilde{G}_{m-1} + 0.5\tilde{G}_m\end{aligned}$$

Finally, for all $k = 1, 2, \dots, m-1$,

$$\begin{aligned}\hat{G}_k &= 2h \left[\hat{R}_0 + \sum_{r=1}^{m-1} \hat{R}_r \left(1 + \cos \frac{\pi r}{m} \right) \cos \frac{\pi r k}{m} \right] \\ &= 2h \left\{ \hat{R}_0 + \sum_{r=1}^{m-1} \hat{R}_r \left[\frac{1}{2} \cos \frac{\pi r}{m} (k+1) + \cos \frac{\pi r k}{m} + \frac{1}{2} \cos \frac{\pi r}{m} (k+1) \right] \right\} \\ &= 0.25 \tilde{G}_{k-1} + 0.5 \tilde{G}_k + 0.25 \tilde{G}_{k+1}\end{aligned}$$

Thus Equation (9.117) is identical to Equation (9.115).

When the computation of frequency response functions and the associated coherence functions is the main goal, certain variations in the above formulas may sometimes be appropriate. In particular, the use of a different lag weighting function, attributed to Parzen, may occasionally be more useful. This lag weighting function is given by the formula

$$\begin{aligned}D_r' &= 1 - 6 \left(\frac{r}{m} \right)^2 + 6 \left(\frac{r}{m} \right)^3 & r = 0, 1, 2, \dots, \frac{m}{2} \\ &= 2 \left(1 - \left[\frac{r}{m} \right] \right)^3 & r = \frac{m}{2} + 1, \dots, m \\ &= 0 & r > m\end{aligned} \quad (9.118)$$

where the prime is used merely to distinguish Equation (9.118) from Equation (9.116). Note here that $D_0' = 1$ and $D_m' = 0$. The use of this Parzen window maintains the sample coherence function between its theoretical values of ± 1 , as opposed to the Hanning window which allows a sample coherence function to vary over wider limits depending on the form of the power spectrum. The final smooth formula for the power spectral density estimate at harmonic k is of the same form as Equation (9.117). Specifically,

$$\hat{G}_k = \hat{G}_x \left(\frac{k f_c}{m} \right) = 2h \left[\hat{R}_0 + \sum_{r=1}^{m-1} D_r' \hat{R}_r \cos \frac{\pi r k}{m} \right] \quad (9.119)$$

where D_r' is given here by Equation (9.118).

To conclude these leakage reduction ideas, consider the Hanning weighting function of Equation (9.116) again, but written in a form for comparison with the two-sided boxcar function $u_{\tau_m}(\tau)$ in Equation (9.108). For both positive and negative τ , by setting $\tau = rh$ and $\tau_m = mh$, the quantity D_r becomes $d_{\tau_m}(\tau)$ where

$$\begin{aligned}d_{\tau_m}(\tau) &= 0 & \tau < -\tau_m \\ &= \frac{1}{2} \left[1 + \cos \left(\frac{\pi \tau}{\tau_m} \right) \right] & -\tau_m \leq \tau \leq \tau_m \\ &= 0 & \tau > \tau_m\end{aligned} \quad (9.120)$$

Then the Fourier transform of $d_{\tau_m}(\tau)$ gives the Hanning window function $D_{\tau_m}(f)$ where

$$D_{\tau_m}(f) = \frac{1}{4}U_{\tau_m}\left(f - \frac{1}{2\tau_m}\right) + \frac{1}{2}U_{\tau_m}(f) + \frac{1}{4}U_{\tau_m}\left(f + \frac{1}{2\tau_m}\right) \quad (9.121)$$

Note that the coefficients $\frac{1}{4}$ $\frac{1}{2}$ $\frac{1}{4}$ are the same as those used in Equation (9.115). Equation (9.121) is the sum of three $(\sin f)/f$ type functions displaced in frequency as indicated by Equations (9.121) and (9.109). This Hanning window function $D_{\tau_m}(f)$ produces considerably less leakage than the previous window function $U_{\tau_m}(f)$ as illustrated in Figure 9.11. A similar comparison can be made for the Parzen window function versus the Hanning window function.

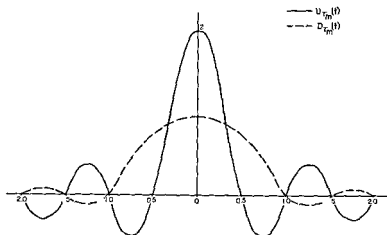


Figure 9.11 Comparison of $U_{\tau_m}(f)$ and $D_{\tau_m}(f)$

PARAMETER CONSIDERATIONS Formulas relating the various parameters involved in the computation of power spectra estimates via autocorrelation estimates will now be summarized. Rules are stated for selecting the sampling interval h , the sample size N , and other parameters so as to achieve digital results which are statistically equivalent to the corresponding analog results and errors listed in Chapters 6 and 8.

Sampling Interval Choose the sampling interval $h = \Delta t$ such that

$$h = \frac{1}{2f_c} \leq \frac{1}{2f_d} \quad (9.122)$$

where f_c is the Nyquist folding frequency of Equation (9.1) and $f_d \leq f_c$ is the highest frequency anticipated in the data. As previously discussed in Section 7.3.1, the surest way to establish an upper bound f_d on the data frequencies is to lowpass filter the original continuous time history record (before digitization) with a high frequency cutoff of f_d . Having established f_d , the sampling interval h should be selected to produce a Nyquist folding frequency f_c which is somewhat higher than f_d , as indicated by Equation (9.122). Two cases are of interest here. For the accurate estimation of autocorrelation functions alone, it is suggested that one choose $f_c = 2f_d$, which corresponds to $h = 1/4f_d$. If power spectra estimates are the ultimate goal, then choosing $h = 2/5f_d$ should be sufficient. Of course, for reasons of economy, one should select h as close to $1/2f_d$ as can be justified.

Number of Correlation Lag Values. Choose the maximum number of correlation lag values m such that

$$m = \frac{1}{B_e h} \quad (9.123)$$

where B_e is the desired equivalent resolution bandwidth for power spectra calculations. Note that

$$B_e = \frac{1}{mh} \quad (9.124)$$

Thus B_e will be small for a given h when m is large.

Sample Size N and Record Length T_r . Choose the sample size N such that

$$N = \frac{m}{\epsilon_r^2} \quad (9.125)$$

where ϵ_r is the normalized standard error desired for spectral calculations. The associated minimum record length T_r is

$$T_r = Nh \quad (9.126)$$

Degrees of Freedom and Standard Error. The number of degrees of freedom for spectral calculations is

$$n = 2B_e T_r = \frac{2N}{m} \quad (9.127)$$

The normalized standard error is given by

$$\epsilon_r = \sqrt{1/B_e T_r} = \sqrt{m/N} \quad (9.128)$$

Thus ϵ_r will be small for a given N when m is small.

Example 9.4 Illustration of Standard Parameter Formulas (a) Suppose it is desired that $\epsilon_r = 0.10$ when $f_c = 2000$ cps and $m = 50$. Compute values for h , B_c , N , and T_r .

$$h = \frac{1}{2f_c} = 0.25 \text{ msec}$$

$$B_c = \frac{1}{mh} = 80 \text{ cps}$$

$$N = \frac{m}{\epsilon_r^2} = 5000$$

$$T_r = Nh = 1.25 \text{ sec}$$

(b) Suppose it is desired that $B_c = 20$ cps when $f_c = 1000$ cps and $\epsilon_r = 0.10$. Compute values for h , m , N , and T_r .

$$h = \frac{1}{2f_c} = 0.50 \text{ msec}$$

$$m = \frac{1}{B_c h} = 100$$

$$N = \frac{m}{\epsilon_r^2} = 10,000$$

$$T_r = Nh = 5.0 \text{ sec}$$

(c) Suppose it is desired that $B_c = 25$ cps when $f_c = 500$ cps and $T_r = 20$ sec. Compute values for h , m , N , and ϵ_r .

$$h = \frac{1}{2f_c} = 1.0 \text{ msec}$$

$$m = \frac{1}{B_c h} = 40$$

$$N = \frac{T_r}{h} = 2000$$

$$\epsilon_r = \sqrt{\frac{m}{N}} \approx 0.14$$

9.6.2 Power Spectra Estimates via FFT Computations

The fast Fourier transform (FFT) procedures discussed in Section 9.3.2 will now be used to compute estimates of power spectral density functions

directly from the original data values. In principle, any sample size N can be handled, but in practice, programs are usually for digital records of length $N = 2^p$. Hence data sequences must either be truncated or have zeros added to obtain the required number of data points. Also, for the same reasons pointed out in the preceding section, it is desirable to modify the time history record by smoothing either in the frequency or time domain so as to suppress the leakage from large side lobes in the effective filter being used to obtain the raw transform. Finally, parameter choices must be made to achieve the desired bias and variability results.

In the equations to follow, it is convenient to let $x(t)$ be defined over the time interval $(-T/2, T/2)$ instead of $(0, T)$. Then the finite range Fourier transform formula of Equation (9.43) can be viewed as a transformation of an infinitely long record $y(t)$ defined over $(-\infty, \infty)$, multiplied by a finite length boxcar function $u_{T/2}(t)$ defined over $(-T/2, T/2)$. That is

$$X(f, T) = \int_{-T/2}^{T/2} x(t) e^{-j2\pi ft} dt = \int_{-\infty}^{\infty} y(t) u_{T/2}(t) e^{-j2\pi ft} dt \quad (9.129)$$

where $y(t)$ is the same as $x(t)$ in the range $(-T/2, T/2)$ and

$$\begin{aligned} u_{T/2}(t) &= 0 & t < -T/2 \\ &= 1 & -T/2 \leq t \leq T/2 \\ &= 0 & t > T/2 \end{aligned} \quad (9.130)$$

This boxcar function $u_{T/2}(t)$ is of length T rather than $2\tau_m$ as shown for $u_{\tau_m}(\tau)$ in Figure 9.9. The Fourier transform of $u_{T/2}(t)$ given by

$$U_{T/2}(f) = T \left(\frac{\sin \pi f T}{\pi f T} \right) \quad (9.131)$$

has its first zero crossing when $f = \pm 1/T$, and represents the effective raw filter shape for FFT estimates. This is illustrated in Figure 9.12. Note the large negative side lobe leakage occurring with this filter.

LEAKAGE REDUCTION. A smooth filter shape for FFT estimates to reduce leakage can be obtained by tapering the original random time series at each end. It is suggested in Reference 4 that a cosine taper be used over $\frac{1}{10}$ of each end of the data in place of the $u_{T/2}(t)$ of Equation (9.130). Such a cosine tapering procedure is shown in Figure 9.13, where the resulting modified boxcar function is labeled $c_{T/2}(t)$. An effect of this tapering is to reduce the variance of the tapered data relative to the original data. Specifically, this variance change, as computed by the ratio of the areas of $u_{T/2}(t)$ to $c_{T/2}(t)$, is $(1/0.875)$. This quantity is the amount by which later smooth spectrum estimates should be multiplied so as to obtain the correct scale factor when $c_{T/2}(t)$ is used.

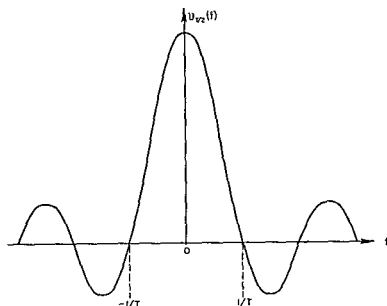


Figure 9.12 Raw filter for FFT estimates

The corresponding smooth filter shape for FFT estimates will now be roughly triangular in shape as shown in Figure 9.14. This result, which is labeled $C_{T/2}(f)$, is found by taking the Fourier transform of the $c_{T/2}(t)$ in Figure 9.13. Note that the width of the main lobe for $C_{T/2}(f)$ is greater than for $U_{T/2}(f)$ in Figure 9.12. Also, the half power bandwidth for $C_{T/2}(f)$ is approximately $B_s = 1/T$, and side lobe leakage is suppressed over that from $U_{T/2}(f)$.

EFFECT OF ADDING ZEROS In the evaluation of $X(f, T)$ at discrete frequency values, see Equation (9.46), the usual spacing is given by

$$\Delta f = \frac{1}{T} = \frac{1}{Nh} \quad (9.132)$$

This is the same as the spacing to the first zero crossing on either side of the main lobe of $U_{T/2}(f)$. When zeros are added to a data sequence, no change takes place in the filter shape of $U_{T/2}(f)$. However, because of the nature of the FFT computational formulas, the spacing of the estimates is based on the

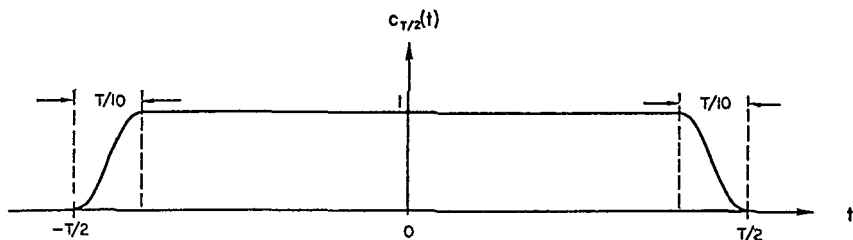


Figure 9.13 Cosine taper data window.

augmented record length and becomes

$$\Delta f' = \frac{1}{(N + N_z)h} \quad (9.133)$$

where N_z represents the number of added zeros. Thus the main effect of making $N_z = N$, for example, is to halve the spacing and obtain twice the number of estimates for a given record length. This is shown in Figure 9.15. Undesirable side lobe effects (not shown in this figure) can also occur.

POWER SPECTRUM ESTIMATES. From the discussions in Section 3.2.3, it is clear that for a single record $x(t)$, a raw estimate of the power spectral

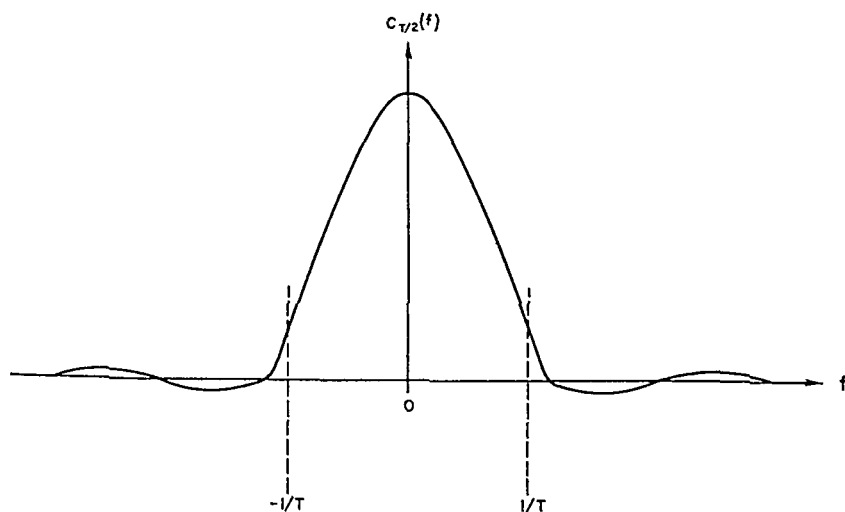
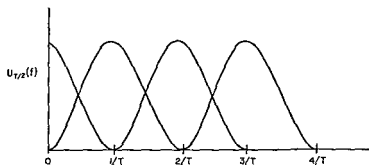
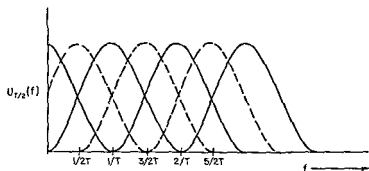


Figure 9.14 Smooth filter for FFT estimates.



N TERMS WITH NO ZEROS



2N TERMS WHEN N ZEROS ARE ADDED

Figure 9.15 Effect on spacing of spectral estimates when N zeros are added

density function at any frequency f is given by the formula

$$\hat{G}_x(f) = \frac{2}{T} |X(f, T)|^2 \quad (9.134)$$

Here, $T = Nh$ and from Equation (9.45),

$$X(f, T) = h \sum_{n=0}^{N-1} x_n \exp[-j2\pi fnh] \quad (9.135)$$

At the usual FFT discrete frequency values

$$f_k = \frac{k}{T} = \frac{k}{Nh} \quad k = 0, 1, 2, \dots, N-1 \quad (9.136)$$

the Fourier components are defined in Equation (9.47) by

$$X_k = \frac{X(f_k, T)}{h} = \sum_{n=0}^{N-1} x_n \exp \left[-j \frac{2\pi kn}{N} \right] \quad (9.137)$$

Hence, from Equations (9.134) and (9.137), the power spectrum estimate becomes

$$\tilde{G}_k = \tilde{G}_x(f_k) = \frac{2}{Nh} |X(f_k, T)|^2 = \frac{2h}{N} |X_k|^2 \quad (9.138)$$

Note that the discrete frequencies in Equation (9.136) are *not* the same discrete frequency values previously used in Equation (9.112) except when $N = 2m$. Hence Equations (9.138) and (9.113) for the same k will usually represent spectrum estimates at different frequencies.

PROCEDURAL STEPS. The following steps are recommended to compute power spectra estimates via FFT procedures. Assume that the sample size for the data sequence x_n is initially of arbitrary size N .

1. Truncate the data sequence or add zeros so that $N = 2^n$.
2. Taper the resulting sequence using the cosine taper data window of Figure 9.13, or some other appropriate tapering.
3. Compute the X_k of Equation (9.137) for $k = 0, 1, \dots, N-1$, using the FFT procedure of Equation (9.75).
4. Compute the $\tilde{G}_k$ of Equation (9.138) for $k = 0, 1, \dots, N-1$.
5. Adjust these estimates for the scale factor due to tapering, e.g., by replacing $\tilde{G}_k$ by $(1/0.875)\tilde{G}_k$ if the cosine tapering of Figure 9.13 is used.

These adjusted estimates are raw estimates of the power spectral density function where $B_e = 1/T$ so that each estimate is governed by a χ^2 distribution with 2 degrees of freedom. Further averaging will now be indicated to obtain final practical smooth estimates which have more suitable statistical variability properties.

FREQUENCY SMOOTHING. If the spectrum is of a bandwidth limited white noise nature, then estimates at a frequency spacing of $1/T_r$ will be essentially uncorrelated. Hence if ℓ neighboring frequency components of the raw spectral estimates are averaged, then the final smooth spectral estimate $\hat{G}_k$,

where the $\wedge$ replaces the $\sim$, is given by

$$\hat{G}_k = \frac{1}{\ell} [\tilde{G}_k + \tilde{G}_{k+1} + \dots + \tilde{G}_{k+\ell-1}] \quad (9.139)$$

The quantity $\hat{G}_k$ will be a χ^2 variable with roughly $n = 2\ell$ degrees of freedom by the χ^2 addition theorem for independent variables. The final effective resolution bandwidth will be approximately $B_s' = \ell B_s$, where each $B_s = 1/T_r$. Hence by this frequency smoothing,

$$B_s' = \ell B_s = \frac{\ell}{T_r} \quad (9.140)$$

$$n = 2B_s' T_r = 2\ell \quad (9.141)$$

The normalized standard error is then given by

$$\epsilon_r = \sqrt{1/B_s' T_r} = \sqrt{1/\ell} \quad (9.142)$$

After smoothing, the filter shape for B_s' will be trapezoidal in form when each B_s is triangular in form. This is shown in Figure 9.16. The estimate $\hat{G}_k$ may be considered as representing the midpoint of the frequency interval from f_k to $f_{k+\ell-1}$. A total of N/ℓ such estimates can be obtained.

SEGMENT AVERAGING A second way to proceed is to average the results from q separate time slices, where each time slice is of length T_r' such that the original $T_r = qT_r'$. Here, the final smooth spectral estimate is given by

$$\hat{G}_k = \frac{1}{q} [\tilde{G}_{k,1} + \tilde{G}_{k,2} + \dots + \tilde{G}_{k,q}] \quad (9.143)$$

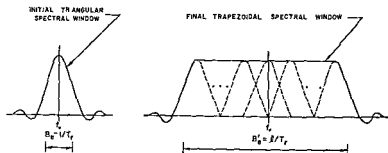


Figure 9.16 Filter shape before and after frequency smoothing

where $\tilde{G}_{k,q}$ is the raw estimate at frequency f_k of the q th time slice. This quantity $\tilde{G}_k$ in Equation (9.143) will be a χ^2 variable with roughly $2q$ degrees of freedom. The effective resolution bandwidth will be approximately $1/T_r'$. Hence by this segment averaging,

$$B_e'' = \frac{1}{T_r'} = \frac{q}{T_r} \quad (9.144)$$

$$n = 2B_e''T_r = 2q \quad (9.145)$$

$$\epsilon_r = \sqrt{1/q} \quad (9.146)$$

The filter shape for B_e'' will be triangular as previously illustrated in Figure 9.14, except that it will be wider because $\pm q/T_r$ replaces $\pm 1/T_r$. This wider filter shape will be the same for one segment as for the average of q segments since T' is unchanged by the segment averaging. The estimate $\tilde{G}_k$ here may be considered as representing the midpoint of the frequency interval covered by B_e'' . A total of N/q such estimates can be obtained.

To apply this procedure using the Cooley-Tukey algorithm, it is required that the total sample size $N = q \cdot 2^n$ so that each time slice is of length 2^n . It is not necessary for q to be a power of 2. If zeros have to be added to achieve N , then a proportionate number of zeros should be added to each segment to maintain the same number of degrees of freedom for each segment.

COMBINED AVERAGING. For combined averaging using both frequency smoothing and segment averaging, the final effective resolution bandwidth will be approximately

$$B_e''' = \frac{\ell q}{T_r} \quad (9.147)$$

and the resulting $\hat{G}_k$ will be a χ^2 variable with the number of degrees of freedom n and the normalized standard error ϵ_r given by

$$n = 2B_e'''T_r = 2\ell q \quad (9.148)$$

$$\epsilon_r = \sqrt{1/\ell q} \quad (9.149)$$

SPEED RATIO. The computation time for the discrete power spectrum estimates from a single record, as computed via the standard autocorrelation-transform method of Section 9.6.1, involves about Nm real multiply-add operations. This time is dictated mainly by the autocorrelation computations. The associated Fourier transform times are not significant because the maximum correlation lag is usually small compared to the record length.

For the discrete power spectrum estimates from a single record, as computed via the FFT procedures in Section 9.6.2, the computation time is

dictated mainly by the basic FFT computations and involves about $4Np$ real multiply add operations. Hence, the speed ratio for computing a single power spectral density function by FFT procedures relative to the standard method is approximately

$$\text{Speed Ratio} = \frac{Nm}{4Np} = \frac{m}{4p} \quad (9.150)$$

For example, if $N = 2^{13} = 8192$, and $m = 0.1N = 819$, then the speed ratio is $(819/52) \approx 16$.

Example 9.5. Illustration of FFT Parameter Formulas. (a) Suppose it is desired that $\epsilon_r = 0.10$ when $f_c = 2000$ cps and $N = 5000$ by the frequency smoothing method of Equation (9.139). Compute values for h , ℓ , T_r , and B_s' .

$$h = \frac{1}{2f_c} = 0.25 \text{ msec}$$

$$\ell = \frac{1}{\epsilon_r^2} = 100$$

$$T_r = Nh = 1.25 \text{ sec}$$

$$B_s' = \frac{\ell}{T_r} = 80 \text{ cps}$$

(b) Suppose it is desired that $B_s'' = 50$ cps when $f_c = 10,000$ cps and $\epsilon_r = 0.20$ by the segment averaging method of Equation (9.143). Compute values for h , q , T_r , and N .

$$h = \frac{1}{2f_c} = 0.05 \text{ msec}$$

$$q = \frac{1}{\epsilon_r^2} = 25$$

$$T_r = \frac{q}{B_s''} = 0.50 \text{ sec}$$

$$N = \frac{T_r}{h} = 10,000$$

9.7 CALCULATIONS FOR TWO RECORDS

In the formulas to follow, it is assumed that two time history records $u(t)$ and $v(t)$ are from stationary processes and exist only for $t_0 \leq t \leq t_0 + T_r$,

where t_0 is arbitrary and may be zero. Assume the sampling time interval is $\Delta t = h$ which corresponds to a cutoff frequency of $f_c = 1/2h$. Let the respective sample values of $u(t)$ and $v(t)$ be denoted by

$$\begin{aligned} u_n &= u(t_0 + nh) & n &= 1, 2, 3, \dots, N \\ v_n &= v(t_0 + nh) & T_r &= Nh \end{aligned} \quad (9.151)$$

The first quantities to compute are the sample mean values

$$\bar{u} = \frac{1}{N} \sum_{n=1}^N u_n \quad \bar{v} = \frac{1}{N} \sum_{n=1}^N v_n \quad (9.152)$$

Then, compute the transformed data values

$$x_n = u_n - \bar{u} \quad y_n = v_n - \bar{v} \quad n = 1, 2, \dots, N \quad (9.153)$$

corresponding to new time records $x(t) = u(t) - \bar{u}$ and $y(t) = v(t) - \bar{v}$, where $\bar{x} = 0$ and $\bar{y} = 0$. Also, trends in the data should be removed as appropriate by the methods in Section 9.1.3.

9.7.1 Joint Probability Density Functions

It follows from Equation (6.61) that the joint probability density function of two stationary records $x(t)$ and $y(t)$ can be estimated from digitized data by

$$\hat{p}(x, y) = \frac{N_{xy}}{NW_x W_y} \quad (9.154)$$

where W_x and W_y are narrow intervals centered on x and y , respectively, and N_{xy} is the number of pairs of data values which simultaneously fall within these intervals. Hence an estimate $\hat{p}(x, y)$ is obtained by dividing the full ranges of x and y into appropriate numbers of equal width class intervals forming two dimensional rectangular cells, tabulating the number of data values in each cell, and dividing by the product of the cell area $W_x W_y$ and the sample size N . Computer procedures for sorting the data values into appropriate cells are similar to those outlined for probability density estimates in Section 9.4.

9.7.2 Cross-Correlation Functions

As for autocorrelation functions, there are two basic approaches to the estimation of cross-correlation functions, namely, the direct approach and the roundabout FFT approach. Procedures for both cases will now be discussed.

DIRECT PROCEDURES Unbiased estimates of the sample cross correlation functions at lag numbers $r = 0, 1, 2, \dots, m$, are defined by

$$\hat{R}_{xy}(rh) = \frac{1}{N-r} \sum_{n=1}^{N-r} x_n y_{n+r} \quad (9.155)$$

$$\hat{R}_{yx}(rh) = \frac{1}{N-r} \sum_{n=1}^{N-r} y_n x_{n+r} \quad (9.156)$$

Note that the two cross-correlation functions $\hat{R}_{xy}(rh)$ and $\hat{R}_{yx}(rh)$ differ by the interchange of the x_n and y_n data values. Instead of dividing by $N-r$ in this definition, if $N \gg m$, it may be more convenient to divide by N as in Equation (9.98).

The sample cross-correlation function $\hat{R}_{xy}(rh)$ may be normalized to have values between plus and minus one through a division by $\sqrt{\hat{R}_x(0)} \sqrt{\hat{R}_y(0)}$. This defines a sample cross correlation coefficient

$$\hat{\rho}_{xy}(rh) = \frac{\hat{R}_{xy}(rh)}{\sqrt{\hat{R}_x(0)} \sqrt{\hat{R}_y(0)}} \quad r = 0, 1, 2, \dots, m \quad (9.157)$$

which theoretically should satisfy $-1 \leq \hat{\rho}_{xy}(rh) \leq 1$. A similar formula exists for $\hat{\rho}_{yx}(rh)$.

For later determination by standard methods of the cross spectral density function estimate $\hat{G}_{xy}(f)$ between $x(t)$ and $y(t)$, calculate for $r = 0, 1, 2, \dots, m$, the two quantities (even and odd parts of the cross-correlation function)

$$\hat{A}_r = \hat{A}_{xy}(rh) = \frac{1}{2} [\hat{R}_{xy}(rh) + \hat{R}_{yx}(rh)] \quad (9.158)$$

$$\hat{B}_r = \hat{B}_{xy}(rh) = \frac{1}{2} [\hat{R}_{xy}(rh) - \hat{R}_{yx}(rh)] \quad (9.159)$$

These quantities are helpful only when the cross spectral density function is computed via the cross-correlation function. They are not used in the newer FFT procedures. Another special operation sometimes used with the older procedures is to translate these $\hat{A}_r$ and $\hat{B}_r$ terms to the point where the cross-correlation function attains its maximum value. This amounts to a simple form of two-dimensional prewhitening of the cross spectrum, and must be corrected for again at the end of the computations. It is done to guarantee that no significant power is lost because of leakage effects through lag windows.

VIA FAST FOURIER TRANSFORMS Similar to the development outlined in Section 9.5.2, the following steps are recommended to compute the cross correlation function via FFT procedures. The initial sample size for both $x(t)$ and $y(t)$ is assumed to be $N = 2^p$. For these computations, the cross-correlation function is obtained from the cross spectral density function and

involves two separate FFT's, one for $x(t)$ and one for $y(t)$. These two FFT's may be computed simultaneously by using the method in Section 9.3.4.

1. Store x_n in the real part and y_n in the imaginary part of $z_n = x_n + jy_n$, $n = 0, 1, \dots, N - 1$.
2. Augment both the real and imaginary parts with N zeros to obtain a new z_n sequence of $2N$ terms.
3. Compute the $2N$ -point FFT giving Z_k for $k = 0, 1, \dots, 2N - 1$, using the FFT procedure of Equation (9.81).
4. Compute X_k and Y_k for $k = 0, 1, \dots, 2N - 1$, using Equation (9.84).
5. Compute the raw cross-spectral density estimate $\tilde{G}_{xy}(f_k)$ for $k = 0, 1, \dots, 2N - 1$, using Equation (9.175).
6. Compute the inverse FFT of $\tilde{G}_{xy}(k)$ and then multiply by the scale factor $N/(N - r)$ to obtain $\tilde{R}_{xy}(rh)$ for $r = -N, -N + 1, \dots, 0, 1, \dots, N - 1$. This scale factor is required because of a circular cross-correlation effect similar to Equation (9.103).

9.7.3 Cross-Spectral Density Functions

The computations for cross-spectral density functions stated here are extensions of the previous power spectral density procedures detailed in Section 9.6. The cross-spectral density function is a complex-valued quantity defined by

$$G_{xy}(f) = C_{xy}(f) - j Q_{xy}(f) \quad (9.160)$$

where $C_{xy}(f)$ is called the coincident-spectral density function and $Q_{xy}(f)$ is called the quadrature-spectral density function. An equivalent representation for $G_{xy}(f)$ is

$$G_{xy}(f) = |G_{xy}(f)| e^{-j\theta_{xy}(f)} \quad (9.161)$$

where $|G_{xy}(f)|$ is the absolute value of $G_{xy}(f)$, and $\theta_{xy}(f)$ is the phase angle contained in $G_{xy}(f)$. Two resulting relations are

$$|G_{xy}(f)| = [C_{xy}^2(f) + Q_{xy}^2(f)]^{1/2} \quad (9.162)$$

$$\theta_{xy}(f) = \tan^{-1} \left[\frac{Q_{xy}(f)}{C_{xy}(f)} \right] \quad (9.163)$$

VIA CROSS-CORRELATION FUNCTIONS. Raw estimates from sampled data for the coincident-spectral density function and the quadrature-spectral density function may be found as follows. The formulas are for one-sided spectra which are nonzero only for $f \geq 0$.

At an arbitrary value of f in the range $0 \leq f \leq f_c$, raw estimates for $\tilde{C}_{xy}(f)$ and $\tilde{Q}_{xy}(f)$ are

$$\tilde{C}_{xy}(f) = 2h \left[\hat{A}_0 + 2 \sum_{r=1}^{m-1} \hat{A}_r \cos \left(\frac{\pi r f}{f_c} \right) + \hat{A}_m \cos \left(\frac{\pi m f}{f_c} \right) \right] \quad (9.164)$$

$$\tilde{Q}_{xy}(f) = 2h \left[2 \sum_{r=1}^{m-1} \hat{B}_r \sin \left(\frac{\pi r f}{f_c} \right) + \hat{B}_m \sin \left(\frac{\pi m f}{f_c} \right) \right] \quad (9.165)$$

where the $\hat{A}_r$ and $\hat{B}_r$ are given by Equations (9.158) and (9.159). As before, these functions should be calculated only at the $m+1$ special discrete frequencies of harmonic number k where

$$f = \frac{k f_c}{m} \quad k = 0, 1, 2, \dots, m \quad (9.166)$$

At these discrete frequency points,

$$\tilde{C}_k = \tilde{C}_{xy} \left(\frac{k f_c}{m} \right) = 2h \left[\hat{A}_0 + 2 \sum_{r=1}^{m-1} \hat{A}_r \cos \left(\frac{\pi r k}{m} \right) + (-1)^k \hat{A}_m \right] \quad (9.167)$$

$$\tilde{Q}_k = \tilde{Q}_{xy} \left(\frac{k f_c}{m} \right) = 4h \sum_{r=1}^{m-1} \hat{B}_r \sin \left(\frac{\pi r k}{m} \right) \quad (9.168)$$

Smooth estimates for C_k and Q_k at harmonic k may now be calculated as in Equation (9.115) by using the Hanning method. This yields

$$\begin{aligned} C_0 &= 0.5 \tilde{C}_0 + 0.5 \tilde{C}_1 \\ Q_0 &= 0.5 \tilde{Q}_0 + 0.5 \tilde{Q}_1 \end{aligned} \quad (9.169)$$

$$\begin{aligned} C_k &= 0.25 \tilde{C}_{k-1} + 0.5 \tilde{C}_k + 0.25 \tilde{C}_{k+1} \\ &\quad k = 1, 2, \dots, m-1 \end{aligned} \quad (9.170)$$

$$\begin{aligned} Q_k &= 0.25 \tilde{Q}_{k-1} + 0.5 \tilde{Q}_k + 0.25 \tilde{Q}_{k+1} \\ C_m &= 0.5 \tilde{C}_{m-1} + 0.5 \tilde{C}_m \\ Q_m &= 0.5 \tilde{Q}_{m-1} + 0.5 \tilde{Q}_m \end{aligned} \quad (9.171)$$

At the $m+1$ discrete frequencies $f = k f_c / m$, $k = 0, 1, 2, \dots, m$, one obtains the smooth cross spectral density estimates

$$G_{xy} \left(\frac{k f_c}{m} \right) = C_k - j Q_k = \left| G_{xy} \left(\frac{k f_c}{m} \right) \right| e^{-j \theta_{xy}(k f_c / m)} \quad (9.172)$$

where

$$\left| \hat{G}_{xy} \left(\frac{kf_c}{m} \right) \right| = (\hat{C}_k^2 + \hat{Q}_k^2)^{1/2} \quad (9.173)$$

$$\hat{\theta}_{xy} \left(\frac{kf_c}{m} \right) = \tan^{-1} \left(\frac{\hat{Q}_k}{\hat{C}_k} \right) \quad (9.174)$$

VIA FAST FOURIER TRANSFORMS. Similar to the development in Section 9.6.2, the following steps are recommended to compute the cross-spectral density function via FFT procedures. Assume that the sample size for each of the two data sequences x_n and y_n is initially of arbitrary size N .

1. Truncate the two data sequences or add zeros so that each sequence has $N = 2^p$ data values.
2. Taper the two resulting sequences using the cosine taper window of Figure 9.13, or some other appropriate tapering.
3. Store the tapered x_n in the real part and the tapered y_n in the imaginary part of $z_n = x_n + jy_n$, $n = 0, 1, \dots, N - 1$.
4. Compute the N -point FFT giving Z_k for $k = 0, 1, \dots, N - 1$, using the FFT procedure of Equation (9.75).
5. Compute X_k and Y_k for $k = 0, 1, \dots, N - 1$, using Equation (9.84).
6. Compute the raw cross-spectral density estimate $\tilde{G}_{xy}(f_k)$ for $k = 0, 1, \dots, N - 1$ and $f_k = k/Nh$ by the formula

$$\tilde{G}_{xy}(f_k) = \frac{2h}{N} \left| X_k^* Y_k \right| \quad (9.175)$$

where X_k^* is the complex conjugate of X_k .

7. Adjust these estimates for the scale factor due to tapering, e.g., by replacing $\tilde{G}_{xy}(f_k)$ by $(1/0.875)\tilde{G}_{xy}(f_k)$ if the cosine tapering of Figure 9.13 is used.

8. Finally, obtain smooth estimates by either frequency smoothing or segment averaging, or some combined averaging, as indicated previously by Equations (9.139) and (9.143).

SPEED RATIO IMPROVEMENT FOR TWO RECORDS. A guideline rule on the possible overall speed ratio improvement between the FFT method and the correlation method of computing spectra when two records of data are completely analyzed is

$$\text{Speed Ratio} = \frac{4Nm}{4Np} = \frac{m}{p} \quad (9.176)$$

This assumes (1) that the desired output is the power spectral density function of each channel and the cross spectral density function between them, (2) that four correlation functions (R_{xx} , R_{yy} , R_{xy} , R_{yx}) are computed requiring approximately $4Nm$ real multiply-add operations, (3) that $N = 2^p$ and (4) that only one pass through the FFT programs is needed requiring approximately $4Np$ real multiply add operations. For example, if $N = 2^{13} = 8192$, and $m = 0.1N = 819$, then the overall speed ratio improvement to analyze two records is $(819/13) \approx 63$.

Speed ratio formulas are listed in Table 9.1 which compare standard and FFT procedures. In practice, special programming operations can achieve in some cases a further improvement in speeds by a factor of two.

Table 9.1

Comparison of the Number of Real multiply add Operations for Standard and Fast Fourier Transform Procedures When $N = 2^p$ and $m = \text{Maximum Lag Number}$

	Standard	FFT	Speed Ratio
Discrete Fourier Transforms	N^2	$4Np$	$\frac{N}{4p}$
Correlation Functions	Nm	$8Np$	$\frac{m}{8p}$
Spectral Density Functions	Nm	$4Np$	$\frac{m}{4p}$
Correlation and Spectral Functions for Two Records (Frequency Response and Coherence Functions)	$4Nm$	$4Np$	$\frac{m}{p}$

9.8 FREQUENCY RESPONSE FUNCTIONS AND COHERENCE FUNCTIONS

9.8.1 Single-Input Linear Systems

From Chapter 2, the frequency response function $H(f)$ of a constant parameter linear system is a complex-valued function defined by

$$H(f) = |H(f)| e^{-j\phi(f)} \quad (9.177)$$

where $|H(f)|$ is the gain factor of the system and $\phi(f)$ is the phase factor of the system. From Equation (5.8), if a stationary input $x(t)$ produces an output $y(t)$, the system gain factor can be estimated by

$$|\hat{H}(f)| = \left(\frac{\hat{G}_y(f)}{\hat{G}_x(f)} \right)^{1/2} \quad (9.178)$$

Hence, at the discrete frequencies $f = kf_c/m$; $k = 0, 1, 2, \dots, m$, the gain factor can be estimated digitally by

$$|\hat{H}_k| = \left| \hat{H} \left(\frac{kf_c}{m} \right) \right| = \left(\frac{\hat{G}_{k,y}}{\hat{G}_{k,x}} \right)^{1/2} \quad (9.179)$$

where $\hat{G}_{k,x}$ and $\hat{G}_{k,y}$ represent the smooth digital values of the power spectrum estimates at harmonic k for $x(t)$ and $y(t)$, respectively. Note that Equation (9.178) gives an unbiased estimate of $|H(f)|$ only for the ideal case where there is no extraneous noise at either the input or output and no other inputs contribute to $y(t)$. The discrete frequency values in Equation (9.179) arise from the older autocorrelation-transform method of computing power spectra where m is the maximum lag value.

With newer FFT procedures, discrete frequency values are located at

$$f_k = \frac{k}{Nh} \quad k = 0, 1, \dots, N-1 \quad (9.180)$$

Since $f_c = 1/2h$, these FFT discrete frequencies will match the previous discrete frequencies for the same value of k only if $N = 2m$. In computing frequency response functions, smooth estimates of the spectral quantities $\hat{G}_x(f_k)$ and $\hat{G}_y(f_k)$ should always be computed via the FFT procedures before substituting into Equation (9.178) to obtain

$$|\hat{H}_k| = \left| \hat{H} \left(\frac{k}{Nh} \right) \right| \quad k = 0, 1, \dots, N-1 \quad (9.181)$$

Note that, in general, Equations (9.179) and (9.181) yield different results for the same value of k .

For either single input single output linear systems where extraneous noise is present only at the output, or multiple input-single output linear systems where the inputs are uncorrelated, the recommended method for estimating the system frequency response function (including both gain and phase factors) is given by Equation (5.9), namely,

$$\hat{H}(f) = \frac{\hat{G}_{xy}(f)}{\hat{G}_x(f)} \quad (9.182)$$

It follows that

$$|\hat{H}(f)| = \frac{|\hat{G}_{xy}(f)|}{\hat{G}_x(f)} \quad \text{and} \quad \hat{\phi}(f) = \hat{\theta}_{xy}(f) \quad (9.183)$$

Hence, at the discrete frequencies $f = kf_c/m$, $k = 0, 1, 2, \dots, m$, the gain factor and phase factor can be estimated digitally by

$$|\hat{H}_k| = \frac{(\hat{C}_k^2 + \hat{Q}_k^2)^{1/2}}{\hat{G}_{kx}} \quad (9.184)$$

$$\hat{\phi}_k = \tan^{-1} \left(\frac{\hat{Q}_k}{\hat{C}_k} \right) \quad (9.185)$$

where $\hat{G}_{kx}$ represents the digital value of the power spectrum estimate at harmonic k for $x(t)$, and $\hat{C}_k$ and $\hat{Q}_k$ represent the digital values of the co-spectrum estimate and quadspectrum estimate at harmonic k between $x(t)$ and $y(t)$. As noted previously, the above discrete frequency values $f = kf_c/m$, $k = 0, 1, 2, \dots, m$ should not be confused with the usual FFT discrete frequency values located at $f_k = k/Nh$, $k = 0, 1, \dots, N-1$. Using FFT procedures to compute smooth spectral estimates $\hat{G}_x(f_k)$ and $\hat{G}_{xy}(f_k)$, Equation (9.183) yields

$$\begin{aligned} |\hat{H}_k| &= \left| \hat{H} \left(\frac{k}{Nh} \right) \right| \\ \hat{\phi}_k &= \hat{\phi} \left(\frac{k}{Nh} \right) \end{aligned} \quad k = 0, 1, \dots, N-1 \quad (9.186)$$

From Section 5.2, the ordinary coherence function $\gamma_{xy}^2(f)$ between two stationary records $x(t)$ and $y(t)$ is defined by

$$\gamma_{xy}^2(f) = \frac{|\hat{G}_{xy}(f)|^2}{\hat{G}_x(f) \hat{G}_y(f)} \quad (9.187)$$

where $G_x(f)$ and $G_y(f)$ are the power spectral density functions of $x(t)$ and $y(t)$, respectively, and $G_{xy}(f)$ is the cross-spectral density function between $x(t)$ and $y(t)$. The coherence function theoretically should satisfy $0 \leq \gamma_{xy}^2(f) \leq 1$ for all f .

In terms of digital calculations, at the discrete frequencies $f = kf_c/m$; $k = 0, 1, 2, \dots, m$, the coherence function is estimated by

$$\hat{\gamma}_k^2 = \frac{\hat{C}_k^2 + \hat{Q}_k^2}{\hat{G}_{k,x} \hat{G}_{k,y}} \quad (9.188)$$

where $\hat{G}_{k,x}$ and $\hat{G}_{k,y}$ represent the digital values of the power spectrum estimates at harmonic k for $x(t)$ and $y(t)$, respectively, whereas $\hat{C}_k$ and $\hat{Q}_k$ represent the digital values of the cospectrum estimate and quadspectrum estimate at harmonic k between $x(t)$ and $y(t)$. Using FFT procedures to compute smooth spectral estimates of $\hat{G}_x(f)$, $\hat{G}_y(f)$, and $\hat{G}_{xy}(f)$ at the frequencies $f_k = k/Nh$; $k = 0, 1, \dots, N-1$, Equation (9.187) yields

$$\hat{\gamma}_k^2 = \frac{|\hat{G}_{xy}(f_k)|^2}{\hat{G}_x(f_k) \hat{G}_y(f_k)} \quad (9.189)$$

As noted before, the k 's in Equations (9.188) and (9.189) do not, in general, represent the same frequency values. In computing Equation (9.189), it is essential to work with smooth spectral density estimates for the quantities shown. Their replacement by raw spectral density estimates will always give a raw ordinary coherence value of unity as previously demonstrated in Section 6.6.

9.8.2 Multiple-Input Linear Systems

Figure 5.6 in Section 5.3 represents the model for the multiple input-single output linear system under consideration. This same model appears as Figure 6.5 in Section 6.8 where statistical criteria are discussed for obtaining good estimates. From Equation (6.149), the q multiple frequency response functions indexed by $i = 1, 2, \dots, q$ depend upon computation of good estimates of the following basic spectral quantities:

1. Smooth power spectral density function estimates for all inputs.
2. Smooth cross-spectral density function estimates between all inputs.
3. Smooth cross-spectral density function estimates between each input and the output.

These various spectral quantities can be computed either via correlation-transform methods or via fast Fourier transform (FFT) procedures as discussed in Sections 9.6 and 9.7.3. A digital program is also required to handle the data in a set of spectral density matrices at certain discrete frequencies indexed by $k = 0, 1, 2, \dots$. Each spectral matrix must be inverted to later compute in turn the set of multiple frequency response functions indexed by both i and k . A package of complex-valued arithmetic and matrix digital routines should be available to carry out the necessary operations.

The most economical method to compute the desired spectral quantities in the spectral matrices is as follows. Using the FFT procedures, compute the Fourier transforms of all inputs and the output. Do *not* compute at this stage their power spectral density or cross spectral density functions. For q inputs and one output, this requires at most $(q/2) + 1$ passes through the FFT by computing FFT's for two time series simultaneously. Having retained all Fourier transforms, the conjugate products of all possible pairs of transforms should now be calculated and then smoothed to produce the desired power spectral density and cross spectral density functions. No additional passes through the FFT are required by this method. For q time histories, a total of $q(q+1)/2$ different power spectra and cross spectra are obtained. On the other hand, if a total of $q(q+1)/2$ different power spectra and cross spectra are calculated in the usual way by considering the time histories two at a time, then approximately $q^2/2$ passes through the FFT will be required instead of the $(q/2)$ passes described above.

Multiple coherence functions between the output and all of the measured inputs depend upon computation of the same basic smooth spectral quantities as for multiple frequency response functions. From these smooth spectral estimates, at certain discrete frequencies indexed by $k = 0, 1, 2, \dots$, the multiple coherence function estimates are computed from (5.126) by

$$\hat{\gamma}_{yx}^2(f) = 1 - \frac{|\hat{G}_{vzx}(f)|}{\hat{G}_{yy}(f) |\hat{G}_{xx}(f)|} \quad (9.190)$$

Thus, besides the output power spectral density function estimate $\hat{G}_{yy}(f)$, the multiple coherence function depends upon computation of good estimates of the following two determinants

1. The determinant $|\hat{G}_{xx}(f)|$ of the $q \times q$ spectral matrix of the inputs
2. The determinant $|\hat{G}_{vzx}(f)|$ of the $(q+1) \times (q+1)$ augmented spectral matrix of the output with all the inputs

Digital computer programs should be available to compute these determinants in some efficient manner at the discrete frequencies indexed by k .

The determinant ratio $|\hat{G}_{vzx}(f)|/|\hat{G}_{xx}(f)|$ represents the conditional or residual output power remaining at frequency f after all power is subtracted out which can be accounted for by possible linear relationships. This determinant ratio divided by $\hat{G}_{yy}(k)$ then represents the fraction of output power at frequency f not accounted for by linear relationships. Hence the multiple coherence function estimate of Equation (9.190) represents the fraction of output power at frequency f which can be attributed to linear relationships.

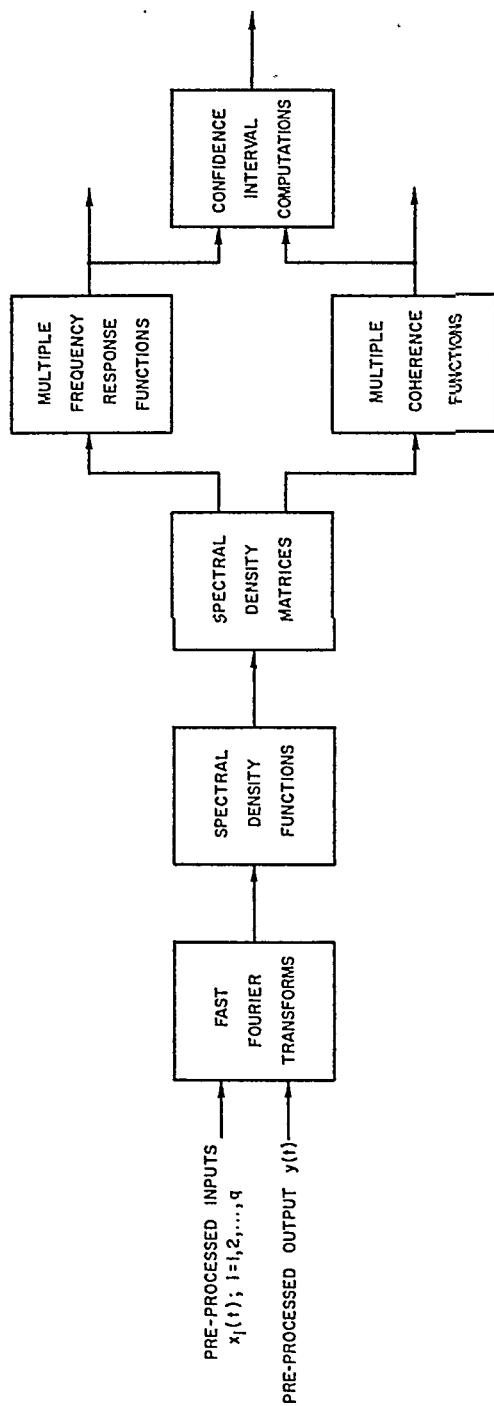


Figure 9.17 Summary of multiple-input linear systems calculations.

in the model of Figure 6.5. Similar formulas and statements can be made for partial coherence functions which may be of interest as discussed in Section 5.4.2.

For frequency response function estimates in multiple input cases, confidence intervals for the true gain factor $|H(f)|$ and true phase factor $\phi(f)$ can be computed for each $i = 1, 2, \dots, q$ at certain discrete frequencies indexed by $k = 0, 1, 2, \dots, N-1$. General coherent input situations are governed by Equations (6.150) through (6.153). These equations also apply to incoherent input situations by setting $\gamma_{i,x}^2(f) = 0$ for each $i = 1, 2, \dots, q$. Confidence intervals for the single input case are obtained by setting $q = 1$. Quantities needed to compute these confidence intervals are as follows:

1. Smooth gain factor estimates for all systems
2. Smooth phase factor estimates for all systems
3. Smooth power spectral density function estimates for all inputs
4. Smooth power spectral density function estimates for the output
5. Smooth estimates of the multiple coherence function between the output and all the inputs
6. Smooth estimates of the multiple coherence function between each input and the other inputs
7. Computation or tabulated values of percentage points of the F distribution at desired values

Confidence interval results are obtained that are indexed by both the system i and discrete frequency k . A block diagram for multiple input linear systems calculations is shown in Figure 9.17.

PROBLEMS

1. Show that distortionless transmission of signals occur with the ideal filters of Figure 9.4 when the gain factor $|H(f)|$ is a constant and the phase factor $\phi(f)$ is a linear function of f .
2. Prove that the coefficient c and the filter weights h_k corresponding to the Butterworth filter of Equation (9.34) are related by the equation

$$c = 1 - \sum_{k=1}^M h_k$$

3. Verify the claims in Section 9.3.2 stated below:
 - (a) Equation (9.57b)
 - (b) Equation (9.62a)
 - (c) Equation (9.67)

4. Estimate the speed ratio in computing autocorrelation functions via fast Fourier transforms versus direct procedures for a 16,000 point data sample if the maximum lag value is 0.2 of the sample size.
5. Prove that power spectra computed using the Parzen window will not be negative by showing that the Parzen window function is given by

$$D_{\tau_m}'(f) = \frac{3}{4}\tau_m \left[\frac{\sin(\pi f \tau_m / 2)}{(\pi f \tau_m / 2)} \right]^4$$

6. Estimate the speed ratio in computing power spectral density functions via fast Fourier transforms versus standard procedures for a 16,000 point data sample if the maximum lag value is 0.2 of the sample size, and discuss how the answer relates to Problem 4.
7. Estimate the leakage reduction properties of Figure (9.14) relative to Figure (9.12) by computing $C_{T/2}(f)$ and comparing with $U_{T/2}(f)$. Assume that $c_{T/2}(t)$ is defined by

$$\begin{aligned} c_{T/2}(t) &= \cos^2 \left(\frac{5\pi t}{T} \right) & -T/2 \leq t \leq -2T/5 \\ &= 1 & -2T/5 \leq t \leq 2T/5 \\ &= \cos^2 \left(\frac{5\pi t}{T} \right) & 2T/5 \leq t \leq T/2 \\ &= 0 & \text{otherwise} \end{aligned}$$

8. Verify the statement made below Equation (9.138) regarding differences between the $\tilde{G}_k$ of Equation (9.138) and the $\tilde{G}_k$ of Equation (9.113).
9. In computing power spectral density functions via FFT procedures, suppose $f_c = 4000$ cps, $T_r = 4.0$ secs and that 60 neighboring in frequency raw power spectral density estimates are averaged to calculate each final smooth spectrum estimate. Determine the final effective spectral resolution and the normalized standard error for these smooth estimates. Estimate how many multiply-add operations will be required to calculate all of the possible smooth power spectral density estimates in this problem.
10. Draw appropriate block diagrams which indicate both the order and the particular computational programs which should be available to carry out a comprehensive digital analysis of:
 - (a) properties of a single ergodic record.
 - (b) joint properties for two ergodic records.

CHAPTER 10

NONSTATIONARY, TRANSIENT, AND MULTIDIMENSIONAL DATA

The material in previous chapters has been restricted largely to the measurement and analysis of stationary data representing one dimensional (time series) random processes. The theoretical ideas, error formulas, and processing techniques do not, for the most part, apply when the data are nonstationary and/or multidimensional. Special considerations are required in these cases. Such considerations are the subject of this chapter. The description and analysis of nonstationary random data are discussed in Section 10.1. Procedures for evaluating a special case of nonstationary data, called transient data, are reviewed in Section 10.2. The description of multidimensional random processes is considered in Section 10.3.

10.1 NONSTATIONARY RANDOM PROCESSES

Nonstationary data represent any class of data whose statistical properties change with time. Consequently, the vast majority of physical data actually fall into this category. It is only for reasons of approximation and simplicity that many data are arbitrarily assumed to be stationary. Nonstationary data are obtained, for example, during transient operating conditions when an environment changes suddenly, or during long range operating periods when system properties change so that a given input will produce a variable output.

A totally adequate methodology does not exist as yet for the analysis of all types of nonstationary data. This is partly due to the fact that a nonstationary

conclusion is generally a negative statement specifying the lack of stationary properties, rather than a positive statement defining the precise nature of the nonstationarity. It follows that special techniques must be developed for nonstationary data which apply only to limited classes of these data. Examples of different types of nonstationary data are shown in Figure 10.1.

The statistical properties of a single nonstationary process at different times can be described in terms of its (1) nonstationary probability density function, (2) nonstationary auto-correlation function, and (3) nonstationary spectral density function. For pairs of nonstationary processes, one might investigate (1) joint nonstationary probability density functions, (2) nonstationary cross-correlation functions, and (3) nonstationary cross-spectral density functions.

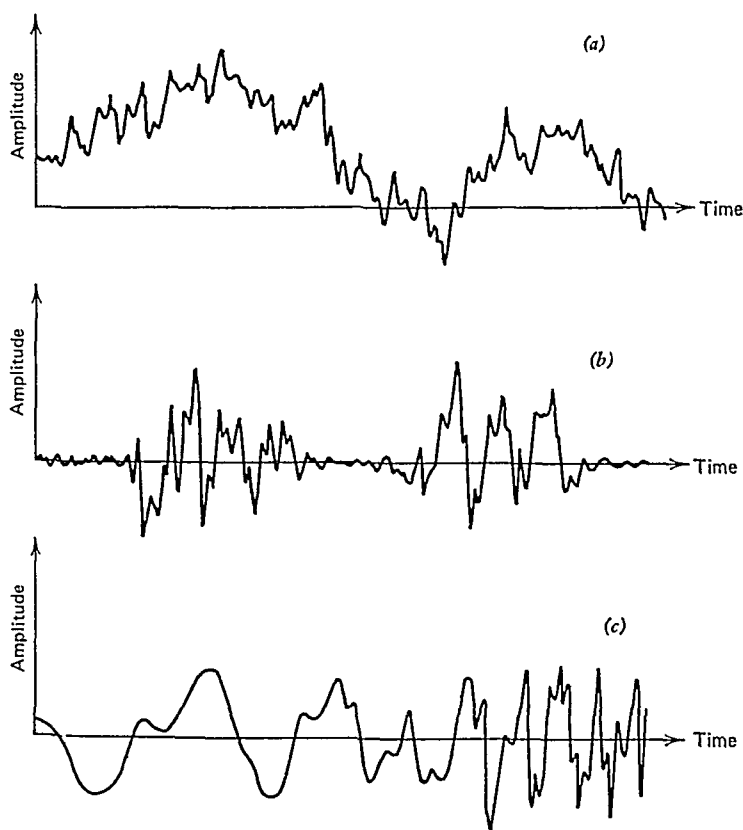


Figure 10.1 Examples of nonstationary data. (a) Time-varying mean value. (b) Time-varying mean square value. (c) Time-varying frequency structure.

It is important also to be able to determine input output relations for passage of nonstationary data through physical systems. These matters will now be discussed.

10.1.1 Probability Structure of Nonstationary Data

For a nonstationary random process $\{x(t)\}$, as discussed in Chapters 1 and 3, statistical properties over the ensemble at any time t are not invariant with respect to translations in t . Hence at any value of $t = t_1$, the probability structure of the random variable $x(t_1)$ would be a function of t_1 . To be precise, the *nonstationary probability density function* $p(x, t_1)$ of $\{x(t_1)\}$ is defined by

$$p(x, t_1) = \lim_{\Delta x \rightarrow 0} \frac{\text{Prob} [x < x(t_1) \leq x + \Delta x]}{\Delta x} \quad (10.1)$$

and has the following basic properties for any t

$$\begin{aligned} 1 &= \int_{-\infty}^{\infty} p(x, t) dx \\ \mu_x(t) &= E\{x(t)\} = \int_{-\infty}^{\infty} x p(x, t) dx \\ \Psi_x^2(t) &= E\{x^2(t)\} = \int_{-\infty}^{\infty} x^2 p(x, t) dx \\ \sigma_x^2(t) &= E\{(x(t) - \mu_x(t))^2\} = \Psi_x^2(t) - \mu_x^2(t) \end{aligned} \quad (10.2)$$

These formulas also apply to stationary cases where $p(x, t) = p(x)$, independent of t . The *nonstationary probability distribution function* $P(x, t_1)$ is defined by

$$P(x, t_1) = \text{Prob} [-\infty < x(t_1) \leq x] \quad (10.3)$$

Similar relationships exist for $P(x, t_1)$ as for the stationary probability distribution function $P(x)$ discussed in Chapter 3.

If the nonstationary random process $\{x(t)\}$ is Gaussian at $t = t_1$, then $p(x, t_1)$ takes the special form

$$p(x, t_1) = [\sigma_x(t_1)\sqrt{2\pi}]^{-1} \exp \left\{ -\frac{[x - \mu_x(t_1)]^2}{2\sigma_x^2(t_1)} \right\} \quad (10.4)$$

which is completely determined by the nonstationary mean and mean square values of $x(t)$ at $t = t_1$. This result indicates that the measurement of these two quantities may be quite significant in many nonstationary applications just as in previous stationary applications.

For a pair of times t_1 and t_2 , the *second-order nonstationary probability density function* of $x(t_1)$ and $x(t_2)$ is defined by

$$p(x_1, t_1; x_2, t_2) = \lim_{\substack{\Delta x_1 \rightarrow 0 \\ \Delta x_2 \rightarrow 0}} \frac{\text{Prob} [x_1 < x(t_1) \leq x_1 + \Delta x_1 \text{ and } x_2 < x(t_2) \leq x_2 + \Delta x_2]}{(\Delta x_1)(\Delta x_2)} \quad (10.5)$$

and has the following basic properties for any t_1, t_2 .

$$\begin{aligned} 1 &= \iint_{-\infty}^{\infty} p(x_1, t_1; x_2, t_2) dx_1 dx_2 \\ p(x_1, t_1) &= \int_{-\infty}^{\infty} p(x_1, t_1; x_2, t_2) dx_2 \\ p(x_2, t_2) &= \int_{-\infty}^{\infty} p(x_1, t_1; x_2, t_2) dx_1 \\ R_x(t_1, t_2) &= E[x(t_1) x(t_2)] = \iint_{-\infty}^{\infty} x_1 x_2 p(x_1, t_1; x_2, t_2) dx_1 dx_2 \end{aligned} \quad (10.6)$$

For stationary cases, $p(x_1, t_1; x_2, t_2) = p(x_1, 0; x_2, t_2 - t_1)$. *Second-order nonstationary probability distribution functions* may be defined analogous to Equation (10.3) by the quantity

$$P(x_1, t_1; x_2, t_2) = \text{Prob} [-\infty < x(t_1) \leq x_1 \text{ and } -\infty < x(t_2) \leq x_2] \quad (10.7)$$

Continuing in this way, higher-order nonstationary probability distribution and density functions may be defined which describe the nonstationary random process $\{x(t)\}$ in greater and greater detail. This procedure supplies a rigorous characterization for the nonstationary random process $\{x(t)\}$.

Consider next two different nonstationary random processes $\{x(t)\}$ and $\{y(t)\}$. For $x(t_1)$ and $y(t_2)$, the *joint (second-order) nonstationary probability density function* is defined by

$$p(x, t_1; y, t_2) = \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{\text{Prob} [x < x(t_1) \leq x + \Delta x \text{ and } y < y(t_2) \leq y + \Delta y]}{(\Delta x)(\Delta y)} \quad (10.8)$$

and has basic properties similar to Equation (10.6). In particular, the nonstationary cross-correlation function, which is discussed in Section 10.1.4,

satisfies the relation

$$R_{xy}(t_1, t_2) = E\{x(t_1)y(t_2)\} = \iint_{-\infty}^{\infty} xy p(x, t_1, y, t_2) dx dy \quad (10.9)$$

For stationary cases, $p(x, t_1, y, t_2) = p(x, 0, y, t_2 - t_1)$

The measurement of nonstationarity probability density functions can be a formidable task. Even for the first-order density function defined in Equation (10.1), all possible combinations of x and t_1 must be considered. This will require analysis of a large collection of sample records. If a Gaussian assumption can be made, Equation (10.4) reduces the problem of measuring $p(x, t_1)$ to measuring $\mu_x(t_1)$ and $\sigma_x^2(t_1)$, which is a much simpler undertaking. Nevertheless, ensemble averaging of a collection of sample records is still generally required as discussed in the next section.

It often occurs in practice that only one or a very few sample records of data are available for a nonstationary random process of interest. There may be a strong temptation in such cases to analyze the data by time averaging procedures, as would be appropriate if the data were a sample record from a stationary (ergodic) random process. For some nonstationary data parameters, time averaging analysis procedures can produce meaningful results in certain special cases, as will be discussed later. For the case of probability density functions, however, time averaging procedures will generally produce severely distorted results. In particular, the probability density function computed by time averaging data with a nonstationary mean square value will tend to exaggerate the probability density of low and high amplitude values at the expense of intermediate values.

The above statement is readily demonstrated by considering a special case. Assume a sample record of data consists of a normally distributed stationary time history with zero mean and variance σ_1^2 over the first half of the record, and a second normally distributed stationary time history with zero mean and variance $\sigma_2^2 > \sigma_1^2$ over the second half of the record. Then the time history, $x(t)$, is given by

$$x(t) = \begin{cases} x_1(t) & 0 \leq t \leq T/2 \\ x_2(t) & T/2 \leq t \leq T \end{cases}$$

and the probability density function of $x(t)$ is given by

$$p(x, t) = \begin{cases} \frac{1}{\sigma_1 \sqrt{2\pi}} e^{-x^2/2\sigma_1^2} & 0 \leq t \leq T/2 \\ \frac{1}{\sigma_2 \sqrt{2\pi}} e^{-x^2/2\sigma_2^2} & T/2 \leq t \leq T \end{cases}$$

Now, if this lack of stationarity is ignored in the computation of a probability

density function for $x(t)$; $0 \leq t \leq T$, the resulting density calculated at any level x will simply be the average of the densities for the two halves of the record at level x . That is,

$$\hat{p}(x) = \frac{1}{2\sqrt{2\pi}} \left[\frac{1}{\sigma_1} e^{-x^2/2\sigma_1^2} + \frac{1}{\sigma_2} e^{-x^2/2\sigma_2^2} \right]$$

For example, let $\sigma_1^2 = 1$ and $\sigma_2^2 = 16$. The nonstationary resultant probability density function which would be obtained for this case is computed and illustrated graphically in Figure 10.2. Observe that $\hat{\sigma} = 2.9$ in $\hat{p}(x)$ since $\hat{\sigma}^2 = (1/2)(\sigma_1^2 + \sigma_2^2) = 8.5$. The equivalent Gaussian probability density function for $\sigma = 2.9$ is shown also in Figure 10.2.

Example 10.1. Illustration of Time Averaged Probability Density Estimate. Consider the response acceleration of the airplane flying through a patch of clear air turbulence as previously discussed in Example 7.2 (Section 7.4.1). Specifically, consider the special case illustrated in Figure 7.11 where the input gust velocity data will not pass a test for stationarity. Then one would also expect the response acceleration of the airplane to be nonstationary. Assume this conclusion is ignored and the probability density function of the response acceleration is computed by time averaging over a long time segment including the 480 second interval covered by the input gust velocity data in Figure 7.11. The resulting probability density function is presented by Figure 10.3(a). Note the strong deviation from the Gaussian form in both general appearance and the magnitude of the sample fourth central moment m_4' (for Gaussian data, $m_4' = 3\sigma^4$ as shown in Section 3.4). These calculations were performed digitally using over ten thousand data points, so that the deviations from the Gaussian form are highly significant in the statistical sense.

Now assume the nonstationarity is observed and only a limited segment of the available data which appears to be stationary is analyzed. From the results in Figure 7.11, the data segment between 200 and 260 seconds appears to be approximately stationary. The probability density function computed digitally by time averaging 700 data points in this segment alone is presented in Figure 10.3(b). Note the strong correlation with Gaussian properties. The data pass a chi-square goodness-of-fit test for normality at the $\alpha = 0.05$ level of significance. Hence, there is no reason to believe the aircraft response is nonGaussian. Failure to consider the nonstationary nature of the data, however, would obscure this conclusion.

10.1.2 Nonstationary Mean Values

Consider the problem of estimating the time varying mean value of nonstationary data. Given a collection of sample records $x_i(t)$; $0 \leq t \leq T$; $i = 1, 2, \dots, N$, from a nonstationary process $\{x(t)\}$, the mean value at any

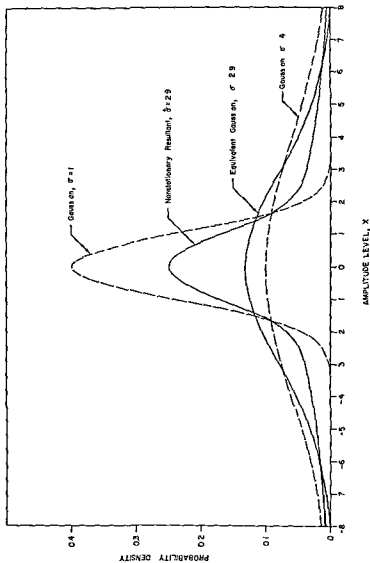


Figure 10.2 Illustration of probability density function of nonstationary data

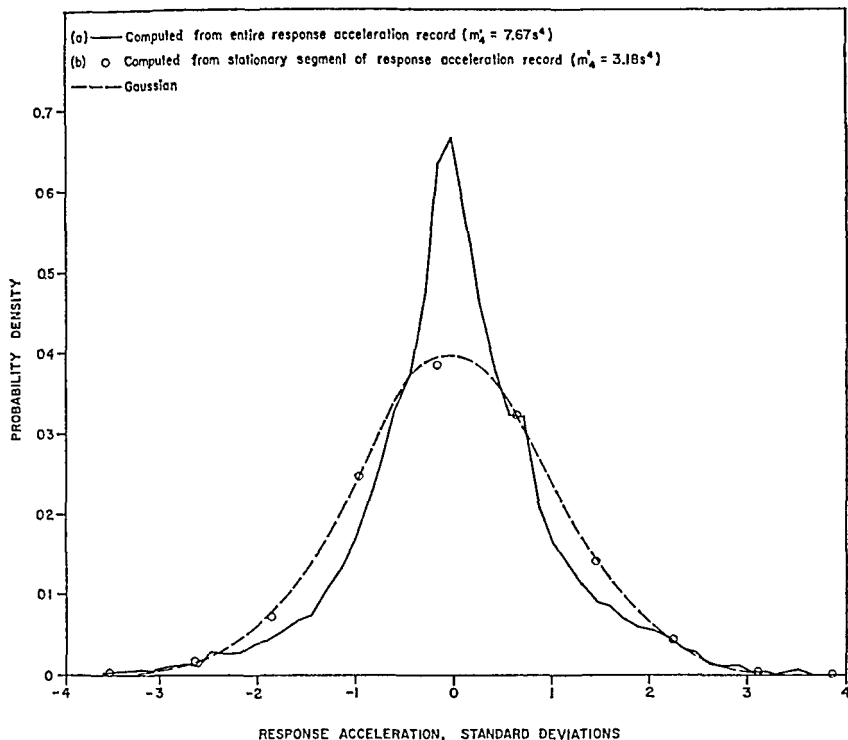


Figure 10.3 Probability density function of response acceleration. These data resulted from studies funded by the NASA Langley Research Center, Hampton, Virginia, under Contract NAS 1-8538.

time t is estimated by the ensemble average

$$\hat{\mu}_x(t) = \frac{1}{N} \sum_{i=1}^N x_i(t) \quad (10.10)$$

The estimate $\hat{\mu}_x(t)$ will differ over different choices of the N samples $\{x_i(t)\}$. Consequently, one must investigate for every t how closely an arbitrary estimate will approximate the true mean value. The expected value of $\hat{\mu}_x(t)$ is given by

$$E[\hat{\mu}_x(t)] = \frac{1}{N} \sum_{i=1}^N E[x_i(t)] = \mu_x(t) \quad (10.11)$$

where

$$\mu_x(t) = E[x_i(t)] \quad (10.12)$$

is the true mean value of the nonstationary process at time t . Hence $\hat{\mu}_x(t)$ is

an unbiased estimate of $\mu_x(t)$ for all t , independent of N . The variance of the estimate $\hat{\mu}_x(t)$ is given by

$$\text{Var} [\hat{\mu}_x(t)] = E\{[\hat{\mu}_x(t) - \mu_x(t)]^2\} \quad (10.13)$$

In most practical applications the N sample records used to compute $\hat{\mu}_x(t)$ will be statistically independent. Hence independence will be assumed here. Upon expanding Equation (10.13), as in the derivation of Equation (4.9), it is seen that the variance of the estimate is

$$\text{Var} [\hat{\mu}_x(t)] = \frac{\sigma_x^2(t)}{N} \quad (10.14)$$

where $\sigma_x^2(t)$ is the variance associated with the underlying nonstationary process $\{x(t)\}$. Thus the $\text{Var} [\hat{\mu}_x(t)] \rightarrow 0$ as $N \rightarrow \infty$ so that $\hat{\mu}_x(t)$ is a consistent estimate of $\mu_x(t)$ for all t .

Mean values of nonstationary random processes can be estimated using a special purpose analog instrument or a digital computer as illustrated in Figure 10.4. Two main steps are involved in the measurement. The first step

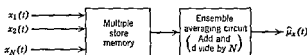


Figure 10.4 Procedure for nonstationary mean value measurement

is to obtain and store each record $x_i(t)$ as a function of t . This may be done continuously for all t in the range $0 \leq t \leq T$, or discretely by some digitizing procedure. After this has been done for N records, the next step is to perform an ensemble average by adding the records together and dividing by N . If each $x_i(t)$ is digitized in, say, M steps, then the total number of stored values would be MN .

As previously noted, it often occurs in practice that only one sample record of data is available for a nonstationary process of interest. In such cases, nonstationary mean values are often estimated from a single sample record by one of several operations equivalent to low pass filtering. This technique can be employed profitably for certain classes of nonstationary data. To be more specific, consider a nonstationary random process of the form

$$\{x(t)\} = A(t) + \{y(t)\} \quad (10.15)$$

where $A(t)$ is a deterministic function and $\{y(t)\}$ is a random process with a stationary mean value of zero. It follows that the mean value of the process $\{x(t)\}$ at any time t is given by

$$E[\{x(t)\}] = E[A(t) + \{y(t)\}] = E[A(t)] + E[\{y(t)\}] = A(t) \quad (10.16)$$

If it is assumed that the variations of $A(t)$ are very slow compared to the lowest frequency in $\{y(t)\}$, then $A(t)$ can be separated from $\{y(t)\}$ by low pass filtering operations on a single sample record $x(t)$. Such filtering operations may be physically accomplished in several ways including:

1. Analog low pass filtering with an RC averaging network as discussed in Section 8.1.
2. Digital low pass filtering with either recursive or nonrecursive filters as discussed in Section 9.2.
3. Polynomial curve fitting (regression analysis) as introduced in Section 4.8 and discussed for trend removal applications in Section 9.1.3.
4. Segmented mean value estimates (short time averaging operations).

In any case, the resulting mean value estimates will involve a bias error which is a function of the low pass filter cutoff frequency (the number of terms in the polynomial fit or the short averaging time) relative to the rate of variation of $A(t)$.

For example, consider the short time averaged estimate of the mean value $\mu_x(t)$ given by

$$\hat{\mu}_x(t) = \int_{t-T/2}^{t+T/2} x(t) dt = \int_{t-T/2}^{t+T/2} [A(t) + y(t)] dt \quad (10.17)$$

where T is a short averaging time. It is readily shown that

$$\begin{aligned} E[\hat{\mu}_x(t)] &= E\left[\int_{t-T/2}^{t+T/2} [A(t) + y(t)] dt\right] = \int_{t-T/2}^{t+T/2} \{E[A(t)] + E[y(t)]\} dt \\ &= \int_{t-T/2}^{t+T/2} A(t) dt \neq A(t) \end{aligned} \quad (10.18)$$

Hence the estimate $\hat{\mu}_x(t)$ will generally be biased. A development similar to that presented in Section 6.3.2 yields a first order approximation for the bias error at any time t as

$$b[\hat{\mu}_x(t)] = \frac{T^2}{24} A''(t) \quad (10.19)$$

where $A''(t)$ is the second derivative of $A(t)$ with respect to t . It is clear from Equation (10.19) that the bias error diminishes as the averaging time T becomes small. However, the random error increases as T becomes small in a manner similar to that developed for stationary data in Section 6.2.1. Thus

the selection of an appropriate averaging time T involves a compromise between random and bias errors. In most cases, this compromise is best arrived at by trial and error procedures.

10.1.3 Nonstationary Mean Square Values

Now consider the problem of estimating the time varying mean square value of nonstationary data. Given a collection of sample records $x_i(t)$, $0 \leq t \leq T$, $i = 1, 2, \dots, N$ from a nonstationary process $\{x(t)\}$ the mean square value at time t is estimated by the ensemble average

$$\hat{\Psi}_x^2(t) = \frac{1}{N} \sum_{i=1}^N x_i^2(t) \quad (10.20)$$

Independent of N , the quantity $\hat{\Psi}_x^2(t)$ is an unbiased estimate of the true mean square value of the nonstationary process $\{x(t)\}$ at any time t since the expected value

$$E[\hat{\Psi}_x^2(t)] = \frac{1}{N} \sum_{i=1}^N E[x_i^2(t)] = \Psi_x^2(t) \quad (10.21)$$

where

$$\Psi_x^2(t) = E[x^2(t)] = \mu_x^2(t) + \sigma_x^2(t) \quad (10.22)$$

is the true mean square value of the nonstationary process at time t .

To arrive at the variance of the estimate $\hat{\Psi}_x^2(t)$ assume the N sample records $x_i(t)$ are independent so that for all i and j

$$E[x_i(t) x_j(t)] = E[x_i(t)] E[x_j(t)] = \mu_x^2(t) \quad (10.23)$$

It follows that

$$\text{Var}[\hat{\Psi}_x^2(t)] = E[(\hat{\Psi}_x^2(t) - \Psi_x^2(t))^2] = E[(\hat{\Psi}_x^2(t))^2] - \Psi_x^4(t) \quad (10.24)$$

where $\Psi_x^2(t)$ is given by Equation (10.22) and

$$\begin{aligned} E[(\Psi_x^2(t))^2] &= \frac{1}{N^2} \sum_{i=1}^N E[x_i^2(t) x_i^2(t)] \\ &= \frac{1}{N^2} \left[\sum_{i=1}^N E[x_i^4(t)] + \sum_{\substack{i,j=1 \\ i \neq j}}^N E[x_i^2(t) x_j^2(t)] \right] \end{aligned} \quad (10.25)$$

Thus the problem reduces to evaluation of the expected values appearing in Equation (10.25).

In order to obtain reasonable closed form answers it will be assumed now that the random process $\{x(t)\}$ at any time t follows a Gaussian distribution

with mean value $\mu_x(t)$ and variance $\sigma_x^2(t)$. One can then derive

$$E[x_i^4(t)] = 3\Psi_x^4(t) - 2\mu_x^4(t) \quad (10.26)$$

$$E[x_i^2(t) x_j^2(t)] = \Psi_x^4(t) \quad \text{for } i \neq j \quad (10.27)$$

The derivation of Equations (10.26) and (10.27) is based upon a nonstationary form for the fourth-order Gaussian relation of Equation (3.132); namely,

$$\begin{aligned} E[x_i(t) x_j(t) x_m(t) x_n(t)] &= E[x_i(t) x_j(t)] E[x_m(t) x_n(t)] \\ &\quad + E[x_i(t) x_m(t)] E[x_j(t) x_n(t)] \\ &\quad + E[x_i(t) x_n(t)] E[x_j(t) x_m(t)] - 2\mu_x^4(t) \end{aligned} \quad (10.28)$$

Substitution into Equations (10.24) and (10.25) yields the result

$$\text{Var} [\hat{\Psi}_x^2(t)] = \frac{2}{N} [\Psi_x^4(t) - \mu_x^4(t)] \quad (10.29)$$

Thus $\text{Var} [\hat{\Psi}_x^2(t)] \rightarrow 0$ as $N \rightarrow \infty$ proving that $\hat{\Psi}_x^2(t)$ is a consistent estimate of $\Psi_x^2(t)$ for all t .

Mean square values of nonstationary random processes can be estimated with the ensemble averaging device illustrated in Figure 10.4 by merely replacing $x_i(t)$ with $x_i^2(t)$. Furthermore, for certain classes of data, the time varying mean square value can be estimated from a single sample record by low pass filtering operations as discussed in Section 10.1.2. For this case, the applicable model is a nonstationary random process given by

$$\{x(t)\} = A(t)\{y(t)\} \quad (10.30)$$

where $A(t)$ is a deterministic function and $\{y(t)\}$ is a random process with a stationary mean and variance of zero and unity, respectively. Hence the mean square value of the process $\{x(t)\}$ at any time t is given by

$$\text{Var} [\{x(t)\}] = \text{Var} [A(t)\{y(t)\}] = A^2(t) \text{Var} [\{y(t)\}] = A^2(t) \quad (10.31)$$

As for mean value estimates, if the variations of $A(t)$ are very slow compared to the lowest frequency of $\{y(t)\}$, then $A^2(t)$ can be separated by low pass filtering operations on a single sample record of $x^2(t)$. This may be physically accomplished by applying any of the techniques discussed in Section 10.1.2 to the squared data signal $x^2(t)$. The resulting mean square value estimates, of course, will generally be biased. For example, if a short time averaging procedure is used

$$\begin{aligned} E[\hat{\Psi}_x^2(t)] &= E \left[\int_{t-T/2}^{t+T/2} A^2(t) y^2(t) dt \right] = \int_{t-T/2}^{t+T/2} E[A^2(t)] E[y^2(t)] dt \\ &= \int_{t-T/2}^{t+T/2} A^2(t) dt \neq A^2(t) \end{aligned} \quad (10.32)$$

Example 10.2. Illustration of Short Time Averaged Mean Square Value Estimate. Consider the structural vibration environment of a spacecraft during lift-off. One would anticipate that the vibration at any point on the spacecraft structure is a nonstationary random process since the excitation forces during lift-off (primarily acoustic noise generated by turbulent mixing of the rocket exhaust gases with the ambient air) are a function of rapidly changing parameters such as vehicle speed and distance from ground reflections. The measurement of a time varying mean square value by ensemble averaging procedures in accordance with Equation (10.20) is not economically feasible since N spacecraft launches would be required to obtain the N sample records for ensemble averaging. Can short time averaging procedures be applied to this problem to obtain meaningful results?

For any given spacecraft configuration the structural vibration during lift-off will display a repeatable time trend since the parameters governing the excitation forces change in the same way from one launch to the next. Furthermore, this time trend is relatively slow (several seconds for a half cycle of variation) relative to the lowest frequency of significant energy in the excitation (several cps). Hence the model of Equation (10.30) should apply to this problem.

To demonstrate its application, consider the results from actual test data presented in Figure 10.5. The data in this figure are short time averaged rms values of the vibration measured at a given structural location on three different spacecraft of similar configuration during lift-off. The averaging time for each computation is about one second and the frequency range of the data is several hundred cps. The close agreement between the results for the three separate launches indicates that the sequence of short time averaged mean square values provides a useful approximation for the time varying mean square value $\Psi^2(t)$ remembering, of course, that a bias error is present as indicated in Equation (10.32).

10.1.4 Correlation Structure of Nonstationary Data

For nonstationary random processes $\{x(t)\}$ and $\{y(t)\}$, as in Section 3.2, the mean values at arbitrary fixed values of time t are defined by the expected values (ensemble averages)

$$\begin{aligned}\mu_x(t) &= E[x(t)] \\ \mu_y(t) &= E[y(t)]\end{aligned}\tag{10.33}$$

The *covariance functions* at arbitrary fixed values of t_1 and t_2 are defined by the expected values (ensemble averages)

$$C_x(t_1, t_2) = E[\{x(t_1) - \mu_x(t_1)\}\{x(t_2) - \mu_x(t_2)\}]\tag{10.34}$$

$$C_y(t_1, t_2) = E[\{y(t_1) - \mu_y(t_1)\}\{y(t_2) - \mu_y(t_2)\}]\tag{10.35}$$

$$C_{xy}(t_1, t_2) = E[\{x(t_1) - \mu_x(t_1)\}\{y(t_2) - \mu_y(t_2)\}]$$

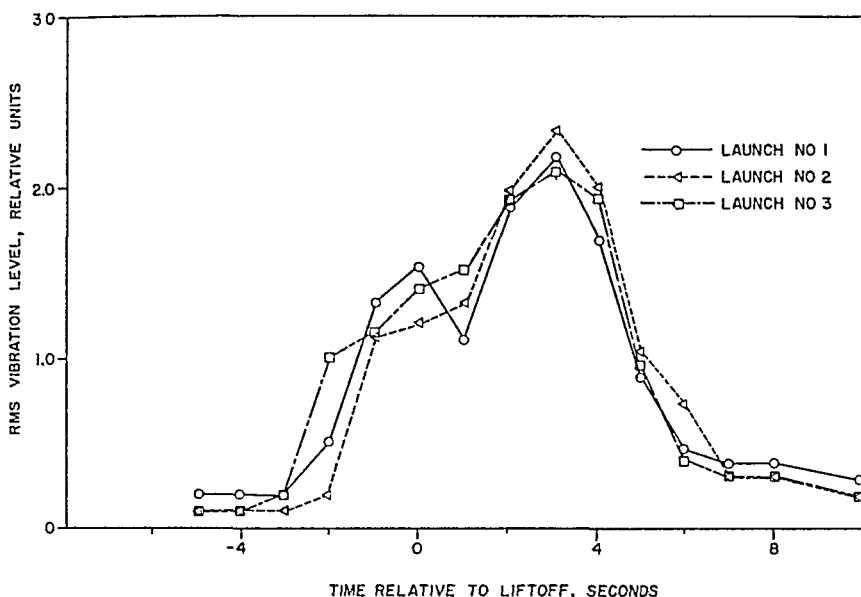


Figure 10.5 Time-varying rms values of spacecraft liftoff vibration data. These data were provided by the Jet Propulsion Laboratory, Pasadena, California.

For stationary random processes, these results would be a function of the time difference $t_2 - t_1$ rather than the particular t_1 and t_2 . For nonstationary processes, however, these quantities will vary with both t_1 and t_2 .

The *correlation functions* of nonstationary processes are defined by the expected values (ensemble averages)

$$R_x(t_1, t_2) = E[x(t_1) x(t_2)] \quad (10.36)$$

$$R_y(t_1, t_2) = E[y(t_1) y(t_2)]$$

$$R_{xy}(t_1, t_2) = E[x(t_1) y(t_2)] \quad (10.37)$$

The quantities $R_x(t_1, t_2)$ and $R_y(t_1, t_2)$ are called *nonstationary autocorrelation functions*, whereas $R_{xy}(t_1, t_2)$ is called a *nonstationary cross-correlation function*. For arbitrary values of $\mu_x(t)$ and $\mu_y(t)$, the covariance functions and correlation functions are related by

$$C_x(t_1, t_2) = R_x(t_1, t_2) - \mu_x(t_1) \mu_x(t_2) \quad (10.38)$$

$$C_y(t_1, t_2) = R_y(t_1, t_2) - \mu_y(t_1) \mu_y(t_2)$$

$$C_{xy}(t_1, t_2) = R_{xy}(t_1, t_2) - \mu_x(t_1) \mu_y(t_2) \quad (10.39)$$

By a derivation similar to Equation (3.50), an upper bound for the

nonstationary cross correlation (or cross covariance) function is given by the inequality

$$|R_{xy}(t_1, t_2)|^2 \leq R_x(t_1, t_1) R_y(t_2, t_2) \quad (10.40)$$

From the original definitions, one sees that the following symmetry properties are satisfied

$$R_x(t_2, t_1) = R_x(t_1, t_2) \quad (10.41)$$

$$R_y(t_2, t_1) = R_y(t_1, t_2)$$

$$R_{xy}(t_2, t_1) = R_{yx}(t_1, t_2) \quad (10.42)$$

Thus the correlation structure of nonstationary random processes $\{x(t)\}$ and $\{y(t)\}$ may be described by the four functions $R_x(t_1, t_2)$, $R_y(t_1, t_2)$, $R_{xy}(t_1, t_2)$, and $R_{yx}(t_1, t_2)$. These need be calculated only for values of $t_1 \leq t_2$ since the symmetry properties yield results for $t_2 < t_1$.

Further insight can be gained by making the following transformations. Let

$$\begin{aligned} \tau &= t_2 - t_1 & t &= \frac{t_1 + t_2}{2} \\ t_1 &= t - \frac{\tau}{2} & t_2 &= t + \frac{\tau}{2} \end{aligned} \quad (10.43)$$

Then, through the above change of variables, it follows that

$$\begin{aligned} E[x(t_1) y(t_2)] &= R_{xy}(t_1, t_2) = R_{xy}\left(t - \frac{\tau}{2}, t + \frac{\tau}{2}\right) \\ &= \mathcal{R}_{xy}(\tau, t) = E\left[x\left(t - \frac{\tau}{2}\right) y\left(t + \frac{\tau}{2}\right)\right] \end{aligned} \quad (10.44)$$

Note that the definition for $\mathcal{R}_{xy}(\tau, t)$ is quite different from $R_{xy}(\tau, t)$. The quantity $\mathcal{R}_{xy}(\tau, t)$ is not a conventional correlation function. The reason for introducing this transformation from the (t_1, t_2) plane to the (τ, t) plane is to permit the nonstationary correlation function to be separated, when possible, into stationary and nonstationary portions. Specifically, assume that the nonstationary correlation function $\mathcal{R}(\tau, t)$ can be separated into the products

$$\mathcal{R}(\tau, t) = R_2(\tau) R_1(t) = R_1\left(\frac{t_1 + t_2}{2}\right) R_2(t_2 - t_1) \quad (10.45)$$

where $R_2(\tau) = R_2(t_2 - t_1)$ is a stationary correlation function, and $R_1(t) = R_1[(t_1 + t_2)/2]$ is a slowly varying scale factor defined at the midpoint (average) of the points t_1 and t_2 . For stationary random processes, $R_1(t)$ is a constant. If $R_2(\tau)$ is normalized so that $R_2(0) = 1$, then $R_1(t) = \Psi^2(t)$, the mean square

value of the process as a function of time. Such a correlation function would apply, for example, to the nonstationary random process defined by Equation (10.30) where $\{y(t)\}$ is a stationary process.

Random processes with correlation functions of the type defined in Equation (10.45) are said to be *locally stationary processes* [Ref. 48]. The correlation functions of certain nonstationary physical phenomena in practice often can be approximated by the locally stationary model. Examples are random processes representing turbulence pressure fields in fluid flow problems, as illustrated later in Example 10.3.

Now consider the problem of estimating $R_x(t_1, t_2)$ using a set of N sample records, $x_i(t)$; $i = 1, 2, \dots, N$, from the nonstationary random process. In place of Equation (10.36), one should compute the ensemble averaged estimate

$$\hat{R}_x(t_1, t_2) = \frac{1}{N} \sum_{i=1}^N x_i(t_1) x_i(t_2) \quad (10.46)$$

A recommended procedure is to hold t_1 fixed and to vary t_2 . Let $t_1 = t$ and let $t_2 = t - \tau$ where τ is a fixed time delay value. This yields

$$\hat{R}_x(t, t - \tau) = \frac{1}{N} \sum_{i=1}^N x_i(t) x_i(t - \tau) \quad (10.47)$$

which for stationary processes would be a function of τ only, but for nonstationary processes would be a function of both t and τ . For each fixed delay value τ and each record $x_i(t)$, calculate and store the product $x_i(t)x_i(t - \tau)$. Repeat for all N records and then perform an ensemble average to yield the estimate of Equation (10.47). This whole operation must be repeated for every different τ of concern. Figure 10.6 illustrates this procedure for computing nonstationary autocorrelation functions. A similar procedure may be followed for nonstationary cross-correlation function estimates.

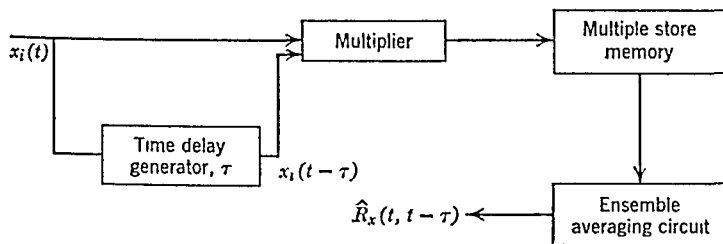


Figure 10.6 Procedure for nonstationary autocorrelation measurement.

For the case of nonstationary data with a separable correlation function as defined in Equation (10.45) the estimation problem is greatly simplified. In particular, it is possible in such cases to calculate meaningful correlation estimates from single sample records. Noting that the time dependent term $R_1(t)$ is simply a slowly varying mean square value, it can be separately estimated by short time averaging procedures as outlined in Section 10.1.3. Since the $R_2(\tau)$ term is a stationary correlation function, it can be separately estimated by averaging over the entire available record as detailed in Sections 8.3 and 9.5.

10.1.5 Spectral Structure of Nonstationary Data

As the spectral density of stationary data can be defined by the Fourier transform of a stationary correlation function, so the spectral density of nonstationary data can be defined by the double Fourier transform of the nonstationary correlation function. To be more specific, given the correlation function $R_x(t_1, t_2)$ of a nonstationary random process, as defined by Equation (10.36), the *double frequency (generalized) power spectral density function* of the process is defined by

$$S_x(f_1, f_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} R_x(t_1, t_2) e^{j2\pi(f_1 t_1 - f_2 t_2)} dt_1 dt_2 \quad (10.48)$$

Note that $S_x(f_1, f_2)$ is complex valued and the frequencies f_1 and f_2 can take on arbitrary positive and negative values in the range $[-\infty, \infty]$. The inverse double Fourier transform yields

$$R_x(t_1, t_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} S_x(f_1, f_2) e^{-j2\pi(f_1 t_1 - f_2 t_2)} df_1 df_2 \quad (10.49)$$

Similarly, one can define the double Fourier transform relations

$$S_{xy}(f_1, f_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} R_{xy}(t_1, t_2) e^{j2\pi(f_1 t_1 - f_2 t_2)} dt_1 dt_2 \quad (10.50)$$

$$R_{xy}(t_1, t_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} S_{xy}(f_1, f_2) e^{-j2\pi(f_1 t_1 - f_2 t_2)} df_1 df_2 \quad (10.51)$$

where the quantity $S_{xy}(f_1, f_2)$ is the *double frequency (generalized) cross spectral density function* between the pair of nonstationary random processes $\{x(t)\}$ and $\{y(t)\}$.

For stationary processes, at the special point $t = t_1 = t_2$, Equation (10.51) becomes

$$R_{xy}(t, t) = R_{xy}(0, 0) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} S_{xy}(f_1, f_2) df_1 df_2 \quad (10.52)$$

It follows that for stationary processes,

$$R_{xy}(0, 0) = \int_{-\infty}^{\infty} S_{xy}(f_2) df_2 \quad (10.53a)$$

where

$$S_{xy}(f_2) = \int_{-\infty}^{\infty} S_{xy}(f_1, f_2) df_1 \quad (10.53b)$$

is the two-sided stationary cross-spectral density function. Hence for stationary processes,

$$S_{xy}(f_1, f_2) = S_{xy}(f_1) \delta(f_2 - f_1) \quad (10.54)$$

where $\delta(f_2 - f_1)$ is the usual delta function at $f_2 = f_1$. In geometrical terms, $S_{xy}(f_1, f_2)$ is concentrated only on the line $f_2 = f_1$ in the (f_1, f_2) plane. Note also that $S_{xy}(f_1, f_2) = 0$ for $f_2 \neq f_1$. Conversely, suppose that a generalized cross-spectral density function has the form of Equation (10.54). One can then derive from Equation (10.51), letting $\tau = t_2 - t_1$,

$$R_{xy}(t_1, t_2) = R_{xy}(t_2 - t_1) = R_{xy}(\tau) = \int_{-\infty}^{\infty} S_{xy}(f) e^{j2\pi f\tau} df$$

where

$$S_{xy}(f) = \int_{-\infty}^{\infty} R_{xy}(\tau) e^{-j2\pi f\tau} d\tau$$

Thus the cross-correlation function $R_{xy}(t_1, t_2)$ corresponding to Equation (10.54) is stationary.

Another approach to describing the spectral structure of nonstationary data is to take the Fourier transform of the nonstationary autocorrelation function $\mathcal{R}_x(\tau, t)$ of Equation (10.45) with respect to τ . This defines a *frequency-time spectral density function*

$$S_x(f, t) = \int_{-\infty}^{\infty} \mathcal{R}_x(\tau, t) e^{-j2\pi f\tau} d\tau \quad (10.55)$$

which is similar to the instantaneous power spectrum of Reference 40. The function $S_x(f, t)$ is real valued and can be negative for particular values of f and t , unlike a stationary $S_x(f)$. The inverse transform of Equation (10.55) yields

$$\mathcal{R}_x(\tau, t) = \int_{-\infty}^{\infty} S_x(f, t) e^{j2\pi f\tau} df = E \left[x \left(t - \frac{\tau}{2} \right) x \left(t + \frac{\tau}{2} \right) \right] \quad (10.56)$$

At the special point $\tau = 0$,

$$E[x^2(t)] = \mathcal{R}_x(0, t) = \int_{-\infty}^{\infty} S_x(f, t) df \geq 0 \quad (10.57)$$

Thus $S_x(f, t)$ indicates how the mean square value $E[x^2(t)]$ is distributed in the (f, t) plane, providing a possible physical interpretation of $S_x(f, t)$.

Equation (10 57) shows that integrating $S_x(f, t)$ over all f gives a mean square value at time t . For the case of stationary processes where there is no dependence on t , Equations (10 55) through (10 57) yield results consistent with those developed for stationary data in Chapter 3.

Both the double frequency spectral density function defined by Equation (10 48) and the frequency time spectral density function of Equation (10 55) must be calculated from Fourier transform operations on nonstationary autocorrelation functions. In general, $S_x(f_1, f_2)$ and $S_x(f, t)$ cannot be estimated by time averaging operations on individual sample records as may be done for stationary data. There is one important exception, however, which arises for nonstationary data of the locally stationary form. Specifically, if the data have a separable nonstationary correlation function as defined in Equation (10 45), then

$$S_x(f, t) = R_1(t) \int_{-\infty}^{\infty} R_2(\tau) e^{-j2\pi f\tau} d\tau = R_1(t) S_0(f) \quad (10 58)$$

By normalizing $R_2(\tau)$ so that $R_2(0) = 1$ and folding over the spectral values at negative frequencies, a one-sided time-varying power spectrum is obtained as follows:

$$G_x(f, t) = \Psi_x^2(t) G_x(f) \quad \int_0^{\infty} G_x(f) df = 1 \quad (10 59)$$

The nonstationary spectral density function given by Equation (10 59) can be estimated from a single sample record by first estimating $\Psi_x^2(t)$ with short time averaging operations as outlined in Section 10 1 3, and then estimating $G_x(f)$ by averaging over the entire available record length as would be done for stationary data.

Note that short time averaging operations on individual sample records are often performed to estimate time-varying power spectra of nonstationary data which do not fit the locally stationary model. This is done by segmenting the sample record into a series of very short subrecords, and then computing a power spectrum for each subrecord as if it were stationary. Although an analysis of this type can produce useful qualitative information about the time-varying spectral properties of the data, quantitative applications of the results should be approached with caution. In order to suppress the time interval bias error, a short averaging time (subrecord length) T is required. To obtain the desired spectral decomposition, a narrow resolution bandwidth B_r is required. The result is a relatively small $B_r T$ product and hence, a large random error.

Example 10.3 Illustration of Time Average Power Spectrum Estimate. Consider the structural vibration of a spacecraft during launch as the spacecraft passes through maximum dynamic pressure ($\frac{1}{2}\rho V^2$ where ρ is air density

and V is velocity). The vibration during this phase of the launch is due primarily to pressure fluctuations in the turbulent boundary layer generated by the spacecraft moving through the air. Since both the velocity and altitude of the spacecraft are changing with time, it would be anticipated that the boundary layer pressures and, hence, structural vibration are nonstationary random processes. Assume that the nonstationary random process representing the vibration at any point on the structure is of the locally stationary form. A time varying power spectrum of the vibration can then be calculated from a single sample record by separate time averaging operations as indicated in Equation (10.59). The results of such calculations are presented in Figure 10.7.

Figure 10.7(a) shows the estimate for $\Psi^2(t)$ computed with a 2 second averaging time over a 20 second interval covering the time of maximum dynamic pressure. Note that the mean square vibration level varies by a factor of 4 to 1 over this interval. Figure 10.7(b) shows the estimate for $G_x(f)$ computed by time averaging over the entire 20 second interval. If the

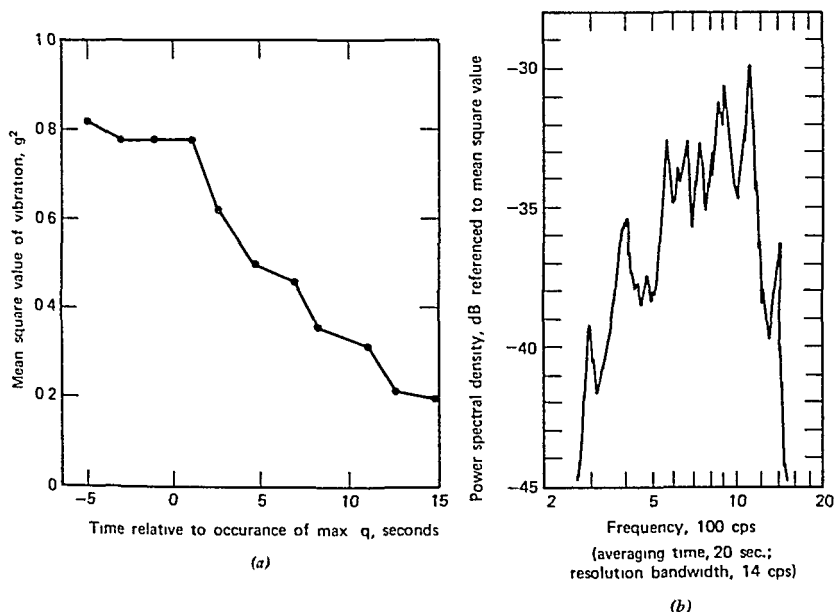


Figure 10.7 Time-varying power spectrum of spacecraft launch vibration data. (a) Time-varying mean square value. (b) Average power spectrum. These data resulted from studies funded by the NASA Goddard Space Flight Center, Greenbelt, Maryland, under Contract NAS 5-4590.

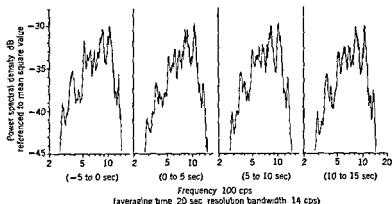


Figure 10.8 Power spectra of spacecraft launch vibration data segments. These data resulted from studies funded by the NASA Goddard Space Flight Center, Greenbelt, Maryland, under Contract NAS 5-4590.

locally stationary assumption is valid for these data, Figure 10.7 provides an acceptable estimate of the time varying power spectrum $G_x(f, t)$. The $B_e T$ product for each portion of the estimate is well over 200 while the time and frequency resolutions are sufficient to avoid unreasonable bias errors.

The suitability of the locally stationary assumption for this problem can be investigated by comparing independent values of $G_x(f)$ computed from individual segments of the 20 second long sample record. Such results are presented in Figure 10.8. Remembering that $B_e T \approx 70$ for each of the estimates ($\epsilon_r \approx 0.12$), it is seen that the results of the individual analyses in Figure 10.8 are not significantly different from the overall result for $G_x(f)$ in Figure 10.7(b).

10.1.6 Input-Output Relations for Nonstationary Data

Consider sample functions from a nonstationary random process $\{x(t)\}$ acting as the input to a constant parameter linear system with weighting function $h(\tau)$ and frequency response function $H(f)$. For an arbitrary input $x(t)$ belonging to $\{x(t)\}$, the output $y(t)$ belonging to $\{y(t)\}$ is given by Equation (2.2) as

$$y(t) = \int_{-\infty}^{\infty} h(\tau) x(t - \tau) d\tau \quad (10.60)$$

It is clear that $\{y(t)\}$ will be a nonstationary random process since its statistical

properties will be a function of t . For a pair of times t_1, t_2 , the product $y(t_1) y(t_2)$ is given by

$$y(t_1) y(t_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(\tau) h(\xi) x(t_1 - \tau) x(t_2 - \xi) d\tau d\xi \quad (10.61)$$

Taking the expected value of Equation (10.61) yields

$$R_y(t_1, t_2) = E[y(t_1) y(t_2)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(\tau) h(\xi) R_x(t_1 - \tau, t_2 - \xi) d\tau d\xi \quad (10.62)$$

Finally, computing the double Fourier transform of Equation (10.62) defines the nonstationary power spectrum of the output as

$$S_y(f_1, f_2) = H^*(f_1) H(f_2) S_x(f_1, f_2) \quad (10.63)$$

where $H^*(f)$ is the complex conjugate of $H(f)$.

For the special case of stationary data where $S_x(f_1, f_2)$ is nonzero only for $f = f_1 = f_2$, Equation (10.63) reduces to the familiar result

$$S_y(f) = |H(f)|^2 S_x(f) \quad (10.64)$$

as previously derived in Section 5.1.

Further interesting input-output relations can be found by examining the nonstationary cross-correlation function and the nonstationary cross-spectral density function between the input process $\{x(t)\}$ and the output process $\{y(t)\}$. From Equation (10.60) the product of $x(t_1)$ by $y(t_2)$ is

$$x(t_1) y(t_2) = \int_{-\infty}^{\infty} h(\tau) x(t_1) x(t_2 - \tau) d\tau \quad (10.65)$$

The taking of expected values now yields

$$R_{xy}(t_1, t_2) = E[x(t_1) y(t_2)] = \int_{-\infty}^{\infty} h(\tau) R_x(t_1, t_2 - \tau) d\tau \quad (10.66)$$

and the double Fourier transform gives

$$S_{xy}(f_1, f_2) = H(f_2) S_x(f_1, f_2) \quad (10.67)$$

Again for the special case of stationary data where $S_x(f_1, f_2)$ is nonzero only for $f = f_1 = f_2$, Equation (10.67) becomes

$$S_{xy}(f) = H(f) S_x(f) \quad (10.68)$$

as previously derived in Section 5.1.

10.2 TRANSIENT (SHOCK) RANDOM PROCESSES

It is often desirable to consider separately a special class of nonstationary random processes which are of short duration and have a clearly defined beginning and end. Examples would include the excitations and system responses associated with short duration earthquakes, automobile collisions, and explosions of all types. Such nonstationary processes are called *transients*, or in mechanical engineering terminology, *shocks*. In some cases, transients can be modeled and analyzed in a deterministic manner, as discussed in Section 1.1.4. In many cases, transients may be too stochastic in character to be modeled by a deterministic function. Nevertheless, they still can be analyzed in a similar way if the results are properly interpreted.

10.2.1 Classification and Analysis of Transient Data

The distinction between transient and other types of nonstationary random processes is purely a practical one. Specifically, consider a nonstationary random process $\{x(t)\}$ which acts as an input (excitation) to a physical system with a weighting function $h(\tau)$. The process $\{x(t)\}$ is considered a transient with respect to the system $h(\tau)$ if $\{x(t)\}$ takes on nonzero values only in the range $(0 \leq t \leq T_1)$ while $h(\tau)$ takes on values significantly different from zero for some τ in the range $(\tau > T_1)$, that is,

$$\begin{aligned} \{x(t)\} &= 0 & \text{for } 0 > t > T_1 \\ h(\tau) &> \Delta & \text{for some } \tau > T_1 \end{aligned} \quad (10.69)$$

where Δ is a small but physically significant nonzero value. Note that the classification of any given nonstationary process as a transient is a function of the physical system associated with the process.

In theory, transient processes can be analyzed as any nonstationary process by the ensemble averaging techniques summarized in Section 10.1. In practice, such techniques are generally required to arrive at estimates of probability density functions, time varying mean and mean square values, and nonstationary correlation functions. However, the spectral structure of transients can be evaluated by procedures somewhat simpler than those outlined in Section 10.1.5, as will now be discussed.

10.2.2 Spectral Structure of Transient Data

Consider a sample record $x_k(t)$ from a transient process $\{x_k(t)\}$ which takes on nonzero values only in the range $(0 \leq t \leq T_1)$. The Fourier transform of $x_k(t)$ is given by

$$X_k(f) = \int_{-\infty}^{\infty} x_k(t) e^{-i2\pi ft} dt = \int_0^{T_1} x_k(t) e^{-i2\pi ft} dt \quad (10.70)$$

which will generally exist since

$$\int_{-\infty}^0 x_k(t) e^{-j2\pi ft} dt = \int_{T_1}^{\infty} x_k(t) e^{-j2\pi ft} dt = 0 \quad (10.71)$$

for transient processes as defined here. The quantity $X_k(f)$ can be expressed in complex polar notation as

$$X_k(f) = |X_k(f)| e^{-j\theta_k(f)} \quad (10.72)$$

where $|X_k(f)|$ is the magnitude of $X_k(f)$ and $\theta_k(f)$ is the argument. The quantity $X_k(f)$ provides a spectral description of $x_k(t)$ called the *Fourier spectrum*. This is the same quantity used to evaluate the spectral properties of deterministic transients, as discussed in Section 1.1.4. For the case of random transients, however, it must be remembered that $X_k(f)$ is a random variable which can take on different values for different sample records from the process $\{x_k(t)\}$.

It is convenient from the viewpoint of both data analysis and engineering applications to define a quantity

$$\begin{aligned} \mathfrak{G}_x(f) &= 2E[|X_k(f)|^2] & f \geq 0 \\ &= 0 & f < 0 \end{aligned} \quad (10.73)$$

where the values at negative frequencies are folded over to obtain a quantity defined for nonnegative frequencies only. The function $\mathfrak{G}_x(f)$ is often called the *energy spectral density function* of the transient process $\{x(t)\}$. A comparison of this quantity to the power spectral density function as defined by Equation (3.101) quickly reveals the strong similarity between $\mathfrak{G}_x(f)$ and $G_x(f)$. The only difference is that $\mathfrak{G}_x(f)$ does not involve a division by the record length T and the limiting operation as $T \rightarrow \infty$. This is true because $|X_k(f)|^2$ for transient data is generally finite and constant for values of $T \geq T_1$. Nevertheless, the two quantities are estimated in a similar way. Specifically, given a sample record $x(t)$ of length $T \geq T_1$ from a transient process $\{x(t)\}$, the energy spectrum can be estimated by

$$\hat{\mathfrak{G}}_x(f) = 2 |X(f)|^2 = T \hat{G}_x(f) \quad (10.74)$$

Hence, the equipment and procedures appropriate for the estimation of power spectra of stationary random data apply directly to the estimation of energy spectra of transient random data. The estimation errors are somewhat more involved for the transient case, but the interpretation of results is otherwise similar.

It follows that a (transient) cross-spectral density function of two transient processes $\{x(t)\}$ and $\{y(t)\}$ may be defined as

$$\mathfrak{G}_{xy}(f) = 2E[X_k^*(f) Y_k(f)] \quad (10.75)$$

where $X_k(f)$ and $Y_k(f)$ are the Fourier transforms of sample functions $x_k(t)$ and $y_k(t)$, respectively. Given two sample records $x(t)$ and $y(t)$ of length $T \geq T_1$, the (transient) cross-spectrum is estimated by

$$\hat{G}_{xy}(f) = 2X^*(f)Y(f) = T\hat{G}_{xy}(f) \quad (10.76)$$

All of the input/output relationships developed in Chapter 5 now apply directly to transient random processes. The statistical error formulations in Sections 6.7 and 6.8, however, do not directly apply. Specifically, consider the random error expressions for frequency response function estimates, as given by Equations (6.146) and (6.151). These equations suggest that the random error decreases as the degrees of freedom n increases. For stationary data, the degrees of freedom $n = 2B_f T$ can always be increased by increasing the record length T for the spectral density estimates. For transient data, this clearly is not the case since the estimated energy spectral density functions of transients (excluding extraneous noise contributions) are independent of the record length assuming $T \geq T_1$. Hence increasing the record length beyond $T \geq T_1$ will not reduce the random error in frequency response function estimates computed from transient data. To the contrary, increasing the record length may increase the random error due to increased contributions of extraneous noise in the calculations for $\hat{G}_x(f)$ and $\hat{G}_{xy}(f)$. The suppression of random errors in frequency response function estimates computed from transient data is best achieved by averaging over an ensemble of estimates computed from a collection of independent transient input-output measurements.

It should be mentioned also that mechanical engineers often employ a special tool for evaluating transient data called the *shock or response spectrum*. In gross terms, the shock spectrum is defined as the maximum response of a linear spring-mass system to a transient motion of the foundation as a function of the natural frequency of the system. The system response is measured in terms of either the strain of the spring or the acceleration of the mass so that the resulting spectrum can be directly associated with damage potential. The shock spectrum concept is really more of an engineering design tool than a strict data analysis technique and, hence, will not be pursued further here. References 28 and 46 contain detailed discussions of this subject.

Example 10.4. Illustration of Energy Spectrum Estimate Consider an automobile traveling at a speed of 30 mph which collides head-on with a second automobile traveling at the same speed in the opposite direction. The automobile is carrying a simulated passenger restrained by a conventional seat belt and shoulder harness in the front passenger seat (right side). Figure 10.9 shows the acceleration time histories measured, (a) on the vehicle frame

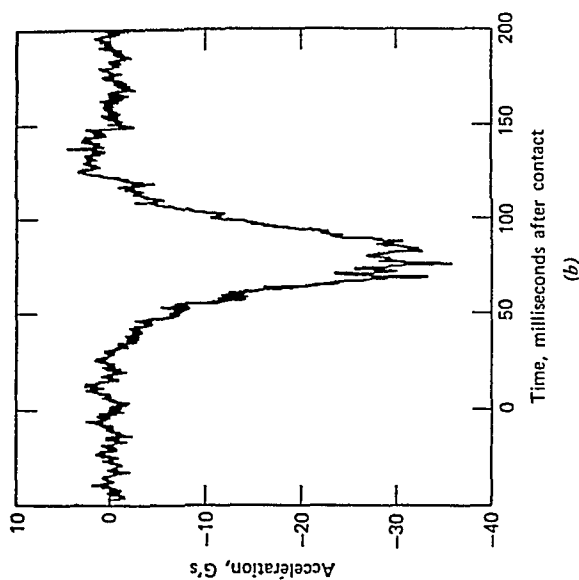
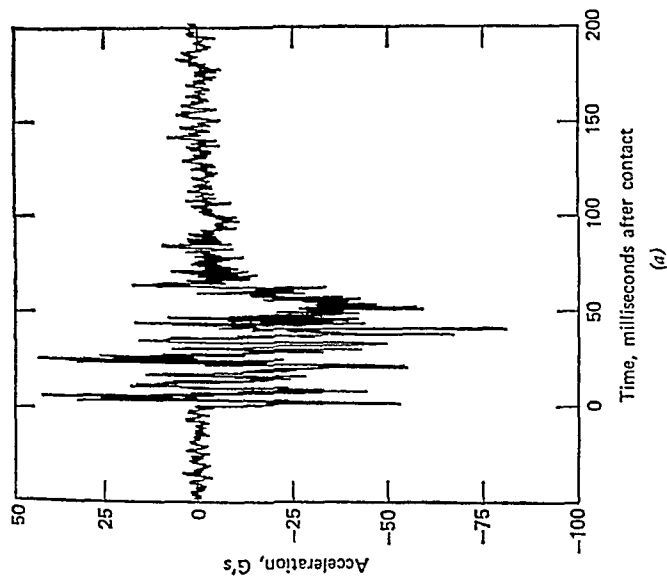


Figure 10.9 Acceleration time histories during automobile collision. (a) Automobile frame. (b) Chest of simulated front seat passenger. These data resulted from studies funded by the FHWA National Highway Safety Bureau, Washington, D.C., under contract FH-11-7218.

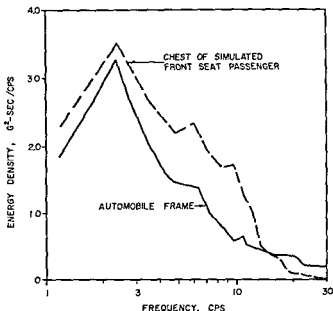


Figure 10.10 Energy density spectra of automobile collision data. These data resulted from studies funded by the FHWA National Highway Safety Bureau, Washington D C, under Contract FH 11 7218

just right of the passenger seat and, (b) in the chest of the simulated passenger. The energy spectra for these time histories are presented in Figure 10.10. These spectra were calculated digitally using FFT procedures on 4096 data points spread over a record length of $T_r = 0.861$ seconds (sampling rate = 4755 sps). No frequency smoothing was employed. Hence, the resolution bandwidth is $B_r = 1.16$ cps.

From Equation (5.8), an acceleration gain factor between the vehicle frame and the simulated passenger's chest is given by $|H(f)| = [\mathcal{G}_p(f)/\mathcal{G}_x(f)]^{1/2}$ where $\mathcal{G}_p(f)$ is the energy spectrum of the passenger acceleration and $\mathcal{G}_x(f)$ is the energy spectrum of the vehicle acceleration. The results of this calculation are presented in Figure 10.11. A comparison of these results with the transmissibility function illustrated in Figure 2.5 shows that the restrained simulated passenger responds to the vehicle impact much like a highly damped spring supported mass with a natural frequency of about 10 cps.

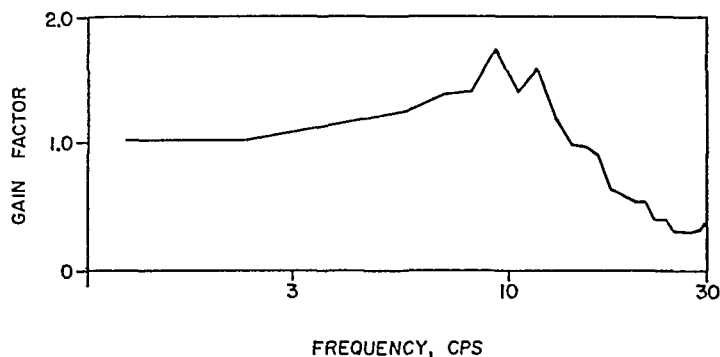


Figure 10.11 Gain factor of simulated passenger response to automobile collision. These data resulted from studies funded by the FHWA National Highway Safety Bureau, Washington, D.C., under Contract FH-11-7218.

10.3 MULTIDIMENSIONAL RANDOM PROCESSES

Analysis formulas and measurement procedures constituting the main body of this text have dealt with physical data such as voltage, pressure, and acceleration quantities which are functions of time alone, or some other single variable which can replace time. However, a number of other problems exist in nature where the random phenomena of interest can be functions of space location as well as time. For example, ocean wave heights in a body of water due to wind effects, pressure fields generated in fluid flow boundary layers, acoustical noise in auditoriums, or the shape of clouds moving across the sky, all generate random records which can vary over both space and time. These situations are typical of the large class of *multidimensional random processes* for which some descriptive ideas and analysis methods are now reviewed.

10.3.1 Vibrating String Problem

To aid the development to follow, consider first the familiar problem of a vibrating string of length L which is fixed at both ends $x = 0$ and $x = L$, and set vibrating from a rest position. The vibration $u(x, t)$ at any location x and time t is determined by solving the wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \quad 0 < x < L \quad t > 0 \quad (10.77)$$

where c is a constant, subject to the boundary conditions

$$u(0, t) = 0 \quad u(L, t) = 0 \quad (10.78)$$

Assume the string has initial values

$$u(x, 0) = f(x) \quad 0 \leq x \leq L \quad (10.79a)$$

$$\frac{\partial u}{\partial t}(x, 0) = 0 \quad 0 \leq x \leq L \quad (10.79b)$$

where $f(x)$ is required only to be an integrable function. The solution to this problem by classical methods of separating variables leads to a finite or infinite series

$$u(x, t) = \sum_{n=1}^{\infty} \alpha_n(x) \beta_n(t) \quad (10.80)$$

where

$$\alpha_n(x) = \sin 2\pi k_n x \quad n = 1, 2, 3, \quad (10.81a)$$

$$\beta_n(t) = b_n \cos 2\pi f_n t \quad n = 1, 2, 3, \quad (10.81b)$$

with

$$k_n = \frac{n}{2L} \quad n = 1, 2, 3, \quad (10.82)$$

and

$$f_n = ck_n \quad n = 1, 2, 3, \quad (10.83)$$

Here, the coefficients b_n can be determined from the Fourier sine series

$$u(x, 0) = \sum_{n=1}^{\infty} b_n \sin 2\pi k_n x = f(x) \quad (10.84)$$

which is periodic of period $2L$, by carrying out the integration

$$b_n = \frac{2}{L} \int_0^L f(x) \sin 2\pi k_n x \, dx \quad n = 1, 2, 3, \quad (10.85)$$

The quantity k_n is called the *wave number* and has similar properties in the wave domain to the properties of f_n in the frequency domain. In particular, observe from Equation (10.82) that k_n is quantized such that

$$\Delta k = k_{n+1} - k_n = k_{\min} = \frac{1}{2L} = \frac{1}{\lambda_{\max}} \quad (10.86)$$

where $\lambda_{\max} = 2L$ is the maximum wavelength in the wave.

From Equation (10.83), the allowable frequencies f_n are also quantized in this application. If T is the time period corresponding to the minimum frequency in the wave, then

$$f_{\min} = \frac{1}{T} = c k_{\min} = \frac{c}{\lambda_{\max}} \quad (10.87)$$

showing that the wavelength $\lambda_{\max} = cT$

10.3.2 Definitions of Fields and Functions

The vibrating string problem illustrates how both a set of discrete frequencies f_n and a set of discrete wave numbers k_n can arise in practice, and that both are needed to describe the detailed properties of records varying in both location and time. Generalization of these ideas will now be carried out for multidimensional random processes $\{u(\mathbf{x}, t)\}$ where $\mathbf{x}$ denotes a space vector which can be of one, two, or three dimensions, and t denotes the time. Let the random records at two different points in some field S be denoted by $u_1(\mathbf{x}, t)$ and $u_2(\mathbf{x}', t')$ as pictured in Figure 10.12. Two types of averages can be

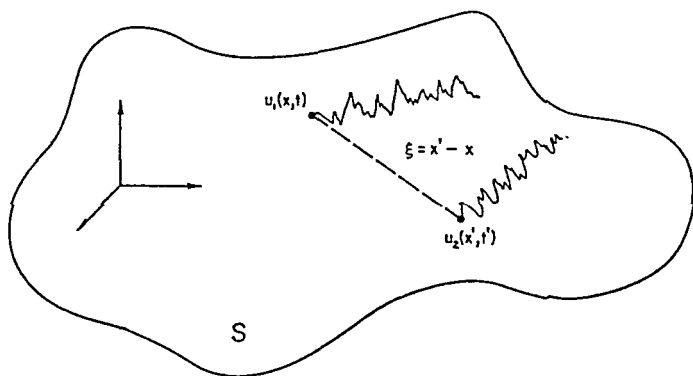


Figure 10.12 Two multidimensional records in a field.

performed over these random records: *time averages* and *spatial averages*. When the field is *stationary*, as previously defined in Section 3.2, time averaged results are independent of the specific times t, t' and a function only of their time separation $\tau = t' - t$. Of course, these results generally still depend upon the specific locations $\mathbf{x}, \mathbf{x}'$. If the field is such that space averaged results are independent of the space locations $\mathbf{x}, \mathbf{x}'$ and a function only of their vector separation $\boldsymbol{\xi} = \mathbf{x}' - \mathbf{x}$, the field is said to be *homogeneous*. If the results further depend only upon the magnitude $|\boldsymbol{\xi}|$, independent of direction, then the field is said to be *isotropic*. For homogenous fields, the results can still depend upon the specific times t, t' . Thus, to obtain time averaged and spatial averaged results which are not dependent upon specific times and specific locations, the field must be both stationary and homogeneous.

For a stationary field, time averaged quantities such as the *temporal* mean

value and *temporal* cross-correlation function can be defined by

$$\mu_u(x) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T u(x, t) dt \quad (10\ 88)$$

$$R_{1\ 2}(x, x', \tau) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T u_1(x, t) u_2(x', t + \tau) dt \quad (10\ 89)$$

For a homogeneous field, corresponding space averaged quantities such as the *spatial* mean value and *spatial* cross-correlation function can be defined by

$$\mu_u(t) = \lim_{S \rightarrow \infty} \frac{1}{S} \int_S u(x, t) dS(x) \quad (10\ 90)$$

$$R_{1\ 2}(\xi, t, t') = \lim_{S \rightarrow \infty} \frac{1}{S} \int_S u_1(x, t) u_2(x + \xi, t') dS(x) \quad (10\ 91)$$

where $dS(x) = dx\ dy\ dz$ or whatever is required to cover the field S . Hence if the field is both stationary and homogeneous, the combined *space-time averaged* mean values and cross-correlation functions are defined by

$$\mu_u = \lim_{\substack{T \rightarrow \infty \\ S \rightarrow \infty}} \frac{1}{2TS} \int_{-T}^T \int_S u(x, t) dS(x) dt \quad (10\ 92)$$

$$R_{1\ 2}(\xi, \tau) = \lim_{\substack{T \rightarrow \infty \\ S \rightarrow \infty}} \frac{1}{2TS} \int_{-T}^T \int_S u_1(x, t) u_2(x + \xi, t + \tau) dS(x) dt \quad (10\ 93)$$

Space time averaged mean square values represent a special case of Equation (10 93) when $u_1 = u_2$ and also $\xi = \tau = 0$

Example 10.5. Cross-Correlation Function of Moving Harmonic Wave. Consider a moving harmonic wave in a stationary homogeneous field which travels only in the positive x direction with constant velocity v . This can be expressed by the formula

$$u(x, t) = A \sin 2\pi k_0(x - vt) \quad (10\ 94)$$

where the wave number k_0 and frequency f_0 satisfy $f_0 = vk_0$, the amplitude A is a constant, and the wavelength $\lambda = (1/k_0)$. For this wave, the time averaged cross-correlation function, from Equation (10 89), is

$$\begin{aligned} R_u(x, x', \tau) &= \lim_{T \rightarrow \infty} \frac{A^2}{2T} \int_{-T}^T \sin 2\pi k_0(x - vt) \sin 2\pi k_0[x' - v(t + \tau)] dt \\ &= \frac{A^2}{2} \cos 2\pi k_0[(x' - x) - v\tau] \end{aligned}$$

Hence, letting $\xi = x' - x$,

$$R_u(x, x', \tau) = R_u(\xi, \tau) = \frac{A^2}{2} \cos 2\pi k_0(\xi - v\tau)$$

This gives the space-time averaged cross-correlation function corresponding to Equation (10.93). Note that for fixed ξ , the peak occurs at $\tau = (\xi/v)$. The possible shape of this space-time cross-correlation function as it moves in the x direction is shown in Figure 10.13. For actual data, there would usually be a change in shape and reduction in amplitude as ξ increases, that is, as the two points become more separated. This figure concludes Example 10.5.

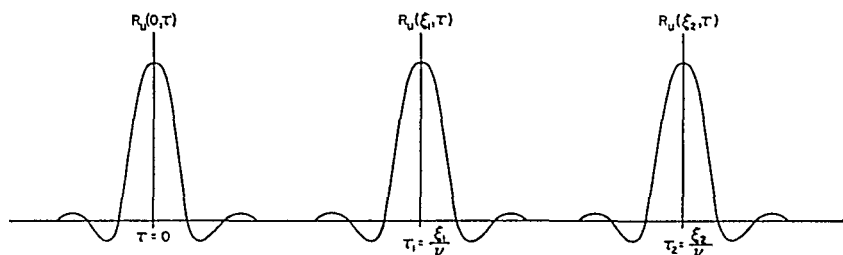


Figure 10.13 Space-time cross-correlation function.

By taking Fourier transforms of these correlation functions with respect to the time separation τ or the space separation ξ , holding the other variable constant, two distinct spectra can be obtained, one in frequency and the other in wave number. In equation form, the *frequency spectrum* is defined by

$$S(\xi, f) = \int_{-\infty}^{\infty} R(\xi, \tau) e^{-j2\pi f\tau} d\tau \quad (10.95)$$

while the *wave number spectrum* is defined by

$$S(k, \tau) = \int_{-\infty}^{\infty} R(\xi, \tau) e^{-j2\pi k \cdot \xi} d\xi \quad (10.96)$$

where the dot product

$$\mathbf{k} \cdot \boldsymbol{\xi} = k_1\xi_1 + k_2\xi_2 + k_3\xi_3 \quad (10.97)$$

when $d\boldsymbol{\xi} = d\xi_1 d\xi_2 d\xi_3$, with fewer terms for one and two dimensions. The wave number spectrum in Equation (10.96) is a function of the wave numbers $\mathbf{k}$, holding τ fixed. This spectrum describes certain important wave properties of a record varying in space, such as the cross section of a rough road.

A combined wave number-frequency spectrum is expressed by

$$S(\mathbf{k}, f) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} R(\xi, \tau) e^{-j2\pi(\mathbf{k} \cdot \xi + f\tau)} d\xi d\tau \quad (10.98)$$

and its inverse by

$$R(\xi, \tau) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} S(\mathbf{k}, f) e^{j2\pi(\mathbf{k} \cdot \xi + f\tau)} d\mathbf{k} df \quad (10.99)$$

Observe that Equation (10.98) can be obtained by taking the wave number Fourier transform of Equation (10.95), or by taking the frequency Fourier transform of Equation (10.96). Also the quantity $R(\xi, \tau)$ in Equation (10.99) can be found by taking appropriate inverse Fourier transforms of Equations (10.95) or (10.96). The reason for using any particular representation should be for the convenience of a desired application since the same information is merely being viewed in different ways.

10.3.3 Measurement Considerations

Consider the measurement problem of where to locate transducers so as to obtain data which will describe a two-dimensional isotropic field, as shown in Figure 10.14. Practical problems in collecting spatial data are much more difficult, in general, than the previously treated problems dealing only with time data. Where one transducer might be sufficient to give useful time data, now each point in space requires its own transducer and other support equipment. The cost for implementing and operating N transducers for one minute, for example, is usually considerably more expensive than operating one transducer for N minutes. Thus it is essential that transducers be located so as to cover the anticipated wave numbers, but not be too numerous so as to provide redundant information.

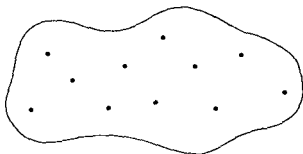


Figure 10.14 Transducers in a spatial region

If ΔL is the separation interval between transducers, then the Nyquist maximum cutoff wave number is given by

$$k_{\max} = \frac{1}{2\Delta L} = \frac{1}{\lambda_{\min}} \quad (10.100)$$

where $\lambda_{\min}$ is the smallest resolvable wavelength in the wave. Wave numbers above $k_{\max}$ are "aliased" so as to appear to be in the range $(0, k_{\max})$. This is the highest resolvable wave number that can be obtained by sampling the random isotropic field with transducers spaced ΔL apart. In practice, it is recommended that ΔL be selected so that there will be at least 2.5 transducers per highest wave number of interest. This is equivalent to selecting ΔL so that there will be at least 2.5 transducers per smallest wavelength of interest.

The physical size and shape of the transducer is also of concern in making measurements [Ref. 50]. Clearly, actual data collected at any point must be the result of averaging the data over some finite region around the point. Information at small wavelengths (large wave numbers) cannot be resolved with measuring devices that are comparable in size to the desired wavelengths. To show how to evaluate possible measurement errors due to transducer properties, consider the one-dimensional case where

$u(s, t)$ = the true multidimensional random process at any point s and time t in a stationary homogeneous field

$K(x, s) = K(x - s)$ = the response kernel of the transducer

$m(x, t)$ = the measured multidimensional random process at point x and time t .

Then, the measured process relative to the true process can be expressed by an input-output equation of the form

$$\begin{aligned} m(x, t) &= \int_{-\infty}^{\infty} K(x - s) u(s, t) ds \\ &= \int_{-\infty}^{\infty} K(\alpha) u(x - \alpha, t) d\alpha \end{aligned} \quad (10.101)$$

The second equation follows from the first by a simple substitution $\alpha = x - s$, replacing ds by $-d\alpha$, and changing the limits of integration as required. Similarly,

$$\begin{aligned} m(x + \xi, t + \tau) &= \int_{-\infty}^{\infty} K(\beta) u(x + \xi - \beta, t + \tau) d\beta \\ &= \int_{-\infty}^{\infty} K(\alpha + \gamma) u(x + \xi - \alpha - \gamma, t + \tau) d\gamma \end{aligned} \quad (10.102)$$

Then, the measured multidimensional correlation function $R_m(\xi, \tau)$, corresponding to Equation (10.99) will here be given by

$$R_m(\xi, \tau) = E[m(x, t) m(x + \xi, t + \tau)] \\ = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} K(\alpha) K(\alpha + \gamma) R_u(\xi - \gamma, \tau) d\alpha d\gamma \quad (10.103)$$

where $R_u(\xi - \gamma, \tau)$ is the true multidimensional correlation function at the indicated parameters. To simplify this equation, let

$$A(\gamma) = \int_{-\infty}^{\infty} K(\alpha) K(\alpha + \gamma) d\alpha \quad (10.104)$$

This represents a basic autocorrelation type of property of the transducer characteristics. Now, Equation (10.103) becomes a convolution integral

$$R_m(\xi, \tau) = \int_{-\infty}^{\infty} A(\gamma) R_u(\xi - \gamma, \tau) d\gamma \quad (10.105)$$

and the Fourier transform of both sides yields

$$S_m(k, f) = S_A(k) S_u(k, f) \quad (10.106)$$

The quantity $S_A(k)$ is the wave number spectrum of $A(\gamma)$ computed by

$$S_A(k) = \int_{-\infty}^{\infty} A(\gamma) e^{-i2\pi k\gamma} d\gamma \quad (10.107)$$

and represents a correction factor to be applied to the measured data. Substitution of Equation (10.104) into Equation (10.107) also readily proves that

$$S_A(k) = \left| \int_{-\infty}^{\infty} K(\alpha) e^{-i2\pi k\alpha} d\alpha \right|^2 \quad (10.108)$$

providing a more direct way to compute this result. The quantities $S_m(k, f)$ and $S_u(k, f)$ are the combined wave number-frequency spectra of $R_m(\xi, \tau)$ and $R_u(\xi, \tau)$, respectively, as computed by Equation (10.98). Thus, from Equation (10.106), by knowing the measured spectral properties and the transducer characteristics, the true spectral properties can be determined by the relation

$$S_u(k, f) = \frac{S_m(k, f)}{S_A(k)} \quad (10.109)$$

assuming $S_A(k)$ is different from zero.

Example 10.5. Wave Number Corrections for Spatial Linear Averaging. Consider the measurement of a plane wave $u(x, t)$ of arbitrary form by using

some device which performs a spatial linear average over a distance d . Here, the measured signal $m(x, t)$ is

$$m(x, t) = \frac{1}{d} \int_0^d u(s, t) ds$$

where x represents some point along the line $[0, d]$. For this example, the response kernel is

$$K(\alpha) = \frac{1}{d} \quad 0 \leq \alpha \leq d$$

$$= 0 \quad \text{otherwise}$$

From Equation (10.108), the correction factor is

$$S_A(k) = \frac{1}{d^2} \left| \int_0^d e^{-j2\pi k\alpha} d\alpha \right|^2 = \frac{\sin^2 \pi kd}{(\pi kd)^2} \quad (10.110)$$

Substitution in Equation (10.109) now shows how measured spectral results should be modified to yield the desired true spectral results. It indicates also that this device should not be used for values of $k = (n/d)$ where n is an integer.

PROBLEMS

1. Suppose an ensemble of N independent nonstationary records $\{x_i(t)\}$ has both a nonstationary mean value and a nonstationary mean square value. Show how to represent an underlying (possible stationary) ensemble of records $\{z_i(t)\}$ where $\mu_z(t) = 0$ and $\Psi_z^2(t) = 1$.
2. Using the Tchebycheff Inequality, prove that for any t and any constant $c > 0$,

$$\text{Prob} \left[|\hat{\mu}_x(t) - \mu_x(t)| \leq c \frac{\sigma_x(t)}{\sqrt{N}} \right] \geq 1 - \frac{1}{c^2}$$

where $\hat{\mu}_x(t)$, $\mu_x(t)$, and $\sigma_x(t)$ are defined in Equations (10.10), (10.12), and (10.14), respectively. Interpret this result for the case when $c = 3$ and $N = 9$, and compare with the result obtained by assuming a Gaussian sampling distribution of $\hat{\mu}_x(t)$.

3. Carry out the derivation of Equations (10.14) and (10.29). Discuss how these results would change if the N sample records are (a) correlated or (b) completely dependent.
4. Consider a nonstationary process as defined by $\{x(t)\} = A(t)\{y(t)\}$ where $A(t) = \cos 2\pi f_0 t$ and $\{y(t)\}$ is stationary with zero mean and unit variance.

Compute the following quantities

- (a) $R_x(t_1, t_2)$ as per Equation (10.36)
- (b) $R_x(\tau, t)$ as per Equation (10.44)
- 5 For the quantities in Problem 4, compute
 - (a) $S_x(f_1, f_2)$ as per Equation (10.48)
 - (b) $S_x(f, t)$ as per Equation (10.55)
- 6 Derive Equations (10.63) and (10.67)
- 7 Assume a transient physical process is measured in terms of force as a function of time. Determine the relationship between the energy spectral density function of the measured transient and the momentum of the physical process which it describes
- 8 For the moving harmonic wave in Example 10.5, compute
 - (a) $S(\xi, f)$ as per Equation (10.95)
 - (b) $S(k, \tau)$ as per Equation (10.96)
 - (c) $S(k, f)$ as per Equation (10.98)
- 9 Derive Equations (10.108) and (10.110)
- 10 Discuss the bias and variability errors expected to occur in estimates of the combined wave number-frequency spectrum of a one dimensional space-time random process $\{u(x, t)\}$ if $0 \leq x \leq L$ and $0 \leq t \leq T$

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APPENDIX

STATISTICAL TABLES

Table A.1

Ordinates of the Standardized Normal Density Function

$$p(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.3989	0.3989	0.3989	0.3988	0.3986	0.3986	0.3982	0.3980	0.3977	0.3973
0.1	0.3970	0.3966	0.3961	0.3956	0.3951	0.3945	0.3939	0.3932	0.3925	0.3918
0.2	0.3910	0.3902	0.3894	0.3884	0.3876	0.3867	0.3857	0.3847	0.3836	0.3825
0.3	0.3814	0.3802	0.3790	0.3778	0.3765	0.3752	0.3739	0.3725	0.3712	0.3697
0.4	0.3683	0.3668	0.3653	0.3637	0.3621	0.3605	0.3589	0.3572	0.3555	0.3538
0.5	0.3521	0.3503	0.3485	0.3467	0.3448	0.3429	0.3410	0.3391	0.3372	0.3352
0.6	0.3332	0.3312	0.3292	0.3271	0.3251	0.3230	0.3209	0.3187	0.3166	0.3144
0.7	0.3123	0.3101	0.3079	0.3056	0.3034	0.3011	0.2989	0.2966	0.2943	0.2920
0.8	0.2897	0.2874	0.2850	0.2827	0.2803	0.2780	0.2756	0.2732	0.2709	0.2685
0.9	0.2661	0.2637	0.2613	0.2589	0.2565	0.2541	0.2516	0.2492	0.2468	0.2444
1.0	0.2420	0.2396	0.2371	0.2347	0.2323	0.2299	0.2275	0.2251	0.2227	0.2203
1.1	0.2179	0.2155	0.2131	0.2107	0.2083	0.2059	0.2036	0.2012	0.1989	0.1965
1.2	0.1942	0.1919	0.1895	0.1872	0.1849	0.1826	0.1804	0.1781	0.1758	0.1736
1.3	0.1714	0.1691	0.1669	0.1647	0.1626	0.1605	0.1582	0.1561	0.1539	0.1518
1.4	0.1497	0.1476	0.1456	0.1435	0.1415	0.1394	0.1374	0.1354	0.1334	0.1315
1.5	0.1295	0.1276	0.1257	0.1238	0.1219	0.1200	0.1282	0.1163	0.1145	0.1127
1.6	0.1109	0.1092	0.1074	0.1057	0.1040	0.1023	0.1006	0.0989	0.0973	0.0957
1.7	0.0940	0.0925	0.0909	0.0893	0.0878	0.0863	0.0848	0.0833	0.0818	0.0804
1.8	0.0790	0.0775	0.0761	0.0748	0.0734	0.0721	0.0707	0.0694	0.0681	0.0669
1.9	0.0656	0.0644	0.0632	0.0620	0.0608	0.0596	0.0584	0.0573	0.0562	0.0551
2.0	0.0540	0.0529	0.0519	0.0508	0.0498	0.0488	0.0478	0.0468	0.0459	0.0449
2.1	0.0440	0.0431	0.0422	0.0413	0.0404	0.0396	0.0387	0.0379	0.0371	0.0363
2.2	0.0355	0.0347	0.0339	0.0332	0.0325	0.0317	0.0310	0.0303	0.0297	0.0290
2.3	0.0283	0.0277	0.0270	0.0264	0.0258	0.0252	0.0246	0.0241	0.0235	0.0229
2.4	0.0224	0.0219	0.0213	0.0208	0.0203	0.0198	0.0194	0.0189	0.0184	0.0180
2.5	0.0175	0.0171	0.0167	0.0163	0.0158	0.0154	0.0151	0.0147	0.0143	0.0139
2.6	0.0136	0.0132	0.0129	0.0126	0.0122	0.0119	0.0116	0.0113	0.0110	0.0107
2.7	0.0104	0.0101	0.0099	0.0096	0.0093	0.0091	0.0088	0.0086	0.0084	0.0081
2.8	0.0079	0.0077	0.0075	0.0073	0.0071	0.0069	0.0067	0.0065	0.0063	0.0061
2.9	0.0060	0.0058	0.0056	0.0055	0.0053	0.0051	0.0050	0.0048	0.0047	0.0046

Table A.1 (continued)

z	0 00	0 01	0 02	0 03	0 04	0 05	0 06	0 07	0 08	0 09
3 0	0 0044	0 0043	0 0042	0 0040	0 0039	0 0038	0 0037	0 0036	0 0035	0 0034
3 1	0 0033	0 0032	0 0031	0 0030	0 0029	0 0028	0 0027	0 0026	0 0025	0 0025
3 2	0 0024	0 0023	0 0022	0 0022	0 0021	0 0020	0 0020	0 0019	0 0018	0 0018
3 3	0 0017	0 0017	0 0016	0 0016	0 0015	0 0015	0 0014	0 0014	0 0013	0 0013
3 4	0 0012	0 0012	0 0012	0 0011	0 0011	0 0010	0 0010	0 0010	0 0009	0 0009
3 5	0 0009	0 0008	0 0008	0 0008	0 0008	0 0007	0 0007	0 0007	0 0007	0 0006
3 6	0 0006	0 0006	0 0006	0 0005	0 0005	0 0005	0 0005	0 0005	0 0005	0 0004
3 7	0 0004	0 0004	0 0004	0 0004	0 0004	0 0004	0 0003	0 0003	0 0003	0 0003
3 8	0 0003	0 0003	0 0003	0 0003	0 0003	0 0002	0 0002	0 0002	0 0002	0 0002
3 9	0 0002	0 0002	0 0002	0 0002	0 0002	0 0002	0 0002	0 0002	0 0001	0 0001

Table A.2
Areas under Standardized Normal Density Function

$$\text{Value of } \alpha = \int_{z_\alpha}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} = \text{Prob}[z > z_\alpha]$$

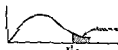


z_α	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
2.3	0.0107	0.0104	0.0102	0.00990	0.00964	0.00939	0.00914	0.00889	0.00866	0.00842
2.4	0.00820	0.00798	0.00776	0.00755	0.00734	0.00714	0.00695	0.00676	0.00657	0.00639
2.5	0.00621	0.00604	0.00587	0.00570	0.00554	0.00539	0.00523	0.00508	0.00494	0.00480
2.6	0.00466	0.00453	0.00440	0.00427	0.00415	0.00402	0.00391	0.00379	0.00368	0.00357
2.7	0.00347	0.00336	0.00326	0.00317	0.00307	0.00298	0.00289	0.00280	0.00272	0.00264
2.8	0.00256	0.00248	0.00240	0.00233	0.00226	0.00219	0.00212	0.00205	0.00199	0.00193
2.9	0.00187	0.00181	0.00175	0.00169	0.00164	0.00159	0.00154	0.00149	0.00144	0.00139

The data in Tables A.2 thru A.6 are extracted from *Handbook of Statistical Tables* by Donald B. Owen with the permission of the publisher, Addison-Wesley Publishing Company, Reading, Mass.

Table A.3
Percentage Points of Chi Square Distribution

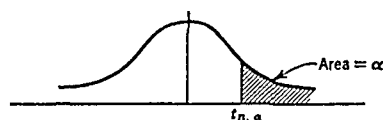
Value of $\chi^2_{\alpha, n}$ such that $\text{Prob}[\chi^2_n > \chi^2_{\alpha, n}] = \alpha$



α	n									
α	0.995	0.990	0.975	0.950	0.900	0.10	0.05	0.025	0.010	0.005
1	0.000039	0.00016	0.00098	0.0039	0.0158	2.71	3.84	5.02	6.63	7.88
2	0.0100	0.0201	0.0506	0.103	0.211	4.61	5.99	7.38	9.21	10.60
3	0.0717	0.113	0.216	0.352	0.594	6.25	7.81	9.35	11.34	12.84
4	0.207	0.297	0.484	0.711	1.06	7.78	9.49	11.14	13.28	14.86
5	0.412	0.554	0.831	1.15	1.61	9.24	11.07	12.83	15.09	16.75
6	0.676	0.872	1.24	1.64	2.20	10.64	12.59	14.45	16.81	18.55
7	0.989	1.24	1.69	2.17	2.83	12.02	14.07	16.01	18.48	20.28
8	1.34	1.65	2.18	2.73	3.49	13.36	15.51	17.53	20.09	21.96
9	1.73	2.09	2.70	3.33	4.17	14.68	16.92	19.02	21.67	23.59
10	2.16	2.56	3.25	3.94	4.87	15.99	18.31	20.48	23.21	25.19
11	2.60	3.05	3.82	4.57	5.58	17.28	19.68	21.92	24.73	26.76
12	3.07	3.57	4.40	5.23	6.30	18.55	21.03	23.34	26.22	28.30
13	3.57	4.11	5.01	5.89	7.04	19.81	22.36	24.74	27.69	29.82
14	4.07	4.66	5.63	6.57	7.79	21.06	23.68	26.12	29.14	31.32
15	4.60	5.23	6.26	7.26	8.55	22.31	25.00	27.49	30.58	32.80
16	5.14	5.81	6.91	7.96	9.31	23.54	26.30	28.85	32.00	34.27
17	5.70	6.41	7.56	8.67	10.08	24.77	27.59	30.19	33.41	35.72
18	6.26	7.01	8.23	9.39	10.86	25.99	28.87	31.53	34.81	37.16
19	6.84	7.63	8.91	10.12	11.65	27.20	30.14	32.85	36.19	38.58
20	7.43	8.26	9.59	10.85	12.44	28.41	31.41	34.17	37.57	40.00
21	8.03	8.90	10.28	11.59	13.24	29.62	32.67	35.48	38.93	41.40
22	8.64	9.54	10.98	12.34	14.04	30.81	33.92	36.78	40.29	42.80
23	9.26	10.20	11.69	13.09	14.85	32.01	35.17	38.08	41.64	44.18
24	9.89	10.86	12.40	13.85	15.66	33.20	36.42	39.36	42.98	45.56
25	10.52	11.52	13.12	14.61	16.47	34.38	37.65	40.65	44.31	46.93
26	11.16	12.20	13.84	15.38	17.29	35.56	38.88	41.92	45.64	48.29
27	11.81	12.88	14.57	16.15	18.11	36.74	40.11	43.19	46.96	49.64
28	12.46	13.56	15.31	16.93	18.94	37.92	41.34	44.46	48.28	50.99
29	13.12	14.26	16.05	17.71	19.77	39.09	42.56	45.72	49.59	52.34
30	13.79	14.95	16.79	18.49	20.60	40.26	43.77	46.96	50.89	53.67
40	20.71	22.16	24.43	26.51	29.05	51.81	55.76	59.34	63.69	66.77
60	35.53	37.48	40.48	43.19	46.46	74.40	79.08	81.30	88.58	91.95
120	83.85	86.92	91.58	95.70	100.62	140.23	146.57	152.21	158.95	163.85

For $n > 120$, $\chi^2_{\alpha, n} \approx n \left[1 - \frac{2}{9n} + z_{\alpha} \sqrt{\frac{2}{9n}} \right]^3$ where z_{α} is the desired percentage point for a standard normal distribution.

Table A.4

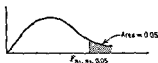
Percentage Points of Student t DistributionValue of $t_{n;\alpha}$ such that $\text{Prob}[t_n > t_{n;\alpha}] = \alpha$ 

n	α				
	0.10	0.050	0.025	0.010	0.005
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845
21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819
23	1.319	1.714	2.069	2.500	2.807
24	1.318	1.711	2.064	2.492	2.797
25	1.316	1.708	2.060	2.485	2.787
26	1.315	1.706	2.056	2.479	2.779
27	1.314	1.703	2.052	2.473	2.771
28	1.313	1.701	2.048	2.467	2.763
29	1.311	1.699	2.045	2.462	2.756
30	1.310	1.697	2.042	2.457	2.750
40	1.303	1.684	2.021	2.423	2.704
60	1.296	1.671	2.000	2.390	2.660
120	1.289	1.658	1.980	2.358	2.617

$\alpha = 0.995, 0.990, 0.975, 0.950$, and 0.900 follow from $t_{n;1-\alpha} = -t_{n;\alpha}$

Table A.5(a)
Percentage Points of F Distribution

Values of $F_{n_1, n_2, \alpha}$ such that $\text{Prob}(F_{n_1, n_2} > F_{n_1, n_2, \alpha}) = \alpha$



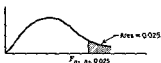
$n_1 \backslash n_2$	1	2	3	4	5	6	7	8	9	10	11	12	13	14	16
1	61	200	216	225	230	234	237	239	241	242	243	244	245	245	246
2	18.5	19.0	19.2	19.3	19.3	19.3	19.4	19.4	19.4	19.4	19.4	19.4	19.4	19.4	19.4
3	10.1	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.76	8.74	8.73	8.71	8.69
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.94	5.91	5.89	5.87	5.84
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.73	4.70	4.68	4.66	4.64	4.60
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.03	4.00	3.98	3.96	3.92
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.60	3.57	3.55	3.53	3.49
8	5.31	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.31	3.28	3.26	3.24	3.20
9	5.12	4.26	3.87	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.10	3.07	3.05	3.03	2.99
10	4.96	4.10	3.71	3.47	3.33	3.22	3.14	3.07	3.02	2.98	2.94	2.91	2.89	2.86	2.83
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.82	2.79	2.76	2.74	2.70
12	4.73	3.87	3.48	3.25	3.11	3.00	2.91	2.85	2.80	2.75	2.72	2.69	2.66	2.64	2.60
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.63	2.60	2.58	2.55	2.51
14	4.60	3.74	3.35	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.57	2.53	2.51	2.48	2.44
16	4.49	3.63	3.24	3.01	2.85	2.74	2.65	2.59	2.54	2.49	2.46	2.42	2.40	2.37	2.33
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.37	2.34	2.31	2.29	2.25
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.31	2.28	2.25	2.22	2.18
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.26	2.23	2.20	2.17	2.13
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.21	2.18	2.15	2.13	2.09
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.18	2.15	2.12	2.09	2.05
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.15	2.12	2.09	2.06	2.02
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.13	2.09	2.06	2.04	1.99
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.04	2.00	1.97	1.95	1.90
50	4.03	3.18	2.79	2.56	2.40	2.29	2.20	2.13	2.07	2.03	1.99	1.95	1.92	1.89	1.85
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.95	1.92	1.89	1.86	1.82
80	3.96	3.11	2.72	2.49	2.33	2.21	2.13	2.06	2.00	1.95	1.91	1.88	1.84	1.82	1.77
100	3.94	3.09	2.70	2.46	2.31	2.19	2.10	2.03	1.97	1.93	1.89	1.85	1.82	1.79	1.75
200	3.89	3.04	2.65	2.42	2.26	2.14	2.06	1.98	1.93	1.88	1.84	1.80	1.77	1.74	1.69
500	3.86	3.01	2.62	2.39	2.23	2.12	2.03	1.96	1.90	1.85	1.81	1.77	1.74	1.71	1.66
∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.79	1.75	1.72	1.69	1.64

Table A.5(a) (continued)

18	20	22	24	26	28	30	40	50	60	80	100	200	500	α	n_1/n_2
247	248	249	249	249	250	250	251	252	252	252	253	254	254	254	1
19.4	19.5	19.5	19.5	19.5	19.5	19.5	19.5	19.5	19.5	19.5	19.5	19.5	19.5	19.5	2
8.67	8.66	8.65	8.64	8.63	8.62	8.62	8.59	8.59	8.57	8.56	8.55	8.54	8.53	8.53	3
5.82	5.80	5.79	5.77	5.76	5.75	5.75	5.72	5.70	5.69	5.67	5.66	5.65	5.64	5.63	4
4.58	3.56	4.54	4.53	4.52	4.50	4.50	4.46	4.44	4.43	4.41	4.41	4.39	4.37	4.37	5
3.90	3.87	3.86	3.84	3.83	3.82	3.81	3.77	3.75	3.74	3.72	3.71	3.69	3.68	3.67	6
3.47	3.44	3.43	3.41	3.40	3.39	3.38	3.34	3.32	3.30	3.29	3.27	3.25	3.24	3.23	7
3.17	3.15	3.13	3.12	3.10	3.09	3.08	3.04	3.02	3.01	2.99	2.97	2.95	2.94	2.93	8
2.96	2.94	2.92	2.90	2.89	2.87	2.86	2.83	2.80	2.79	2.77	2.76	2.73	2.72	2.71	9
2.80	2.77	2.75	2.74	2.72	2.71	2.70	2.66	2.64	2.62	2.60	2.59	2.56	2.55	2.54	10
2.67	2.65	2.63	2.61	2.59	2.58	2.57	2.53	2.51	2.49	2.47	2.46	2.43	2.42	2.40	11
2.57	2.54	2.52	2.51	2.49	2.48	2.47	2.43	2.40	2.38	2.36	2.35	2.32	2.31	2.30	12
2.48	2.46	2.44	2.42	2.41	2.39	2.38	2.34	2.31	2.30	2.27	2.26	2.23	2.22	2.21	13
2.41	2.38	2.37	2.35	2.33	2.32	2.31	2.27	2.24	2.22	2.20	2.19	2.16	2.14	2.13	14
2.30	2.28	2.25	2.24	2.22	2.21	2.19	2.15	2.12	2.11	2.08	2.07	2.04	2.02	2.01	16
2.22	2.19	2.17	2.15	2.13	2.12	2.11	2.06	2.04	2.02	1.99	1.98	1.95	1.93	1.92	18
2.15	2.12	2.10	2.08	2.07	2.05	2.04	1.99	1.97	1.95	1.92	1.91	1.88	1.86	1.84	20
2.10	2.07	2.05	2.03	2.01	2.00	1.98	1.94	1.91	1.89	1.86	1.85	1.82	1.80	1.78	22
2.05	2.03	2.00	1.98	1.97	1.95	1.94	1.89	1.86	1.84	1.82	1.80	1.77	1.75	1.73	24
2.02	1.99	1.97	1.95	1.93	1.91	1.90	1.84	1.82	1.80	1.78	1.76	1.73	1.71	1.69	26
1.99	1.96	1.93	1.91	1.90	1.88	1.87	1.82	1.79	1.77	1.74	1.73	1.69	1.67	1.65	28
1.96	1.93	1.91	1.89	1.87	1.85	1.84	1.79	1.76	1.74	1.71	1.70	1.66	1.64	1.62	30
1.87	1.84	1.81	1.79	1.77	1.76	1.74	1.69	1.66	1.64	1.61	1.59	1.55	1.53	1.51	40
1.81	1.78	1.76	1.74	1.72	1.70	1.69	1.63	1.60	1.58	1.54	1.52	1.48	1.46	1.44	50
1.78	1.75	1.72	1.70	1.68	1.66	1.65	1.59	1.56	1.53	1.50	1.48	1.44	1.41	1.39	60
1.73	1.70	1.68	1.65	1.63	1.62	1.60	1.54	1.51	1.48	1.45	1.43	1.38	1.35	1.32	80
1.71	1.68	1.65	1.63	1.61	1.59	1.57	1.52	1.48	1.45	1.41	1.39	1.34	1.31	1.28	100
1.66	1.62	1.60	1.57	1.55	1.53	1.52	1.46	1.41	1.39	1.35	1.32	1.26	1.22	1.19	200
1.62	1.59	1.56	1.54	1.52	1.50	1.48	1.42	1.38	1.34	1.30	1.28	1.21	1.16	1.11	500
1.60	1.57	1.54	1.52	1.50	1.48	1.46	1.39	1.35	1.32	1.27	1.24	1.17	1.11	1.00	∞

Table A 5(b)
Percentage Points of F Distribution

Values of $F_{\alpha_1, n_1, n_2, 0.025}$ such that $\text{Prob}[F_{\alpha_1, n_1, n_2} > F_{\alpha_1, n_1, n_2, 0.025}] = 0.025$



$n_2 \backslash n_1$	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	648	800	854	900	9 2	937	948	957	963	969	973	977	980	983	987
2	18.5	19.0	19.2	19.2	19.3	19.3	19.4	19.4	19.4	19.4	19.4	19.4	19.4	19.4	19.4
3	17.4	18.0	18.4	18.6	18.9	19.0	19.1	19.2	19.2	19.3	19.3	19.3	19.3	19.3	19.3
4	16.7	17.2	17.6	17.8	18.1	18.2	18.3	18.4	18.4	18.5	18.5	18.5	18.5	18.5	18.5
5	16.2	16.6	17.0	17.2	17.5	17.6	17.7	17.8	17.8	17.9	17.9	17.9	17.9	17.9	17.9
6	15.8	16.2	16.6	16.8	17.1	17.2	17.3	17.4	17.4	17.5	17.5	17.5	17.5	17.5	17.5
7	15.5	15.9	16.3	16.5	16.8	16.9	17.0	17.1	17.1	17.2	17.2	17.2	17.2	17.2	17.2
8	15.3	15.7	16.1	16.3	16.6	16.7	16.8	16.9	16.9	17.0	17.0	17.0	17.0	17.0	17.0
9	15.1	15.5	15.9	16.1	16.4	16.5	16.6	16.7	16.7	16.8	16.8	16.8	16.8	16.8	16.8
10	15.0	15.4	15.8	16.0	16.3	16.4	16.5	16.6	16.6	16.7	16.7	16.7	16.7	16.7	16.7
11	14.9	15.3	15.7	15.9	16.2	16.3	16.4	16.5	16.5	16.6	16.6	16.6	16.6	16.6	16.6
12	14.8	15.2	15.6	15.8	16.1	16.2	16.3	16.4	16.4	16.5	16.5	16.5	16.5	16.5	16.5
13	14.7	15.1	15.5	15.7	16.0	16.1	16.2	16.3	16.3	16.4	16.4	16.4	16.4	16.4	16.4
14	14.6	15.0	15.4	15.6	15.9	16.0	16.1	16.2	16.2	16.3	16.3	16.3	16.3	16.3	16.3
15	14.5	14.9	15.3	15.5	15.8	15.9	16.0	16.1	16.1	16.2	16.2	16.2	16.2	16.2	16.2
16	14.4	14.8	15.2	15.4	15.7	15.8	15.9	16.0	16.0	16.1	16.1	16.1	16.1	16.1	16.1
17	14.3	14.7	15.1	15.3	15.6	15.7	15.8	15.9	15.9	16.0	16.0	16.0	16.0	16.0	16.0
18	14.2	14.6	15.0	15.2	15.5	15.6	15.7	15.8	15.8	15.9	15.9	15.9	15.9	15.9	15.9
19	14.1	14.5	14.9	15.1	15.4	15.5	15.6	15.7	15.7	15.8	15.8	15.8	15.8	15.8	15.8
20	14.0	14.4	14.8	15.0	15.3	15.4	15.5	15.6	15.6	15.7	15.7	15.7	15.7	15.7	15.7
21	13.9	14.3	14.7	14.9	15.2	15.3	15.4	15.5	15.5	15.6	15.6	15.6	15.6	15.6	15.6
22	13.8	14.2	14.6	14.8	15.1	15.2	15.3	15.4	15.4	15.5	15.5	15.5	15.5	15.5	15.5
23	13.7	14.1	14.5	14.7	15.0	15.1	15.2	15.3	15.3	15.4	15.4	15.4	15.4	15.4	15.4
24	13.6	14.0	14.4	14.6	14.9	15.0	15.1	15.2	15.2	15.3	15.3	15.3	15.3	15.3	15.3
25	13.5	13.9	14.3	14.5	14.8	14.9	15.0	15.1	15.1	15.2	15.2	15.2	15.2	15.2	15.2
26	13.4	13.8	14.2	14.4	14.7	14.8	14.9	15.0	15.0	15.1	15.1	15.1	15.1	15.1	15.1
27	13.3	13.7	14.1	14.3	14.6	14.7	14.8	14.9	14.9	15.0	15.0	15.0	15.0	15.0	15.0
28	13.2	13.6	14.0	14.2	14.5	14.6	14.7	14.8	14.8	14.9	14.9	14.9	14.9	14.9	14.9
29	13.1	13.5	13.9	14.1	14.4	14.5	14.6	14.7	14.7	14.8	14.8	14.8	14.8	14.8	14.8
30	13.0	13.4	13.8	14.0	14.3	14.4	14.5	14.6	14.6	14.7	14.7	14.7	14.7	14.7	14.7
40	12.6	13.0	13.4	13.6	13.9	14.0	14.1	14.2	14.2	14.3	14.3	14.3	14.3	14.3	14.3
50	12.3	12.7	13.1	13.3	13.6	13.7	13.8	13.9	13.9	14.0	14.0	14.0	14.0	14.0	14.0
60	12.1	12.5	12.9	13.1	13.4	13.5	13.6	13.7	13.7	13.8	13.8	13.8	13.8	13.8	13.8
80	11.8	12.2	12.6	12.8	13.1	13.2	13.3	13.4	13.4	13.5	13.5	13.5	13.5	13.5	13.5
100	11.6	12.0	12.4	12.6	12.9	13.0	13.1	13.2	13.2	13.3	13.3	13.3	13.3	13.3	13.3
200	11.2	11.6	12.0	12.2	12.5	12.6	12.7	12.8	12.8	12.9	12.9	12.9	12.9	12.9	12.9
500	10.8	11.2	11.6	11.8	12.1	12.2	12.3	12.4	12.4	12.5	12.5	12.5	12.5	12.5	12.5
∞	10.6	11.0	11.4	11.6	11.9	12.0	12.1	12.2	12.2	12.3	12.3	12.3	12.3	12.3	12.3

Table A.5(b) (continued)

18	20	22	24	26	28	30	40	50	60	80	100	200	500	∞	n_1/n_2
990	993	995	997	999	1000	1001	1006	1008	1010	1012	1013	1016	1017	1018	1
39.4	39.4	39.5	39.5	39.5	39.5	39.5	39.5	39.5	39.5	39.5	39.5	39.5	39.5	39.5	2
14.2	14.2	14.1	14.1	14.1	14.1	14.1	14.1	14.0	14.0	14.0	14.0	13.9	13.9	13.9	3
8.60	8.56	8.53	8.51	8.49	8.48	8.46	8.41	8.38	8.36	8.33	8.32	8.29	8.27	8.26	4
6.37	6.33	6.30	6.28	6.26	6.24	6.23	6.18	6.14	6.12	6.10	6.08	6.05	6.03	6.01	5
5.21	5.17	5.14	5.12	5.10	5.08	5.07	5.01	4.98	4.96	4.93	4.92	4.88	4.86	4.85	6
4.50	4.47	4.44	4.42	4.39	4.38	4.36	4.31	4.28	4.25	4.23	4.21	4.18	4.16	4.14	7
4.03	4.00	3.97	3.95	3.93	3.91	3.89	3.84	3.81	3.78	3.76	3.74	3.70	3.68	3.67	8
3.70	3.67	3.64	3.61	3.59	3.58	3.56	3.51	3.47	3.45	3.42	3.40	3.37	3.35	3.33	9
3.45	3.42	3.39	3.37	3.34	3.33	3.31	3.26	3.22	3.20	3.17	3.15	3.12	3.09	3.08	10
3.26	3.23	3.20	3.17	3.15	3.13	3.12	3.06	3.03	3.00	2.97	2.96	2.92	2.90	2.88	11
3.11	3.07	3.04	3.02	3.00	2.98	2.96	2.91	2.87	2.85	2.82	2.80	2.76	2.74	2.72	12
2.98	2.95	2.92	2.89	2.87	2.85	2.84	2.78	2.74	2.72	2.69	2.67	2.63	2.61	2.60	13
2.88	2.84	2.81	2.79	2.77	2.75	2.73	2.67	2.64	2.61	2.58	2.56	2.53	2.50	2.49	14
2.72	2.68	2.65	2.63	2.60	2.58	2.57	2.51	2.47	2.45	2.42	2.40	2.36	2.33	2.32	16
2.60	2.56	2.53	2.50	2.48	2.46	2.44	2.38	2.35	2.32	2.29	2.27	2.23	2.20	2.19	18
2.50	2.46	2.43	2.41	2.39	2.37	2.35	2.29	2.25	2.22	2.19	2.17	2.13	2.10	2.09	20
2.43	2.39	2.36	2.33	2.31	2.29	2.27	2.21	2.17	2.14	2.11	2.09	2.05	2.02	2.00	22
2.36	2.33	2.30	2.27	2.25	2.23	2.21	2.15	2.11	2.08	2.05	2.02	1.98	1.95	1.94	24
2.31	2.28	2.24	2.22	2.19	2.17	2.16	2.09	2.05	2.03	1.99	1.97	1.92	1.90	1.88	26
2.27	2.23	2.20	2.17	2.15	2.13	2.11	2.05	2.01	1.98	1.94	1.92	1.88	1.85	1.83	28
2.23	2.20	2.16	2.14	2.11	2.09	2.07	2.01	1.97	1.94	1.90	1.88	1.84	1.81	1.79	30
2.11	2.07	2.03	2.01	1.98	1.96	1.94	1.88	1.83	1.80	1.76	1.74	1.69	1.66	1.64	40
2.03	1.99	1.96	1.93	1.91	1.88	1.87	1.80	1.75	1.72	1.68	1.66	1.60	1.57	1.55	50
1.98	1.94	1.91	1.88	1.86	1.83	1.82	1.74	1.70	1.67	1.62	1.60	1.54	1.51	1.48	60
1.93	1.88	1.85	1.82	1.79	1.77	1.75	1.68	1.63	1.60	1.55	1.53	1.47	1.43	1.40	80
1.89	1.85	1.81	1.78	1.76	1.74	1.71	1.64	1.59	1.56	1.51	1.48	1.42	1.38	1.35	100
1.82	1.78	1.74	1.71	1.68	1.66	1.64	1.56	1.51	1.47	1.42	1.39	1.32	1.27	1.23	200
1.78	1.74	1.70	1.67	1.64	1.62	1.60	1.51	1.46	1.42	1.37	1.34	1.25	1.19	1.14	500
1.75	1.71	1.67	1.64	1.61	1.59	1.57	1.48	1.43	1.39	1.33	1.30	1.21	1.13	1.00	∞

Table A.5(c)
Percentage Points of F Distribution

Values of $F_{\alpha; n_1, n_2}$ such that $\text{Prob}\{F_{n_1, n_2} > F_{\alpha; n_1, n_2}\} = \alpha$



$n_1 \backslash n_2$	1	2	3	4	5	6	7	8	9	10	11	12	13	14	16
*1	405	500	540	563	576	586	593	598	602	606	608	611	613	614	617
2	98.5	99.0	99.2	99.2	99.3	99.3	99.4	99.4	99.4	99.4	99.4	99.4	99.4	99.4	99.4
3	34.1	30.8	29.5	28.7	28.2	27.9	27.7	27.5	27.3	27.2	27.1	27.1	27.0	26.9	26.8
4	21.2	18.0	16.7	16.0	15.5	15.2	15.0	14.8	14.7	14.5	14.4	14.4	14.3	14.2	14.2
5	16.3	13.3	12.1	11.4	11.0	10.7	10.5	10.3	10.2	10.1	9.95	9.89	9.82	9.77	9.68
6	13.7	10.9	9.78	9.15	8.75	8.47	8.26	8.10	7.98	7.87	7.79	7.72	7.66	7.60	7.52
7	12.2	9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72	6.62	6.54	6.47	6.41	6.36	6.27
8	11.3	8.65	7.59	7.01	6.63	6.37	6.16	6.03	5.91	5.81	5.73	5.67	5.61	5.56	5.48
9	10.6	8.02	6.99	6.42	6.06	5.80	5.61	5.47	5.35	5.26	5.18	5.11	5.05	5.00	4.92
10	10.0	7.56	6.55	5.99	5.64	5.39	5.20	5.06	4.94	4.85	4.77	4.71	4.65	4.60	4.52
11	9.65	7.21	6.22	5.67	5.32	5.07	4.89	4.74	4.63	4.54	4.46	4.40	4.34	4.29	4.21
12	9.33	6.93	5.95	5.41	5.06	4.82	4.64	4.50	4.39	4.30	4.22	4.16	4.10	4.05	3.97
13	9.07	6.70	5.74	5.21	4.86	4.62	4.44	4.30	4.19	4.10	4.02	3.96	3.91	3.86	3.78
14	8.86	6.51	5.56	5.04	4.70	4.46	4.28	4.14	4.03	3.94	3.86	3.80	3.75	3.70	3.62
16	8.53	6.23	5.29	4.77	4.44	4.20	4.03	3.89	3.78	3.69	3.62	3.55	3.50	3.45	3.37
18	8.29	6.01	5.09	4.58	4.25	4.01	3.84	3.71	3.60	3.51	3.43	3.37	3.32	3.27	3.19
20	8.10	5.85	4.94	4.43	4.10	3.87	3.70	3.56	3.46	3.37	3.29	3.23	3.18	3.13	3.05
22	7.95	5.72	4.82	4.31	3.99	3.76	3.59	3.45	3.35	3.26	3.18	3.12	3.07	3.02	2.94
24	7.81	5.61	4.72	4.22	3.90	3.67	3.50	3.36	3.26	3.17	3.09	3.03	2.98	2.93	2.85
26	7.72	5.53	4.64	4.14	3.82	3.59	3.42	3.29	3.18	3.09	3.02	2.96	2.90	2.86	2.78
28	7.64	5.45	4.57	4.07	3.75	3.53	3.36	3.23	3.12	3.03	2.96	2.90	2.84	2.79	2.72
30	7.56	5.39	4.51	4.02	3.70	3.47	3.30	3.17	3.07	2.98	2.91	2.84	2.79	2.74	2.66
40	7.31	5.18	4.31	3.83	3.51	3.29	3.12	2.99	2.89	2.80	2.73	2.66	2.61	2.56	2.48
50	7.17	5.06	4.20	3.72	3.41	3.19	3.02	2.89	2.79	2.70	2.63	2.56	2.51	2.46	2.38
60	7.08	4.98	4.13	3.65	3.34	3.12	2.95	2.82	2.72	2.63	2.56	2.50	2.44	2.39	2.31
80	6.96	4.88	4.04	3.56	3.26	3.04	2.87	2.74	2.64	2.55	2.48	2.42	2.36	2.31	2.23
100	6.90	4.82	3.98	3.51	3.21	2.99	2.82	2.69	2.59	2.50	2.43	2.37	2.31	2.26	2.19
200	6.76	4.71	3.88	3.41	3.11	2.89	2.73	2.60	2.50	2.41	2.34	2.27	2.22	2.17	2.09
500	6.69	4.65	3.82	3.36	3.05	2.84	2.68	2.55	2.44	2.36	2.28	2.22	2.17	2.12	2.04
∞	6.65	4.61	3.78	3.32	3.02	2.80	2.64	2.51	2.41	2.32	2.25	2.18	2.13	2.08	2.00

Table A.5(c) (continued)

18	20	22	24	26	28	30	40	50	60	80	100	200	500	∞	$\frac{n_1}{n_2}$
619	621	622	623	624	625	626	629	630	631	633	633	635	636	637	1
99.4	99.4	99.5	99.5	99.5	99.5	99.5	99.5	99.5	99.5	99.5	99.5	99.5	99.5	99.5	2
26.8	26.7	26.6	26.6	26.6	26.5	26.5	26.4	26.4	26.3	26.3	26.2	26.2	26.1	26.1	3
14.1	14.0	14.0	13.9	13.9	13.9	13.8	13.7	13.7	13.7	13.6	13.6	13.5	13.5	13.5	4
9.61	9.55	9.51	9.47	9.43	9.40	9.38	9.29	9.24	9.20	9.16	9.13	9.08	9.04	9.02	5
7.45	7.40	7.35	7.31	7.28	7.25	7.23	7.14	7.09	7.06	7.01	6.99	6.93	6.90	6.88	6
6.21	6.16	6.11	6.07	6.04	6.02	5.99	5.91	5.86	5.82	5.78	5.75	5.70	5.67	5.65	7
5.41	5.36	5.32	5.28	5.25	5.22	5.20	5.12	5.07	5.03	4.99	4.96	4.91	4.88	4.85	8
4.86	4.81	4.77	4.73	4.70	4.67	4.65	4.57	4.52	4.48	4.44	4.42	4.36	4.33	4.31	9
4.46	4.41	4.36	4.33	4.30	4.27	4.25	4.17	4.12	4.08	4.04	4.01	3.96	3.93	3.91	10
4.15	4.10	4.06	4.02	3.99	3.96	3.94	3.86	3.81	3.78	3.73	3.71	3.66	3.62	3.60	11
3.91	3.86	3.82	3.78	3.75	3.72	3.70	3.62	3.57	3.54	3.49	3.47	3.41	3.38	3.36	12
3.72	3.66	3.62	3.59	3.56	3.53	3.51	3.43	3.38	3.34	3.30	3.27	3.22	3.19	3.16	13
3.56	3.51	3.46	3.43	3.40	3.37	3.35	3.27	3.22	3.18	3.14	3.11	3.06	3.03	3.00	14
3.31	3.26	3.22	3.18	3.15	3.12	3.10	3.02	2.97	2.93	2.89	2.86	2.81	2.78	2.75	16
3.13	3.08	3.03	3.00	2.97	2.94	2.92	2.84	2.78	2.75	2.70	2.68	2.62	2.59	2.57	18
2.99	2.94	2.90	2.86	2.83	2.80	2.78	2.69	2.64	2.61	2.56	2.54	2.48	2.44	2.42	20
2.88	2.83	2.78	2.75	2.72	2.69	2.67	2.58	2.53	2.50	2.45	2.42	2.36	2.33	2.31	22
2.79	2.74	2.70	2.66	2.63	2.60	2.58	2.49	2.44	2.40	2.36	2.33	2.27	2.24	2.21	24
2.72	2.66	2.62	2.58	2.55	2.53	2.50	2.42	2.36	2.33	2.28	2.25	2.19	2.16	2.13	26
2.65	2.60	2.56	2.52	2.49	2.46	2.44	2.35	2.30	2.26	2.22	2.19	2.13	2.09	2.06	28
2.60	2.55	2.51	2.47	2.44	2.41	2.39	2.30	2.25	2.21	2.16	2.13	2.07	2.03	2.01	30
2.42	2.37	2.33	2.29	2.26	2.23	2.20	2.11	2.06	2.02	1.97	1.94	1.87	1.83	1.80	40
2.32	2.27	2.22	2.18	2.15	2.12	2.10	2.01	1.95	1.91	1.86	1.82	1.76	1.71	1.68	50
2.25	2.20	2.15	2.12	2.08	2.05	2.03	1.94	1.88	1.84	1.78	1.75	1.68	1.63	1.60	60
2.17	2.12	2.07	2.03	2.00	1.97	1.94	1.85	1.79	1.75	1.69	1.66	1.58	1.53	1.49	80
2.12	2.07	2.02	1.98	1.94	1.92	1.89	1.80	1.73	1.69	1.63	1.60	1.52	1.47	1.43	100
2.02	1.97	1.93	1.89	1.85	1.82	1.79	1.69	1.63	1.58	1.52	1.48	1.39	1.33	1.28	200
1.97	1.92	1.87	1.83	1.79	1.76	1.74	1.63	1.56	1.52	1.45	1.41	1.31	1.23	1.16	500
1.93	1.88	1.83	1.79	1.76	1.72	1.70	1.59	1.52	1.47	1.40	1.36	1.25	1.15	1.00	∞

* Multiply the number of the first row ($n_2 = 1$) by 10.

Table A 6

Percentage Points of Run Distribution

Values of $r_{n, \alpha}$ such that $\text{Prob} \{r_n > r_{n, \alpha}\} = \alpha$ where $n = N_1 = N_2 = N/2$

$n = N/2$	α					
	0.99	0.975	0.95	0.05	0.025	0.01
5	2	2	3	8	9	9
6	2	3	3	10	10	11
7	3	3	4	11	12	12
8	4	4	5	12	13	13
9	4	5	6	13	14	15
10	5	6	6	15	15	16
11	6	7	7	16	16	17
12	7	7	8	17	18	18
13	7	8	9	18	19	20
14	8	9	10	19	20	21
15	9	10	11	20	21	22
16	10	11	11	22	22	23
18	11	12	13	24	25	26
20	13	14	15	26	27	28
25	17	18	19	32	33	34
30	21	22	24	37	39	40
35	25	27	28	43	44	46
40	30	31	33	48	50	51
45	34	36	37	54	55	57
50	38	40	42	59	61	63
55	43	45	46	65	66	68
60	47	49	51	70	72	74
65	52	54	56	75	77	79
70	56	58	60	81	83	85
75	61	63	65	86	88	90
80	65	68	70	91	93	96
85	70	72	74	97	99	101
90	74	77	79	102	104	107
95	79	82	84	107	109	112
100	84	86	88	113	115	117

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